

COMPENDIUM AMC 12

2008 – 2025

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Toomates Coolección vol. 44



Toomates Colección

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Índice.

2008	4	
2009	18	
2010	34	
2011	48	
2012	62	
2013	78	
2014	93	
2015	108	
2016	123	
2017	138	
2018	AMC12A Enunciados en inglés	154
	AMC12A Enunciados en español	283
	AMC12A Soluciones (Letra)	289
	AMC12A Soluciones desarrolladas	290
	AMC12B Enunciados en inglés	160
	AMC12B Enunciados en español	303
	AMC12B Soluciones (Letra)	310
	AMC12B Soluciones desarrolladas	311
2019	Enunciados en inglés (web)	168
	AMC12A Enunciados en español	238
	AMC12A Soluciones (Letra)	243
	AMC12A Soluciones desarrolladas	244
	AMC12B Enunciados en español	260
	AMC12B Soluciones (Letra)	266
	AMC12B Soluciones desarrolladas	267
2020	AMC12A Enunciados en inglés (web)	184
	AMC12A Enunciados en español	198
	AMC12A Soluciones (Letra)	203
	AMC12A Soluciones desarrolladas	204
	AMC12B Enunciados en español	219
	AMC12B Soluciones (Letra)	225
	AMC12B Soluciones desarrolladas	226
2021	AMC 12A Spring Enunciados	324
	Soluciones (Letra)	330
	AMC 12B Spring Enunciados	331
	Soluciones (Letra)	337
2021	AMC 12A Fall Enunciados	338
	Soluciones (Letra)	344
	AMC 12B Fall Enunciados	345
	Soluciones (Letra)	352
2022	AMC 12A Enunciados	353
	Soluciones (Letra)	364
	AMC 12B Enunciados	365
	Soluciones (Letra)	376
2023	AMC12A Enunciados en inglés (web)	377
	AMC12A Enunciados en español	382
	AMC12A Soluciones (Letra)	388
	AMC12A Soluciones desarrolladas en español	389
	AMC12A Links a los libros Toomates Coolección	403
	AMC12A Enunciados en inglés (web)	405
	AMC12B Enunciados en español	411
	AMC12B Soluciones (Letra)	417
	AMC12B Soluciones desarrolladas en español	418
	AMC12B Links a los libros Toomates Coolección	434
2024	AMC12A Enunciados	435
	Soluciones (Letra)	443
	AMC 12B Enunciados	444
	Soluciones (Letra)	452
2025	AMC12A Enunciados	453
	Soluciones (Letra)	462
	AMC 12B Enunciados	463
	Soluciones (Letra)	471

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2008 AMC 12A Problems

2008 AMC 12A (Answer Key)

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Instructions

1. This is a 25-question, multiple choice test. Each question is followed by answers marked A, B, C, D and E. Only one of these is correct.
2. You will receive 6 points for each correct answer, 2.5 points for each problem left unanswered if the year is before 2006, 1.5 points for each problem left unanswered if the year is after 2006, and 0 points for each incorrect answer.
3. No aids are permitted other than scratch paper, graph paper, ruler, compass, protractor and erasers (and calculators that are accepted for use on the test if before 2006. No problems on the test will *require* the use of a calculator).
4. Figures are not necessarily drawn to scale.
5. You will have **75 minutes** working time to complete the test.

1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19
• 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

A bakery owner turns on his doughnut machine at 8:30 AM. At 11:10 AM the machine has completed one third of the day's job. At what time will the doughnut machine complete the job?

- (A) 1:50 PM (B) 3:00 PM (C) 3:30 PM (D) 4:30 PM (E) 5:50 PM

Solution

Problem 2

What is the reciprocal of $\frac{1}{2} + \frac{2}{3}$?

- (A) $\frac{6}{7}$ (B) $\frac{7}{6}$ (C) $\frac{5}{3}$ (D) 3 (E) $\frac{7}{2}$

Solution

Problem 3

Suppose that $\frac{2}{3}$ of 10 bananas are worth as much as 8 oranges. How many oranges are worth as much as $\frac{1}{2}$ of 5 bananas?

- (A) 2 (B) $\frac{5}{2}$ (C) 3 (D) $\frac{7}{2}$ (E) 4

Solution

Problem 4

Which of the following is equal to the product

$$\frac{8}{4} \cdot \frac{12}{8} \cdot \frac{16}{12} \cdots \frac{4n+4}{4n} \cdots \frac{2008}{2004}?$$

- (A) 251 (B) 502 (C) 1004 (D) 2008 (E) 4016

Solution

Problem 5

Suppose that

$$\frac{2x}{3} - \frac{x}{6}$$

is an integer. Which of the following statements must be true about x ?

- (A) It is negative. (B) It is even, but not necessarily a multiple of 3.
 (C) It is a multiple of 3, but not necessarily even.
 (D) It is a multiple of 6, but not necessarily a multiple of 12.
 (E) It is a multiple of 12.

Solution

Problem 6

Heather compares the price of a new computer at two different stores. Store A offers 15% off the sticker price followed by a \$90 rebate, and store B offers 25% off the same sticker price with no rebate. Heather saves \$15 by buying the computer at store A instead of store B . What is the sticker price of the computer, in dollars?

- (A) 750 (B) 900 (C) 1000 (D) 1050 (E) 1500

Solution

Problem 7

While Steve and LeRoy are fishing 1 mile from shore, their boat springs a leak, and water comes in at a constant rate of 10 gallons per minute. The boat will sink if it takes in more than 30 gallons of water. Steve starts rowing toward the shore at a constant rate of 4 miles per hour while LeRoy bails water out of the boat. What is the slowest rate, in gallons per minute, at which LeRoy can bail if they are to reach the shore without sinking?

- (A) 2 (B) 4 (C) 6 (D) 8 (E) 10

Solution

Problem 8

What is the volume of a cube whose surface area is twice that of a cube with volume 1?

- (A) $\sqrt{2}$ (B) 2 (C) $2\sqrt{2}$ (D) 4 (E) 8

Solution

Problem 9

Older television screens have an aspect ratio of $4 : 3$. That is, the ratio of the width to the height is $4 : 3$. The aspect ratio of many movies is not $4 : 3$, so they are sometimes shown on a television screen by "letterboxing" - darkening strips of equal height at the top and bottom of the screen, as shown. Suppose a movie has an aspect ratio of $2 : 1$ and is shown on an older television screen with a 27-inch diagonal. What is the height, in inches, of each darkened strip?



- (A) 2 (B) 2.25 (C) 2.5 (D) 2.7 (E) 3

Solution

Problem 10

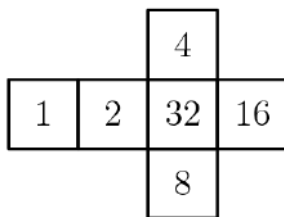
Doug can paint a room in 5 hours. Dave can paint the same room in 7 hours. Doug and Dave paint the room together and take a one-hour break for lunch. Let t be the total time, in hours, required for them to complete the job working together, including lunch. Which of the following equations is satisfied by t ?

- (A) $\left(\frac{1}{5} + \frac{1}{7}\right)(t + 1) = 1$ (B) $\left(\frac{1}{5} + \frac{1}{7}\right)t + 1 = 1$ (C) $\left(\frac{1}{5} + \frac{1}{7}\right)t = 1$
 (D) $\left(\frac{1}{5} + \frac{1}{7}\right)(t - 1) = 1$ (E) $(5 + 7)t = 1$

Solution

Problem 11

Three cubes are each formed from the pattern shown. They are then stacked on a table one on top of another so that the 13 visible numbers have the greatest possible sum. What is that sum?



- (A) 154 (B) 159 (C) 164 (D) 167 (E) 189

Solution

Problem 12

A function f has domain $[0, 2]$ and range $[0, 1]$ (The notation $[a, b]$ denotes $\{x : a \leq x \leq b\}$.) What are the domain and range, respectively, of the function g defined by $g(x) = 1 - f(x + 1)$?

- (A) $[-1, 1], [-1, 0]$ (B) $[-1, 1], [0, 1]$ (C) $[0, 2], [-1, 0]$ (D) $[1, 3], [-1, 0]$ (E) $[1, 3], [0, 1]$

Solution

Problem 13

Points A and B lie on a circle centered at O , and $\angle AOB = 60^\circ$. A second circle is internally tangent to the first and tangent to both $\overline{OA}$ and $\overline{OB}$. What is the ratio of the area of the smaller circle to that of the larger circle?

- (A) $\frac{1}{16}$ (B) $\frac{1}{9}$ (C) $\frac{1}{8}$ (D) $\frac{1}{6}$ (E) $\frac{1}{4}$

Solution

Problem 14

What is the area of the region defined by the inequality $|3x - 18| + |2y + 7| \leq 3$?

- (A) 3 (B) $\frac{7}{2}$ (C) 4 (D) $\frac{9}{2}$ (E) 5

Solution

Problem 15

Let $k = 2008^2 + 2^{2008}$. What is the units digit of $k^2 + 2^k$?

- (A) 0 (B) 2 (C) 4 (D) 6 (E) 8

Solution

Problem 16

The numbers $\log(a^3b^7)$, $\log(a^5b^{12})$, and $\log(a^8b^{15})$ are the first three terms of an arithmetic sequence, and the 12^{th} term of the sequence is $\log(b^n)$. What is n ?

- (A) 40 (B) 56 (C) 76 (D) 112 (E) 143

Solution

Problem 17

Let $a_1, a_2, \dots$ be a sequence determined by the rule $a_n = a_{n-1}/2$ if a_{n-1} is even and $a_n = 3a_{n-1} + 1$ if a_{n-1} is odd. For how many positive integers $a_1 \leq 2008$ is it true that a_1 is less than each of a_2, a_3 , and a_4 ?

- (A) 250 (B) 251 (C) 501 (D) 502 (E) 1004

Solution

Problem 18

A triangle $\triangle ABC$ with sides 5, 6, 7 is placed in the three-dimensional plane with one vertex on the positive x axis, one on the positive y axis, and one on the positive z axis. Let O be the origin. What is the volume of $OABC$?

- (A) $\sqrt{85}$ (B) $\sqrt{90}$ (C) $\sqrt{95}$ (D) 10 (E) $\sqrt{105}$

Solution

Problem 19

In the expansion of

$$(1 + x + x^2 + \dots + x^{27})(1 + x + x^2 + \dots + x^{14})^2,$$

what is the coefficient of x^{28} ?

- (A) 195 (B) 196 (C) 224 (D) 378 (E) 405

Solution

Problem 20

Triangle ABC has $AC = 3$, $BC = 4$, and $AB = 5$. Point D is on $\overline{AB}$, and $\overline{CD}$ bisects the right angle. The inscribed circles of $\triangle ADC$ and $\triangle BCD$ have radii r_a and r_b respectively. What is r_a/r_b ?

- (A) $\frac{1}{28}(10 - \sqrt{2})$ (B) $\frac{3}{56}(10 - \sqrt{2})$ (C) $\frac{1}{14}(10 - \sqrt{2})$ (D) $\frac{5}{56}(10 - \sqrt{2})$
 (E) $\frac{3}{28}(10 - \sqrt{2})$

Solution

Problem 21

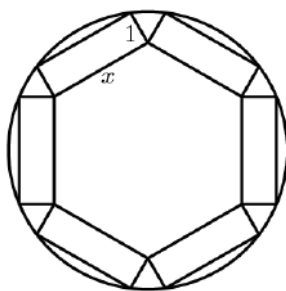
A permutation $(a_1, a_2, a_3, a_4, a_5)$ of $(1, 2, 3, 4, 5)$ is heavy-tailed if $a_1 + a_2 < a_4 + a_5$. What is the number of heavy-tailed permutations?

- (A) 36 (B) 40 (C) 44 (D) 48 (E) 52

Solution

Problem 22

A round table has radius 4. Six rectangular place mats are placed on the table. Each place mat has width 1 and length x as shown. They are positioned so that each mat has two corners on the edge of the table, these two corners being end points of the same side of length x . Further, the mats are positioned so that the inner corners each touch an inner corner of an adjacent mat. What is x ?



- (A) $2\sqrt{5} - \sqrt{3}$ (B) 3 (C) $\frac{3\sqrt{7} - \sqrt{3}}{2}$ (D) $2\sqrt{3}$ (E) $\frac{5 + 2\sqrt{3}}{2}$

Solution

Problem 23

The solutions of the equation $z^4 + 4z^3i - 6z^2 - 4zi - i = 0$ are the vertices of a convex polygon in the complex plane. What is the area of the polygon?

- (A) $2^{\frac{5}{8}}$ (B) $2^{\frac{3}{4}}$ (C) 2 (D) $2^{\frac{5}{4}}$ (E) $2^{\frac{3}{2}}$

Solution

Problem 24

Triangle ABC has $\angle C = 60^\circ$ and $BC = 4$. Point D is the midpoint of BC . What is the largest possible value of $\tan \angle BAD$?

- (A) $\frac{\sqrt{3}}{6}$ (B) $\frac{\sqrt{3}}{3}$ (C) $\frac{\sqrt{3}}{2\sqrt{2}}$ (D) $\frac{\sqrt{3}}{4\sqrt{2} - 3}$ (E) 1

Solution

Problem 25

A sequence $(a_1, b_1), (a_2, b_2), (a_3, b_3), \dots$ of points in the coordinate plane satisfies

$$(a_{n+1}, b_{n+1}) = (\sqrt{3}a_n - b_n, \sqrt{3}b_n + a_n) \text{ for } n = 1, 2, 3, \dots$$

Suppose that $(a_{100}, b_{100}) = (2, 4)$. What is $a_1 + b_1$?

- (A) $-\frac{1}{2^{97}}$ (B) $-\frac{1}{2^{99}}$ (C) 0 (D) $\frac{1}{2^{98}}$ (E) $\frac{1}{2^{96}}$

Solution

See also

2008 AMC 12A (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/Forum/resources.php?cid=182&cid=44&year=2008))	
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1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
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1. D
2. A
3. C
4. B
5. B
6. A
7. D
8. C
9. D
10. D
11. C
12. B
13. B
14. A
15. D
16. D
17. D
18. C
19. C
20. E
21. D
22. C
23. D
24. D
25. D

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2008 AMC 12B Problems

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

A basketball player made 5 baskets during a game. Each basket was worth either 2 or 3 points. How many different numbers could represent the total points scored by the player?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Solution

Problem 2

A 4×4 block of calendar dates is shown. The order of the numbers in the second row is to be reversed. Then the order of the numbers in the fourth row is to be reversed. Finally, the numbers on each diagonal are to be added. What will be the positive difference between the two diagonal sums?

1	2	3	4
8	9	10	11
15	16	17	18
22	23	24	25

- (A) 2 (B) 4 (C) 6 (D) 8 (E) 10

Solution

Problem 3

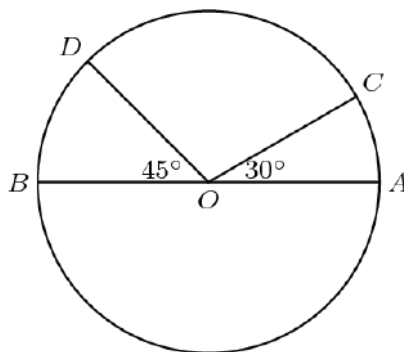
A semipro baseball league has teams with 21 players each. League rules state that a player must be paid at least 15,000 dollars, and that the total of all players' salaries for each team cannot exceed 700,000 dollars. What is the maximum possible salary, in dollars, for a single player?

- (A) 270,000 (B) 385,000 (C) 400,000 (D) 430,000 (E) 700,000

Solution

Problem 4

On circle O , points C and D are on the same side of diameter $\overline{AB}$, $\angle AOC = 30^\circ$, and $\angle DOB = 45^\circ$. What is the ratio of the area of the smaller sector COD to the area of the circle?



- (A) $\frac{2}{9}$ (B) $\frac{1}{4}$ (C) $\frac{5}{18}$ (D) $\frac{7}{24}$ (E) $\frac{3}{10}$

(Solution)

Problem 5

A class collects 50 dollars to buy flowers for a classmate who is in the hospital. Roses cost 3 dollars each, and carnations cost 2 dollars each. No other flowers are to be used. How many different bouquets could be purchased for exactly 50 dollars?

- (A) 1 (B) 7 (C) 9 (D) 16 (E) 17

(Solution)

Problem 6

Postman Pete has a pedometer to count his steps. The pedometer records up to 99999 steps, then flips over to 00000 on the next step. Pete plans to determine his mileage for a year. On January 1 Pete sets the pedometer to 00000. During the year, the pedometer flips from 99999 to 00000 forty-four times. On December 31 the pedometer reads 50000. Pete takes 1800 steps per mile. Which of the following is closest to the number of miles Pete walked during the year?

- (A) 2500 (B) 3000 (C) 3500 (D) 4000 (E) 4500

(Solution)

Problem 7

For real numbers a and b , define $a\$b = (a - b)^2$. What is $(x - y)^2\$ (y - x)^2$?

- (A) 0 (B) $x^2 + y^2$ (C) $2x^2$ (D) $2y^2$ (E) $4xy$

(Solution)

Problem 8

Points B and C lie on $\overline{AD}$. The length of $\overline{AB}$ is 4 times the length of $\overline{BD}$, and the length of $\overline{AC}$ is 9 times the length of $\overline{CD}$. The length of $\overline{BC}$ is what fraction of the length of $\overline{AD}$?

- (A) $\frac{1}{36}$ (B) $\frac{1}{13}$ (C) $\frac{1}{10}$ (D) $\frac{5}{36}$ (E) $\frac{1}{5}$

(Solution)

Problem 9

Points A and B are on a circle of radius 5 and $AB = 6$. Point C is the midpoint of the minor arc AB . What is the length of the line segment AC ?

- (A) $\sqrt{10}$ (B) $\frac{7}{2}$ (C) $\sqrt{14}$ (D) $\sqrt{15}$ (E) 4

(Solution)

Problem 10

Bricklayer Brenda would take 9 hours to build a chimney alone, and bricklayer Brandon would take 10 hours to build it alone. When they work together they talk a lot, and their combined output is decreased by 10 bricks per hour. Working together, they build the chimney in 5 hours. How many bricks are in the chimney?

- (A) 500 (B) 900 (C) 950 (D) 1000 (E) 1900

(Solution)

Problem 11

A cone-shaped mountain has its base on the ocean floor and has a height of 8000 feet. The top $\frac{1}{8}$ of the volume of the mountain is above water. What is the depth of the ocean at the base of the mountain in feet?

- (A) 4000 (B) $2000(4 - \sqrt{2})$ (C) 6000 (D) 6400 (E) 7000

Solution

Problem 12

For each positive integer n , the mean of the first n terms of a sequence is n . What is the 2008th term of the sequence?

- (A) 2008 (B) 4015 (C) 4016 (D) 4030056 (E) 4032064

Solution

Problem 13

Vertex E of equilateral triangle $\triangle ABE$ is in the interior of unit square $ABCD$. Let R be the region consisting of all points inside $ABCD$ and outside $\triangle ABE$ whose distance from $\overline{AD}$ is between $\frac{1}{3}$ and $\frac{2}{3}$. What is the area of R ?

- (A) $\frac{12 - 5\sqrt{3}}{72}$ (B) $\frac{12 - 5\sqrt{3}}{36}$ (C) $\frac{\sqrt{3}}{18}$ (D) $\frac{3 - \sqrt{3}}{9}$ (E) $\frac{\sqrt{3}}{12}$

(Solution)

Problem 14

A circle has a radius of $\log_{10}(a^2)$ and a circumference of $\log_{10}(b^4)$. What is $\log_a b$?

- (A) $\frac{1}{4\pi}$ (B) $\frac{1}{\pi}$ (C) π (D) 2π (E) $10^{2\pi}$

(Solution)

Problem 15

On each side of a unit square, an equilateral triangle of side length 1 is constructed. On each new side of each equilateral triangle, another equilateral triangle of side length 1 is constructed. The interiors of the square and the 12 triangles have no points in common. Let R be the region formed by the union of the square and all the triangles, and S be the smallest convex polygon that contains R . What is the area of the region that is inside S but outside R ?

- (A) $\frac{1}{4}$ (B) $\frac{\sqrt{2}}{4}$ (C) 1 (D) $\sqrt{3}$ (E) $2\sqrt{3}$

(Solution)

Problem 16

A rectangular floor measures a by b feet, where a and b are positive integers with $b > a$. An artist paints a rectangle on the floor with the sides of the rectangle parallel to the sides of the floor. The unpainted part of the floor forms a border of width 1 foot around the painted rectangle and occupies half of the area of the entire floor. How many possibilities are there for the ordered pair (a, b) ?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

(Solution)

Problem 17

Let A, B and C be three distinct points on the graph of $y = x^2$ such that line AB is parallel to the x -axis and $\triangle ABC$ is a right triangle with area 2008. What is the sum of the digits of the y -coordinate of C ?

- (A) 16 (B) 17 (C) 18 (D) 19 (E) 20

(Solution)

Problem 18

A pyramid has a square base $ABCD$ and vertex E . The area of square $ABCD$ is 196, and the areas of $\triangle ABE$ and $\triangle CDE$ are 105 and 91, respectively. What is the volume of the pyramid?

- (A) 392 (B) $196\sqrt{6}$ (C) $392\sqrt{2}$ (D) $392\sqrt{3}$ (E) 784

(Solution)

Problem 19

A function f is defined by $f(z) = (4 + i)z^2 + \alpha z + \gamma$ for all complex numbers z , where α and γ are complex numbers and $i^2 = -1$. Suppose that $f(1)$ and $f(i)$ are both real. What is the smallest possible value of $|\alpha| + |\gamma|$?

- (A) 1 (B) $\sqrt{2}$ (C) 2 (D) $2\sqrt{2}$ (E) 4

(Solution)

Problem 20

Michael walks at the rate of 5 feet per second on a long straight path. Trash pails are located every 200 feet along the path. A garbage truck travels at 10 feet per second in the same direction as Michael and stops for 30 seconds at each pail. As Michael passes a pail, he notices the truck ahead of him just leaving the next pail. How many times will Michael and the truck meet?

- (A) 4 (B) 5 (C) 6 (D) 7 (E) 8

(Solution)

Problem 21

Two circles of radius 1 are to be constructed as follows. The center of circle A is chosen uniformly and at random from the line segment joining $(0, 0)$ and $(2, 0)$. The center of circle B is chosen uniformly and at random, and independently of the first choice, from the line segment joining $(0, 1)$ to $(2, 1)$. What is the probability that circles A and B intersect?

- (A) $\frac{2 + \sqrt{2}}{4}$ (B) $\frac{3\sqrt{3} + 2}{8}$ (C) $\frac{2\sqrt{2} - 1}{2}$ (D) $\frac{2 + \sqrt{3}}{4}$ (E) $\frac{4\sqrt{3} - 3}{4}$

(Solution)

Problem 22

A parking lot has 16 spaces in a row. Twelve cars arrive, each of which requires one parking space, and their drivers chose spaces at random from among the available spaces. Auntie Em then arrives in her SUV, which requires 2 adjacent spaces. What is the probability that she is able to park?

- (A) $\frac{11}{20}$ (B) $\frac{4}{7}$ (C) $\frac{81}{140}$ (D) $\frac{3}{5}$ (E) $\frac{17}{28}$

(Solution)

Problem 23

The sum of the base-10 logarithms of the divisors of 10^n is 792. What is n ?

- (A) 11 (B) 12 (C) 13 (D) 14 (E) 15

(Solution)

Problem 24

Let $A_0 = (0, 0)$. Distinct points $A_1, A_2, \dots$ lie on the x -axis, and distinct points $B_1, B_2, \dots$ lie on the graph of $y = \sqrt{x}$. For every positive integer n , $A_{n-1}B_nA_n$ is an equilateral triangle. What is the least n for which the length $A_0A_n \geq 100$?

- (A) 13 (B) 15 (C) 17 (D) 19 (E) 21

(Solution)

Problem 25

Let $ABCD$ be a trapezoid with $AB \parallel CD$, $AB = 11$, $BC = 5$, $CD = 19$, and $DA = 7$. Bisectors of $\angle A$ and $\angle D$ meet at P , and bisectors of $\angle B$ and $\angle C$ meet at Q . What is the area of hexagon $ABQCDP$?

- (A) $28\sqrt{3}$ (B) $30\sqrt{3}$ (C) $32\sqrt{3}$ (D) $35\sqrt{3}$ (E) $36\sqrt{3}$

(Solution)

See also

2008 AMC 12B (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2008))	
Preceded by 2008 AMC 12A Problems	Followed by 2009 AMC 12A Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
All AMC 12 Problems and Solutions	

- AMC 12
- AMC 12 Problems and Solutions
- 2008 AMC 10B
- 2008 AMC B Math Jam Transcript (http://www.artofproblemsolving.com/Community/AoPS_Y_MJ_Transcripts.php?mj_id=219)
- Mathematics competition resources

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2008 AMC 12B Answer Key

1. E
2. B
3. C
4. D
5. C
6. A
7. A
8. C
9. A
10. B
11. A
12. B
13. B
14. C
15. C
16. B
17. C
18. E
19. B
20. B
21. E
22. E
23. A
24. C
25. B

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2009

2009 AMC 12A Problems

2009 AMC 12A (Answer Key)

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Instructions

1. This is a 25-question, multiple choice test. Each question is followed by answers marked A, B, C, D and E. Only one of these is correct.
2. You will receive 6 points for each correct answer, 2.5 points for each problem left unanswered if the year is before 2006, 1.5 points for each problem left unanswered if the year is after 2006, and 0 points for each incorrect answer.
3. No aids are permitted other than scratch paper, graph paper, ruler, compass, protractor and erasers (and calculators that are accepted for use on the test if before 2006. No problems on the test will *require* the use of a calculator).
4. Figures are not necessarily drawn to scale.
5. You will have **75 minutes** working time to complete the test.

1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

Kim's flight took off from Newark at 10:34 AM and landed in Miami at 1:18 PM. Both cities are in the same time zone. If her flight took h hours and m minutes, with $0 < m < 60$, what is $h + m$?

- (A) 46 (B) 47 (C) 50 (D) 53 (E) 54

Solution

Problem 2

Which of the following is equal to $1 + \frac{1}{1 + \frac{1}{1+1}}$?

- (A) $\frac{5}{4}$ (B) $\frac{3}{2}$ (C) $\frac{5}{3}$ (D) 2 (E) 3

Solution

Problem 3

What number is one third of the way from $\frac{1}{4}$ to $\frac{3}{4}$?

- (A) $\frac{1}{3}$ (B) $\frac{5}{12}$ (C) $\frac{1}{2}$ (D) $\frac{7}{12}$ (E) $\frac{2}{3}$

Solution

Problem 4

Four coins are picked out of a piggy bank that contains a collection of pennies, nickels, dimes, and quarters. Which of the following could *not* be the total value of the four coins, in cents?

- (A) 15 (B) 25 (C) 35 (D) 45 (E) 55

Solution

Problem 5

One dimension of a cube is increased by 1, another is decreased by 1, and the third is left unchanged. The volume of the new rectangular solid is 5 less than that of the cube. What was the volume of the cube?

- (A) 8 (B) 27 (C) 64 (D) 125 (E) 216

Solution

Problem 6

Suppose that $P = 2^m$ and $Q = 3^n$. Which of the following is equal to 12^{mn} for every pair of integers (m, n) ?

- (A) P^2Q (B) P^nQ^m (C) P^nQ^{2m} (D) $P^{2m}Q^n$ (E) $P^{2n}Q^m$

Solution

Problem 7

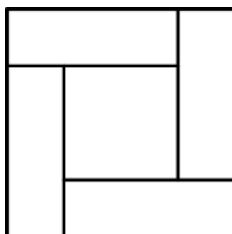
The first three terms of an arithmetic sequence are $2x - 3$, $5x - 11$, and $3x + 1$ respectively. The n th term of the sequence is 2009. What is n ?

- (A) 255 (B) 502 (C) 1004 (D) 1506 (E) 8037

Solution

Problem 8

Four congruent rectangles are placed as shown. The area of the outer square is 4 times that of the inner square. What is the ratio of the length of the longer side of each rectangle to the length of its shorter side?



- (A) 3 (B) $\sqrt{10}$ (C) $2 + \sqrt{2}$ (D) $2\sqrt{3}$ (E) 4

Solution

Problem 9

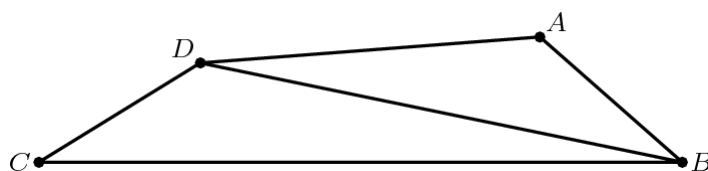
Suppose that $f(x + 3) = 3x^2 + 7x + 4$ and $f(x) = ax^2 + bx + c$. What is $a + b + c$?

- (A) -1 (B) 0 (C) 1 (D) 2 (E) 3

Solution

Problem 10

In quadrilateral $ABCD$, $AB = 5$, $BC = 17$, $CD = 5$, $DA = 9$, and BD is an integer. What is BD ?

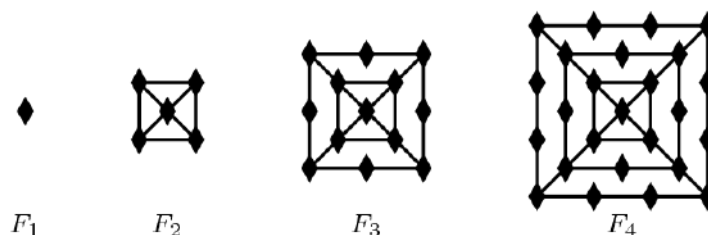


- (A) 11 (B) 12 (C) 13 (D) 14 (E) 15

Solution

Problem 11

The figures F_1 , F_2 , F_3 , and F_4 shown are the first in a sequence of figures. For $n \geq 3$, F_n is constructed from F_{n-1} by surrounding it with a square and placing one more diamond on each side of the new square than F_{n-1} had on each side of its outside square. For example, figure F_3 has 13 diamonds. How many diamonds are there in figure F_{20} ?



- (A) 401 (B) 485 (C) 585 (D) 626 (E) 761

Solution

Problem 12

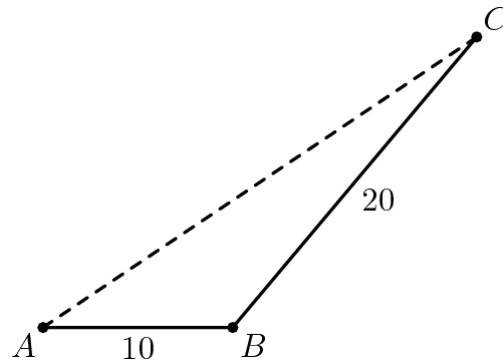
How many positive integers less than 1000 are 6 times the sum of their digits?

- (A) 0 (B) 1 (C) 2 (D) 4 (E) 12

Solution

Problem 13

A ship sails 10 miles in a straight line from A to B , turns through an angle between 45° and 60° , and then sails another 20 miles to C . Let AC be measured in miles. Which of the following intervals contains AC^2 ?



- (A) $[400, 500]$ (B) $[500, 600]$ (C) $[600, 700]$ (D) $[700, 800]$ (E) $[800, 900]$

Solution

Problem 14

A triangle has vertices $(0, 0)$, $(1, 1)$, and $(6m, 0)$, and the line $y = mx$ divides the triangle into two triangles of equal area. What is the sum of all possible values of m ?

- (A) $-\frac{1}{3}$ (B) $-\frac{1}{6}$ (C) $\frac{1}{6}$ (D) $\frac{1}{3}$ (E) $\frac{1}{2}$

Solution

Problem 15

For what value of n is $i + 2i^2 + 3i^3 + \cdots + ni^n = 48 + 49i$?

Note: here $i = \sqrt{-1}$.

- (A) 24 (B) 48 (C) 49 (D) 97 (E) 98

Solution

Problem 16

A circle with center C is tangent to the positive x and y -axes and externally tangent to the circle centered at $(3, 0)$ with radius 1. What is the sum of all possible radii of the circle with center C ?

- (A) 3 (B) 4 (C) 6 (D) 8 (E) 9

Solution

Problem 17

Let $a + ar_1 + ar_1^2 + ar_1^3 + \cdots$ and $a + ar_2 + ar_2^2 + ar_2^3 + \cdots$ be two different infinite geometric series of positive numbers with the same first term. The sum of the first series is r_1 , and the sum of the second series is r_2 . What is $r_1 + r_2$?

- (A) 0 (B) $\frac{1}{2}$ (C) 1 (D) $\frac{1 + \sqrt{5}}{2}$ (E) 2

Solution

Problem 18

For $k > 0$, let $I_k = 10 \dots 064$, where there are k zeros between the 1 and the 6. Let $N(k)$ be the number of factors of 2 in the prime factorization of I_k . What is the maximum value of $N(k)$?

- (A) 6 (B) 7 (C) 8 (D) 9 (E) 10

Solution

Problem 19

Andrea inscribed a circle inside a regular pentagon, circumscribed a circle around the pentagon, and calculated the area of the region between the two circles. Bethany did the same with a regular heptagon (7 sides). The areas of the two regions were A and B , respectively. Each polygon had a side length of 2. Which of the following is true?

- (A) $A = \frac{25}{49}B$ (B) $A = \frac{5}{7}B$ (C) $A = B$ (D) $A = \frac{7}{5}B$ (E) $A = \frac{49}{25}B$

Solution

Problem 20

Convex quadrilateral $ABCD$ has $AB = 9$ and $CD = 12$. Diagonals AC and BD intersect at E , $AC = 14$, and $\triangle AED$ and $\triangle BEC$ have equal areas. What is AE ?

- (A) $\frac{9}{2}$ (B) $\frac{50}{11}$ (C) $\frac{21}{4}$ (D) $\frac{17}{3}$ (E) 6

Solution

Problem 21

Let $p(x) = x^3 + ax^2 + bx + c$, where a , b , and c are complex numbers. Suppose that

$$p(2009 + 9002\pi i) = p(2009) = p(9002) = 0$$

What is the number of nonreal zeros of $x^{12} + ax^8 + bx^4 + c$?

- (A) 4 (B) 6 (C) 8 (D) 10 (E) 12

Solution

Problem 22

A regular octahedron has side length 1. A plane parallel to two of its opposite faces cuts the octahedron into the two congruent solids. The polygon formed by the intersection of the plane and the octahedron has area $\frac{a\sqrt{b}}{c}$, where a , b , and c are positive integers, a and c are relatively prime, and b is not divisible by the square of any prime. What is $a + b + c$?

- (A) 10 (B) 11 (C) 12 (D) 13 (E) 14

Solution

Problem 23

Functions f and g are quadratic, $g(x) = -f(100 - x)$, and the graph of g contains the vertex of the graph of f . The four x -intercepts on the two graphs have x -coordinates x_1, x_2, x_3 , and x_4 , in increasing order, and $x_3 - x_2 = 150$. The value of $x_4 - x_1$ is $m + n\sqrt{p}$, where m , n , and p are positive integers, and p is not divisible by the square of any prime. What is $m + n + p$?

- (A) 602 (B) 652 (C) 702 (D) 752 (E) 802

Solution

Problem 24

The tower function of twos is defined recursively as follows: $T(1) = 2$ and $T(n+1) = 2^{T(n)}$ for $n \geq 1$. Let $A = (T(2009))^{T(2009)}$ and $B = (T(2009))^A$. What is the largest integer k such that

$$\underbrace{\log_2 \log_2 \log_2 \dots \log_2 B}_{k \text{ times}}$$

is defined?

(A) 2009 (B) 2010 (C) 2011 (D) 2012 (E) 2013

Solution

Problem 25

The first two terms of a sequence are $a_1 = 1$ and $a_2 = \frac{1}{\sqrt{3}}$. For $n \geq 1$,

$$a_{n+2} = \frac{a_n + a_{n+1}}{1 - a_n a_{n+1}}.$$

What is $|a_{2009}|$?

(A) 0 (B) $2 - \sqrt{3}$ (C) $\frac{1}{\sqrt{3}}$ (D) 1 (E) $2 + \sqrt{3}$

Solution

See also

2009 AMC 12A (Problems • Answer Key • Resources) (http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2009)	
Preceded by 2008 AMC 12B Problems	Followed by 2009 AMC 12B Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
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2009 AMC 12A Answer Key

1. A
2. C
3. B
4. A
5. D
6. A
7. B
8. A
9. D
10. C
11. E
12. B
13. D
14. B
15. D
16. D
17. C
18. B
19. C
20. E
21. C
22. E
23. D
24. E
25. A

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2009 AMC 12B Problems

2009 AMC 12B (Answer Key)

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1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

Each morning of her five-day workweek, Jane bought either a 50-cent muffin or a 75-cent bagel. Her total cost for the week was a whole number of dollars. How many bagels did she buy?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Solution

Problem 2

Paula the painter had just enough paint for 30 identically sized rooms. Unfortunately, on the way to work, three cans of paint fell off her truck, so she had only enough paint for 25 rooms. How many cans of paint did she use for the 25 rooms?

- (A) 10 (B) 12 (C) 15 (D) 18 (E) 25

Solution

Problem 3

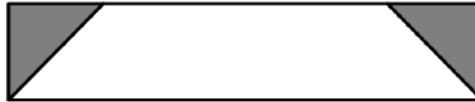
Twenty percent off 60 is one-third more than what number?

- (A) 16 (B) 30 (C) 32 (D) 36 (E) 48

Solution

Problem 4

A rectangular yard contains two flower beds in the shape of congruent isosceles right triangles. The remainder of the yard has a trapezoidal shape, as shown. The parallel sides of the trapezoid have lengths 15 and 25 meters. What fraction of the yard is occupied by the flower beds?



- (A) $\frac{1}{8}$ (B) $\frac{1}{6}$ (C) $\frac{1}{5}$ (D) $\frac{1}{4}$ (E) $\frac{1}{3}$

Solution

Problem 5

Kiana has two older twin brothers. The product of their ages is 128. What is the sum of their three ages?

- (A) 10 (B) 12 (C) 16 (D) 18 (E) 24

Solution

Problem 6

By inserting parentheses, it is possible to give the expression

$$2 \times 3 + 4 \times 5$$

several values. How many different values can be obtained?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Solution

Problem 7

In a certain year the price of gasoline rose by 20% during January, fell by 20% during February, rose by 25% during March, and fell by $x\%$ during April. The price of gasoline at the end of April was the same as it had been at the beginning of January. To the nearest integer, what is x ?

- (A) 12 (B) 17 (C) 20 (D) 25 (E) 35

Solution

Problem 8

When a bucket is two-thirds full of water, the bucket and water weigh a kilograms. When the bucket is one-half full of water the total weight is b kilograms. In terms of a and b , what is the total weight in kilograms when the bucket is full of water?

(A) $\frac{2}{3}a + \frac{1}{3}b$ (B) $\frac{3}{2}a - \frac{1}{2}b$ (C) $\frac{3}{2}a + b$ (D) $\frac{3}{2}a + 2b$ (E) $3a - 2b$

Solution

Problem 9

Triangle ABC has vertices $A = (3, 0)$, $B = (0, 3)$, and C , where C is on the line $x + y = 7$. What is the area of $\triangle ABC$?

(A) 6 (B) 8 (C) 10 (D) 12 (E) 14

Solution

Problem 10

A particular 12-hour digital clock displays the hour and minute of a day. Unfortunately, whenever it is supposed to display a 1, it mistakenly displays a 9. For example, when it is 1:16 PM the clock incorrectly shows 9:96 PM. What fraction of the day will the clock show the correct time?

(A) $\frac{1}{2}$ (B) $\frac{5}{8}$ (C) $\frac{3}{4}$ (D) $\frac{5}{6}$ (E) $\frac{9}{10}$

Solution

Problem 11

On Monday, Millie puts a quart of seeds, 25% of which are millet, into a bird feeder. On each successive day she adds another quart of the same mix of seeds without removing any seeds that are left. Each day the birds eat only 25% of the millet in the feeder, but they eat all of the other seeds. On which day, just after Millie has placed the seeds, will the birds find that more than half the seeds in the feeder are millet?

(A) Tuesday (B) Wednesday (C) Thursday (D) Friday (E) Saturday

Solution

Problem 12

The fifth and eighth terms of a geometric sequence of real numbers are $7!$ and $8!$ respectively. What is the first term?

(A) 60 (B) 75 (C) 120 (D) 225 (E) 315

Solution

Problem 13

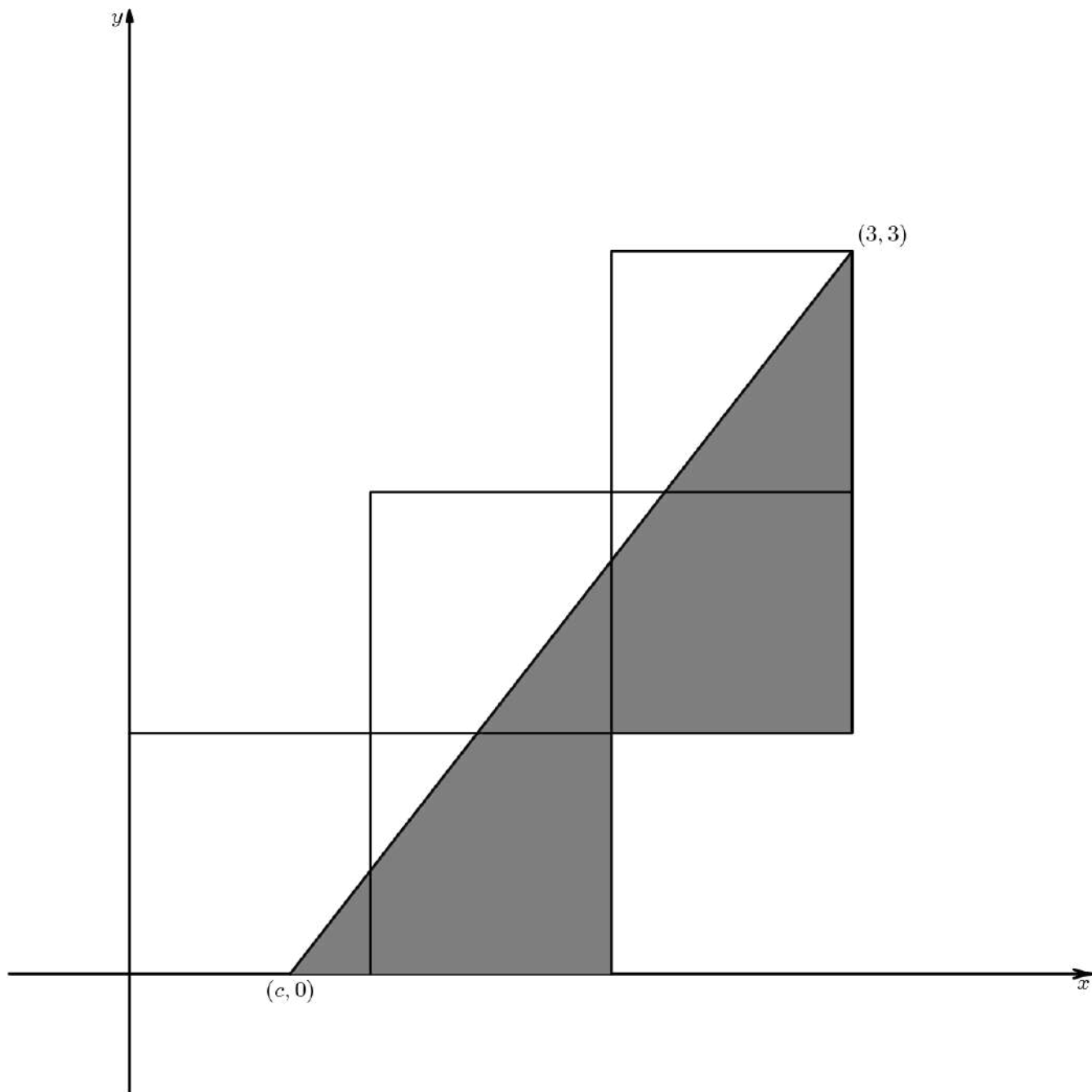
Triangle ABC has $AB = 13$ and $AC = 15$, and the altitude to $\overline{BC}$ has length 12. What is the sum of the two possible values of BC ?

(A) 15 (B) 16 (C) 17 (D) 18 (E) 19

Solution

Problem 14

Five unit squares are arranged in the coordinate plane as shown, with the lower left corner at the origin. The slanted line, extending from $(c, 0)$ to $(3, 3)$, divides the entire region into two regions of equal area. What is c ?



- (A) $\frac{1}{2}$ (B) $\frac{3}{5}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$ (E) $\frac{4}{5}$

Solution

Problem 15

Assume $0 < r < 3$. Below are five equations for x . Which equation has the largest solution x ?

- (A) $3(1+r)^x = 7$ (B) $3(1+r/10)^x = 7$ (C) $3(1+2r)^x = 7$
 (D) $3(1+\sqrt{r})^x = 7$ (E) $3(1+1/r)^x = 7$

Solution

Problem 16

Trapezoid $ABCD$ has $AD \parallel BC$, $BD = 1$, $\angle DBA = 23^\circ$, and $\angle BDC = 46^\circ$. The ratio $BC : AD$ is $9 : 5$. What is CD ?

- (A) $\frac{7}{9}$ (B) $\frac{4}{5}$ (C) $\frac{13}{15}$ (D) $\frac{8}{9}$ (E) $\frac{14}{15}$

Solution

Problem 17

Each face of a cube is given a single narrow stripe painted from the center of one edge to the center of its opposite edge. The choice of the edge pairing is made at random and independently for each face. What is the probability that there is a continuous stripe encircling the cube?

- (A) $\frac{1}{8}$ (B) $\frac{3}{16}$ (C) $\frac{1}{4}$ (D) $\frac{3}{8}$ (E) $\frac{1}{2}$

Solution

Problem 18

Rachel and Robert run on a circular track. Rachel runs counterclockwise and completes a lap every 90 seconds, and Robert runs clockwise and completes a lap every 80 seconds. Both start from the start line at the same time. At some random time between 10 minutes and 11 minutes after they begin to run, a photographer standing inside the track takes a picture that shows one-fourth of the track, centered on the starting line. What is the probability that both Rachel and Robert are in the picture?

- (A) $\frac{1}{16}$ (B) $\frac{1}{8}$ (C) $\frac{3}{16}$ (D) $\frac{1}{4}$ (E) $\frac{5}{16}$

Solution

Problem 19

For each positive integer n , let $f(n) = n^4 - 360n^2 + 400$. What is the sum of all values of $f(n)$ that are prime numbers?

- (A) 794 (B) 796 (C) 798 (D) 800 (E) 802

Solution

Problem 20

A convex polyhedron Q has vertices $V_1, V_2, \dots, V_n$ and 100 edges. The polyhedron is cut by planes $P_1, P_2, \dots, P_n$ in such a way that plane P_k cuts only those edges that meet at vertex V_k . In addition, no two planes intersect inside or on Q . The cuts produce n pyramids and a new polyhedron R . How many edges does R have?

- (A) 200 (B) $2n$ (C) 300 (D) 400 (E) $4n$

Solution

Problem 21

Ten women sit in 10 seats in a line. All of the 10 get up and then reseal themselves using all 10 seats, each sitting in the seat she was in before or a seat next to the one she occupied before. In how many ways can the women be reseated?

- (A) 89 (B) 90 (C) 120 (D) 2^{10} (E) $2^2 3^8$

Solution

Problem 22

Parallelogram $ABCD$ has area 1,000,000. Vertex A is at $(0, 0)$ and all other vertices are in the first quadrant. Vertices B and D are lattice points on the lines $y = x$ and $y = kx$ for some integer $k > 1$, respectively. How many such parallelograms are there?

- (A) 49 (B) 720 (C) 784 (D) 2009 (E) 2048

Solution

Problem 23

A region S in the complex plane is defined by

$$S = \{x + iy : -1 \leq x \leq 1, -1 \leq y \leq 1\}.$$

A complex number $z = x + iy$ is chosen uniformly at random from S . What is the probability that $\left(\frac{3}{4} + \frac{3}{4}i\right)z$ is also in S ?

- (A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{7}{9}$ (E) $\frac{7}{8}$

Solution

Problem 24

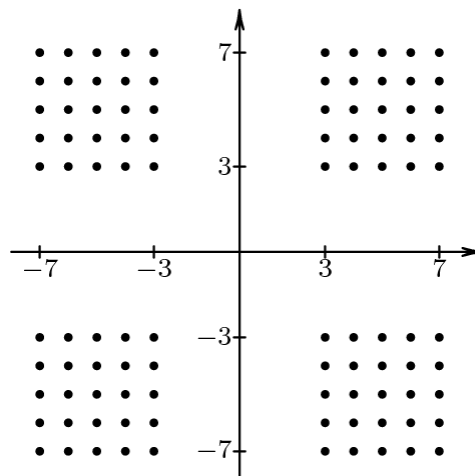
For how many values of x in $[0, \pi]$ is $\sin^{-1}(\sin 6x) = \cos^{-1}(\cos x)$? Note: The functions $\sin^{-1} = \arcsin$ and $\cos^{-1} = \arccos$ denote inverse trigonometric functions.

- (A) 3 (B) 4 (C) 5 (D) 6 (E) 7

Solution

Problem 25

The set G is defined by the points (x, y) with integer coordinates, $3 \leq |x| \leq 7$, $3 \leq |y| \leq 7$. How many squares of side at least 6 have their four vertices in G ?



- (A) 125 (B) 150 (C) 175 (D) 200 (E) 225

Solution

See also

2009 AMC 12B (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2009))	
Preceded by 2009 AMC 12A Problems	Followed by 2010 AMC 12A Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
All AMC 12 Problems and Solutions	

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2009 AMC 12B Answer Key

1. B
2. C
3. D
4. C
5. D
6. C
7. B
8. E
9. A
10. A
11. D
12. E
13. D
14. C
15. B
16. B
17. B
18. C
19. E
20. C
21. A
22. C
23. D
24. B
25. E

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2010

2010 AMC 12A Problems

2010 AMC 12A (Answer Key)

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Instructions

1. This is a 25-question, multiple choice test. Each question is followed by answers marked A, B, C, D and E. Only one of these is correct.
2. You will receive 6 points for each correct answer, 2.5 points for each problem left unanswered if the year is before 2006, 1.5 points for each problem left unanswered if the year is after 2006, and 0 points for each incorrect answer.
3. No aids are permitted other than scratch paper, graph paper, ruler, compass, protractor and erasers (and calculators that are accepted for use on the test if before 2006. No problems on the test will *require* the use of a calculator).
4. Figures are not necessarily drawn to scale.
5. You will have **75 minutes** working time to complete the test.

1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

What is $(20 - (2010 - 201)) + (2010 - (201 - 20))$?

(A) -4020 (B) 0 (C) 40 (D) 401 (E) 4020

Solution

Problem 2

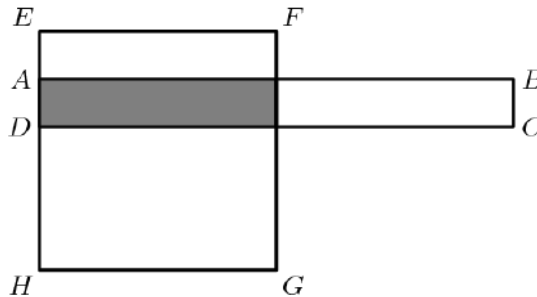
A ferry boat shuttles tourists to an island every hour starting at 10 AM until its last trip, which starts at 3 PM. One day the boat captain notes that on the 10 AM trip there were 100 tourists on the ferry boat, and that on each successive trip, the number of tourists was 1 fewer than on the previous trip. How many tourists did the ferry take to the island that day?

- (A) 585 (B) 594 (C) 672 (D) 679 (E) 694

Solution

Problem 3

Rectangle $ABCD$, pictured below, shares 50% of its area with square $EFGH$. Square $EFGH$ shares 20% of its area with rectangle $ABCD$. What is $\frac{AB}{AD}$?



- (A) 4 (B) 5 (C) 6 (D) 8 (E) 10

Solution

Problem 4

If $x < 0$, then which of the following must be positive?

- (A) $\frac{x}{|x|}$ (B) $-x^2$ (C) -2^x (D) $-x^{-1}$ (E) $\sqrt[3]{x}$

Solution

Problem 5

Halfway through a 100-shot archery tournament, Chelsea leads by 50 points. For each shot a bullseye scores 10 points, with other possible scores being 8, 4, 2, and 0 points. Chelsea always scores at least 4 points on each shot. If Chelsea's next n shots are bullseyes she will be guaranteed victory. What is the minimum value for n ?

- (A) 38 (B) 40 (C) 42 (D) 44 (E) 46

Solution

Problem 6

A *palindrome*, such as 83438, is a number that remains the same when its digits are reversed. The numbers x and $x + 32$ are three-digit and four-digit palindromes, respectively. What is the sum of the digits of x ?

- (A) 20 (B) 21 (C) 22 (D) 23 (E) 24

Solution

Problem 7

Logan is constructing a scaled model of his town. The city's water tower stands 40 meters high, and the top portion is a sphere that holds 100,000 liters of water. Logan's miniature water tower holds 0.1 liters. How tall, in meters, should Logan make his tower?

- (A) 0.04 (B) $\frac{0.4}{\pi}$ (C) 0.4 (D) $\frac{4}{\pi}$ (E) 4

Solution

Problem 8

Triangle ABC has $AB = 2 \cdot AC$. Let D and E be on $\overline{AB}$ and $\overline{BC}$, respectively, such that $\angle BAE = \angle ACD$. Let F be the intersection of segments AE and CD , and suppose that $\triangle CFE$ is equilateral. What is $\angle ACB$?

- (A) 60° (B) 75° (C) 90° (D) 105° (E) 120°

Solution

Problem 9

A solid cube has side length 3 inches. A 2-inch by 2-inch square hole is cut into the center of each face. The edges of each cut are parallel to the edges of the cube, and each hole goes all the way through the cube. What is the volume, in cubic inches, of the remaining solid?

- (A) 7 (B) 8 (C) 10 (D) 12 (E) 15

Solution

Problem 10

The first four terms of an arithmetic sequence are p , 9 , $3p - q$, and $3p + q$. What is the 2010^{th} term of this sequence?

- (A) 8041 (B) 8043 (C) 8045 (D) 8047 (E) 8049

Solution

Problem 11

The solution of the equation $7^{x+7} = 8^x$ can be expressed in the form $x = \log_b 7^7$. What is b ?

- (A) $\frac{7}{15}$ (B) $\frac{7}{8}$ (C) $\frac{8}{7}$ (D) $\frac{15}{8}$ (E) $\frac{15}{7}$

Solution

Problem 12

In a magical swamp there are two species of talking amphibians: toads, whose statements are always true, and frogs, whose statements are always false. Four amphibians, Brian, Chris, LeRoy, and Mike live together in this swamp, and they make the following statements.

Brian: "Mike and I are different species."

Chris: "LeRoy is a frog."

LeRoy: "Chris is a frog."

Mike: "Of the four of us, at least two are toads."

How many of these amphibians are frogs?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Solution

Problem 13

For how many integer values of k do the graphs of $x^2 + y^2 = k^2$ and $xy = k$ not intersect?

- (A) 0 (B) 1 (C) 2 (D) 4 (E) 8

Solution

Problem 14

Nondegenerate $\triangle ABC$ has integer side lengths, $\overline{BD}$ is an angle bisector, $AD = 3$, and $DC = 8$. What is the smallest possible value of the perimeter?

(A) 30 (B) 33 (C) 35 (D) 36 (E) 37

Solution

Problem 15

A coin is altered so that the probability that it lands on heads is less than $\frac{1}{2}$ and when the coin is flipped four times, the probability of an equal number of heads and tails is $\frac{1}{6}$. What is the probability that the coin lands on heads?

(A) $\frac{\sqrt{15}-3}{6}$ (B) $\frac{6-\sqrt{6\sqrt{6}+2}}{12}$ (C) $\frac{\sqrt{2}-1}{2}$ (D) $\frac{3-\sqrt{3}}{6}$ (E) $\frac{\sqrt{3}-1}{2}$

Solution

Problem 16

Bernardo randomly picks 3 distinct numbers from the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and arranges them in descending order to form a 3-digit number. Silvia randomly picks 3 distinct numbers from the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$ and also arranges them in descending order to form a 3-digit number. What is the probability that Bernardo's number is larger than Silvia's number?

(A) $\frac{47}{72}$ (B) $\frac{37}{56}$ (C) $\frac{2}{3}$ (D) $\frac{49}{72}$ (E) $\frac{39}{56}$

Solution

Problem 17

Equiangular hexagon $ABCDEF$ has side lengths $AB = CD = EF = 1$ and $BC = DE = FA = r$. The area of $\triangle ACE$ is 70% of the area of the hexagon. What is the sum of all possible values of r ?

(A) $\frac{4\sqrt{3}}{3}$ (B) $\frac{10}{3}$ (C) 4 (D) $\frac{17}{4}$ (E) 6

Solution

Problem 18

A 16-step path is to go from $(-4, -4)$ to $(4, 4)$ with each step increasing either the x -coordinate or the y -coordinate by 1. How many such paths stay outside or on the boundary of the square $-2 \leq x \leq 2, -2 \leq y \leq 2$ at each step?

(A) 92 (B) 144 (C) 1568 (D) 1698 (E) 12,800

Solution

Problem 19

Each of 2010 boxes in a line contains a single red marble, and for $1 \leq k \leq 2010$, the box in the k th position also contains k white marbles. Isabella begins at the first box and successively draws a single marble at random from each box, in order. She stops when she first draws a red marble. Let $P(n)$ be the probability that Isabella stops after drawing exactly n marbles. What is the smallest value of n for which $P(n) < \frac{1}{2010}$?

(A) 45 (B) 63 (C) 64 (D) 201 (E) 1005

Solution

Problem 20

Arithmetic sequences (a_n) and (b_n) have integer terms with $a_1 = b_1 = 1 < a_2 \leq b_2$ and $a_n b_n = 2010$ for some n . What is the largest possible value of n ?

(A) 2 (B) 3 (C) 8 (D) 288 (E) 2009

Solution

Problem 21

The graph of $y = x^6 - 10x^5 + 29x^4 - 4x^3 + ax^2$ lies above the line $y = bx + c$ except at three values of x , where the graph and the line intersect. What is the largest of these values?

- (A) 4 (B) 5 (C) 6 (D) 7 (E) 8

Solution

Problem 22

What is the minimum value of $|x - 1| + |2x - 1| + |3x - 1| + \cdots + |119x - 1|$?

- (A) 49 (B) 50 (C) 51 (D) 52 (E) 53

Solution

Problem 23

The number obtained from the last two nonzero digits of $90!$ is equal to n . What is n ?

- (A) 12 (B) 32 (C) 48 (D) 52 (E) 68

Solution

Problem 24

Let $f(x) = \log_{10}(\sin(\pi x) \cdot \sin(2\pi x) \cdot \sin(3\pi x) \cdots \sin(8\pi x))$. The intersection of the domain of $f(x)$ with the interval $[0, 1]$ is a union of n disjoint open intervals. What is n ?

- (A) 2 (B) 12 (C) 18 (D) 22 (E) 36

Solution

Problem 25

Two quadrilaterals are considered the same if one can be obtained from the other by a rotation and a translation. How many different convex cyclic quadrilaterals are there with integer sides and perimeter equal to 32?

- (A) 560 (B) 564 (C) 568 (D) 1498 (E) 2255

Solution

See also

2010 AMC 12A (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2010))	
Preceded by 2009 AMC 12B Problems	Followed by 2010 AMC 12B Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
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2010 AMC 12A Answer Key

1. C
2. A
3. E
4. D
5. C
6. E
7. C
8. C
9. A
10. A
11. C
12. D
13. C
14. B
15. D
16. B
17. E
18. D
19. A
20. C
21. A
22. A
23. A
24. B
25. C

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2010 AMC 12B Problems

2010 AMC 12B (Answer Key)

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Instructions

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3. No aids are permitted other than scratch paper, graph paper, ruler, compass, protractor and erasers (and calculators that are accepted for use on the test if before 2006. No problems on the test will *require* the use of a calculator).
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1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

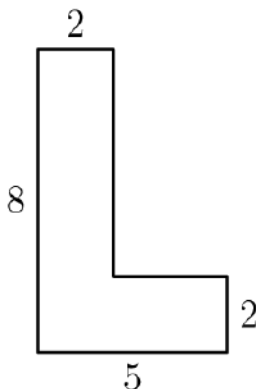
Makarla attended two meetings during her 9-hour work day. The first meeting took 45 minutes and the second meeting took twice as long. What percent of her work day was spent attending meetings?

- (A) 15 (B) 20 (C) 25 (D) 30 (E) 35

Solution

Problem 2

A big L is formed as shown. What is its area?



- (A) 22 (B) 24 (C) 26 (D) 28 (E) 30

Solution

Problem 3

A ticket to a school play cost x dollars, where x is a whole number. A group of 9th graders buys tickets costing a total of \$48, and a group of 10th graders buys tickets costing a total of \$64. How many values for x are possible?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Solution

Problem 4

A month with 31 days has the same number of Mondays and Wednesdays. How many of the seven days of the week could be the first day of this month?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Solution

Problem 5

Lucky Larry's teacher asked him to substitute numbers for a, b, c, d , and e in the expression $a - (b - (c - (d + e)))$ and evaluate the result. Larry ignored the parentheses but added and subtracted correctly and obtained the correct result by coincidence. The numbers Larry substituted for a, b, c , and d were 1, 2, 3, and 4, respectively. What number did Larry substitute for e ?

- (A) -5 (B) -3 (C) 0 (D) 3 (E) 5

Solution

Problem 6

At the beginning of the school year, 50% of all students in Mr. Wells' math class answered "Yes" to the question "Do you love math", and 50% answered "No." At the end of the school year, 70% answered "Yes" and 30% answered "No." Altogether, $x\%$ of the students gave a different answer at the beginning and end of the school year. What is the difference between the maximum and the minimum possible values of x ?

- (A) 0 (B) 20 (C) 40 (D) 60 (E) 80

Solution

Problem 7

Shelby drives her scooter at a speed of 30 miles per hour if it is not raining, and 20 miles per hour if it is raining. Today she drove in the sun in the morning and in the rain in the evening, for a total of 16 miles in 40 minutes. How many minutes did she drive in the rain?

(A) 18 (B) 21 (C) 24 (D) 27 (E) 30

Solution

Problem 8

Every high school in the city of Euclid sent a team of 3 students to a math contest. Each participant in the contest received a different score. Andrea's score was the median among all students, and hers was the highest score on her team. Andrea's teammates Beth and Carla placed 37th and 64th, respectively. How many schools are in the city?

(A) 22 (B) 23 (C) 24 (D) 25 (E) 26

Solution

Problem 9

Let n be the smallest positive integer such that n is divisible by 20, n^2 is a perfect cube, and n^3 is a perfect square. What is the number of digits of n ?

(A) 3 (B) 4 (C) 5 (D) 6 (E) 7

Solution

Problem 10

The average of the numbers 1, 2, 3, ..., 98, 99, and x is $100x$. What is x ?

(A) $\frac{49}{101}$ (B) $\frac{50}{101}$ (C) $\frac{1}{2}$ (D) $\frac{51}{101}$ (E) $\frac{50}{99}$

Solution

Problem 11

A palindrome between 1000 and 10,000 is chosen at random. What is the probability that it is divisible by 7?

(A) $\frac{1}{10}$ (B) $\frac{1}{9}$ (C) $\frac{1}{7}$ (D) $\frac{1}{6}$ (E) $\frac{1}{5}$

Solution

Problem 12

For what value of x does

$$\log_{\sqrt{2}} \sqrt{x} + \log_2 x + \log_4 x^2 + \log_8 x^3 + \log_{16} x^4 = 40?$$

(A) 8 (B) 16 (C) 32 (D) 256 (E) 1024

Solution

Problem 13

In $\triangle ABC$, $\cos(2A - B) + \sin(A + B) = 2$ and $AB = 4$. What is BC ?

(A) $\sqrt{2}$ (B) $\sqrt{3}$ (C) 2 (D) $2\sqrt{2}$ (E) $2\sqrt{3}$

Solution

Problem 14

Let a, b, c, d , and e be positive integers with $a + b + c + d + e = 2010$ and let M be the largest of the sums $a + b$, $b + c$, $c + d$ and $d + e$. What is the smallest possible value of M ?

(A) 670 (B) 671 (C) 802 (D) 803 (E) 804

Solution

Problem 15

For how many ordered triples (x, y, z) of nonnegative integers less than 20 are there exactly two distinct elements in the set $\{i^x, (1+i)^y, z\}$, where $i = \sqrt{-1}$?

(A) 149 (B) 205 (C) 215 (D) 225 (E) 235

Solution

Problem 16

Positive integers a , b , and c are randomly and independently selected with replacement from the set $\{1, 2, 3, \dots, 2010\}$. What is the probability that $abc + ab + a$ is divisible by 3?

(A) $\frac{1}{3}$ (B) $\frac{29}{81}$ (C) $\frac{31}{81}$ (D) $\frac{11}{27}$ (E) $\frac{13}{27}$

Solution

Problem 17

The entries in a 3×3 array include all the digits from 1 through 9, arranged so that the entries in every row and column are in increasing order. How many such arrays are there?

(A) 18 (B) 24 (C) 36 (D) 42 (E) 60

Solution

Problem 18

A frog makes 3 jumps, each exactly 1 meter long. The directions of the jumps are chosen independently at random. What is the probability that the frog's final position is no more than 1 meter from its starting position?

(A) $\frac{1}{6}$ (B) $\frac{1}{5}$ (C) $\frac{1}{4}$ (D) $\frac{1}{3}$ (E) $\frac{1}{2}$

Solution

Problem 19

A high school basketball game between the Raiders and Wildcats was tied at the end of the first quarter. The number of points scored by the Raiders in each of the four quarters formed an increasing geometric sequence, and the number of points scored by the Wildcats in each of the four quarters formed an increasing arithmetic sequence. At the end of the fourth quarter, the Raiders had won by one point. Neither team scored more than 100 points. What was the total number of points scored by the two teams in the first half?

(A) 30 (B) 31 (C) 32 (D) 33 (E) 34

Solution

Problem 20

A geometric sequence (a_n) has $a_1 = \sin x$, $a_2 = \cos x$, and $a_3 = \tan x$ for some real number x . For what value of n does $a_n = 1 + \cos x$?

(A) 4 (B) 5 (C) 6 (D) 7 (E) 8

Solution

Problem 21

Let $a > 0$, and let $P(x)$ be a polynomial with integer coefficients such that

$$P(1) = P(3) = P(5) = P(7) = a, \text{ and} \\ P(2) = P(4) = P(6) = P(8) = -a.$$

What is the smallest possible value of a ?

- (A) 105 (B) 315 (C) 945 (D) 7! (E) 8!

Solution

Problem 22

Let $ABCD$ be a cyclic quadrilateral. The side lengths of $ABCD$ are distinct integers less than 15 such that $BC \cdot CD = AB \cdot DA$. What is the largest possible value of BD ?

- (A) $\sqrt{\frac{325}{2}}$ (B) $\sqrt{185}$ (C) $\sqrt{\frac{389}{2}}$ (D) $\sqrt{\frac{425}{2}}$ (E) $\sqrt{\frac{533}{2}}$

Solution

Problem 23

Monic quadratic polynomials $P(x)$ and $Q(x)$ have the property that $P(Q(x))$ has zeros at $x = -23, -21, -17$, and -15 , and $Q(P(x))$ has zeros at $x = -59, -57, -51$ and -49 . What is the sum of the minimum values of $P(x)$ and $Q(x)$?

- (A) -100 (B) -82 (C) -73 (D) -64 (E) 0

Solution

Problem 24

The set of real numbers x for which

$$\frac{1}{x-2009} + \frac{1}{x-2010} + \frac{1}{x-2011} \geq 1$$

is the union of intervals of the form $a < x \leq b$. What is the sum of the lengths of these intervals?

- (A) $\frac{1003}{335}$ (B) $\frac{1004}{335}$ (C) 3 (D) $\frac{403}{134}$ (E) $\frac{202}{67}$

Solution

Problem 25

For every integer $n \geq 2$, let $\text{pow}(n)$ be the largest power of the largest prime that divides n . For example $\text{pow}(144) = \text{pow}(2^4 \cdot 3^2) = 3^2$. What is the largest integer m such that 2010^m divides

$$\prod_{n=2}^{5300} \text{pow}(n)?$$

- (A) 74 (B) 75 (C) 76 (D) 77 (E) 78

Solution

See also

2010 AMC 12B (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2010))	
Preceded by 2010 AMC 12A Problems	Followed by 2011 AMC 12A Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
All AMC 12 Problems and Solutions	

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2010 AMC 12B Answer Key

1. C
2. A
3. E
4. B
5. D
6. D
7. C
8. B
9. E
10. B
11. E
12. D
13. C
14. B
15. D
16. E
17. D
18. C
19. E
20. E
21. B
22. D
23. A
24. C
25. D

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2011

2011 AMC 12A Problems

2011 AMC 12A (Answer Key)

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Instructions

1. This is a 25-question, multiple choice test. Each question is followed by answers marked A, B, C, D and E. Only one of these is correct.
2. You will receive 6 points for each correct answer, 2.5 points for each problem left unanswered if the year is before 2006, 1.5 points for each problem left unanswered if the year is after 2006, and 0 points for each incorrect answer.
3. No aids are permitted other than scratch paper, graph paper, ruler, compass, protractor and erasers (and calculators that are accepted for use on the test if before 2006. No problems on the test will *require* the use of a calculator).
4. Figures are not necessarily drawn to scale.
5. You will have **75 minutes** working time to complete the test.

1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

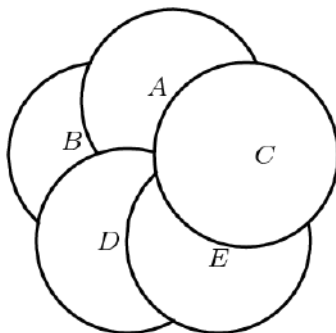
Problem 1

A cell phone plan costs \$20 dollars each month, plus 5 cents per text message sent, plus 10 cents for each minute used over 30 hours. In January Michelle sent 100 text messages and talked for 30.5 hours. How much did she have to pay?
(A) 24.00 **(B)** 24.50 **(C)** 25.50 **(D)** 28.00 **(E)** 30.00

Solution

Problem 2

There are 5 coins placed flat on a table according to the figure. What is the order of the coins from top to bottom?



- (A) (C, A, E, D, B) (B) (C, A, D, E, B) (C) (C, D, E, A, B) (D) (C, E, A, D, B)
 (E) (C, E, D, A, B)

Solution

Problem 3

A small bottle of shampoo can hold 35 milliliters of shampoo, whereas a large bottle can hold 500 milliliters of shampoo. Jasmine wants to buy the minimum number of small bottles necessary to completely fill a large bottle. How many bottles must she buy?

- (A) 11 (B) 12 (C) 13 (D) 14 (E) 15

Solution

Problem 4

At an elementary school, the students in third grade, fourth grade, and fifth grade run an average of 12, 15, and 10 minutes per day, respectively. There are twice as many third graders as fourth graders, and twice as many fourth graders as fifth graders. What is the average number of minutes run per day by these students?

- (A) 12 (B) $\frac{37}{3}$ (C) $\frac{88}{7}$ (D) 13 (E) 14

Solution

Problem 5

Last summer 30% of the birds living on Town Lake were geese, 25% were swans, 10% were herons, and 35% were ducks. What percent of the birds that were not swans were geese?

- (A) 20 (B) 30 (C) 40 (D) 50 (E) 60

Solution

Problem 6

The players on a basketball team made some three-point shots, some two-point shots, and some one-point free throws. They scored as many points with two-point shots as with three-point shots. Their number of successful free throws was one more than their number of successful two-point shots. The team's total score was 61 points. How many free throws did they make?

- (A) 13 (B) 14 (C) 15 (D) 16 (E) 17

Solution

Problem 7

A majority of the 30 students in Ms. Demeanor's class bought pencils at the school bookstore. Each of these students bought the same number of pencils, and this number was greater than 1. The cost of a pencil in cents was greater than the number of pencils each student bought, and the total cost of all the pencils was 17.71. What was the cost of a pencil in cents?

- (A) 7 (B) 11 (C) 17 (D) 23 (E) 77

Solution

Problem 8

In the eight term sequence A, B, C, D, E, F, G, H , the value of C is 5 and the sum of any three consecutive terms is 30. What is $A + H$?

- (A) 17 (B) 18 (C) 25 (D) 26 (E) 43

Solution

Problem 9

At a twins and triplets convention, there were 9 sets of twins and 6 sets of triplets, all from different families. Each twin shook hands with all the twins except his/her siblings and with half the triplets. Each triplet shook hands with all the triplets except his/her siblings and with half the twins. How many handshakes took place?

- (A) 324 (B) 441 (C) 630 (D) 648 (E) 882

Solution

Problem 10

A pair of standard 6-sided dice is rolled once. The sum of the numbers rolled determines the diameter of a circle. What is the probability that the numerical value of the area of the circle is less than the numerical value of the circle's circumference?

- (A) $\frac{1}{36}$ (B) $\frac{1}{12}$ (C) $\frac{1}{6}$ (D) $\frac{1}{4}$ (E) $\frac{5}{18}$

Solution

Problem 11

Circles A , B , and C each have radius 1. Circles A and B share one point of tangency. Circle C has a point of tangency with the midpoint of $\overline{AB}$. What is the area inside circle C but outside circle A and circle B ?

- (A) $3 - \frac{\pi}{2}$ (B) $\frac{\pi}{2}$ (C) 2 (D) $\frac{3\pi}{4}$ (E) $1 + \frac{\pi}{2}$

Solution

Problem 12

A power boat and a raft both left dock A on a river and headed downstream. The raft drifted at the speed of the river current. The power boat maintained a constant speed with respect to the river. The power boat reached dock B downriver, then immediately turned and traveled back upriver. It eventually met the raft on the river 9 hours after leaving dock A . How many hours did it take the power boat to go from A to B ?

- (A) 3 (B) 3.5 (C) 4 (D) 4.5 (E) 5

Solution

Problem 13

Triangle ABC has side-lengths $AB = 12$, $BC = 24$, and $AC = 18$. The line through the incenter of $\triangle ABC$ parallel to $\overline{BC}$ intersects $\overline{AB}$ at M and $\overline{AC}$ at N . What is the perimeter of $\triangle AMN$?

- (A) 27 (B) 30 (C) 33 (D) 36 (E) 42

Solution

Problem 14

Suppose a and b are single-digit positive integers chosen independently and at random. What is the probability that the point (a, b) lies above the parabola $y = ax^2 - bx$?

- (A) $\frac{11}{81}$ (B) $\frac{13}{81}$ (C) $\frac{5}{27}$ (D) $\frac{17}{81}$ (E) $\frac{19}{81}$

Solution

Problem 15

The circular base of a hemisphere of radius 2 rests on the base of a square pyramid of height 6. The hemisphere is tangent to the other four faces of the pyramid. What is the edge-length of the base of the pyramid?

- (A) $3\sqrt{2}$ (B) $\frac{13}{3}$ (C) $4\sqrt{2}$ (D) 6 (E) $\frac{13}{2}$

Solution

Problem 16

Each vertex of convex polygon $ABCDE$ is to be assigned a color. There are 6 colors to choose from, and the ends of each diagonal must have different colors. How many different colorings are possible?

- (A) 2520 (B) 2880 (C) 3120 (D) 3250 (E) 3750

Solution

Problem 17

Circles with radii 1, 2, and 3 are mutually externally tangent. What is the area of the triangle determined by the points of tangency?

- (A) $\frac{3}{5}$ (B) $\frac{4}{5}$ (C) 1 (D) $\frac{6}{5}$ (E) $\frac{4}{3}$

Solution

Problem 18

Suppose that $|x + y| + |x - y| = 2$. What is the maximum possible value of $x^2 - 6x + y^2$?

- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

Solution

Problem 19

At a competition with N players, the number of players given elite status is equal to $2^{1+\lfloor \log_2(N-1) \rfloor} - N$. Suppose that 19 players are given elite status. What is the sum of the two smallest possible values of N ?

- (A) 38 (B) 90 (C) 154 (D) 406 (E) 1024

Solution

Problem 20

Let $f(x) = ax^2 + bx + c$, where a , b , and c are integers. Suppose that $f(1) = 0$, $50 < f(7) < 60$, $70 < f(8) < 80$, $5000k < f(100) < 5000(k+1)$ for some integer k . What is k ?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Solution

Problem 21

Let $f_1(x) = \sqrt{1-x}$, and for integers $n \geq 2$, let $f_n(x) = f_{n-1}(\sqrt{n^2-x})$. If N is the largest value of n for which the domain of f_n is nonempty, the domain of f_N is $[c]$. What is $N + c$?

- (A) -226 (B) -144 (C) -20 (D) 20 (E) 144

Solution

Problem 22

Let R be a square region and $n \geq 4$ an integer. A point X in the interior of R is called n -ray *partitional* if there are n rays emanating from X that divide R into n triangles of equal area. How many points are 100-ray partitional but not 60-ray partitional?

- (A) 1500 (B) 1560 (C) 2320 (D) 2480 (E) 2500

Solution

Problem 23

Let $f(z) = \frac{z+a}{z+b}$ and $g(z) = f(f(z))$, where a and b are complex numbers. Suppose that $|a| = 1$ and $g(g(z)) = z$ for all z for which $g(g(z))$ is defined. What is the difference between the largest and smallest possible values of $|b|$?

- (A) 0 (B) $\sqrt{2} - 1$ (C) $\sqrt{3} - 1$ (D) 1 (E) 2

Solution

Problem 24

Consider all quadrilaterals $ABCD$ such that $AB = 14$, $BC = 9$, $CD = 7$, and $DA = 12$. What is the radius of the largest possible circle that fits inside or on the boundary of such a quadrilateral?

- (A) $\sqrt{15}$ (B) $\sqrt{21}$ (C) $2\sqrt{6}$ (D) 5 (E) $2\sqrt{7}$

Solution

Problem 25

Triangle ABC has $\angle BAC = 60^\circ$, $\angle CBA \leq 90^\circ$, $BC = 1$, and $AC \geq AB$. Let H , I , and O be the orthocenter, incenter, and circumcenter of $\triangle ABC$, respectively. Assume that the area of pentagon $BCOIH$ is the maximum possible. What is $\angle CBA$?

- (A) 60° (B) 72° (C) 75° (D) 80° (E) 90°

Solution

See also

2011 AMC 12A (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2011))	
Preceded by 2010 AMC 12B Problems	Followed by 2011 AMC 12B Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
All AMC 12 Problems and Solutions	

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2011 AMC 12A Answer Key

1. D
2. E
3. E
4. C
5. C
6. A
7. B
8. C
9. B
10. B
11. C
12. D
13. B
14. E
15. A
16. C
17. D
18. D
19. C
20. C
21. A
22. C
23. C
24. C
25. D

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2011 AMC 12B Problems

2011 AMC 12B (Answer Key)

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1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

What is

$$\frac{2 + 4 + 6}{1 + 3 + 5} - \frac{1 + 3 + 5}{2 + 4 + 6}?$$

(A) -1 (B) $\frac{5}{36}$ (C) $\frac{7}{12}$ (D) $\frac{147}{60}$ (E) $\frac{43}{3}$

Solution

Problem 2

Josanna's test scores to date are 90, 80, 70, 60, and 85. Her goal is to raise her test average at least 3 points with her next test. What is the minimum test score she would need to accomplish this goal?

(A) 80 (B) 82 (C) 85 (D) 90 (E) 95

Solution

Problem 3

LeRoy and Bernardo went on a week-long trip together and agreed to share the costs equally. Over the week, each of them paid for various joint expenses such as gasoline and car rental. At the end of the trip it turned out that LeRoy had paid A dollars and Bernardo had paid B dollars, where $A < B$. How many dollars must LeRoy give to Bernardo so that they share the costs equally?

(A) $\frac{A+B}{2}$ (B) $\frac{A-B}{2}$ (C) $\frac{B-A}{2}$ (D) $B-A$ (E) $A+B$

Solution

Problem 4

In multiplying two positive integers a and b , Ron reversed the digits of the two-digit number a . His erroneous product was 161. What is the correct value of the product of a and b ?

(A) 116 (B) 161 (C) 204 (D) 214 (E) 224

Solution

Problem 5

Let N be the second smallest positive integer that is divisible by every positive integer less than 7. What is the sum of the digits of N ?

(A) 3 (B) 4 (C) 5 (D) 6 (E) 9

Solution

Problem 6

Two tangents to a circle are drawn from a point A . The points of contact B and C divide the circle into arcs with lengths in the ratio 2 : 3. What is the degree measure of $\angle BAC$?

(A) 24 (B) 30 (C) 36 (D) 48 (E) 60

Solution

Problem 7

Let x and y be two-digit positive integers with mean 60. What is the maximum value of the ratio $\frac{x}{y}$?

(A) 3 (B) $\frac{33}{7}$ (C) $\frac{39}{7}$ (D) 9 (E) $\frac{99}{10}$

Solution

Problem 8

Keiko walks once around a track at exactly the same constant speed every day. The sides of the track are straight, and the ends are semicircles. The track has width 6 meters, and it takes her 36 seconds longer to walk around the outside edge of the track than around the inside edge. What is Keiko's speed in meters per second?

- (A) $\frac{\pi}{3}$ (B) $\frac{2\pi}{3}$ (C) π (D) $\frac{4\pi}{3}$ (E) $\frac{5\pi}{3}$

Solution

Problem 9

Two real numbers are selected independently and at random from the interval $[-20, 10]$. What is the probability that the product of those numbers is greater than zero?

- (A) $\frac{1}{9}$ (B) $\frac{1}{3}$ (C) $\frac{4}{9}$ (D) $\frac{5}{9}$ (E) $\frac{2}{3}$

Solution

Problem 10

Rectangle $ABCD$ has $AB = 6$ and $BC = 3$. Point M is chosen on side AB so that $\angle AMD = \angle CMD$. What is the degree measure of $\angle AMD$?

- (A) 15 (B) 30 (C) 45 (D) 60 (E) 75

Solution

Problem 11

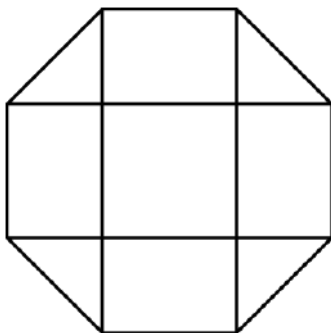
A frog located at (x, y) , with both x and y integers, makes successive jumps of length 5 and always lands on points with integer coordinates. Suppose that the frog starts at $(0, 0)$ and ends at $(1, 0)$. What is the smallest possible number of jumps the frog makes?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Solution

Problem 12

A dart board is a regular octagon divided into regions as shown below. Suppose that a dart thrown at the board is equally likely to land anywhere on the board. What is the probability that the dart lands within the center square?



- (A) $\frac{\sqrt{2}-1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{2-\sqrt{2}}{2}$ (D) $\frac{\sqrt{2}}{4}$ (E) $2-\sqrt{2}$

Solution

Problem 13

Brian writes down four integers $w > x > y > z$ whose sum is 44. The pairwise positive differences of these numbers are 1, 3, 4, 5, 6 and 9. What is the sum of the possible values of w ?

- (A) 16 (B) 31 (C) 48 (D) 62 (E) 93

Solution

Problem 14

A segment through the focus F of a parabola with vertex V is perpendicular to $\overline{FV}$ and intersects the parabola in points A and B . What is $\cos(\angle AVB)$?

- (A) $-\frac{3\sqrt{5}}{7}$ (B) $-\frac{2\sqrt{5}}{5}$ (C) $-\frac{4}{5}$ (D) $-\frac{3}{5}$ (E) $-\frac{1}{2}$

Solution

Problem 15

How many positive two-digit integers are factors of $2^{24} - 1$?

- (A) 4 (B) 8 (C) 10 (D) 12 (E) 14

Solution

Problem 16

Rhombus $ABCD$ has side length 2 and $\angle B = 120^\circ$. Region R consists of all points inside of the rhombus that are closer to vertex B than any of the other three vertices. What is the area of R ?

- (A) $\frac{\sqrt{3}}{3}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{2\sqrt{3}}{3}$ (D) $1 + \frac{\sqrt{3}}{3}$ (E) 2

Solution

Problem 17

Let $f(x) = 10^{10x}$, $g(x) = \log_{10}\left(\frac{x}{10}\right)$, $h_1(x) = g(f(x))$, and $h_n(x) = h_1(h_{n-1}(x))$ for integers $n \geq 2$. What is the sum of the digits of $h_{2011}(1)$?

- (A) 16081 (B) 16089 (C) 18089 (D) 18098 (E) 18099

Solution

Problem 18

A pyramid has a square base with side of length 1 and has lateral faces that are equilateral triangles. A cube is placed within the pyramid so that one face is on the base of the pyramid and its opposite face has all its edges on the lateral faces of the pyramid. What is the volume of this cube?

- (A) $5\sqrt{2} - 7$ (B) $7 - 4\sqrt{3}$ (C) $\frac{2\sqrt{2}}{27}$ (D) $\frac{\sqrt{2}}{9}$ (E) $\frac{\sqrt{3}}{9}$

Solution

Problem 19

A lattice point in an xy -coordinate system is any point (x, y) where both x and y are integers. The graph of $y = mx + 2$ passes through no lattice point with $0 < x \leq 100$ for all m such that $\frac{1}{2} < m < a$. What is the maximum possible value of a ?

- (A) $\frac{51}{101}$ (B) $\frac{50}{99}$ (C) $\frac{51}{100}$ (D) $\frac{52}{101}$ (E) $\frac{13}{25}$

Solution

Problem 20

Triangle ABC has $AB = 13$, $BC = 14$, and $AC = 15$. The points D , E , and F are the midpoints of $\overline{AB}$, $\overline{BC}$, and $\overline{AC}$ respectively. Let $X \neq E$ be the intersection of the circumcircles of $\triangle BDE$ and $\triangle CEF$. What is $XA + XB + XC$?

- (A) 24 (B) $14\sqrt{3}$ (C) $\frac{195}{8}$ (D) $\frac{129\sqrt{7}}{14}$ (E) $\frac{69\sqrt{2}}{4}$

Solution

Problem 21

The arithmetic mean of two distinct positive integers x and y is a two-digit integer. The geometric mean of x and y is obtained by reversing the digits of the arithmetic mean. What is $|x - y|$?

- (A) 24 (B) 48 (C) 54 (D) 66 (E) 70

Solution

Problem 22

Let T_1 be a triangle with side lengths 2011, 2012, and 2013. For $n \geq 1$, if $T_n = \triangle ABC$ and D , E , and F are the points of tangency of the incircle of $\triangle ABC$ to the sides AB , BC , and AC , respectively, then T_{n+1} is a triangle with side lengths AD , BE , and CF , if it exists. What is the perimeter of the last triangle in the sequence (T_n) ?

- (A) $\frac{1509}{8}$ (B) $\frac{1509}{32}$ (C) $\frac{1509}{64}$ (D) $\frac{1509}{128}$ (E) $\frac{1509}{256}$

Solution

Problem 23

A bug travels in the coordinate plane, moving only along the lines that are parallel to the x -axis or y -axis. Let $A = (-3, 2)$ and $B = (3, -2)$. Consider all possible paths of the bug from A to B of length at most 20. How many points with integer coordinates lie on at least one of these paths?

- (A) 161 (B) 185 (C) 195 (D) 227 (E) 255

Solution

Problem 24

Let $P(z) = z^8 + (4\sqrt{3} + 6)z^4 - (4\sqrt{3} + 7)$. What is the minimum perimeter among all the 8-sided polygons in the complex plane whose vertices are precisely the zeros of $P(z)$?

- (A) $4\sqrt{3} + 4$ (B) $8\sqrt{2}$ (C) $3\sqrt{2} + 3\sqrt{6}$ (D) $4\sqrt{2} + 4\sqrt{3}$ (E) $4\sqrt{3} + 6$

Solution

Problem 25

For every m and k integers with k odd, denote by $\left[\frac{m}{k}\right]$ the integer closest to $\frac{m}{k}$. For every odd integer k , let $P(k)$ be the probability that

$$\left[\frac{n}{k}\right] + \left[\frac{100-n}{k}\right] = \left[\frac{100}{k}\right]$$

for an integer n randomly chosen from the interval $1 \leq n \leq 99$. What is the minimum possible value of $P(k)$ over the odd integers k in the interval $1 \leq k \leq 99$?

- (A) $\frac{1}{2}$ (B) $\frac{50}{99}$ (C) $\frac{44}{87}$ (D) $\frac{34}{67}$ (E) $\frac{7}{13}$

See also

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Preceded by 2011 AMC 12A Problems	Followed by 2012 AMC 12A Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
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2011 AMC 12B Answer Key

1. C
2. E
3. C
4. E
5. A
6. C
7. B
8. A
9. D
10. E
11. B
12. A
13. B
14. D
15. D
16. C
17. B
18. A
19. B
20. C
21. D
22. D
23. C
24. B
25. D

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2012

2012 AMC 12A Problems

2012 AMC 12A (Answer Key)

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Instructions

1. This is a 25-question, multiple choice test. Each question is followed by answers marked A, B, C, D and E. Only one of these is correct.
2. You will receive 6 points for each correct answer, 2.5 points for each problem left unanswered if the year is before 2006, 1.5 points for each problem left unanswered if the year is after 2006, and 0 points for each incorrect answer.
3. No aids are permitted other than scratch paper, graph paper, ruler, compass, protractor and erasers (and calculators that are accepted for use on the test if before 2006. No problems on the test will *require* the use of a calculator).
4. Figures are not necessarily drawn to scale.
5. You will have **75 minutes** working time to complete the test.

1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

A bug crawls along a number line, starting at -2 . It crawls to -6 , then turns around and crawls to 5 . How many units does the bug crawl altogether?

- (A) 9 (B) 11 (C) 13 (D) 14 (E) 15

Solution

Problem 2

Cagney can frost a cupcake every 20 seconds and Lacey can frost a cupcake every 30 seconds. Working together, how many cupcakes can they frost in 5 minutes?

- (A) 10 (B) 15 (C) 20 (D) 25 (E) 30

Solution

Problem 3

A box 2 centimeters high, 3 centimeters wide, and 5 centimeters long can hold 40 grams of clay. A second box with twice the height, three times the width, and the same length as the first box can hold n grams of clay. What is n ?

- (A) 120 (B) 160 (C) 200 (D) 240 (E) 280

Solution

Problem 4

In a bag of marbles, $\frac{3}{5}$ of the marbles are blue and the rest are red. If the number of red marbles is doubled and the number of blue marbles stays the same, what fraction of the marbles will be red?

- (A) $\frac{2}{5}$ (B) $\frac{3}{7}$ (C) $\frac{4}{7}$ (D) $\frac{3}{5}$ (E) $\frac{4}{5}$

Solution

Problem 5

A fruit salad consists of blueberries, raspberries, grapes, and cherries. The fruit salad has a total of 280 pieces of fruit. There are twice as many raspberries as blueberries, three times as many grapes as cherries, and four times as many cherries as raspberries. How many cherries are there in the fruit salad?

- (A) 8 (B) 16 (C) 25 (D) 64 (E) 96

Solution

Problem 6

The sums of three whole numbers taken in pairs are 12, 17, and 19. What is the middle number?

- (A) 4 (B) 5 (C) 6 (D) 7 (E) 8

Solution

Problem 7

Mary divides a circle into 12 sectors. The central angles of these sectors, measured in degrees, are all integers and they form an arithmetic sequence. What is the degree measure of the smallest possible sector angle?

- (A) 5 (B) 6 (C) 8 (D) 10 (E) 12

Solution

Problem 8

An *iterative average* of the numbers 1, 2, 3, 4, and 5 is computed in the following way. Arrange the five numbers in some order. Find the mean of the first two numbers, then find the mean of that with the third number, then the mean of that with the fourth number, and finally the mean of that with the fifth number. What is the difference between the largest and smallest possible values that can be obtained using this procedure?

- (A) $\frac{31}{16}$ (B) 2 (C) $\frac{17}{8}$ (D) 3 (E) $\frac{65}{16}$

Solution

Problem 9

A year is a leap year if and only if the year number is divisible by 400 (such as 2000) or is divisible by 4 but not by 100 (such as 2012). The 200th anniversary of the birth of novelist Charles Dickens was celebrated on February 7, 2012, a Tuesday. On what day of the week was Dickens born?

- (A) Friday (B) Saturday (C) Sunday (D) Monday (E) Tuesday

Solution

Problem 10

A triangle has area 30, one side of length 10, and the median to that side of length 9. Let θ be the acute angle formed by that side and the median. What is $\sin \theta$?

- (A) $\frac{3}{10}$ (B) $\frac{1}{3}$ (C) $\frac{9}{20}$ (D) $\frac{2}{3}$ (E) $\frac{9}{10}$

Solution

Problem 11

Alex, Mel, and Chelsea play a game that has 6 rounds. In each round there is a single winner, and the outcomes of the rounds are independent. For each round the probability that Alex wins is $\frac{1}{2}$, and Mel is twice as likely to win as Chelsea. What is the probability that Alex wins three rounds, Mel wins two rounds, and Chelsea wins one round?

- (A) $\frac{5}{72}$ (B) $\frac{5}{36}$ (C) $\frac{1}{6}$ (D) $\frac{1}{3}$ (E) 1

Solution

Problem 12

A square region $ABCD$ is externally tangent to the circle with equation $x^2 + y^2 = 1$ at the point $(0, 1)$ on the side CD . Vertices A and B are on the circle with equation $x^2 + y^2 = 4$. What is the side length of this square?

- (A) $\frac{\sqrt{10} + 5}{10}$ (B) $\frac{2\sqrt{5}}{5}$ (C) $\frac{2\sqrt{2}}{3}$ (D) $\frac{2\sqrt{19} - 4}{5}$ (E) $\frac{9 - \sqrt{17}}{5}$

Solution

Problem 13

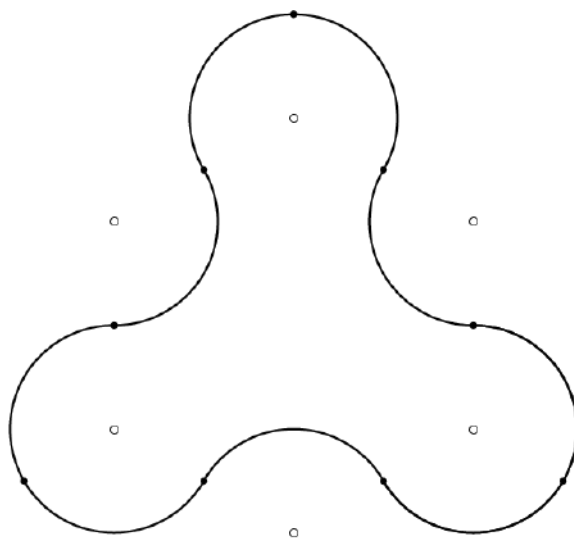
Paula the painter and her two helpers each paint at constant, but different, rates. They always start at 8:00 AM, and all three always take the same amount of time to eat lunch. On Monday the three of them painted 50% of a house, quitting at 4:00 PM. On Tuesday, when Paula wasn't there, the two helpers painted only 24% of the house and quit at 2:12 PM. On Wednesday Paula worked by herself and finished the house by working until 7:12 PM. How long, in minutes, was each day's lunch break?

- (A) 30 (B) 36 (C) 42 (D) 48 (E) 60

Solution

Problem 14

The closed curve in the figure is made up of 9 congruent circular arcs each of length $\frac{2\pi}{3}$, where each of the centers of the corresponding circles is among the vertices of a regular hexagon of side 2. What is the area enclosed by the curve?



- (A) $2\pi + 6$ (B) $2\pi + 4\sqrt{3}$ (C) $3\pi + 4$ (D) $2\pi + 3\sqrt{3} + 2$ (E) $\pi + 6\sqrt{3}$

Solution

Problem 15

A 3×3 square is partitioned into 9 unit squares. Each unit square is painted either white or black with each color being equally likely, chosen independently and at random. The square is then rotated 90° clockwise about its center, and every white square in a position formerly occupied by a black square is painted black. The colors of all other squares are left unchanged. What is the probability that the grid is now entirely black?

- (A) $\frac{49}{512}$ (B) $\frac{7}{64}$ (C) $\frac{121}{1024}$ (D) $\frac{81}{512}$ (E) $\frac{9}{32}$

Solution

Problem 16

Circle C_1 has its center O lying on circle C_2 . The two circles meet at X and Y . Point Z in the exterior of C_1 lies on circle C_2 and $XZ = 13$, $OZ = 11$, and $YZ = 7$. What is the radius of circle C_1 ?

- (A) 5 (B) $\sqrt{26}$ (C) $3\sqrt{3}$ (D) $2\sqrt{7}$ (E) $\sqrt{30}$

Solution

Problem 17

Let S be a subset of $\{1, 2, 3, \dots, 30\}$ with the property that no pair of distinct elements in S has a sum divisible by 5. What is the largest possible size of S ?

- (A) 10 (B) 13 (C) 15 (D) 16 (E) 18

Solution

Problem 18

Triangle ABC has $AB = 27$, $AC = 26$, and $BC = 25$. Let I denote the intersection of the internal angle bisectors of $\triangle ABC$. What is BI ?

- (A) 15 (B) $5 + \sqrt{26} + 3\sqrt{3}$ (C) $3\sqrt{26}$ (D) $\frac{2}{3}\sqrt{546}$ (E) $9\sqrt{3}$

Solution

Problem 19

Adam, Benin, Chiang, Deshawn, Esther, and Fiona have internet accounts. Some, but not all, of them are internet friends with each other, and none of them has an internet friend outside this group. Each of them has the same number of internet friends. In how many different ways can this happen?

- (A) 60 (B) 170 (C) 290 (D) 320 (E) 660

Solution

Problem 20

Consider the polynomial

$$P(x) = \prod_{k=0}^{10} (x^{2^k} + 2^k) = (x+1)(x^2+2)(x^4+4) \cdots (x^{1024}+1024)$$

The coefficient of x^{2012} is equal to 2^a . What is a ?

- (A) 5 (B) 6 (C) 7 (D) 10 (E) 24

Solution

Problem 21

Let a , b , and c be positive integers with $a \geq b \geq c$ such that

$$\begin{aligned} a^2 - b^2 - c^2 + ab &= 2011 \text{ and} \\ a^2 + 3b^2 + 3c^2 - 3ab - 2ac - 2bc &= -1997 \end{aligned}$$

What is a ?

- (A) 249 (B) 250 (C) 251 (D) 252 (E) 253

Solution

Problem 22

Distinct planes $p_1, p_2, \dots, p_k$ intersect the interior of a cube Q . Let S be the union of the faces of Q and let $P = \bigcup_{j=1}^k p_j$.

The intersection of P and S consists of the union of all segments joining the midpoints of every pair of edges belonging to the same face of Q . What is the difference between the maximum and minimum possible values of k ?

- (A) 8 (B) 12 (C) 20 (D) 23 (E) 24

Solution

Problem 23

Let S be the square one of whose diagonals has endpoints $(0.1, 0.7)$ and $(-0.1, -0.7)$. A point $v = (x, y)$ is chosen uniformly at random over all pairs of real numbers x and y such that $0 \leq x \leq 2012$ and $0 \leq y \leq 2012$. Let $T(v)$ be a translated copy of S centered at v . What is the probability that the square region determined by $T(v)$ contains exactly two points with integer coordinates in its interior?

- (A) 0.125 (B) 0.14 (C) 0.16 (D) 0.25 (E) 0.32

Solution

Problem 24

Let $\{a_k\}_{k=1}^{2011}$ be the sequence of real numbers defined by $a_1 = 0.201$, $a_2 = (0.2011)^{a_1}$, $a_3 = (0.20101)^{a_2}$, $a_4 = (0.201011)^{a_3}$, and in general,

$$a_k = \begin{cases} (0.\underbrace{20101 \dots 0101}_{k+2 \text{ digits}})^{a_{k-1}} & \text{if } k \text{ is odd,} \\ (0.\underbrace{20101 \dots 01011}_{k+2 \text{ digits}})^{a_{k-1}} & \text{if } k \text{ is even.} \end{cases}$$

Rearranging the numbers in the sequence $\{a_k\}_{k=1}^{2011}$ in decreasing order produces a new sequence $\{b_k\}_{k=1}^{2011}$. What is the sum of all integers k , $1 \leq k \leq 2011$, such that $a_k = b_k$?

- (A) 671 (B) 1006 (C) 1341 (D) 2011 (E) 2012

Solution

Problem 25

Let $f(x) = |2\{x\} - 1|$ where $\{x\}$ denotes the fractional part of x . The number n is the smallest positive integer such that the equation

$$nf(xf(x)) = x$$

has at least 2012 real solutions. What is n ? **Note:** the fractional part of x is a real number $y = \{x\}$ such that $0 \leq y < 1$ and $x - y$ is an integer.

- (A) 30 (B) 31 (C) 32 (D) 62 (E) 64

Solution

See also

2012 AMC 12A (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2012))	
Preceded by 2011 AMC 12B Problems	Followed by 2012 AMC 12B Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
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2012 AMC 12A Answer Key

1. E
2. D
3. D
4. C
5. D
6. D
7. C
8. C
9. A
10. D
11. B
12. D
13. D
14. E
15. A
16. E
17. B
18. A
19. B
20. B
21. E
22. C
23. C
24. C
25. C

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2012 AMC 12B Problems

2012 AMC 12B (Answer Key)

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1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

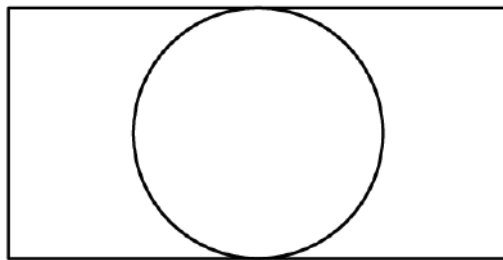
Each third-grade classroom at Pearl Creek Elementary has 18 students and 2 pet rabbits. How many more students than rabbits are there in all 4 of the third-grade classrooms?

(A) 48 (B) 56 (C) 64 (D) 72 (E) 80

Solution

Problem 2

A circle of radius 5 is inscribed in a rectangle as shown. The ratio of the length of the rectangle to its width is 2:1. What is the area of the rectangle?



- (A) 50 (B) 100 (C) 125 (D) 150 (E) 200

Solution

Problem 3

For a science project, Sammy observed a chipmunk and squirrel stashing acorns in holes. The chipmunk hid 3 acorns in each of the holes it dug. The squirrel hid 4 acorns in each of the holes it dug. They each hid the same number of acorns, although the squirrel needed 4 fewer holes. How many acorns did the chipmunk hide?

- (A) 30 (B) 36 (C) 42 (D) 48 (E) 54

Solution

Problem 4

Suppose that the euro is worth 1.30 dollars. If Diana has 500 dollars and Etienne has 400 euros, by what percent is the value of Etienne's money greater than the value of Diana's money?

- (A) 2 (B) 4 (C) 6.5 (D) 8 (E) 13

Solution

Problem 5

Two integers have a sum of 26. When two more integers are added to the first two, the sum is 41. Finally, when two more integers are added to the sum of the previous 4 integers, the sum is 57. What is the minimum number of even integers among the 6 integers?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Solution

Problem 6

In order to estimate the value of $x - y$ where x and y are real numbers with $x > y > 0$, Xiaoli rounded x up by a small amount, rounded y down by the same amount, and then subtracted her rounded values. Which of the following statements is necessarily correct?

- (A) Her estimate is larger than $x - y$
(B) Her estimate is smaller than $x - y$
(C) Her estimate equals $x - y$
(D) Her estimate equals $y - x$
(E) Her estimate is 0

Solution

Problem 7

Small lights are hung on a string 6 inches apart in the order red, red, green, green, green, red, red, green, green, green, and so on continuing this pattern of 2 red lights followed by 3 green lights. How many feet separate the 3rd red light and the 21st red light?

Note: 1 foot is equal to 12 inches.

- (A) 18 (B) 18.5 (C) 20 (D) 20.5 (E) 22.5

Solution

Problem 8

A dessert chef prepares the dessert for every day of a week starting with Sunday. The dessert each day is either cake, pie, ice cream, or pudding. The same dessert may not be served two days in a row. There must be cake on Friday because of a birthday. How many different dessert menus for the week are possible?

- (A) 729 (B) 972 (C) 1024 (D) 2187 (E) 2304

Solution

Problem 9

It takes Clea 60 seconds to walk down an escalator when it is not moving, and 24 seconds when it is moving. How many seconds would it take Clea to ride the escalator down when she is not walking?

- (A) 36 (B) 40 (C) 42 (D) 48 (E) 52

Solution

Problem 10

What is the area of the polygon whose vertices are the points of intersection of the curves $x^2 + y^2 = 25$ and $(x - 4)^2 + 9y^2 = 81$?

- (A) 24 (B) 27 (C) 36 (D) 37.5 (E) 42

Solution

Problem 11

In the equation below, A and B are consecutive positive integers, and A , B , and $A + B$ represent number bases:

$$132_A + 43_B = 69_{A+B}.$$

What is $A + B$?

- (A) 9 (B) 11 (C) 13 (D) 15 (E) 17

Solution

Problem 12

How many sequences of zeros and ones of length 20 have all the zeros consecutive, or all the ones consecutive, or both?

- (A) 190 (B) 192 (C) 211 (D) 380 (E) 382

Solution

Problem 13

Two parabolas have equations $y = x^2 + ax + b$ and $y = x^2 + cx + d$, where a , b , c , and d are integers, each chosen independently by rolling a fair six-sided die. What is the probability that the parabolas will have at least one point in common?

- (A) $\frac{1}{2}$ (B) $\frac{25}{36}$ (C) $\frac{5}{6}$ (D) $\frac{31}{36}$ (E) 1

Solution

Problem 14

Bernardo and Silvia play the following game. An integer between 0 and 999 inclusive is selected and given to Bernardo. Whenever Bernardo receives a number, he doubles it and passes the result to Silvia. Whenever Silvia receives a number, she adds 50 to it and passes the result to Bernardo. The winner is the last person who produces a number less than 1000. Let N be the smallest initial number that results in a win for Bernardo. What is the sum of the digits of N ?

(A) 7 (B) 8 (C) 9 (D) 10 (E) 11

Solution

Problem 15

Jesse cuts a circular paper disk of radius 12 along two radii to form two sectors, the smaller having a central angle of 120 degrees. He makes two circular cones, using each sector to form the lateral surface of a cone. What is the ratio of the volume of the smaller cone to that of the larger?

(A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{\sqrt{10}}{10}$ (D) $\frac{\sqrt{5}}{6}$ (E) $\frac{\sqrt{5}}{5}$

Solution

Problem 16

Amy, Beth, and Jo listen to four different songs and discuss which ones they like. No song is liked by all three. Furthermore, for each of the three pairs of the girls, there is at least one song liked by those girls but disliked by the third. In how many different ways is this possible?

(A) 108 (B) 132 (C) 671 (D) 846 (E) 1105

Solution

Problem 17

Square $PQRS$ lies in the first quadrant. Points $(3, 0)$, $(5, 0)$, $(7, 0)$, and $(13, 0)$ lie on lines SP , RQ , PQ , and SR , respectively. What is the sum of the coordinates of the center of the square $PQRS$?

(A) 6 (B) 6.2 (C) 6.4 (D) 6.6 (E) 6.8

Solution

Problem 18

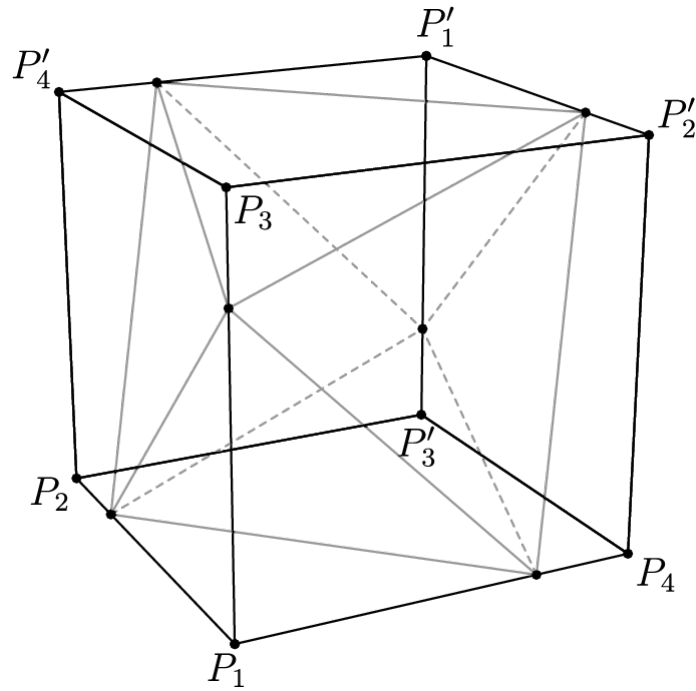
Let $(a_1, a_2, \dots, a_{10})$ be a list of the first 10 positive integers such that for each $2 \leq i \leq 10$ either $a_i + 1$ or $a_i - 1$ or both appear somewhere before a_i in the list. How many such lists are there?

(A) 120 (B) 512 (C) 1024 (D) 181,440 (E) 362,880

Solution

Problem 19

A unit cube has vertices $P_1, P_2, P_3, P_4, P'_1, P'_2, P'_3$, and P'_4 . Vertices P_2, P_3 , and P_4 are adjacent to P_1 , and for $1 \leq i \leq 4$, vertices P_i and P'_i are opposite to each other. A regular octahedron has one vertex in each of the segments P_1P_2 , P_1P_3 , P_1P_4 , $P'_1P'_2$, $P'_1P'_3$, and $P'_1P'_4$. What is the octahedron's side length?



- (A) $\frac{3\sqrt{2}}{4}$ (B) $\frac{7\sqrt{6}}{16}$ (C) $\frac{\sqrt{5}}{2}$ (D) $\frac{2\sqrt{3}}{3}$ (E) $\frac{\sqrt{6}}{2}$

Solution

Problem 20

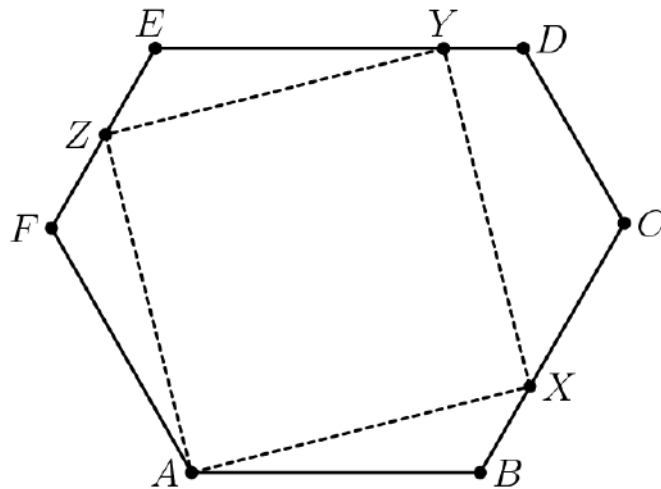
A trapezoid has side lengths 3, 5, 7, and 11. The sums of all the possible areas of the trapezoid can be written in the form of $r_1\sqrt{n_1} + r_2\sqrt{n_2} + r_3$, where r_1, r_2 and r_3 are rational numbers and n_1 and n_2 are positive integers not divisible by the square of any prime. What is the greatest integer less than or equal to $r_1 + r_2 + r_3 + n_1 + n_2$?

- (A) 57 (B) 59 (C) 61 (D) 63 (E) 65

Solution

Problem 21

Square $AXYZ$ is inscribed in equiangular hexagon $ABCDEF$ with X on $\overline{BC}$, Y on $\overline{DE}$, and Z on $\overline{EF}$. Suppose that $AB = 40$, and $EF = 41(\sqrt{3} - 1)$. What is the side-length of the square?

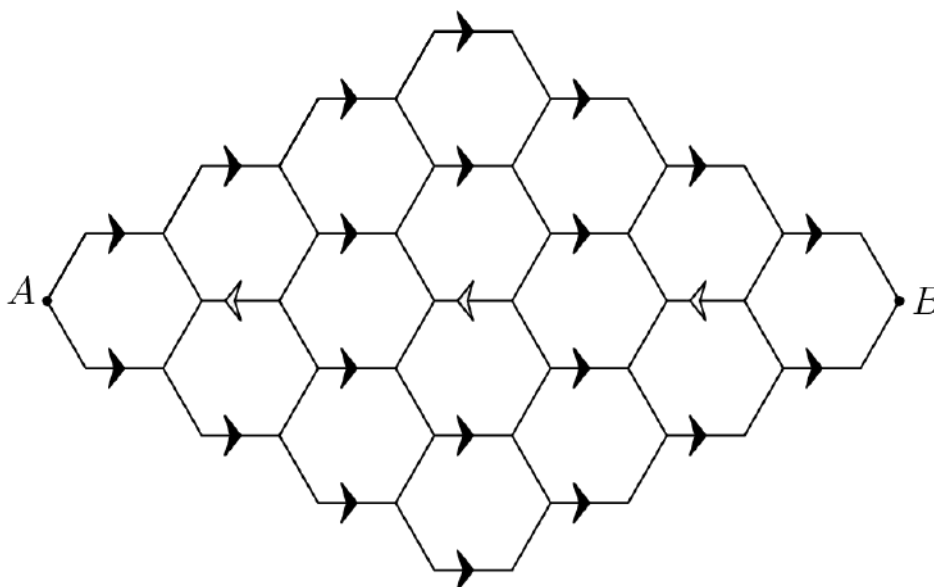


- (A) $29\sqrt{3}$ (B) $\frac{21}{2}\sqrt{2} + \frac{41}{2}\sqrt{3}$ (C) $20\sqrt{3} + 16$
 (D) $20\sqrt{2} + 13\sqrt{3}$ (E) $21\sqrt{6}$

Solution

Problem 22

A bug travels from A to B along the segments in the hexagonal lattice pictured below. The segments marked with an arrow can be traveled only in the direction of the arrow, and the bug never travels the same segment more than once. How many different paths are there?



- (A) 2112 (B) 2304 (C) 2368 (D) 2384 (E) 2400

Solution

Problem 23

Consider all polynomials of a complex variable, $P(z) = 4z^4 + az^3 + bz^2 + cz + d$, where a, b, c , and d are integers, $0 \leq d \leq c \leq b \leq a \leq 4$, and the polynomial has a zero z_0 with $|z_0| = 1$. What is the sum of all values $P(1)$ over all the polynomials with these properties?

- (A) 84 (B) 92 (C) 100 (D) 108 (E) 120

Solution

Problem 24

Define the function f_1 on the positive integers by setting $f_1(1) = 1$ and if $n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$ is the prime factorization of $n > 1$, then

$$f_1(n) = (p_1 + 1)^{e_1 - 1} (p_2 + 1)^{e_2 - 1} \cdots (p_k + 1)^{e_k - 1}.$$

For every $m \geq 2$, let $f_m(n) = f_1(f_{m-1}(n))$. For how many N in the range $1 \leq N \leq 400$ is the sequence $(f_1(N), f_2(N), f_3(N), \dots)$ unbounded?

Note: A sequence of positive numbers is unbounded if for every integer B , there is a member of the sequence greater than B .

- (A) 15 (B) 16 (C) 17 (D) 18 (E) 19

Solution

Problem 25

Let $S = \{(x, y) : x \in \{0, 1, 2, 3, 4\}, y \in \{0, 1, 2, 3, 4, 5\}, \text{ and } (x, y) \neq (0, 0)\}$. Let T be the set of all right triangles whose vertices are in S . For every right triangle $t = \triangle ABC$ with vertices A, B , and C in counter-clockwise order and right angle at A , let $f(t) = \tan(\angle CBA)$. What is

$$\prod_{t \in T} f(t)?$$

- (A) 1
- (B) $\frac{625}{144}$
- (C) $\frac{125}{24}$
- (D) 6
- (E) $\frac{625}{24}$

Solution

See also

2012 AMC 12B (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2012))	
Preceded by 2012 AMC 12A Problems	Followed by 2013 AMC 12A Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
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2012 AMC 12B Answer Key

1. C
2. E
3. D
4. B
5. A
6. A
7. E
8. A
9. B
10. B
11. C
12. D or E (both were accepted)
13. D
14. A
15. C
16. B
17. C
18. B
19. A
20. D
21. A
22. E
23. B
24. D
25. B

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2013

2013 AMC 12A Problems

2013 AMC 12A (Answer Key)

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Instructions

1. This is a 25-question, multiple choice test. Each question is followed by answers marked A, B, C, D and E. Only one of these is correct.
2. You will receive 6 points for each correct answer, 2.5 points for each problem left unanswered if the year is before 2006, 1.5 points for each problem left unanswered if the year is after 2006, and 0 points for each incorrect answer.
3. No aids are permitted other than scratch paper, graph paper, ruler, compass, protractor and erasers (and calculators that are accepted for use on the test if before 2006. No problems on the test will *require* the use of a calculator).
4. Figures are not necessarily drawn to scale.
5. You will have **75 minutes** working time to complete the test.

1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

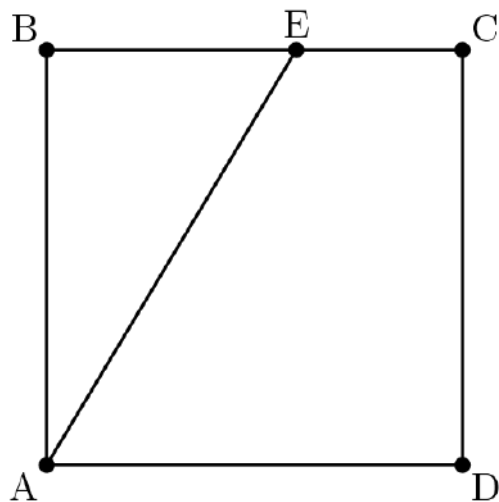
Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

Square $ABCD$ has side length 10. Point E is on $\overline{BC}$, and the area of $\triangle ABE$ is 40. What is BE ?

- (A) 4 (B) 5 (C) 6 (D) 7 (E) 8



Solution

Problem 2

A softball team played ten games, scoring 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10 runs. They lost by one run in exactly five games. In each of the other games, they scored twice as many runs as their opponent. How many total runs did their opponents score?

- (A) 35 (B) 40 (C) 45 (D) 50 (E) 55

Solution

Problem 3

A flower bouquet contains pink roses, red roses, pink carnations, and red carnations. One third of the pink flowers are roses, three fourths of the red flowers are carnations, and six tenths of the flowers are pink. What percent of the flowers are carnations?

- (A) 15 (B) 30 (C) 40 (D) 60 (E) 70

Solution

Problem 4

What is the value of

$$\frac{2^{2014} + 2^{2012}}{2^{2014} - 2^{2012}}?$$

- (A) -1 (B) 1 (C) $\frac{5}{3}$ (D) 2013 (E) 2^{4024}

Solution

Problem 5

Tom, Dorothy, and Sammy went on a vacation and agreed to split the costs evenly. During their trip Tom paid \$105, Dorothy paid \$125, and Sammy paid \$175. In order to share the costs equally, Tom gave Sammy t dollars, and Dorothy gave Sammy d dollars. What is $t - d$?

- (A) 15 (B) 20 (C) 25 (D) 30 (E) 35

Solution

Problem 6

In a recent basketball game, Shenille attempted only three-point shots and two-point shots. She was successful on 20% of her three-point shots and 30% of her two-point shots. Shenille attempted 30 shots. How many points did she score?

- (A) 12 (B) 18 (C) 24 (D) 30 (E) 36

Solution

Problem 7

The sequence $S_1, S_2, S_3, \dots, S_{10}$ has the property that every term beginning with the third is the sum of the previous two. That is,

$$S_n = S_{n-2} + S_{n-1} \text{ for } n \geq 3.$$

Suppose that $S_9 = 110$ and $S_7 = 42$. What is S_4 ?

- (A) 4 (B) 6 (C) 10 (D) 12 (E) 16

Solution

Problem 8

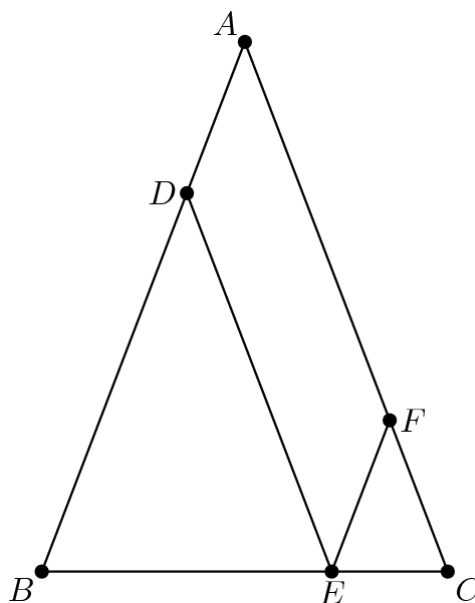
Given that x and y are distinct nonzero real numbers such that $x + \frac{2}{x} = y + \frac{2}{y}$, what is xy ?

- (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 1 (D) 2 (E) 4

Solution

Problem 9

In $\triangle ABC$, $AB = AC = 28$ and $BC = 20$. Points D , E , and F are on sides $\overline{AB}$, $\overline{BC}$, and $\overline{AC}$, respectively, such that $\overline{DE}$ and $\overline{EF}$ are parallel to $\overline{AC}$ and $\overline{AB}$, respectively. What is the perimeter of parallelogram $ADEF$?



- (A) 48 (B) 52 (C) 56 (D) 60 (E) 72

Solution

Problem 10

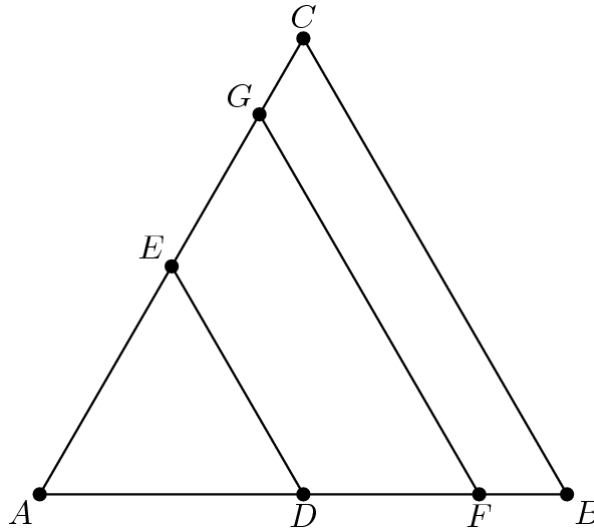
Let S be the set of positive integers n for which $\frac{1}{n}$ has the repeating decimal representation $0.\overline{ab} = 0.ababab\dots$, with a and b different digits. What is the sum of the elements of S ?

- (A) 11 (B) 44 (C) 110 (D) 143 (E) 155

Solution

Problem 11

Triangle ABC is equilateral with $AB = 1$. Points E and G are on $\overline{AC}$ and points D and F are on $\overline{AB}$ such that both $\overline{DE}$ and $\overline{FG}$ are parallel to $\overline{BC}$. Furthermore, triangle ADE and trapezoids $DFGE$ and $FBCG$ all have the same perimeter. What is $DE + FG$?



- (A) 1 (B) $\frac{3}{2}$ (C) $\frac{21}{13}$ (D) $\frac{13}{8}$ (E) $\frac{5}{3}$

Solution

Problem 12

The angles in a particular triangle are in arithmetic progression, and the side lengths are $4, 5, x$. The sum of the possible values of x equals $a + \sqrt{b} + \sqrt{c}$ where a, b , and c are positive integers. What is $a + b + c$?

- (A) 36 (B) 38 (C) 40 (D) 42 (E) 44

Solution

Problem 13

Let points $A = (0, 0)$, $B = (1, 2)$, $C = (3, 3)$, and $D = (4, 0)$. Quadrilateral $ABCD$ is cut into equal area pieces by a line passing through A . This line intersects $\overline{CD}$ at point $\left(\frac{p}{q}, \frac{r}{s}\right)$ where these fractions are in lowest terms. What is $p + q + r + s$?

- (A) 54 (B) 58 (C) 62 (D) 70 (E) 75

Solution

Problem 14

The sequence

$$\log_{12} 162, \log_{12} x, \log_{12} y, \log_{12} z, \log_{12} 1250$$

is an arithmetic progression. What is x ?

- (A) $125\sqrt{3}$ (B) 270 (C) $162\sqrt{5}$ (D) 434 (E) $225\sqrt{6}$

Solution

Problem 15

Rabbits Peter and Pauline have three offspring—Flopsie, Mopsie, and Cotton-tail. These five rabbits are to be distributed to four different pet stores so that no store gets both a parent and a child. It is not required that every store gets a rabbit. In how many different ways can this be done?

- (A) 96 (B) 108 (C) 156 (D) 204 (E) 372

Solution

Problem 16

A, B, C are three piles of rocks. The mean weight of the rocks in A is 40 pounds, the mean weight of the rocks in B is 50 pounds, the mean weight of the rocks in the combined piles A and B is 43 pounds, and the mean weight of the rocks in the combined piles A and C is 44 pounds. What is the greatest possible integer value for the mean in pounds of the rocks in the combined piles B and C ?

- (A) 55 (B) 56 (C) 57 (D) 58 (E) 59

Solution

Problem 17

A group of 12 pirates agree to divide a treasure chest of gold coins among themselves as follows. The k^{th} pirate to take a share takes $\frac{k}{12}$ of the coins that remain in the chest. The number of coins initially in the chest is the smallest number for which this arrangement will allow each pirate to receive a positive whole number of coins. How many coins does the 12th pirate receive?

- (A) 720 (B) 1296 (C) 1728 (D) 1925 (E) 3850

Solution

Problem 18

Six spheres of radius 1 are positioned so that their centers are at the vertices of a regular hexagon of side length 2. The six spheres are internally tangent to a larger sphere whose center is the center of the hexagon. An eighth sphere is externally tangent to the six smaller spheres and internally tangent to the larger sphere. What is the radius of this eighth sphere?

- (A) $\sqrt{2}$ (B) $\frac{3}{2}$ (C) $\frac{5}{3}$ (D) $\sqrt{3}$ (E) 2

Solution

Problem 19

In $\triangle ABC$, $AB = 86$, and $AC = 97$. A circle with center A and radius AB intersects $\overline{BC}$ at points B and X . Moreover $\overline{BX}$ and $\overline{CX}$ have integer lengths. What is BC ?

- (A) 11 (B) 28 (C) 33 (D) 61 (E) 72

Solution

Problem 20

Let S be the set $\{1, 2, 3, \dots, 19\}$. For $a, b \in S$, define $a \succ b$ to mean that either $0 < a - b \leq 9$ or $b - a > 9$. How many ordered triples (x, y, z) of elements of S have the property that $x \succ y$, $y \succ z$, and $z \succ x$?

- (A) 810 (B) 855 (C) 900 (D) 950 (E) 988

Solution

Problem 21

Consider $A = \log(2013 + \log(2012 + \log(2011 + \log(\dots + \log(3 + \log 2) \dots))))$. Which of the following intervals contains A ?

- (A) $(\log 2016, \log 2017)$ (B) $(\log 2017, \log 2018)$ (C) $(\log 2018, \log 2019)$
(D) $(\log 2019, \log 2020)$ (E) $(\log 2020, \log 2021)$

Solution

Problem 22

A palindrome is a nonnegative integer number that reads the same forwards and backwards when written in base 10 with no leading zeros. A 6-digit palindrome n is chosen uniformly at random. What is the probability that $\frac{n}{11}$ is also a palindrome?

- (A) $\frac{8}{25}$ (B) $\frac{33}{100}$ (C) $\frac{7}{20}$ (D) $\frac{9}{25}$ (E) $\frac{11}{30}$

Solution

Problem 23

$ABCD$ is a square of side length $\sqrt{3} + 1$. Point P is on $\overline{AC}$ such that $AP = \sqrt{2}$. The square region bounded by $ABCD$ is rotated 90° counterclockwise with center P , sweeping out a region whose area is $\frac{1}{c}(a\pi + b)$, where a , b , and c are positive integers and $\gcd(a, b, c) = 1$. What is $a + b + c$?

- (A) 15 (B) 17 (C) 19 (D) 21 (E) 23

Solution

Problem 24

Three distinct segments are chosen at random among the segments whose end-points are the vertices of a regular 12-gon. What is the probability that the lengths of these three segments are the three side lengths of a triangle with positive area?

- (A) $\frac{553}{715}$ (B) $\frac{443}{572}$ (C) $\frac{111}{143}$ (D) $\frac{81}{104}$ (E) $\frac{223}{286}$

Solution

Problem 25

Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be defined by $f(z) = z^2 + iz + 1$. How many complex numbers z are there such that $\text{Im}(z) > 0$ and both the real and the imaginary parts of $f(z)$ are integers with absolute value at most 10?

- (A) 399 (B) 401 (C) 413 (D) 431 (E) 441

Solution

See also

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2013 AMC 12A Answer Key

1. E
2. C
3. E
4. C
5. B
6. B
7. C
8. D
9. C
10. D
11. C
12. A
13. B
14. B
15. D
16. E
17. D
18. B
19. D
20. B
21. A
22. E
23. C
24. E
25. A

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2013 AMC 12B Problems

2013 AMC 12B (Answer Key)

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1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

On a particular January day, the high temperature in Lincoln, Nebraska, was 16 degrees higher than the low temperature, and the average of the high and low temperatures was 3° . In degrees, what was the low temperature in Lincoln that day?

(A) -13 (B) -8 (C) -5 (D) -3 (E) 11

Solution

Problem 2

Mr. Green measures his rectangular garden by walking two of the sides and finds that it is 15 steps by 20 steps. Each of Mr. Green's steps is 2 feet long. Mr. Green expects a half a pound of potatoes per square foot from his garden. How many pounds of potatoes does Mr. Green expect from his garden?

- (A) 600 (B) 800 (C) 1000 (D) 1200 (E) 1400

Solution

Problem 3

When counting from 3 to 201, 53 is the 51st number counted. When counting backwards from 201 to 3, 53 is the n^{th} number counted. What is n ?

- (A) 146 (B) 147 (C) 148 (D) 149 (E) 150

Solution

Problem 4

Ray's car averages 40 miles per gallon of gasoline, and Tom's car averages 10 miles per gallon of gasoline. Ray and Tom each drive the same number of miles. What is the cars' combined rate of miles per gallon of gasoline?

- (A) 10 (B) 16 (C) 25 (D) 30 (E) 40

Solution

Problem 5

The average age of 33 fifth-graders is 11. The average age of 55 of their parents is 33. What is the average age of all of these parents and fifth-graders?

- (A) 22 (B) 23.25 (C) 24.75 (D) 26.25 (E) 28

Solution

Problem 6

Real numbers x and y satisfy the equation $x^2 + y^2 = 10x - 6y - 34$. What is $x + y$?

- (A) 1 (B) 2 (C) 3 (D) 6 (E) 8

Solution

Problem 7

Jo and Blair take turns counting from 1 to one more than the last number said by the other person. Jo starts by saying "1", so Blair follows by saying "1, 2". Jo then says "1, 2, 3", and so on. What is the 53rd number said?

- (A) 2 (B) 3 (C) 5 (D) 6 (E) 8

Solution

Problem 8

Line l_1 has equation $3x - 2y = 1$ and goes through $A = (-1, -2)$. Line l_2 has equation $y = 1$ and meets line l_1 at point B . Line l_3 has positive slope, goes through point A , and meets l_2 at point C . The area of $\triangle ABC$ is 3. What is the slope of l_3 ?

- (A) $\frac{2}{3}$ (B) $\frac{3}{4}$ (C) 1 (D) $\frac{4}{3}$ (E) $\frac{3}{2}$

Solution

Problem 9

What is the sum of the exponents of the prime factors of the square root of the largest perfect square that divides 12!?

- (A) 5 (B) 7 (C) 8 (D) 10 (E) 12

Solution

Problem 10

Alex has 75 red tokens and 75 blue tokens. There is a booth where Alex can give two red tokens and receive in return a silver token and a blue token, and another booth where Alex can give three blue tokens and receive in return a silver token and a red token. Alex continues to exchange tokens until no more exchanges are possible. How many silver tokens will Alex have at the end?

- (A) 62 (B) 82 (C) 83 (D) 102 (E) 103

Solution

Problem 11

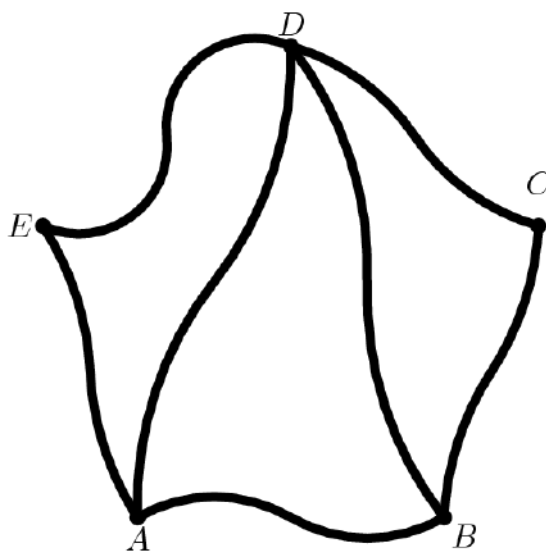
Two bees start at the same spot and fly at the same rate in the following directions. Bee A travels 1 foot north, then 1 foot east, then 1 foot upwards, and then continues to repeat this pattern. Bee B travels 1 foot south, then 1 foot west, and then continues to repeat this pattern. In what directions are the bees traveling when they are exactly 10 feet away from each other?

- (A) A east, B west
 (B) A north, B south
 (C) A north, B west
 (D) A up, B south
 (E) A up, B west

Solution

Problem 12

Cities A, B, C, D , and E are connected by roads $\widetilde{AB}, \widetilde{AD}, \widetilde{AE}, \widetilde{BC}, \widetilde{BD}, \widetilde{CD}$, and $\widetilde{DE}$. How many different routes are there from A to B that use each road exactly once? (Such a route will necessarily visit some cities more than once.)



- (A) 7 (B) 9 (C) 12 (D) 16 (E) 18

Solution

Problem 13

The internal angles of quadrilateral $ABCD$ form an arithmetic progression. Triangles ABD and DCB are similar with $\angle DBA = \angle DCB$ and $\angle ADB = \angle CBD$. Moreover, the angles in each of these two triangles also form an arithmetic progression. In degrees, what is the largest possible sum of the two largest angles of $ABCD$?

- (A) 210 (B) 220 (C) 230 (D) 240 (E) 250

Solution

Problem 14

Two non-decreasing sequences of nonnegative integers have different first terms. Each sequence has the property that each term beginning with the third is the sum of the previous two terms, and the seventh term of each sequence is N . What is the smallest possible value of N ?

- (A) 55 (B) 89 (C) 104 (D) 144 (E) 273

Solution

Problem 15

The number 2013 is expressed in the form

$$2013 = \frac{a_1!a_2!\dots a_m!}{b_1!b_2!\dots b_n!},$$

where $a_1 \geq a_2 \geq \dots \geq a_m$ and $b_1 \geq b_2 \geq \dots \geq b_n$ are positive integers and $a_1 + b_1$ is as small as possible. What is $|a_1 - b_1|$?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Solution

Problem 16

Let $ABCDE$ be an equiangular convex pentagon of perimeter 1. The pairwise intersections of the lines that extend the sides of the pentagon determine a five-pointed star polygon. Let S be the perimeter of this star. What is the difference between the maximum and the minimum possible values of S ?

- (A) 0 (B) $\frac{1}{2}$ (C) $\frac{\sqrt{5}-1}{2}$ (D) $\frac{\sqrt{5}+1}{2}$ (E) $\sqrt{5}$

Solution

Problem 17

Let a , b , and c be real numbers such that

$$a + b + c = 2, \text{ and}$$

$$a^2 + b^2 + c^2 = 12$$

What is the difference between the maximum and minimum possible values of c ?

- (A) 2 (B) $\frac{10}{3}$ (C) 4 (D) $\frac{16}{3}$ (E) $\frac{20}{3}$

Solution

Problem 18

Barbara and Jenna play the following game, in which they take turns. A number of coins lie on a table. When it is Barbara's turn, she must remove 2 or 4 coins, unless only one coin remains, in which case she loses her turn. When it is Jenna's turn, she must remove 1 or 3 coins. A coin flip determines who goes first. Whoever removes the last coin wins the game. Assume both players use their best strategy. Who will win when the game starts with 2013 coins and when the game starts with 2014 coins?

- (A) Barbara will win with 2013 coins and Jenna will win with 2014 coins.
 (B) Jenna will win with 2013 coins, and whoever goes first will win with 2014 coins.
 (C) Barbara will win with 2013 coins, and whoever goes second will win with 2014 coins.
 (D) Jenna will win with 2013 coins, and Barbara will win with 2014 coins.
 (E) Whoever goes first will win with 2013 coins, and whoever goes second will win with 2014 coins.

Solution

Problem 19

In triangle ABC , $AB = 13$, $BC = 14$, and $CA = 15$. Distinct points D , E , and F lie on segments $\overline{BC}$, $\overline{CA}$, and $\overline{DE}$, respectively, such that $\overline{AD} \perp \overline{BC}$, $\overline{DE} \perp \overline{AC}$, and $\overline{AF} \perp \overline{BF}$. The length of segment $\overline{DF}$ can be written as $\frac{m}{n}$, where m and n are relatively prime positive integers. What is $m + n$?

- (A) 18 (B) 21 (C) 24 (D) 27 (E) 30

Solution

Problem 20

For $135^\circ < x < 180^\circ$, points $P = (\cos x, \cos^2 x)$, $Q = (\cot x, \cot^2 x)$, $R = (\sin x, \sin^2 x)$ and $S = (\tan x, \tan^2 x)$ are the vertices of a trapezoid. What is $\sin(2x)$?

- (A) $2 - 2\sqrt{2}$ (B) $3\sqrt{3} - 6$ (C) $3\sqrt{2} - 5$ (D) $-\frac{3}{4}$ (E) $1 - \sqrt{3}$

Solution

Problem 21

Consider the set of 30 parabolas defined as follows: all parabolas have as focus the point $(0, 0)$ and the directrix lines have the form $y = ax + b$ with a and b integers such that $a \in \{-2, -1, 0, 1, 2\}$ and $b \in \{-3, -2, -1, 1, 2, 3\}$. No three of these parabolas have a common point. How many points in the plane are on two of these parabolas?

- (A) 720 (B) 760 (C) 810 (D) 840 (E) 870

Solution

Problem 22

Let $m > 1$ and $n > 1$ be integers. Suppose that the product of the solutions for x of the equation

$$8(\log_n x)(\log_m x) - 7\log_n x - 6\log_m x - 2013 = 0$$

is the smallest possible integer. What is $m + n$?

- (A) 12 (B) 20 (C) 24 (D) 48 (E) 272

Solution

Problem 23

Bernardo chooses a three-digit positive integer N and writes both its base-5 and base-6 representations on a blackboard. Later LeRoy sees the two numbers Bernardo has written. Treating the two numbers as base-10 integers, he adds them to obtain an integer S . For example, if $N = 749$, Bernardo writes the numbers 10, 444 and 3, 245, and LeRoy obtains the sum $S = 13, 689$. For how many choices of N are the two rightmost digits of S , in order, the same as those of $2N$?

- (A) 5 (B) 10 (C) 15 (D) 20 (E) 25

Solution

Problem 24

Let ABC be a triangle where M is the midpoint of $\overline{AC}$, and $\overline{CN}$ is the angle bisector of $\angle ACB$ with N on $\overline{AB}$. Let X be the intersection of the median $\overline{BM}$ and the bisector $\overline{CN}$. In addition $\triangle BXN$ is equilateral with $AC = 2$. What is BN^2 ?

- (A) $\frac{10 - 6\sqrt{2}}{7}$ (B) $\frac{2}{9}$ (C) $\frac{5\sqrt{2} - 3\sqrt{3}}{8}$ (D) $\frac{\sqrt{2}}{6}$ (E) $\frac{3\sqrt{3} - 4}{5}$

Solution

Problem 25

Let G be the set of polynomials of the form

$$P(z) = z^n + c_{n-1}z^{n-1} + \cdots + c_2z^2 + c_1z + 50,$$

where $c_1, c_2, \dots, c_{n-1}$ are integers and $P(z)$ has distinct roots of the form $a + ib$ with a and b integers. How many polynomials are in G ?

- (A) 288 (B) 528 (C) 576 (D) 992 (E) 1056

Solution

See also

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12. D
13. D
14. C
15. B
16. A
17. D
18. B
19. B
20. A
21. C
22. A
23. E
24. A
25. B

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2014

2014 AMC 12A Problems

2014 AMC 12A (Answer Key)

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Instructions

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1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

What is $10 \cdot \left(\frac{1}{2} + \frac{1}{5} + \frac{1}{10}\right)^{-1}$?

- (A) 3 (B) 8 (C) $\frac{25}{2}$ (D) $\frac{170}{3}$ (E) 170

Solution

Problem 2

At the theater children get in for half price. The price for 5 adult tickets and 4 child tickets is 24.50. How much would 8 adult tickets and 6 child tickets cost?

- (A) 35 (B) 38.50 (C) 40 (D) 42 (E) 42.50

Solution

Problem 3

Walking down Jane Street, Ralph passed four houses in a row, each painted a different color. He passed the orange house before the red house, and he passed the blue house before the yellow house. The blue house was not next to the yellow house. How many orderings of the colored houses are possible?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Solution

Problem 4

Suppose that a cows give b gallons of milk in c days. At this rate, how many gallons of milk will d cows give in e days?

- (A) $\frac{bde}{ac}$ (B) $\frac{ac}{bde}$ (C) $\frac{abde}{c}$ (D) $\frac{bcde}{a}$ (E) $\frac{abc}{de}$

Solution

Problem 5

On an algebra quiz, 10% of the students scored 70 points, 35% scored 80 points, 30% scored 90 points, and the rest scored 100 points. What is the difference between the mean and median score of the students' scores on this quiz?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Solution

Problem 6

The difference between a two-digit number and the number obtained by reversing its digits is 5 times the sum of the digits of either number. What is the sum of the two digit number and its reverse?

- (A) 44 (B) 55 (C) 77 (D) 99 (E) 110

Solution

Problem 7

The first three terms of a geometric progression are $\sqrt{3}$, $\sqrt[3]{3}$, and $\sqrt[6]{3}$. What is the fourth term?

- (A) 1 (B) $\sqrt[7]{3}$ (C) $\sqrt[8]{3}$ (D) $\sqrt[9]{3}$ (E) $\sqrt[10]{3}$

Solution

Problem 8

A customer who intends to purchase an appliance has three coupons, only one of which may be used:

Coupon 1: 10% off the listed price if the listed price is at least 50

Coupon 2: 20 dollars off the listed price if the listed price is at least 100

Coupon 3: 18% off the amount by which the listed price exceeds 100

For which of the following listed prices will coupon 1 offer a greater price reduction than either coupon 2 or coupon 3?

- (A) 179.95 (B) 199.95 (C) 219.95 (D) 239.95 (E) 259.95

Solution

Problem 9

Five positive consecutive integers starting with a have average b . What is the average of 5 consecutive integers that start with b ?

- (A) $a + 3$ (B) $a + 4$ (C) $a + 5$ (D) $a + 6$ (E) $a + 7$

Solution

Problem 10

Three congruent isosceles triangles are constructed with their bases on the sides of an equilateral triangle of side length 1. The sum of the areas of the three isosceles triangles is the same as the area of the equilateral triangle. What is the length of one of the two congruent sides of one of the isosceles triangles?

- (A) $\frac{\sqrt{3}}{4}$ (B) $\frac{\sqrt{3}}{3}$ (C) $\frac{2}{3}$ (D) $\frac{\sqrt{2}}{2}$ (E) $\frac{\sqrt{3}}{2}$

Solution

Problem 11

David drives from his home to the airport to catch a flight. He drives 35 miles in the first hour, but realizes that he will be 1 hour late if he continues at this speed. He increases his speed by 15 miles per hour for the rest of the way to the airport and arrives 30 minutes early. How many miles is the airport from his home?

- (A) 140 (B) 175 (C) 210 (D) 245 (E) 280

Solution

Problem 12

Two circles intersect at points A and B . The minor arcs AB measure 30° on one circle and 60° on the other circle. What is the ratio of the area of the larger circle to the area of the smaller circle?

- (A) 2 (B) $1 + \sqrt{3}$ (C) 3 (D) $2 + \sqrt{3}$ (E) 4

Solution

Problem 13

A fancy bed and breakfast inn has 5 rooms, each with a distinctive color-coded decor. One day 5 friends arrive to spend the night. There are no other guests that night. The friends can room in any combination they wish, but with no more than 2 friends per room. In how many ways can the innkeeper assign the guests to the rooms?

- (A) 2100 (B) 2220 (C) 3000 (D) 3120 (E) 3125

Solution

Problem 14

Let $a < b < c$ be three integers such that a, b, c is an arithmetic progression and a, c, b is a geometric progression. What is the smallest possible value of c ?

- (A) -2 (B) 1 (C) 2 (D) 4 (E) 6

Solution

Problem 15

A five-digit palindrome is a positive integer with respective digits $abcba$, where a is non-zero. Let S be the sum of all five-digit palindromes. What is the sum of the digits of S ?

- (A) 9 (B) 18 (C) 27 (D) 36 (E) 45

Solution

Problem 16

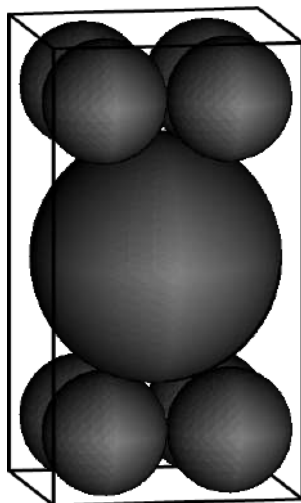
The product $(8)(888 \dots 8)$, where the second factor has k digits, is an integer whose digits have a sum of 1000. What is k ?

- (A) 901 (B) 911 (C) 919 (D) 991 (E) 999

Solution

Problem 17

A $4 \times 4 \times h$ rectangular box contains a sphere of radius 2 and eight smaller spheres of radius 1. The smaller spheres are each tangent to three sides of the box, and the larger sphere is tangent to each of the smaller spheres. What is h ?



- (A) $2 + 2\sqrt{7}$ (B) $3 + 2\sqrt{5}$ (C) $4 + 2\sqrt{7}$ (D) $4\sqrt{5}$ (E) $4\sqrt{7}$

Solution

Problem 18

The domain of the function $f(x) = \log_{\frac{1}{2}}(\log_4(\log_{\frac{1}{4}}(\log_{16}(\log_{\frac{1}{16}} x))))$ is an interval of length $\frac{m}{n}$, where m and n are relatively prime positive integers. What is $m + n$?

- (A) 19 (B) 31 (C) 271 (D) 319 (E) 511

Solution

Problem 19

There are exactly N distinct rational numbers k such that $|k| < 200$ and

$$5x^2 + kx + 12 = 0$$

has at least one integer solution for x . What is N ?

- (A) 6 (B) 12 (C) 24 (D) 48 (E) 78

Solution

Problem 20

In $\triangle BAC$, $\angle BAC = 40^\circ$, $AB = 10$, and $AC = 6$. Points D and E lie on $\overline{AB}$ and $\overline{AC}$ respectively. What is the minimum possible value of $BE + DE + CD$?

- (A) $6\sqrt{3} + 3$ (B) $\frac{27}{2}$ (C) $8\sqrt{3}$ (D) 14 (E) $3\sqrt{3} + 9$

Solution

Problem 21

For every real number x , let $\lfloor x \rfloor$ denote the greatest integer not exceeding x , and let

$$f(x) = \lfloor x \rfloor (2014^{x - \lfloor x \rfloor} - 1).$$

The set of all numbers x such that $1 \leq x < 2014$ and $f(x) \leq 1$ is a union of disjoint intervals. What is the sum of the lengths of those intervals?

- (A) 1 (B) $\frac{\log 2015}{\log 2014}$ (C) $\frac{\log 2014}{\log 2013}$ (D) $\frac{2014}{2013}$ (E) $2014^{\frac{1}{2014}}$

Solution

Problem 22

The number 5^{867} is between 2^{2013} and 2^{2014} . How many pairs of integers (m, n) are there such that $1 \leq m \leq 2012$ and

$$5^n < 2^m < 2^{m+2} < 5^{n+1}?$$

- (A) 278 (B) 279 (C) 280 (D) 281 (E) 282

Solution

Problem 23

The fraction

$$\frac{1}{99^2} = 0.\overline{b_{n-1}b_{n-2} \dots b_2b_1b_0},$$

where n is the length of the period of the repeating decimal expansion. What is the sum $b_0 + b_1 + \dots + b_{n-1}$?

- (A) 874 (B) 883 (C) 887 (D) 891 (E) 892

Solution

Problem 24

Let $f_0(x) = x + |x - 100| - |x + 100|$ and for $n \geq 1$, let $f_n(x) = |f_{n-1}(x)| - 1$. For how many values of x is $f_{100}(x) = 0$?

- (A) 299 (B) 300 (C) 301 (D) 302 (E) 303

Solution

Problem 25

The parabola P has focus $(0, 0)$ and goes through the points $(4, 3)$ and $(-4, -3)$. For how many points $(x, y) \in P$ with integer coordinates is it true that $|4x + 3y| \leq 1000$?

- (A) 38 (B) 40 (C) 42 (D) 44 (E) 46

Solution

See also

2014 AMC 12A (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2014))	
Preceded by 2013 AMC 12B Problems	Followed by 2014 AMC 12B Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
All AMC 12 Problems and Solutions	

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2014 AMC 12A Answer Key

1. C
2. B
3. B
4. A
5. C
6. D
7. A
8. C
9. B
10. B
11. C
12. D
13. B
14. C
15. B
16. D
17. A
18. C
19. E
20. D
21. A
22. B
23. B
24. C
25. B

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2014 AMC 12B Problems

2014 AMC 12B (Answer Key)

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1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

Leah has 13 coins, all of which are pennies and nickels. If she had one more nickel than she has now, then she would have the same number of pennies and nickels. In cents, how much are Leah's coins worth?

(A) 33 (B) 35 (C) 37 (D) 39 (E) 41

Solution

Problem 2

Orvin went to the store with just enough money to buy 30 balloons. When he arrived he discovered that the store had a special sale on balloons: buy 1 balloon at the regular price and get a second at $\frac{1}{3}$ off the regular price. What is the greatest number of balloons Orvin could buy?

- (A) 33 (B) 34 (C) 36 (D) 38 (E) 39

Solution

Problem 3

Randy drove the first third of his trip on a gravel road, the next 20 miles on pavement, and the remaining one-fifth on a dirt road. In miles, how long was Randy's trip?

- (A) 30 (B) $\frac{400}{11}$ (C) $\frac{75}{2}$ (D) 40 (E) $\frac{300}{7}$

Solution

Problem 4

Susie pays for 4 muffins and 3 bananas. Calvin spends twice as much paying for 2 muffins and 16 bananas. A muffin is how many times as expensive as a banana?

- (A) $\frac{3}{2}$ (B) $\frac{5}{3}$ (C) $\frac{7}{4}$ (D) 2 (E) $\frac{13}{4}$

Solution

Problem 5

Doug constructs a square window using 8 equal-size panes of glass, as shown. The ratio of the height to width for each pane is $5 : 2$, and the borders around and between the panes are 2 inches wide. In inches, what is the side length of the square window?



- (A) 26 (B) 28 (C) 30 (D) 32 (E) 34

Solution

Problem 6

Ed and Ann both have lemonade with their lunch. Ed orders the regular size. Ann gets the large lemonade, which is 50% more than the regular. After both consume $\frac{3}{4}$ of their drinks, Ann gives Ed a third of what she has left, and 2 additional ounces. When they finish their lemonades they realize that they both drank the same amount. How many ounces of lemonade did they drink together?

- (A) 30 (B) 32 (C) 36 (D) 40 (E) 50

Solution

Problem 7

For how many positive integers n is $\frac{n}{30-n}$ also a positive integer?

- (A) 4 (B) 5 (C) 6 (D) 7 (E) 8

Solution

Problem 8

In the addition shown below A, B, C , and D are distinct digits. How many different values are possible for D ?

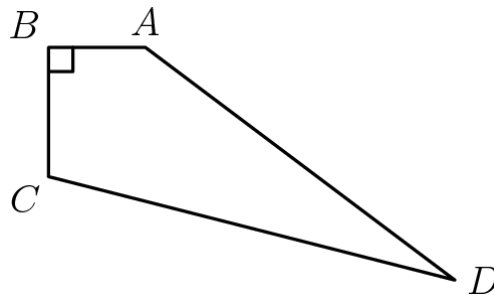
$$\begin{array}{r} A B B C B \\ + B C A D A \\ \hline D B D D D \end{array}$$

- (A) 2 (B) 4 (C) 7 (D) 8 (E) 9

Solution

Problem 9

Convex quadrilateral $ABCD$ has $AB = 3, BC = 4, CD = 13, AD = 12$, and $\angle ABC = 90^\circ$, as shown. What is the area of the quadrilateral?



- (A) 30 (B) 36 (C) 40 (D) 48 (E) 58.5

Solution

Problem 10

Danica drove her new car on a trip for a whole number of hours, averaging 55 miles per hour. At the beginning of the trip, abc miles was displayed on the odometer, where abc is a 3-digit number with $a \geq 1$ and $a + b + c \leq 7$. At the end of the trip, the odometer showed cba miles. What is $a^2 + b^2 + c^2$?

- (A) 26 (B) 27 (C) 36 (D) 37 (E) 41

Solution

Problem 11

A list of 11 positive integers has a mean of 10, a median of 9, and a unique mode of 8. What is the largest possible value of an integer in the list?

- (A) 24 (B) 30 (C) 31 (D) 33 (E) 35

Solution

Problem 12

A set S consists of triangles whose sides have integer lengths less than 5, and no two elements of S are congruent or similar. What is the largest number of elements that S can have?

- (A) 8 (B) 9 (C) 10 (D) 11 (E) 12

Solution

Problem 13

Real numbers a and b are chosen with $1 < a < b$ such that no triangles with positive area has side lengths 1 , a , and b or $\frac{1}{b}$, $\frac{1}{a}$, and 1 . What is the smallest possible value of b ?

- (A) $\frac{3 + \sqrt{3}}{2}$ (B) $\frac{5}{2}$ (C) $\frac{3 + \sqrt{5}}{2}$ (D) $\frac{3 + \sqrt{6}}{2}$ (E) 3

Solution

Problem 14

A rectangular box has a total surface area of 94 square inches. The sum of the lengths of all its edges is 48 inches. What is the sum of the lengths in inches of all of its interior diagonals?

- (A) $8\sqrt{3}$ (B) $10\sqrt{2}$ (C) $16\sqrt{3}$ (D) $20\sqrt{2}$ (E) $40\sqrt{2}$

Solution

Problem 15

When $p = \sum_{k=1}^6 k \ln k$, the number e^p is an integer. What is the largest power of 2 that is a factor of e^p ?

- (A) 2^{12} (B) 2^{14} (C) 2^{16} (D) 2^{18} (E) 2^{20}

Solution

Problem 16

Let P be a cubic polynomial with $P(0) = k$, $P(1) = 2k$, and $P(-1) = 3k$. What is $P(2) + P(-2)$?

- (A) 0 (B) k (C) $6k$ (D) $7k$ (E) $14k$

Solution

Problem 17

Let P be the parabola with equation $y = x^2$ and let $Q = (20, 14)$. There are real numbers r and s such that the line through Q with slope m does not intersect P if and only if $r < m < s$. What is $r + s$?

- (A) 1 (B) 26 (C) 40 (D) 52 (E) 80

Solution

Problem 18

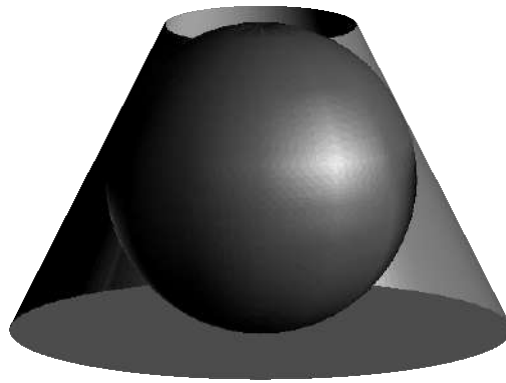
The numbers $1, 2, 3, 4, 5$ are to be arranged in a circle. An arrangement is *bad* if it is not true that for every n from 1 to 15 one can find a subset of the numbers that appear consecutively on the circle that sum to n . Arrangements that differ only by a rotation or a reflection are considered the same. How many different bad arrangements are there?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Solution

Problem 19

A sphere is inscribed in a truncated right circular cone as shown. The volume of the truncated cone is twice that of the sphere. What is the ratio of the radius of the bottom base of the truncated cone to the radius of the top base of the truncated cone?



- (A) $\frac{3}{2}$ (B) $\frac{1+\sqrt{5}}{2}$ (C) $\sqrt{3}$ (D) 2 (E) $\frac{3+\sqrt{5}}{2}$

Solution

Problem 20

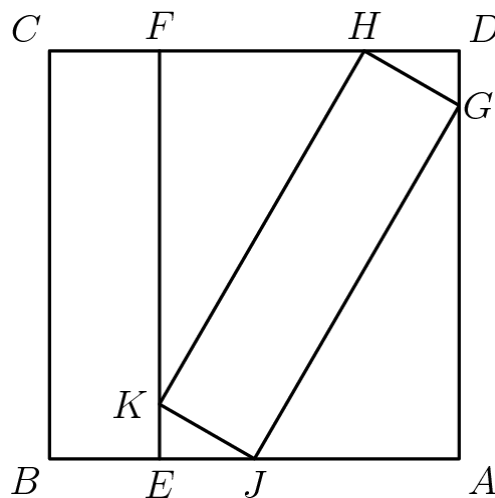
For how many positive integers x is $\log_{10}(x - 40) + \log_{10}(60 - x) < 2$?

- (A) 10 (B) 18 (C) 19 (D) 20 (E) infinitely many

Solution

Problem 21

In the figure, $ABCD$ is a square of side length 1. The rectangles $JKHG$ and $EBCF$ are congruent. What is BE ?



- (A) $\frac{1}{2}(\sqrt{6} - 2)$ (B) $\frac{1}{4}$ (C) $2 - \sqrt{3}$ (D) $\frac{\sqrt{3}}{6}$ (E) $1 - \frac{\sqrt{2}}{2}$

Solution

Problem 22

In a small pond there are eleven lily pads in a row labeled 0 through 10. A frog is sitting on pad 1. When the frog is on pad N , $0 < N < 10$, it will jump to pad $N - 1$ with probability $\frac{N}{10}$ and to pad $N + 1$ with probability $1 - \frac{N}{10}$. Each jump is independent of the previous jumps. If the frog reaches pad 0 it will be eaten by a patiently waiting snake. If the frog reaches pad 10 it will exit the pond, never to return. What is the probability that the frog will escape without being eaten by the snake?

- (A) $\frac{32}{79}$ (B) $\frac{161}{384}$ (C) $\frac{63}{146}$ (D) $\frac{7}{16}$ (E) $\frac{1}{2}$

Solution

Problem 23

The number 2017 is prime. Let $S = \sum_{k=0}^{62} \binom{2014}{k}$. What is the remainder when S is divided by 2017?

- (A) 32 (B) 684 (C) 1024 (D) 1576 (E) 2016

Solution

Problem 24

Let $ABCDE$ be a pentagon inscribed in a circle such that $AB = CD = 3$, $BC = DE = 10$, and $AE = 14$. The sum of the lengths of all diagonals of $ABCDE$ is equal to $\frac{m}{n}$, where m and n are relatively prime positive integers. What is $m + n$?

- (A) 129 (B) 247 (C) 353 (D) 391 (E) 421

Solution

Problem 25

What is the sum of all positive real solutions x to the equation

$$2 \cos 2x \left(\cos 2x - \cos \left(\frac{2014\pi^2}{x} \right) \right) = \cos 4x - 1?$$

- (A) π (B) 810π (C) 1008π (D) 1080π (E) 1800π

Solution

See also

2014 AMC 12B (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2014))	
Preceded by 2014 AMC 12A Problems	Followed by 2015 AMC 12A Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
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2014 AMC 12B Answer Key

1. C
2. C
3. E
4. B
5. A
6. D
7. D
8. C
9. B
10. D
11. E
12. B
13. C
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15. C
16. E
17. E
18. B
19. E
20. B
21. C
22. C
23. C
24. D
25. D

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2015

2015 AMC 12A Problems

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1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

What is the value of $(2^0 - 1 + 5^2 - 0)^{-1} \times 5$?

- (A) -125 (B) -120 (C) $\frac{1}{5}$ (D) $\frac{5}{24}$ (E) 25

Solution

Problem 2

Two of the three sides of a triangle are 20 and 15. Which of the following numbers is not a possible perimeter of the triangle?

- (A) 52 (B) 57 (C) 62 (D) 67 (E) 72

Solution

Problem 3

Mr. Patrick teaches math to 15 students. He was grading tests and found that when he graded everyone's test except Payton's, the average grade for the class was 80. After he graded Payton's test, the class average became 81. What was Payton's score on the test?

- (A) 81 (B) 85 (C) 91 (D) 94 (E) 95

Solution

Problem 4

The sum of two positive numbers is 5 times their difference. What is the ratio of the larger number to the smaller?

- (A) $\frac{5}{4}$ (B) $\frac{3}{2}$ (C) $\frac{9}{5}$ (D) 2 (E) $\frac{5}{2}$

Solution

Problem 5

Amelia needs to estimate the quantity $\frac{a}{b} - c$, where a , b , and c are large positive integers. She rounds each of the integers so that the calculation will be easier to do mentally. In which of these situations will her answer necessarily be greater than the exact value of $\frac{a}{b} - c$?

- (A) She rounds all three numbers up.
(B) She rounds a and b up, and she rounds c down.
(C) She rounds a and c up, and she rounds b down.
(D) She rounds a up, and she rounds b and c down.
(E) She rounds c up, and she rounds a and b down.

Solution

Problem 6

Two years ago Pete was three times as old as his cousin Claire. Two years before that, Pete was four times as old as Claire. In how many years will the ratio of their ages be 2 : 1?

- (A) 2 (B) 4 (C) 5 (D) 6 (E) 8

Solution

Problem 7

Two right circular cylinders have the same volume. The radius of the second cylinder is 10% more than the radius of the first. What is the relationship between the heights of the two cylinders?

- (A) The second height is 10% less than the first.
(B) The first height is 10% more than the second.
(C) The second height is 21% less than the first.
(D) The first height is 21% more than the second.
(E) The second height is 80% of the first.

Solution

Problem 8

The ratio of the length to the width of a rectangle is $4:3$. If the rectangle has diagonal of length d , then the area may be expressed as kd^2 for some constant k . What is k ?

- (A) $\frac{2}{7}$ (B) $\frac{3}{7}$ (C) $\frac{12}{25}$ (D) $\frac{16}{26}$ (E) $\frac{3}{4}$

Solution

Problem 9

A box contains 2 red marbles, 2 green marbles, and 2 yellow marbles. Carol takes 2 marbles from the box at random; then Claudia takes 2 of the remaining marbles at random; and then Cheryl takes the last 2 marbles. What is the probability that Cheryl gets 2 marbles of the same color?

- (A) $\frac{1}{10}$ (B) $\frac{1}{6}$ (C) $\frac{1}{5}$ (D) $\frac{1}{3}$ (E) $\frac{1}{2}$

Solution

Problem 10

Integers x and y with $x > y > 0$ satisfy $x + y + xy = 80$. What is x ?

- (A) 8 (B) 10 (C) 15 (D) 18 (E) 26

Solution

Problem 11

On a sheet of paper, Isabella draws a circle of radius 2, a circle of radius 3, and all possible lines simultaneously tangent to both circles. Isabella notices that she has drawn exactly $k \geq 0$ lines. How many different values of k are possible?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Solution

Problem 12

The parabolas $y = ax^2 - 2$ and $y = 4 - bx^2$ intersect the coordinate axes in exactly four points, and these four points are the vertices of a kite of area 12. What is $a + b$?

- (A) 1 (B) 1.5 (C) 2 (D) 2.5 (E) 3

Solution

Problem 13

A league with 12 teams holds a round-robin tournament, with each team playing every other team exactly once. Games either end with one team victorious or else end in a draw. A team scores 2 points for every game it wins and 1 point for every game it draws. Which of the following is NOT a true statement about the list of 12 scores?

- (A) There must be an even number of odd scores.
(B) There must be an even number of even scores.
(C) There cannot be two scores of 0.
(D) The sum of the scores must be at least 100.
(E) The highest score must be at least 12.

Solution

Problem 14

What is the value of a for which $\frac{1}{\log_2 a} + \frac{1}{\log_3 a} + \frac{1}{\log_4 a} = 1$?

- (A) 9 (B) 12 (C) 18 (D) 24 (E) 36

Solution

Problem 15

What is the minimum number of digits to the right of the decimal point needed to express the fraction $\frac{123456789}{2^{26} \cdot 5^4}$ as a decimal?

- (A) 4 (B) 22 (C) 26 (D) 30 (E) 104

Solution

Problem 16

Tetrahedron $ABCD$ has $AB = 5$, $AC = 3$, $BC = 4$, $BD = 4$, $AD = 3$, and $CD = \frac{12}{5}\sqrt{2}$. What is the volume of the tetrahedron?

- (A) $3\sqrt{2}$ (B) $2\sqrt{5}$ (C) $\frac{24}{5}$ (D) $3\sqrt{3}$ (E) $\frac{24}{5}\sqrt{2}$

Solution

Problem 17

Eight people are sitting around a circular table, each holding a fair coin. All eight people flip their coins and those who flip heads stand while those who flip tails remain seated. What is the probability that no two adjacent people will stand?

- (A) $\frac{47}{256}$ (B) $\frac{3}{16}$ (C) $\frac{49}{256}$ (D) $\frac{25}{128}$ (E) $\frac{51}{256}$

Solution

Problem 18

The zeros of the function $f(x) = x^2 - ax + 2a$ are integers. What is the sum of the possible values of a ?

- (A) 7 (B) 8 (C) 16 (D) 17 (E) 18

Solution

Problem 19

For some positive integers p , there is a quadrilateral $ABCD$ with positive integer side lengths, perimeter p , right angles at B and C , $AB = 2$, and $CD = AD$. How many different values of $p < 2015$ are possible?

- (A) 30 (B) 31 (C) 61 (D) 62 (E) 63

Solution

Problem 20

Isosceles triangles T and T' are not congruent but have the same area and the same perimeter. The sides of T have lengths of 5, 5, and 8, while those of T' have lengths of a , a , and b . Which of the following numbers is closest to b ?

- (A) 3 (B) 4 (C) 5 (D) 6 (E) 8

Solution

Problem 21

A circle of radius r passes through both foci of, and exactly four points on, the ellipse with equation $x^2 + 16y^2 = 16$. The set of all possible values of r is an interval $[a, b]$. What is $a + b$?

- (A) $5\sqrt{2} + 4$ (B) $\sqrt{17} + 7$ (C) $6\sqrt{2} + 3$ (D) $\sqrt{15} + 8$ (E) 12

Solution

Problem 22

For each positive integer n , let $S(n)$ be the number of sequences of length n consisting solely of the letters A and B , with no more than three A s in a row and no more than three B s in a row. What is the remainder when $S(2015)$ is divided by 12?

- (A) 0 (B) 4 (C) 6 (D) 8 (E) 10

Solution

Problem 23

Let S be a square of side length 1. Two points are chosen independently at random on the sides of S . The probability that the straight-line distance between the points is at least $\frac{1}{2}$ is $\frac{a - b\pi}{c}$, where a , b , and c are positive integers and $\gcd(a, b, c) = 1$. What is $a + b + c$?

- (A) 59 (B) 60 (C) 61 (D) 62 (E) 63

Solution

Problem 24

Rational numbers a and b are chosen at random among all rational numbers in the interval $[0, 2)$ that can be written as fractions $\frac{n}{d}$ where n and d are integers with $1 \leq d \leq 5$. What is the probability that

$$(\cos(a\pi) + i\sin(b\pi))^4$$

is a real number?

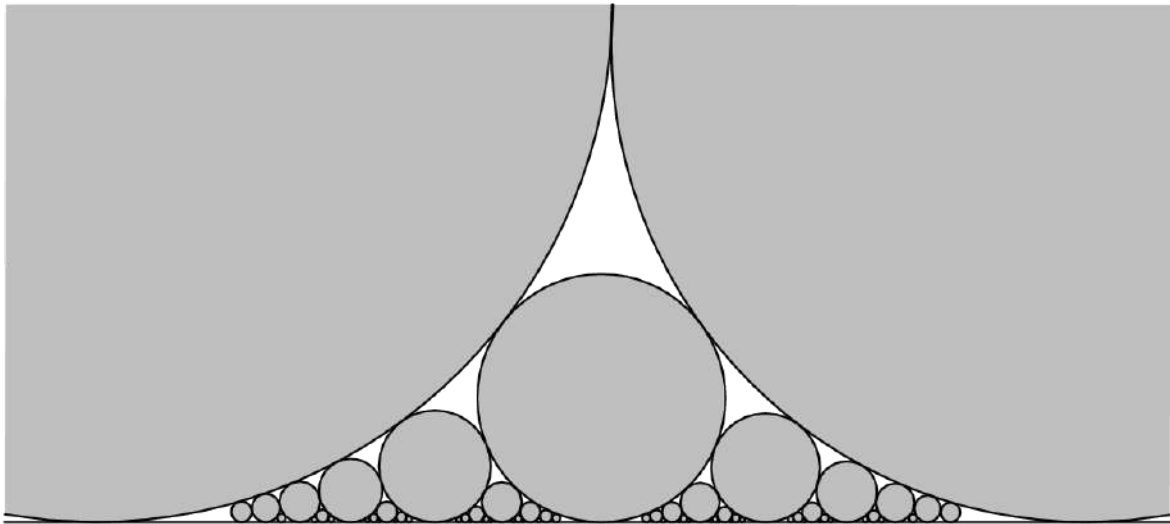
- (A) $\frac{3}{50}$ (B) $\frac{4}{25}$ (C) $\frac{41}{200}$ (D) $\frac{6}{25}$ (E) $\frac{13}{50}$

Solution

Problem 25

A collection of circles in the upper half-plane, all tangent to the x -axis, is constructed in layers as follows. Layer L_0 consists of two circles of radii 70^2 and 73^2 that are externally tangent. For $k \geq 1$, the circles in $\bigcup_{j=0}^{k-1} L_j$ are ordered according to their points of tangency with the x -axis. For every pair of consecutive circles in this order, a new circle is constructed externally tangent to each of the two circles in the pair. Layer L_k consists of the 2^{k-1} circles constructed in this way. Let $S = \bigcup_{j=0}^6 L_j$ and for every circle C denote by $r(C)$ its radius. What is

$$\sum_{C \in S} \frac{1}{\sqrt{r(C)}}?$$



- (A) $\frac{286}{35}$ (B) $\frac{583}{70}$ (C) $\frac{715}{73}$ (D) $\frac{143}{14}$ (E) $\frac{1573}{146}$

Solution

See also

2015 AMC 12A (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2015))	
Preceded by 2014 AMC 12B Problems	Followed by 2015 AMC 12B Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
All AMC 12 Problems and Solutions	

- AMC Problems and Solutions

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2015 AMC 12A Answer Key

1. C
2. E
3. E
4. B
5. D
6. B
7. D
8. C
9. C
10. E
11. D
12. B
13. E
14. D
15. C
16. C
17. A
18. C
19. B
20. A
21. D
22. D
23. A
24. D
25. D

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2015 AMC 12B Problems

2015 AMC 12B (Answer Key)

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Instructions

1. This is a 25-question, multiple choice test. Each question is followed by answers marked A, B, C, D and E. Only one of these is correct.
2. You will receive 6 points for each correct answer, 2.5 points for each problem left unanswered if the year is before 2006, 1.5 points for each problem left unanswered if the year is after 2006, and 0 points for each incorrect answer.
3. No aids are permitted other than scratch paper, graph paper, ruler, compass, protractor and erasers (and calculators that are accepted for use on the test if before 2006. No problems on the test will *require* the use of a calculator).
4. Figures are not necessarily drawn to scale.
5. You will have **75 minutes** working time to complete the test.

1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

What is the value of $2 - (-2)^{-2}$?

- (A) -2 (B) $\frac{1}{16}$ (C) $\frac{7}{4}$ (D) $\frac{9}{4}$ (E) 6

Solution

Problem 2

Marie does three equally time-consuming tasks in a row without taking breaks. She begins the first task at 1:00 PM and finishes the second task at 2:40 PM. When does she finish the third task?

- (A) 3:10 PM (B) 3:30 PM (C) 4:00 PM (D) 4:10 PM (E) 4:30 PM

Solution

Problem 3

Isaac has written down one integer two times and another integer three times. The sum of the five numbers is 100, and one of the numbers is 28. What is the other number?

- (A) 8 (B) 11 (C) 14 (D) 15 (E) 18

Solution

Problem 4

David, Hikmet, Jack, Marta, Rand, and Todd were in a 12-person race with 6 other people. Rand finished 6 places ahead of Hikmet. Marta finished 1 place behind Jack. David finished 2 places behind Hikmet. Jack finished 2 places behind Todd. Todd finished 1 place behind Rand. Marta finished in 6th place. Who finished in 8th place?

- (A) David (B) Hikmet (C) Jack (D) Rand (E) Todd

Solution

Problem 5

The Tigers beat the Sharks 2 out of the 3 times they played. They then played N more times, and the Sharks ended up winning at least 95% of all the games played. What is the minimum possible value for N ?

- (A) 35 (B) 37 (C) 39 (D) 41 (E) 43

Solution

Problem 6

Back in 1930, Tillie had to memorize her multiplication facts from 0×0 to 12×12 . The multiplication table she was given had rows and columns labeled with the factors, and the products formed the body of the table. To the nearest hundredth, what fraction of the numbers in the body of the table are odd?

- (A) 0.21 (B) 0.25 (C) 0.46 (D) 0.50 (E) 0.75

Solution

Problem 7

A regular 15-gon has L lines of symmetry, and the smallest positive angle for which it has rotational symmetry is R degrees. What is $L + R$?

- (A) 24 (B) 27 (C) 32 (D) 39 (E) 54

Solution

Problem 8

What is the value of $(625^{\log_5 2015})^{\frac{1}{4}}$?

- (A) 5 (B) $\sqrt[4]{2015}$ (C) 625 (D) 2015 (E) $\sqrt[4]{5^{2015}}$

Solution

Problem 9

Larry and Julius are playing a game, taking turns throwing a ball at a bottle sitting on a ledge. Larry throws first. The winner is the first person to knock the bottle off the ledge. At each turn the probability that a player knocks the bottle off the ledge is $\frac{1}{2}$, independently of what has happened before. What is the probability that Larry wins the game?

- (A) $\frac{1}{2}$ (B) $\frac{3}{5}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$ (E) $\frac{4}{5}$

Solution

Problem 10

How many noncongruent integer-sided triangles with positive area and perimeter less than 15 are neither equilateral, isosceles, nor right triangles?

- (A) 3 (B) 4 (C) 5 (D) 6 (E) 7

Solution

Problem 11

The line $12x + 5y = 60$ forms a triangle with the coordinate axes. What is the sum of the lengths of the altitudes of this triangle?

- (A) 20 (B) $\frac{360}{17}$ (C) $\frac{107}{5}$ (D) $\frac{43}{2}$ (E) $\frac{281}{13}$

Solution

Problem 12

Let a , b , and c be three distinct one-digit numbers. What is the maximum value of the sum of the roots of the equation $(x - a)(x - b) + (x - b)(x - c) = 0$?

- (A) 15 (B) 15.5 (C) 16 (D) 16.5 (E) 17

Solution

Problem 13

Quadrilateral $ABCD$ is inscribed in a circle with $\angle BAC = 70^\circ$, $\angle ADB = 40^\circ$, $AD = 4$, and $BC = 6$. What is AC ?

- (A) $3 + \sqrt{5}$ (B) 6 (C) $\frac{9}{2}\sqrt{2}$ (D) $8 - \sqrt{2}$ (E) 7

Solution

Problem 14

A circle of radius 2 is centered at A . An equilateral triangle with side 4 has a vertex at A . What is the difference between the area of the region that lies inside the circle but outside the triangle and the area of the region that lies inside the triangle but outside the circle?

- (A) $8 - \pi$ (B) $\pi + 2$ (C) $2\pi - \frac{\sqrt{2}}{2}$ (D) $4(\pi - \sqrt{3})$ (E) $2\pi - \frac{\sqrt{3}}{2}$

Solution

Problem 15

At Rachel's school an A counts 4 points, a B 3 points, a C 2 points, and a D 1 point. Her GPA on the four classes she is taking is computed as the total sum of points divided by 4. She is certain that she will get As in both Mathematics and Science, and at least a C in each of English and History. She thinks she has a $\frac{1}{6}$ chance of getting an A in English, and a $\frac{1}{4}$ chance of getting a B. In History, she has a $\frac{1}{4}$ chance of getting an A, and a $\frac{1}{3}$ chance of getting a B, independently of what she gets in English. What is the probability that Rachel will get a GPA of at least 3.5?

- (A) $\frac{11}{72}$ (B) $\frac{1}{6}$ (C) $\frac{3}{16}$ (D) $\frac{11}{24}$ (E) $\frac{1}{2}$

Solution

Problem 16

A regular hexagon with sides of length 6 has an isosceles triangle attached to each side. Each of these triangles has two sides of length 8. The isosceles triangles are folded to make a pyramid with the hexagon as the base of the pyramid. What is the volume of the pyramid?

- (A) 18 (B) 162 (C) $36\sqrt{21}$ (D) $18\sqrt{138}$ (E) $54\sqrt{21}$

Solution

Problem 17

An unfair coin lands on heads with a probability of $\frac{1}{4}$. When tossed n times, the probability of exactly two heads is the same as the probability of exactly three heads. What is the value of n ?

- (A) 5 (B) 8 (C) 10 (D) 11 (E) 13

Solution

Problem 18

For every composite positive integer n , define $r(n)$ to be the sum of the factors in the prime factorization of n . For example, $r(50) = 12$ because the prime factorization of 50 is 2×5^2 , and $2 + 5 + 5 = 12$. What is the range of the function r , $\{r(n) : n \text{ is a composite positive integer}\}$?

- (A) the set of positive integers
 (B) the set of composite positive integers
 (C) the set of even positive integers
 (D) the set of integers greater than 3
 (E) the set of integers greater than 4

Solution

Problem 19

In $\triangle ABC$, $\angle C = 90^\circ$ and $AB = 12$. Squares $ABXY$ and $ACWZ$ are constructed outside of the triangle. The points X, Y, Z , and W lie on a circle. What is the perimeter of the triangle?

- (A) $12 + 9\sqrt{3}$ (B) $18 + 6\sqrt{3}$ (C) $12 + 12\sqrt{2}$ (D) 30 (E) 32

Solution

Problem 20

For every positive integer n , let $\text{mod}_5(n)$ be the remainder obtained when n is divided by 5. Define a function $f : \{0, 1, 2, 3, \dots\} \times \{0, 1, 2, 3, 4\} \rightarrow \{0, 1, 2, 3, 4\}$ recursively as follows:

$$f(i, j) = \begin{cases} \text{mod}_5(j + 1) & \text{if } i = 0 \text{ and } 0 \leq j \leq 4, \\ f(i - 1, 1) & \text{if } i \geq 1 \text{ and } j = 0, \text{ and} \\ f(i - 1, f(i, j - 1)) & \text{if } i \geq 1 \text{ and } 1 \leq j \leq 4. \end{cases}$$

What is $f(2015, 2)$?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Solution

Problem 21

Cozy the Cat and Dash the Dog are going up a staircase with a certain number of steps. However, instead of walking up the steps one at a time, both Cozy and Dash jump. Cozy goes two steps up with each jump (though if necessary, he will just jump the last step). Dash goes five steps up with each jump (though if necessary, he will just jump the last steps if there are fewer than 5 steps

left). Suppose that Dash takes 19 fewer jumps than Cozy to reach the top of the staircase. Let S denote the sum of all possible numbers of steps this staircase can have. What is the sum of the digits of S ?

- (A) 9 (B) 11 (C) 12 (D) 13 (E) 15

Solution

Problem 22

Six chairs are evenly spaced around a circular table. One person is seated in each chair. Each person gets up and sits down in a chair that is not the same chair and is not adjacent to the chair he or she originally occupied, so that again one person is seated in each chair. In how many ways can this be done?

- (A) 14 (B) 16 (C) 18 (D) 20 (E) 24

Solution

Problem 23

A rectangular box measures $a \times b \times c$, where a, b , and c are integers and $1 \leq a \leq b \leq c$. The volume and the surface area of the box are numerically equal. How many ordered triples (a, b, c) are possible?

- (A) 4 (B) 10 (C) 12 (D) 21 (E) 26

Solution

Problem 24

Four circles, no two of which are congruent, have centers at A, B, C , and D , and points P and Q lie on all four circles. The radius of circle A is $\frac{5}{8}$ times the radius of circle B , and the radius of circle C is $\frac{5}{8}$ times the radius of circle D . Furthermore, $AB = CD = 39$ and $PQ = 48$. Let R be the midpoint of $\overline{PQ}$. What is $AR + BR + CR + DR$?

- (A) 180 (B) 184 (C) 188 (D) 192 (E) 196

Solution

Problem 25

A bee starts flying from point P_0 . She flies 1 inch due east to point P_1 . For $j \geq 1$, once the bee reaches point P_j , she turns 30° counterclockwise and then flies $j + 1$ inches straight to point P_{j+1} . When the bee reaches P_{2015} she is exactly $a\sqrt{b} + c\sqrt{d}$ inches away from P_0 , where a, b, c and d are positive integers and b and d are not divisible by the square of any prime. What is $a + b + c + d$?

- (A) 2016 (B) 2024 (C) 2032 (D) 2040 (E) 2048

Solution

See also

2015 AMC 12B (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2015))	
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1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
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- AMC 10
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2015 AMC 12B Answer Key

1. C
2. B
3. A
4. B
5. B
6. A
7. D
8. D
9. C
10. C
11. E
12. D
13. B
14. D
15. D
16. C
17. D
18. D
19. C
20. B
21. D
22. D
23. B
24. D
25. B

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2016

2016 AMC 12A Problems

2016 AMC 12A (Answer Key)

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1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

What is the value of $\frac{11! - 10!}{9!}$?

- (A) 99 (B) 100 (C) 110 (D) 121 (E) 132

Solution

Problem 2

For what value of x does $10^x \cdot 100^{2x} = 1000^5$?

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Solution

Problem 3

The remainder can be defined for all real numbers x and y with $y \neq 0$ by

$$\text{rem}(x, y) = x - y \left\lfloor \frac{x}{y} \right\rfloor$$

where $\left\lfloor \frac{x}{y} \right\rfloor$ denotes the greatest integer less than or equal to $\frac{x}{y}$. What is the value of $\text{rem}(\frac{3}{8}, -\frac{2}{5})$?

(A) $-\frac{3}{8}$ (B) $-\frac{1}{40}$ (C) 0 (D) $\frac{3}{8}$ (E) $\frac{31}{40}$

Solution

Problem 4

The mean, median, and mode of the 7 data values 60, 100, x , 40, 50, 200, 90 are all equal to x . What is the value of x ?

(A) 50 (B) 60 (C) 75 (D) 90 (E) 100

Solution

Problem 5

Goldbach's conjecture states that every even integer greater than 2 can be written as the sum of two prime numbers (for example, $2016 = 13 + 2003$). So far, no one has been able to prove that the conjecture is true, and no one has found a counterexample to show that the conjecture is false. What would a counterexample consist of?

- (A) an odd integer greater than 2 that can be written as the sum of two prime numbers
- (B) an odd integer greater than 2 that cannot be written as the sum of two prime numbers
- (C) an even integer greater than 2 that can be written as the sum of two numbers that are not prime
- (D) an even integer greater than 2 that can be written as the sum of two prime numbers
- (E) an even integer greater than 2 that cannot be written as the sum of two prime numbers

Solution

Problem 6

A triangular array of 2016 coins has 1 coin in the first row, 2 coins in the second row, 3 coins in the third row, and so on up to N coins in the N th row. What is the sum of the digits of N ?

(A) 6 (B) 7 (C) 8 (D) 9 (E) 10

Solution

Problem 7

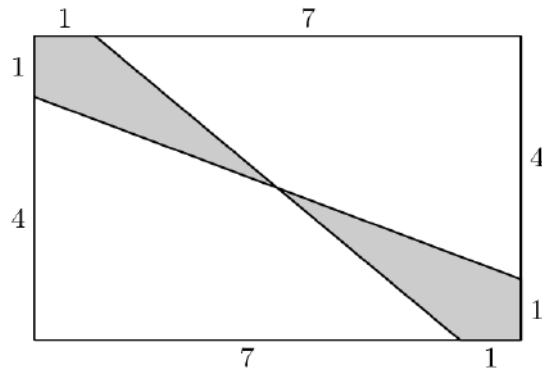
Which of these describes the graph of $x^2(x + y + 1) = y^2(x + y + 1)$?

- (A) two parallel lines
- (B) two intersecting lines
- (C) three lines that all pass through a common point
- (D) three lines that do not all pass through a common point
- (E) a line and a parabola

Solution

Problem 8

What is the area of the shaded region of the given 8×5 rectangle?

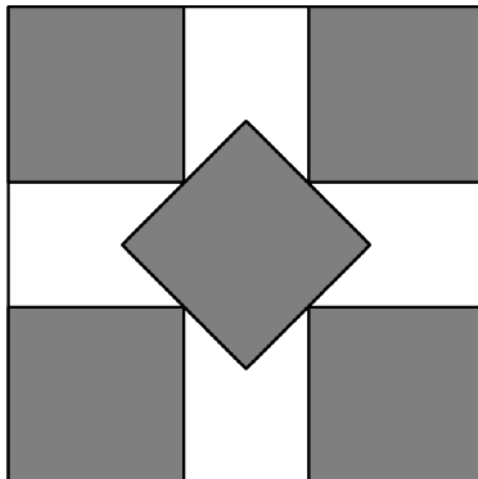


- (A) $4\frac{3}{4}$ (B) 5 (C) $5\frac{1}{4}$ (D) $6\frac{1}{2}$ (E) 8

Solution

Problem 9

The five small shaded squares inside this unit square are congruent and have disjoint interiors. The midpoint of each side of the middle square coincides with one of the vertices of the other four small squares as shown. The common side length is $\frac{a-\sqrt{2}}{b}$, where a and b are positive integers. What is $a + b$?



- (A) 7 (B) 8 (C) 9 (D) 10 (E) 11

Solution

Problem 10

Five friends sat in a movie theater in a row containing 5 seats, numbered 1 to 5 from left to right. (The directions "left" and "right" are from the point of view of the people as they sit in the seats.) During the movie Ada went to the lobby to get some popcorn. When she returned, she found that Bea had moved two seats to the right, Ceci had moved one seat to the left, and Dee and Edie had switched seats, leaving an end seat for Ada. In which seat had Ada been sitting before she got up?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Solution

Problem 11

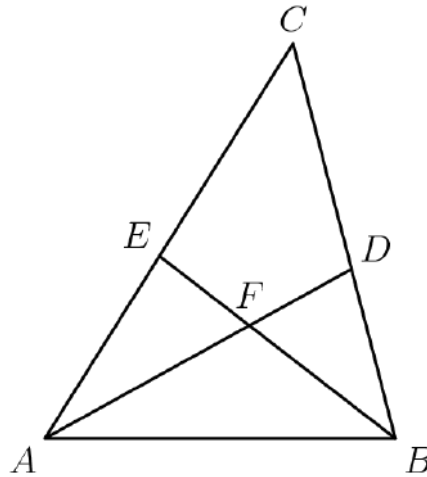
Each of the 100 students in a certain summer camp can either sing, dance, or act. Some students have more than one talent, but no student has all three talents. There are 42 students who cannot sing, 65 students who cannot dance, and 29 students who cannot act. How many students have two of these talents?

- (A) 16 (B) 25 (C) 36 (D) 49 (E) 64

Solution

Problem 12

In $\triangle ABC$, $AB = 6$, $BC = 7$, and $CA = 8$. Point D lies on $\overline{BC}$, and $\overline{AD}$ bisects $\angle BAC$. Point E lies on $\overline{AC}$, and $\overline{BE}$ bisects $\angle ABC$. The bisectors intersect at F . What is the ratio $AF : FD$?



- (A) 3 : 2 (B) 5 : 3 (C) 2 : 1 (D) 7 : 3 (E) 5 : 2

Solution

Problem 13

Let N be a positive multiple of 5. One red ball and N green balls are arranged in a line in random order. Let $P(N)$ be the probability that at least $\frac{3}{5}$ of the green balls are on the same side of the red ball. Observe that $P(5) = 1$ and that $P(N)$ approaches $\frac{4}{5}$ as N grows large. What is the sum of the digits of the least value of N such that $P(N) < \frac{321}{400}$?

- (A) 12 (B) 14 (C) 16 (D) 18 (E) 20

Solution

Problem 14

Each vertex of a cube is to be labeled with an integer from 1 through 8, with each integer being used once, in such a way that the sum of the four numbers on the vertices of a face is the same for each face. Arrangements that can be obtained from each other through rotations of the cube are considered to be the same. How many different arrangements are possible?

- (A) 1 (B) 3 (C) 6 (D) 12 (E) 24

Solution

Problem 15

Circles with centers P , Q and R , having radii 1, 2 and 3, respectively, lie on the same side of line l and are tangent to l at P' , Q' and R' , respectively, with Q' between P' and R' . The circle with center Q is externally tangent to each of the other two circles. What is the area of triangle PQR ?

- (A) 0 (B) $\sqrt{6}/3$ (C) 1 (D) $\sqrt{6} - \sqrt{2}$ (E) $\sqrt{6}/2$

Solution

Problem 16

The graphs of $y = \log_3 x$, $y = \log_x 3$, $y = \log_{\frac{1}{3}} x$, and $y = \log_x \frac{1}{3}$ are plotted on the same set of axis. How many points in the plane with positive x -coordinates lie on two or more of the graphs?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Solution

Problem 17

Let $ABCD$ be a square. Let E , F , G and H be the centers, respectively, of equilateral triangles with bases $\overline{AB}$, $\overline{BC}$, $\overline{CD}$, and $\overline{DA}$, each exterior to the square. What is the ratio of the area of square $EFGH$ to the area of square $ABCD$?

(A) 1 (B) $\frac{2 + \sqrt{3}}{3}$ (C) $\sqrt{2}$ (D) $\frac{\sqrt{2} + \sqrt{3}}{2}$ (E) $\sqrt{3}$

Solution

Problem 18

For some positive integer n , the number $110n^3$ has 110 positive integer divisors, including 1 and the number $110n^3$. How many positive integer divisors does the number $81n^4$ have?

(A) 110 (B) 191 (C) 261 (D) 325 (E) 425

Solution

Problem 19

Jerry starts at 0 on the real number line. He tosses a fair coin 8 times. When he gets heads, he moves 1 unit in the positive direction; when he gets tails, he moves 1 unit in the negative direction. The probability that he reaches 4 at some time during this process is $\frac{a}{b}$, where a and b are relatively prime positive integers. What is $a + b$? (For example, he succeeds if his sequence of tosses is *HTHHHHHH*.)

(A) 69 (B) 151 (C) 257 (D) 293 (E) 313

Solution

Problem 20

A binary operation $\diamond$ has the properties that $a \diamond (b \diamond c) = (a \diamond b) \cdot c$ and that $a \diamond a = 1$ for all nonzero real numbers a , b and c . (Here the dot $\cdot$ represents the usual multiplication operation.) The solution to the equation $2016 \diamond (6 \diamond x) = 100$ can be written as $\frac{p}{q}$, where p and q are relatively prime positive integers. What is $p + q$?

(A) 109 (B) 201 (C) 301 (D) 3049 (E) 33,601

Solution

Problem 21

A quadrilateral is inscribed in a circle of radius $200\sqrt{2}$. Three of the sides of this quadrilateral have length 200. What is the length of its fourth side?

(A) 200 (B) $200\sqrt{2}$ (C) $200\sqrt{3}$ (D) $300\sqrt{2}$ (E) 500

Solution

Problem 22

How many ordered triples (x, y, z) of positive integers satisfy $\text{lcm}(x, y) = 72$, $\text{lcm}(x, z) = 600$ and $\text{lcm}(y, z) = 900$?

(A) 15 (B) 16 (C) 24 (D) 27 (E) 64

Solution

Problem 23

Three numbers in the interval $[0, 1]$ are chosen independently and at random. What is the probability that the chosen numbers are the side lengths of a triangle with positive area?

(A) $\frac{1}{6}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$ (E) $\frac{5}{6}$

Solution

Problem 24

There is a smallest positive real number a such that there exists a positive real number b such that all the roots of the polynomial $x^3 - ax^2 + bx - a$ are real. In fact, for this value of a the value of b is unique. What is the value of b ?

(A) 8 (B) 9 (C) 10 (D) 11 (E) 12

Solution

Problem 25

Let k be a positive integer. Bernardo and Silvia take turns writing and erasing numbers on a blackboard as follows: Bernardo starts by writing the smallest perfect square with $k + 1$ digits. Every time Bernardo writes a number, Silvia erases the last k digits of it. Bernardo then writes the next perfect square, Silvia erases the last k digits of it, and this process continues until the last two numbers that remain on the board differ by at least 2. Let $f(k)$ be the smallest positive integer not written on the board. For example, if $k = 1$, then the numbers that Bernardo writes are 16, 25, 36, 49, 64, and the numbers showing on the board after Silvia erases are 1, 2, 3, 4, and 6, and thus $f(1) = 5$. What is the sum of the digits of $f(2) + f(4) + f(6) + \dots + f(2016)$?

(A) 7986 (B) 8002 (C) 8030 (D) 8048 (E) 8064

Solution

See also

2016 AMC 12A (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2016))	
Preceded by 2015 AMC 12A Problems	Followed by 2017 AMC 12A Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
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2016 AMC 12A Answer Key

1. B
2. C
3. B
4. D
5. E
6. D
7. D
8. D
9. E
10. B
11. E
12. C
13. A
14. C
15. D
16. D
17. B
18. D
19. B
20. A
21. E
22. A
23. C
24. B
25. E

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2016 AMC 12B Problems

2016 AMC 12B (Answer Key)

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Instructions

1. This is a 25-question, multiple choice test. Each question is followed by answers marked A, B, C, D and E. Only one of these is correct.
2. You will receive 6 points for each correct answer, 2.5 points for each problem left unanswered if the year is before 2006, 1.5 points for each problem left unanswered if the year is after 2006, and 0 points for each incorrect answer.
3. No aids are permitted other than scratch paper, graph paper, ruler, compass, protractor and erasers (and calculators that are accepted for use on the test if before 2006. No problems on the test will *require* the use of a calculator).
4. Figures are not necessarily drawn to scale.
5. You will have **75 minutes** working time to complete the test.

1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

What is the value of $\frac{2a^{-1} + \frac{a^{-1}}{2}}{a}$ when $a = \frac{1}{2}$?

- (A) 1 (B) 2 (C) $\frac{5}{2}$ (D) 10 (E) 20

Solution

Problem 2

The harmonic mean of two numbers can be calculated as twice their product divided by their sum. The harmonic mean of 1 and 2016 is closest to which integer?

- (A) 2 (B) 45 (C) 504 (D) 1008 (E) 2015

Solution

Problem 3

Let $x = -2016$. What is the value of $\left| \left| |x| - x \right| - |x| \right| - x$?

- (A) -2016 (B) 0 (C) 2016 (D) 4032 (E) 6048

Solution

Problem 4

The ratio of the measures of two acute angles is $5 : 4$, and the complement of one of these two angles is twice as large as the complement of the other. What is the sum of the degree measures of the two angles?

- (A) 75 (B) 90 (C) 135 (D) 150 (E) 270

Solution

Problem 5

The War of 1812 started with a declaration of war on Thursday, June 18, 1812. The peace treaty to end the war was signed 919 days later, on December 24, 1814. On what day of the week was the treaty signed?

- (A) Friday (B) Saturday (C) Sunday (D) Monday (E) Tuesday

Solution

Problem 6

All three vertices of $\triangle ABC$ lie on the parabola defined by $y = x^2$, with A at the origin and $\overline{BC}$ parallel to the x -axis. The area of the triangle is 64. What is the length of BC ?

- (A) 4 (B) 6 (C) 8 (D) 10 (E) 16

Solution

Problem 7

Josh writes the numbers 1, 2, 3, ..., 99, 100. He marks out 1, skips the next number (2), marks out 3, and continues skipping and marking out the next number to the end of the list. Then he goes back to the start of his list, marks out the first remaining number (2), skips the next number (4), marks out 6, skips 8, marks out 10, and so on to the end. Josh continues in this manner until only one number remains. What is that number?

- (A) 13 (B) 32 (C) 56 (D) 64 (E) 96

Solution

Problem 8

A thin piece of wood of uniform density in the shape of an equilateral triangle with side length 3 inches weighs 12 ounces. A second piece of the same type of wood, with the same thickness, also in the shape of an equilateral triangle, has side length of 5 inches. Which of the following is closest to the weight, in ounces, of the second piece?

- (A) 14.0 (B) 16.0 (C) 20.0 (D) 33.3 (E) 55.6

Solution

Problem 9

Carl decided to fence in his rectangular garden. He bought 20 fence posts, placed one on each of the four corners, and spaced out the rest evenly along the edges of the garden, leaving exactly 4 yards between neighboring posts. The longer side of his garden, including the corners, has twice as many posts as the shorter side, including the corners. What is the area, in square yards, of Carl's garden?

- (A) 256 (B) 336 (C) 384 (D) 448 (E) 512

Solution

Problem 10

A quadrilateral has vertices $P(a, b)$, $Q(b, a)$, $R(-a, -b)$, and $S(-b, -a)$, where a and b are integers with $a > b > 0$. The area of $PQRS$ is 16. What is $a + b$?

- (A) 4 (B) 5 (C) 6 (D) 12 (E) 13

Solution

Problem 11

How many squares whose sides are parallel to the axes and whose vertices have coordinates that are integers lie entirely within the region bounded by the line $y = \pi x$, the line $y = -0.1$ and the line $x = 5.1$?

- (A) 30 (B) 41 (C) 45 (D) 50 (E) 57

Solution

Problem 12

All the numbers 1, 2, 3, 4, 5, 6, 7, 8, 9 are written in a 3×3 array of squares, one number in each square, in such a way that if two numbers are consecutive then they occupy squares that share an edge. The numbers in the four corners add up to 18. What is the number in the center?

- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

Solution

Problem 13

Alice and Bob live 10 miles apart. One day Alice looks due north from her house and sees an airplane. At the same time Bob looks due west from his house and sees the same airplane. The angle of elevation of the airplane is 30° from Alice's position and 60° from Bob's position. Which of the following is closest to the airplane's altitude, in miles?

- (A) 3.5 (B) 4 (C) 4.5 (D) 5 (E) 5.5

Solution

Problem 14

The sum of an infinite geometric series is a positive number S , and the second term in the series is 1. What is the smallest possible value of S ?

- (A) $\frac{1 + \sqrt{5}}{2}$ (B) 2 (C) $\sqrt{5}$ (D) 3 (E) 4

Solution

Problem 15

All the numbers 2, 3, 4, 5, 6, 7 are assigned to the six faces of a cube, one number to each face. For each of the eight vertices of the cube, a product of three numbers is computed, where the three numbers are the numbers assigned to the three faces that include that vertex. What is the greatest possible value of the sum of these eight products?

- (A) 312 (B) 343 (C) 625 (D) 729 (E) 1680

Solution

Problem 16

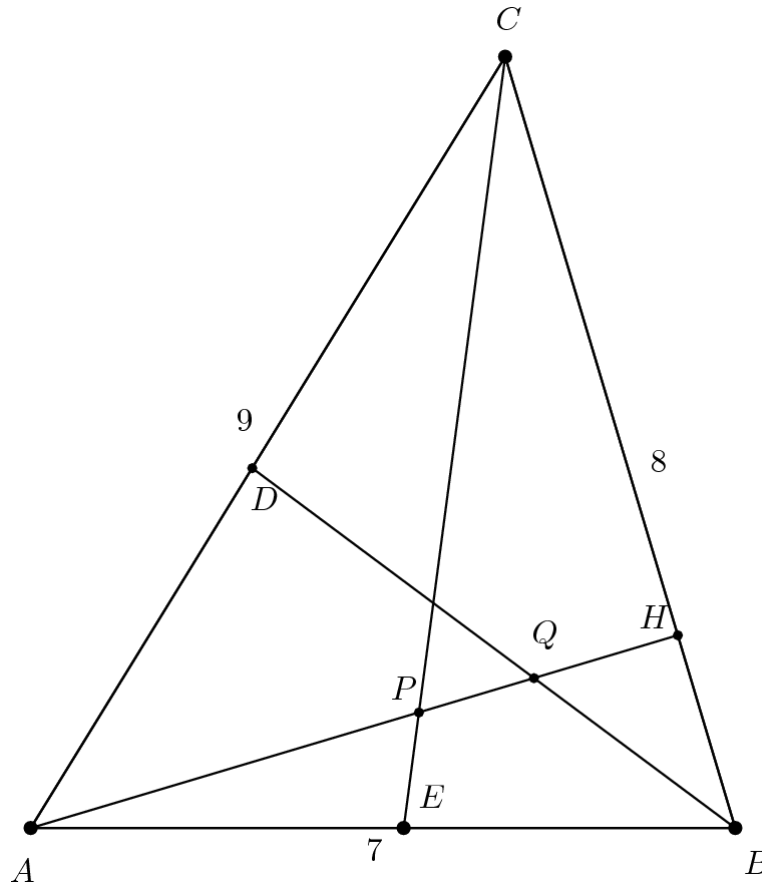
In how many ways can 345 be written as the sum of an increasing sequence of two or more consecutive positive integers?

- (A) 1 (B) 3 (C) 5 (D) 6 (E) 7

Solution

Problem 17

In $\triangle ABC$ shown in the figure, $AB = 7$, $BC = 8$, $CA = 9$, and $\overline{AH}$ is an altitude. Points D and E lie on sides $\overline{AC}$ and $\overline{AB}$, respectively, so that $\overline{BD}$ and $\overline{CE}$ are angle bisectors, intersecting $\overline{AH}$ at Q and P , respectively. What is PQ ?



- (A) 1 (B) $\frac{5}{8}\sqrt{3}$ (C) $\frac{4}{5}\sqrt{2}$ (D) $\frac{8}{15}\sqrt{5}$ (E) $\frac{6}{5}$

Solution

Problem 18

What is the area of the region enclosed by the graph of the equation $x^2 + y^2 = |x| + |y|$?

- (A) $\pi + \sqrt{2}$ (B) $\pi + 2$ (C) $\pi + 2\sqrt{2}$ (D) $2\pi + \sqrt{2}$ (E) $2\pi + 2\sqrt{2}$

Solution

Problem 19

Tom, Dick, and Harry are playing a game. Starting at the same time, each of them flips a fair coin repeatedly until he gets his first head, at which point he stops. What is the probability that all three flip their coins the same number of times?

- (A) $\frac{1}{8}$ (B) $\frac{1}{7}$ (C) $\frac{1}{6}$ (D) $\frac{1}{4}$ (E) $\frac{1}{3}$

Solution

Problem 20

A set of teams held a round-robin tournament in which every team played every other team exactly once. Every team won 10 games and lost 10 games; there were no ties. How many sets of three teams $\{A, B, C\}$ were there in which A beat B , B beat C , and C beat A ?

- (A) 385 (B) 665 (C) 945 (D) 1140 (E) 1330

Solution

Problem 21

Let $ABCD$ be a unit square. Let Q_1 be the midpoint of $\overline{CD}$. For $i = 1, 2, \dots$, let P_i be the intersection of $\overline{AQ_i}$ and $\overline{BD}$, and let Q_{i+1} be the foot of the perpendicular from P_i to $\overline{CD}$. What is

$$\sum_{i=1}^{\infty} \text{Area of } \triangle DQ_iP_i?$$

- (A) $\frac{1}{6}$ (B) $\frac{1}{4}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$ (E) 1

Solution

Problem 22

For a certain positive integer n less than 1000, the decimal equivalent of $\frac{1}{n}$ is $0.\overline{abcdef}$, a repeating decimal of period 6, and the decimal equivalent of $\frac{1}{n+6}$ is $0.\overline{wxyz}$, a repeating decimal of period 4. In which interval does n lie?

- (A) $[1, 200]$ (B) $[201, 400]$ (C) $[401, 600]$ (D) $[601, 800]$ (E) $[801, 999]$

Solution

Problem 23

What is the volume of the region in three-dimensional space defined by the inequalities $|x| + |y| + |z| \leq 1$ and $|x| + |y| + |z - 1| \leq 1$?

- (A) $\frac{1}{6}$ (B) $\frac{1}{4}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$ (E) 1

Solution

Problem 24

There are exactly 77,000 ordered quadruplets (a, b, c, d) such that $\gcd(a, b, c, d) = 77$ and $\text{lcm}(a, b, c, d) = n$. What is the smallest possible value for n ?

- (A) 13,860 (B) 20,790 (C) 21,560 (D) 27,720 (E) 41,580

Solution

Problem 25

The sequence (a_n) is defined recursively by $a_0 = 1$, $a_1 = \sqrt[19]{2}$, and $a_n = a_{n-1}a_{n-2}^2$ for $n \geq 2$. What is the smallest positive integer k such that the product $a_1a_2 \cdots a_k$ is an integer?

- (A) 17 (B) 18 (C) 19 (D) 20 (E) 21

Solution

See also

2016 AMC 12B (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2016))	
Preceded by 2016 AMC 12A Problems	Followed by 2017 AMC 12A Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
All AMC 12 Problems and Solutions	

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2016 AMC 12B Answer Key

1. D
2. A
3. D
4. C
5. B
6. C
7. D
8. D
9. B
10. A
11. D
12. C
13. E
14. E
15. D
16. E
17. D
18. B
19. B
20. A
21. B
22. B
23. A
24. D
25. A

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2017

2017 AMC 12A Problems

2017 AMC 12A (Answer Key)

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Instructions

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1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 •
18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

Pablo buys popsicles for his friends. The store sells single popsicles for \$1 each, 3-popsicle boxes for \$2, and 5-popsicle boxes for \$3. What is the greatest number of popsicles that Pablo can buy with \$8?

(A) 8 (B) 11 (C) 12 (D) 13 (E) 15

Solution

Problem 2

The sum of two nonzero real numbers is 4 times their product. What is the sum of the reciprocals of the two numbers?

- (A) 1 (B) 2 (C) 4 (D) 8 (E) 12

Solution

Problem 3

Ms. Carroll promised that anyone who got all the multiple choice questions right on the upcoming exam would receive an A on the exam. Which one of these statements necessarily follows logically?

- (A) If Lewis did not receive an A, then he got all of the multiple choice questions wrong.
(B) If Lewis did not receive an A, then he got at least one of the multiple choice questions wrong.
(C) If Lewis got at least one of the multiple choice questions wrong, then he did not receive an A.
(D) If Lewis received an A, then he got all of the multiple choice questions right.
(E) If Lewis received an A, then he got at least one of the multiple choice questions right.

Solution

Problem 4

Jerry and Silvia wanted to go from the southwest corner of a square field to the northeast corner. Jerry walked due east and then due north to reach the goal, but Silvia headed northeast and reached the goal walking in a straight line. Which of the following is closest to how much shorter Silvia's trip was, compared to Jerry's trip?

- (A) 30% (B) 40% (C) 50% (D) 60% (E) 70%

Solution

Problem 5

At a gathering of 30 people, there are 20 people who all know each other and 10 people who know no one. People who know each other hug, and people who do not know each other shake hands. How many handshakes occur?

- (A) 240 (B) 245 (C) 290 (D) 480 (E) 490

Solution

Problem 6

Joy has 30 thin rods, one each of every integer length from 1 cm through 30 cm. She places the rods with lengths 3 cm, 7 cm, and 15 cm on a table. She then wants to choose a fourth rod that she can put with these three to form a quadrilateral with positive area. How many of the remaining rods can she choose as the fourth rod?

- (A) 16 (B) 17 (C) 18 (D) 19 (E) 20

Solution

Problem 7

Define a function on the positive integers recursively by $f(1) = 2$, $f(n) = f(n-1) + 1$ if n is even, and $f(n) = f(n-2) + 2$ if n is odd and greater than 1. What is $f(2017)$?

- (A) 2017 (B) 2018 (C) 4034 (D) 4035 (E) 4036

Solution

Problem 8

The region consisting of all points in three-dimensional space within 3 units of line segment $\overline{AB}$ has volume 216π . What is the length AB ?

- (A) 6 (B) 12 (C) 18 (D) 20 (E) 24

Solution

Problem 9

Let S be the set of points (x, y) in the coordinate plane such that two of the three quantities 3 , $x + 2$, and $y - 4$ are equal and the third of the three quantities is no greater than the common value. Which of the following is a correct description of S ?

- (A) a single point (B) two intersecting lines
 (C) three lines whose pairwise intersections are three distinct points
 (D) a triangle (E) three rays with a common point

Solution

Problem 10

Chloé chooses a real number uniformly at random from the interval $[0, 2017]$. Independently, Laurent chooses a real number uniformly at random from the interval $[0, 4034]$. What is the probability that Laurent's number is greater than Chloe's number?

- (A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{5}{6}$ (E) $\frac{7}{8}$

Solution

Problem 11

Claire adds the degree measures of the interior angles of a convex polygon and arrives at a sum of 2017 . She then discovers that she forgot to include one angle. What is the degree measure of the forgotten angle?

- (A) 37 (B) 63 (C) 117 (D) 143 (E) 163

Solution

Problem 12

There are 10 horses, named Horse 1, Horse 2, . . . , Horse 10. They get their names from how many minutes it takes them to run one lap around a circular race track: Horse k runs one lap in exactly k minutes. At time 0 all the horses are together at the starting point on the track. The horses start running in the same direction, and they keep running around the circular track at their constant speeds. The least time $S > 0$, in minutes, at which all 10 horses will again simultaneously be at the starting point is $S = 2520$. Let $T > 0$ be the least time, in minutes, such that at least 5 of the horses are again at the starting point. What is the sum of the digits of T ?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Solution

Problem 13

Driving at a constant speed, Sharon usually takes 180 minutes to drive from her house to her mother's house. One day Sharon begins the drive at her usual speed, but after driving $\frac{1}{3}$ of the way, she hits a bad snowstorm and reduces her speed by 20 miles per hour. This time the trip takes her a total of 276 minutes. How many miles is the drive from Sharon's house to her mother's house?

- (A) 132 (B) 135 (C) 138 (D) 141 (E) 144

Solution

Problem 14

Alice refuses to sit next to either Bob or Carla. Derek refuses to sit next to Eric. How many ways are there for the five of them to sit in a row of 5 chairs under these conditions?

- (A) 12 (B) 16 (C) 28 (D) 32 (E) 40

Solution

Problem 15

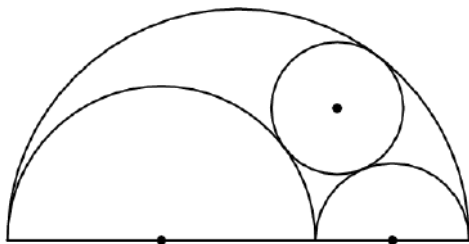
Let $f(x) = \sin x + 2 \cos x + 3 \tan x$, using radian measure for the variable x . In what interval does the smallest positive value of x for which $f(x) = 0$ lie?

- (A) $(0, 1)$ (B) $(1, 2)$ (C) $(2, 3)$ (D) $(3, 4)$ (E) $(4, 5)$

Solution

Problem 16

In the figure below, semicircles with centers at A and B and with radii 2 and 1, respectively, are drawn in the interior of, and sharing bases with, a semicircle with diameter JK . The two smaller semicircles are externally tangent to each other and internally tangent to the largest semicircle. A circle centered at P is drawn externally tangent to the two smaller semicircles and internally tangent to the largest semicircle. What is the radius of the circle centered at P ?



- (A) $\frac{3}{4}$ (B) $\frac{6}{7}$ (C) $\frac{1}{2}\sqrt{3}$ (D) $\frac{5}{8}\sqrt{2}$ (E) $\frac{11}{12}$

Solution

Problem 17

There are 24 different complex numbers z such that $z^{24} = 1$. For how many of these is z^6 a real number?

- (A) 0 (B) 4 (C) 6 (D) 12 (E) 24

Solution

Problem 18

Let $S(n)$ equal the sum of the digits of positive integer n . For example, $S(1507) = 13$. For a particular positive integer n , $S(n) = 1274$. Which of the following could be the value of $S(n+1)$?

- (A) 1 (B) 3 (C) 12 (D) 1239 (E) 1265

Solution

Problem 19

A square with side length x is inscribed in a right triangle with sides of length 3, 4, and 5 so that one vertex of the square coincides with the right-angle vertex of the triangle. A square with side length y is inscribed in another right triangle with sides of length 3, 4, and 5 so that one side of the square lies on the hypotenuse of the triangle. What is $\frac{x}{y}$?

- (A) $\frac{12}{13}$ (B) $\frac{35}{37}$ (C) 1 (D) $\frac{37}{35}$ (E) $\frac{13}{12}$

Solution

Problem 20

How many ordered pairs (a, b) such that a is a positive real number and b is an integer between 2 and 200, inclusive, satisfy the equation $(\log_b a)^{2017} = \log_b(a^{2017})$?

- (A) 198 (B) 199 (C) 398 (D) 399 (E) 597

Solution

Problem 21

A set S is constructed as follows. To begin, $S = \{0, 10\}$. Repeatedly, as long as possible, if x is an integer root of some polynomial $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ for some $n \geq 1$, all of whose coefficients a_i are elements of S , then x is put into S . When no more elements can be added to S , how many elements does S have?

- (A) 4 (B) 5 (C) 7 (D) 9 (E) 11

Solution

Problem 22

A square is drawn in the Cartesian coordinate plane with vertices at $(2, 2)$, $(-2, 2)$, $(-2, -2)$, $(2, -2)$. A particle starts at $(0, 0)$. Every second it moves with equal probability to one of the eight lattice points (points with integer coordinates) closest to its current position, independently of its previous moves. In other words, the probability is $1/8$ that the particle will move from (x, y) to each of $(x, y + 1)$, $(x + 1, y + 1)$, $(x + 1, y)$, $(x + 1, y - 1)$, $(x, y - 1)$, $(x - 1, y - 1)$, $(x - 1, y)$, or $(x - 1, y + 1)$. The particle will eventually hit the square for the first time, either at one of the 4 corners of the square or at one of the 12 lattice points in the interior of one of the sides of the square. The probability that it will hit at a corner rather than at an interior point of a side is m/n , where m and n are relatively prime positive integers. What is $m + n$?

- (A) 4 (B) 5 (C) 7 (D) 15 (E) 39

Solution

Problem 23

For certain real numbers a , b , and c , the polynomial

$$g(x) = x^3 + ax^2 + x + 10$$

has three distinct roots, and each root of $g(x)$ is also a root of the polynomial

$$f(x) = x^4 + x^3 + bx^2 + 100x + c.$$

What is $f(1)$?

- (A) -9009 (B) -8008 (C) -7007 (D) -6006 (E) -5005

Solution

Problem 24

Quadrilateral $ABCD$ is inscribed in circle O and has side lengths $AB = 3$, $BC = 2$, $CD = 6$, and $DA = 8$. Let X and Y be points on $\overline{BD}$ such that $\frac{DX}{BD} = \frac{1}{4}$ and $\frac{BY}{BD} = \frac{11}{36}$. Let E be the intersection of line AX and the line through Y parallel to $\overline{AD}$. Let F be the intersection of line CX and the line through E parallel to $\overline{AC}$. Let G be the point on circle O other than C that lies on line CX . What is $XF \cdot XG$?

- (A) 17 (B) $\frac{59 - 5\sqrt{2}}{3}$ (C) $\frac{91 - 12\sqrt{3}}{4}$ (D) $\frac{67 - 10\sqrt{2}}{3}$ (E) 18

Solution

Problem 25

The vertices V of a centrally symmetric hexagon in the complex plane are given by

$$V = \left\{ \sqrt{2}i, -\sqrt{2}i, \frac{1}{\sqrt{8}}(1 + i), \frac{1}{\sqrt{8}}(-1 + i), \frac{1}{\sqrt{8}}(1 - i), \frac{1}{\sqrt{8}}(-1 - i) \right\}.$$

For each j , $1 \leq j \leq 12$, an element z_j is chosen from V at random, independently of the other choices. Let $P = \prod_{j=1}^{12} z_j$ be the product of the 12 numbers selected. What is the probability that $P = -1$?

- (A) $\frac{5 \cdot 11}{3^{10}}$ (B) $\frac{5^2 \cdot 11}{2 \cdot 3^{10}}$ (C) $\frac{5 \cdot 11}{3^9}$ (D) $\frac{5 \cdot 7 \cdot 11}{2 \cdot 3^{10}}$ (E) $\frac{2^2 \cdot 5 \cdot 11}{3^{10}}$

Solution

See also

2017 AMC 12A (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2017))	
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1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
All AMC 12 Problems and Solutions	

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2017 AMC 12A Answer Key

1. D
2. C
3. B
4. A
5. B
6. B
7. B
8. D
9. E
10. C
11. D
12. B
13. B
14. C
15. D
16. B
17. D
18. D
19. D
20. E
21. D
22. E
23. C
24. A
25. E

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2017 AMC 12B Problems

2017 AMC 12B (Answer Key)

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Instructions

1. This is a 25-question, multiple choice test. Each question is followed by answers marked A, B, C, D and E. Only one of these is correct.
2. You will receive 6 points for each correct answer, 2.5 points for each problem left unanswered if the year is before 2006, 1.5 points for each problem left unanswered if the year is after 2006, and 0 points for each incorrect answer.
3. No aids are permitted other than scratch paper, graph paper, ruler, compass, protractor and erasers (and calculators that are accepted for use on the test if before 2006. No problems on the test will *require* the use of a calculator).
4. Figures are not necessarily drawn to scale.
5. You will have **75 minutes** working time to complete the test.

1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

Kymbrea's comic book collection currently has 30 comic books in it, and she is adding to her collection at the rate of 2 comic books per month. LaShawn's collection currently has 10 comic books in it, and he is adding to his collection at the rate of 6 comic books per month. After how many months will LaShawn's collection have twice as many comic books as Kymbrea's?

- (A) 1 (B) 4 (C) 5 (D) 20 (E) 25

Solution

Problem 2

Real numbers x , y , and z satisfy the inequalities $0 < x < 1$, $-1 < y < 0$, and $1 < z < 2$. Which of the following numbers is necessarily positive?

- (A) $y + x^2$ (B) $y + xz$ (C) $y + y^2$ (D) $y + 2y^2$ (E) $y + z$

Solution

Problem 3

Supposed that x and y are nonzero real numbers such that $\frac{3x + y}{x - 3y} = -2$. What is the value of $\frac{x + 3y}{3x - y}$?

- (A) -3 (B) -1 (C) 1 (D) 2 (E) 3

Solution

Problem 4

Samia set off on her bicycle to visit her friend, traveling at an average speed of 17 kilometers per hour. When she had gone half the distance to her friend's house, a tire went flat, and she walked the rest of the way at 5 kilometers per hour. In all it took her 44 minutes to reach her friend's house. In kilometers rounded to the nearest tenth, how far did Samia walk?

- (A) 2.0 (B) 2.2 (C) 2.8 (D) 3.4 (E) 4.4

Solution

Problem 5

The data set $[6, 19, 33, 33, 39, 41, 41, 43, 51, 57]$ has median $Q_2 = 40$, first quartile $Q_1 = 33$, and third quartile $Q_3 = 43$. An outlier in a data set is a value that is more than 1.5 times the interquartile range below the first quartile (Q_1) or more than 1.5 times the interquartile range above the third quartile (Q_3), where the interquartile range is defined as $Q_3 - Q_1$. How many outliers does this data set have?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Solution

Problem 6

The circle having $(0, 0)$ and $(8, 6)$ as the endpoints of a diameter intersects the x -axis at a second point. What is the x -coordinate of this point?

- (A) $4\sqrt{2}$ (B) 6 (C) $5\sqrt{2}$ (D) 8 (E) $6\sqrt{2}$

Solution

Problem 7

The functions $\sin(x)$ and $\cos(x)$ are periodic with least period 2π . What is the least period of the function $\cos(\sin(x))$?

- (A) $\frac{\pi}{2}$ (B) π (C) 2π (D) 4π (E) It's not periodic.

Solution

Problem 8

The ratio of the short side of a certain rectangle to the long side is equal to the ratio of the long side to the diagonal. What is the square of the ratio of the short side to the long side of this rectangle?

- (A) $\frac{\sqrt{3} - 1}{2}$ (B) $\frac{1}{2}$ (C) $\frac{\sqrt{5} - 1}{2}$ (D) $\frac{\sqrt{2}}{2}$ (E) $\frac{\sqrt{6} - 1}{2}$

Solution

Problem 9

A circle has center $(-10, -4)$ and radius 13 . Another circle has center $(3, 9)$ and radius $\sqrt{65}$. The line passing through the two points of intersection of the two circles has equation $x + y = c$. What is c ?

- (A) 3 (B) $3\sqrt{3}$ (C) $4\sqrt{2}$ (D) 6 (E) $\frac{13}{2}$

Solution

Problem 10

At Typico High School, 60% of the students like dancing, and the rest dislike it. Of those who like dancing, 80% say that they like it, and the rest say that they dislike it. Of those who dislike dancing, 90% say that they dislike it, and the rest say that they like it. What fraction of students who say they dislike dancing actually like it?

- (A) 10% (B) 12% (C) 20% (D) 25% (E) $33\frac{1}{3}\%$

Solution

Problem 11

Call a positive integer *monotonous* if it is a one-digit number or its digits, when read from left to right, form either a strictly increasing or a strictly decreasing sequence. For example, 3 , 23578 , and 987620 are *monotonous*, but 88 , 7434 , and 23557 are not. How many *monotonous* positive integers are there?

- (A) 1024 (B) 1524 (C) 1533 (D) 1536 (E) 2048

Solution

Problem 12

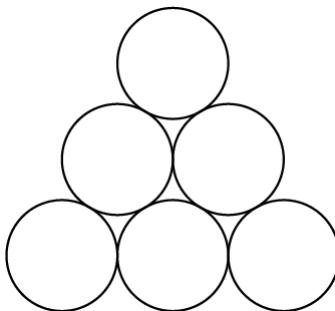
What is the sum of the roots of $z^{12} = 64$ that have a positive real part?

- (A) 2 (B) 4 (C) $\sqrt{2} + 2\sqrt{3}$ (D) $2\sqrt{2} + \sqrt{6}$ (E) $(1 + \sqrt{3}) + (1 + \sqrt{3})i$

Solution

Problem 13

In the figure below, 3 of the 6 disks are to be painted blue, 2 are to be painted red, and 1 is to be painted green. Two paintings that can be obtained from one another by a rotation or a reflection of the entire figure are considered the same. How many different paintings are possible?



- (A) 6 (B) 8 (C) 9 (D) 12 (E) 15

Solution

Problem 14

An ice-cream novelty item consists of a cup in the shape of a 4-inch-tall frustum of a right circular cone, with a 2-inch-diameter base at the bottom and a 4-inch-diameter base at the top, packed solid with ice cream, together with a solid cone of ice cream of height 4 inches, whose base, at the bottom, is the top base of the frustum. What is the total volume of the ice cream, in cubic inches?

- (A) 8π (B) $\frac{28\pi}{3}$ (C) 12π (D) 14π (E) $\frac{44\pi}{3}$

Solution

Problem 15

Let $\triangle ABC$ be an equilateral triangle. Extend side $\overline{AB}$ beyond B to a point B' so that $BB' = 3 \cdot AB$. Similarly, extend side $\overline{BC}$ beyond C to a point C' so that $CC' = 3 \cdot BC$, and extend side $\overline{CA}$ beyond A to a point A' so that $AA' = 3 \cdot CA$. What is the ratio of the area of $\triangle A'B'C'$ to the area of $\triangle ABC$?

- (A) 9 (B) 16 (C) 25 (D) 36 (E) 37

Solution

Problem 16

The number $21! = 51,090,942,171,709,440,000$ has over 60,000 positive integer divisors. One of them is chosen at random. What is the probability that it is odd?

- (A) $\frac{1}{21}$ (B) $\frac{1}{19}$ (C) $\frac{1}{18}$ (D) $\frac{1}{2}$ (E) $\frac{11}{21}$

Solution

Problem 17

A coin is biased in such a way that on each toss the probability of heads is $\frac{2}{3}$ and the probability of tails is $\frac{1}{3}$. The outcomes of the tosses are independent. A player has the choice of playing Game A or Game B. In Game A she tosses the coin three times and wins if all three outcomes are the same. In Game B she tosses the coin four times and wins if both the outcomes of the first and second tosses are the same and the outcomes of the third and fourth tosses are the same. How do the chances of winning Game A compare to the chances of winning Game B?

- (A) The probability of winning Game A is $\frac{4}{81}$ less than the probability of winning Game B.
 (B) The probability of winning Game A is $\frac{2}{81}$ less than the probability of winning Game B.
 (C) The probabilities are the same.
 (D) The probability of winning Game A is $\frac{2}{81}$ greater than the probability of winning Game B.
 (E) The probability of winning Game A is $\frac{4}{81}$ greater than the probability of winning Game B.

Solution

Problem 18

The diameter AB of a circle of radius 2 is extended to a point D outside the circle so that $BD = 3$. Point E is chosen so that $ED = 5$ and line ED is perpendicular to line AD . Segment AE intersects the circle at a point C between A and E . What is the area of $\triangle ABC$?

- (A) $\frac{120}{37}$ (B) $\frac{140}{39}$ (C) $\frac{145}{39}$ (D) $\frac{140}{37}$ (E) $\frac{120}{31}$

Solution

Problem 19

Let $N = 123456789101112 \dots 4344$ be the 79-digit number that is formed by writing the integers from 1 to 44 in order, one after the other. What is the remainder when N is divided by 45?

- (A) 1 (B) 4 (C) 9 (D) 18 (E) 44

Solution

Problem 20

Real numbers x and y are chosen independently and uniformly at random from the interval $(0, 1)$. What is the probability that $\lfloor \log_2 x \rfloor = \lfloor \log_2 y \rfloor$?

- (A) $\frac{1}{8}$ (B) $\frac{1}{6}$ (C) $\frac{1}{4}$ (D) $\frac{1}{3}$ (E) $\frac{1}{2}$

Solution

Problem 21

Last year Isabella took 7 math tests and received 7 different scores, each an integer between 91 and 100, inclusive. After each test she noticed that the average of her test scores was an integer. Her score on the seventh test was 95. What was her score on the sixth test?

- (A) 92 (B) 94 (C) 96 (D) 98 (E) 100

Solution

Problem 22

Abby, Bernardo, Carl, and Debra play a game in which each of them starts with four coins. The game consists of four rounds. In each round, four balls are placed in an urn—one green, one red, and two white. The players each draw a ball at random without replacement. Whoever gets the green ball gives one coin to whoever gets the red ball. What is the probability that, at the end of the fourth round, each of the players has four coins?

- (A) $\frac{7}{576}$ (B) $\frac{5}{192}$ (C) $\frac{1}{36}$ (D) $\frac{5}{144}$ (E) $\frac{7}{48}$

Solution

Problem 23

The graph of $y = f(x)$, where $f(x)$ is a polynomial of degree 3, contains points $A(2, 4)$, $B(3, 9)$, and $C(4, 16)$. Lines AB , AC , and BC intersect the graph again at points D , E , and F , respectively, and the sum of the x -coordinates of D , E , and F is 24. What is $f(0)$? (A) -2 (B) 0 (C) 2 (D) $\frac{24}{5}$ (E) 8

Solution

Problem 24

Quadrilateral $ABCD$ has right angles at B and C , $\triangle ABC \sim \triangle BCD$, and $AB > BC$. There is a point E in the interior of $ABCD$ such that $\triangle ABC \sim \triangle CEB$ and the area of $\triangle AED$ is 17 times the area of $\triangle CEB$. What is $\frac{AB}{BC}$?

- (A) $1 + \sqrt{2}$ (B) $2 + \sqrt{2}$ (C) $\sqrt{17}$ (D) $2 + \sqrt{5}$ (E) $1 + 2\sqrt{3}$

Solution

Problem 25

A set of n people participate in an online video basketball tournament. Each person may be a member of any number of 5-player teams, but no teams may have exactly the same 5 members. The site statistics show a curious fact: The average, over all subsets of size 9 of the set of n participants, of the number of complete teams whose members are among those 9 people is equal to the reciprocal of the average, over all subsets of size 8 of the set of n participants, of the number of complete teams whose members are among those 8 people. How many values n , $9 \leq n \leq 2017$, can be the number of participants?

(A) 477 (B) 482 (C) 487 (D) 557 (E) 562

Solution

See also

2017 AMC 12B (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2017))	
Preceded by 2017 AMC 12A Problems	Followed by 2018 AMC 12A Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
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2017 AMC 12B Answer Key

1. E
2. E
3. D
4. C
5. B
6. D
7. B
8. C
9. A
10. D
11. B
12. D
13. D
14. E
15. E
16. B
17. D
18. D
19. C
20. D
21. E
22. B
23. D
24. D
25. D

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2018

2018 AMC 12A Problems

2018 AMC 12A (Answer Key)

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1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

A large urn contains 100 balls, of which 36% are red and the rest are blue. How many of the blue balls must be removed so that the percentage of red balls in the urn will be 72%? (No red balls are to be removed.)

- (A) 28 (B) 32 (C) 36 (D) 50 (E) 64

Solution

Problem 2

While exploring a cave, Carl comes across a collection of 5-pound rocks worth \$14 each, 4-pound rocks worth \$11 each, and 1-pound rocks worth \$2 each. There are at least 20 of each size. He can carry at most 18 pounds. What is the maximum value, in dollars, of the rocks he can carry out of the cave?

- (A) 48 (B) 49 (C) 50 (D) 51 (E) 52

Solution

Problem 3

How many ways can a student schedule 3 mathematics courses – algebra, geometry, and number theory – in a 6-period day if no two mathematics courses can be taken in consecutive periods? (What courses the student takes during the other 3 periods is of no concern here.)

- (A) 3 (B) 6 (C) 12 (D) 18 (E) 24

Solution

Problem 4

Alice, Bob, and Charlie were on a hike and were wondering how far away the nearest town was. When Alice said, "We are at least 6 miles away," Bob replied, "We are at most 5 miles away." Charlie then remarked, "Actually the nearest town is at most 4 miles away." It turned out that none of the three statements were true. Let d be the distance in miles to the nearest town. Which of the following intervals is the set of all possible values of d ?

- (A) $(0, 4)$ (B) $(4, 5)$ (C) $(4, 6)$ (D) $(5, 6)$ (E) $(5, \infty)$

Solution

Problem 5

What is the sum of all possible values of k for which the polynomials $x^2 - 3x + 2$ and $x^2 - 5x + k$ have a root in common?

- (A) 3 (B) 4 (C) 5 (D) 6 (E) 10

Solution

Problem 6

For positive integers m and n such that $m + 10 < n + 1$, both the mean and the median of the set $\{m, m + 4, m + 10, n + 1, n + 2, 2n\}$ are equal to n . What is $m + n$?

- (A) 20 (B) 21 (C) 22 (D) 23 (E) 24

Solution

Problem 7

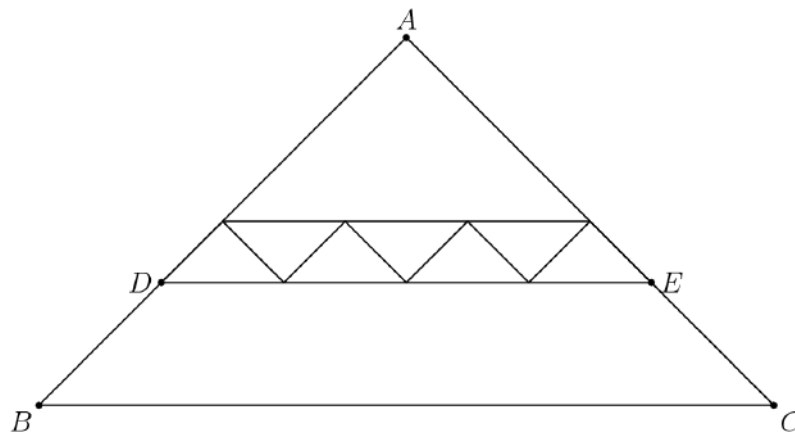
For how many (not necessarily positive) integer values of n is the value of $4000 \cdot \left(\frac{2}{5}\right)^n$ an integer?

- (A) 3 (B) 4 (C) 6 (D) 8 (E) 9

Solution

Problem 8

All of the triangles in the diagram below are similar to isosceles triangle ABC , in which $AB = AC$. Each of the 7 smallest triangles has area 1, and $\triangle ABC$ has area 40. What is the area of trapezoid $DBCE$?



- (A) 16 (B) 18 (C) 20 (D) 22 (E) 24

Solution

Problem 9

Which of the following describes the largest subset of values of y within the closed interval $[0, \pi]$ for which

$$\sin(x + y) \leq \sin(x) + \sin(y)$$

for every x between 0 and π , inclusive?

- (A) $y = 0$ (B) $0 \leq y \leq \frac{\pi}{4}$ (C) $0 \leq y \leq \frac{\pi}{2}$ (D) $0 \leq y \leq \frac{3\pi}{4}$ (E) $0 \leq y \leq \pi$

Solution

Problem 10

How many ordered pairs of real numbers (x, y) satisfy the following system of equations?

$$x + 3y = 3$$

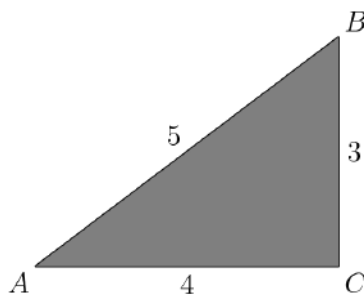
$$||x| - |y|| = 1$$

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 8

Solution

Problem 11

A paper triangle with sides of lengths 3, 4, and 5 inches, as shown, is folded so that point A falls on point B . What is the length in inches of the crease?



- (A) $1 + \frac{1}{2}\sqrt{2}$ (B) $\sqrt{3}$ (C) $\frac{7}{4}$ (D) $\frac{15}{8}$ (E) 2

Solution

Problem 12

Let S be a set of 6 integers taken from $\{1, 2, \dots, 12\}$ with the property that if a and b are elements of S with $a < b$, then b is not a multiple of a . What is the least possible value of an element in S ?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 7

Solution

Problem 13

How many nonnegative integers can be written in the form

$$a_7 \cdot 3^7 + a_6 \cdot 3^6 + a_5 \cdot 3^5 + a_4 \cdot 3^4 + a_3 \cdot 3^3 + a_2 \cdot 3^2 + a_1 \cdot 3^1 + a_0 \cdot 3^0,$$

where $a_i \in \{-1, 0, 1\}$ for $0 \leq i \leq 7$?

- (A) 512 (B) 729 (C) 1094 (D) 3281 (E) 59,048

Solution

Problem 14

The solutions to the equation $\log_{3x} 4 = \log_{2x} 8$, where x is a positive real number other than $\frac{1}{3}$ or $\frac{1}{2}$, can be written as $\frac{p}{q}$ where p and q are relatively prime positive integers. What is $p + q$?

- (A) 5 (B) 13 (C) 17 (D) 31 (E) 35

Solution

Problem 15

A scanning code consists of a 7×7 grid of squares, with some of its squares colored black and the rest colored white. There must be at least one square of each color in this grid of 49 squares. A scanning code is called *symmetric* if its look does not change when the entire square is rotated by a multiple of 90° counterclockwise around its center, nor when it is reflected across a line joining opposite corners or a line joining midpoints of opposite sides. What is the total number of possible symmetric scanning codes?

- (A) 510 (B) 1022 (C) 8190 (D) 8192 (E) 65,534

Solution

Problem 16

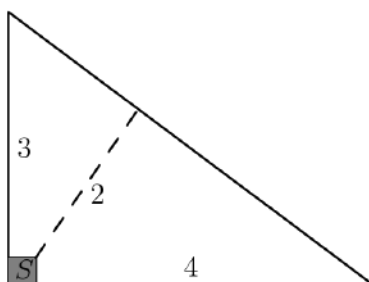
Which of the following describes the set of values of a for which the curves $x^2 + y^2 = a^2$ and $y = x^2 - a$ in the real xy -plane intersect at exactly 3 points?

- (A) $a = \frac{1}{4}$ (B) $\frac{1}{4} < a < \frac{1}{2}$ (C) $a > \frac{1}{4}$ (D) $a = \frac{1}{2}$ (E) $a > \frac{1}{2}$

Solution

Problem 17

Farmer Pythagoras has a field in the shape of a right triangle. The right triangle's legs have lengths 3 and 4 units. In the corner where those sides meet at a right angle, he leaves a small unplanted square S so that from the air it looks like the right angle symbol. The rest of the field is planted. The shortest distance from S to the hypotenuse is 2 units. What fraction of the field is planted?



- (A) $\frac{25}{27}$ (B) $\frac{26}{27}$ (C) $\frac{73}{75}$ (D) $\frac{145}{147}$ (E) $\frac{74}{75}$

Solution

Problem 18

Triangle ABC with $AB = 50$ and $AC = 10$ has area 120. Let D be the midpoint of $\overline{AB}$, and let E be the midpoint of $\overline{AC}$. The angle bisector of $\angle BAC$ intersects $\overline{DE}$ and $\overline{BC}$ at F and G , respectively. What is the area of quadrilateral $FDBG$?

- (A) 60 (B) 65 (C) 70 (D) 75 (E) 80

Solution

Problem 19

Let A be the set of positive integers that have no prime factors other than 2, 3, or 5. The infinite sum

$$\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{8} + \frac{1}{9} + \frac{1}{10} + \frac{1}{12} + \frac{1}{15} + \frac{1}{16} + \frac{1}{18} + \frac{1}{20} + \cdots$$

of the reciprocals of the elements of A can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. What is $m + n$?

- (A) 16 (B) 17 (C) 19 (D) 23 (E) 36

Solution

Problem 20

Triangle ABC is an isosceles right triangle with $AB = AC = 3$. Let M be the midpoint of hypotenuse $\overline{BC}$. Points I and E lie on sides $\overline{AC}$ and $\overline{AB}$, respectively, so that $AI > AE$ and $AIME$ is a cyclic quadrilateral. Given that triangle EMI has area 2, the length CI can be written as $\frac{a - \sqrt{b}}{c}$, where a , b , and c are positive integers and b is not divisible by the square of any prime. What is the value of $a + b + c$?

- (A) 9 (B) 10 (C) 11 (D) 12 (E) 13

Solution

Problem 21

Which of the following polynomials has the greatest real root?

- (A) $x^{19} + 2018x^{11} + 1$ (B) $x^{17} + 2018x^{11} + 1$ (C) $x^{19} + 2018x^{13} + 1$ (D) $x^{17} + 2018x^{13} + 1$ (E) $2019x + 2018$

Solution

Problem 22

The solutions to the equations $z^2 = 4 + 4\sqrt{15}i$ and $z^2 = 2 + 2\sqrt{3}i$, where $i = \sqrt{-1}$, form the vertices of a parallelogram in the complex plane. The area of this parallelogram can be written in the form $p\sqrt{q} - r\sqrt{s}$, where p , q , r , and s are positive integers and neither q nor s is divisible by the square of any prime number. What is $p + q + r + s$?

- (A) 20 (B) 21 (C) 22 (D) 23 (E) 24

Solution

Problem 23

In $\triangle PAT$, $\angle P = 36^\circ$, $\angle A = 56^\circ$, and $PA = 10$. Points U and G lie on sides $\overline{TP}$ and $\overline{TA}$, respectively, so that $PU = AG = 1$. Let M and N be the midpoints of segments $\overline{PA}$ and $\overline{UG}$, respectively. What is the degree measure of the acute angle formed by lines MN and PA ?

- (A) 76 (B) 77 (C) 78 (D) 79 (E) 80

Solution

Problem 24

Alice, Bob, and Carol play a game in which each of them chooses a real number between 0 and 1. The winner of the game is the one whose number is between the numbers chosen by the other two players. Alice announces that she will choose her number uniformly at random from all the numbers between 0 and 1, and Bob announces that he will choose his number uniformly at random from all the numbers between $\frac{1}{2}$ and $\frac{2}{3}$. Armed with this information, what number should Carol choose to maximize her chance of winning?

- (A) $\frac{1}{2}$ (B) $\frac{13}{24}$ (C) $\frac{7}{12}$ (D) $\frac{5}{8}$ (E) $\frac{2}{3}$

Solution

Problem 25

For a positive integer n and nonzero digits a , b , and c , let A_n be the n -digit integer each of whose digits is equal to a ; let B_n be the n -digit integer each of whose digits is equal to b , and let C_n be the $2n$ -digit (not n -digit) integer each of whose digits is equal to c . What is the greatest possible value of $a + b + c$ for which there are at least two values of n such that $C_n - B_n = A_n^2$?

- (A) 12 (B) 14 (C) 16 (D) 18 (E) 20

Solution

See also

2018 AMC 12A (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2018))	
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1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
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2018 AMC 12A Answer Key

1. D
2. C
3. E
4. D
5. E
6. B
7. E
8. E
9. E
10. C
11. D
12. C
13. D
14. D
15. B
16. E
17. D
18. D
19. C
20. D
21. B
22. A
23. E
24. B
25. D

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2018 AMC 12B Problems

2018 AMC 12B (Answer Key)

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Instructions

1. This is a 25-question, multiple choice test. Each question is followed by answers marked A, B, C, D and E. Only one of these is correct.
2. You will receive 6 points for each correct answer, 2.5 points for each problem left unanswered if the year is before 2006, 1.5 points for each problem left unanswered if the year is after 2006, and 0 points for each incorrect answer.
3. No aids are permitted other than scratch paper, graph paper, ruler, compass, protractor and erasers (and calculators that are accepted for use on the test if before 2006. No problems on the test will *require* the use of a calculator).
4. Figures are not necessarily drawn to scale.
5. You will have **75 minutes** working time to complete the test.

1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

Kate bakes 20-inch by 18-inch pan of cornbread. The cornbread is cut into pieces that measure 2 inches by 2 inches. How many pieces of cornbread does the pan contain?

- (A) 90 (B) 100 (C) 180 (D) 200 (E) 360

Solution

Problem 2

Sam drove 96 miles in 90 minutes. His average speed during the first 30 minutes was 60 mph (miles per hour), and his average speed during the second 30 minutes was 65 mph. What was his average speed, in mph, during the last 30 minutes?

- (A) 64 (B) 65 (C) 66 (D) 67 (E) 68

Solution

Problem 3

A line with slope 2 intersects a line with slope 6 at the point $(40, 30)$. What is the distance between the x -intercepts of these two lines?

- (A) 5 (B) 10 (C) 20 (D) 25 (E) 50

Solution

Problem 4

A circle has a chord of length 10, and the distance from the center of the circle to the chord is 5. What is the area of the circle?

- (A) 25π (B) 50π (C) 75π (D) 100π (E) 125π

Solution

Problem 5

How many subsets of $\{2, 3, 4, 5, 6, 7, 8, 9\}$ contain at least one prime number?

- (A) 128 (B) 192 (C) 224 (D) 240 (E) 256

Solution

Problem 6

Suppose S cans of soda can be purchased from a vending machine for Q quarters. Which of the following expressions describes the number of cans of soda that can be purchased for D dollars, where 1 dollar is worth 4 quarters?

- (A) $\frac{4DQ}{S}$ (B) $\frac{4DS}{Q}$ (C) $\frac{4Q}{DS}$ (D) $\frac{DQ}{4S}$ (E) $\frac{DS}{4Q}$

Solution

Problem 7

What is the value of

$$\log_3 7 \cdot \log_5 9 \cdot \log_7 11 \cdot \log_9 13 \cdots \log_{21} 25 \cdot \log_{23} 27?$$

- (A) 3 (B) $3 \log_7 23$ (C) 6 (D) 9 (E) 10

Solution

Problem 8

Line segment $\overline{AB}$ is a diameter of a circle with $AB = 24$. Point C , not equal to A or B , lies on the circle. As point C moves around the circle, the centroid (center of mass) of $\triangle ABC$ traces out a closed curve missing two points. To the nearest positive integer, what is the area of the region bounded by this curve?

- (A) 25 (B) 38 (C) 50 (D) 63 (E) 75

Solution

Problem 9

What is

$$\sum_{i=1}^{100} \sum_{j=1}^{100} (i + j)?$$

- (A) 100, 100 (B) 500, 500 (C) 505, 000 (D) 1, 001, 000 (E) 1, 010, 000

Solution

Problem 10

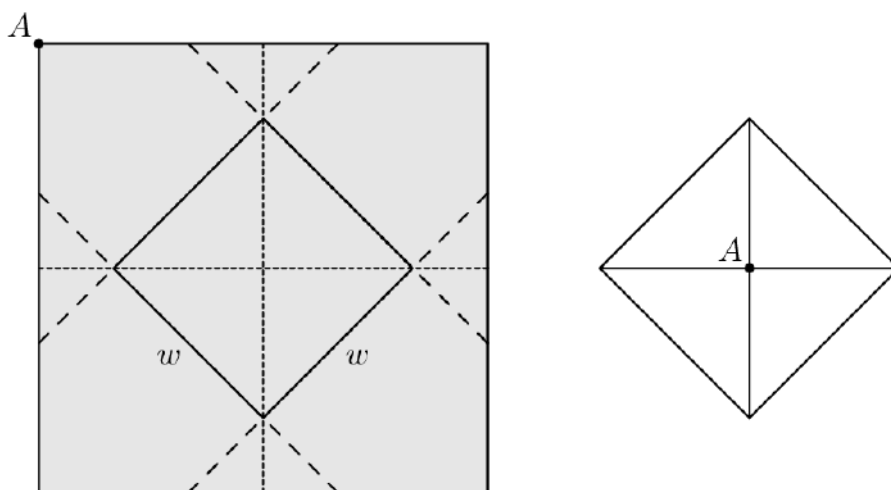
A list of 2018 positive integers has a unique mode, which occurs exactly 10 times. What is the least number of distinct values that can occur in the list?

- (A) 202 (B) 223 (C) 224 (D) 225 (E) 234

Solution

Problem 11

A closed box with a square base is to be wrapped with a square sheet of wrapping paper. The box is centered on the wrapping paper with the vertices of the base lying on the midlines of the square sheet of paper, as shown in the figure on the left. The four corners of the wrapping paper are to be folded up over the sides and brought together to meet at the center of the top of the box, point A in the figure on the right. The box has base length w and height h . What is the area of the sheet of wrapping paper?



- (A) $2(w + h)^2$ (B) $\frac{(w + h)^2}{2}$ (C) $2w^2 + 4wh$ (D) $2w^2$ (E) w^2h

Solution

Problem 12

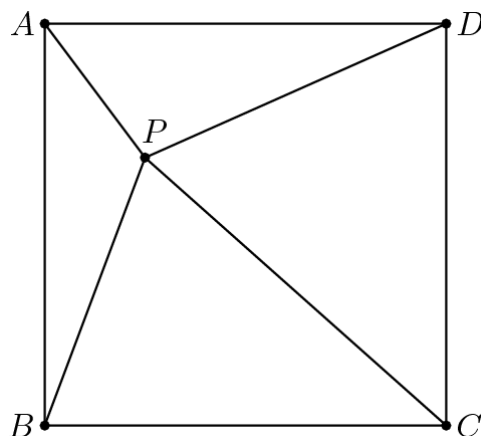
Side $\overline{AB}$ of $\triangle ABC$ has length 10. The bisector of angle A meets $\overline{BC}$ at D , and $CD = 3$. The set of all possible values of AC is an open interval (m, n) . What is $m + n$?

- (A) 16 (B) 17 (C) 18 (D) 19 (E) 20

Solution

Problem 13

Square $ABCD$ has side length 30. Point P lies inside the square so that $AP = 12$ and $BP = 26$. The centroids of $\triangle ABP$, $\triangle BCP$, $\triangle CDP$, and $\triangle DAP$ are the vertices of a convex quadrilateral. What is the area of that quadrilateral?



- (A) $100\sqrt{2}$ (B) $100\sqrt{3}$ (C) 200 (D) $200\sqrt{2}$ (E) $200\sqrt{3}$

Solution

Problem 14

Joey, Chloe, and their daughter Zoe all have the same birthday. Joey is 1 year older than Chloe, and Zoe is exactly 1 year old today. Today is the first of the 9 birthdays on which Chloe's age will be an integral multiple of Zoe's age. What will be the sum of the two digits of Joey's age the next time his age is a multiple of Zoe's age?

- (A) 7 (B) 8 (C) 9 (D) 10 (E) 11

Solution

Problem 15

How many odd positive 3-digit integers are divisible by 3 but do not contain the digit 3?

- (A) 96 (B) 97 (C) 98 (D) 102 (E) 120

Solution

Problem 16

The solutions to the equation $(z + 6)^8 = 81$ are connected in the complex plane to form a convex regular polygon, three of whose vertices are labeled A , B , and C . What is the least possible area of $\triangle ABC$?

- (A) $\frac{1}{6}\sqrt{6}$ (B) $\frac{3}{2}\sqrt{2} - \frac{3}{2}$ (C) $2\sqrt{3} - 3\sqrt{2}$ (D) $\frac{1}{2}\sqrt{2}$ (E) $\sqrt{3} - 1$

Solution

Problem 17

Let p and q be positive integers such that

$$\frac{5}{9} < \frac{p}{q} < \frac{4}{7}$$

and q is as small as possible. What is $q - p$?

- (A) 7 (B) 11 (C) 13 (D) 17 (E) 19

Solution

Problem 18

A function f is defined recursively by $f(1) = f(2) = 1$ and

$$f(n) = f(n-1) - f(n-2) + n$$

for all integers $n \geq 3$. What is $f(2018)$?

- (A) 2016 (B) 2017 (C) 2018 (D) 2019 (E) 2020

Solution

Problem 19

Mary chose an even 4-digit number n . She wrote down all the divisors of n in increasing order from left to right: $1, 2, \dots, \frac{n}{2}, n$. At some moment Mary wrote 323 as a divisor of n . What is the smallest possible value of the next divisor written to the right of 323 ?

- (A) 324 (B) 330 (C) 340 (D) 361 (E) 646

Solution

Problem 20

Let $ABCDEF$ be a regular hexagon with side length 1. Denote by X , Y , and Z the midpoints of sides $\overline{AB}$, $\overline{CD}$, and $\overline{EF}$, respectively. What is the area of the convex hexagon whose interior is the intersection of the interiors of $\triangle ACE$ and $\triangle XYZ$?

- (A) $\frac{3}{8}\sqrt{3}$ (B) $\frac{7}{16}\sqrt{3}$ (C) $\frac{15}{32}\sqrt{3}$ (D) $\frac{1}{2}\sqrt{3}$ (E) $\frac{9}{16}\sqrt{3}$

Solution

Problem 21

In $\triangle ABC$ with side lengths $AB = 13$, $AC = 12$, and $BC = 5$, let O and I denote the circumcenter and incenter, respectively. A circle with center M is tangent to the legs AC and BC and to the circumcircle of $\triangle ABC$. What is the area of $\triangle MOI$?

- (A) $5/2$ (B) $11/4$ (C) 3 (D) $13/4$ (E) $7/2$

Solution

Problem 22

Consider polynomials $P(x)$ of degree at most 3, each of whose coefficients is an element of $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. How many such polynomials satisfy $P(-1) = -9$?

- (A) 110 (B) 143 (C) 165 (D) 220 (E) 286

Solution

Problem 23

Ajay is standing at point A near Pontianak, Indonesia, 0° latitude and 110° E longitude. Billy is standing at point B near Big Baldy Mountain, Idaho, USA, 45° N latitude and 115° W longitude. Assume that Earth is a perfect sphere with center C . What is the degree measure of $\angle ACB$?

- (A) 105 (B) $112\frac{1}{2}$ (C) 120 (D) 135 (E) 150

Solution

Problem 24

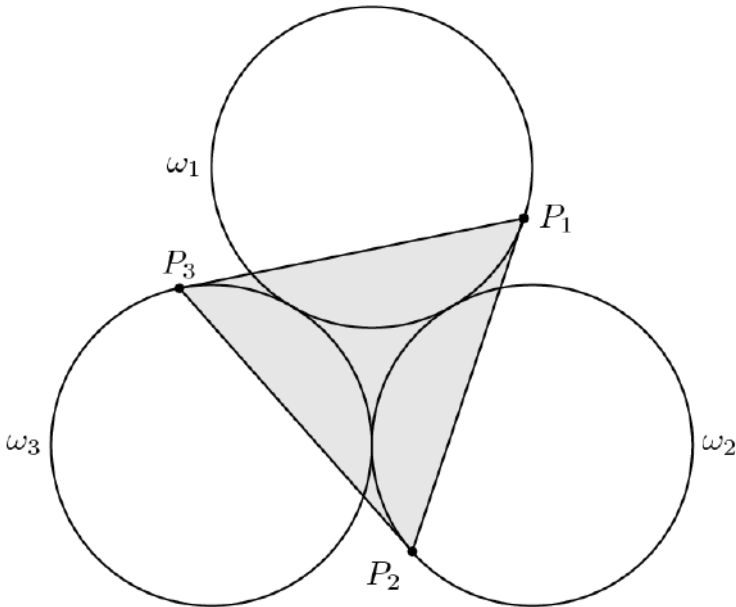
Let $\lfloor x \rfloor$ denote the greatest integer less than or equal to x . How many real numbers x satisfy the equation $x^2 + 10,000\lfloor x \rfloor = 10,000x$?

- (A) 197 (B) 198 (C) 199 (D) 200 (E) 201

Solution

Problem 25

Circles ω_1, ω_2 , and ω_3 each have radius 4 and are placed in the plane so that each circle is externally tangent to the other two. Points P_1, P_2 , and P_3 lie on ω_1, ω_2 , and ω_3 respectively such that $P_1P_2 = P_2P_3 = P_3P_1$ and line P_iP_{i+1} is tangent to ω_i for each $i = 1, 2, 3$, where $P_4 = P_1$. See the figure below. The area of $\triangle P_1P_2P_3$ can be written in the form $\sqrt{a} + \sqrt{b}$ for positive integers a and b . What is $a + b$?



- (A) 546
- (B) 548
- (C) 550
- (D) 552
- (E) 554

Solution

See also

2018 AMC 12B (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2018))	
Preceded by 2018 AMC 12A Problems	Followed by 2019 AMC 12A Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
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2018 AMC 12B Answer Key

1. A
2. D
3. B
4. B
5. D
6. B
7. C
8. C
9. E
10. D
11. A
12. C
13. C
14. E
15. A
16. B
17. A
18. B
19. C
20. C
21. E
22. D
23. C
24. C
25. D

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1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 •
19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

The area of a pizza with radius 4 is N percent larger than the area of a pizza with radius 3 inches. What is the integer closest to N ?

- (A) 25 (B) 33 (C) 44 (D) 66 (E) 78

Solution

Problem 2

Suppose a is 150% of b . What percent of a is $3b$?

- (A) 50 (B) $66\frac{2}{3}$ (C) 150 (D) 200 (E) 450

Solution

Problem 3

A box contains 28 red balls, 20 green balls, 19 yellow balls, 13 blue balls, 11 white balls, and 9 black balls. What is the minimum number of balls that must be drawn from the box without replacement to guarantee that at least 15 balls of a single color will be drawn?

- (A) 75 (B) 76 (C) 79 (D) 84 (E) 91

Solution

Problem 4

What is the greatest number of consecutive integers whose sum is 45?

- (A) 9 (B) 25 (C) 45 (D) 90 (E) 120

Solution

Problem 5

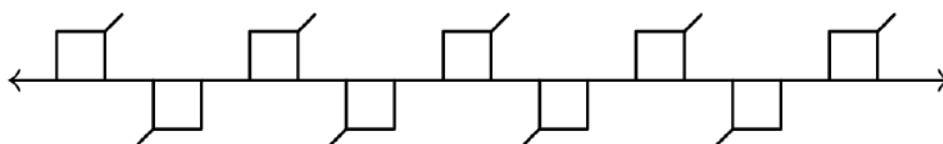
Two lines with slopes $\frac{1}{2}$ and 2 intersect at $(2, 2)$. What is the area of the triangle enclosed by these two lines and the line $x + y = 10$?

- (A) 4 (B) $4\sqrt{2}$ (C) 6 (D) 8 (E) $6\sqrt{2}$

Solution

Problem 6

The figure below shows line ℓ with a regular, infinite, recurring pattern of squares and line segments.



How many of the following four kinds of rigid motion transformations of the plane in which this figure is drawn, other than the identity transformation, will transform this figure into itself?

- some rotation around a point of line ℓ
- some translation in the direction parallel to line ℓ
- the reflection across line ℓ
- some reflection across a line perpendicular to line ℓ

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Solution

Problem 7

Melanie computes the mean μ , the median M , and the modes of the 365 values that are the dates in the months of 2019. Thus her data consist of 12 1s, 12 2s, ..., 12 28s, 11 29s, 11 30s, and 7 31s. Let d be the median of the modes. Which of the following statements is true?

- (A) $\mu < d < M$ (B) $M < d < \mu$ (C) $d = M = \mu$ (D) $d < M < \mu$ (E) $d < \mu < M$

Solution

Problem 8

For a set of four distinct lines in a plane, there are exactly N distinct points that lie on two or more of the lines. What is the sum of all possible values of N ?

- (A) 14 (B) 16 (C) 18 (D) 19 (E) 21

Solution

Problem 9

A sequence of numbers is defined recursively by $a_1 = 1$, $a_2 = \frac{3}{7}$ and

$$a_n = \frac{a_{n-2} \cdot a_{n-1}}{2a_{n-2} - a_{n-1}}$$

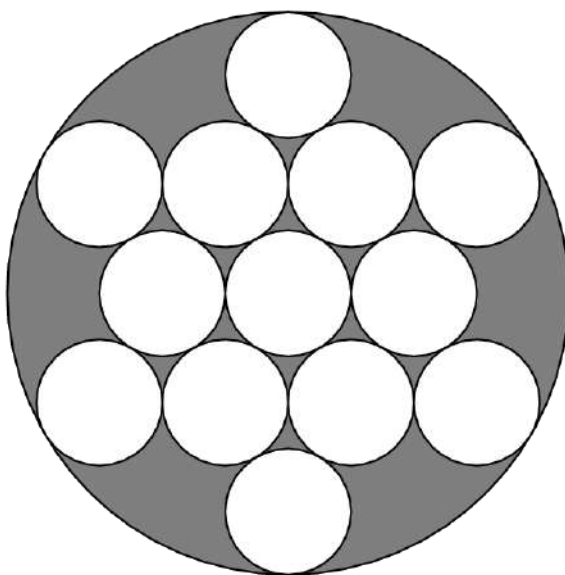
for all $n \geq 3$. Then a_{2019} can be written as $\frac{p}{q}$, where p and q are relatively prime positive integers. What is $p + q$?

- (A) 2020 (B) 4039 (C) 6057 (D) 6061 (E) 8078

Solution

Problem 10

The figure below shows 13 circles of radius 1 within a larger circle. All the intersections occur at points of tangency. What is the area of the region, shaded in the figure, inside the larger circle but outside all the circles of radius 1?



- (A) $4\pi\sqrt{3}$ (B) 7π (C) $\pi(3\sqrt{3} + 2)$ (D) $10\pi(\sqrt{3} - 1)$ (E) $\pi(\sqrt{3} + 6)$

Solution

Problem 11

For some positive integer k , the repeating base- k representation of the (base-ten) fraction $\frac{7}{51}$ is $0.\overline{23}_k = 0.232323\dots_k$. What is k ?

- (A) 13 (B) 14 (C) 15 (D) 16 (E) 17

Solution

Problem 12

Positive real numbers $x \neq 1$ and $y \neq 1$ satisfy $\log_2 x = \log_y 16$ and $xy = 64$. What is $(\log_2 \frac{x}{y})^2$?

- (A) $\frac{25}{2}$ (B) 20 (C) $\frac{45}{2}$ (D) 25 (E) 32

Solution

Problem 13

How many ways are there to paint each of the integers $2, 3, \dots, 9$ either red, green, or blue so that each number has a different color from each of its proper divisors?

- (A) 144 (B) 216 (C) 256 (D) 384 (E) 432

Solution

Problem 14

For a certain complex number c , the polynomial

$$P(x) = (x^2 - 2x + 2)(x^2 - cx + 4)(x^2 - 4x + 8)$$

has exactly 4 distinct roots. What is $|c|$?

- (A) 2 (B) $\sqrt{6}$ (C) $2\sqrt{2}$ (D) 3 (E) $\sqrt{10}$

Solution

Problem 15

Positive real numbers a and b have the property that

$$\sqrt{\log a} + \sqrt{\log b} + \log \sqrt{a} + \log \sqrt{b} = 100$$

and all four terms on the left are positive integers, where $\log$ denotes the base-10 logarithm. What is ab ?

- (A) 10^{52} (B) 10^{100} (C) 10^{144} (D) 10^{164} (E) 10^{200}

Solution

Problem 16

The numbers $1, 2, \dots, 9$ are randomly placed into the 9 squares of a 3×3 grid. Each square gets one number, and each of the numbers is used once. What is the probability that the sum of the numbers in each row and each column is odd?

- (A) $\frac{1}{21}$ (B) $\frac{1}{14}$ (C) $\frac{5}{63}$ (D) $\frac{2}{21}$ (E) $\frac{1}{7}$

Solution

Problem 17

Let s_k denote the sum of the k th powers of the roots of the polynomial $x^3 - 5x^2 + 8x - 13$. In particular, $s_0 = 3$, $s_1 = 5$, and $s_2 = 9$. Let a , b , and c be real numbers such that $s_{k+1} = a s_k + b s_{k-1} + c s_{k-2}$ for $k = 2, 3, \dots$. What is $a + b + c$?

- (A) -6 (B) 0 (C) 6 (D) 10 (E) 26

Solution

Problem 18

A sphere with center O has radius 6. A triangle with sides of length 15, 15, and 24 is situated in space so that each of its sides is tangent to the sphere. What is the distance between O and the plane determined by the triangle?

- (A) $2\sqrt{3}$ (B) 4 (C) $3\sqrt{2}$ (D) $2\sqrt{5}$ (E) 5

Solution

Problem 19

In $\triangle ABC$ with integer side lengths,

$$\cos A = \frac{11}{16}, \quad \cos B = \frac{7}{8}, \quad \text{and} \quad \cos C = -\frac{1}{4}.$$

What is the least possible perimeter for $\triangle ABC$?

- (A) 9 (B) 12 (C) 23 (D) 27 (E) 44

Solution

Problem 20

Real numbers between 0 and 1, inclusive, are chosen in the following manner. A fair coin is flipped. If it lands heads, then it is flipped again and the chosen number is 0 if the second flip is heads and 1 if the second flip is tails. On the other hand, if the first coin flip is tails, then the number is chosen uniformly at random from the closed interval $[0, 1]$. Two random numbers x and y are chosen independently in this manner. What is the probability that $|x - y| > \frac{1}{2}$?

- (A) $\frac{1}{3}$ (B) $\frac{7}{16}$ (C) $\frac{1}{2}$ (D) $\frac{9}{16}$ (E) $\frac{2}{3}$

Solution

Problem 21

Let

$$z = \frac{1+i}{\sqrt{2}}.$$

What is

$$\left(z^{1^2} + z^{2^2} + z^{3^2} + \cdots + z^{12^2}\right) \cdot \left(\frac{1}{z^{1^2}} + \frac{1}{z^{2^2}} + \frac{1}{z^{3^2}} + \cdots + \frac{1}{z^{12^2}}\right)?$$

- (A) 18 (B) $72 - 36\sqrt{2}$ (C) 36 (D) 72 (E) $72 + 36\sqrt{2}$

Solution

Problem 22

Circles ω and γ , both centered at O , have radii 20 and 17, respectively. Equilateral triangle ABC , whose interior lies in the interior of ω but in the exterior of γ , has vertex A on ω , and the line containing side $\overline{BC}$ is tangent to γ . Segments $\overline{AO}$ and $\overline{BC}$ intersect at P , and $\frac{BP}{CP} = 3$. Then AB can be written in the form $\frac{m}{\sqrt{n}} - \frac{p}{\sqrt{q}}$ for positive integers m, n, p, q with $\gcd(m, n) = \gcd(p, q) = 1$. What is $m + n + p + q$?

- (A) 42 (B) 86 (C) 92 (D) 114 (E) 130

Solution

Problem 23

Define binary operations $\diamond$ and $\heartsuit$ by

$$a \diamond b = a^{\log_7(b)} \quad \text{and} \quad a \heartsuit b = a^{\frac{1}{\log_7(b)}}$$

for all real numbers a and b for which these expressions are defined. The sequence (a_n) is defined recursively by $a_3 = 3 \heartsuit 2$ and

$$a_n = (n \heartsuit (n-1)) \diamond a_{n-1}$$

for all integers $n \geq 4$. To the nearest integer, what is $\log_7(a_{2019})$?

- (A) 8 (B) 9 (C) 10 (D) 11 (E) 12

Solution

Problem 24

For how many integers n between 1 and 50, inclusive, is

$$\frac{(n^2 - 1)!}{(n!)^n}$$

an integer? (Recall that $0! = 1$.)

- (A) 31 (B) 32 (C) 33 (D) 34 (E) 35

Solution

Problem 25

Let $\triangle A_0B_0C_0$ be a triangle whose angle measures are exactly 59.999° , 60° , and 60.001° . For each positive integer n , define A_n to be the foot of the altitude from A_{n-1} to line $B_{n-1}C_{n-1}$. Likewise, define B_n to be the foot of the altitude from B_{n-1} to line $A_{n-1}C_{n-1}$, and C_n to be the foot of the altitude from C_{n-1} to line $A_{n-1}B_{n-1}$. What is the least positive integer n for which $\triangle A_nB_nC_n$ is obtuse?

- (A) 10 (B) 11 (C) 13 (D) 14 (E) 15

Solution

See also

2019 AMC 12A (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2019))	
Preceded by 2018 AMC 12B Problems	Followed by 2019 AMC 12B Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
All AMC 12 Problems and Solutions	

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2019 AMC 12A Answer Key

1. E
2. D
3. B
4. D
5. C
6. C
7. E
8. D
9. E
10. A
11. D
12. B
13. E
14. E
15. D
16. B
17. D
18. D
19. A
20. B
21. C
22. E
23. D
24. D
25. E

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Instructions

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4. Figures are not necessarily drawn to scale.
5. You will have **75 minutes** working time to complete the test.

1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24

- 25 Problem 25
- 26 See also

Problem 1

Alicia had two containers. The first was $\frac{5}{6}$ full of water and the second was empty. She poured all the water from the first container into the second container, at which point the second container was $\frac{3}{4}$ full of water. What is the ratio of the volume of the first container to the volume of the second container?

- (A) $\frac{5}{8}$ (B) $\frac{4}{5}$ (C) $\frac{7}{8}$ (D) $\frac{9}{10}$ (E) $\frac{11}{12}$

Solution

Problem 2

Consider the statement, "If n is not prime, then $n - 2$ is prime." Which of the following values of n is a counterexample to this statement?

- (A) 11 (B) 15 (C) 19 (D) 21 (E) 27

Solution

Problem 3

Which one of the following rigid transformations (isometries) maps the line segment $\overline{AB}$ onto the line segment $\overline{A'B'}$ so that the image of $A(-2, 1)$ is $A'(2, -1)$ and the image of $B(-1, 4)$ is $B'(1, -4)$?

- (A) reflection in the y -axis
(B) counterclockwise rotation around the origin by 90°
(C) translation by 3 units to the right and 5 units down
(D) reflection in the x -axis
(E) clockwise rotation about the origin by 180°

Solution

Problem 4

A positive integer n satisfies the equation $(n + 1)! + (n + 2)! = 440 \cdot n!$. What is the sum of the digits of n ?

- (A) 2 (B) 5 (C) 10 (D) 12 (E) 15

Solution

Problem 5

Each piece of candy in a store costs a whole number of cents. Casper has exactly enough money to buy either 12 pieces of red candy, 14 pieces of green candy, 15 pieces of blue candy, or n pieces of purple candy. A piece of purple candy costs 20 cents. What is the smallest possible value of n ?

- (A) 18 (B) 21 (C) 24 (D) 25 (E) 28

Solution

Problem 6

In a given plane, points A and B are 10 units apart. How many points C are there in the plane such that the perimeter of $\triangle ABC$ is 50 units and the area of $\triangle ABC$ is 100 square units?

- (A) 0 (B) 2 (C) 4 (D) 8 (E) infinitely many

Solution

Problem 7

What is the sum of all real numbers x for which the median of the numbers 4, 6, 8, 17, and x is equal to the mean of those five numbers?

- (A) -5 (B) 0 (C) 5 (D) $\frac{15}{4}$ (E) $\frac{35}{4}$

Solution

Problem 8

Let $f(x) = x^2(1 - x)^2$. What is the value of the sum

$$f\left(\frac{1}{2019}\right) - f\left(\frac{2}{2019}\right) + f\left(\frac{3}{2019}\right) - f\left(\frac{4}{2019}\right) + \cdots + f\left(\frac{2017}{2019}\right) - f\left(\frac{2018}{2019}\right)?$$

- (A) 0 (B) $\frac{1}{2019^4}$ (C) $\frac{2018^2}{2019^4}$ (D) $\frac{2020^2}{2019^4}$ (E) 1

Solution

Problem 9

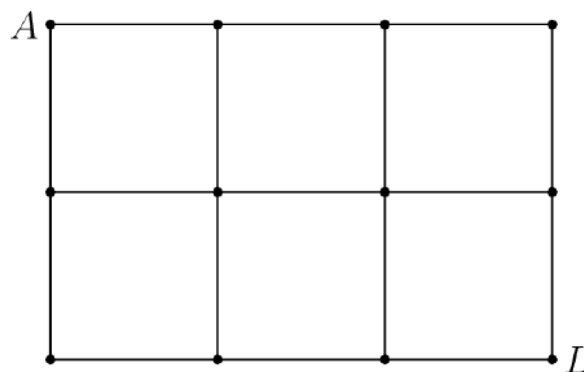
For how many integral values of x can a triangle of positive area be formed having side lengths $\log_2 x$, $\log_4 x$, 3 ?

- (A) 57 (B) 59 (C) 61 (D) 62 (E) 63

Solution

Problem 10

The figure below is a map showing 12 cities and 17 roads connecting certain pairs of cities. Paula wishes to travel along exactly 13 of those roads, starting at city A and ending at city L , without traveling along any portion of a road more than once. (Paula is allowed to visit a city more than once.)



How many different routes can Paula take?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Solution

Problem 11

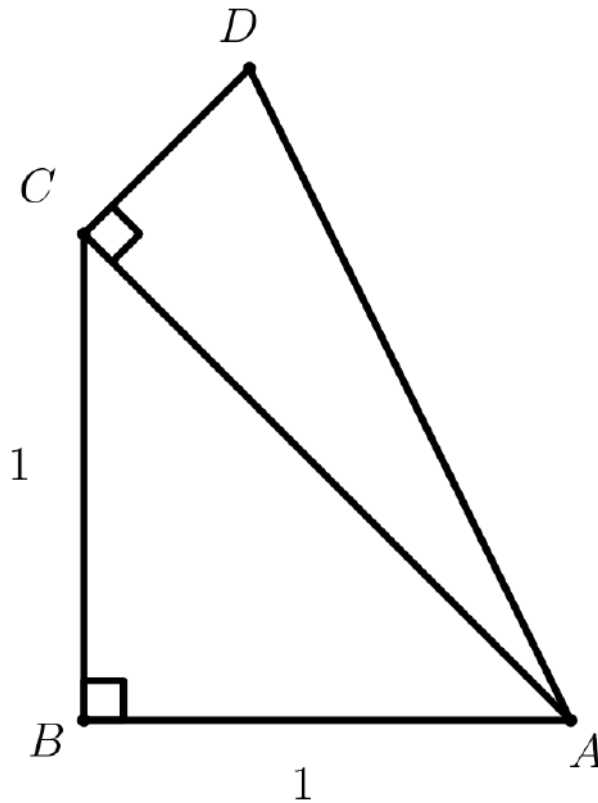
How many unordered pairs of edges of a given cube determine a plane?

- (A) 12 (B) 28 (C) 36 (D) 42 (E) 66

Solution

Problem 12

Right triangle ACD with right angle at C is constructed outwards on the hypotenuse $\overline{AC}$ of isosceles right triangle ABC with leg length 1, as shown, so that the two triangles have equal perimeters. What is $\sin(2\angle BAD)$?



- (A) $\frac{1}{3}$ (B) $\frac{\sqrt{2}}{2}$ (C) $\frac{3}{4}$ (D) $\frac{7}{9}$ (E) $\frac{\sqrt{3}}{2}$

Solution

Problem 13

A red ball and a green ball are randomly and independently tossed into bins numbered with positive integers so that for each ball, the probability that it is tossed into bin k is 2^{-k} for $k = 1, 2, 3, \dots$. What is the probability that the red ball is tossed into a higher-numbered bin than the green ball?

- (A) $\frac{1}{4}$ (B) $\frac{2}{7}$ (C) $\frac{1}{3}$ (D) $\frac{3}{8}$ (E) $\frac{3}{7}$

Solution

Problem 14

Let S be the set of all positive integer divisors of 100,000. How many numbers are the product of two distinct elements of S ?

- (A) 98 (B) 100 (C) 117 (D) 119 (E) 121

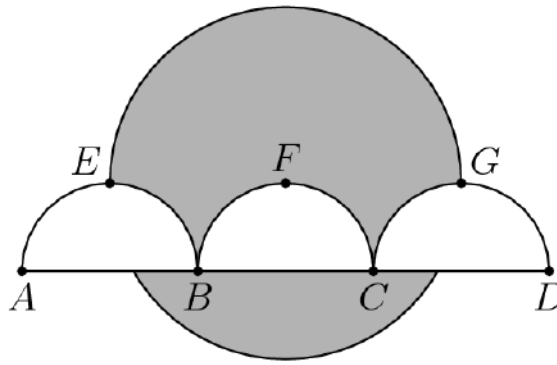
Solution

Problem 15

As shown in the figure, line segment $\overline{AD}$ is trisected by points B and C so that $AB = BC = CD = 2$. Three semicircles of radius 1, $\overline{AEB}$, $\overline{BFC}$, and $\overline{CGD}$, have their diameters on $\overline{AD}$, and are tangent to line EG at E , F , and G , respectively. A circle of radius 2 has its center on F . The area of the region inside the circle but outside the three semicircles, shaded in the figure, can be expressed in the form

$$\frac{a}{b} \cdot \pi - \sqrt{c} + d,$$

where a , b , c , and d are positive integers and a and b are relatively prime. What is $a + b + c + d$?



- (A) 13 (B) 14 (C) 15 (D) 16 (E) 17

Solution

Problem 16

There are lily pads in a row numbered 0 to 11, in that order. There are predators on lily pads 3 and 6, and a morsel of food on lily pad 10. Fiona the frog starts on pad 0, and from any given lily pad, has a $\frac{1}{2}$ chance to hop to the next pad, and an equal chance to jump 2 pads. What is the probability that Fiona reaches pad 10 without landing on either pad 3 or pad 6?

- (A) $\frac{15}{256}$ (B) $\frac{1}{16}$ (C) $\frac{15}{128}$ (D) $\frac{1}{8}$ (E) $\frac{1}{4}$

Solution

Problem 17

How many nonzero complex numbers z have the property that 0, z , and z^3 , when represented by points in the complex plane, are the three distinct vertices of an equilateral triangle?

- (A) 0 (B) 1 (C) 2 (D) 4 (E) infinitely many

Solution

Problem 18

Square pyramid $ABCDE$ has base $ABCD$, which measures 3 cm on a side, and altitude $\overline{AE}$ perpendicular to the base, which measures 6 cm. Point P lies on $\overline{BE}$, one third of the way from B to E ; point Q lies on $\overline{DE}$, one third of the way from D to E ; and point R lies on $\overline{CE}$, two thirds of the way from C to E . What is the area, in square centimeters, of $\triangle PQR$?

- (A) $\frac{3\sqrt{2}}{2}$ (B) $\frac{3\sqrt{3}}{2}$ (C) $2\sqrt{2}$ (D) $2\sqrt{3}$ (E) $3\sqrt{2}$

Solution

Problem 19

Raashan, Sylvia, and Ted play the following game. Each starts with \$1. A bell rings every 15 seconds, at which time each of the players who currently have money simultaneously chooses one of the other two players independently and at random and gives \$1 to that player. What is the probability that after the bell has rung 2019 times, each player will have \$1? (For example, Raashan and Ted may each decide to give \$1 to Sylvia, and Sylvia may decide to give her her dollar to Ted, at which point Raashan will have \$0, Sylvia will have \$2, and Ted will have \$1, and that is the end of the first round of play. In the second round Raashan has no money to give, but Sylvia and Ted might choose each other to give their \$1 to, and the holdings will be the same at the end of the second round.)

- (A) $\frac{1}{7}$ (B) $\frac{1}{4}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$ (E) $\frac{2}{3}$

Solution

Problem 20

Points $A(6, 13)$ and $B(12, 11)$ lie on circle ω in the plane. Suppose that the tangent lines to ω at A and B intersect at a point on the x -axis. What is the area of ω ?

- (A) $\frac{83\pi}{8}$ (B) $\frac{21\pi}{2}$ (C) $\frac{85\pi}{8}$ (D) $\frac{43\pi}{4}$ (E) $\frac{87\pi}{8}$

Solution

Problem 21

How many quadratic polynomials with real coefficients are there such that the set of roots equals the set of coefficients? (For clarification: If the polynomial is $ax^2 + bx + c$, $a \neq 0$, and the roots are r and s , then the requirement is that $\{a, b, c\} = \{r, s\}$.)

- (A) 3 (B) 4 (C) 5 (D) 6 (E) infinitely many

Solution

Problem 22

Define a sequence recursively by $x_0 = 5$ and

$$x_{n+1} = \frac{x_n^2 + 5x_n + 4}{x_n + 6}$$

for all nonnegative integers n . Let m be the least positive integer such that

$$x_m \leq 4 + \frac{1}{2^{20}}.$$

In which of the following intervals does m lie?

- (A) $[9, 26]$ (B) $[27, 80]$ (C) $[81, 242]$ (D) $[243, 728]$ (E) $[729, \infty]$

Solution

Problem 23

How many sequences of 0s and 1s of length 19 are there that begin with a 0, end with a 0, contain no two consecutive 0s, and contain no three consecutive 1s?

- (A) 55 (B) 60 (C) 65 (D) 70 (E) 75

Solution

Problem 24

Let $\omega = -\frac{1}{2} + \frac{1}{2}i\sqrt{3}$. Let S denote all points in the complex plane of the form $a + b\omega + c\omega^2$, where $0 \leq a \leq 1$, $0 \leq b \leq 1$, and $0 \leq c \leq 1$. What is the area of S ?

- (A) $\frac{1}{2}\sqrt{3}$ (B) $\frac{3}{4}\sqrt{3}$ (C) $\frac{3}{2}\sqrt{3}$ (D) $\frac{1}{2}\pi\sqrt{3}$ (E) π

Solution

Problem 25

Let $ABCD$ be a convex quadrilateral with $BC = 2$ and $CD = 6$. Suppose that the centroids of $\triangle ABC$, $\triangle BCD$, and $\triangle ACD$ form the vertices of an equilateral triangle. What is the maximum possible value of the area of $ABCD$?

- (A) 27 (B) $16\sqrt{3}$ (C) $12 + 10\sqrt{3}$ (D) $9 + 12\sqrt{3}$ (E) 30

Solution

See also

2019 AMC 12B (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2019))	
Preceded by 2019 AMC 12A Problems	Followed by 2020 AMC 12A Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
All AMC 12 Problems and Solutions	

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2019 AMC 12B Answer Key

1. D
2. E
3. E
4. C
5. B
6. A
7. A
8. A
9. B
10. E
11. D
12. D
13. C
14. C
15. E
16. A
17. D
18. C
19. B
20. C
21. B
22. C
23. C
24. C
25. C

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2020

2020 AMC 12A Problems

2020 AMC 12A (Answer Key)

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1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

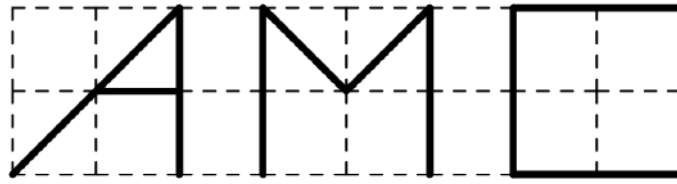
Carlos took 70% of a whole pie. Maria took one third of the remainder. What portion of the whole pie was left?

(A) 10% (B) 15% (C) 20% (D) 30% (E) 35%

Solution

Problem 2

The acronym AMC is shown in the rectangular grid below with grid lines spaced 1 unit apart. In units, what is the sum of the lengths of the line segments that form the acronym AMC?



- (A) 17 (B) $15 + 2\sqrt{2}$ (C) $13 + 4\sqrt{2}$ (D) $11 + 6\sqrt{2}$ (E) 21

Solution

Problem 3

A driver travels for 2 hours at 60 miles per hour, during which her car gets 30 miles per gallon of gasoline. She is paid \$0.50 per mile, and her only expense is gasoline at \$2.00 per gallon. What is her net rate of pay, in dollars per hour, after this expense?

- (A) 20 (B) 22 (C) 24 (D) 25 (E) 26

Solution

Problem 4

How many 4-digit positive integers (that is, integers between 1000 and 9999, inclusive) having only even digits are divisible by 5?

- (A) 80 (B) 100 (C) 125 (D) 200 (E) 500

Solution

Problem 5

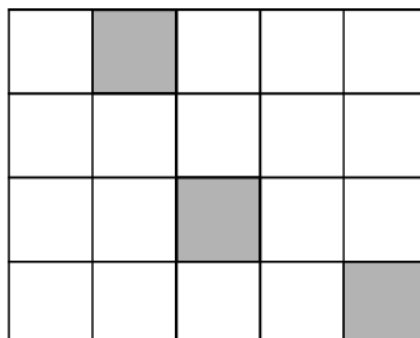
The 25 integers from -10 to 14 , inclusive, can be arranged to form a 5-by-5 square in which the sum of the numbers in each row, the sum of the numbers in each column, and the sum of the numbers along each of the main diagonals are all the same. What is the value of this common sum?

- (A) 2 (B) 5 (C) 10 (D) 25 (E) 50

Solution

Problem 6

In the plane figure shown below, 3 of the unit squares have been shaded. What is the least number of additional unit squares that must be shaded so that the resulting figure has two lines of symmetry?



(A) 4 (B) 5 (C) 6 (D) 7 (E) 8

Solution

Problem 7

Seven cubes, whose volumes are 1, 8, 27, 64, 125, 216, and 343 cubic units, are stacked vertically to form a tower in which the volumes of the cubes decrease from bottom to top. Except for the bottom cube, the bottom face of each cube lies completely on top of the cube below it. What is the total surface area of the tower (including the bottom) in square units?

(A) 644 (B) 658 (C) 664 (D) 720 (E) 749

Solution

Problem 8

What is the median of the following list of 4040 numbers?

$$1, 2, 3, \dots, 2020, 1^2, 2^2, 3^2, \dots, 2020^2$$

(A) 1974.5 (B) 1975.5 (C) 1976.5 (D) 1977.5 (E) 1978.5

Solution

Problem 9

How many solutions does the equation $\tan(2x) = \cos\left(\frac{x}{2}\right)$ have on the interval $[0, 2\pi]$?

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Solution

Problem 10

There is a unique positive integer n such that

$$\log_2(\log_{16} n) = \log_4(\log_4 n).$$

What is the sum of the digits of n ?

(A) 4 (B) 7 (C) 8 (D) 11 (E) 13

Solution

Problem 11

A frog sitting at the point $(1, 2)$ begins a sequence of jumps, where each jump is parallel to one of the coordinate axes and has length 1, and the direction of each jump (up, down, right, or left) is chosen independently at random. The sequence ends when the frog reaches a side of the square with vertices $(0, 0)$, $(0, 4)$, $(4, 4)$, and $(4, 0)$. What is the probability that the sequence of jumps ends on a vertical side of the square?

(A) $\frac{1}{2}$ (B) $\frac{5}{8}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$ (E) $\frac{7}{8}$

Solution

Problem 12

Line ℓ in the coordinate plane has the equation $3x - 5y + 40 = 0$. This line is rotated 45° counterclockwise about the point $(20, 20)$ to obtain line k . What is the x -coordinate of the x -intercept of line k ?

- (A) 10 (B) 15 (C) 20 (D) 25 (E) 30

Solution

Problem 13

There are integers a , b , and c , each greater than 1, such that

$$\sqrt[a]{N \sqrt[b]{N \sqrt[c]{N}}} = \sqrt[36]{N^{25}}$$

for all $N > 1$. What is b ?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Solution

Problem 14

Regular octagon $ABCDEFGH$ has area n . Let m be the area of quadrilateral $ACEG$. What is $\frac{m}{n}$?

- (A) $\frac{\sqrt{2}}{4}$ (B) $\frac{\sqrt{2}}{2}$ (C) $\frac{3}{4}$ (D) $\frac{3\sqrt{2}}{5}$ (E) $\frac{2\sqrt{2}}{3}$

Solution

Problem 15

In the complex plane, let A be the set of solutions to $z^3 - 8 = 0$ and let B be the set of solutions to $z^3 - 8z^2 - 8z + 64 = 0$. What is the greatest distance between a point of A and a point of B ?

- (A) $2\sqrt{3}$ (B) 6 (C) 9 (D) $2\sqrt{21}$ (E) $9 + \sqrt{3}$

Solution

Problem 16

A point is chosen at random within the square in the coordinate plane whose vertices are $(0, 0)$, $(2020, 0)$, $(2020, 2020)$, and $(0, 2020)$. The probability that the point is within d units of a lattice point is $\frac{1}{2}$. (A point (x, y) is a lattice point if x and y are both integers.) What is d to the nearest tenth?

- (A) 0.3 (B) 0.4 (C) 0.5 (D) 0.6 (E) 0.7

Solution

Problem 17

The vertices of a quadrilateral lie on the graph of $y = \ln x$, and the x -coordinates of these vertices are consecutive positive integers. The area of the quadrilateral is $\ln \frac{91}{90}$. What is the x -coordinate of the leftmost vertex?

- (A) 6 (B) 7 (C) 10 (D) 12 (E) 13

Solution

Problem 18

Quadrilateral $ABCD$ satisfies $\angle ABC = \angle ACD = 90^\circ$, $AC = 20$, and $CD = 30$. Diagonals $\overline{AC}$ and $\overline{BD}$ intersect at point E , and $AE = 5$. What is the area of quadrilateral $ABCD$?

- (A) 330 (B) 340 (C) 350 (D) 360 (E) 370

Solution

Problem 19

There exists a unique strictly increasing sequence of nonnegative integers $a_1 < a_2 < \dots < a_k$ such that

$$\frac{2^{289} + 1}{2^{17} + 1} = 2^{a_1} + 2^{a_2} + \dots + 2^{a_k}.$$

What is k ?

- (A) 117 (B) 136 (C) 137 (D) 273 (E) 306

Solution

Problem 20

Let T be the triangle in the coordinate plane with vertices $(0, 0)$, $(4, 0)$, and $(0, 3)$. Consider the following five isometries (rigid transformations) of the plane: rotations of 90° , 180° , and 270° counterclockwise around the origin, reflection across the x -axis, and reflection across the y -axis. How many of the 125 sequences of three of these transformations (not necessarily distinct) will return T to its original position? (For example, a 180° rotation, followed by a reflection across the x -axis, followed by a reflection across the y -axis will return T to its original position, but a 90° rotation, followed by a reflection across the x -axis, followed by another reflection across the x -axis will not return T to its original position.)

- (A) 12 (B) 15 (C) 17 (D) 20 (E) 25

Solution

Problem 21

How many positive integers n are there such that n is a multiple of 5 , and the least common multiple of $5!$ and n equals 5 times the greatest common divisor of $10!$ and n ?

- (A) 12 (B) 24 (C) 36 (D) 48 (E) 72

Solution

Problem 22

Let (a_n) and (b_n) be the sequences of real numbers such that

$$(2 + i)^n = a_n + b_n i$$

for all integers $n \geq 0$, where $i = \sqrt{-1}$. What is

$$\sum_{n=0}^{\infty} \frac{a_n b_n}{7^n} ?$$

- (A) $\frac{3}{8}$ (B) $\frac{7}{16}$ (C) $\frac{1}{2}$ (D) $\frac{9}{16}$ (E) $\frac{4}{7}$

Solution

Problem 23

Jason rolls three fair standard six-sided dice. Then he looks at the rolls and chooses a subset of the dice (possibly empty, possibly all three dice) to reroll. After rerolling, he wins if and only if the sum of the numbers face up on the three dice is exactly 7. Jason always plays to optimize his chances of winning. What is the probability that he chooses to reroll exactly two of the dice?

- (A) $\frac{7}{36}$ (B) $\frac{5}{24}$ (C) $\frac{2}{9}$ (D) $\frac{17}{72}$ (E) $\frac{1}{4}$

Solution

Problem 24

Suppose that $\triangle ABC$ is an equilateral triangle of side length s , with the property that there is a unique point P inside the triangle such that $AP = 1$, $BP = \sqrt{3}$, and $CP = 2$. What is s ?

- (A) $1 + \sqrt{2}$ (B) $\sqrt{7}$ (C) $\frac{8}{3}$ (D) $\sqrt{5 + \sqrt{5}}$ (E) $2\sqrt{2}$

Solution

Problem 25

The number $a = \frac{p}{q}$, where p and q are relatively prime positive integers, has the property that the sum of all real numbers x satisfying

$$\lfloor x \rfloor \cdot \{x\} = a \cdot x^2$$

is 420, where $\lfloor x \rfloor$ denotes the greatest integer less than or equal to x and $\{x\} = x - \lfloor x \rfloor$ denotes the fractional part of x . What is $p + q$?

- (A) 245 (B) 593 (C) 929 (D) 1331 (E) 1332

Solution

See also

2020 AMC 12A (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2020))	
Preceded by 2019 AMC 12B Problems	Followed by 2020 AMC 12B Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
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2020 AMC 12A Answer Key

1. C
2. C
3. E
4. B
5. C
6. D
7. B
8. C
9. E
10. E
11. B
12. B
13. B
14. B
15. D
16. B
17. D
18. D
19. C
20. A
21. D
22. B
23. A
24. B
25. C

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2020 AMC 12B Problems

2020 AMC 12B (Answer Key)

Printable version: | AoPS Resources (<http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2020>) • PDF (http://www.artofproblemsolving.com/Forum/resources/files/usa/USA-AMC_12-AHSME-2020-44)

Instructions

1. This is a 25-question, multiple choice test. Each question is followed by answers marked A, B, C, D and E. Only one of these is correct.
2. You will receive 6 points for each correct answer, 2.5 points for each problem left unanswered if the year is before 2006, 1.5 points for each problem left unanswered if the year is after 2006, and 0 points for each incorrect answer.
3. No aids are permitted other than scratch paper, graph paper, ruler, compass, protractor and erasers (and calculators that are accepted for use on the test if before 2006. No problems on the test will *require* the use of a calculator).
4. Figures are not necessarily drawn to scale.
5. You will have **75 minutes** working time to complete the test.

1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

What is the value in simplest form of the following expression?

$$\sqrt{1} + \sqrt{1+3} + \sqrt{1+3+5} + \sqrt{1+3+5+7}$$

- (A) 5 (B) $4 + \sqrt{7} + \sqrt{10}$ (C) 10 (D) 15 (E) $4 + 3\sqrt{3} + 2\sqrt{5} + \sqrt{7}$

Solution

Problem 2

What is the value of the following expression?

$$\frac{100^2 - 7^2}{70^2 - 11^2} \cdot \frac{(70 - 11)(70 + 11)}{(100 - 7)(100 + 7)}$$

- (A) 1 (B) $\frac{9951}{9950}$ (C) $\frac{4780}{4779}$ (D) $\frac{108}{107}$ (E) $\frac{81}{80}$

Solution

Problem 3

The ratio of w to x is $4 : 3$, the ratio of y to z is $3 : 2$, and the ratio of z to x is $1 : 6$. What is the ratio of w to y ?

- (A) $4 : 3$ (B) $3 : 2$ (C) $8 : 3$ (D) $4 : 1$ (E) $16 : 3$

Solution

Problem 4

The acute angles of a right triangle are a° and b° , where $a > b$ and both a and b are prime numbers. What is the least possible value of b ?

- (A) 2 (B) 3 (C) 5 (D) 7 (E) 11

Solution

Problem 5

Teams A and B are playing in a basketball league where each game results in a win for one team and a loss for the other team. Team A has won $\frac{2}{3}$ of its games and team B has won $\frac{5}{8}$ of its games. Also, team B has won 7 more games and lost 7 more games than team A . How many games has team A played?

- (A) 21 (B) 27 (C) 42 (D) 48 (E) 63

Solution

Problem 6

For all integers $n \geq 9$, the value of

$$\frac{(n+2)! - (n+1)!}{n!}$$

is always which of the following?

- (A) a multiple of 4 (B) a multiple of 10 (C) a prime number
(D) a perfect square (E) a perfect cube

Solution

Problem 7

Two nonhorizontal, non vertical lines in the xy -coordinate plane intersect to form a 45° angle. One line has slope equal to 6 times the slope of the other line. What is the greatest possible value of the product of the slopes of the two lines?

- (A) $\frac{1}{6}$ (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) 3 (E) 6

Solution

Problem 8

How many ordered pairs of integers (x, y) satisfy the equation

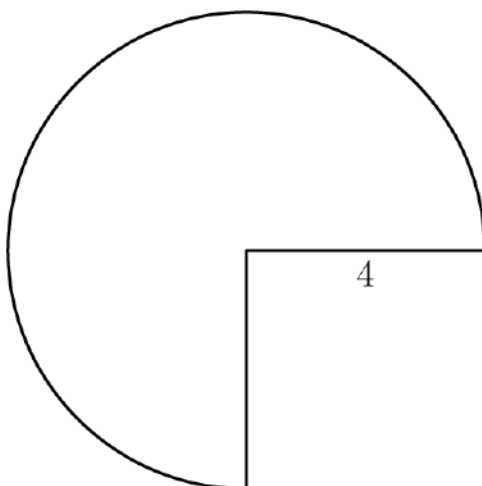
$$x^{2020} + y^2 = 2y?$$

- (A) 1 (B) 2 (C) 3 (D) 4 (E) infinitely many

Solution

Problem 9

A three-quarter sector of a circle of radius 4 inches together with its interior can be rolled up to form the lateral surface area of a right circular cone by taping together along the two radii shown. What is the volume of the cone in cubic inches?



- (A) $3\pi\sqrt{5}$ (B) $4\pi\sqrt{3}$ (C) $3\pi\sqrt{7}$ (D) $6\pi\sqrt{3}$ (E) $6\pi\sqrt{7}$

Solution

Problem 10

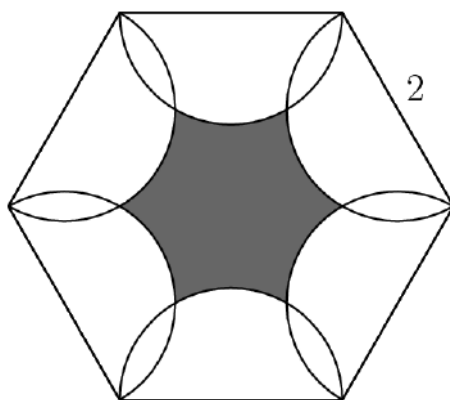
In unit square $ABCD$, the inscribed circle ω intersects $\overline{CD}$ at M , and $\overline{AM}$ intersects ω at a point P different from M . What is AP ?

- (A) $\frac{\sqrt{5}}{12}$ (B) $\frac{\sqrt{5}}{10}$ (C) $\frac{\sqrt{5}}{9}$ (D) $\frac{\sqrt{5}}{8}$ (E) $\frac{2\sqrt{5}}{15}$

Solution

Problem 11

As shown in the figure below, six semicircles lie in the interior of a regular hexagon with side length 2 so that the diameters of the semicircles coincide with the sides of the hexagon. What is the area of the shaded region—inside the hexagon but outside all of the semicircles?



- (A) $6\sqrt{3} - 3\pi$ (B) $\frac{9\sqrt{3}}{2} - 2\pi$ (C) $\frac{3\sqrt{3}}{2} - \frac{\pi}{3}$ (D) $3\sqrt{3} - \pi$ (E) $\frac{9\sqrt{3}}{2} - \pi$

Solution

Problem 12

Let $\overline{AB}$ be a diameter in a circle of radius $5\sqrt{2}$. Let $\overline{CD}$ be a chord in the circle that intersects $\overline{AB}$ at a point E such that $BE = 2\sqrt{5}$ and $\angle AEC = 45^\circ$. What is $CE^2 + DE^2$?

- (A) 96 (B) 98 (C) $44\sqrt{5}$ (D) $70\sqrt{2}$ (E) 100

Solution

Problem 13

Which of the following is the value of $\sqrt{\log_2 6 + \log_3 6}$?

- (A) 1 (B) $\sqrt{\log_5 6}$ (C) 2 (D) $\sqrt{\log_2 3} + \sqrt{\log_3 2}$ (E) $\sqrt{\log_2 6} + \sqrt{\log_3 6}$

Solution

Problem 14

Bela and Jenn play the following game on the closed interval $[0, n]$ of the real number line, where n is a fixed integer greater than 4. They take turns playing, with Bela going first. At his first turn, Bela chooses any real number in the interval $[0, n]$. Thereafter, the player whose turn it is chooses a real number that is more than one unit away from all numbers previously chosen by either player. A player unable to choose such a number loses. Using optimal strategy, which player will win the game?

- (A) Bela will always win. (B) Jenn will always win. (C) Bela will win if and only if n is odd.
(D) Jenn will win if and only if n is odd. (E) Jenn will win if and only if $n > 8$.

Solution

Problem 15

There are 10 people standing equally spaced around a circle. Each person knows exactly 3 of the other 9 people: the 2 people standing next to her or him, as well as the person directly across the circle. How many ways are there for the 10 people to split up into 5 pairs so that the members of each pair know each other?

- (A) 11 (B) 12 (C) 13 (D) 14 (E) 15

Solution

Problem 16

An urn contains one red ball and one blue ball. A box of extra red and blue balls lie nearby. George performs the following operation four times: he draws a ball from the urn at random and then takes a ball of the same color from the box and returns those two matching balls to the urn. After the four iterations the urn contains six balls. What is the probability that the urn contains three balls of each color?

- (A) $\frac{1}{6}$ (B) $\frac{1}{5}$ (C) $\frac{1}{4}$ (D) $\frac{1}{3}$ (E) $\frac{1}{2}$

Solution

Problem 17

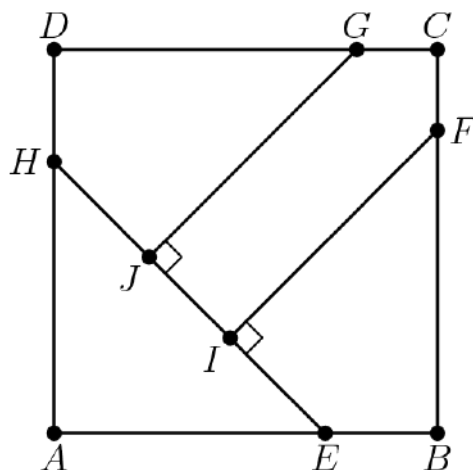
How many polynomials of the form $x^5 + ax^4 + bx^3 + cx^2 + dx + 2020$, where a, b, c , and d are real numbers, have the property that whenever r is a root, so is $\frac{-1 + i\sqrt{3}}{2} \cdot r$? (Note that $i = \sqrt{-1}$)

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Solution

Problem 18

In square $ABCD$, points E and H lie on $\overline{AB}$ and $\overline{DA}$, respectively, so that $AE = AH$. Points F and G lie on $\overline{BC}$ and $\overline{CD}$, respectively, and points I and J lie on $\overline{EH}$ so that $FI \perp \overline{EH}$ and $GJ \perp \overline{EH}$. See the figure below. Triangle AEH , quadrilateral $BFIE$, quadrilateral $DHJG$, and pentagon $FCGJI$ each has area 1. What is FI^2 ?



- (A) $\frac{7}{3}$ (B) $8 - 4\sqrt{2}$ (C) $1 + \sqrt{2}$ (D) $\frac{7}{4}\sqrt{2}$ (E) $2\sqrt{2}$

Solution

Problem 19

Square $ABCD$ in the coordinate plane has vertices at the points $A(1, 1)$, $B(-1, 1)$, $C(-1, -1)$, and $D(1, -1)$. Consider the following four transformations: L , a rotation of 90° counterclockwise around the origin; R , a rotation of 90° clockwise around the origin; H , a reflection across the x -axis; and V , a reflection across the y -axis.

Each of these transformations maps the squares onto itself, but the positions of the labeled vertices will change. For example, applying R and then V would send the vertex A at $(1, 1)$ to $(-1, -1)$ and would send the vertex B at $(-1, 1)$ to itself. How many sequences of 20 transformations chosen from $\{L, R, H, V\}$ will send all of the labeled vertices back to their original positions? (For example, R, R, V, H is one sequence of 4 transformations that will send the vertices back to their original positions.)

- (A) 2^{37} (B) $3 \cdot 2^{36}$ (C) 2^{38} (D) $3 \cdot 2^{37}$ (E) 2^{39}

Solution

Problem 20

Two different cubes of the same size are to be painted, with the color of each face being chosen independently and at random to be either black or white. What is the probability that after they are painted, the cubes can be rotated to be identical in appearance?

- (A) $\frac{9}{64}$ (B) $\frac{289}{2048}$ (C) $\frac{73}{512}$ (D) $\frac{147}{1024}$ (E) $\frac{589}{4096}$

Solution

Problem 21

How many positive integers n satisfy

$$\frac{n + 1000}{70} = \lfloor \sqrt{n} \rfloor?$$

(Recall that $\lfloor x \rfloor$ is the greatest integer not exceeding x .)

- (A) 2 (B) 4 (C) 6 (D) 30 (E) 32

Solution

Problem 22

What is the maximum value of $\frac{(2^t - 3t)t}{4^t}$ for real values of t ?

- (A) $\frac{1}{16}$ (B) $\frac{1}{15}$ (C) $\frac{1}{12}$ (D) $\frac{1}{10}$ (E) $\frac{1}{9}$

Solution

Problem 23

How many integers $n \geq 2$ are there such that whenever $z_1, z_2, \dots, z_n$ are complex numbers such that

$$|z_1| = |z_2| = \dots = |z_n| = 1 \text{ and } z_1 + z_2 + \dots + z_n = 0,$$

then the numbers $z_1, z_2, \dots, z_n$ are equally spaced on the unit circle in the complex plane?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Solution

Problem 24

Let $D(n)$ denote the number of ways of writing the positive integer n as a product

$$n = f_1 \cdot f_2 \cdots f_k,$$

where $k \geq 1$, the f_i are integers strictly greater than 1, and the order in which the factors are listed matters (that is, two representations that differ only in the order of the factors are counted as distinct). For example, the number 6 can be written as $6, 2 \cdot 3$, and $3 \cdot 2$, so $D(6) = 3$. What is $D(96)$?

- (A) 112 (B) 128 (C) 144 (D) 172 (E) 184

Solution

Problem 25

For each real number a with $0 \leq a \leq 1$, let numbers x and y be chosen independently at random from the intervals $[0, a]$ and $[0, 1]$ respectively, and let $P(a)$ be the probability that

$$\sin^2(\pi x) + \sin^2(\pi y) > 1$$

What is the maximum value of $P(a)$?

- (A) $\frac{7}{12}$ (B) $2 - \sqrt{2}$ (C) $\frac{1 + \sqrt{2}}{4}$ (D) $\frac{\sqrt{5} - 1}{2}$ (E) $\frac{5}{8}$

Solution

See also

2020 AMC 12B (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2020))	
Preceded by 2020 AMC 12A Problems	Followed by 2021 AMC 12A Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
All AMC 12 Problems and Solutions	

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2020 AMC 12B Answer Key

1. C
2. A
3. E
4. D
5. C
6. D
7. C
8. D
9. C
10. B
11. D
12. E
13. D
14. A
15. C
16. B
17. C
18. B
19. C
20. D
21. C
22. C
23. B
24. A
25. B

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AMC12A 2020 Enunciados

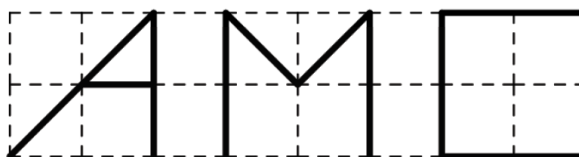
1

Carlos toma el 70% de un pastel entero. María toma una tercera parte del resto. ¿Qué porción del pastel queda sin tomar?

- (A) 10% (B) 15% (C) 20% (D) 30% (E) 35%

2

El acrónimo AMC se muestra en la siguiente cuadrícula con cuadrados de lado 1. ¿Cuál es la suma de las longitudes de todos los segmentos de este acrónimo?



- (A) 17 (B) $15 + 2\sqrt{2}$ (C) $13 + 4\sqrt{2}$ (D) $11 + 6\sqrt{2}$ (E) 21

3

Un conductor conduce durante 2 horas a 60 millas por hora, durante las cuales su coche consume un galón de gasolina cada 30 millas. Además, ella cobra 0.50\$ cada milla, y su único gasto es la gasolina, a 2.00\$ por galón. ¿Cuál es la ganancia del viaje, en dólares por hora, gastos incluidos?

- (A) 20 (B) 22 (C) 24 (D) 25 (E) 26

4

Determina la cantidad de números enteros positivos de cuatro cifras (es decir, enteros entre 1000 y 9999, inclusive) que tienen solo cifras pares y son divisibles entre 5.

- (A) 80 (B) 100 (C) 125 (D) 200 (E) 500

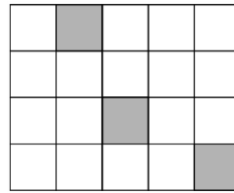
5

Los 25 enteros entre -10 y 14, inclusive, se pueden organizar para formar un cuadrado de 5 por 5 en el que la suma de los números de cada fila, de cada columna y de las dos diagonales sumen lo mismo. ¿Cuál es el valor de esta suma común?

- (A) 2 (B) 5 (C) 10 (D) 25 (E) 50

6

En la siguiente figura se han sombreado 3 recuadros. ¿Cuál es el mínimo número de recuadros adicionales que tenemos que sombrar para que la figura resultante tenga dos rectas de simetría?



(A) 4 (B) 5 (C) 6 (D) 7 (E) 8

7

Siete cubos, cuyos volúmenes son 1, 8, 27, 64, 125, 216 y 343 unidades cúbicas, se apilan en vertical formando una torre en la que los volúmenes van en orden decreciente de abajo a arriba. Excepto el cubo de la base, la cara inferior de cada cubo queda en el interior de la cara superior del cubo que tiene debajo. Determina la superficie total de la torre (incluyendo la base) en unidades cuadradas.

(A) 644 (B) 658 (C) 664 (D) 720 (E) 749

8

Determina la mediana de los siguientes 4040 números:

$$1, 2, 3, \dots, 2020, 1^2, 2^2, 3^2, \dots, 2020^2$$

(A) 1974.5 (B) 1975.5 (C) 1976.5 (D) 1977.5 (E) 1978.5

9

¿Cuántas soluciones tiene la ecuación $\tan(2x) = \cos\left(\frac{x}{2}\right)$ en el intervalo $[0, 2\pi]$?

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

10

Existe un único entero positivo n tal que

$$\log_2(\log_{16} n) = \log_4(\log_4 n)$$

Determina la suma de los dígitos de n .

(A) 4 (B) 7 (C) 8 (D) 11 (E) 13

11

Una rana situada en el punto $(1,2)$ empieza una secuencia de saltos, cada uno paralelo a los ejes de coordenadas y de longitud 1 y con dirección (arriba, abajo, derecha o izquierda) de forma aleatoria e independiente. La secuencia acaba cuando la rana alcanza uno de los lados del cuadrado con vértices $(0,0)$, $(0,4)$, $(4,4)$ y $(4,0)$. ¿Cuál es la probabilidad de que la secuencia de saltos llegue a uno de los dos lados verticales del cuadrado?

(A) $1/2$ (B) $5/8$ (C) $2/3$ (D) $3/4$ (E) $7/8$

12

La recta l tiene por ecuación $3x - 5y + 40 = 0$. Esta recta se somete a una rotación de 45° en el sentido contrario de las agujas del reloj alrededor del punto $(20,20)$ para obtener la recta k . Determina la coordenada x del punto de intersección entre k y el eje X .

A) 10 B) 15 C) 20 D) 25 E) 30

13

Existen enteros a , b y c , todos ellos mayores que 1, tales que

$$\sqrt[a]{N} \sqrt[b]{N} \sqrt[c]{N} = \sqrt[36]{N^{25}}$$

para todo $N > 1$. Determina b .

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

14

Dado un octógono regular $ABCDEFGH$ de área n , y siendo m el área del cuadrilátero $ACEG$, determina m/n .

(A) $\frac{\sqrt{2}}{4}$ (B) $\frac{\sqrt{2}}{2}$ (C) $\frac{3}{4}$ (D) $\frac{3\sqrt{2}}{5}$ (E) $\frac{2\sqrt{2}}{3}$

15

En el plano complejo, sea A el conjunto de soluciones de $z^3 - 8 = 0$ y sea B el conjunto de soluciones de $z^3 - 8z^2 - 8z + 64 = 0$. Determina la máxima distancia entre un punto de A y un punto de B .

(A) $2\sqrt{3}$ (B) 6 (C) 9 (D) $2\sqrt{21}$ (E) $9 + \sqrt{3}$

16

Se toma un punto aleatorio en el plano coordenado en el interior del cuadrado de vértices $(0,0)$, $(2020,0)$, $(2020,2020)$ y $(0,2020)$. Supongamos que la probabilidad de que dicho punto esté a d unidades de un punto “cruceta” es $1/2$ (llamaremos punto “cruceta” a todo punto (x,y) del plano cuyas coordenadas x,y sean enteras). Determina la aproximación de d a las décimas.

(A) 0.3 (B) 0.4 (C) 0.5 (D) 0.6 (E) 0.7

17

Los vértices de un cuadrilátero pertenecen a la gráfica de $y = \ln x$, y las abscisas de estos vértices son enteros positivos consecutivos. El área del cuadrilátero es $\ln \frac{91}{90}$. Determina la abscisa del punto situado más a la izquierda.

(A) 6 (B) 7 (C) 10 (D) 12 (E) 13

18

Determina el área del cuadrilátero ABCD en el que $\angle ABC = \angle ACD = 90^\circ$, $AC = 20$, $CD = 30$ y las diagonales $\overline{AC}$ y $\overline{BD}$ se cortan en el punto E, con $AE = 5$.

(A) 330 (B) 340 (C) 350 (D) 360 (E) 370

19

Existe una única secuencia de enteros no negativos $a_1 < a_2 < \dots < a_k$ tal que

$$\frac{2^{289} + 1}{2^{17} + 1} = 2^{a_1} + 2^{a_2} + \dots + 2^{a_k}$$

Determina el número k .

(A) 117 (B) 136 (C) 137 (D) 273 (E) 306

20

Sea T el triángulo del plano cartesiano con vértices $(0,0)$, $(4,0)$ y $(0,3)$. Consideremos las siguientes cinco isometrías (transformaciones rígidas) del plano: rotaciones de 90° , 180° y 270° en el sentido contrario de las agujas del reloj, simetría respecto del eje X y simetría respecto del eje Y. ¿Cuántas de las 125 secuencias de tres de estas transformaciones (no necesariamente distintas) dejarán T en su posición original? (Por ejemplo, una rotación de 180° , seguida de una reflexión respecto del eje X, y seguida de una reflexión respecto del eje X volverá T a su posición original, pero una rotación de 90° , seguida de una reflexión respecto del eje X y seguida de otra reflexión respecto del eje X no volverá T a su posición original).

(A) 12 (B) 15 (C) 17 (D) 20 (E) 25

21

Determina el número de enteros positivos n múltiplos de 5 tales que el mínimo común múltiplo de $5!$ y n sea igual a 5 veces el máximo común divisor de $10!$ y n .

(A) 12 (B) 24 (C) 36 (D) 48 (E) 72

22

Sean (a_n) y (b_n) las sucesiones de números reales tales que

$$(2+i)^n = a_n + b_n i$$

para todo entero $n \geq 0$, donde $i = \sqrt{-1}$. Determina $\sum_{n=0}^{\infty} \frac{a_n b_n}{7^n}$

(A) $\frac{3}{8}$ (B) $\frac{7}{16}$ (C) $\frac{1}{2}$ (D) $\frac{9}{16}$ (E) $\frac{4}{7}$

23

Jason lanza tres dados perfectos de seis caras, mira las puntuaciones obtenidas y después decide volver a tirar los tres, dos, uno o ninguno de ellos. Gana si, después de la segunda tirada opcional, la suma de los tres dados es igual a 7. Jason siempre juega optimizando sus posibilidades de ganar. ¿Cuál es la probabilidad de que decida volver a tirar exactamente dos dados?

(A) $7/36$ (B) $5/24$ (C) $2/9$ (D) $17/72$ (E) $1/4$

24

Sea $\triangle ABC$ un triángulo equilátero de lado s , con la propiedad de que existe un único punto P en su interior tal que $AP = 1$, $BP = \sqrt{3}$ y $CP = 2$. Determina s .

(A) $1 + \sqrt{2}$ (B) $\sqrt{7}$ (C) $\frac{8}{3}$ (D) $\sqrt{5 + \sqrt{5}}$ (E) $2\sqrt{2}$

25

El número $a = \frac{p}{q}$, donde p y q son enteros coprimos positivos, cumple la propiedad de que la suma de todos los números reales x satisfaciendo

$$\lfloor x \rfloor \cdot \{x\} = a \cdot x^2$$

es 420, donde $\lfloor x \rfloor$ denota el mayor entero menor o igual que x y $\{x\} = x - \lfloor x \rfloor$ denota la parte fraccionaria de x . Determina $p + q$.

(A) 245 (B) 593 (C) 929 (D) 1331 (E) 1332

AMC12A 2020 Soluciones (Letra)

1. C
2. C
3. E
4. B
5. C
6. D
7. B
8. C
9. E
10. E
11. B
12. B
13. B
14. B
15. D
16. B
17. D
18. D
19. C
20. A
21. D
22. B
23. A
24. B
25. C

AMC12A 2020 Soluciones desarrolladas

1

$$\frac{1}{3}(100\% - 70\%) = \frac{1}{3}30\% = \frac{1}{3} \frac{30}{100} = \frac{1}{3} \frac{3}{10} = \frac{1}{10} = 10\%. \text{ Queda } 70\% - 10\% = 20\% \quad (C)$$

2

Tenemos $3 + 4 + 6 = 13$ segmentos de 1 unidad y $2 + 2 + 0 = 4$ segmentos de longitud $\sqrt{2}$, luego en total, $13 + 4\sqrt{2}$ (C)

3

2 horas a 60 millas por hora = 120 millas = $4 \cdot 30$ millas = 4 galones de gasolina = 8\$ en gasolina = 4\$/hora
 $120 \text{ millas} \cdot 0.50\$ = 60\$$, $60\$ / 2 \text{ horas} = 30\$ / \text{hora}$.

$30\$ / \text{hora} - 4\$ / \text{hora} = 26\$ / \text{hora}$ de beneficio. (E)

4

Todo número que cumpla las condiciones del enunciado se podrá escribir como ABCD con $D \in \{0\}$, $B, C \in \{0, 2, 4, 6, 8\}$, $A \in \{2, 4, 6, 8\}$

En total hay $4 \cdot 5 \cdot 5 \cdot 1 = 100$ (B)

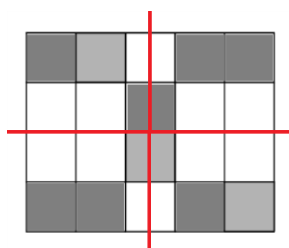
5

$$\begin{aligned} &(-10) + (-9) + (-8) + (-7) + (-6) + (-5) + (-4) + (-3) + (-2) + (-1) + 0 + \\ &1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 + 11 + 12 + 13 + 14 = \\ &= 11 + 12 + 13 + 14 = 50 \end{aligned}$$

Luego en cada fila, columna y diagonal la suma será 10 (C)

6

Los ejes de simetría deben ser una recta horizontal y una recta vertical que pasen por el centro de la figura. Completando los recuadros que faltan llegamos a la siguiente figura:



En la que hemos añadido 7 recuadros más (D).

7

Vemos que son cubos de arista 1, 2, 3, 4, 5, 6, 7. Luego sus áreas laterales son

$$4(1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2) = 4 \frac{7 \cdot 8 \cdot 15}{6} = 560$$

Las caras superiores se van encajando unas con otras hasta completar un cuadrado de área 7^2 .
 En total: $540 + 2 \cdot 49 = 638$ (B)

8

Queremos determinar el valor medio entre las posiciones 2020 y 2021.

Observamos que $44^2 = 1936$ y $45^2 = 2025$, luego $1^2, 2^2, 3^2, \dots, 44^2$ están repetidos.

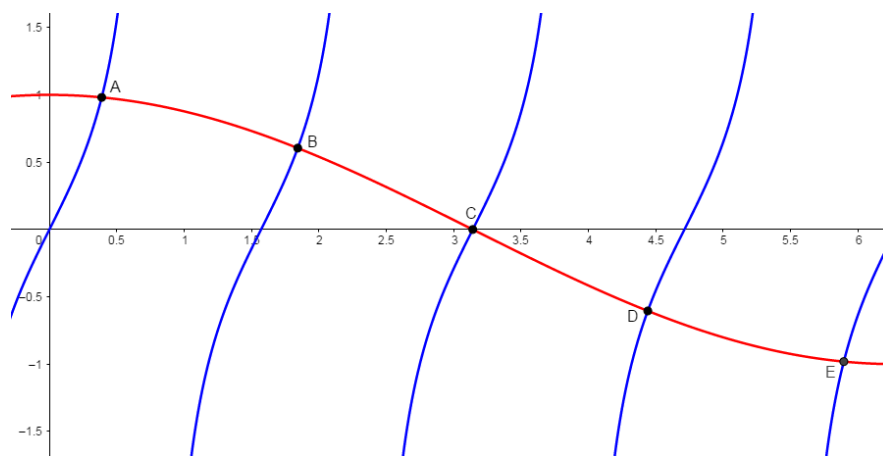
Luego serán los números $2020 - 44 = 1976$ y 1977 , es decir, 1976.5 (C)

9

Basta tener en cuenta las gráficas de las funciones $f(x) = \tan(x)$ y $f(x) = \cos(x)$ y las

modificaciones que producen los cambios $f(x) \rightarrow f(2x)$ y $f(x) \rightarrow f\left(\frac{x}{2}\right)$ y representarlas

esquemáticamente para ver que son cinco los puntos de corte en dicho intervalo:



Luego la solución es (E).

10

De la igualdad $\log_a b = \frac{\log_c b}{\log_c a}$ se deduce $\log_2 x = \frac{\log_4 x}{\log_4 2} = \frac{\log_4 x}{1/2} \Rightarrow \log_2 x = 2 \log_4 x$

(En general se puede deducir la igualdad $\log_{a^b} c = \frac{1}{b} \log_a c$)

Luego

$$\log_2(\log_{16} n) = \log_4(\log_4 n) \Leftrightarrow 2 \log_4(\log_{16} n) = \log_4(\log_4 n) \Leftrightarrow$$

$$\log_4((\log_{16} n)^2) = \log_4(\log_4 n) \Leftrightarrow (\log_{16} n)^2 = \log_4 n$$

De nuevo

$$\log_{16} n = \frac{\log_4 n}{\log_4 16} = \frac{\log_4 n}{2}$$

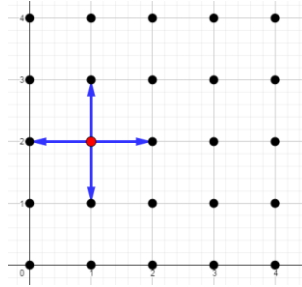
Luego

$$(*) \Leftrightarrow \left(\frac{\log_4 n}{2}\right)^2 = \log_4 n \Leftrightarrow \frac{(\log_4 n)^2}{4} = \log_4 n \Leftrightarrow (\log_4 n)^2 = 4 \log_4 n$$

$$\Leftrightarrow \log_4 n = 4 \Leftrightarrow \log_4 n = 4 \Leftrightarrow n = 4^4 = 256 \rightarrow 2 + 5 + 6 = 13 \quad (E)$$

11

Si la rana está en una posición diagonal (1,1), (2,2), (3,3), (1,3), (3,1), la probabilidad será 1/2, por simetría. Si la rana está en la posición central (2,2) la probabilidad será también de 1/2, por simetría.



Si la rana va hacia la izquierda (con probabilidad 1/4) seguro tocará el lado izquierdo, luego la probabilidad total es

$$P = \frac{1}{4} + \frac{1}{4} \cdot \frac{1}{2} + \frac{1}{4} \cdot \frac{1}{2} + \frac{1}{4} \cdot \frac{1}{2} = \frac{5}{8} \quad (B)$$

12

Primera versión. Mediante trigonometría.

Sean α y β los ángulos respectivos entre las rectas l y k con el eje X.

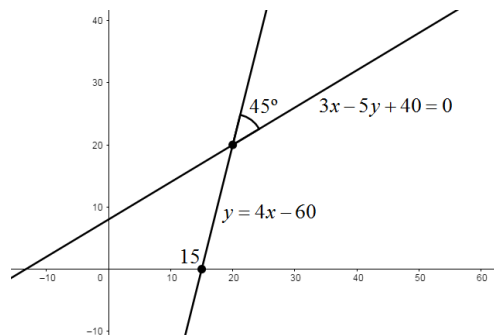
$$3x - 5y + 40 = 0 \Leftrightarrow y = \frac{3}{5}x + 8 \Rightarrow \tan \alpha = \frac{3}{5}$$

Luego

$$\tan \beta = \frac{\tan \alpha + \tan 45^\circ}{1 - \tan \alpha \tan 45^\circ} = \frac{3/5 + 1}{1 - 3/5 \cdot 1} = \frac{8/5}{2/5} = 4$$

Luego la recta k tiene por ecuación $y = 4x + b$, y puesto que pasa por el punto (20,20), $20 = 4 \cdot 20 + b \Rightarrow b = -60 \Rightarrow y = 4x - 60$

Su punto de corte con el eje X es $0 = 4x - 60 \Leftrightarrow 4x = 60 \Leftrightarrow x = 60/4 = 15 \quad (B)$

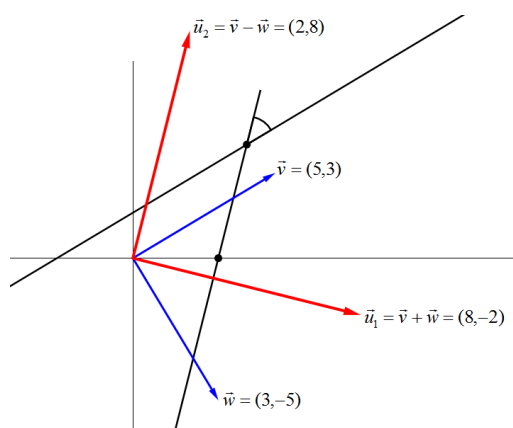


Segunda versión. Sin trigonometría, mediante vectores.

Sea $\vec{v} = (5,3)$. Este vector determina un ángulo α con $\tan \alpha = \frac{3}{5}$.

Sea $\vec{w} = (3,-5)$. Este vector es perpendicular a $\vec{v}$ y tiene su mismo módulo.

Luego $\vec{u}_1 = \vec{v} + \vec{w} = (8, -2)$ y $\vec{u}_2 = \vec{v} - \vec{w} = (2, 8)$ serán sus bisectrices, generando ángulos de 45° respecto de la recta l. En particular, $\vec{u}_2 = \vec{v} - \vec{w} = (2, 8) \Rightarrow m = \frac{8}{2} = 4$ es la pendiente de la recta k buscada.



13

Pasando a forma exponencial tenemos

$$\sqrt[36]{N^{25}} = N^{25/36}$$

$$\sqrt[a]{N \sqrt[b]{N \sqrt[c]{N}}} = \sqrt[a]{N \sqrt[b]{N N^{1/c}}} = \sqrt[a]{N \sqrt[b]{N^{1/c+1}}} = \sqrt[a]{N N^{(1/c+1)/b}} = \sqrt[a]{N^{(1/c+1)/b+1}} = N^{((1/c+1)/b+1)/a}$$

Luego llegamos a la igualdad

$$((1/c+1)/b+1)/a = 25/36 \Leftrightarrow \left(\frac{1+c}{bc} + 1 \right) / a = 25/36 \Leftrightarrow \frac{1+c+bc}{abc} = \frac{25}{36}$$

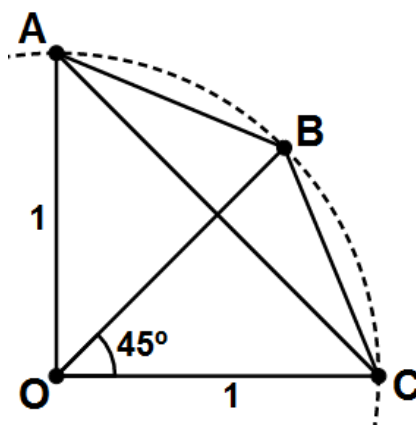
El sistema

$$\begin{cases} 1+c+bc = 25 \\ abc = 36 \end{cases} \Leftrightarrow \begin{cases} c(b+1) = 24 = 2^3 \cdot 3 \\ abc = 36 \end{cases}$$

Probando diferentes combinaciones de c y $b+1$ llegamos a la única solución con enteros positivos: $c = 6, b = 3, a = 2$ (B).

14

Sea O el centro del octógono. Estudiando la figura, este problema se reduce fácilmente a encontrar la razón $[\Delta OAC]/[OABC]$:



$$[\Delta OAC] = \frac{1 \cdot 1}{2} = \frac{1}{2}$$

$$[OABC] = 2[\Delta OBC] = 2 \frac{1 \cdot 1 \cdot \sin 45^\circ}{2} = \sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$\frac{[\Delta OAC]}{[OABC]} = \frac{1/2}{1/\sqrt{2}} = \frac{\sqrt{2}}{2} \quad (B)$$

15

El conjunto A consta de tres puntos:

$$A_1 = (2, 0)$$

$$A_2 = 2(\cos 120^\circ + i \sin 120^\circ)$$

$$A_3 = 2(\cos 240^\circ + i \sin 240^\circ)$$

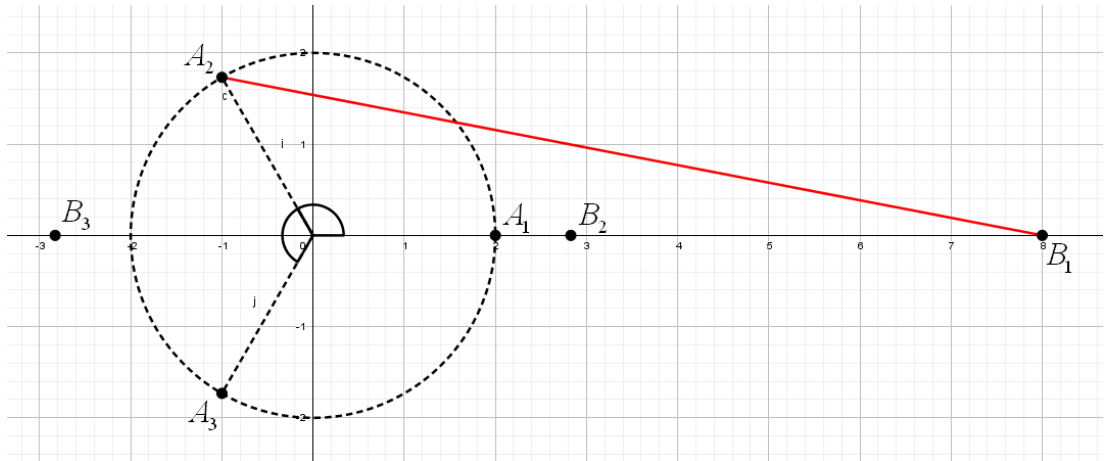
Observamos, por Ruffini, que 8 es solución de la ecuación $z^3 - 8z^2 - 8z + 64 = 0$. Las otras dos soluciones son $\pm \sqrt{8}$. Las tres son números reales. Luego el conjunto B consta de tres puntos:

$$B_1 = (8, 0)$$

$$B_2 = (\sqrt{8}, 0)$$

$$B_3 = (-\sqrt{8}, 0)$$

Representando estos puntos en un plano cartesiano vemos que la distancia máxima está entre A_2 y B_1 (la misma que hay entre A_3 y B_1)



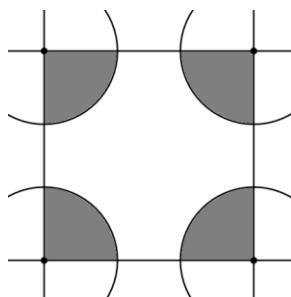
$$A_2 = 2(\cos 120^\circ + i \sin 120^\circ) = 2\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right) = (-1, \sqrt{3})$$

Luego

$$\text{dist}(A_2, B_1) = \sqrt{(8 - (-1))^2 + (0 - \sqrt{3})^2} = \sqrt{9^2 + 3} = \sqrt{84} = 2\sqrt{21} \quad (D)$$

16

El cuadrado que contiene el punto está dividido en $2020 \cdot 2020$ cuadrados de área unidad, cada uno de ellos tiene una zona digamos “favorable” cuya distancia a un vértice es igual o menor que d (aquí estamos suponiendo tácitamente que $d \leq 1$, y por tanto no hay solapamientos):



Las unión de estas zonas favorables equivale a un círculo de radio d , luego

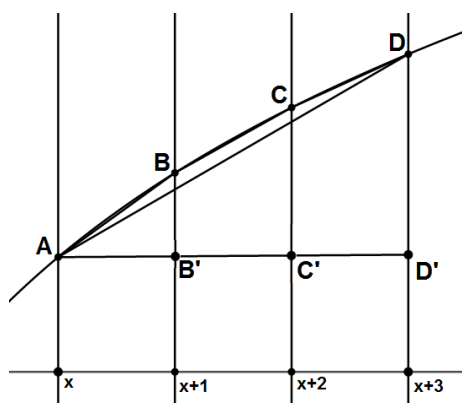
$$P = \frac{\pi \cdot d^2}{1^2 = 1} = \pi \cdot d^2 = \frac{1}{2} \Leftrightarrow d = \sqrt{\frac{1}{2\pi}}$$

Luego el problema se reduce a calcular el valor $d = \sqrt{\frac{1}{2\pi}}$ a mano, sin calculadora.

$$d \cong 0.4 \quad (B)$$

Nota: En las soluciones oficiales de AoPS se indican algunas estrategias para realizar dicho cálculo manualmente.

17



$$[\triangle ADD'] = \frac{1}{2} (\ln(x+3) - \ln(x)) \cdot 3 = \frac{3}{2} \ln\left(\frac{x+3}{x}\right)$$

$$[\triangle ABB'] = \frac{1}{2} (\ln(x+1) - \ln(x)) = \frac{1}{2} \ln\left(\frac{x+1}{x}\right)$$

$$[\triangle BCC'B'] = \frac{1}{2} (\ln(x+2) - \ln(x+1)) + \ln(x+1) - \ln(x) = \frac{1}{2} \ln\left(\frac{x+2}{x+1}\right) + \ln\left(\frac{x+1}{x}\right)$$

$$[\triangle CDD'C'] = \frac{1}{2} (\ln(x+3) - \ln(x+2)) + \ln(x+2) - \ln(x) = \frac{1}{2} \ln\left(\frac{x+3}{x+2}\right) + \ln\left(\frac{x+2}{x}\right)$$

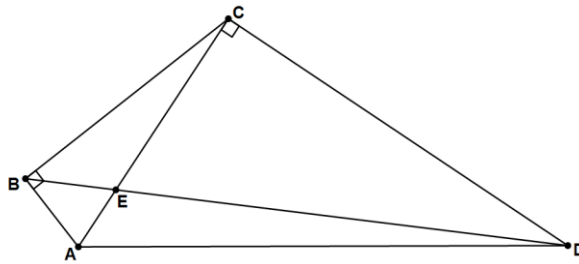
$$\begin{aligned}
[ABCD] &= [\triangle ABB'] + [BCC'B'] + [CDD'C'] - [\triangle ADD'] = \\
&= \frac{1}{2} \ln\left(\frac{x+1}{x}\right) + \frac{1}{2} \ln\left(\frac{x+2}{x+1}\right) + \ln\left(\frac{x+1}{x}\right) + \frac{1}{2} \ln\left(\frac{x+3}{x+2}\right) + \ln\left(\frac{x+2}{x}\right) - \frac{3}{2} \ln\left(\frac{x+3}{x}\right) \Rightarrow \\
2[ABCD] &= \ln\left(\frac{x+1}{x}\right) + \ln\left(\frac{x+2}{x+1}\right) + 2 \ln\left(\frac{x+1}{x}\right) + \ln\left(\frac{x+3}{x+2}\right) + 2 \ln\left(\frac{x+2}{x}\right) - 3 \ln\left(\frac{x+3}{x}\right) = \\
&= \ln\left(\frac{x+1}{x} \cdot \frac{x+2}{x+1} \cdot \left(\frac{x+1}{x}\right)^2 \cdot \frac{x+3}{x+2} \cdot \left(\frac{x+2}{x}\right)^2\right) - 3 \ln\left(\frac{x+3}{x}\right) \\
&= \ln\left(\frac{(x+1)^2(x+2)^2(x+3)}{x^3}\right) - \ln\left(\frac{(x+3)^3}{x^3}\right) = \\
&= \ln\left(\frac{(x+1)^2(x+2)^2(x+3)}{x^3} \cdot \frac{x^3}{(x+3)^3}\right) = \ln\left(\frac{(x+1)^2(x+2)^2}{(x+3)^2}\right) = 2 \ln\left(\frac{(x+1)(x+2)}{x+3}\right) \Rightarrow \\
\ln\left(\frac{(x+1)(x+2)}{x+3}\right) &= \ln \frac{91}{90} \Rightarrow \frac{(x+1)(x+2)}{x+3} = \frac{91}{90} = \frac{7 \cdot 13}{90}
\end{aligned}$$

$$\frac{(x+1)(x+2)}{x+3} = \frac{91}{90} \Leftrightarrow x = \begin{cases} -15 \\ 12 \end{cases}$$

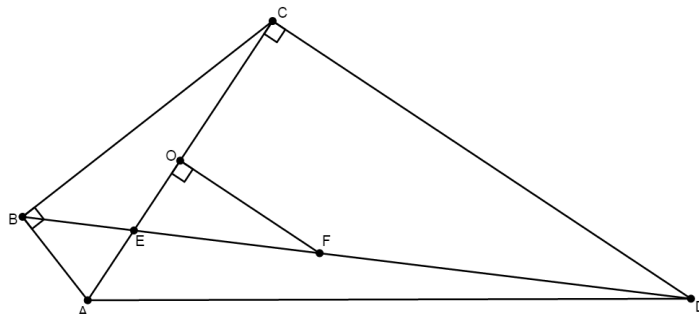
Descartando el valor negativo, llegamos finalmente a $x=12$ (D).

Nota: Al disponer de cinco posibles soluciones, podemos saltarnos la resolución de esta ecuación comprobando las cinco posibles respuestas correctas y determinando la que es aceptable.

18



Sea O el punto medio de la diagonal $\overline{AC}$. Trazamos la perpendicular a AC por O, que cortará la diagonal $\overline{BD}$ en el punto F.



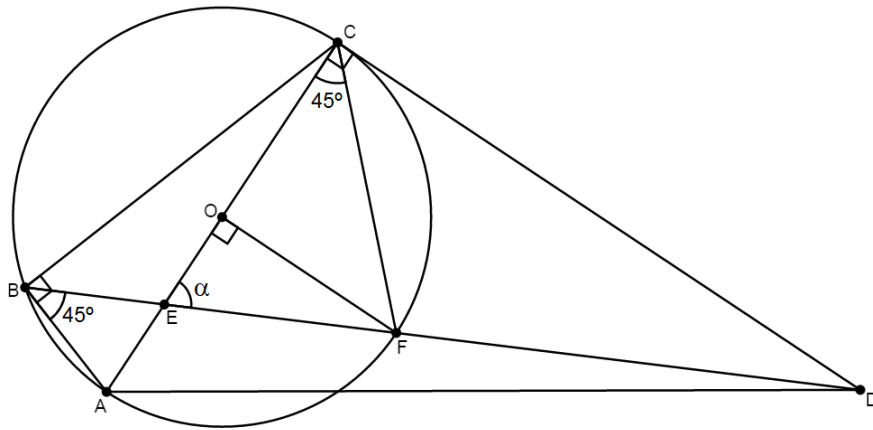
$$\left. \begin{array}{l} AC = 20 \Rightarrow AO = OC = 10 \\ AE = 5 \Rightarrow EO = 5 \end{array} \right\} \Rightarrow EC = EO + OC = 5 + 10 = 15$$

Los triángulos $\triangle ECD$ y $\triangle EOF$ son semejantes pues $OF \parallel CD$, luego

$$\frac{CD}{OF} = \frac{OE}{OF} \Leftrightarrow \frac{30}{OF} = \frac{15}{5} = 3 \Rightarrow OF = 10$$

Vemos que $OF = OC = AO$, luego $\triangle COF$ es un triángulo rectángulo isósceles, y por tanto $\angle OCF = \angle OAF = 45^\circ$

La circunferencia de diámetro AC contiene el punto B, pues $\angle ABC = 90^\circ$, y también el punto F, pues $OF = OC = AO$.



Por Pitágoras, $DE = \sqrt{15^2 + 30^2} = \sqrt{1125}$

$$\angle ABD = \angle ACF = 45^\circ$$

$$\alpha = \angle CEF = \angle BEA, \sin \alpha = \frac{CD}{ED} = \frac{30}{\sqrt{1125}}$$

Aplicando el teorema del seno en $\triangle ABE$:

$$\frac{\sin 45^\circ}{AE} = \frac{\sin \alpha}{AB} \Leftrightarrow \frac{1/\sqrt{2}}{5} = \frac{30/\sqrt{1125}}{AB} \Rightarrow AB = \frac{5 \cdot 30 / \sqrt{1125}}{1/\sqrt{2}} = \frac{5 \cdot 30 \sqrt{2}}{\sqrt{1125}} = 2\sqrt{10}$$

Por Pitágoras:

$$\begin{aligned} BC^2 &= AC^2 - AB^2 = 20^2 - \left(\frac{5 \cdot 30 \sqrt{2}}{\sqrt{1125}} \right)^2 = 20^2 - \frac{150^2 \cdot 2}{1125} = 20^2 - 40 = 360 \\ \Rightarrow BC &= \sqrt{360} = 6\sqrt{10} \end{aligned}$$

Finalmente:

$$[ABCD] = [\triangle ABC] + [\triangle ACD] = \frac{2\sqrt{10} \cdot 6\sqrt{10}}{2} + \frac{20 \cdot 30}{2} = 6 \cdot 10 + 300 = 360 \quad (D)$$

19

Teniendo en cuenta que $289 = 17^2$, hacemos el cambio de variable $u = 2^{17}$, y por tanto

$$\frac{2^{289} + 1}{2^{17} + 1} = \frac{u^{17} + 1}{u + 1}$$

Realizamos la división sintética:

$$\begin{aligned} \frac{u^{17} + 1}{u + 1} &= u^{16} - u^{15} + u^{14} - u^{13} + \dots + u^2 - u + 1 = \\ &= u^{15}(u - 1) + u^{13}(u - 1) + u^{11}(u - 1) + \dots + u(u - 1) + 1 = \\ &= (u - 1)(u^{15} + u^{13} + \dots + u) + 1 \end{aligned}$$

Ahora observamos que tenemos una suma de una sucesión geométrica de razón 2:

$$u - 1 = 2^{17} - 1 = 2^{16} + 2^{15} + \dots + 2 + 1$$

Luego, finalmente:

$$\frac{2^{289} + 1}{2^{17} + 1} = (2^{16} + 2^{15} + \dots + 2^1 + 2^0) (2^{17 \cdot 15} + 2^{17 \cdot 13} + 2^{17 \cdot 11} + \dots + 2^{17 \cdot 3} + 2^{17 \cdot 1}) + 2^0$$

Es una suma de $17 \cdot 8 + 1 = 137$ potencias de 2 (C)

20

Denotamos las rotaciones por $R = \{R_{90}, R_{180}, R_{270}\}$, y denotamos las simetrías por $S = \{S_x, S_y\}$. Sea I la transformación identidad, es decir, la que deja el triángulo invariante. Podemos estructurar nuestro estudio en función del número de simetrías que contiene. Sin simetrías, es decir, solo con rotaciones, solo hay la siguiente (y las seis permutaciones correspondientes):

$$R_{90} \circ R_{180} \circ R_{90} = I$$

pues entre las tres deben sumar 360° , y esto solo se consigue como $90 + 180 + 90$.

Con una sola simetría no hay ninguna, pues las simetrías cambian la orientación y las rotaciones no.

Con dos simetrías.

$S_x \circ S_y$ equivale a una rotación R_{180} , luego tenemos

$$S_x \circ S_y \circ R_{180} = I$$

Y sus seis permutaciones correspondientes.

Dos permutaciones iguales equivalen a la identidad, y por tanto no podemos añadir una tercera que deje el triángulo invariante.

Con tres simetrías.

Cada simetría cambia la orientación del triángulo, luego con tres simetrías el triángulo ha cambiado la orientación y por lo tanto no puede quedar igual.

Por lo tanto, finalmente, solo hay $6 + 6 = 12$ (A).

21

Queremos determinar los enteros positivos n cumpliendo $[5!, n] = 5(10!, n)$ con $5 | n$
 $[5!, n] = 5(10!, n) \Leftrightarrow [5 \cdot 3 \cdot 2^3, n] = 5(7 \cdot 5^2 \cdot 3^4 \cdot 2^8, n)$

Aplicaremos AR/4.30:

Sea $n = 2^{a_2} 3^{a_3} 5^{a_5} 7^{a_7} p_8^{a_8} p_9^{a_9} \dots p_k^{a_k}$ la descomposición factorial de n , con $7 < p_8, p_9, \dots, p_k$,
 cumpliendo $a_2, a_3, a_7 \geq 0$ y $a_5 \geq 1$ pues $5 | n$.

$$5(7 \cdot 5^2 \cdot 3^4 \cdot 2^8, n) = 2^{b_2} 3^{b_3} 5^{b_5+1} 7^{b_7} \text{ con } b_2 = \min(a_2, 8), b_3 = \min(a_3, 4), b_5 = \min(a_5, 2) \text{ y } b_7 = \min(a_7, 1)$$

Por otro lado,

$$[5 \cdot 3 \cdot 2^3, n] = 2^{c_2} 3^{c_3} 5^{c_5} 7^{c_7} p_8^{a_8} p_9^{a_9} \dots p_k^{a_k} \text{ con } c_2 = \max(a_2, 3), c_3 = \max(a_3, 1), c_5 = \max(a_5, 1), c_7 = \max(a_7, 0)$$

Y de la igualdad $[5 \cdot 3 \cdot 2^3, n] = 5(7 \cdot 5^2 \cdot 3^4 \cdot 2^8, n)$ deducimos que $a_8 = a_9 = \dots = a_k = 0$

Y también se deduce que:

$$\min(a_2, 8) = \max(a_2, 3) \Leftrightarrow 3 \leq a_2 \leq 8$$

$$\min(a_3, 4) = \max(a_3, 1) \Leftrightarrow 1 \leq a_3 \leq 4$$

$$\min(a_5, 2) + 1 = \max(a_5, 1) \Leftrightarrow a_5 = 3$$

$$\min(a_7, 1) = \max(a_7, 0) \Leftrightarrow 0 \leq a_7 \leq 1$$

En total hay $6 \cdot 4 \cdot 1 \cdot 2 = 48$ (D)

22

Primera versión. Mediante una serie geométrica compleja.

Elevamos al cuadrado la igualdad del enunciado:

$$(2+i)^n = a_n + b_n i \Rightarrow (2+i)^{2n} = (a_n + b_n i)^2 = a_n^2 - b_n^2 + 2a_n b_n i \Rightarrow$$

$$a_n b_n = \frac{1}{2} \operatorname{Im}((2+i)^{2n}) = \frac{1}{2} \operatorname{Im}(((2+i)^2)^n) = \frac{1}{2} \operatorname{Im}((3+4i)^n)$$

Y por tanto

$$\sum_{n=0}^{\infty} \frac{a_n b_n}{7^n} = \sum_{n=0}^{\infty} \frac{\frac{1}{2} \operatorname{Im}((3+4i)^n)}{7^n} = \frac{1}{2} \operatorname{Im} \left(\sum_{n=0}^{\infty} \frac{(3+4i)^n}{7^n} \right) = \frac{1}{2} \operatorname{Im} \left(\frac{1}{1 - \frac{3+4i}{7}} \right) = \frac{7}{16} \quad (B)$$

Segunda versión. Aplicando la fórmula de Moire y serie geométrica compleja.

Aplicamos la fórmula de Moire:

$$a = r(\cos \alpha + \sin \alpha i) \Rightarrow a^n = (r^n)(\cos(n\alpha) + \sin(n\alpha)i)$$

En nuestro caso:

$$r = \sqrt{2^2 + 1^2} = \sqrt{5}$$

$$\alpha = \arctan\left(\frac{1}{2}\right)$$

$$\begin{aligned}
(2+i)^n &= (\sqrt{5})^n (\cos(n\alpha) + \sin(n\alpha)i) \Rightarrow \\
\Rightarrow a_n b_n &= (\sqrt{5})^n (\sqrt{5})^n \cos(n\alpha) \cdot \sin(n\alpha) = 5^n \sin(2n\alpha) \\
\Rightarrow \frac{a_n b_n}{7^n} &= \frac{1}{2} \left(\frac{5}{7}\right)^n \sin(2n\alpha)
\end{aligned}$$

En donde hemos aplicado la identidad trigonométrica $\sin(2x) = 2\sin(x)\cos(x)$

Ahora observamos que $\sin(2n\alpha) = \text{Im}(e^{2in\alpha})$

$$\text{Con lo cual } \sum_{n=0}^{\infty} \frac{a_n b_n}{7^n} = \sum_{n=0}^{\infty} \frac{1}{2} \left(\frac{5}{7}\right)^n \sin(2n\alpha) = \frac{1}{2} \text{Im} \left(\sum_{n=0}^{\infty} \left(\frac{5}{7}\right)^n e^{2in\alpha} \right)$$

En donde $\sum_{n=0}^{\infty} \left(\frac{5}{7}\right)^n e^{2in\alpha}$ es una serie compleja convergente igual a $\frac{1}{1 - \frac{5}{7}e^{2\alpha i}}$

$$\sin(2\alpha) = 2\cos(\alpha)\sin(\alpha) = 2 \frac{2}{\sqrt{5}} \frac{1}{\sqrt{5}} = \frac{4}{5}$$

$$\cos(2\alpha) = \cos^2(\alpha) - \sin^2(\alpha) = \frac{4}{5} - \frac{1}{5} = \frac{3}{5}$$

$$e^{2\alpha i} = \cos(2\alpha) + \sin(2\alpha)i = \frac{3}{5} + \frac{4}{5}i$$

$$\frac{1}{1 - \frac{5}{7}e^{2\alpha i}} = \frac{1}{1 - \frac{5}{7}\left(\frac{3}{5} + \frac{4}{5}i\right)} = \frac{7}{8} + \frac{7}{8}i, \text{ cuya parte imaginaria es } \frac{7}{8}.$$

$$\text{Finalmente, la soluci3n es } \frac{1}{2} \frac{7}{8} = \frac{7}{16} \quad (B)$$

Fuente de estas dos versiones: www.artofproblemsolving.com

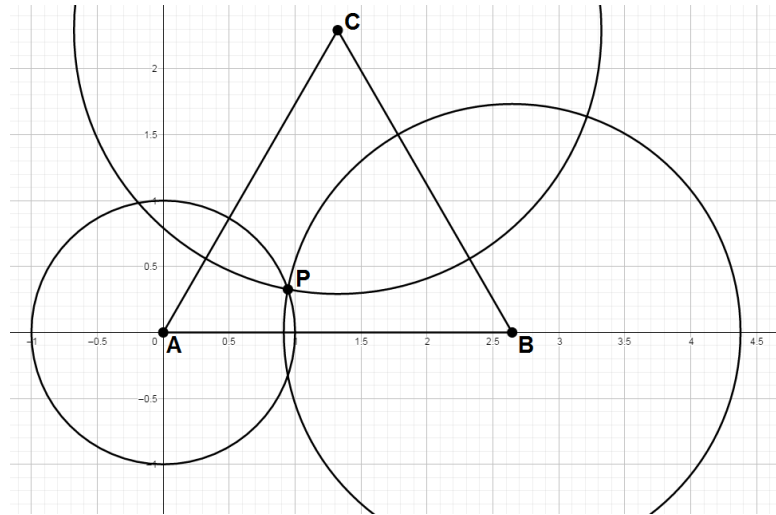
23

Soluci3n: https://artofproblemsolving.com/wiki/index.php/2020_AMC_12A_Problems/Problem_23

24

Nota: Vamos a resolver este problema mediante geometría cartesiana, y resolviendo el sistema de ecuaciones que aparece implícito en el enunciado. Esta estrategia no es en absoluto la más adecuada en el contexto de limitación de tiempo de una competición matemática, en las soluciones oficiales aparecen indicados planteamientos mejores y más elegantes.

Sea $A = (0,0)$, $B = (s,0)$ y $c = \left(\frac{s}{2}, \frac{\sqrt{3}s}{2}\right)$. Sea $P = (x,y)$.



Nuestro problema consiste en resolver el siguiente sistema en x, y, s

$$\begin{cases} x^2 + y^2 = 1^2 \\ (x-s)^2 + y^2 = (\sqrt{3})^2 \\ \left(x - \frac{s}{2}\right)^2 + \left(y - \frac{\sqrt{3}s}{2}\right)^2 = 2^2 \end{cases}$$

Sabiendo, además, que la solución existe y es única.

$$\begin{aligned} (x-s)^2 + y^2 &= (\sqrt{3})^2 \Leftrightarrow x^2 - 2xs + s^2 + y^2 = 3 \Leftrightarrow -2xs + s^2 + 1 = 3 \Leftrightarrow \\ s^2 - 2xs &= 2 \Leftrightarrow s(s - 2x) = 2 \end{aligned}$$

$$\begin{aligned} \left(x - \frac{s}{2}\right)^2 + \left(y - \frac{\sqrt{3}s}{2}\right)^2 &= 2^2 \Leftrightarrow x^2 + \frac{s^2}{4} - 2x\frac{s}{2} + y^2 + \frac{3s^2}{4} - 2y\frac{\sqrt{3}s}{2} = 4 \Leftrightarrow \\ 1 + \frac{s^2}{4} - xs + \frac{3s^2}{4} - \sqrt{3}ys &= 4 \Leftrightarrow s^2 - xs - \sqrt{3}ys = 3 \Leftrightarrow s(s - x - \sqrt{3}y) = 3 \end{aligned}$$

Llegamos al sistema:

$$\begin{cases} x^2 + y^2 = 1 \\ s(s - 2x) = 2 \\ s(s - x - \sqrt{3}y) = 3 \end{cases} \Leftrightarrow \begin{cases} x^2 + y^2 = 1 \\ s^2 - 2xs = 2 \Rightarrow s^2 = 2 + 2xs \Rightarrow 2 + 2xs - xs - \sqrt{3}ys = 3 \\ s^2 - xs - \sqrt{3}ys = 3 \end{cases}$$

$$\Rightarrow 2 + xs - \sqrt{3}ys = 3 \Rightarrow xs - \sqrt{3}ys = 1 \Rightarrow s(x - \sqrt{3}y) = 1 \Rightarrow s = \frac{1}{x - \sqrt{3}y}$$

$$\begin{aligned}
2 &= \frac{1}{x - \sqrt{3}y} \left(\frac{1}{x - \sqrt{3}y} - 2x \right) = \frac{1}{(x - \sqrt{3}y)^2} - \frac{2x}{x - \sqrt{3}y} = \frac{1 - 2x(x - \sqrt{3}y)}{(x - \sqrt{3}y)^2} = \\
&= \frac{1 - 2x^2 + 2\sqrt{3}xy}{x^2 + 3y^2 - 2\sqrt{3}xy} = \frac{1 - 2(1 - y^2) + 2\sqrt{3}xy}{1 - y^2 + 3y^2 - 2\sqrt{3}xy} = \frac{-1 + 2y^2 + 2\sqrt{3}xy}{1 + 2y^2 - 2\sqrt{3}xy} \Leftrightarrow \\
2(1 + 2y^2 - 2\sqrt{3}xy) &= -1 + 2y^2 + 2\sqrt{3}xy \Leftrightarrow 2 + 4y^2 - 4\sqrt{3}xy = -1 + 2y^2 + 2\sqrt{3}xy \Leftrightarrow \\
3 + 2y^2 - 6\sqrt{3}xy &= 0
\end{aligned}$$

Sustituyendo el valor $x = \sqrt{1 - y^2}$

$$3 + 2y^2 - 6\sqrt{3}\sqrt{1 - y^2}y = 0 \Leftrightarrow 3 + 2y^2 - \sqrt{3 \cdot 6^2(1 - y^2)}y^2 = 0 \Leftrightarrow$$

$$3 + 2y^2 = \sqrt{3 \cdot 6^2(1 - y^2)}y^2 \Leftrightarrow (3 + 2y^2)^2 = 3 \cdot 6^2(1 - y^2)y^2 \Leftrightarrow$$

$$9 + 4y^4 + 12y^2 = 108y^2 - 108y^4 \Leftrightarrow$$

$$0 = 108y^2 - 108y^4 - 9 - 4y^4 - 12y^2 = 96y^2 - 112y^4 - 9$$

Llegando a una ecuación bicuadrada, que resolvemos por el método habitual:

$$\begin{aligned}
y^2 &= \frac{-96 \pm \sqrt{96^2 - 4 \cdot (-112) \cdot (-9)}}{2 \cdot (-112)} = \frac{-96 \pm \sqrt{9216 - 4032}}{-224} = \frac{96 \pm \sqrt{5184}}{224} = \\
&= \frac{96 \pm \sqrt{2^6 \cdot 3^4}}{224} = \frac{96 \pm 2^3 \cdot 3^2}{224} = \frac{96 \pm 72}{224} = \begin{cases} 3/4 \\ 3/28 \end{cases}
\end{aligned}$$

Puesto que en todo momento estamos suponiendo $x, y \geq 0$, llegamos a los siguientes resultados posibles:

$$y^2 = \frac{3}{4} \Rightarrow x^2 = 1 - y^2 = \frac{1}{4} \Rightarrow x = \frac{1}{2}, y = \frac{\sqrt{3}}{2} \Rightarrow$$

$$\Rightarrow x - \sqrt{3}y = \frac{1}{2} - \sqrt{3} \frac{\sqrt{3}}{2} = \frac{1}{2} - \frac{3}{2} = -1 < 0 \Rightarrow s = \frac{1}{x - \sqrt{3}y} < 0 \text{ absurdo.}$$

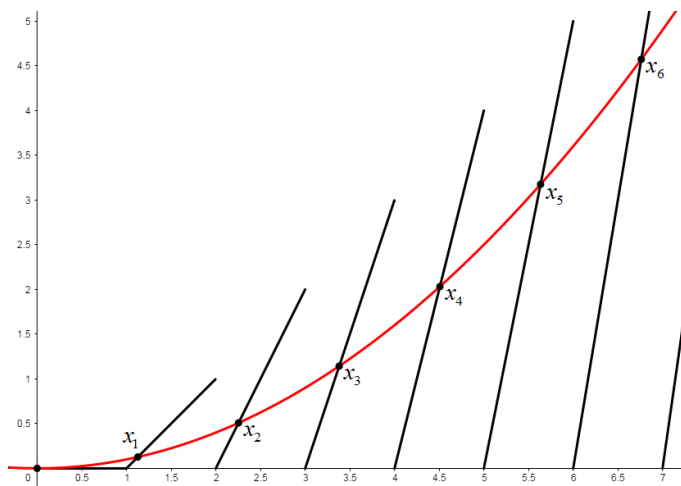
$$y^2 = \frac{3}{28} \Rightarrow x^2 = 1 - y^2 = \frac{25}{28} \Rightarrow x = \frac{5}{2\sqrt{7}}, y = \frac{\sqrt{3}}{2\sqrt{7}} \Rightarrow$$

$$\Rightarrow x - \sqrt{3}y = \frac{5}{2\sqrt{7}} - \sqrt{3} \frac{\sqrt{3}}{2\sqrt{7}} = \frac{5}{2\sqrt{7}} - \frac{3}{2\sqrt{7}} = \frac{2}{2\sqrt{7}} = \frac{1}{\sqrt{7}} \Rightarrow$$

$$s = \frac{1}{x - \sqrt{3}y} = \sqrt{7} \quad (B)$$

25

Estudiando la función $f(x) = \lfloor x \rfloor \cdot \{x\}$ vemos que es una función definida a trozos, y en cada intervalo $[n, n+1)$ es una función lineal entre 0 y n . Luego se cortará con la gráfica de ax^2 en los puntos $k \leq x_k \leq k+1$



Sabemos que suman 420:

$$420 = x_1 + x_2 + \dots + x_n \geq 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

Con $n = 27 \Rightarrow \frac{n(n+1)}{2} = 378$ y no podríamos llegar a 420.

Con $n = 29 \Rightarrow \frac{n(n+1)}{2} = 435$ y nos pasamos, luego seguro que $n = 28$.

Por otro lado, sabemos que $\lfloor kx \rfloor = k \lfloor x \rfloor$ para todo entero k , luego

$$\{kx\} = kx - \lfloor kx \rfloor = kx - k \lfloor x \rfloor = k(x - \lfloor x \rfloor) = k\{x\}$$

Y por tanto, si x es solución de $\lfloor x \rfloor \cdot \{x\} = a \cdot x^2$, también lo es kx :

$$\lfloor kx \rfloor \cdot \{kx\} = k \cdot k \lfloor x \rfloor \cdot \{x\} = k^2 \lfloor x \rfloor \cdot \{x\} = k^2 a \cdot (x)^2 = a \cdot (kx)^2$$

Con todo esto podemos suponer que $x_k = k \cdot x_1$ para cierto $1 \leq x_1 \leq 2$, y por tanto

$$\begin{aligned} 420 &= x_1 + x_2 + \dots + x_{28} = x_1 + 2x_1 + 3x_1 + \dots + 28x_1 = x_1(1 + 2 + 3 + 4 + \dots + 28) = \\ &= x_1 \cdot 406 \Rightarrow x_1 = \frac{420}{406} = \frac{30}{29} \end{aligned}$$

Y por tanto, sustituyendo en la ecuación:

$$\begin{aligned} \left\lfloor \frac{30}{29} \right\rfloor \cdot \left\{ \frac{30}{29} \right\} &= a \cdot \left(\frac{30}{29} \right)^2 \Rightarrow 1 \cdot \left\{ \frac{30}{29} \right\} = a \cdot \left(\frac{30}{29} \right)^2 \Rightarrow \frac{30}{29} - 1 = a \cdot \left(\frac{30}{29} \right)^2 \\ \Rightarrow \frac{1}{29} &= a \cdot \left(\frac{30}{29} \right)^2 \Rightarrow \frac{1}{29} = a \cdot \frac{30^2}{29^2} \Rightarrow a = \frac{29^2}{30^2 \cdot 29} = \frac{29}{30^2} = \frac{29}{900} \Rightarrow 29 + 900 = 929 \quad (C) \end{aligned}$$

AMC12B 2020 Enunciados

1

Determina el valor simplificado de la siguiente expresión:

$$\sqrt{1} + \sqrt{1+3} + \sqrt{1+3+5} + \sqrt{1+3+5+7}$$

(A) 5 (B) $4 + \sqrt{7} + \sqrt{10}$ (C) 10 (D) 15 (E) $4 + 3\sqrt{3} + 2\sqrt{5} + \sqrt{7}$

2

Determina el valor de la siguiente expresión:

$$\frac{100^2 - 7^2}{70^2 - 11^2} \cdot \frac{(70-11)(70+11)}{(100-7)(100+7)}$$

(A) 1 (B) $\frac{9951}{9950}$ (C) $\frac{4780}{4779}$ (D) $\frac{108}{107}$ (E) $\frac{81}{80}$

3

La razón de w a x es 4:3, la razón de y a z es 3:2, y la razón de z a x es 1:6. ¿Cuál es la razón de w a y ?

(A) 4:3 (B) 3:2 (C) 8:3 (D) 4:1 (E) 16:3

4

Sean a° y b° los ángulos agudos de un triángulo rectángulo, con $a > b$ y ambos números primos. ¿Cuál es el menor valor posible de b ?

(A) 2 (B) 3 (C) 5 (D) 7 (E) 11

5

Los equipos A y B juegan en una liga de baloncesto. Cada partido se gana o se pierde, no hay empates. El equipo A ha ganado los $\frac{2}{3}$ de los partidos que ha disputado, y el equipo B ha ganado las $\frac{5}{8}$ partes de los partidos que ha disputado. Además, el equipo B ha ganado 7 partidos más y ha perdido 7 partidos más que A. ¿Cuántos partidos ha disputado el equipo A?

(A) 21 (B) 27 (C) 42 (D) 48 (E) 63

6

Para todos los enteros $n \geq 9$, el valor de

$$\frac{(n+2)! - (n+1)!}{n!}$$

(A) es un múltiplo de 4 (B) es un múltiplo de 10 (C) es un número primo
(D) es un cuadrado perfecto (E) es un cubo perfecto.

7

Dos rectas ni horizontales ni verticales en el plano cartesiano se cortan formando un ángulo de 45° . Una de ellas tiene una pendiente igual a 6 veces la pendiente de la otra. ¿Cuál es el mayor valor posible del producto de las pendientes de dichas rectas?

- (A) $1/6$ (B) $2/3$ (C) $3/2$ (D) 3 (E) 6

8

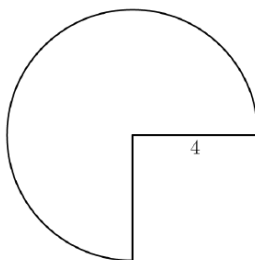
Determina el número de pares ordenados (x, y) de enteros que satisfacen la ecuación

$$x^{2020} + y^2 = 2y$$

- (A) 1 (B) 2 (C) 3 (D) 4 (E) hay infinitos pares

9

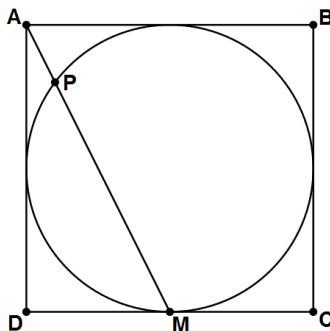
Mediante un sector circular de tres cuartos de círculo de radio 4 construimos la superficie lateral de un cono recto juntando sus dos lados rectos. ¿Cuál es el volumen del cono generado?



- (A) $3\pi\sqrt{5}$ (B) $4\pi\sqrt{3}$ (C) $3\pi\sqrt{7}$ (D) $6\pi\sqrt{3}$ (E) $6\pi\sqrt{7}$

10

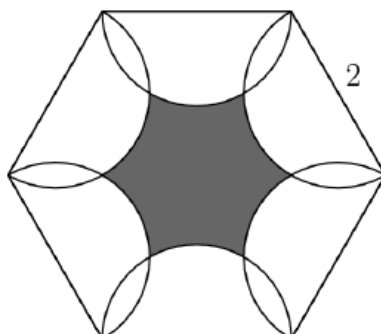
En un cuadrado ABCD de lado unidad, trazamos la circunferencia inscrita ω que toca el lado $\overline{CD}$ en M, y sea $P \neq M$ el segundo punto de corte entre ω y $\overline{AM}$. Determina la longitud AP .



- (A) $\frac{\sqrt{5}}{12}$ (B) $\frac{\sqrt{5}}{10}$ (C) $\frac{\sqrt{5}}{9}$ (D) $\frac{\sqrt{5}}{8}$ (E) $\frac{2\sqrt{5}}{15}$

11

La siguiente figura representa seis semicírculos en el interior de un hexágono regular de lado 2, de forma que los diámetros de los semicírculos coinciden con los lados del hexágono. Determina el área de la región sombreada (el interior del hexágono pero fuera de los semicírculos).



- (A) $6\sqrt{3} - 3\pi$ (B) $\frac{9\sqrt{3}}{2} - 2\pi$ (C) $\frac{3\sqrt{3}}{2} - \frac{\pi}{3}$ (D) $3\sqrt{3} - \pi$ (E) $\frac{9\sqrt{3}}{2} - \pi$

12

Sea $\overline{AB}$ el diámetro de una circunferencia de radio $5\sqrt{2}$. Sea $\overline{CD}$ una cuerda que corta $\overline{AB}$ en el punto E tal que $BE = 2\sqrt{5}$ y $\angle AEC = 45^\circ$. Determina $CE^2 + DE^2$.

- (A) 96 (B) 98 (C) $44\sqrt{5}$ (D) $70\sqrt{2}$ (E) 100

13

Determina el valor de $\sqrt{\log_2 6 + \log_3 6}$.

- (A) 1 (B) $\sqrt{\log_5 6}$ (C) 2 (D) $\sqrt{\log_2 3} + \sqrt{\log_3 2}$ (E) $\sqrt{\log_2 6} + \sqrt{\log_3 6}$

14

Bela y Jenn juegan al siguiente juego en el intervalo cerrado $[0, n]$ de la recta real, donde n es un entero fijo mayor que 4. Se van turnando, y empieza Bela. En su primer turno, Bela escoge un número real en el intervalo $[0, n]$, y después cada jugador, en su turno, debe escoger un número real que esté a más de una unidad de distancia de los números escogidos previamente por cualquiera de los dos jugadores. Pierde el primer jugador que no pueda escoger ninguno. Utilizando siempre estrategias optimizadoras, ¿Qué jugador ganará la partida?

- (A) Bela siempre ganará (B) Jenn siempre ganará (C) Bela ganará si y solo si n es par
(D) Jenn ganará si y solo si n es par (E) Jenn ganará si y solo si $n > 8$.

15

Tenemos diez personas sentadas alrededor de un círculo. Cada persona conoce exactamente otras tres del resto: Las dos personas que se sientan a sus lados y la persona que está en la posición diametralmente opuesta. ¿De cuántas formas podemos agrupar estas 10 personas en 5 parejas, de forma que en cada pareja se conozcan entre ambos?

(A) 11 (B) 12 (C) 13 (D) 14 (E) 15

16

Una urna contiene una bola roja y una bola azul. Disponemos, además, de una caja con un montón de bolas rojas y azules. George realiza la siguiente operación cuatro veces: Toma aleatoriamente una bola de la urna, coge una bola del mismo color de la caja y deposita estas dos bolas a la urna. Después de cuatro veces, en la urna hay seis bolas. ¿Cuál es la probabilidad que contenga tres bolas de cada color?

(A) $1/6$ (B) $1/5$ (C) $1/4$ (D) $1/3$ (E) $1/2$

17

Determina el número de polinomios de la forma $x^5 + ax^4 + bx^3 + cx^2 + dx + 2020$

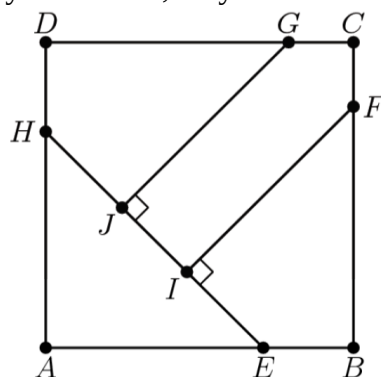
donde a, b, c, d son números reales, que cumplan la siguiente propiedad:

Si r es una raíz, también lo es $\frac{-1+i\sqrt{3}}{2} \cdot r$, en donde $i = \sqrt{-1}$.

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

18

En un cuadrado ABCD, los puntos E y H pertenecen a $\overline{AB}$ y $\overline{DA}$, respectivamente, y $AE = AH$. Los puntos F y G pertenecen a $\overline{BC}$ y $\overline{CD}$, y los puntos I y J pertenecen a $\overline{EH}$ de forma que $\overline{FI} \perp \overline{EH}$ y $\overline{GJ} \perp \overline{EH}$, tal y como se muestra en la figura:



El triángulo $\triangle AEH$, los cuadriláteros BFIE y DHJG y el pentágono FCGJI todos tienen área 1. Determina FI^2 .

(A) $\frac{7}{3}$ (B) $8 - 4\sqrt{2}$ (C) $1 + \sqrt{2}$ (D) $\frac{7}{4}\sqrt{2}$ (E) $2\sqrt{2}$

19

Sea ABCD el cuadrado del plano cartesiano de vértices A(1,1), B(-1,1), C(-1,-1) y D(1,-1). Consideremos las siguientes cuatro transformaciones: L, una rotación de 90° en el sentido contrario de las agujas del reloj alrededor del origen; R, la rotación de 90° en el sentido de las agujas del reloj alrededor del origen; H, la simetría respecto del eje X, y V, la simetría respecto del eje Y.

Cada una de estas cuatro transformaciones convierte el cuadrado en sí mismo, pero las posiciones de los vértices cambian. Por ejemplo, aplicando R y después V se envía el vértice A a (-1,-1), y se envía el vértice B a (-1,1). Determina el número de secuencias de 20 transformaciones tomadas de $\{L, R, H, V\}$ que envíen los vértices a sus posiciones originales. (Por ejemplo, R, R, V, H es una secuencia de 4 transformaciones que envían los vértices a sus posiciones originales).

(A) 2^{37} (B) $3 \cdot 2^{36}$ (C) 2^{38} (D) $3 \cdot 2^{37}$ (E) 2^{39}

20

Se pintan dos cubos del mismo tamaño, cada cara blanca o negra, de forma aleatoria e independiente. ¿Cuál es la probabilidad de que, una vez pintados los cubos, estos puedan ser rotados para ser idénticos en apariencia?

(A) $9/64$ (B) $289/2048$ (C) $73/512$ (D) $147/1024$ (E) $589/4096$

21

Determina el número de posibles enteros n que satisfacen $\frac{n+1000}{70} = \lfloor \sqrt{n} \rfloor$

(recuerda que $\lfloor x \rfloor$ es el mayor entero que no excede x)

(A) 2 (B) 4 (C) 6 (D) 30 (E) 32

22

Determina el valor máximo de la expresión $\frac{(2^t - 3t)t}{4^t}$ con t real.

(A) $\frac{1}{16}$ (B) $\frac{1}{15}$ (C) $\frac{1}{12}$ (D) $\frac{1}{10}$ (E) $\frac{1}{9}$

23

Determina la cantidad de enteros $n \geq 2$ para los cuales se cumple la siguiente condición: Dados $z_1, z_2, \dots, z_n$ números enteros cumpliendo

$$|z_1| = |z_2| = \dots = |z_n| = 1 \text{ y } z_1 + z_2 + \dots + z_n = 0$$

Entonces $z_1, z_2, \dots, z_n$ están en posiciones equiespaciadas en la circunferencia unidad del plano complejo.

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

24

Denotamos por $D(n)$ el número de formas de escribir un entero positivo n como producto

$$n = f_1 \cdot f_2 \cdot \dots \cdot f_k$$

donde $k \geq 1$, los f_i son enteros estrictamente mayores que 1 y el orden de los factores sí importa (es decir, dos representaciones que difieren solo en el orden se consideran diferentes). Por ejemplo, el número 6 se puede escribir como 6, $2 \cdot 3$ y $3 \cdot 2$, y por tanto $D(6) = 3$.

Determina $D(96)$.

- (A) 112 (B) 128 (C) 144 (D) 172 (E) 184

25

Fijado un número real a , con $0 \leq a \leq 1$, se toman aleatoriamente números reales x, y en los intervalos $[0, a]$ y $[0, 1]$, respectivamente, y sea $P(a)$ la probabilidad de que

$$\sin^2(\pi x) + \sin^2(\pi y) > 1$$

Determina el valor máximo de $P(a)$.

- (A) $\frac{7}{12}$ (B) $2 - \sqrt{2}$ (C) $\frac{1 + \sqrt{2}}{4}$ (D) $\frac{\sqrt{5} - 1}{2}$ (E) $\frac{5}{8}$

AMC12B 2020 Soluciones (Letra)

1. C
2. A
3. E
4. D
5. C
6. D
7. C
8. D
9. C
10. B
11. D
12. E
13. D
14. A
15. C
16. B
17. C
18. B
19. C
20. D
21. C
22. C
23. B
24. A
25. B

AMC12B 2020 Soluciones desarrolladas

1

$$\sqrt{1} + \sqrt{1+3} + \sqrt{1+3+5} + \sqrt{1+3+5+7} = 1+2+3+4=10 \text{ (C)}$$

2

Aplicando la igualdad “Diferencia de cuadrados”:

$$100^2 - 7^2 = (100 - 7)(100 + 7)$$

$$70^2 - 11^2 = (70 - 11)(70 + 11)$$

Está claro que el resultado es 1 (A)

3

$$\left. \begin{array}{l} 4w = 3x \Rightarrow 8w = 6x \\ 1z = 6x \end{array} \right\} \Rightarrow 8w = 1z, \quad \left. \begin{array}{l} 8w = 1z \Rightarrow 16w = 2z \\ 3y = 2z \end{array} \right\} \Rightarrow 16w = 3y \Rightarrow 16:3 \text{ (E)}$$

4

$$a = 2 \Rightarrow b = 88 \text{ y no es primo.}$$

$$a = 3 \Rightarrow b = 87 \text{ y no es primo.}$$

$$a = 5 \Rightarrow b = 85 \text{ y no es primo.}$$

$$a = 7 \Rightarrow b = 83 \text{ y sí es primo. (D)}$$

5

Sean A el número de partidos disputados por el equipo A y sea B el número de partidos disputados por el equipo B.

$$\left\{ \begin{array}{l} \frac{5}{8}B = \frac{2}{3}A + 7 \\ \frac{3}{8}B = \frac{1}{3}A + 7 \end{array} \right. \Leftrightarrow \left\{ \begin{array}{l} 15B = 16A + 7 \cdot 24 \\ 9B = 8A + 7 \cdot 24 \end{array} \right. \Leftrightarrow A = 42, B = 56 \text{ (C)}$$

6

$$\frac{(n+2)! - (n+1)!}{n!} = \frac{(n+1)!(n+2-1)}{n!} = \frac{(n+1)!(n+1)}{n!} = (n+1)(n+1) = (n+1)^2$$

7

Utilizamos la fórmula de GN/2.5: $\tan \alpha = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$, suponiendo $m_2 > m_1$:

$$1 = \tan(45^\circ) = \tan \alpha = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right| = \frac{6m_1 - m_1}{1 + m_1 6m_1} = \frac{5m_1}{1 + 6m_1^2} \Leftrightarrow 1 + 6m_1^2 = 5m_1 \Leftrightarrow$$

$$m_1 = \begin{cases} \frac{1}{3} \Rightarrow m_1 m_2 = 6m_1^2 = \frac{6}{9} = \frac{2}{3} \\ 1/2 \Rightarrow m_1 m_2 = 6m_1^2 = \frac{6}{4} = \frac{3}{2} \end{cases}$$

El valor máximo es 3/2 (C)

8

Interpretando esta ecuación como una ecuación de segundo grado en y, tenemos

$$y^2 - 2y + x^{2020} = 0 \Leftrightarrow y = \frac{2 \pm \sqrt{4 - 4 \cdot 1 \cdot x^{2020}}}{2} = \frac{2 \pm \sqrt{4(1 - x^{2020})}}{2} = \frac{2 \pm 2\sqrt{1 - x^{2020}}}{2} = 1 \pm \sqrt{1 - x^{2020}}$$

Que tendrá solución siempre que $1 - x^{2020} \geq 0 \Leftrightarrow \begin{cases} x=1 \\ x=0 \end{cases}$

$x=1 \Rightarrow y=1 \pm \sqrt{0}=1$ una solución

$x=0 \Rightarrow y=1 \pm \sqrt{1}=0,1$ dos soluciones

En total, hay 3 soluciones (C)

9

La longitud de la circunferencia será $\frac{3}{4} 2\pi r = \frac{3}{4} 2\pi 4 = 2\pi 3$, es decir, el cono tendrá como

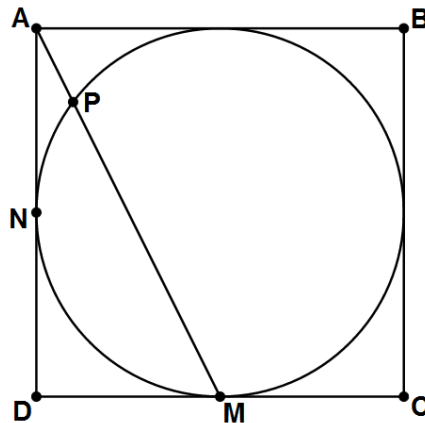
base un círculo de radio 3 y apotema 4. Luego su altura será $\sqrt{4^2 - 3^2} = \sqrt{16 - 9} = \sqrt{7}$

Y por lo tanto su volumen será $\frac{1}{3} \pi r^2 h = \frac{1}{3} \pi 3^2 \sqrt{7} = 3\pi \sqrt{7}$ (C)

10

Está claro que $DM = 1/2$ y por Pitágoras $AM^2 = AD^2 + DM^2 = 1^2 + (1/2)^2 = 5/4$

Sea N el punto de contacto entre ω y $\overline{AD}$.



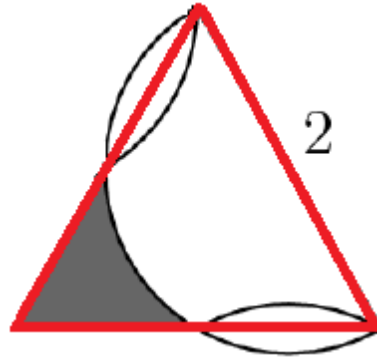
Aplicando Potencia de un punto,

$$\frac{1}{4} = \left(\frac{1}{2}\right)^2 = AN^2 = AP \cdot AM = AP \cdot \frac{\sqrt{5}}{2} \Rightarrow AP = \frac{1}{4} \cdot \frac{2}{\sqrt{5}} = \frac{1}{2\sqrt{5}} = \frac{\sqrt{5}}{10} \quad (B)$$

Nota: En la web www.artofproblemsolving.com se encuentran soluciones alternativas sin utilizar “Potencia de un punto”.

11

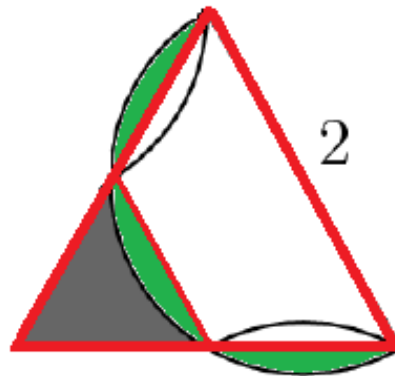
Podemos ver la figura como compuesta de 6 triángulos equiláteros de lado 2, y por tanto de altura $h = \sqrt{2^2 - 1^2} = \sqrt{3}$, y por tanto área $A_1 = \frac{2\sqrt{3}}{2} = \sqrt{3}$



Con un semicírculo de área $A_2 = \frac{\pi 1^2}{2} = \frac{\pi}{2}$

Estos triángulos se dividen en su interior en cuatro triángulos iguales, de área $A_3 = \frac{A_1}{4} = \frac{\sqrt{3}}{4}$

Luego el área verde será $A_4 = A_2 - 3A_3 = \frac{\pi}{2} - \frac{3\sqrt{3}}{4}$



Y por último el área gris será

$$A_5 = A_3 - \frac{A_4}{3} = \frac{\sqrt{3}}{4} - \frac{1}{3} \left(\frac{\pi}{2} - \frac{3\sqrt{3}}{4} \right) = \frac{\sqrt{3}}{4} - \frac{\pi}{6} + \frac{\sqrt{3}}{4} = \frac{2\sqrt{3}}{4} - \frac{\pi}{6} = \frac{\sqrt{3}}{2} - \frac{\pi}{6}$$

Y el área buscada será seis veces ésta:

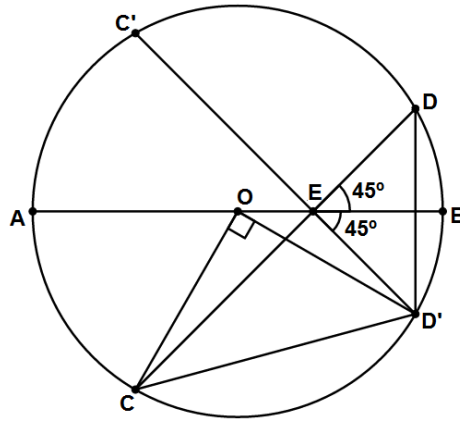
$$A_6 = 6A_5 = 6 \left(\frac{\sqrt{3}}{2} - \frac{\pi}{6} \right) = 3\sqrt{3} - \pi \quad (\text{D})$$

12

Determinamos la cuerda $\overline{C'D'}$ determinada por los puntos simétricos de C y D respecto del diámetro AB.

$DD' \perp AB$ y por tanto $\angle EDD' = 180 - 90 - 45 = 45^\circ$

Luego, por el teorema del ángulo central, $\angle COD' = 2\angle CDD' = 2\angle EDD' = 90^\circ$



Luego $CD'^2 = r^2 + r^2 = 2r^2 = 2 \cdot (5\sqrt{2})^2 = 2 \cdot 25 \cdot 2 = 100$

Y finalmente $CE^2 + DE^2 = CE^2 + D'E^2 = CD'^2 = 100$ (E)

13

Primera versión.

$$\begin{aligned} \log_2 6 + \log_3 6 &= \log_2 2 \cdot 3 + \log_3 2 \cdot 3 = \log_2 2 + \log_2 3 + \log_3 2 + \log_3 3 = \\ &= 1 + \log_2 3 + \log_3 2 + 1 = \log_2 3 + \log_3 2 + 2 = \log_2 3 + \log_3 2 + 2\sqrt{\log_2 3 \log_3 2} = \\ &= \log_2 3 + \log_3 2 + 2\sqrt{\log_2 3} \sqrt{\log_3 2} = \left(\sqrt{\log_2 3} + \sqrt{\log_3 2}\right)^2 \Rightarrow \\ \sqrt{\log_2 6 + \log_3 6} &= \sqrt{\log_2 3} + \sqrt{\log_3 2} \quad (D) \end{aligned}$$

En donde hemos utilizado la identidad

$$\log_2 3 \log_3 2 = 1 \Rightarrow \sqrt{\log_2 3} \sqrt{\log_3 2} = \sqrt{\log_2 3 \log_3 2} = 1$$

Que se deduce directamente de la fórmula del cambio de variable: $\log_a b = \frac{\log_c b}{\log_c a}$

Segunda versión.

$$\begin{aligned} \log_2 6 + \log_3 6 &= \log_2 2 \cdot 3 + \log_3 2 \cdot 3 = \log_2 2 + \log_2 3 + \log_3 2 + \log_3 3 = \\ &= 1 + \log_2 3 + \log_3 2 + 1 = \log_2 3 + \log_3 2 + 2 \Rightarrow \\ \sqrt{\log_2 6 + \log_3 6} &= \sqrt{\log_2 3 + \log_3 2 + 2} \end{aligned}$$

Definiendo $x = \log_2 3$, sabemos que $\log_2 3 = \frac{1}{x}$

Y aplicamos la identidad $\sqrt{2 + x + \frac{1}{x}} = \sqrt{x} + \sqrt{\frac{1}{x}}$, en efecto:

$$\sqrt{2 + x + \frac{1}{x}} = \sqrt{\frac{2x + x^2 + 1}{x}} = \sqrt{\frac{(x+1)^2}{x}} = \frac{x+1}{\sqrt{x}} = \frac{x}{\sqrt{x}} + \frac{1}{\sqrt{x}} = \sqrt{x} + \frac{1}{\sqrt{x}}$$

14

Existe una estrategia ganadora para Bela: En su primera jugada, Bela debe seleccionar el número central del segmento $[0, n]$, y después, limitarse a repetir la jugada que haga Jenn, pero reflejada, como si hubiera un espejo en el centro del segmento. Así pues, si Jenn puede hacer una jugada, seguro que también la podrá hacer después Bela, y por tanto Jenn será forzosamente el primero en quedarse sin números para escoger. La respuesta es (A).

Fuente: https://artofproblemsolving.com/wiki/index.php/2020_AMC_12B_Problems/Problem_14

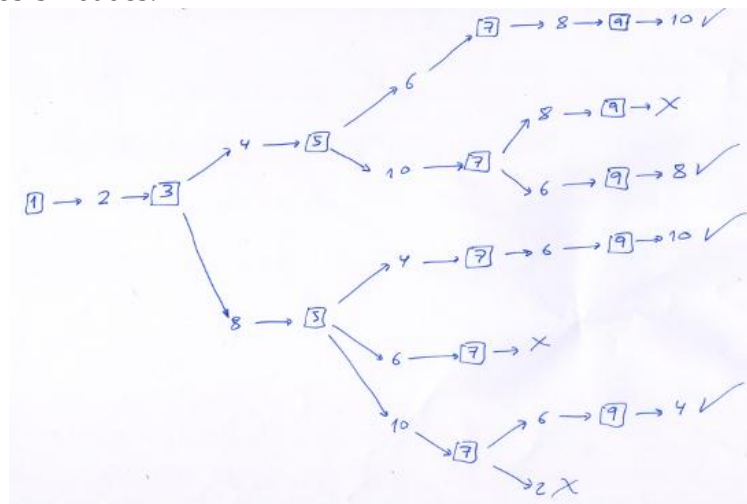
15

Haciendo un esquema vemos las diferentes posibilidades, y vemos que las parejas se pueden organizar por las posiciones impares: 1, 3, 5, 7, 9 pues estas posiciones no compartirán nunca parejas entre ellas.

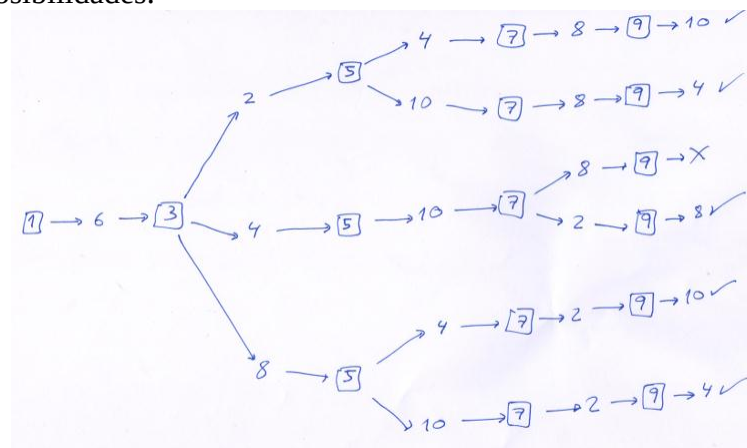
$$1 \rightarrow \{2, 10, 6\}, 3 \rightarrow \{2, 4, 7\}, 5 \rightarrow \{4, 6, 10\}, 7 \rightarrow \{6, 8, 2\}, 9 \rightarrow \{8, 10, 4\}$$

Podemos ir escribiendo una a una todas las posibilidades.

Si $1 \rightarrow 2$, hay 4 posibilidades:



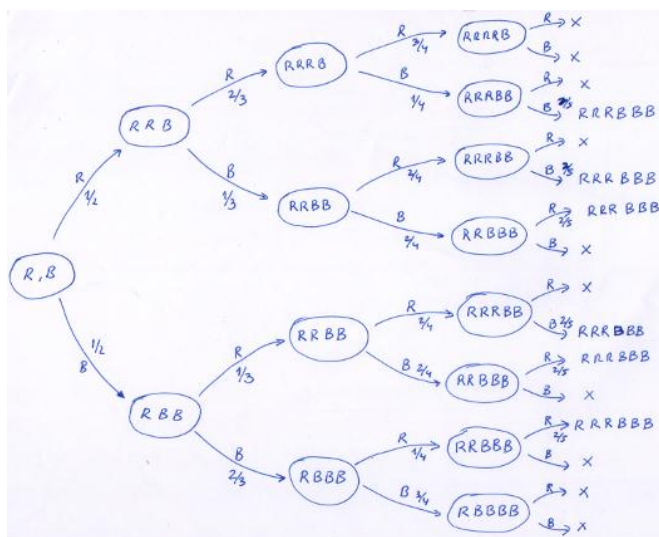
Si $1 \rightarrow 6$, hay 5 posibilidades:



Si $1 \rightarrow 10$, es un caso simétrico al primero, luego habrán 4 posibilidades también. En total $4 + 4 + 5 = 13$ posibilidades. (C)

16

Estamos ante un problema que se puede resolver por probabilidad condicional. Puesto que el número de operaciones no es muy alto, se puede exhibir todo el árbol de posibilidades condicionadas:



Para ahorrarnos cálculos, y por simetría, podemos considerar una de las dos ramas principales y multiplicar por 2 el resultado:

$$2P = \frac{1}{2} \frac{2}{3} \frac{1}{4} \frac{2}{5} + \frac{1}{2} \frac{1}{3} \frac{2}{4} \frac{2}{5} + \frac{1}{2} \frac{1}{3} \frac{2}{4} \frac{2}{5} = \frac{4}{120} + \frac{4}{120} + \frac{4}{120} = \frac{12}{120} = \frac{1}{10} \rightarrow P = \frac{2}{10} = \frac{1}{5} \quad (B)$$

17

Sea $p(x) = x^5 + ax^4 + bx^3 + cx^2 + dx + 2020$. Observamos que $\frac{-1+i\sqrt{3}}{2} = e^{2\pi/3i}$.

Luego, si r es una raíz, también lo serán $re^{2\pi/3i}$ y $re^{4\pi/3i}$.

Puesto que el polinomio es de grado 5, tendrá también dos raíces más.

Sea w una de estas dos raíces. Pero entonces $we^{2\pi/3i}$ y $we^{4\pi/3i}$ también serían raíces, lo que no puede pasar con un polinomio de grado 5 a menos que w sea r , $re^{2\pi/3i}$ o $re^{4\pi/3i}$.

Así pues, el polinomio no puede tener más raíces que estas, y por tanto se puede escribir como

$$p(x) = (x-r)^m (x-re^{2\pi/3i})^n (x-re^{4\pi/3i})^p$$

Además, por Vieta sabemos que existe un único posible valor para la magnitud de r como

$\|r\|^5 = 2020$ (????), luego el conjunto de polinomios equivale al conjunto de posibles ternas (m, n, p) , cumpliendo $m + n + p = 5$.

Además, para que el polinomio resultante tenga coeficientes reales se deberá cumplir $n = p$, puesto que $re^{2\pi/3i}$ y $re^{4\pi/3i}$ son conjugados.

Con todo esto llegamos a las dos únicas posibles soluciones de (m, n, p) , que son $(1, 2, 2)$ y $(3, 1, 1)$, es decir, hay dos posibles polinomios (C).

18

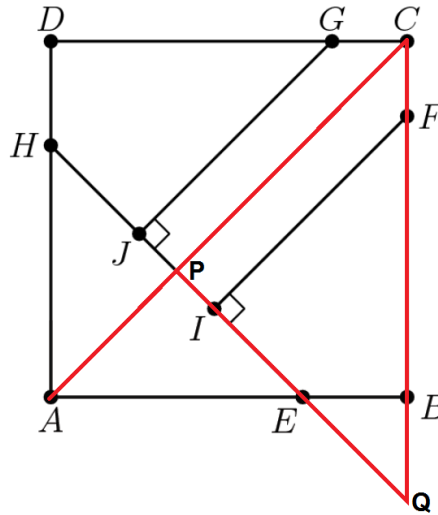
Tenemos un cuadrado de área

$$1+1+1+1=4=AB^2 \Rightarrow AB=2 \Rightarrow AC=\sqrt{2^2+2^2}=\sqrt{8}=2\sqrt{2}$$

$$1=[\triangle AEH]=\frac{AE^2}{2} \Rightarrow AE=\sqrt{2}$$

$$EB=AB-AE=2-\sqrt{2}$$

Prolongamos los segmentos HE y BC hasta su intersección Q. Sea $P=AC \cap HE$.



$$\frac{1}{2}=[\triangle APE]=\frac{AP^2}{2} \Rightarrow AP^2=1 \Rightarrow AP=EP=1 \Rightarrow PC=AC-AP=2\sqrt{2}-1$$

$$[\triangle PCQ]=\frac{PC^2}{2}=\frac{(2\sqrt{2}-1)^2}{2}$$

$$[\triangle EBQ]=\frac{EB^2}{2}=\frac{(2-\sqrt{2})^2}{2}$$

$$\frac{IF^2}{2}=[\triangle IFQ]=IFBE+[\triangle EBQ]=1+\frac{(2-\sqrt{2})^2}{2}=\frac{2+(2-\sqrt{2})^2}{2} \Rightarrow$$

$$IF^2=2+(2-\sqrt{2})^2=2+4+2-2\cdot 2\sqrt{2}=8-4\sqrt{2} \quad (B)$$

19

https://artofproblemsolving.com/wiki/index.php/2020_AMC_12B_Problems/Problem_19

20

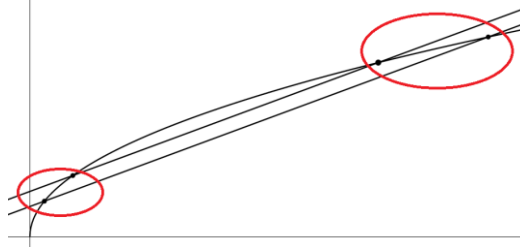
https://artofproblemsolving.com/wiki/index.php/2020_AMC_12B_Problems/Problem_20

21

$$\frac{n+1000}{70} = \lfloor \sqrt{n} \rfloor \Leftrightarrow \frac{n+1000}{70} \leq \sqrt{n} < \frac{n+1000}{70} + 1 = \frac{n+1000+70}{70}$$

$$\Leftrightarrow n+1000 \leq 70\sqrt{n} < n+1070$$

Analizando las familias de las gráficas de las funciones anteriores, vemos que la zona solución será dos intervalos:



$$n+1000 = 70\sqrt{n} \Leftrightarrow \begin{cases} n = 400 \\ n = 2500 \end{cases} \Rightarrow n+1000 \leq 70\sqrt{n} \Leftrightarrow n \in [400, 2500]$$

$$n+1070 = 70\sqrt{n} \Leftrightarrow \begin{cases} n = 508.507 \\ n = 2251.49 \end{cases} \Rightarrow 70\sqrt{n} < n+1070 \Leftrightarrow n \notin [508, 2251)$$

Luego la desigualdad se cumplirá para todos los enteros en $[400, 508)$ y $[2252, 2500]$

Además se cumple que $\frac{n+1000}{70}$ es entero, es decir, $n+1000$ es múltiplo de 70.

En $[400, 508]$:

$$n = 400 \Rightarrow n+1000 = 1400 \text{ es múltiplo de } 70$$

$$n+1000 = 1470 \Rightarrow n = 470$$

$$n+1000 = 1540 \Rightarrow n = 540 \text{ queda fuera del intervalo. Hay dos posibles valores de } n.$$

En $[2252, 2500]$:

$$n = 2290 \Rightarrow n+1000 = 3290 \text{ es múltiplo de } 70.$$

$$n = 2290 + 70 = 2360 \Rightarrow n+1000 = 3360 \text{ es múltiplo de } 70.$$

$$n = 2360 + 70 = 2430 \Rightarrow n+1000 = 3430 \text{ es múltiplo de } 70.$$

$$n = 2430 + 70 = 2500 \Rightarrow n+1000 = 3500 \text{ es múltiplo de } 70.$$

Luego hay 6 posibles valores.

22

$$u = \frac{(2^t - 3t)t}{4^t} = \frac{1}{3} \frac{(2^t - 3t)3t}{(2^t)^2} = \frac{1}{3} \frac{(2^t - 3t)3t}{(2^t)^2} \Rightarrow \sqrt{u} = \frac{1}{\sqrt{3}} \frac{\sqrt{(2^t - 3t)3t}}{2^t} \quad (*)$$

Aplicando la desigualdad AM-GM:

$$(2^t - 3t)3t \leq \frac{1}{2}(2^t - 3t + 3t) = \frac{2^t}{2}$$

Luego

$$(*) \leq \frac{1}{\sqrt{3}} \frac{2^t}{2 \cdot 2^t} = \frac{1}{2\sqrt{3}} \Rightarrow u \leq \frac{1}{2^2 \cdot 3} = \frac{1}{12} \quad (C)$$

La igualdad se alcanza cuando $2^t - 3t = 3t \Leftrightarrow 2^t = 6t$, que es una ecuación que tiene solución entre 0 y 1, pues aplicando Bolzano a la función continua $f(x) = 2^x - 6x$:

$$t = 0 \Rightarrow f(x) = 2^0 - 6 \cdot 0 = 1$$

$$t = 1 \Rightarrow f(1) = 2^1 - 6 \cdot 1 = -4$$

23

https://artofproblemsolving.com/wiki/index.php/2020_AMC_12B_Problems/Problem_23

24

Resolveremos este problema “por fuerza bruta”.

Vemos que $96 = 2^5 \cdot 3$, por lo que el número de posibilidades quedará determinado por las diferentes formas de agrupar los “2”.

En un único grupo:

1 posibilidad.

En dos grupos:

En todo momento, escribimos los exponentes de las potencias de 2, y multiplicamos por las posiciones del “3”

0 5 $\times$ 1

1 4 $\times$ 2

2 3 $\times$ 2

3 2 $\times$ 2

4 1 $\times$ 2

5 0 $\times$ 1

10 posibilidades.

En tres grupos:

0 0 5 $\times$ 0

0 1 4 $\times$ 1

0 2 3 $\times$ 1

0 3 2 $\times$ 1

0 4 1 $\times$ 1

0 5 0 $\times$ 0

1 0 4 $\times$ 1

1 1 3 $\times$ 3

1 2 2 $\times$ 3

1 3 1 $\times$ 3

1 4 0 $\times$ 1

2 0 3 $\times$ 1

2 1 2 $\times$ 3

2 2 1 $\times$ 3

2 3 0 $\times$ 1

3 0 2 $\times$ 1

3 1 1 × 3
3 2 0 × 1
4 0 1 × 1
4 1 0 × 1
5 0 0 × 0

30 posibilidades en total.

En cuatro grupos:

0 0 x x × 0
0 1 0 4 × 0
0 1 1 3 × 1
0 1 2 2 × 1
0 1 3 1 × 1
0 1 4 0 × 0
0 2 1 2 × 1
0 2 2 1 × 1
0 2 3 0 × 0
0 3 1 1 × 1
0 3 2 0 × 0
0 4 x x × 0
1 0 1 3 × 1
1 0 2 2 × 1
1 0 3 1 × 1
1 0 4 0 × 0
1 1 0 3 × 1
1 1 1 2 × 4
1 1 2 1 × 4
1 1 3 0 × 1
1 2 0 2 × 1
1 2 1 1 × 4
1 2 2 0 × 1
1 3 0 1 × 1
1 3 1 0 × 1
1 4 0 0 × 0
2 0 0 3 × 0
2 0 1 2 × 1
2 0 2 1 × 1
2 0 3 0 × 0
2 1 0 2 × 1
2 1 1 1 × 4
2 1 2 0 × 1
2 2 0 1 × 1
2 2 1 0 × 1
2 3 0 0 × 0
3 0 0 2 × 0
3 0 1 1 × 1
3 0 2 0 × 0
3 1 0 1 × 1
3 1 1 0 × 1
3 2 0 0 × 0

4 x x x × 0

5 x x x × 0

40 posibilidades en total.

En cinco grupos:

0 0 x x x × 0

0 1 0 x x × 0

0 1 1 1 2 × 1

0 1 1 2 1 × 1

0 1 1 3 0 × 0

0 1 2 0 2 × 0

0 1 2 1 1 × 1

0 1 2 2 0 × 0

0 1 3 x x × 0

0 1 4 x x × 0

0 2 0 x x × 0

0 2 1 1 1 × 1

0 3 x x x × 0

0 4 x x x × 0

1 0 0 x x × 0

1 0 1 1 2 × 1

1 0 1 2 1 × 1

1 0 2 1 1 × 1

1 0 3 x x × 0

1 1 0 1 2 × 1

1 1 0 2 1 × 1

1 1 1 0 2 × 1

1 1 1 1 1 × 5

1 1 1 2 0 × 1

1 1 2 0 1 × 1

1 1 2 1 0 × 1

1 2 0 1 1 × 1

1 2 1 1 0 × 1

1 2 1 0 1 × 1

2 0 1 1 1 × 1

2 1 0 1 1 × 1

2 1 1 0 1 × 1

2 1 1 1 0 × 1

3 x x x x × 0

4 x x x x × 0

5 x x x x × 0

25 posibilidades en total.

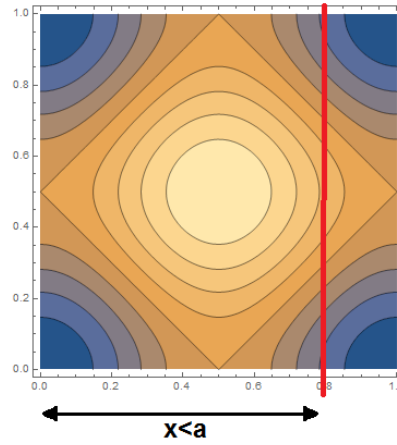
En seis grupos:

Hay 6 posibilidades: $96 = 3 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$ y sus otras 5 posiciones del “3”.

Total: $6 + 25 + 40 + 30 + 10 + 1 = 112$ (A)

25

En primer lugar interpretamos la probabilidad $P(a)$ del enunciado en términos de áreas. Estamos trabajando en un rectángulo de base a y altura 1. Luego su área es $a \cdot 1 = a$.



$$P(a) = \frac{\text{Área}(\sin^2(\pi x) + \sin^2(\pi y) > 1)}{\text{Área Total}} = \frac{\text{Área}(\sin^2(\pi x) + \sin^2(\pi y) > 1)}{a}$$

Para determinar la región del plano determinada por

$$\sin^2(\pi x) + \sin^2(\pi y) > 1$$

Procedemos a determinar su frontera:

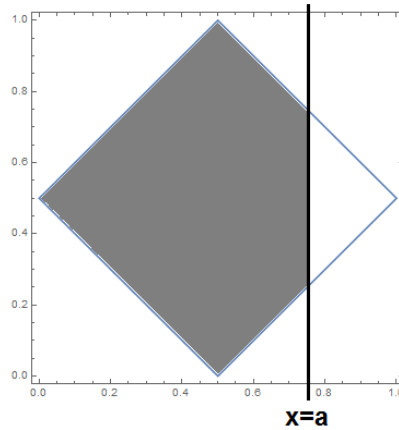
$$\sin^2(\pi x) + \sin^2(\pi y) = 1$$

$$\sin^2(\pi x) + \sin^2(\pi y) = 1 \Leftrightarrow \sin^2(\pi x) = 1 - \sin^2(\pi y) = \cos^2(\pi y) \Leftrightarrow \sin(\pi x) = \pm \cos(\pi y)$$

$$\text{Primer caso: } \sin(\pi x) = \cos(\pi y) = \sin\left(\frac{\pi}{2} \pm \pi y\right) \Leftrightarrow \begin{cases} y = \frac{1}{2} - x \\ y = x - \frac{1}{2} \end{cases}$$

$$\text{Segundo caso: } \sin(\pi x) = -\cos(\pi y) \Leftrightarrow \begin{cases} y = x + \frac{1}{2} \\ y = \frac{3}{2} - x \end{cases}$$

Tenemos una región en forma de rombo, y probando con algún punto auxiliar vemos que la región que satisface la desigualdad es su interior:



Ahora ya podemos calcular su área A . Suponiendo $a \geq 1/2$, $A = \frac{1}{4} + \frac{(a-1/2)(3-2a)}{2}$ y por lo tanto

$$P(a) = \frac{\text{Área}(\sin^2(\pi x) + \sin^2(\pi y) > 1)}{a} = \frac{1}{a} \left(\frac{1}{4} + \frac{(a-1/2)(3-2a)}{2} \right) =$$

$$= \frac{-4a^2 + 8a - 2}{4a} = 2 - \left(a + \frac{1}{2a} \right)$$

El máximo de esta función lo encontraremos en el mínimo de $a + \frac{1}{2a}$, que determinaremos aplicando la desigualdad AM-GM:

$$a + \frac{1}{2a} \geq 2\sqrt{a \cdot \frac{1}{2a}} = \frac{2}{\sqrt{2}} = \sqrt{2}$$

Que se encuentra cuando $a = \frac{1}{2a} \Leftrightarrow 2a^2 = 1 \Leftrightarrow a = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

Así pues, el máximo de esta función es:

$$P\left(\frac{\sqrt{2}}{2}\right) = 2 - \left(\frac{\sqrt{2}}{2} + \frac{1}{2\left(\frac{\sqrt{2}}{2}\right)} \right) = 2 - \left(\frac{\sqrt{2}}{2} + \frac{1}{\sqrt{2}} \right) = 2 - \sqrt{2} \quad (B)$$

Fuente: https://artofproblemsolving.com/wiki/index.php/2020_AMC_12B_Problems/Problem_25

AMC 12A 2019 Enunciados

1

El área de una pizza de radio 4 pulgadas es un N por ciento mayor que el área de una pizza de radio 3 pulgadas. ¿Cuál es el entero más próximo a N ?

(A) 25 (B) 33 (C) 44 (D) 66 (E) 78

2

Supongamos que a es el 150 % de b . ¿Qué porcentaje de a es $3b$?

(A) 50 (B) $66\frac{2}{3}$ (C) 150 (D) 200 (E) 450

3

Una caja contiene 20 bolas rojas, 20 bolas verdes, 19 bolas amarillas, 13 bolas azules, 11 bolas blancas y 9 bolas negras. ¿Cuál es el mínimo de bolas que debemos coger de la caja, sin reemplazamiento, para garantizar que saquemos como mínimo 15 bolas del mismo color?

(A) 75 (B) 76 (C) 79 (D) 84 (E) 91

4

¿Cuál es la mayor cantidad de números enteros consecutivos cuya suma sea 45?

(A) 9 (B) 25 (C) 45 (D) 90 (E) 120

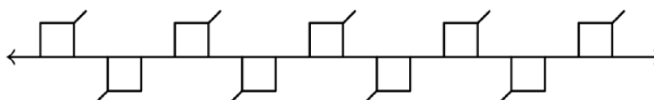
5

Dos rectas con pendiente 2 y $\frac{1}{2}$ se cortan en $(2, 2)$. ¿Cuál es el área del triángulo determinado por estas dos rectas y la recta $x + y = 10$?

(A) 4 (B) $4\sqrt{2}$ (C) 6 (D) 8 (E) $6\sqrt{2}$

6

La siguiente figura muestra una recta l con una pauta regular, infinita y recurrente de cuadrados y segmentos.



¿Cuántas, de entre las siguientes transformaciones rígidas del plano, diferentes de la identidad, transformarán esta figura en sí misma?

- Cierta rotación alrededor de un punto de la recta l .
- Cierta traslación en la dirección paralela a la recta l .
- La reflexión respecto de la recta l .
- Cierta reflexión respecto de una recta perpendicular a la recta l .

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

7

Melanie calcula la media μ , la mediana M y las modas de los 365 números que son los días del mes de los meses de 2019. Es decir, los datos consisten en

12 “1”, 12 “2”, ..., 12 “28”, 11 “29”, 11 “30” y 7 “31”.

Sea d la mediana de las modas. ¿Cuál de las siguientes afirmaciones es cierta?

- (A) $\mu < d < M$ (B) $M < d < \mu$ (C) $d = M = \mu$ (D) $d < M < \mu$ (E) $d < \mu < M$

8

Dado un conjunto de cuatro rectas diferentes en el plano, sea N el número de puntos diferentes que pertenecen a dos o más rectas de dicho conjunto. ¿Cuál es la suma de todos los posibles valores de N ?

- (A) 14 (B) 16 (C) 18 (D) 19 (E) 21

9

Definimos recursivamente la siguiente sucesión de números por $a_1 = 1$, $a_2 = \frac{3}{7}$, y

$$a_n = \frac{a_{n-2} \cdot a_{n-1}}{2a_{n-2} - a_{n-1}}.$$

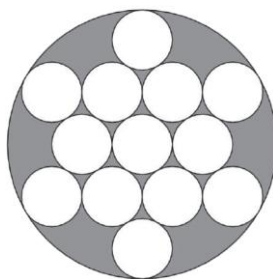
Entonces a_{2019} se puede escribir como $\frac{p}{q}$, con p y q enteros positivos coprimos.

Determina $p + q$.

- (A) 2020 (B) 4039 (C) 6057 (D) 6061 (E) 8078

10

En la siguiente figura se muestran 13 círculos de radio 1 en el interior de un círculo más grande. Todos los puntos de corte son puntos de tangencia. Determina el área de la región sombreada, que se encuentra en el interior del círculo mayor y en el exterior de los círculos de radio 1.



- (A) $4\pi\sqrt{3}$ (B) 7π (C) $\pi(3\sqrt{3} + 2)$ (D) $10\pi(\sqrt{3} - 1)$ (E) $\pi(\sqrt{3} + 6)$

11

Para cierto entero positivo k , la representación en base k de la fracción (en base 10) $\frac{7}{51}$ es $0.\overline{23}_k = 0.232323\dots_k$. Determina k .

- (A) 13 (B) 14 (C) 15 (D) 16 (E) 17

12

Sean $x \neq 1$, $y \neq 1$ números reales positivos satisfaciendo $\log_2 x = \log_y 16$, $xy = 64$.

Determina $\left(\log_2 \frac{x}{y}\right)^2$

(A) $\frac{25}{2}$ (B) 20 (C) $\frac{45}{2}$ (D) 25 (E) 32

13

Determina el número de formas diferentes de pintar los enteros 2,3,...,9 de rojo, verde y azul, de forma que cada número tenga un color diferente de todos y cada uno de sus divisores propios.

(A) 144 (B) 216 (C) 256 (D) 384 (E) 432

14

Supongamos que para cierto número complejo c , el polinomio

$$P(x) = (x^2 - 2x + 2)(x^2 - cx + 4)(x^2 - 4x + 8)$$

Tiene exactamente cuatro raíces diferentes. Determina $|z|$.

(A) 2 (B) $\sqrt{6}$ (C) $2\sqrt{2}$ (D) 3 (E) $\sqrt{10}$

15

Sean a y b números reales positivos cumpliendo la condición

$$\sqrt{\log a} + \sqrt{\log b} + \log \sqrt{a} + \log \sqrt{b} = 100$$

y en donde los cuatro términos de la izquierda son enteros positivos, denotando por $\log$ el logaritmo en base 10. Determina ab .

(A) 10^{52} (B) 10^{100} (C) 10^{144} (D) 10^{164} (E) 10^{200}

16

Colocamos aleatoriamente los números 1, 2, ..., 9 en una cuadrícula de 3 filas y 3 columnas. En cada casilla ponemos un número, y cada uno de los números se usa una única vez. Determina la probabilidad de que la suma de los números en cada fila y en cada columna sean números impares.

(A) $\frac{1}{21}$ (B) $\frac{1}{14}$ (C) $\frac{5}{63}$ (D) $\frac{2}{21}$ (E) $\frac{1}{7}$

17

Denotamos por s_k la suma de las potencias k -ésimas de las raíces del polinomio $x^3 - 5x^2 + 8x - 13$. En particular, $s_0 = 3$, $s_1 = 5$, y $s_2 = 9$. Sean a, b, c números reales tales que

$$s_{k+1} = as_k + bs_{k-1} + cs_{k-2} \text{ para todo } k = 2, 3, \dots \text{ Determina } a + b + c.$$

(A) -6 (B) 0 (C) 6 (D) 10 (E) 26

18

Dada una esfera de centro O y radio 6, un triángulo de lados de longitud 15, 15 y 24 está situado en el espacio de forma que sus tres lados son tangentes a la circunferencia. Determina la distancia entre O y el plano determinado por el triángulo.

(A) $2\sqrt{3}$ (B) 4 (C) $3\sqrt{2}$ (D) $2\sqrt{5}$ (E) 5

19

Determina el mínimo perímetro de un triángulo $\triangle ABC$ con lados de longitudes enteras cumpliendo

$$\cos A = \frac{11}{16}, \cos B = \frac{7}{8}, \cos C = -\frac{1}{4}$$

(A) 9 (B) 12 (C) 23 (D) 27 (E) 44

20

Tomamos aleatoriamente números reales entre 0 y 1 de la siguiente manera: Se lanza una moneda, y si sale cara, se lanza una segunda moneda y tomaremos 0 si la segunda moneda es cara y 1 si la segunda moneda es cruz. Si la primera moneda es cruz, el número es tomado aleatoriamente en el intervalo cerrado $[0, 1]$ mediante una distribución uniforme. Dos números x, y han sido escogidos independientemente mediante el procedimiento anterior. Determina la probabilidad de que $|x - y| > 1/2$.

(A) $\frac{1}{3}$ (B) $\frac{7}{16}$ (C) $\frac{1}{2}$ (D) $\frac{9}{16}$ (E) $\frac{2}{3}$

21

Sea

$$z = \frac{1+i}{\sqrt{2}}$$

Determina

$$\left(z^{1^2} + z^{2^2} + z^{3^2} + \dots + z^{12^2} \right) \cdot \left(\frac{1}{z^{1^2}} + \frac{1}{z^{2^2}} + \frac{1}{z^{3^2}} + \dots + \frac{1}{z^{12^2}} \right)$$

(A) 18 (B) $72 - 36\sqrt{2}$ (C) 36 (D) 72 (E) $72 + 36\sqrt{2}$

22

Las circunferencias ω y γ , ambas con centro O, tienen radio 20 y 17, respectivamente. Sea $\triangle ABC$ un triángulo equilátero, cuyo interior está en el interior de ω pero en el exterior de γ , tiene el vértice A en ω y la recta que contiene el segmento $\overline{BC}$ es tangente a γ . Los segmentos $\overline{AO}$ y $\overline{BC}$ se cortan en P, cumpliendo $\frac{BP}{CP} = 3$. La

distancia AB se puede escribir de la forma $\frac{m}{\sqrt{n}} - \frac{p}{\sqrt{q}}$ para ciertos enteros positivos m, n, p, q con $\text{Mcd}(m, n) = \text{Mcd}(p, q) = 1$. Determina $m + n + p + q$.

(A) 42 (B) 86 (C) 92 (D) 114 (E) 130

23

Definimos las siguientes operaciones binarias $\diamond$ y ∇ por:

$$a \diamond b = a^{\log_7(b)}, \quad a \nabla b = a^{\frac{1}{\log_7(b)}}$$

para todos los reales a, b para los cuales tengan sentido estas expresiones. Definimos la sucesión (a_n) de forma recursiva por $a_3 = 3 \nabla 2$ y

$$a_n = (n \nabla (n-1)) \diamond a_{n-1}$$

para todo entero $n \geq 4$. Determina el entero más próximo a $\log_7(a_{2019})$.

(A) 8 (B) 9 (C) 10 (D) 11 (E) 12

24

Determina el número de enteros n entre 1 y 50, inclusive, tales que

$$\frac{(n^2 - 1)!}{(n!)^n}$$

es un entero. (Recuerda que $0! = 1$).

(A) 31 (B) 32 (C) 33 (D) 34 (E) 35

25

Sea $\triangle A_0 B_0 C_0$ un triángulo cuyos ángulos miden exactamente 59.999° , 60° y 60.001° .

Para cada entero positivo n , definimos A_n como el pie de la altura de A_{n-1} a la recta $B_{n-1}C_{n-1}$. De la misma manera, se define B_n como el pie de la altura de B_{n-1} a la recta $A_{n-1}C_{n-1}$, y C_n como el pie de la altura de C_{n-1} a la recta $A_{n-1}B_{n-1}$. Determina el menor valor de n para el cual $\triangle A_n B_n C_n$ es obtuso.

(A) 10 (B) 11 (C) 13 (D) 14 (E) 15

AMC 12A 2019 Soluciones (letra)

1. E
2. D
3. B
4. D
5. C
6. C
7. E
8. D
9. E
10. A
11. D
12. B
13. E
14. E
15. D
16. B
17. D
18. D
19. A
20. B
21. C
22. E
23. D
24. D
25. E

AMC 12A 2019 Soluciones desarrolladas

1

$$\left. \begin{array}{l} A_1 = \pi \cdot 4^2 \\ A_2 = \pi \cdot 3^2 \end{array} \right\} \Rightarrow N = \frac{A_1 - A_2}{A_2} = \frac{16\pi - 9\pi}{9\pi} = \frac{7\pi}{9\pi} = \frac{7}{9} \cong 0.77 \rightarrow 78 (E)$$

2

$$a = \frac{150}{100}b = \frac{3}{2}b \Leftrightarrow 2a = 3b \Leftrightarrow 3b \text{ es el doble de } a, \text{ es decir, un } 200 \% (D).$$

3

Poniéndonos en el peor de los casos, podríamos sacar 13 bolas azules, 11 bolas blancas y 9 bolas negras, 33 bolas en total, y no habríamos sacado 15 del mismo color. Además, podríamos sacar 14 bolas rojas, 14 bolas verdes y 14 bolas amarillas (en total 75) y seguiríamos sin haber sacado 15 bolas del mismo color. Pero entonces, con una más, seguro que hay 15 rojas, verdes o amarillas. La solución es 76 (B).

4

En principio podríamos creer que la respuesta correcta es 9, puesto que

$$1 + 2 + 3 + \dots + 9 = 45$$

Pero debemos observar que podemos trabajar también con números negativos, y ver que

$$(-89) + (-88) + (-87) + \dots + 89 + 90 = 90$$

Luego podemos encontrar una suma de 90 términos.

Aunque no es apropiado en un contexto de una competición matemática, se puede demostrar que 90 es la solución máxima:

$$\begin{aligned} 45 &= x + x + 1 + x + 2 + \dots + x + (n-1) = nx + 1 + 2 + 3 + \dots + (n-2) + (n-1) = \\ &= nx + \frac{n(n-1)}{2} \Rightarrow 90 = 2nx + n(n-1) = n(2x + n - 1) \Rightarrow n \mid 90 \Rightarrow 90 \leq 90 \end{aligned}$$

5

La primera recta tiene por ecuación

$$y = 2x + b \Rightarrow 2 = 2 \cdot 2 + b \Rightarrow b = -2 \rightarrow y = 2x - 2$$

Y su punto de corte con $x + y = 10$ es:

$$\left. \begin{array}{l} y = 2x - 2 \\ x + y = 10 \end{array} \right\} \Rightarrow x + 2x - 2 = 10 \Rightarrow 3x = 12 \Rightarrow x = \frac{12}{3} = 4 \rightarrow y = 10 - 4 = 6$$

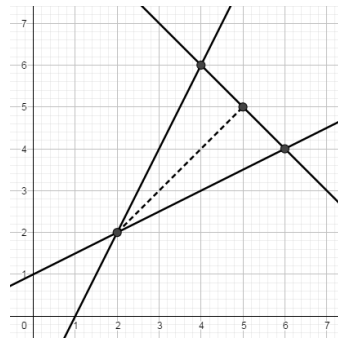
La segunda recta tiene por ecuación

$$y = \frac{1}{2}x + b \Rightarrow 2 = \frac{1}{2} \cdot 2 + b = 1 + b \Rightarrow b = 2 - 1 = 1 \rightarrow y = \frac{1}{2}x + 1$$

Y su punto de corte con $x + y = 10$ es:

$$\left. \begin{array}{l} y = \frac{1}{2}x + 1 \\ x + y = 10 \end{array} \right\} \Rightarrow x + \frac{1}{2}x + 1 = 10 \Rightarrow \frac{3}{2}x = 9 \Rightarrow x = \frac{18}{3} = 6 \rightarrow y = 10 - 6 = 4$$

Vemos que forman un triángulo isósceles de base $2\sqrt{2}$ y altura $3\sqrt{2}$, y por tanto área $\frac{2\sqrt{2} \cdot 3\sqrt{2}}{2} = 6$ (C)



Nota: En [link](#) se presentan hasta 11 soluciones diferentes que pueden ser muy interesantes para repasar todas las posibles formas de determinar el área de un triángulo determinado por tres puntos.

6

Analizando la forma de la figura y el efecto de las transformaciones, vemos que solo las dos primeras son ciertas, luego la solución es (C).

7

La mediana es el número que hace $182 = 12 \cdot 15 + 2$, luego es el segundo “16”, luego $M = 16$. Las modas son del “1” al “28”, y su mediana es $d = 14.5$

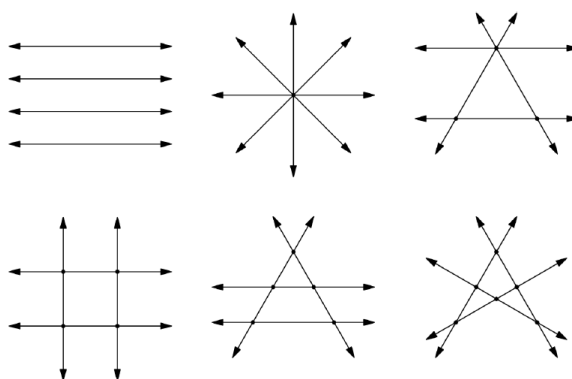
La media es

$$\begin{aligned} \mu &= \frac{12 \cdot 1 + 12 \cdot 2 + \dots + 12 \cdot 28 + 11 \cdot 29 + 11 \cdot 30 + 7 \cdot 31}{365} = \\ &= \frac{12(1 + 2 + 3 + \dots + 29) + 11 \cdot 30 + 7 \cdot 31}{365} = \\ &= \frac{12 \cdot 29 \cdot 30 / 2 + 11 \cdot 30 + 7 \cdot 31}{365} = \frac{12 \cdot 29 \cdot 15 + 11 \cdot 30 + 7 \cdot 31}{365} = \frac{5767}{365} = 15,8 \end{aligned}$$

Luego la respuesta correcta es $d < \mu < M$ (E)

8

Vemos que cuatro rectas diferentes en el plano se pueden cortar en 0, 1, 3, 4, 5 o 6 puntos, como se muestra en la siguiente figura:



pero no en 2, luego la solución es $0 + 1 + 3 + 4 + 5 + 6 = 19$ (D)

Observación: En [Link](#) podemos encontrar un estudio más riguroso de este problema, en el que se demuestra el máximo de 6 y la imposibilidad del 2.

9

Primera versión.

Calculamos a mano los primeros valores de esta sucesión:

$$a_1 = 1, a_2 = \frac{3}{7}, a_3 = \frac{3}{11}, a_4 = \frac{1}{5}, a_5 = \frac{3}{19}, a_6 = \frac{3}{23}, a_7 = \frac{1}{9}, a_8 = \frac{3}{31}, a_9 = \frac{3}{35}$$

Y observamos que siguen una pauta en grupos de tres en tres: Sea $N = \lfloor (n-1)/3 \rfloor$,

$$\left(\frac{1}{4N+1}, \frac{3}{12N+7}, \frac{3}{12N+11} \right)$$

$2019 - 1 = 672 \cdot 3 + 2$, luego nos piden el tercer elemento del grupo $N = 672$, es decir:

$$a_{2019} = \frac{3}{12N+11} = \frac{3}{12 \cdot 672 + 11} = \frac{3}{8075}$$

Y la respuesta correcta es $8075 + 3 = 8078$ (E)

Segunda versión.

Con un poco más de observación podríamos haber visto que $\frac{1}{4N+1} = \frac{3}{12N+3}$

Y que por tanto la sucesión del enunciado se reduce a:

$$a_n = \frac{3}{4n-1} \rightarrow a_{2019} = \frac{3}{4 \cdot 2019 - 1} = \frac{3}{8075}$$

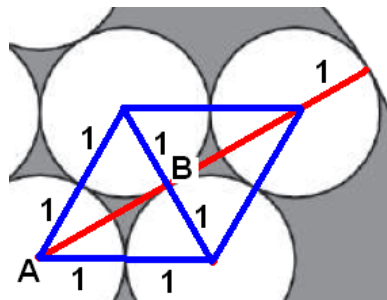
Esto se puede demostrar por inducción:

Supongamos que es cierto hasta n , entonces

$$\begin{aligned}
 a_{n+1} &= \frac{a_{n-1} \cdot a_n}{2a_{n-1} - a_n} = \frac{\frac{3}{4(n-1)-1} \cdot \frac{3}{4n-1}}{2 \frac{3}{4(n-1)-1} - \frac{3}{4n-1}} = \frac{\frac{9}{(4(n-1)-1)(4n-1)}}{\frac{6(4(n-1)-1) - 3(4(n-1)-1)}{(4(n-1)-1)(4n-1)}} = \\
 &= \frac{9}{6(4n-1) - 3(4(n-1)-1)} = \frac{9}{24n-6-12n+15} = \frac{9}{12n+9} = \frac{3}{4n+3} = \frac{3}{4n+4-1} = \\
 &= \frac{3}{4(n+1)-1}
 \end{aligned}$$

10

Determinaremos el radio de la circunferencia exterior. Observamos que los radios y los puntos de tangencia determinan triángulos equiláteros de radio 2:



La distancia AB es la longitud de la mediana de un triángulo equilátero de lado 2, pero en un triángulo isósceles la mediana coincide con la altura, y por tanto se puede determinar por

Pitágoras: $AB = \sqrt{2^2 - 1^2} = \sqrt{3}$

Luego el radio de la circunferencia exterior es $R = 2AB + 1 = 2\sqrt{3} + 1$.

Y por tanto el área de la región sombreada es

$$\begin{aligned}
 \pi(2\sqrt{3} + 1)^2 - 13\pi 1^2 &= \pi \left[(2\sqrt{3} + 1)^2 - 13 \right] = \pi \left[4 \cdot 3 + 4\sqrt{3} + 1 - 13 \right] = \\
 &= \pi 4\sqrt{3} \quad (A)
 \end{aligned}$$

11

$$0.\overline{23}_k = 0.232323..._k = 2 \cdot k^{-1} + 3 \cdot k^{-2} + 2 \cdot k^{-3} + 3 \cdot k^{-4} + 2 \cdot k^{-5} + 3 \cdot k^{-6} + \dots =$$

$$(2k+3)k^{-2} + (2k+3)k^{-4} + (2k+3)k^{-6} + \dots = (2k+3) \sum_{n=1}^{\infty} (k^2)^{-n} =$$

$$(2k+3) \left(\sum_{n=0}^{\infty} (k^{-2})^n - 1 \right) = (2k+3) \left(\frac{1}{1-k^{-2}} - 1 \right) = (2k+3) \left(\frac{k^{-2}}{1-k^{-2}} \right) =$$

$$(2k+3) \left(\frac{1}{k^2-1} \right) = \frac{2k+3}{k^2-1}$$

Así pues, llegamos a la ecuación

$$\frac{7}{51} = \frac{2k+3}{k^2-1} \Leftrightarrow 7(k^2-1) = 51(2k+3) \Leftrightarrow 7k^2 - 7 = 102k + 153 \Leftrightarrow$$

$$7k^2 - 102k - 160 = 0 \Leftrightarrow k = \frac{102 \pm \sqrt{102^2 - 4 \cdot 7 \cdot (-160)}}{2 \cdot 7} = \frac{102 \pm \sqrt{14884}}{2 \cdot 7} =$$

$$= \frac{102 \pm 122}{2 \cdot 7} = \begin{cases} 16 \\ 10/7 \end{cases}$$

La única solución entera es $k = 16$.

Observación. Nos podemos “saltar” esta última ecuación, muy larga sin calculadora, aprovechando que la pregunta es de tipo multirespuesta, con lo que todo se reduce a determinar qué opción de entre las cinco propuestas satisface dicha ecuación.

12

$$\left(\log_2 \frac{x}{y} \right)^2 = (\log_2 x - \log_2 y)^2$$

Por otro lado, $xy = 64 \Rightarrow 6 = \log_2 64 = \log_2 xy = \log_2 x + \log_2 y$

Y aprovechamos la identidad $(A - B)^2 = (A + B)^2 - 4AB$

Luego todo se reduce a determinar $\log_2 x \cdot \log_2 y$

Aquí utilizamos la identidad del cambio de base:

$$\log_2 y = \frac{\log_2 16}{\log_y 16} = \frac{4}{\log_y 16} = \frac{4}{\log_2 x} \Rightarrow \log_2 x \cdot \log_2 y = 4$$

Finalmente:

$$\left(\log_2 \frac{x}{y} \right)^2 = (\log_2 x - \log_2 y)^2 = (\log_2 x + \log_2 y)^2 - 4 \log_2 x \log_2 y =$$

$$6^2 - 4 \cdot 4 = 36 - 16 = 20$$

13

En primer lugar, vemos que los números 5 y 7 se pueden pintar de cualquier color, pues ni son divisibles ni son divisores de ningún otro.

Por lo que parece, el 2 es el elemento con más restricciones. Fijamos un color para el 2, por ejemplo, $2 = R$.

$$2 = R \Rightarrow \begin{cases} 4 = G \Rightarrow 8 = B \\ 4 = B \Rightarrow 8 = G \end{cases}$$

$$\text{Por otro lado, } 2 = R \Rightarrow \begin{cases} 6 = B \Rightarrow \begin{cases} 3 = G \\ 3 = R \end{cases} \\ 6 = G \Rightarrow \begin{cases} 3 = R \\ 3 = B \end{cases} \end{cases}$$

Y con esto quedan determinados todos los números, excepto el 9, que no puede ser del mismo color que el 3.

En total $3 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 48$

Y con las combinaciones de los números 5 y 7, en total $48 \cdot 3 \cdot 3 = 432$ (E)

14

Primera versión.

$$x^2 - 2x + 2 = 0 \Leftrightarrow x = \frac{2 \pm \sqrt{(-2)^2 - 4 \cdot 1 \cdot 2}}{2} = \frac{2 \pm \sqrt{-4}}{2} = \frac{2 \pm 2i}{2} = 1 \pm i$$

$$x^2 - 4x + 8 = 0 \Leftrightarrow x = \frac{4 \pm \sqrt{(-4)^2 - 4 \cdot 1 \cdot 8}}{2} = \frac{4 \pm \sqrt{-16}}{2} = \frac{4 \pm 4i}{2} = 2 \pm 2i$$

Las raíces de $P(x)$ son estas cuatro anteriores más las raíces de $x^2 - cx + 4$, luego para que tenga exactamente cuatro raíces diferentes, las raíces de $x^2 - cx + 4$ deben ser alguna de estas cuatro.

$$0 = x^2 - cx + 4 \Leftrightarrow cx = x^2 + 4 \Leftrightarrow c = \frac{x^2 + 4}{x} \Rightarrow |c| = \left| \frac{x^2 + 4}{x} \right| = \frac{|x^2 + 4|}{|x|}$$

Si $x = 1 + i$:

$$x^2 = (1 + i)(1 + i) = 1 - 1 + 2i = 2i \Rightarrow x^2 + 4 = 4 + 2i \Rightarrow$$

$$|x^2 + 4| = |4 + 2i| = \sqrt{4^2 + 2^2} = \sqrt{16 + 4} = \sqrt{20}$$

$$|x| = |1 + i| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$|c| = \frac{\sqrt{20}}{\sqrt{2}} = \sqrt{10}$$

Si $x = 2 + 2i$:

$$x^2 = (2 + 2i)(2 + 2i) = 4 - 4 + 8i = 8i \Rightarrow x^2 + 4 = 4 + 8i \Rightarrow$$

$$|x^2 + 4| = |4 + 8i| = \sqrt{4^2 + 8^2} = \sqrt{16 + 64} = \sqrt{80} = 4\sqrt{5}$$

$$|x| = |2 + 2i| = \sqrt{2^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$$

$$|c| = \frac{4\sqrt{5}}{2\sqrt{2}} = \sqrt{10}$$

Con los casos $x = 1 - i$, $x = 2 - 2i$ se demuestra igualmente que $|c| = \sqrt{10}$ (E)

Segunda versión.

Siguiendo el desarrollo anterior, vemos que el polinomio $x^2 - cx + 4$ debe descomponer en dos de los factores

$$x - (1+i), x - (1-i), x - (2+2i), x - (2-2i)$$

Puesto que el término independiente es 4, vemos que las dos únicas combinaciones aceptables son:

$$x^2 - cx + 4 = (x - (1+i))(x - (2-2i)) \Rightarrow c = -(1+i+2-2i) = -(3-i) = i-3 \Rightarrow |c| = \sqrt{1^2 + 3^2} = \sqrt{10}$$

$$x^2 - cx + 4 = (x - (1-i))(x - (2+2i)) \Rightarrow c = -(1-i+2+2i) = -(3+i) = -i-3 \Rightarrow |c| = \sqrt{1^2 + 3^2} = \sqrt{10}$$

15

Vemos que $\log \sqrt{a}$ es un entero positivo si y solo si $\sqrt{a} = 10^c \Leftrightarrow a = (10^c)^2 = 10^{2c}$ para cierto entero positivo c .

Luego $\sqrt{\log a} = \sqrt{\log(10^{2c})} = \sqrt{2c}$ y debe ser un entero, luego $c = 2e^2$ para cierto entero e , y

$$\sqrt{\log a} = \sqrt{\log(10^{2c})} = \sqrt{2c} = 2e$$

Por otro lado, $\log \sqrt{a} = c = 2e^2$.

De la misma forma vemos que $\sqrt{b} = 10^d \Leftrightarrow b = (10^d)^2 = 10^{2d}$ para cierto entero positivo d .

Luego $\sqrt{\log b} = \sqrt{\log(10^{2d})} = \sqrt{2d}$ debe ser un entero, luego $d = 2f^2$ para cierto entero f , y

$$\sqrt{\log b} = \sqrt{2d} = 2f$$

Por otro lado, $\log \sqrt{b} = d = 2f^2$.

Así pues, la ecuación del enunciado se transforma en

$$100 = \sqrt{\log a} + \sqrt{\log b} + \log \sqrt{a} + \log \sqrt{b} = 2e + 2f + 2e^2 + 2f^2 \Leftrightarrow$$

$$50 = e + f + e^2 + f^2 = e(e+1) + f(f+1)$$

Buscamos posibilidades:

$$1 \cdot 2 = 2, 2 \cdot 3 = 6, 3 \cdot 4 = 12, 4 \cdot 5 = 20, 5 \cdot 6 = 30, 6 \cdot 7 = 42$$

$$50 = 2 + 48, 50 = 6 + 44, 50 = 12 + 38, 50 = 20 + 30$$

$$50 = 32 + 20, 50 = 42 + 8$$

Las únicas combinaciones aceptables son $e = 4, f = 5$ y $e = 5, f = 4$.

Luego

$$\left. \begin{aligned} a &= (10^c)^2 = 10^{2c} = 10^{2 \cdot 2e^2} = 10^{4e^2} \\ b &= (10^d)^2 = 10^{2d} = 10^{2 \cdot 2f^2} = 10^{4f^2} \end{aligned} \right\} \Rightarrow ab = 10^{4e^2} \cdot 10^{4f^2} = 10^{4e^2 + 4f^2} = 10^{4(e^2 + f^2)} = 10^{4(4^2 + 5^2)} = 10^{164} \quad (D)$$

Observación: En las soluciones oficiales

https://artofproblemsolving.com/wiki/index.php?title=2019_AMC_12A_Problems/Problem_15&oldid=122937
se presentan otros desarrollos alternativos a este.

16

Para que la suma de tres números dados sea impar es necesario que sean los tres impares o solo uno de ellos. Puesto que disponemos de cinco números impares (1, 3, 5, 7 y 9) las únicas posibilidades son las siguientes nueve:

I	I	I	I	P	P	I	P	P
I	P	P	I	I	I	I	P	P
I	P	P	I	P	P	I	I	I

I	I	I	P	I	P	P	I	P
P	I	P	I	I	I	P	I	P
P	I	P	P	I	P	I	I	I

I	I	I	P	P	I	P	P	I
P	P	I	I	I	I	P	P	I
P	P	I	P	P	I	I	I	I

En cada una de ellas podemos colocar 5 números impares y 4 números pares, luego los casos favorables son

$$F = 9 \cdot 5! \cdot 4!$$

Los casos totales son $T = 9!$, luego la probabilidad es

$$P = \frac{F}{T} = \frac{9 \cdot 5! \cdot 4!}{9!} = \frac{4 \cdot 3 \cdot 2}{8 \cdot 7 \cdot 6} = \frac{1}{14} \quad (\text{B})$$

17

Sean p, q, r las tres raíces del polinomio. Entonces se cumple

$$p^3 - 5p^2 + 8p - 13 = 0$$

$$q^3 - 5q^2 + 8q - 13 = 0$$

$$r^3 - 5r^2 + 8r - 13 = 0$$

Y por tanto

$$p^3 + q^3 + r^3 - 5(p^2 + q^2 + r^2) + 8(p + q + r) - 39 = 0$$

Es decir:

$$s_3 - 5s_2 + 8s_1 - 39 = 0 \Rightarrow s_3 = 5s_2 - 8s_1 + 13s_0$$

Y por tanto $a = 5, b = -8, c = 13 \Rightarrow a + b + c = 10$ (D)

Observación: Este problema se resuelve fácilmente aplicando las llamadas “Sumas de Newton” ([link](#))

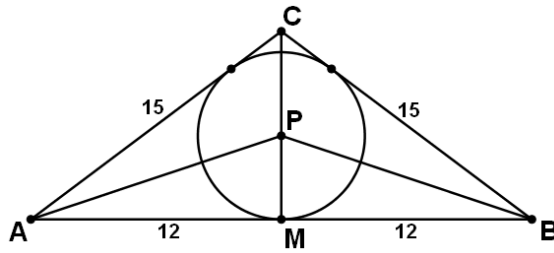
Fuente de la solución:

https://artofproblemsolving.com/wiki/index.php?title=2019_AMC_12A_Problems/Problem_17&oldid=132067

18

La intersección entre la esfera y el triángulo es el círculo inscrito del triángulo.

Sea $\triangle ABC$ dicho triángulo, con $AB = 24$ y $AC = BC = 15$. Sea M el punto medio del lado AB. Sea P su centro. Necesitamos calcular el radio de dicho círculo inscrito, es decir, la distancia PM.



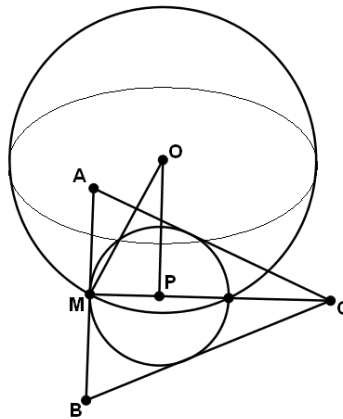
$$MC = \sqrt{15^2 - 12^2} = \sqrt{3^2 5^2 - 3^2 4^2} = \sqrt{3^2(5^2 - 4^2)} = 3\sqrt{5^2 - 4^2} = 3\sqrt{9} = 9$$

$$[\triangle ABC] = \frac{9 \cdot 24}{2} = 108$$

$$s = \frac{15 + 15 + 24}{2} = 27$$

$$\text{Aplicamos ahora 11.4.8: } [\triangle ABC] = r \cdot s \Leftrightarrow 108 = 27r \Rightarrow r = 4$$

Con estos datos ya podemos calcular la distancia OP:



$$OP = \sqrt{6^2 - 4^2} = \sqrt{36 - 16} = \sqrt{20} = 2\sqrt{5} \quad (\text{D})$$

19

Primera versión.

Aplicamos el Teorema del Coseno a cada uno de los lados, obteniendo el sistema de ecuaciones:

$$\begin{cases} a^2 = b^2 + c^2 - 2bc \frac{11}{16} \\ b^2 = a^2 + c^2 - 2ac \frac{7}{8} \\ c^2 = a^2 + b^2 - 2ab \left(-\frac{1}{4}\right) \end{cases} \Leftrightarrow \begin{cases} a^2 = b^2 + c^2 - \frac{11}{8}bc \\ b^2 = a^2 + c^2 - \frac{7}{4}ac \\ c^2 = a^2 + b^2 + \frac{1}{2}ab \end{cases}$$

$$\Leftrightarrow \begin{cases} a^2 = b^2 + a^2 + b^2 + \frac{1}{2}ab - \frac{11}{8}bc \\ b^2 = a^2 + a^2 + b^2 + \frac{1}{2}ab - \frac{7}{4}ac \end{cases} \Leftrightarrow \begin{cases} 0 = 2b^2 + \frac{1}{2}ab - \frac{11}{8}bc \\ 0 = 2a^2 + \frac{1}{2}ab - \frac{7}{4}ac \end{cases}$$

Puesto que estamos suponiendo en todo momento que $a, b, c \neq 0$, podemos simplificar:

$$\Leftrightarrow \begin{cases} 0 = 2b + \frac{1}{2}a - \frac{11}{8}c \\ 0 = 2a + \frac{1}{2}b - \frac{7}{4}c \end{cases} \Rightarrow \begin{cases} 0 = 2b + \frac{1}{2}a - \frac{11}{8}c \\ 0 = 8a + 2b - 7c \end{cases}$$

$$0 = \frac{15}{2}a - \frac{45}{8}c \Rightarrow c = \frac{8}{45} \frac{15}{2}a = \frac{4}{3}a \Rightarrow 2b = 7\frac{4}{3}a - 8a = \frac{4}{3}a \Rightarrow b = \frac{2}{3}a$$

Llegando a: $c = \frac{4}{3}a, b = \frac{2}{3}a$

La solución entera positiva mínima posible la obtendremos tomando $a = 3, b = 2, c = 4$, (se comprueba que estos valores satisfacen el sistema de ecuaciones inicial), y por tanto el perímetro será $3 + 2 + 4 = 9$ (A)

Segunda versión.

Podemos pasar los cosenos a senos mediante la identidad $\sin^2 a + \cos^2 a = 1$, puesto que $\sin \alpha \geq 0$ para todo ángulo $0 \leq \alpha \leq 180^\circ$, obteniendo:

$$\sin A = \frac{3\sqrt{15}}{16}, \sin B = \frac{\sqrt{15}}{8}, \sin C = \frac{\sqrt{15}}{4}$$

Ahora, aplicando el Teorema del seno, sabemos que los lados son proporcionales a los senos de los ángulos respectivos:

$$a : b : c = \sin A : \sin B : \sin C$$

Y observamos que estos senos están en proporción $3 : 2 : 4$, luego los menores valores enteros para los lados serán $3, 2, 4$, y por tanto el perímetro será $3 + 2 + 4 = 9$.

Fuente de la segunda versión: https://artofproblemsolving.com/wiki/index.php?title=2019_AMC_12A_Problems/Problem_19&oldid=115282

20

Llamaremos A al suceso $|x - y| > 1/2$. Queremos determinar $P(A)$.

Vamos a resolver este problema mediante probabilidad condicional. Para simplificar los esquemas, vamos a suponer que lanzamos la segunda moneda salga lo que salga en la primera, con lo cual nos encontramos una probabilidad que depende del lanzamiento de cuatro monedas.

Sea C "sacar cara", y sea X "sacar cruz".

Caso 1.

$$C_1 C_2 C_3 C_4 \Rightarrow \begin{cases} x = 0 \\ y = 0 \end{cases} \Rightarrow P(A | C_1 C_2 C_3 C_4) = 0$$

$$C_1 C_2 C_3 X_4 \Rightarrow \begin{cases} x=0 \\ y=1 \end{cases} \Rightarrow P(A | C_1 C_2 C_3 C_4) = 1$$

$$C_1 C_2 X_3^* \Rightarrow \begin{cases} x=0 \\ y \in [0,1] \end{cases} \Rightarrow P(A | C_1 C_2 X_3^*) = 1/2$$

$$C_1 X_2 C_3 C_4 \Rightarrow \begin{cases} x=1 \\ y=0 \end{cases} \Rightarrow P(A | C_1 X_2 C_3 C_4) = 1$$

$$C_1 X_2 C_3 X_4 \Rightarrow \begin{cases} x=1 \\ y=1 \end{cases} \Rightarrow P(A | C_1 X_2 C_3 X_4) = 0$$

$$C_1 X_2 X_3^* \Rightarrow \begin{cases} x=1 \\ y \in [0,1] \end{cases} \Rightarrow P(A | C_1 X_2 X_3^*) = 1/2$$

$$X_1^* C_3 C_4 \Rightarrow \begin{cases} x \in [0,1] \\ y=0 \end{cases} \Rightarrow P(A | X_1^* C_3 C_4) = 1/2$$

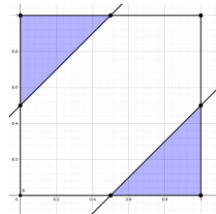
$$X_1^* C_3 X_4 \Rightarrow \begin{cases} x \in [0,1] \\ y=1 \end{cases} \Rightarrow P(A | X_1^* C_3 X_4) = 1/2$$

$$X_1^* X_3^* \Rightarrow \begin{cases} x \in [0,1] \\ y \in [0,1] \end{cases}$$

Este caso requiere un poco de análisis.

Representamos el conjunto $|x - y| > 1/2$ donde $x, y \in [0,1]$, mediante las rectas frontera

$x - y = 1/2$ y $x - y = -1/2$:



$$\text{Y su área es } (1/2)^2 = 1/4, \text{ así pues, } X_1^* X_3^* \Rightarrow \begin{cases} x \in [0,1] \\ y \in [0,1] \end{cases} \Rightarrow P(A | X_1^* X_3^*) = 1/4$$

Puesto que los lanzamientos son independientes, la probabilidad de cada uno será $\frac{1}{2^4} = \frac{1}{16}$,

luego el esquema de casos es

$C_1 C_2 C_3 C_4$	0
$C_1 C_2 C_3 X_4$	1
$C_1 C_2 X_3 C_4$	1/2
$C_1 C_2 X_3 X_4$	1/2
$C_1 X_2 C_3 C_4$	1
$C_1 X_2 C_3 X_4$	0
$C_1 X_2 X_3 C_4$	1/2
$C_1 X_2 X_3 X_4$	1/2
$X_1 C_2 C_3 C_4$	1/2
$X_1 C_2 C_3 X_4$	1/2
$X_1 C_2 X_3 C_4$	1/4
$X_1 C_2 X_3 X_4$	1/4

$X_1X_2C_3C_4$	$1/2$
$X_1X_2C_3X_4$	$1/2$
$X_1X_2X_3C_4$	$1/4$
$X_1X_2X_3X_4$	$1/4$

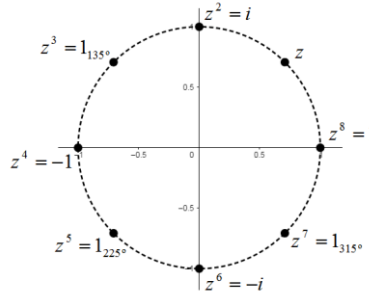
$$P(A) = \frac{1}{16} \left(0 + 1 + \frac{1}{2} + \frac{1}{2} + 1 + 0 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{4} + \frac{1}{4} + \frac{1}{2} + \frac{1}{2} + \frac{1}{4} + \frac{1}{4} \right) = \frac{7}{16} \quad (B)$$

Observación. En las soluciones oficiales encontramos un desarrollo mucho más simple, reducido a 4 casos solamente.

21

En primer lugar vemos que $z = \frac{1+i}{\sqrt{2}} = 1_{45^\circ}$ es la unidad principal de orden 8, es decir:

$$z^2 = i, \quad z^3 = 1_{135^\circ}, \quad z^4 = -1, \quad z^5 = 1_{225^\circ}, \quad z^6 = -i, \quad z^7 = 1_{315^\circ}, \quad z^8 = 1.$$



$$\begin{aligned} z^{1^2} &= z, z^{2^2} = z^4 = -1, z^{3^2} = z^9 = z, z^{4^2} = z^{16} = 1, z^{5^2} = z^{25} = z, \\ z^{6^2} &= z^{36} = z^4 = -1, z^{7^2} = z^{49} = z, z^{8^2} = z^{64} = 1, z^{9^2} = z^{81} = z, \\ z^{10^2} &= z^{100} = z^4 = -1, z^{11^2} = z^{121} = z, z^{12^2} = z^{144} = 1 \end{aligned}$$

Y por tanto:

$$z^{1^2} + z^{2^2} + z^{3^2} + \dots + z^{12^2} = z - 1 + z + 1 + z - 1 + z + 1 + z - 1 + z + 1 = 6z$$

Por otro lado, puesto que z pertenece a la circunferencia unidad, se cumple $z^{-1} = \bar{z}$ y por tanto

$$\begin{aligned} z^{-1^2} &= \bar{z}, z^{-2^2} = -1, z^{-3^2} = \bar{z}, z^{-4^2} = 1, z^{-5^2} = \bar{z}, z^{-6^2} = -1 \\ z^{-7^2} &= \bar{z}, z^{-8^2} = 1, z^{-9^2} = \bar{z}, z^{-10^2} = -1, z^{-11^2} = \bar{z}, z^{-12^2} = 1 \end{aligned}$$

Luego:

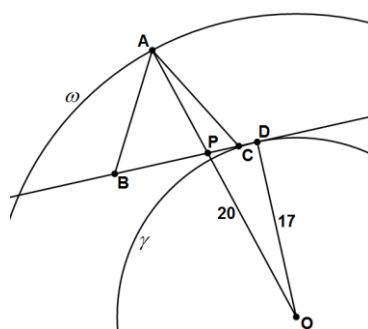
$$\frac{1}{z^{1^2}} + \frac{1}{z^{2^2}} + \frac{1}{z^{3^2}} + \dots + \frac{1}{z^{12^2}} = 6\bar{z}$$

Y finalmente:

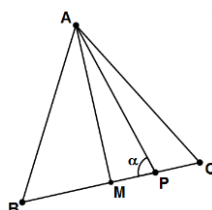
$$\left(z^{1^2} + z^{2^2} + z^{3^2} + \dots + z^{12^2} \right) \cdot \left(\frac{1}{z^{1^2}} + \frac{1}{z^{2^2}} + \frac{1}{z^{3^2}} + \dots + \frac{1}{z^{12^2}} \right) = 6z \cdot 6\bar{z} = 36z\bar{z} = 36|z|^2 = 36 \quad (C)$$

22

Sea D el punto de tangencia entre BC y γ .



En primer lugar, estudiamos el triángulo $\triangle ABC$.



Suponiendo un lado $AC = 2$, $AM = \sqrt{2^2 - 1^2} = \sqrt{3}$, y por tanto, si $\alpha = \angle APM$,

$$\tan \alpha = \frac{\sqrt{3}}{1/2} = 2\sqrt{3}$$

$$PA = \sqrt{3 + (1/2)^2} = \frac{\sqrt{13}}{2} \Rightarrow \cos \alpha = \frac{1/2}{\sqrt{13}/2} = \frac{1}{\sqrt{13}}$$

Volviendo al esquema del problema, vemos que $\alpha = \angle APM = \angle DPO$, luego

$$\frac{DO}{PD} = \frac{17}{PD} = \tan \alpha = 2\sqrt{3} \Rightarrow PD = \frac{17}{2\sqrt{3}} \Rightarrow$$

$$PO = \sqrt{\left(\frac{17}{2\sqrt{3}}\right)^2 + 17^2} = 17\sqrt{\left(\frac{1}{2\sqrt{3}}\right)^2 + 1} = 17\sqrt{\frac{1}{12} + 1} = 17\sqrt{\frac{13}{12}} \Rightarrow$$

$$PA = 20 - 17\sqrt{\frac{13}{12}}$$

Puesto que los triángulos rectángulos $\triangle APM$ y $\triangle OPD$ son semejantes, tenemos

$$\cos \alpha = \frac{1}{\sqrt{13}} = \frac{PM}{AP} = \frac{PM}{20 - 17\sqrt{\frac{13}{12}}} \Rightarrow PM = \frac{20 - 17\sqrt{13}/\sqrt{12}}{\sqrt{13}} = \frac{20}{\sqrt{13}} - \frac{17}{\sqrt{12}} \Rightarrow$$

$$AB = BC = 4PM = \frac{80}{\sqrt{13}} - \frac{68}{\sqrt{12}} = \frac{80}{\sqrt{13}} - \frac{34}{\sqrt{3}}$$

La respuesta es $80 + 13 + 34 + 3 = 130$ (E)

Nota: En la página web

https://artofproblemsolving.com/wiki/index.php?title=2019_AMC_12A_Problems/Problem_22&oldid=103269

se presenta una solución alternativa sin trigonometría.

23

Analizamos la operación binaria del enunciado:

$$a_n = (n \nabla (n-1)) \diamond a_{n-1} = n^{1/\log_7(n-1)} \diamond a_{n-1} = \left(n^{1/\log_7(n-1)}\right)^{\log_7(a_{n-1})} = n^{\log_7(a_{n-1})/\log_7(n-1)}$$

Aplicamos la fórmula del cambio de base:

$$\log_7(n-1) \cdot \log_{n-1}(a_{n-1}) = \log_7(a_{n-1}) \Rightarrow \log_{n-1}(a_{n-1}) = \log_7(a_{n-1})/\log_7(n-1)$$

Luego

$$a_n = n^{\log_7(a_{n-1})/\log_7(n-1)} = n^{\log_{n-1}(a_{n-1})}$$

Y por tanto

$$\log_n(a_n) = \log_{n-1}(a_{n-1})$$

Así pues, $\log_{2019}(a_{2019}) = \log_{2018}(a_{2018}) = \log_{2017}(a_{2017}) = \dots = \log_3(a_3)$

Y este último lo podemos calcular:

$$a_3 = 3 \nabla 2 = 3^{1/\log_7(2)} \Rightarrow \log_3(a_3) = 1/\log_7(2)$$

Así pues,

$$\log_{2019}(a_{2019}) = \frac{1}{\log_7(2)} = \log_2(7)$$

En donde hemos aplicado la fórmula del cambio de base:

$$\log_2(7) \log_7(2) = \log_2(2) = 1 \Rightarrow \log_2(7) = \frac{1}{\log_7(2)}$$

Aplicamos de nuevo la fórmula del cambio de base:

$$\begin{aligned} \log_7(a_{2019}) &= \log_7 2019 \cdot \log_{2019}(a_{2019}) = \log_7 2019 \cdot \log_2 7 = \\ &= \frac{\log 2019}{\log 7} \cdot \frac{\log 7}{\log 2} = \frac{\log 2019}{\log 2} = \log_2 2019 \end{aligned}$$

Finalmente, para calcular aproximadamente $\log_2 2019$ vemos que

$$\left. \begin{array}{l} 2^{10} = 1024 \\ 2^{11} = 2048 \end{array} \right\} \Rightarrow \log_2 2019 \cong 11 \quad (\text{D})$$

Fuente de la solución: https://artofproblemsolving.com/wiki/index.php?title=2019_AMC_12A_Problems/Problem_23&oldid=122483

24

$\frac{(n^2-1)!}{(n!)^n}$ es un entero si y solo si $(n!)^n \mid (n^2-1)!$.

Siguiendo con las definiciones introducidas en AR/16.4 sobre orden p-ádico de un número, tenemos que

$$v_p(a) \leq v_p(b) \Rightarrow a \mid b$$

Luego vamos a ver para qué enteros n entre 1 y 50 se cumple:

$$v_p((n!)^n) \leq v_p((n^2 - 1)!)$$

Por otro lado, se cumple

$$v_p(a^b) = b \cdot v_p(a)$$

Así pues, queremos comprobar la desigualdad

$$n \cdot v_p(n!) \leq v_p((n^2 - 1)!) \quad (*)$$

Podemos reducir nuestro estudio a las potencias de números primos.

Primer caso: Si $n = p$, un número primo. Aplicando la fórmula de Polignac:

$$v_p((n^2 - 1)!) = \sum_{k=1}^{\infty} \left\lfloor \frac{n^2 - 1}{p^k} \right\rfloor = (p - 1) + 0 + \dots + 0 = p - 1$$

$$v_p(n!) = 1 + 0 + 0 + \dots = 1 \quad (\text{ver 16.14b})$$

Y por lo tanto la desigualdad (*) se convierte en

$$p \leq p - 1$$

Lo cual no es cierto para ningún p .

Segundo caso: Si $n = p^2$, el cuadrado de un primo. Aplicando la fórmula de Polignac:

$$v_p((n^2 - 1)!) = v_p((p^4 - 1)!) = p^3 + p^2 + p - 3$$

$$v_p(n!) = p + 1 \quad (\text{ver 16.14c})$$

Y por lo tanto la desigualdad (*) se convierte en

$$p^2(p + 1) \leq p^3 + p^2 + p - 3 \Leftrightarrow 0 \leq p - 3 \Leftrightarrow 3 \leq p$$

Lo cual es cierto para todo primo excepto el 2.

Caso general: Se puede demostrar que para todo $n = p^i$ con $i > 2$ se satisface la desigualdad (*)

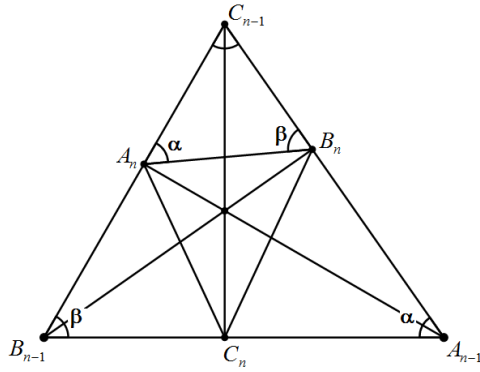
Así pues, hay 16 números para los cuales no se verifica la desigualdad, y $50 - 16 = 34$ números para los que sí se satisface (D).

Fuente de esta solución: https://artofproblemsolving.com/wiki/index.php?title=2019_AMC_10A_Problems/Problem_25&oldid=132200

25

$\Delta A_n B_n C_n$ es el triángulo órtico de $\Delta A_{n-1} B_{n-1} C_{n-1}$, introducido en GA/11.3.3. En particular sabemos (GA/11.3.4d) que la recta $A_n B_n$ es la antiparalela de $A_{n-1} B_{n-1}$, es decir:

$$\alpha = \angle C_{n-1} A_n B_n = \angle C_{n-1} A_{n-1} B_{n-1} \quad \text{y} \quad \beta = \angle C_{n-1} B_n A_n = \angle C_{n-1} B_{n-1} A_{n-1}$$



En nuestro caso, suponiendo $\angle A_0 = 59.999^\circ$, $\angle B_0 = 60^\circ$ y $\angle C_0 = 60.001^\circ$, tendremos:

$$\varepsilon = 0.001^\circ$$

$$\angle A_0 = 59.999^\circ = 60^\circ - \varepsilon$$

$$\angle A_1 = 180^\circ - \angle A_0 - \angle B_0 = 180^\circ - 2\angle A_0 = 180^\circ - 2(60^\circ - \varepsilon) = 180^\circ - 120^\circ - 2\varepsilon = 60^\circ - 2\varepsilon$$

$$\angle A_2 = 180^\circ - 2\angle A_1 = 180^\circ - 2(60^\circ - 2\varepsilon) = 180^\circ - 120^\circ + 4\varepsilon = 60^\circ + 4\varepsilon$$

$$\angle A_3 = 180^\circ - 2\angle A_2 = 180^\circ - 2(60^\circ + 4\varepsilon) = 180^\circ - 120^\circ - 8\varepsilon = 60^\circ - 8\varepsilon = 60^\circ - 2^3 \varepsilon$$

En general, vemos que los ángulos de $\Delta A_n B_n C_n$ serán 60° , $60^\circ - 2^n \varepsilon$ y $60^\circ + 2^n \varepsilon$, luego nuestro problema se reduce a encontrar el menor n para el cual

$$60^\circ + 2^n \varepsilon > 90^\circ \Leftrightarrow 2^n \varepsilon > 30^\circ \Leftrightarrow 2^n \frac{1}{1000} > 30 \Leftrightarrow 2^n > 30000$$

Que es claramente 15, pues $2^{14} = 16384$ y $2^{15} = 32768$ (E).

AMC 12B 2019 Enunciados

1

Alicia dispone de dos recipientes. El primero está lleno de agua hasta sus $\frac{5}{6}$ partes y el segundo está vacío. Vierte todo el agua del primer recipiente en el segundo, y observa que el segundo recipiente queda lleno hasta sus $\frac{3}{4}$ partes. Determina la razón entre el volumen del primer recipiente y del segundo.

(A) $\frac{5}{8}$ (B) $\frac{4}{5}$ (C) $\frac{7}{8}$ (D) $\frac{9}{10}$ (E) $\frac{11}{12}$

2

Consideremos la afirmación "Si n no es primo, entonces $n - 2$ es primo". ¿Cuáles de los siguientes valores de n es un contraejemplo de esta afirmación?

(A) 11 (B) 15 (C) 19 (D) 21 (E) 27

3

Determina, entre todas las siguientes transformaciones rígidas (isometrías), la que transforma el segmento $\overline{AB}$ en el segmento $\overline{A'B'}$, de forma que la imagen de $A = (-2, 1)$ es $A' = (2, -1)$ y la imagen de $B = (-1, 4)$ es $B' = (1, -4)$.

- (A) Simetría respecto del eje Y.
- (B) Rotación de 90° alrededor del origen y en el sentido contrahorario.
- (C) Traslación de 3 unidades hacia la derecha y 5 unidades hacia abajo.
- (D) Simetría respecto del eje X.
- (E) Rotación de 180° alrededor del origen y en el sentido horario.

4

Sea n un entero positivo que satisface la ecuación

$$(n+1)! + (n+2)! = 440 \cdot n!$$

¿Cuál es la suma de los dígitos de n ?

(A) 2 (B) 5 (C) 10 (D) 12 (E) 15

5

Los caramelos de una tienda cuestan valores enteros en céntimos. Casper tiene exactamente suficiente dinero para comprar 12 caramelos rojos, 14 caramelos verdes, 15 caramelos azules, o n caramelos púrpura (una de las cuatro opciones). Un caramelo púrpura cuesta 20 céntimos. Determina el menor valor posible de n .

(A) 18 (B) 21 (C) 24 (D) 25 (E) 28

6

Sean A y B dos puntos del plano separados en 10 unidades. ¿Cuántos puntos hay en el plano tales que el perímetro de $\triangle ABC$ sea de 50 unidades y el área de $\triangle ABC$ sea de 100 unidades cuadradas?

(A) 0 (B) 2 (C) 4 (D) 8 (E) una cantidad infinita.

7

Determina la suma de todos los números reales x para los cuales la mediana de los números 4, 6, 8, 17, x es igual a la media de estos cinco números.

(A) -5 (B) 0 (C) 5 (D) $\frac{15}{4}$ (E) $\frac{35}{4}$

8

Sea $f(x) = x^2(1-x)^2$. Determina el valor de la suma

$$f\left(\frac{1}{2019}\right) - f\left(\frac{2}{2019}\right) + f\left(\frac{3}{2019}\right) - f\left(\frac{4}{2019}\right) + \dots + f\left(\frac{2017}{2019}\right) - f\left(\frac{2018}{2019}\right)$$

(A) 0 (B) $\frac{1}{2019^4}$ (C) $\frac{2018^2}{2019^4}$ (D) $\frac{2020^2}{2019^4}$ (E) 1

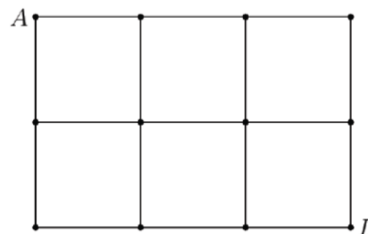
9

¿Para cuantos valores enteros de x se puede construir un triángulo cuyos lados tengan longitudes $\log_2 x$, $\log_4 x$, 3?

(A) 57 (B) 59 (C) 61 (D) 62 (E) 63

10

La siguiente figura es un mapa en el que se muestran 12 ciudades y 17 carreteras que conectan ciertos pares de ciudades. Paula desea recorrer exactamente 13 de estas carreteras, empezando en la ciudad A y finalizando en la ciudad L, sin recorrer ninguna porción de carretera más de una vez (Paula puede visitar una ciudad más de una vez).



¿Cuántas rutas diferentes puede tomar Paula?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

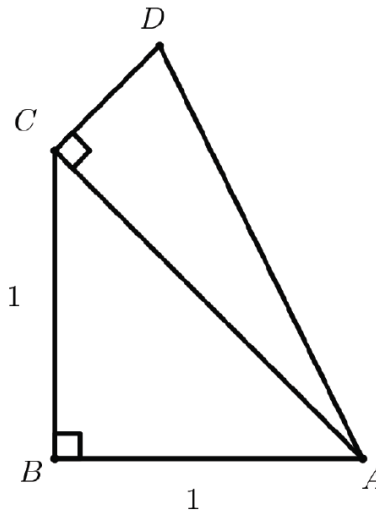
11

¿Cuántos pares no ordenados de aristas de un cubo determinan un plano?

- (A) 12 (B) 28 (C) 36 (D) 42 (E) 66

12

Construimos el triángulo $\triangle ACD$, con ángulo recto en C, sobre la hipotenusa $\overline{AC}$ de un triángulo recto isósceles $\triangle ABC$ con catetos de longitud 1, de forma que los dos triángulos tengan perímetros iguales. Determina $\sin(2\angle BAD)$.



- (A) $\frac{1}{3}$ (B) $\frac{\sqrt{2}}{2}$ (C) $\frac{3}{4}$ (D) $\frac{7}{9}$ (E) $\frac{\sqrt{3}}{2}$

13

Tiramos de forma aleatoria e independiente una bola roja y una bola verde en una sucesión de botes numerados con enteros positivos, de forma que la probabilidad de que una bola entre en el bote k es 2^{-k} , para todo $k = 1, 2, 3, \dots$ ¿Cuál es la probabilidad de que la bola roja entre en un bote numerado con un número mayor que la bola verde?

- (A) $\frac{1}{4}$ (B) $\frac{2}{7}$ (C) $\frac{1}{3}$ (D) $\frac{3}{8}$ (E) $\frac{3}{7}$

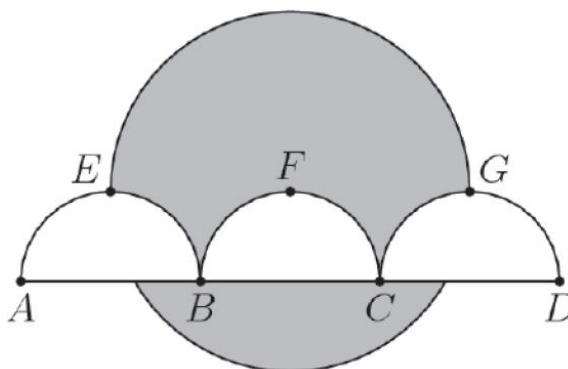
14

Sea S el conjunto de todos los divisores enteros positivos de 100 000. ¿Cuántos números son el producto de dos elementos diferentes de S?

- (A) 98 (B) 100 (C) 117 (D) 119 (E) 121

15

Como se muestra en la figura, un segmento $\overline{AD}$ se divide en tres segmentos iguales mediante los puntos B y C, de forma que $AB = BC = CD = 2$. Trazamos tres semicírculos de radio 1, AEB, BFC y CGD, con sus diámetros en $\overline{AD}$, que son tangentes a la recta EG en E, F y G, respectivamente. Un círculo de radio 2 tiene su centro en F.



El área de la región sombreada en la figura, interna de este círculo pero externa a los tres semicírculos, se puede expresar de la forma

$$\frac{a}{b} \cdot \pi - \sqrt{c} + d$$

Donde a, b, c y d son enteros positivos y a y b son coprimos. Determina $a + b + c + d$.

(A) 13 (B) 14 (C) 15 (D) 16 (E) 17

16

Tenemos una fila de nenúfares numerados del 0 al 12, en este orden. En las posiciones 3 y 6 hay depredadores, y en la posición 10 una apetitosa mosca. La rana Fiona empieza en la posición 0, y desde una posición puede saltar una posición o dos, con una probabilidad de $1/2$ y siempre hacia delante (por ejemplo, de la 7 puede saltar a la 8 o a la 9). Determina la probabilidad de que Fiona alcance la posición 10 sin caer en las posiciones 3 o 6.

(A) $\frac{15}{256}$ (B) $\frac{1}{16}$ (C) $\frac{15}{128}$ (D) $\frac{1}{8}$ (E) $\frac{1}{4}$

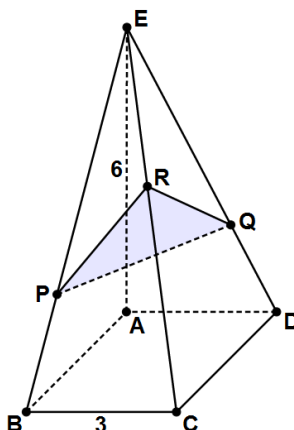
17

Determina la cantidad de números complejos z tales que 0 , z y z^3 , representados por puntos en el plano complejo, son los tres vértices de un triángulo equilátero.

(A) 0 (B) 1 (C) 2 (D) 4 (E) una cantidad infinita.

18

Una pirámide de base cuadrada $ABCD$ tiene base $ABCD$, con 3 cm de lado, y una altura $\overline{AE}$ perpendicular a la base, que mide 6 cm. Sea P el punto en $\overline{BE}$, a un tercio de la distancia de B a E ; sea Q el punto en $\overline{DE}$, a un tercio de la distancia de D a E ; y sea R el punto en $\overline{CE}$, a dos tercios de la distancia de C a E . Determina el área, en centímetros cuadrados, de $\triangle PQR$.



- (A) $\frac{3\sqrt{2}}{2}$ (B) $\frac{3\sqrt{3}}{2}$ (C) $2\sqrt{2}$ (D) $2\sqrt{3}$ (E) $3\sqrt{2}$

19

Raashan, Sylvia y Ted juegan al siguiente juego: Cada uno empieza con 1\$. Una campana suena cada 15 segundos, y cada vez que suena cada uno de los tres jugadores que tenga algo de dinero elige a uno de los otros dos jugadores, al azar, de forma simultánea e independiente, y le da 1\$. Por ejemplo, Raashan y Ted pueden dar 1\$ a Sylvia cada uno, y Sylvia puede dar su dólar a Ted, de forma que Raashan tendrá 0\$, Sylvia tendrá 2\$ y Ted tendrá 1\$. En el siguiente toque de campana, Raashan no tiene dinero para dar, pero Sylvia y Ted pueden, por ejemplo, darse 1\$ mutuamente, y por lo tanto quedarán los tres con la misma cantidad que en la partida anterior. Determina la probabilidad de que, una vez haya sonado la campana 2019 veces, cada jugador tenga 1\$.

- (A) $\frac{1}{7}$ (B) $\frac{1}{4}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$ (E) $\frac{2}{3}$

20

Supongamos que los puntos $A = (6,13)$ y $B = (12,11)$ pertenecen a una circunferencia ω . Supongamos, además, que las tangentes a ω por A y por B se cortan en un punto del eje x . Determina el área de ω .

- (A) $\frac{83\pi}{8}$ (B) $\frac{21\pi}{2}$ (C) $\frac{85\pi}{8}$ (D) $\frac{43\pi}{4}$ (E) $\frac{87\pi}{8}$

21

Determina el número de polinomios cuadráticos con coeficientes reales tales que su conjunto de raíces es igual a su conjunto de coeficientes. Es decir: si el polinomio es $ax^2 + bx + c$, $a \neq 0$, y las raíces son r y s , entonces la condición es $\{a, b, c\} = \{r, s\}$.

(A) 3 (B) 4 (C) 5 (D) 6 (E) una cantidad infinita

22

Definimos una secuencia recursivamente por $x_0 = 5$ y

$$x_{n+1} = \frac{x_n^2 + 5x_n + 4}{x_n + 6}$$

para todo entero no negativo n . Sea m el menor entero positivo tal que

$$x_m \leq 4 + \frac{1}{2^{20}}$$

(A) $[9, 26]$ (B) $[27, 80]$ (C) $[81, 242]$ (D) $[243, 728]$ (E) $[729, \infty)$

23

Determina el número de secuencias de “0” y “1” de longitud 19, que empiezan y terminan con “0”, no contienen dos “0” consecutivos, y no contienen tres “1” consecutivos.

(A) 55 (B) 60 (C) 65 (D) 70 (E) 75

24

Sea $\omega = -\frac{1}{2} + \frac{1}{2}i\sqrt{3}$. Denotamos por S el conjunto de todos los puntos del plano

complejo de la forma $a + b\omega + c\omega^2$, donde $0 \leq a \leq 1$, $0 \leq b \leq 1$, $0 \leq c \leq 1$. Determina el área de S .

(A) $\frac{1}{2}\sqrt{3}$ (B) $\frac{3}{4}\sqrt{3}$ (C) $\frac{3}{2}\sqrt{3}$ (D) $\frac{1}{2}\pi\sqrt{3}$ (E) π

25

Sea $ABCD$ un cuadrilátero convexo tal que $BC = 2$ y $CD = 6$. Supongamos que los baricentros de $\triangle ABC$, $\triangle BCD$ y $\triangle ACD$ forman los vértices de un triángulo equilátero. Determina el máximo valor posible del área de $ABCD$.

(A) 27 (B) $16\sqrt{3}$ (C) $12 + 10\sqrt{3}$ (D) $9 + 12\sqrt{3}$ (E) 30

AMC 12B 2020 Soluciones (letra)

1. C
2. A
3. E
4. D
5. C
6. D
7. C
8. D
9. C
10. B
11. D
12. E
13. D
14. A
15. C
10. B
17. C
11. B
19. C
20. D
21. C
22. C
23. B
24. A
25. B

AMC 12B 2020 Soluciones desarrolladas

1

Vemos que la ecuación que relaciona los volúmenes de los recipientes es

$$\frac{5}{6}A = \frac{3}{4}B \Rightarrow A = \frac{3 \cdot 6}{4 \cdot 5}B = \frac{3 \cdot 3}{2 \cdot 5}B = \frac{9}{10}B \Rightarrow \frac{A}{B} = \frac{9}{10}$$

2

Claramente (E), pues 27 no es primo y $27 - 2 = 25$ tampoco lo es.

3

Por simple observación, está claro que es la (E).

4

$$440 \cdot n! = (n+1)! + (n+2)! = n!(n+1) + n!(n+1)(n+2) = n!(n+1)(1+n+2) = n!(n+1)(n+3) \Rightarrow$$

$$440 \cdot n! = n!(n+1)(n+3) \Rightarrow 440 = (n+1)(n+3)$$

$$(n+1)(n+3) = 440 = 2^3 \cdot 5 \cdot 11$$

Probando posibles combinaciones (o resolviendo la ecuación cuadrática resultante) vemos que una que satisface esta condición es

$$\left. \begin{array}{l} n+1 = 2^2 \cdot 5 = 20 \\ n+3 = 2 \cdot 11 = 22 \end{array} \right\} \Rightarrow n = 19$$

Y por tanto la respuesta correcta es $1+9=10$ (C).

5

Sea x la cantidad de dinero de Casper, en céntimos.

Luego x debe ser múltiple de 12, 14 y 15, y por tanto el valor mínimo es

$$x = \text{mcm}(12,14,15) = 420 \text{ cent.}$$

Para esta cantidad, puede comprar $\frac{420}{20} = 21$ caramelos púrpura. (B).

6

Sean $a = \overline{BC}$, $b = \overline{AC}$, $c = \overline{AB} = 10$.

$$50 = a + b + c = a + b + 10 \Rightarrow a + b = 40$$

Aplicando la fórmula de Heron:

$$s = p/2 = 25$$

$$100 = \sqrt{s(s-a)(s-b)(s-c)} \Rightarrow$$

$$100^2 = 25(25-a)(25-b)(25-10) \Rightarrow (25-a)(25-b) = \frac{100^2}{25 \cdot 15} = \frac{80}{3}$$

Llegamos a la ecuación

$$(25 - a)(25 - (40 - a)) = \frac{80}{3} \Leftrightarrow$$

$$(25 - a)(a - 15) = \frac{80}{3} \Leftrightarrow$$

$$(a - 25)(3a - 45) = -80 \Leftrightarrow$$

$$3a^2 - 120a + 1125 = -80 \Leftrightarrow$$

$$3a^2 - 120a + 1205 = 0$$

El discriminante de esta ecuación es $\Delta = 120^2 - 4 \cdot 3 \cdot 1205 = 14400 - 14460 < 0$, negativo, luego no tiene solución: No hay ningún triángulo que cumpla las condiciones del enunciado (A).

7

Supongamos que $x \geq 8$. Entonces la mediana es $M = 8$ y por tanto la ecuación resultante es

$$M = 8 = \frac{4 + 6 + 8 + 17 + x}{5} = \frac{35 + x}{5} \Leftrightarrow 40 = 35 + x \Rightarrow x = 5 \text{ absurdo.}$$

Supongamos que $6 < x < 8$. Entonces la mediana es $M = x$ y por tanto la ecuación resultante es

$$x = \frac{35 + x}{5} \Leftrightarrow 5x = 35 + x \Rightarrow x = \frac{35}{4} = 8.75 \text{ absurdo.}$$

Supongamos que $x \leq 6$. Entonces la mediana es $M = 6$ y por tanto la ecuación resultante es

$$6 = \frac{35 + x}{5} \Leftrightarrow 30 = 35 + x \Rightarrow x = -5, \text{ que es un resultado aceptable. (A)}$$

8

$$\text{Vemos que } f\left(\frac{n}{2019}\right) = f\left(\frac{2019-n}{2019}\right)$$

$$f\left(\frac{n}{2019}\right) = \left(\frac{n}{2019}\right)^2 \left(1 - \frac{n}{2019}\right)^2 = \left(\frac{n}{2019}\right)^2 \left(\frac{2019-n}{2019}\right)^2$$

$$f\left(\frac{2019-n}{2019}\right) = \left(\frac{2019-n}{2019}\right)^2 \left(1 - \frac{2019-n}{2019}\right)^2 = \left(\frac{2019-n}{2019}\right)^2 \left(\frac{n}{2019}\right)^2$$

Vemos que los términos se cancelan por parejas, y al haber 1009 parejas en esta suma, el resultado final es cero (A).

9

Las tres desigualdades que se deben cumplir son:

$$\log_2 x + \log_4 x > 3, \log_2 x + 3 > \log_4 x, \log_4 x + 3 > \log_2 x$$

Por otro lado, aplicando la fórmula del cambio de base:

$$\log_2 4 \cdot \log_4 x = \log_2 x \Leftrightarrow 2 \log_4 x = \log_2 x \text{ y por tanto tenemos:}$$

$$\log_2 x + \log_4 x > 3 \Leftrightarrow 2 \log_4 x + \log_4 x > 3 \Leftrightarrow 3 \log_4 x > 3 \Leftrightarrow \log_4 x > 1$$

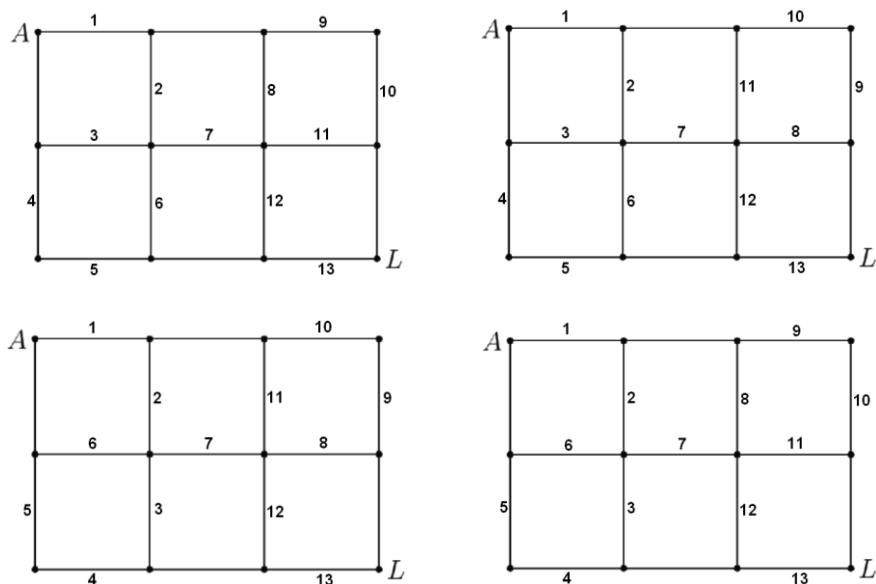
$$\log_2 x + 3 > \log_4 x \Leftrightarrow 2\log_4 x + 3 > \log_4 x \Leftrightarrow \log_4 x + 3 > 0$$

$$\log_4 x + 3 > \log_2 x \Leftrightarrow \log_4 x + 3 > 2\log_4 x \Leftrightarrow 3 > \log_4 x$$

Así pues tenemos $1 < \log_4 x < 3 \Leftrightarrow 4^1 < 4^{\log_4 x} < 4^3 \Leftrightarrow 4 < x < 64$, así pues, los valores enteros posibles van del 5 al 63, es decir, 59 (B).

10

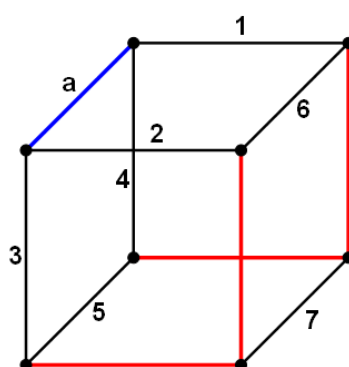
Hay cuatro posibilidades, en función del sentido en el que "tomamos la curva" de las dos esquinas:



Probando se observa que cualquier otra opción no es aceptable. (E)

11

Fijada una arista del cubo, a , hay 7 aristas más con las que puede generar un plano:



Puesto que los pares no están ordenados, deberemos dividir el total entre dos. Así pues, el total de parejas será $\frac{12 \cdot 7}{2} = 42$ (D).

12

Por Pitágoras, $AC = \sqrt{2}$.

El Perímetro de $\triangle ABC$ es $\sqrt{2} + 2$, luego $CD + AD = 2$.

Por Pitágoras,

$$AD^2 = AC^2 + 2 \Rightarrow 2 = AD^2 - AC^2 = (AD - AC)(AD + AC) = 2(AD - AC) \Rightarrow$$

$$AD - AC = \frac{2}{2} = 1$$

Luego

$$\left. \begin{array}{l} CD + AD = 2 \\ AD - AC = 1 \end{array} \right\} \Rightarrow AD = \frac{3}{2}, CD = \frac{1}{2}$$

Así pues:

$$\sin(\angle BAC) = \cos(\angle BAC) = \frac{1}{\sqrt{2}}$$

$$\sin(\angle CAD) = \frac{1/2}{3/2} = \frac{1}{3}, \cos(\angle CAD) = \frac{\sqrt{2}}{3/2} = \frac{2\sqrt{2}}{3}$$

Aplicando las identidades de la suma de ángulos:

$$\sin(\angle BAD) = \sin(\angle BAC)\cos(\angle CAD) + \cos(\angle BAC)\sin(\angle CAD) =$$

$$= \frac{1}{\sqrt{2}} \frac{1}{3} + \frac{1}{\sqrt{2}} \frac{2\sqrt{2}}{3} = \frac{1}{3\sqrt{2}} (1 + 2\sqrt{2})$$

$$\cos(\angle BAD) = \cos(\angle BAC)\cos(\angle CAD) - \sin(\angle BAC)\sin(\angle CAD) =$$

$$= \frac{1}{\sqrt{2}} \frac{2\sqrt{2}}{3} - \frac{1}{\sqrt{2}} \frac{1}{3} = \frac{1}{3\sqrt{2}} (2\sqrt{2} - 1)$$

Aplicando la identidad del ángulo doble:

$$\sin(2\angle BAD) = 2\sin(\angle BAD)\cos(\angle BAD) = 2 \frac{1}{3\sqrt{2}} (1 + 2\sqrt{2}) \frac{1}{3\sqrt{2}} (2\sqrt{2} - 1) =$$

$$= \frac{1}{9} (2\sqrt{2} - 1)(2\sqrt{2} + 1) = \frac{1}{9} (4 \cdot 2 - 1) = \frac{7}{9} \quad (D)$$

Observación:

Un "truco" interesante sería utilizar la identidad $\sin(90 + 2\alpha) = \cos(2\alpha)$, aplicada a $\alpha = \angle DAC$, con lo que (casi) todo se reduce a calcular:

$$\cos(\alpha) = \frac{2\sqrt{2}}{3} \Rightarrow \cos(2\alpha) = 2\left(\frac{8}{9}\right) - 1 = \frac{7}{9}$$

13

Se pueden dar tres casos: Que entre en un número menor, que entre en un número mayor o que entren las dos en el mismo número. Por simetría los dos casos tendrán la misma probabilidad, luego calculemos la probabilidad de que las dos bolas entren en el mismo bote.

Podemos interpretar este problema en términos de probabilidad condicional con una partición infinita, luego la probabilidad de que las bolas entren en el mismo bote será, aplicando la fórmula de la serie geométrica (ver PA/12):

$$P = \sum_{k=1}^{\infty} \frac{1}{2^k} \cdot \frac{1}{2^k} = \sum_{k=1}^{\infty} \frac{1}{4^k} = \sum_{k=0}^{\infty} \frac{1}{4^k} - 1 = \frac{1}{1-1/4} - 1 = \frac{4}{3} - 1 = \frac{1}{3}$$

Luego las otras dos probabilidades serán $1 - \frac{1}{3} = \frac{2}{3}$, y por tanto, por ser ambas equiprobables, cada una tendrá asignada una probabilidad $\frac{2}{3} \div 2 = \frac{1}{3}$ (C)

14

$100000 = 2^5 \cdot 5^5$, luego S está formado por todos los números de la forma $2^a \cdot 5^b$ con $0 \leq a, b \leq 5$, y por tanto los productos de dos números de S serán todos aquellos números de la forma $2^c \cdot 5^d$ con $0 \leq c, d \leq 10$.

En total hay $11 \cdot 11 = 121$ casos.

Pero debemos observar que algunos casos solo se pueden aparecer como el producto de dos elementos iguales (recordemos que, por separado, el exponente máximo es 5).

$$1 = 2^0 \cdot 5^0 = (2^0 \cdot 5^0)(2^0 \cdot 5^0)$$

$$1 = 2^{10} = 2^5 \cdot 2^5$$

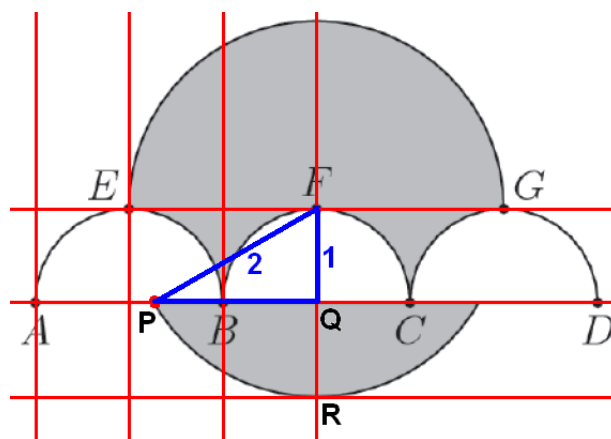
$$1 = 5^{10} = 5^5 \cdot 5^5$$

$$1 = 2^{10} \cdot 5^{10} = (2^5 \cdot 5^5)(2^5 \cdot 5^5)$$

Luego la respuesta correcta es $121 - 4 = 117$ (C).

15

Marcamos los puntos P, Q y R en el esquema, y vemos que estamos ante un triángulo rectángulo $1 - 2 - \sqrt{3}$, conocido, y por tanto $\angle PFQ = \arctan(\sqrt{3}) = 60^\circ$, una sexta parte del círculo.



Luego el sector circular PFR tendrá área

$$A_1 = \frac{1}{6} \pi \cdot 2^2 = \frac{2}{3} \pi$$

El área del triángulo $\triangle PFQ$ será:

$$A_2 = \frac{1}{2} \sqrt{3} \cdot 1 = \frac{\sqrt{3}}{2}$$

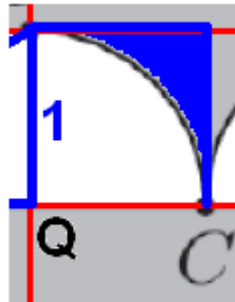
Por lo tanto, el área de la región circular determinada por la recta PQ será

$$A_3 = 2(A_1 - A_2) = 2\left(\frac{2}{3}\pi - \frac{\sqrt{3}}{2}\right) = \frac{4}{3}\pi - \sqrt{3}$$

El semicírculo grande tiene área

$$A_4 = \frac{1}{2}\pi \cdot 2^2 = 2\pi$$

Las regiones sombreadas



Tienen área $A_5 = 1 - \frac{\pi \cdot 1^2}{4} = 1 - \frac{\pi}{4}$

Luego, finalmente, el área buscada es

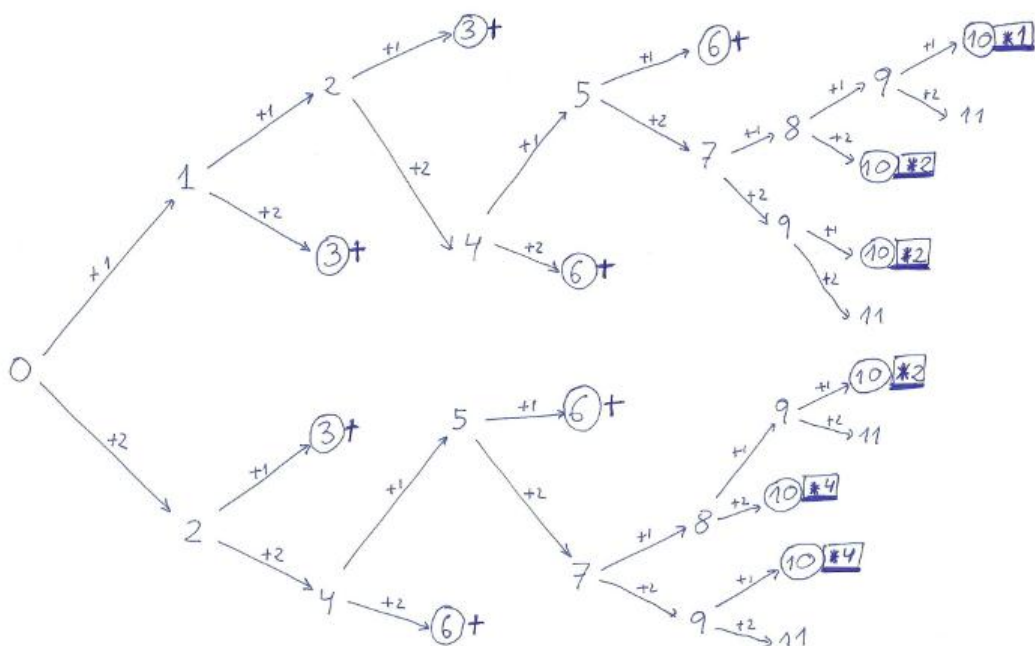
$$A = A_4 + 4A_5 + A_3 = 2\pi + 4\left(1 - \frac{\pi}{4}\right) + \frac{4}{3}\pi - \sqrt{3} = 2\pi + 4 - \pi + \frac{4}{3}\pi - \sqrt{3} = \frac{7}{3}\pi - \sqrt{3} + 4$$

Y por lo tanto, la solución es $7 + 3 + 3 + 4 = 17$ (E)

16

El máximo de saltos son 8, luego existirán un total de 2^8 secuencias de salto posibles.

Resolveremos este problema por puro "bash", viendo todas las posibles opciones que tiene:



Vemos que las secuencias aceptables son 15, y por tanto la probabilidad es

$$P = \frac{15}{2^8} = \frac{15}{256} \quad (\text{A})$$

Observación: En las soluciones oficiales

https://artofproblemsolving.com/wiki/index.php?title=2019_AMC_12B_Problems/Problem_16&oldid=120456

Encontramos otros desarrollos más elegantes, pero como es habitual, no queda claro si serían mejores en el contexto de una competición, donde el tiempo es muy limitado.

17

$$\left. \begin{aligned} |z| &= |z^3| = |z|^3 \\ z &\neq 0 \end{aligned} \right\} \Rightarrow |z|^2 = 1 \Rightarrow |z| = 1 \Rightarrow z = e^{i\pi\alpha} \text{ para cierto ángulo } \alpha.$$

$z = e^{i\pi\alpha} \Rightarrow z^3 = e^{i\pi 3\alpha}$, luego el ángulo determinado por las rectas Oz y Oz^3 será $3\alpha - \alpha = 2\alpha$.

Así pues, puesto que buscamos un ángulo de $60^\circ = \frac{\pi}{3} \text{ rad}$, $2\alpha = \frac{\pi}{3} + 2\pi k$.

Para $0 < \alpha < \frac{\pi}{2} \Rightarrow \alpha = \frac{\pi}{6}$.

Para $\frac{\pi}{2} < \alpha < \pi \Rightarrow \alpha = \frac{5\pi}{6}$.

Por simetría, para $\pi < \alpha < 2\pi$ nos encontraremos con otros dos valores, por lo que resulta un total de 4 complejos diferentes (D).

18

Por Pitágoras,

$$BE = \sqrt{6^2 + 3^2} = \sqrt{45} = 3\sqrt{5}$$

$$AC = BD = \sqrt{3^2 + 3^2} = \sqrt{18} = 3\sqrt{2}$$

$$EC = \sqrt{6^2 + 18} = \sqrt{54} = 3\sqrt{6}$$

Está claro que $\triangle EBD$ y $\triangle EPQ$ son triángulos semejantes con razón $2/3$, luego

$$PQ = \frac{2}{3} BD = \frac{2}{3} 3\sqrt{2} = 2\sqrt{2}$$

Para calcular PR aplicaremos el Teorema del Coseno:

$$BC^2 = BE^2 + CE^2 - 2 \cdot BE \cdot CE \cdot \cos \angle BEC \Leftrightarrow$$

$$9 = 45 + 54 - 2 \cdot 3\sqrt{5} \cdot 3\sqrt{6} \cdot \cos \angle BEC \Leftrightarrow$$

$$9 = 99 - 18\sqrt{30} \cos \angle BEC \Leftrightarrow \cos \angle BEC = \frac{90}{18\sqrt{30}} = \frac{5}{\sqrt{30}} = \frac{\sqrt{5}}{\sqrt{6}}$$

$$PR^2 = EP^2 + ER^2 - 2 \cdot EP \cdot ER \cdot \cos \angle BEC \Leftrightarrow$$

$$PR^2 = \left(\frac{2}{3} 3\sqrt{5}\right)^2 + \left(\frac{1}{3} 3\sqrt{6}\right)^2 - 2 \frac{2}{3} 3\sqrt{5} \frac{1}{3} 3\sqrt{6} \frac{\sqrt{5}}{\sqrt{6}} = 4 \cdot 5 + 6 - 20 = 6 \Rightarrow PR = \sqrt{6}$$

Debemos calcular el área de un triángulo isósceles de lados $a = 2\sqrt{2}$, $b = c = \sqrt{6}$.

Mediante la Fórmula de Heron:

$$s = \frac{a+b+c}{2} = \frac{2\sqrt{2} + \sqrt{6} + \sqrt{6}}{2} = \sqrt{2} + \sqrt{6}$$

$$[\triangle PQR] = \sqrt{s(s-a)(s-b)(s-c)} =$$

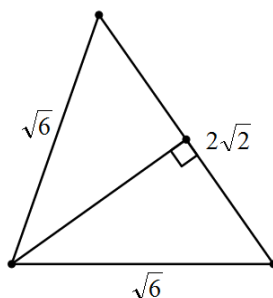
$$= \sqrt{(\sqrt{2} + \sqrt{6})(\sqrt{2} + \sqrt{6} - 2\sqrt{2})(\sqrt{2} + \sqrt{6} - \sqrt{6})(\sqrt{2} + \sqrt{6} - \sqrt{6})} =$$

$$= \sqrt{(\sqrt{2} + \sqrt{6})(\sqrt{6} - \sqrt{2})(\sqrt{2})(\sqrt{2})} =$$

$$= \sqrt{(6-2)2} = \sqrt{4 \cdot 2} = 2\sqrt{2}$$

La respuesta correcta es (C).

Observación: Mucho más inteligente hubiera sido determinar la altura del triángulo $\triangle PQR$ por Pitágoras:



$$h = \sqrt{6-2} = \sqrt{4} = 2$$

$$\text{Y por tanto } [\triangle PQR] = \frac{1}{2} 2\sqrt{2} \cdot 2 = 2\sqrt{2}$$

19

En primer lugar vemos que es imposible que uno de los jugadores tenga los 3\$, porque entonces significaría que en la jugada anterior no dio ninguno de los dólares que tenía, lo cual no es aceptable.

La combinación “1-1-1” tiene $2^3 = 8$ salidas posibles, de las cuales 2 vuelven a la combinación “1-1-1”. Por lo tanto, la probabilidad de ir de “1-1-1” a “1-1-1” es $2/8 = 1/4$.

Por otro lado tenemos seis combinaciones diferentes en las que un jugador tiene 2\$: “2-1-0”, “2-0-1”, “1-0-2”, “1-2-0”, “0-1-2”, “0-2-1”

Tomemos, por ejemplo, la combinación “1-2-0”. Ahora el tercer jugador no tiene nada para dar, luego las salidas posibles son $2^2 = 4$, que son las siguientes:

“1-1-1”, “1-2-0”, “0-2-1” y “0-1-2”

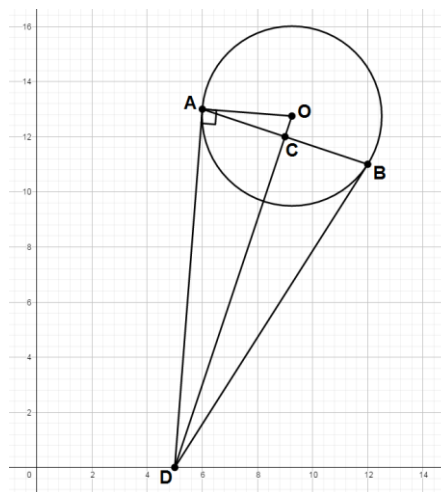
Todas ellas con la misma probabilidad: $1/4$.

Luego, llegar a “1-1-1” desde la posición “1-2-0” tiene probabilidad $1/4$, que es la misma que llegar a “1-1-1” desde la posición “1-1-1”. Así pues, la probabilidad de llegar a “1-1-1” es siempre $1/4$, independientemente de las veces que suene la campana y de la posición de partida. La solución es (B)

Fuente: https://artofproblemsolving.com/wiki/index.php?title=2019_AMC_10B_Problems/Problem_22&oldid=121073

20

Sea C el punto medio del segmento $\overline{AB}$. Sea D el punto de corte de las tangentes. Sea O el centro de la circunferencia.



Determinamos la mediatriz del segmento $\overline{AB}$.

$$C = \frac{A+B}{2} = (9,12)$$

La recta AB tiene pendiente $m_1 = \frac{11-13}{12-6} = \frac{-1}{3}$, luego su mediatriz tendrá pendiente

$m = \frac{-1}{m_1} = 3$, y por tanto su ecuación será

$$y = 3x + b \Rightarrow 12 = 3 \cdot 9 + b \Rightarrow b = 12 - 27 = -15 \Rightarrow y = 3x - 15$$

El punto de corte D de las tangentes pasa por la mediatriz del segmento $\overline{AB}$, luego será

$$0 = 3x - 15 \Rightarrow x = 5 \Rightarrow D = (5, 0)$$

Observamos que $\triangle AOD$ y $\triangle CAD$ son triángulos semejantes, pues son triángulos rectángulos y comparten el ángulo en D. Luego

$$\frac{AO}{AD} = \frac{AC}{CD} \Rightarrow r = AO = \frac{AD \cdot AC}{CD}$$

$$AD = \sqrt{13^2 + 1^2} = \sqrt{170}$$

$$AC = \sqrt{3^2 + 1^2} = \sqrt{10}$$

$$CD = \sqrt{12^2 + 4^2} = \sqrt{160}$$

$$r = AO = \frac{AD \cdot AC}{CD} = \frac{\sqrt{170} \cdot \sqrt{10}}{\sqrt{160}} = \frac{\sqrt{170}}{\sqrt{16}} = \sqrt{\frac{85}{8}}$$

Y por tanto, finalmente,

$$[\omega] = \pi r^2 = \pi \frac{85}{8} \quad (C)$$

Nota: En las soluciones https://artofproblemsolving.com/wiki/index.php/2019_AMC_12B_Problems/Problem_20 se presentan otros cuatro métodos más para determinar el radio de la circunferencia.

21

$$ax^2 + bx + c = 0 \Leftrightarrow x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$

Y aplicando las fórmulas de Vieta tenemos que:

$$\frac{b}{a} = -(r + s), \quad \frac{c}{a} = rs \quad (*)$$

Supongamos que $r = s$. Entonces $a = b = c = r = s \neq 0$ y las ecuaciones (*) se reducen a

$$1 = \frac{s}{s} = -(s + s) = -2s \Rightarrow s = \frac{-1}{2}$$

$$1 = \frac{s}{s} = s^2$$

Que es un sistema incompatible.

Supongamos pues que $r \neq s$. Para que se cumpla $\{a, b, c\} = \{r, s\}$ distinguiremos casos:

a) $a = b = r, c = s$

Las condiciones (*) quedan ahora:

$$1 = \frac{r}{r} = -(r + s), \quad \frac{s}{r} = rs$$

Si $s = 0$ entonces:

$$1 = -r \Rightarrow r = -1, \text{ es decir, } a = b = -1, c = 0, \text{ un caso.}$$

Si $s \neq 0$ entonces:

$$1 = r^2 \Rightarrow \begin{cases} r = 1 \Rightarrow 1 = -(1 + s) \Rightarrow -1 = 1 + s \Rightarrow -2 = s \Rightarrow a = b = 1, c = -2 \\ r = -1 \Rightarrow 1 = -(-1 + s) \Rightarrow s = 0 \text{ absurdo} \end{cases}.$$

Un caso.

b) $a = c = r, b = s$

Las condiciones (*) quedan ahora:

$$\frac{s}{r} = -(r + s), \quad 1 = \frac{r}{r} = rs \quad (*)$$

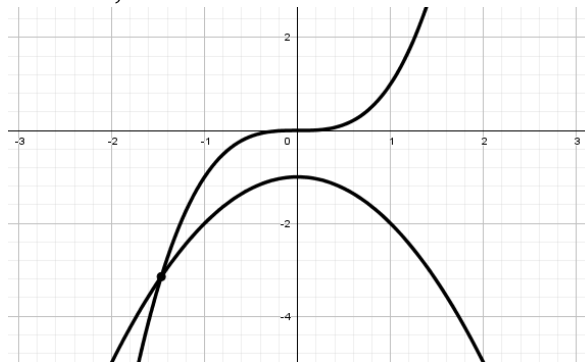
Si $s = 0$ entonces:

$$0 = -r \Rightarrow r = 0 = s \text{ absurdo.}$$

Si $s \neq 0$ entonces: $\frac{1}{s} = r$ y por tanto

$$\frac{s}{r} = -(r + s) \Leftrightarrow s = -(r^2 + rs) \Leftrightarrow s = -\left(\frac{1}{s^2} + 1\right) \Leftrightarrow s^3 = -(1 + s^2) \Leftrightarrow s^3 + s^2 + 1 = 0$$

Trazando las gráficas de las funciones s^3 y $-(1 + s^2)$ vemos que se cortan en un único punto, que da lugar a una única solución, otro caso.



c) $b = c = r, a = s$

Las condiciones (*) quedan ahora:

$$\frac{r}{s} = -(r + s), \quad \frac{r}{s} = rs$$

Si $r = 0$ tenemos

$$0 = \frac{0}{s} = -(0 + s) \Rightarrow s = 0 = r, \text{ absurdo.}$$

Si $r \neq 0$ tenemos

$$\frac{r}{s} = rs \Rightarrow \frac{1}{s} = s \Rightarrow s^2 = 1$$

Estamos ante un caso similar al caso a), luego generará otro caso.

En total hay cuatro casos (B).

22

https://artofproblemsolving.com/wiki/index.php/2019_AMC_12B_Problems/Problem_22

23

Puesto que empiezan y acaban en “0” y no pueden tener dos “0” consecutivos, el segundo número y el que hace 18 serán “1”. Luego reducimos las secuencias a longitud 15, que pueden empezar y acabar con “0” o con “1”.

Dividimos las 15 posiciones en 5 secuencias de longitud 3. Las secuencias pueden ser:

000, 001, 010, 011, 100, 101, 110, 111

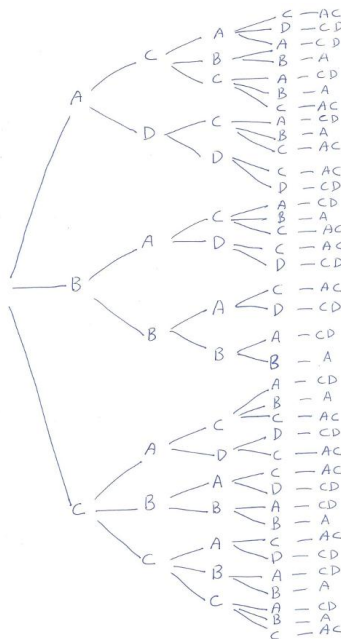
Pero por condiciones del enunciado se reducen a:

A="010", B="011", C="101", D="110"

Ahora podemos construir el árbol de posibilidades:

La primera pieza puede ser cualquiera de las cuatro, pero después de A solo puede ir C o D, después de B solo puede ir A o B, después de C solo puede ir A, B o C, y después de D solo puede ir C o D.

Además, la primera pieza no puede ser D y la quinta pieza no puede ser B.



Hay un total de 65 (C).

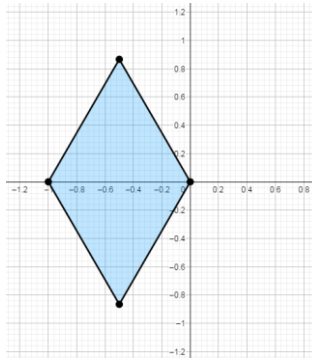
24

Vamos a resolver este problema interpretando los números complejos como vectores en el plano cartesiano.

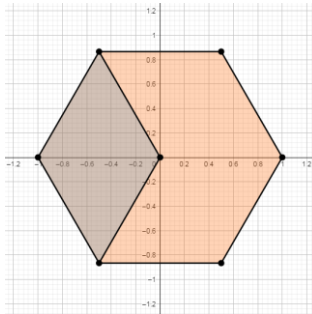
$$\omega = -\frac{1}{2} + \frac{1}{2}i\sqrt{3} \Rightarrow \omega^2 = \left(-\frac{1}{2} + \frac{1}{2}i\sqrt{3}\right)\left(-\frac{1}{2} + \frac{1}{2}i\sqrt{3}\right) = \frac{1}{4} - \frac{1}{4}i\sqrt{3} - \frac{1}{4}i\sqrt{3} - \frac{3}{4} =$$

$$= -\frac{1}{2} - \frac{1}{2}i\sqrt{3} = \bar{\omega}$$

Los puntos de la forma $b\omega + c\omega^2$, con $0 \leq b, c \leq 1$ determinarán un rombo de vértices 0, ω , $\omega^2 = \bar{\omega}$ y $\omega + \omega^2 = \omega + \bar{\omega} = -1$



Si cada uno de estos puntos z los podemos mover, además de la forma $a + z$, con $0 \leq a \leq 1$, todo el rombo anterior se desplaza en horizontal a lo largo de una unidad de longitud formando un hexágono:



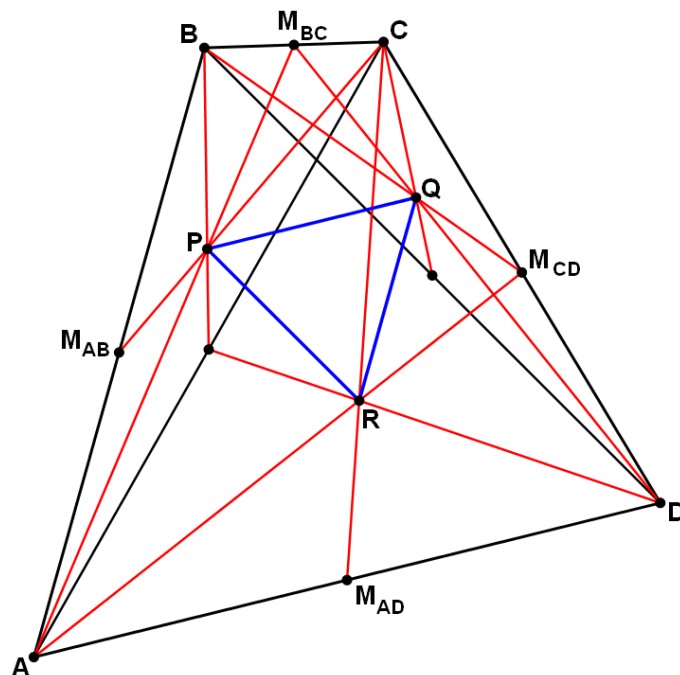
Esta figura se puede interpretar como seis triángulos equiláteros de lado 1 para calcular su área:

$$[S] = 6 \frac{1}{2} \cdot \frac{1}{2} \cdot \sqrt{3} = \frac{3}{2} \sqrt{3} \quad (C)$$

25

Sean P , Q y R los baricentros respectivos de $\triangle ABC$, $\triangle BCD$ y $\triangle ACD$.

En este problema utilizaremos la siguiente notación: Denotaremos por M_{AB} el punto medio del segmento AB .



En primer lugar vamos a demostrar que si ΔPQR es equilátero entonces ΔABC es equilátero.

Veamos que PR es paralelo a BD.

$$\frac{1}{2} = \frac{CM_{BC}}{BC} = \frac{CM_{CD}}{CD} \Rightarrow \Delta CM_{BC}M_{CD} \approx \Delta CBD \Rightarrow M_{BC}M_{CD} \parallel BD$$

Sabemos (ver GA/11.5.3) que las medianas de un triángulo se cortan en razón 2:1. En particular:

$$\frac{2}{3} = \frac{AP}{AM_{BC}} = \frac{AR}{AM_{CD}} \Rightarrow \Delta APR \approx \Delta AM_{BC}M_{CD} \Rightarrow PR \parallel M_{BC}M_{CD}$$

Y por tanto $PR \parallel M_{BC}M_{CD} \parallel BD \Rightarrow PR \parallel BD$

Veamos que QR es paralelo a AB.

Basta aplicar de nuevo GA/11.5.3:

$$\frac{2}{3} = \frac{AR}{AM_{CD}} = \frac{BQ}{BM_{CD}} \Rightarrow \Delta BM_{CD}A \approx \Delta QM_{CD}R \Rightarrow QR \parallel AB$$

Finalmente, veamos que PQ es paralelo a AD.

De nuevo por 11.5.3,

$$\frac{2}{3} = \frac{CP}{CM_{AB}} = \frac{CQ}{CM_{BD}} \Rightarrow \Delta M_{AB}CM_{BD} \approx \Delta PCQ \Rightarrow PQ \parallel M_{AB}M_{BD}$$

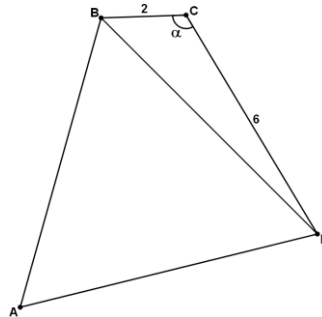
Pero por otro lado,

$$\frac{1}{2} = \frac{BM_{AB}}{BA} = \frac{BM_{BD}}{BD} \Rightarrow \Delta ABD \approx \Delta M_{AB}BM_{BD} \Rightarrow AD \parallel M_{AB}M_{BD}$$

y por tanto $AD \parallel M_{AB}M_{BD} \parallel PQ \Rightarrow AD \parallel PQ$

Así pues, ΔABC es también un triángulo equilátero.

Queremos calcular el área máxima de un cuadrilátero ABCD con $BC = 2$ y $CD = 6$, y en donde ΔABC es equilátero. Sea $\alpha = \angle BCD$.



El área del triángulo ΔBCD será $[\Delta BCD] = \frac{1}{2} \cdot 2 \cdot 6 \cdot \sin \alpha = 6 \sin \alpha$.

Por otro lado, aplicando el Teorema del Coseno,

$$BD^2 = 2^2 + 6^2 - 2 \cdot 2 \cdot 6 \cdot \cos \alpha = 40 - 24 \cos \alpha.$$

El área del triángulo $\triangle ABD$ será

$$[\triangle ABD] = \frac{\sqrt{3}}{4} BD^2 = \frac{\sqrt{3}}{4} (40 - 24 \cos \alpha) = \sqrt{3}(10 - 6 \cos \alpha)$$

Luego la función que tenemos que hacer máxima es

$$\begin{aligned} [\triangle ABCD] &= [\triangle BCD] + [\triangle ABD] = 6 \sin \alpha + \sqrt{3}(10 - 6 \cos \alpha) = 6 \sin \alpha - 6\sqrt{3} \cos \alpha + 10\sqrt{3} = \\ &= 6(\sin \alpha - \sqrt{3} \cos \alpha) + 10\sqrt{3} \end{aligned}$$

Para encontrar el máximo de esta expresión realizamos la siguiente substitución trigonométrica:

$$\begin{aligned} \sin \alpha - \sqrt{3} \cos \alpha &= 2 \left(\frac{1}{2} \sin \alpha - \frac{\sqrt{3}}{2} \cos \alpha \right) = 2 \left(\frac{1}{2} \sin \alpha - \frac{\sqrt{3}}{2} \cos \alpha \right) = \\ &= 2(\cos(60^\circ) \sin \alpha - \sin(60^\circ) \cos \alpha) = 2 \sin(\alpha - 60^\circ) \end{aligned}$$

En donde hemos aplicado la identidad del seno de la diferencia de ángulos (ver TR/5.1),

El valor máximo de $2 \sin(\alpha - 60^\circ)$ se obtendrá cuando:

$$\sin(\alpha - 60^\circ) = 1 \Rightarrow \alpha - 60^\circ = 90^\circ \Rightarrow \alpha = 90^\circ + 60^\circ = 150^\circ$$

Para este valor, el área obtenida es

$$\begin{aligned} [\triangle ABCD] &= 6(\sin \alpha - \sqrt{3} \cos \alpha) + 10\sqrt{3} = 6 \left(\frac{1}{2} - \sqrt{3} \left(\frac{-\sqrt{3}}{2} \right) \right) + 10\sqrt{3} = \\ &= 12 + 10\sqrt{3} \quad (C) \end{aligned}$$

Observación. Si no queremos utilizar trigonometría, podemos determinar el área mediante la fórmula de Heron (GA/10.5.11). Sea x la longitud del triángulo $\triangle ABD$.

$$p = \frac{x+2+6}{2} = \frac{x+8}{2}$$

$$\begin{aligned} [\triangle BCD] &= \sqrt{\left(\frac{x+8}{2}\right)\left(\frac{x+8}{2}-2\right)\left(\frac{x+8}{2}-6\right)\left(\frac{x+8}{2}-x\right)} = \\ &= \sqrt{\left(\frac{x+8}{2}\right)\left(\frac{x+4}{2}\right)\left(\frac{x-4}{2}\right)\left(\frac{8-x}{2}\right)} = \frac{1}{4} \sqrt{(x+8)(x+4)(x-4)(8-x)} = \\ &= \frac{1}{4} \sqrt{(8^2 - x^2)(x^2 - 4^2)} = \frac{1}{4} \sqrt{(64 - x^2)(x^2 - 16)} = \frac{1}{4} \sqrt{-x^4 + 80x^2 - 1024} \end{aligned}$$

Luego la fórmula del área total será:

$$\begin{aligned} [\triangle ABCD] &= [\triangle BCD] + [\triangle ABD] = \frac{1}{4} \sqrt{-x^4 + 80x^2 - 1024} + \sqrt{3}x^2 = \\ &= \frac{1}{4} \sqrt{576 - (x^2 - 40)^2} + \frac{\sqrt{3}}{4} x^2 \end{aligned}$$

Haciendo ahora la substitución $k = x^2 - 40$, llegamos a la expresión

$$\frac{1}{4}\sqrt{576-k^2} + \frac{\sqrt{3}}{4}(k+40) = \frac{1}{4}\sqrt{576-k^2} + \frac{\sqrt{3}}{4}k + 10\sqrt{3}$$

Podemos ignorar el término $10\sqrt{3}$ para concentrarnos en encontrar el máximo de

$$\frac{1}{4}(\sqrt{576-k^2} + \sqrt{3}k)$$

Aplicando la desigualdad de Cauchy-Schwarz:

$$\begin{aligned} (\sqrt{576-k^2} + \sqrt{3}k)^2 &\leq (1^2 + (\sqrt{3})^2) \left((\sqrt{576-k^2})^2 + k^2 \right) = \\ &= 4(576-k^2+k^2) = 4 \cdot 576 = 2304 = 48^2 \Rightarrow \sqrt{576-k^2} + \sqrt{3}k \leq 48 \Rightarrow \\ \frac{1}{4}(\sqrt{576-k^2} + \sqrt{3}k) &\leq \frac{48}{4} = 12 \end{aligned}$$

Y finalmente

$$[\triangle ABCD] \leq 12 + 10\sqrt{3} \quad (C)$$

Fuente de estas versiones: Soluciones oficiales en artofproblemsolving.com

AMC 12A 2018

1

Una gran urna contiene 100 bolas, de las cuales un 36% son rojas y el resto son azules. ¿Cuántas bolas azules hay que quitar para que el porcentaje de bolas rojas en la urna sea del 72%? (No se quita ninguna bola roja)

(A) 28 (B) 32 (C) 36 (D) 50 (E) 64

2

Al explorar una cueva, Carl encuentra unas rocas de 5 quilos de valor 14\$ la unidad, unas rocas de 4 quilos de valor 11\$ la unidad, y unas rocas de 1 quilo de valor 2\$ la unidad. Hay al menos 20 piedras de cada peso. Él puede cargar con 18 quilos como máximo. ¿Cuál es el valor máximo, en dólares, del total de rocas que puede extraer de la cueva?

(A) 48 (B) 49 (C) 50 (D) 51 (E) 52

3

Determina el número de posibilidades que tiene un estudiante para programarse 3 cursos de matemáticas (álgebra, geometría y teoría de números) en un horario de seis períodos si no se pueden tomar dos cursos de matemáticas en períodos consecutivos. (Los cursos que tome el estudiante durante los otros 3 períodos no son relevantes). (Usando las iniciales, dos ejemplos de horarios aceptables serían AXGXTX, o AXXTXG, donde X es cualquier otro curso que no sea A, T o G)

(A) 3 (B) 6 (C) 12 (D) 18 (E) 24

4

Alice, Bob y Charlie van de excursión y se preguntan como de lejos está el pueblo más cercano. Alice dice: “Estamos al menos a 6 millas de distancia”, y Bob replica: “Estamos como mucho a 5 millas de distancia”, y entonces Charlie remarca: “Realmente el pueblo más cercano está a 4 millas como mucho”. Resulta que ninguna de las tres afirmaciones es cierta. Sea d la distancia en millas al pueblo más cercano. ¿Cuál de los siguientes intervalos es el conjunto de todos los posibles valores de d ?

(A) $(0,4)$ (B) $(4,5)$ (C) $(4,6)$ (D) $(5,6)$ (E) $(5,\infty)$

5

Determina la suma de todos los posibles valores de k para los cuales los polinomios $x^2 - 3x + 2$ y $x^2 - 5x + k$ tienen una raíz en común.

(A) 3 (B) 4 (C) 5 (D) 6 (E) 10

6

Sean m y n ciertos enteros positivos tales que $m+10 < n+1$ y que la media aritmética y la mediana del conjunto $\{m, m+4, m+10, n+1, n+2, 2n\}$ son iguales a n .
Determina $m+n$.

- (A) 20 (B) 21 (C) 22 (D) 23 (E) 24

7

Determina el número de enteros n (no necesariamente positivos) para los cuales

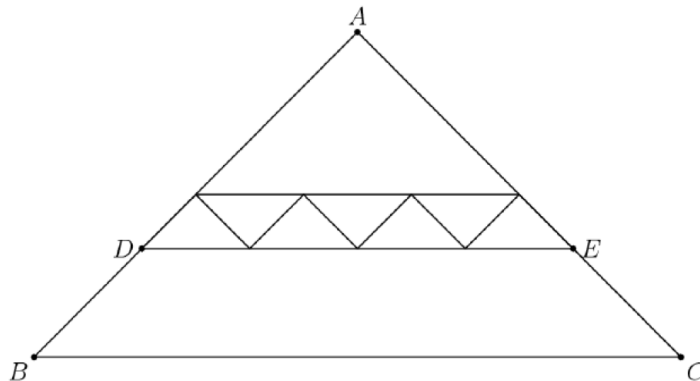
$$4000 \cdot \left(\frac{2}{5}\right)^n$$

es un entero.

- (A) 3 (B) 4 (C) 6 (D) 8 (E) 9

8

Todos los triángulos que parecen en el esquema inferior son semejantes al triángulo isósceles $\triangle ABC$, con $AB = AC$. Cada uno de los 7 triángulos pequeños tiene área 1, y $\triangle ABC$ tiene área 40. Determina el área del trapecio DBCE.



- (A) 16 (B) 18 (C) 20 (D) 22 (E) 24

9

Determina el subconjunto de $[0, \pi]$ más grande que describe los valores de y para los cuales

$$\sin(x+y) \leq \sin(x) + \sin(y)$$

para todo x entre 0 y π , inclusive.

- (A) $y = 0$ (B) $0 \leq y \leq \frac{\pi}{4}$ (C) $0 \leq y \leq \frac{\pi}{2}$ (D) $0 \leq y \leq \frac{3\pi}{4}$ (E) $0 \leq y \leq \pi$

10

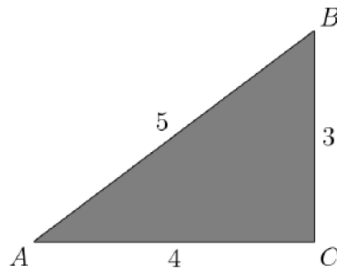
Determina el número de pares (x, y) que satisfacen el siguiente sistema de ecuaciones:

$$\begin{cases} x + 3y = 3 \\ ||x| + |y|| = 1 \end{cases}$$

(A) 1 (B) 2 (C) 3 (D) 4 (E) 8

11

Doblamos un triángulo de papel de longitudes 3, 4 y 5 pulgadas de forma que el punto A coincida con el punto B. ¿Cuál es la longitud en pulgadas del doblez?



(A) $1 + \frac{1}{2}\sqrt{2}$ (B) $\sqrt{3}$ (C) $\frac{7}{4}$ (D) $\frac{15}{8}$ (E) 2

12

Sea S un subconjunto de 6 elementos de $\{1, 2, \dots, 12\}$ con la propiedad de que si a y b son elementos de S y $a < b$, entonces b no es un múltiplo de a . Determina el menor valor posible de los elementos de S.

(A) 2 (B) 3 (C) 4 (D) 5 (E) 7

13

Determina la cantidad de números enteros no negativos que se pueden escribir de la forma

$$a_7 \cdot 3^7 + a_6 \cdot 3^6 + a_5 \cdot 3^5 + a_4 \cdot 3^4 + a_3 \cdot 3^3 + a_2 \cdot 3^2 + a_1 \cdot 3^1 + a_0 \cdot 3^0$$

donde $a_i \in \{-1, 0, 1\}$ para todo $0 \leq i \leq 7$.

(A) 512 (B) 729 (C) 1094 (D) 3281 (E) 59048

14

La solución de la ecuación $\log_{3x} 4 = \log_{2x} 8$, donde x es un número real positivo que no es $1/3$ ni $1/2$, se puede escribir como p/q , donde p y q son enteros coprimos positivos. Determina $p + q$.

(A) 5 (B) 13 (C) 17 (D) 31 (E) 35

15

Un código de scanner consiste en una cuadrícula de 7×7 cuadrados, algunos pintados de negro y el resto de blanco. Debe haber al menos un cuadrado pintado de cada color en esta cuadrícula de 49 cuadrados. Diremos que un código es *simétrico* queda inalterable cuando toda la cuadrícula se rota un múltiplo de 90° alrededor de su centro y en el sentido de las agujas del reloj, o cuando se somete a una reflexión respecto de las rectas que unen las esquinas opuestas, o se somete a una reflexión respecto de las rectas que unen los puntos medios de los lados opuestos. Determina el número de posibles códigos simétricos.

(A) 510 (B) 1022 (C) 8190 (D) 8192 (E) 65534

16

Determina el intervalo que describe el conjunto de valores de a para los cuales las curvas

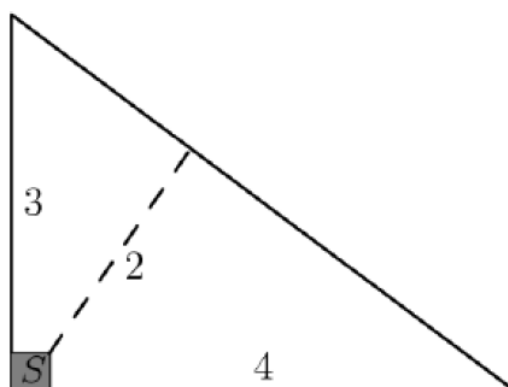
$$x^2 + y^2 = a^2, \quad y = x^2 - a$$

se cortan exactamente en tres puntos del plano cartesiano real.

(A) $a = \frac{1}{4}$ (B) $\frac{1}{4} < a < \frac{1}{2}$ (C) $a > \frac{1}{4}$ (D) $a = \frac{1}{2}$ (E) $a > \frac{1}{2}$

17

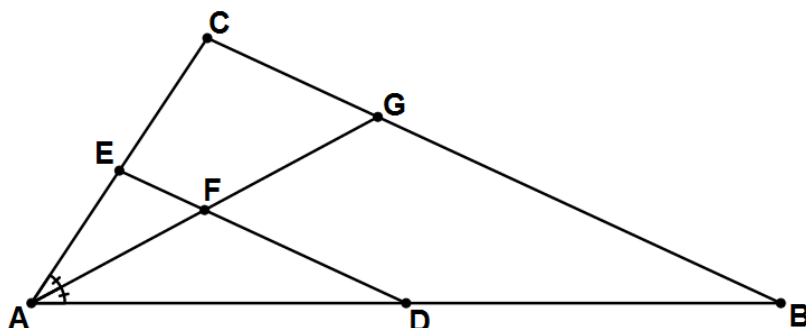
Un granjero llamado Pitágoras cultiva un campo en forma de triángulo rectángulo, con catetos de longitud 3 y 4. En la esquina que forma un ángulo recto ha dejado sin cultivar un pequeño cuadrado S cuya forma, desde el aire, se asemeja al símbolo de ángulo recto. Cultiva el resto del campo. La menor distancia entre S y la hipotenusa es 2. Determina la fracción cultivada del campo.



(A) $\frac{25}{27}$ (B) $\frac{26}{27}$ (C) $\frac{73}{75}$ (D) $\frac{145}{147}$ (E) $\frac{74}{75}$

18

Sea el triángulo $\triangle ABC$ con $AB = 50$ y $AC = 10$ y área 120. Sea D el punto medio de $\overline{AB}$ y E el punto medio de $\overline{AC}$. La bisectriz de $\angle ABC$ corta $\overline{DE}$ y $\overline{BC}$ en F y G, respectivamente. Determina el área del cuadrilátero FDBG.



(A) 60 (B) 65 (C) 70 (D) 75 (E) 80

19

Sea A el conjunto de todos los enteros cuyos únicos factores primos son 2, 3 y 5. La suma infinita

$$\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{8} + \frac{1}{9} + \frac{1}{10} + \frac{1}{12} + \frac{1}{15} + \frac{1}{16} + \frac{1}{18} + \frac{1}{20} + \dots$$

de todos los recíprocos de los elementos de A se puede escribir como $\frac{m}{n}$, donde m y n son enteros positivos relativamente primos. Determina $m + n$.

(A) 16 (B) 17 (C) 19 (D) 23 (E) 36

20

Sea un triángulo $\triangle ABC$ rectángulo isósceles con $AB = AC = 3$. Sea M el punto medio de la hipotenusa $\overline{BC}$. Sean I y E puntos en $\overline{AC}$ y $\overline{AB}$, respectivamente, con $AI > AE$, de forma que AIME sea un cuadrilátero cíclico. Si el triángulo $\triangle EMI$ tiene área 2, la

longitud CI se puede escribir como $\frac{a - \sqrt{b}}{c}$ con a, b, c enteros positivos y b no divisible por el cuadrado de ningún primo. Determina el valor de $a + b + c$.

(A) 9 (B) 10 (C) 11 (D) 12 (E) 13

21

Determina el polinomio que tiene la raíz real mayor:

(A) $x^{19} + 2018x^{11} + 1$ (B) $x^{17} + 2018x^{11} + 1$ (C) $x^{19} + 2018x^{13} + 1$
 (D) $x^{17} + 2018x^{13} + 1$ (E) $2019x + 2018$

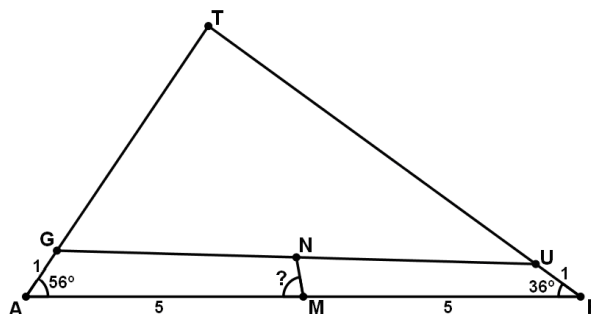
22

Las soluciones de las ecuaciones $z^2 = 4 + 4\sqrt{15}i$ y $z^2 = 2 + 2\sqrt{3}i$, donde $i = \sqrt{-1}$, forman los vértices de un paralelogramo en el plano complejo. El área de este paralelogramo se puede escribir como $p\sqrt{q} - r\sqrt{s}$, donde p, q, r, s son enteros positivos y ni q ni s son divisibles por el cuadrado de ningún número primo. Determina $p + q + r + s$.

(A) 20 (B) 21 (C) 22 (D) 23 (E) 24

23

Sea el triángulo $\triangle PAT$ tal que $\angle P = 36^\circ$, $\angle A = 56^\circ$ y $PA = 10$. Sean los puntos U y G perteneciendo a los lados $\overline{TP}$ y $\overline{TA}$, respectivamente, de forma que $PU = AG = 1$. Sean M y N los respectivos puntos medios de los segmentos $\overline{PA}$ y $\overline{UG}$. Determina la medida en grados del ángulo agudo determinado por las rectas MN y PA .



(A) 76 (B) 77 (C) 78 (D) 79 (E) 80

24

Alicia, Bob y Carol juegan al siguiente juego: Cada uno de ellos elige un número real entre 0 y 1, y gana aquel cuyo número está entre los otros dos. Alicia informa que ella elegirá un número aleatorio entre 0 y 1 de forma uniforme, y Bob informa que él elegirá un número aleatorio entre $1/2$ y $2/3$ de forma uniforme. Sabiendo esta información, ¿Qué número deberá elegir Carol para maximalizar sus posibilidades de ganar?

(A) $1/2$ (B) $13/24$ (C) $7/12$ (D) $5/8$ (E) $2/3$

25

Fijado un entero positivo n , y tres dígitos a, b, c diferentes de cero, sea A_n el entero de n dígitos todos ellos iguales a a , sea B_n el entero de n dígitos todos ellos iguales a b , y sea C_n el entero de $2n$ dígitos (no de n dígitos) todos ellos iguales a c . Determina el mayor valor posible de $a + b + c$ para el que existen al menos dos valores de n tales que $C_n - B_n = A_n^2$.

(A) 12 (B) 14 (C) 16 (D) 18 (E) 20

AMC12A 2018 Soluciones (Letra)

1. D
2. C
3. E
4. D
5. E
6. B
7. E
8. E
9. E
10. C
11. D
12. C
13. D
14. D
15. B
16. E
17. D
18. D
19. C
20. D
21. B
22. A
23. E
24. B
25. D

AMC12A 2018 Soluciones desarrolladas

1

Planteamos la ecuación $\frac{36}{100-x} = \frac{72}{100} \Leftrightarrow 100-x = \frac{36 \cdot 100}{72} = 50 \Leftrightarrow x = 50$ (D)

2

Lo mejor que puede hacer es tomar 2 piedras de 5 kilos y 2 piedras de 4 kilos, con un total de 18 kilos y un valor de $2 \cdot 14 + 2 \cdot 11 = 50\$$ (C)

Otras opciones, como por ejemplo: 3 piedras de 5 kilos y 3 piedras de 1 kilo generan un valor de $3 \cdot 14 + 3 \cdot 2 = 48\$$, que es menor.

3

Los cursos de matemáticas se pueden organizar a lo largo de los 6 períodos de cuatro formas diferentes:

$$1-3-5, 1-3-6, 1-4-6, 2-4-6$$

Para cada una de estas posibilidades, hay $3 \cdot 2 = 6$ formas diferentes de ordenar los tres cursos de matemáticas, haciendo un total de $6 \cdot 4 = 24$ horarios diferentes (E).

4

Las negaciones de las tres frases del enunciado nos dan los intervalos $(0,6)$, $(5,\infty)$ y $(4,\infty)$, cuya intersección es el intervalo $(5,6)$ (D)

5

Factorizando el polinomio $x^2 - 3x + 2$ vemos que tiene raíces 1 y 2. Luego el polinomio $x^2 - 5x + k$ debe tener una de estas raíces.

Si $x = 1$ es raíz de $x^2 - 5x + k$ entonces $1^2 - 5 \cdot 1 + k = 0 \Rightarrow k = 5 - 1 = 4$.

Si $x = 2$ es raíz de $x^2 - 5x + k$ entonces $2^2 - 5 \cdot 2 + k = 0 \Rightarrow k = 10 - 4 = 6$.

Por lo tanto, la suma de los posibles valores de k es $4 + 6 = 10$ (E).

6

Aplicando la definición de media aritmética, tenemos:

$$n = \frac{m + m + 4 + m + 10 + n + 1 + n + 2 + 2n}{6} \Leftrightarrow 6n = 3m + 4n + 17 \Leftrightarrow 2n = 3m + 17$$

Está claro que los elementos de este conjunto están ordenados de menor a mayor, luego, aplicando la definición de mediana, tenemos:

$$n = \frac{m + 10 + n + 1}{2} \Leftrightarrow 2n = m + n + 11 \Leftrightarrow n = m + 11$$

Luego solo nos queda resolver el sistema:

$$2(m + 11) = 3m + 17 \Leftrightarrow 2m + 22 = 3m + 17 \Leftrightarrow 5 = m \Rightarrow n = 5 + 11 = 16$$

Luego la respuesta correcta es $16 + 5 = 21$ (B)

7

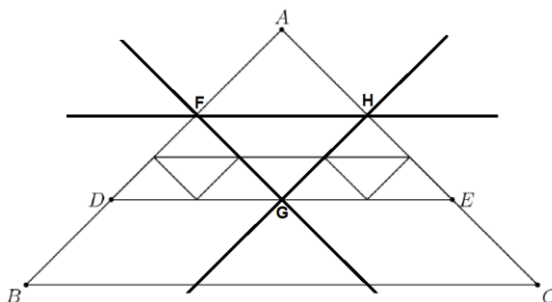
Si n es positivo, $4000 \cdot \left(\frac{2}{5}\right)^n = 2^5 \cdot 5^3 \frac{2^n}{5^n}$ que será entero para $n = 0, 1, 2, 3$.

Si n es negativo, $4000 \cdot \left(\frac{2}{5}\right)^n = 2^5 \cdot 5^3 \frac{5^{-n}}{2^{-n}}$ que será entero para $n = -1, -2, -3, -4, -5$.

En total hay 9 casos (E)

8

Prolongamos uno de los lados de los triángulos pequeños determinando los segmentos FG y GH tal y como se muestra en el siguiente esquema:



Está claro que $FG \parallel AC$ y que G es el punto medio de DE , luego F es el punto medio de AD y H es el punto medio de AE . Así pues, los triángulos $\triangle DFG$, $\triangle GHE$, $\triangle FGH$ y $\triangle FAH$ son todos congruentes y semejantes a $\triangle BAC$. Así pues, $\triangle BAC$ tiene área 16, y el área del trapecio $DBCE$ es $40 - 16 = 24$ (E).

Nota: En https://artofproblemsolving.com/wiki/index.php/2018_AMC_12A_Problems/Problem_8 podemos encontrar hasta seis soluciones alternativas, todas ellas alrededor de estos mismos conceptos.

9

Aplicando la identidad trigonométrica de la suma de ángulos:

$$\sin(x + y) = \sin(x)\cos(y) + \sin(y)\cos(x)$$

Puesto que $x, y \in [0, \pi]$, $0 \leq \sin(x) \leq 1$, $-1 \leq \cos(y) \leq 1$, luego $\sin(x)\cos(y) \leq \sin(x)$

Y de la misma manera $\sin(y)\cos(x) \leq \sin(y)$

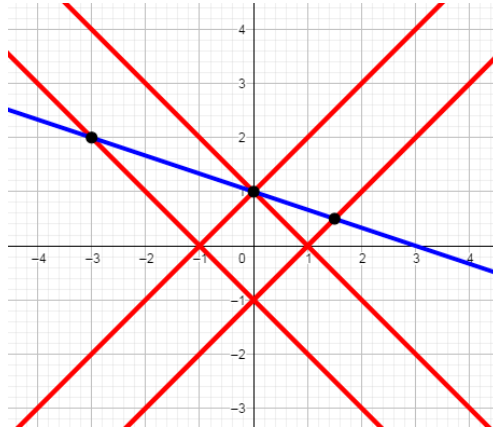
Y por tanto siempre se cumple la desigualdad del enunciado para todo par $x, y \in [0, \pi]$.

La respuesta es (E).

10

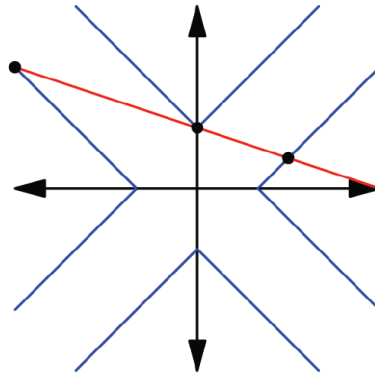
$$||x| + |y|| = 1 \Leftrightarrow \begin{cases} |x| + |y| = 1 \Leftrightarrow |y| = 1 - |x| \Leftrightarrow \begin{cases} y = 1 - |x| \\ y = -(1 - |x|) = |x| - 1 \end{cases} \\ |x| + |y| = -1 \Leftrightarrow |y| = -1 - |x| \Leftrightarrow \begin{cases} y = -1 - |x| \\ y = -(-1 - |x|) = |x| + 1 \end{cases} \end{cases}$$

Si, además, representamos $x + 3y = 3$, vemos que el conjunto de rectas se corta en tres puntos:



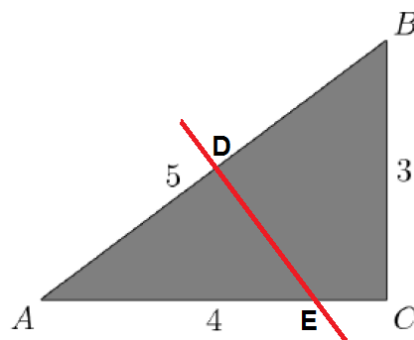
La solución es 3 (C)

Observación: El gráfico que aparece en la solución oficial (https://artofproblemsolving.com/wiki/index.php/2018_AMC_10A_Problems/Problem_12) es ligeramente diferente:



11

Al doblar el papel generamos la mediatriz del lado $\overline{AB}$, que cortará en D a $\overline{AB}$ y en E a $\overline{AC}$.



Está claro que $\triangle ABC$ es un triángulo rectángulo, y puesto que $DE \perp AB$,
 $\triangle ABC \approx \triangle AED \Rightarrow \frac{AD}{AC} = \frac{DE}{BC} \Leftrightarrow \frac{5/2}{4} = \frac{DE}{3} \Rightarrow DE = \frac{3 \cdot 5/2}{4} = \frac{15}{8}$ (D)

12

Está claro que 1 no puede ser mínimo de S , pues sería divisor de cualquier otro elemento posible, llegando a contradicción.

El 2 tampoco puede ser mínimo, pues es incompatible con 4, 6, 8, 10 y 12, y con el resto tenemos un único posible conjunto $S = \{3, 5, 7, 9, 11\}$ que no es aceptable porque 3 y 9 no son compatibles.

El 3 tampoco puede ser mínimo: Los otros candidatos son: 4, 5, 7, 8, 10, 11, y en este caso el 5 es incompatible con el 10, y el 4 es incompatible con el 8 luego eliminando alguno de ellos nos quedan solo 4 candidatos posibles, y no llegamos a los 6 necesarios.

Con el 4 de mínimo tenemos candidatos posibles 5, 6, 7, 9, 10, 11. El 5 es incompatible con el 10, y con el resto podemos hacer un posible conjunto $S = \{4, 6, 7, 9, 10, 11\}$. Así pues, el valor mínimo es 4 (C).

13

Haciendo el listado exhaustivo de todos los valores que vamos obteniendo para valores pequeños de n , vemos que los conjuntos

$$S_n = \{ a_n \cdot 3^n + a_{n-1} \cdot 3^{n-1} + \dots + a_1 \cdot 3^1 + a_0 \cdot 3^0, -1 \leq a_i \leq 1 \}$$

Constan de 3^n enteros, sin que se produzcan repeticiones, y se cumple $0 \in S_n$, y hay simetría: $x \in S_n \Leftrightarrow -x \in S_n$.

Así pues, en particular, para el caso $n = 7$, tendremos $3^8 = 3^4 \cdot 3^4 = 81 \cdot 81 = 6561$ números diferentes: 3280 negativos, 3280 positivos y el cero. El total de elementos no negativos es, por tanto, 3281 (D).

14

Aplicando la identidad del cambio de base: $\log_a b \cdot \log_b c = \log_a c \Rightarrow \log_a b = \frac{\log_a c}{\log_b c}$

Tenemos:

$$\log_{3x} 4 = \log_{2x} 8 \Leftrightarrow \log_{3x} 2^2 = \log_{2x} 2^3 \Leftrightarrow 2 \log_{3x} 2 = 3 \log_{2x} 2 \Leftrightarrow$$

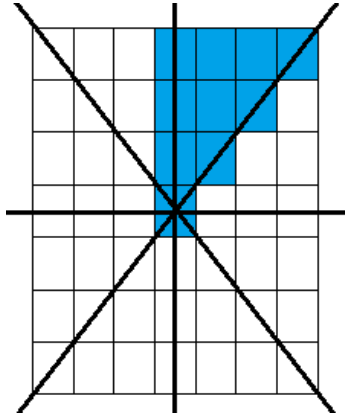
$$\frac{2}{3} = \frac{\log_{2x} 2}{\log_{3x} 2} = \log_{2x} 3x \Leftrightarrow$$

$$(2x)^{2/3} = 3x \Leftrightarrow (2x)^2 = (3x)^3 \Leftrightarrow 4x^2 = 3^3 x^3 \Leftrightarrow 4 = 3^3 x \Leftrightarrow x = \frac{4}{27}$$

Y la respuesta correcta es $4 + 27 = 31$ (D)

15

Observando el efecto de las transformaciones que deben dejar el código invariante deducimos que el código queda determinado por 9 casillas, que pueden ser, por ejemplo, las que aparecen en el siguiente esquema:



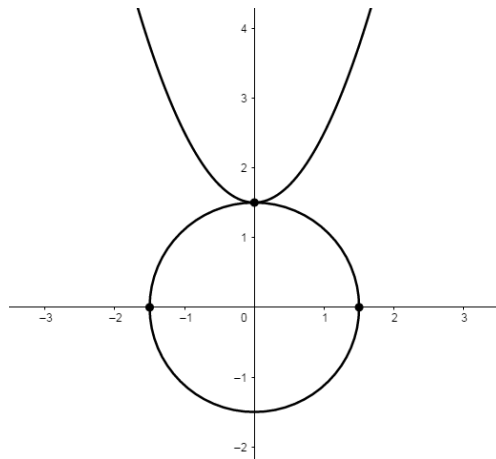
Luego, en total, y teniendo en cuenta que no puede ser totalmente blanca o totalmente negra, habrá $2^{10} - 2 = 1024 - 2 = 1022$ posibilidades distintas.

16

La curva $x^2 + y^2 = a^2$ describe una circunferencia de centro $(0,0)$ y radio $|a|$.

La curva $y = x^2 - a$ describe una parábola con vértice $V = (0, -a)$ y ramas hacia arriba.

Si $a < 0$, la parábola está por encima de la circunferencia y solo tienen el vértice como punto en común.



Si $a = 0$, la circunferencia se reduce al punto $(0,0)$ y por lo tanto es obvio que no puede haber tres puntos en común.

Si $a > 0$

$$\begin{cases} x^2 + y^2 = a^2 \\ y = x^2 - a \end{cases} \Rightarrow x^2 + (x^2 - a)^2 = a^2 \Leftrightarrow x^2 + x^4 + a^2 - 2ax^2 = a^2$$

$$\Leftrightarrow x^2 + x^4 - 2ax^2 = 0 \Leftrightarrow x^2(1 + x^2 - 2a) = 0 \Leftrightarrow \begin{cases} x = 0 \\ 1 + x^2 - 2a = 0 \Leftrightarrow x^2 = 2a - 1 \end{cases}$$

Y la ecuación $x^2 = 2a - 1$ tendrá dos soluciones si y solo si $2a - 1 > 0 \Leftrightarrow a > \frac{1}{2}$ (E)

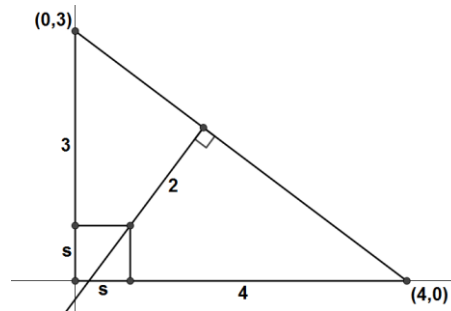
17

Primera versión. Mediante geometría cartesiana y fórmula de la distancia punto-recta. La recta asociada a la hipotenusa tendrá ecuación

$$y = ax + b \Rightarrow \begin{cases} 3 = 0 \cdot a + b \\ 0 = 4 \cdot a + b \end{cases} \Rightarrow b = 3, a = -3/4 \Rightarrow y = \frac{-3}{4}x + 3 \Rightarrow 4y = -3x + 12$$

$$\Rightarrow 3x + 4y = 12$$

Aplicando ahora la fórmula de la distancia entre un punto y una recta, siendo $P = (s, s)$ el vértice del cuadrado S,



Tenemos

$$2 = \frac{|3s + 4s - 12|}{\sqrt{3^2 + 4^2}} = \frac{|3s + 4s - 12|}{5} \Leftrightarrow 10 = |7s - 12| \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} 10 = 7s - 12 \Leftrightarrow 22 = 7s \Leftrightarrow s = 22/7 \\ 10 = -(7s - 12) = -7s + 12 \Leftrightarrow s = 2/7 \end{cases}$$

Puesto que $22/7 > 3$, la única solución aceptable es $s = 2/7$.

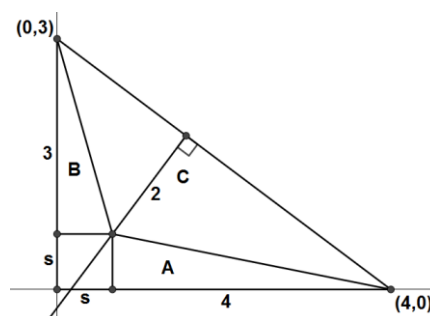
Para este valor, el área del cuadrado S es $s = (2/7)^2$

El área del triángulo es $3 \cdot 4 / 2 = 6$, y por tanto el área cultivada es $6 - (2/7)^2$, y la razón que nos piden determinar es

$$\frac{6 - (2/7)^2}{6} = \frac{6 \cdot 7^2 - 2^2}{6 \cdot 7^2} = \frac{290}{294} = \frac{145}{147} \quad (D)$$

Segunda versión. Descomponiendo la figura en triángulos.

Está claro que la hipotenusa del triángulo es $\sqrt{4^2 + 3^2} = 5$.



Sea s el lado del cuadrado S.

El triángulo grande de área $3 \cdot 4 / 2 = 6$ se puede descomponer en el cuadrado S de área s^2 , el triángulo A de área $\frac{s(4-s)}{2}$, el triángulo B de área $\frac{s(3-s)}{2}$ y el triángulo C de área $\frac{2 \cdot 5}{2} = 5$.

Así pues,

$$6 = s^2 + \frac{s(4-s)}{2} + \frac{s(3-s)}{2} + 5 \Rightarrow 1 = s^2 + 2s - \frac{s^2}{2} + \frac{3s}{2} - \frac{s^2}{2} \Rightarrow$$

$$1 = 2s + \frac{3s}{2} - \frac{7s}{2} \Rightarrow s = \frac{2}{7}$$

Y se continúa como en la primera versión.

Nota: Consulta en https://artofproblemsolving.com/wiki/index.php/2018_AMC_12A_Problems/Problem_17 Hasta seis versiones diferentes para la resolución de este problema.

18

Aplicando el Teorema de la bisectriz, $\frac{CG}{10} = \frac{GB}{50} \Leftrightarrow \frac{CG}{GB} = \frac{10}{50} = \frac{1}{5}$

$$\frac{BC}{CG} = \frac{BG + GC}{CG} = \frac{BG}{CG} + 1 = 5 + 1 = 6 \Rightarrow \frac{CG}{BC} = \frac{1}{6} \Rightarrow \frac{GB}{BC} = \frac{5}{6}$$

Y por tanto, puesto que, por Elementos 6.1, las áreas son proporcionales a las razones de las bases, (ver [GA/8.2.6](#)), tenemos:

$$[\triangle AGB] = \frac{5}{6} [\triangle ABC] = \frac{5}{6} 120 = 100$$

Por otro lado, por semejanza de triángulos, F es el punto medio del segmento $\overline{AG}$ y el triángulo $\triangle AFD$ será semejante a $\triangle AGB$ con una razón de proporcionalidad $1/2$, y por tanto la razón de proporcionalidad del área será su cuadrado, es decir, $1/4$:

$$[\triangle AFD] = \frac{1}{4} [\triangle AGB] = \frac{1}{4} 100 = 25 \Rightarrow [\triangle FDBG] = 100 - 25 = 75 \quad (D)$$

Nota: En https://artofproblemsolving.com/wiki/index.php/2018_AMC_12A_Problems/Problem_18 encontramos hasta 7 desarrollos diferentes de este problema.

19

Queremos calcular la siguiente serie: $\sum_{0 \leq a, b, c < \infty} 2^a 3^b 5^c$

Si fijamos un b y un c , tenemos:

$$\begin{aligned} \sum_{0 \leq a, b, c < \infty} \frac{1}{2^a 3^b 5^c} &= \sum_{0 \leq a < \infty} \frac{1}{2^a 3^b 5^c} = \sum_{0 \leq a < \infty} \frac{1}{2^a 3^b 5^c} = \frac{1}{3^b 5^c} \sum_{0 \leq a < \infty} \frac{1}{2^a} = \frac{1}{3^b 5^c} \left(\frac{1}{1 - (1/2)} \right) = \\ &= \frac{1}{3^b 5^c} (2) = 2 \frac{1}{3^b 5^c} \end{aligned}$$

Donde hemos aplicado la fórmula de la serie geométrica (PA/12.2).

Luego ahora podemos hacer lo mismo fijando b y haciendo la suma infinita en c :

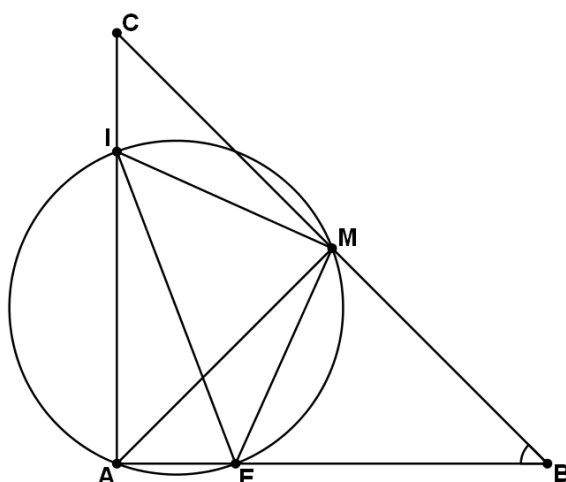
$$\begin{aligned}\sum_{0 \leq a, b, c < \infty} \frac{1}{2^a 3^b 5^c} &= \sum_{0 \leq c < \infty} 2 \frac{1}{3^b 5^c} = 2 \sum_{0 \leq c < \infty} \frac{1}{3^b 5^c} = 2 \frac{1}{3^b} \sum_{0 \leq c < \infty} \frac{1}{5^c} = 2 \frac{1}{3^b} \left(\frac{1}{1 - (1/5)} \right) = \\ &= 2 \frac{1}{3^b} \left(\frac{5}{4} \right) = 2 \frac{5}{4} \frac{1}{3^b} = \frac{5}{2} \frac{1}{3^b}\end{aligned}$$

A ahora haciendo la suma infinita en b :

$$\sum_{0 \leq a, b, c < \infty} \frac{1}{2^a 3^b 5^c} = \sum_{0 \leq b < \infty} \frac{5}{2} \frac{1}{3^b} = \frac{5}{2} \sum_{0 \leq b < \infty} \frac{1}{3^b} = \frac{5}{2} \left(\frac{1}{1 - 1/3} \right) = \frac{5}{2} \left(\frac{3}{2} \right) = \frac{15}{4}$$

Y la solución es $15 + 4 = 19$

20



Por ser AIME cíclico, $\angle IME = 180^\circ - \angle CAB = 180^\circ - 90^\circ = 90^\circ$.

Por otro lado, por ser AM la mediana de un triángulo isósceles, también es su altura, es decir $\angle AMB = 90^\circ$, y por tanto $\triangle AMB \approx \triangle CAB$, de donde se deduce que $AB = AC \Rightarrow AM = MB$, $\angle MAB = \angle ABM = 45^\circ$.

Por ser AIME cíclico, $\angle MIE = \angle MAE = 45^\circ$, y puesto que $\angle IME = 90^\circ$, se deduce que $\angle IEM = 45^\circ$, y por tanto $\triangle IME$ es un triángulo rectángulo isósceles: $IM = ME$.

$$2 = [\triangle EMI] = \frac{IM \cdot ME}{2} = \frac{IM^2}{2} \Rightarrow IM^2 = 4 \Rightarrow IM = ME = 2 \Rightarrow IE^2 = 8$$

De nuevo, por ser AIME cíclico, $\angle CIM = \angle AEM$ y puesto que $\angle EAM = \angle ICM$, $\triangle AEM \approx \triangle CIM$, pero $EM = IM$, y por tanto $\triangle AEM \cong \triangle CIM$, de donde se deduce que $AE = IC$.

Así pues, si $x = IC = AE$, aplicando Pitágoras,

$$AE^2 + AI^2 = IE^2 \Leftrightarrow x^2 + (3 - x)^2 = 8 \Rightarrow x = \frac{3 \pm \sqrt{7}}{2}$$

y puesto que, por hipótesis, $AI > AE$, se deduce, finalmente, que $x = \frac{3 - \sqrt{7}}{2}$

y la respuesta correcta es $3 + 7 + 2 = 12$ (D)

21

Observamos que los cinco polinomios presentados cumplen $p(-1) = -2018 < 0$ y $p(0) = 1 > 0$. Observamos también que todos los polinomios tienen potencias de x de grado impar, y que en el polinomio B se encuentran los exponentes menores (excepto E)

Sabemos que si $a \in (-1, 0)$, entonces $a^n < a^m$ si $n > m$ y n, m son impares, luego las posibles raíces de A, C y D siempre serán menores (más negativas) que las posibles raíces de B. Así pues, solo nos queda comparar B y E.

La raíz de E es $a = \frac{-2018}{2019} \cong -0.99$.

Observamos que $\frac{-1}{2} = -0.50$ es aproximadamente una raíz del polinomio B:

$$\left(\frac{-1}{2}\right)^{17} + 2018\left(\frac{-1}{2}\right)^{11} + 1 = \frac{-1}{2^{17}} - \frac{2018}{2^{11}} + 1 = \frac{-1}{2^{17}} - \frac{2018}{2048} + 1 \cong \frac{-1}{2^{17}} - \frac{1}{2} + 1 \cong 0$$

Y podemos asegurar que la raíz de B será mayor que la raíz de E. Así pues, la respuesta correcta es (B).

Fuente: https://artofproblemsolving.com/wiki/index.php/2018_AMC_12A_Problems/Problem_21

22

Calculamos las raíces de $4 + 4\sqrt{15}i$:

$$4 + 4\sqrt{15}i = z^2 = (a + bi)^2 = a^2 - b^2 + 2abi \Rightarrow \begin{cases} a^2 - b^2 = 4 \\ 2ab = 4\sqrt{15} \end{cases}$$

$$|4 + 4\sqrt{15}i| = \sqrt{4^2 + (4\sqrt{15})^2} = \sqrt{4^2 + 4^2 \cdot 15} = \sqrt{4^2(1+15)} = \sqrt{16^2} = 16$$

$$16 = |z^2| = |z|^2 \Rightarrow 16 = a^2 + b^2$$

Luego

$$20 = 2a^2 \Rightarrow 10 = a^2 \Rightarrow \begin{cases} a = \sqrt{10} \Rightarrow 2\sqrt{10}b = 4\sqrt{15} \Rightarrow b = \frac{4\sqrt{15}}{2\sqrt{10}} = \frac{4\sqrt{3 \cdot 5}}{2\sqrt{2 \cdot 5}} = \frac{2\sqrt{3}}{\sqrt{2}} = \sqrt{6} \\ a = -\sqrt{10} \Rightarrow b = -\sqrt{6} \end{cases}$$

Las dos raíces cuadradas son $z_1 = \sqrt{10} + \sqrt{6}i$, y $z_2 = -\sqrt{10} - \sqrt{6}i$

cumpliendo: $|z_1| = |z_2| = \sqrt{10+6} = \sqrt{16} = 4$

El ángulo α que determina z_1 como punto del plano cumple:

$$\sin \alpha = \frac{\sqrt{6}}{4}, \cos \alpha = \frac{\sqrt{10}}{4}$$

Calculamos las raíces de $2 + 2\sqrt{3}i$:

$$2 + 2\sqrt{3}i = z^2 = (a + bi)^2 = a^2 - b^2 + 2abi \Rightarrow \begin{cases} a^2 - b^2 = 2 \\ 2ab = 2\sqrt{3} \end{cases}$$

$$|2 + 2\sqrt{3}i| = \sqrt{2^2 + (2\sqrt{3})^2} = \sqrt{2^2 + 2^2 \cdot 3} = \sqrt{2^2(1+3)} = \sqrt{16} = 4$$

$$4 = |z^2| = |z|^2 \Rightarrow 4 = a^2 + b^2$$

Luego

$$6 = 2a^2 \Rightarrow 3 = a^2 \Rightarrow \begin{cases} a = \sqrt{3} \Rightarrow 2\sqrt{3}b = 2\sqrt{3} \Rightarrow b = 1 \\ a = -\sqrt{3} \Rightarrow b = -1 \end{cases}$$

Las dos raíces cuadradas son $z_3 = \sqrt{3} + i$, y $z_4 = -\sqrt{3} - i$, cumpliendo:

$$|z_3| = |z_4| = \sqrt{3+1} = \sqrt{4} = 2$$

El ángulo β que determina z_3 como punto del plano cumple:

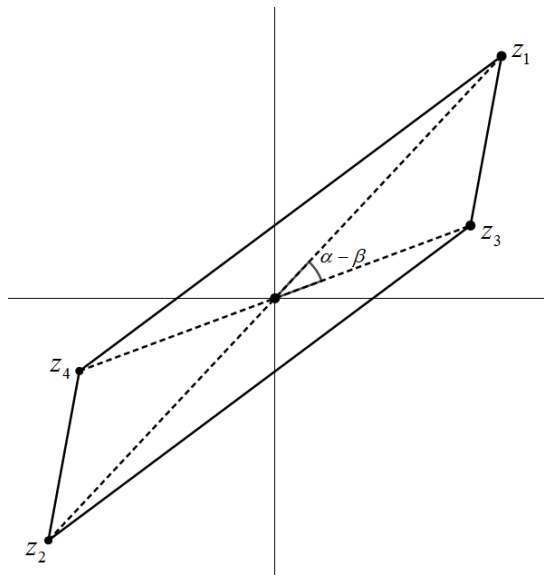
$$\sin \beta = \frac{1}{2}, \cos \beta = \frac{\sqrt{3}}{2}$$

Luego el ángulo entre los puntos z_1 y z_3 cumple

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta = \frac{\sqrt{6}}{4} \frac{\sqrt{3}}{2} - \frac{\sqrt{10}}{4} \frac{1}{2} = \frac{3\sqrt{2} - \sqrt{10}}{8}$$

Y el área del triángulo que determinan los puntos z_1 y z_3 y el origen de coordenadas es:

$$A = \frac{1}{2} |z_1| |z_3| \sin(\alpha - \beta) = \frac{1}{2} 4 \cdot 2 \cdot \frac{3\sqrt{2} - \sqrt{10}}{8} = \frac{3\sqrt{2} - \sqrt{10}}{2}$$



Y el área total será $4A = 4 \left(\frac{3\sqrt{2} - \sqrt{10}}{2} \right) = 6\sqrt{2} - 2\sqrt{10}$, y la respuesta al problema es

$$6 + 2 + 2 + 10 = 20 \quad (\text{A})$$

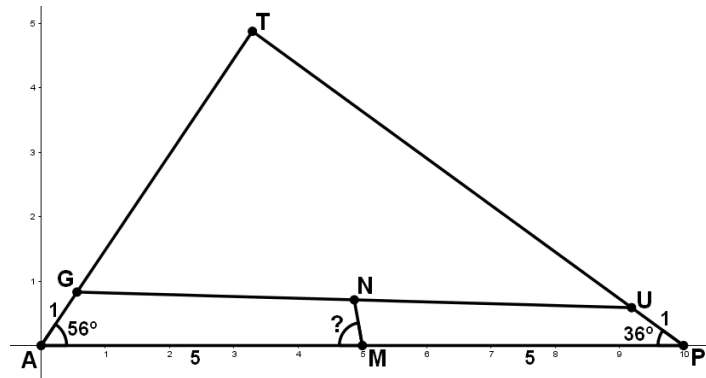
Observación: Una forma mucho más sencilla de calcular el área del triángulo sabiendo las coordenadas es mediante la fórmula “Shoelace” (ver [GA/18.6.6](#)): $(0,0)$, $(\sqrt{10}, \sqrt{6})$, $(\sqrt{3}, 1)$:

$$A = \frac{1}{2} \begin{vmatrix} 1 & 1 & 1 \\ 0 & \sqrt{10} & \sqrt{3} \\ 0 & \sqrt{6} & 1 \end{vmatrix} = \frac{1}{2} |\sqrt{10} \cdot 1 - \sqrt{6} \cdot \sqrt{3}| = \frac{1}{2} (3\sqrt{2} - \sqrt{10})$$

23

Primera versión. Mediante geometría cartesiana y trigonometría.

Sea $A = (0,0)$, $P = (10,0)$, $M = (5,0)$.



Entonces $G = (\cos 56^\circ, \sin 56^\circ)$, $U = (10 - \cos 36^\circ, \sin 36^\circ)$, y por tanto

$$N = \left(\frac{\cos 56^\circ + 10 - \cos 36^\circ}{2}, \frac{\sin 56^\circ + \sin 36^\circ}{2} \right) = \left(\frac{\cos 56^\circ - \cos 36^\circ}{2} + 5, \frac{\sin 56^\circ + \sin 36^\circ}{2} \right)$$

El vector $\overrightarrow{MN}$ será $\overrightarrow{MN} = N - M = \left(\frac{\cos 56^\circ - \cos 36^\circ}{2}, \frac{\sin 56^\circ + \sin 36^\circ}{2} \right)$, y aplicando la identidad de la semisuma de ángulos ([TR/5.1r](#)):

$$\tan \alpha = \frac{(\sin 56^\circ + \sin 36^\circ)/2}{(\cos 56^\circ - \cos 36^\circ)/2} = \frac{\sin 56^\circ + \sin 36^\circ}{\cos 56^\circ - \cos 36^\circ} \stackrel{(*)}{=} \frac{\sin 56^\circ + \sin 144^\circ}{\cos 56^\circ + \cos 144^\circ} = \tan(100^\circ)$$

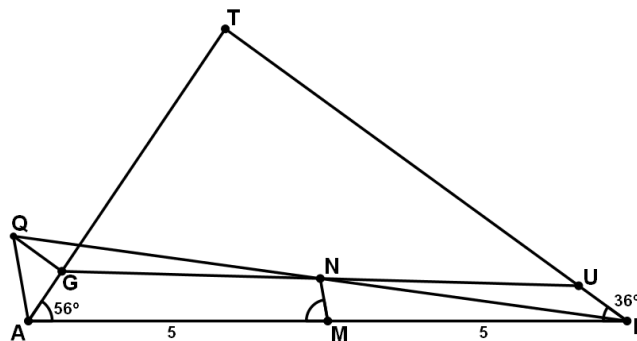
El ángulo agudo que nosotros estamos buscando será $180^\circ - 100^\circ = 80^\circ$ (E)

Nota: Un desarrollo alternativo en (*) aplicando las igualdades de paso de suma a producto es:

$$\frac{\sin 36^\circ + \sin 56^\circ}{\cos 36^\circ - \cos 56^\circ} \stackrel{(*)}{=} \frac{2 \sin 46^\circ \cos 10^\circ}{-2 \sin 46^\circ \sin(-10^\circ)} = \frac{\sin 80^\circ}{\cos 80^\circ} = \tan(80^\circ)$$

Segunda versión. Mediante triángulos semejantes.

Sea Q el punto de la recta NP tal que $QN = NP$. Trazamos los segmentos $\overline{AQ}$ y $\overline{GQ}$.



Por ser N y M puntos medios de dos lados del triángulo ΔAQP tenemos $AQ \parallel MN$ y $\angle PMN = \angle PAQ$.

Está claro que los triángulos $\triangle NUP$ y $\triangle NGQ$ son congruentes, por lo que $QG = UP = 1$, y $\angle GQN = \angle NPU$. De $QG = AG = 1$ se deduce que $\triangle AGQ$ es isósceles en G y por tanto $\angle AQG = \angle GAQ$.

Observemos el triángulo $\triangle APQ$:

$$\angle GQN + \angle QPA = \angle QPU + \angle QPA = \angle P = 36^\circ$$

Luego

$$56^\circ + \angle GAQ + \angle AQG + \angle GQN + \angle QPA = 180^\circ \Rightarrow$$

$$56^\circ + \angle GAQ + \angle AQG + 36^\circ = 180^\circ \Rightarrow$$

$$2\angle GAQ = \angle GAQ + \angle AQG = 180^\circ - 56^\circ - 36^\circ = 88^\circ \Rightarrow \angle GAQ = 44^\circ$$

Y finalmente:

$$\angle PMN = \angle PAQ = \angle PAG + \angle GAQ = 56^\circ + 44^\circ = 100^\circ \Rightarrow \angle AMN = 180^\circ - 100^\circ = 80^\circ$$

Tercera versión. Mediante lugar geométrico.

Trazamos la bisectriz TX del ángulo $\angle T$. $\angle ATX = \angle XTP = 44^\circ$ y por tanto $\angle AXT = 180^\circ - 45^\circ - 44^\circ = 80^\circ$. Vamos a demostrar que $MN \parallel XT$, y por tanto $\angle AMN = \angle AXT = 80^\circ$.

Se puede observar que la recta MN es independiente de la longitud concreta $AG = UP = 1$, es decir, que obtenemos la misma recta MN tomando cualquier distancia $AG = UP = x$ que queramos. El lugar geométrico de los puntos N_x al tomar el punto medio del segmento $\overline{G_x U_x}$ con $AG_x = PU_x = x$ es una recta puesto que los puntos G_x y U_x varían linealmente con x . El punto M aparece tomando $x = 0$, luego M pertenece a dicha recta.

Sean $N' = N_x$, $G' = G_x$, y $U' = U_x$ para el caso concreto $x = AT$.

Entonces $N' \in TP$ (el punto N' pertenece a TP y no a AT porque $PT > AT$)

También está claro que $G' = T$.

Sea $a = AT$, $b = TP$ y $c = PT$. Entonces $AG' = PU' = AT = a$.

$$PN' = \frac{a+b}{2} \text{ (????)}, \text{ y por tanto } \frac{PN'}{PT} = \frac{(a+b)/2}{b} = \frac{a+b}{2}$$

Por otro lado, aplicando el Teorema de la bisectriz:

$$\frac{XP}{AX} = \frac{TP}{AT} = \frac{b}{a} \Rightarrow \frac{XP}{c - XP} = \frac{b}{a} \Rightarrow aXP + bXP = cb \Rightarrow XP = \frac{cb}{a+b}$$

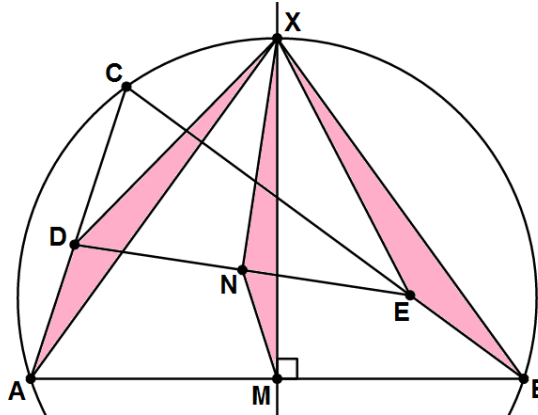
$$PM = c/2 \Rightarrow \frac{PM}{PX} = \frac{c/2}{cb/(a+b)} = \frac{a+b}{2b}$$

Así pues, hemos visto que $\frac{PN'}{PT} = \frac{PM}{PX}$, y por tanto la recta que contiene todos los N_x debe ser paralela a la recta AX , tal y como queríamos ver.

Cuarta versión. Mediante una semejanza espiral y Punto de Miquel.

Nos basamos en el siguiente lema:

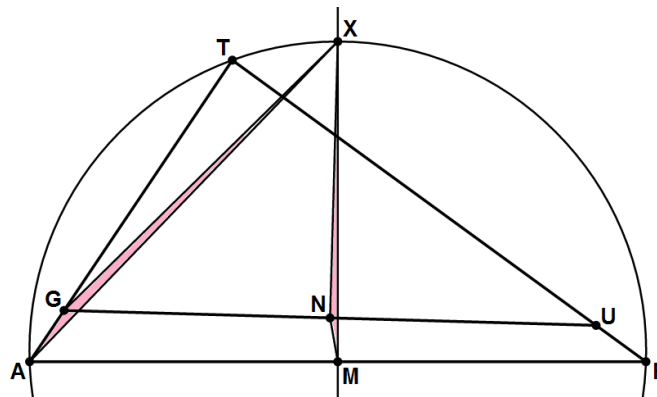
Dado un triángulo $\triangle ABC$ y puntos D y E en $\overline{AC}$ y en $\overline{BC}$ respectivamente, tales que $AD = BE$, el punto medio X del arco mayor de ACB en la circunferencia circunscrita ($\triangle ABC$) es el punto de Miquel asociado al cuadrilátero ADEB.



Luego es el centro de la semejanza espiral que envía el segmento $\overline{DE}$ a $\overline{AB}$. Puesto que una semejanza espiral envía puntos medios a puntos medios, enviará el punto medio N del segmento $\overline{DE}$ al punto medio M del segmento $\overline{AB}$, y por tanto:

$$\angle XMN = \angle XAD = \angle XBE$$

Con este lema ya podemos prescindir los incómodos puntos N y M. Volviendo a nuestro problema:



El triángulo $\triangle PAX$ es isósceles, y por tanto

$$2\angle PAX = 180 - \angle PXA = 180 - \angle PTA = 180 - 88 = 92 \Rightarrow \angle PAX = 46^\circ$$

 Luego

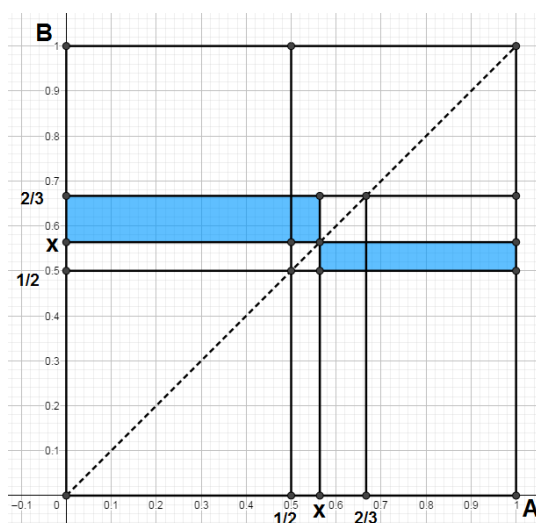
$$\angle XMN = \angle XAG = \angle PAT - \angle PAX = 56^\circ - 46^\circ = 10^\circ$$

Y finalmente: $\angle AMN = 90 - \angle XMN = 90^\circ - 10^\circ = 80^\circ$

Fuente de las versiones 2, 3 y 4: https://artofproblemsolving.com/wiki/index.php/2018_AMC_12A_Problems/Problem_23

24

Interpretando este problema en términos de áreas, la probabilidad de ganar para $C = x$ es proporcional al área $P_1 : A < x < B$ o $P_2 : B < x < A$, que corresponde a los dos rectángulos que aparecen en la siguiente figura:



Con lo que podemos modelizar la probabilidad de ganar mediante la fórmula

$$f(x) = x\left(\frac{2}{3} - x\right) + (1-x)\left(x - \frac{1}{2}\right) = \frac{2x}{3} - x^2 + x - \frac{1}{2} - x^2 + \frac{1}{2}x = -2x^2 + \frac{13}{6}x - \frac{1}{2}$$

Es decir, una parábola con las ramas hacia abajo, cuyo mínimo lo encontramos en su vértice:

$$v = \frac{-b}{2a} = \frac{-13/6}{2(-2)} = \frac{13}{24} \quad (\text{B})$$

Nota: En https://artofproblemsolving.com/wiki/index.php/2018_AMC_12A_Problems/Problem_24 encontramos un razonamiento alternativo interesante mediante probabilidad condicional.

25

Aplicando la suma de la sucesión geométrica:

$$A_n = \underbrace{\overline{a}\overline{a}\dots\overline{a}}_n = a \cdot \underbrace{11\dots1}_n = a(10^0 + 10^1 + \dots + 10^n) = a \cdot \frac{10^{n+1} - 1}{9}$$

$$B_n = \underbrace{\overline{b}\overline{b}\dots\overline{b}}_n = b \cdot \frac{10^n - 1}{9}$$

$$C_n = \underbrace{\overline{c}\overline{c}\dots\overline{c}}_{2n} = c \cdot \underbrace{11\dots1}_{2n} = c \cdot \left(\frac{10^{2n} - 1}{9}\right) = c \cdot \frac{(10^n - 1)(10^n + 1)}{9}$$

$$C_n - B_n = A_n^2 \Leftrightarrow c \cdot \frac{(10^n - 1)(10^n + 1)}{9} - b \cdot \frac{10^n - 1}{9} = \left(a \cdot \frac{10^n - 1}{9}\right)^2 = a^2 \cdot \left(\frac{10^n - 1}{9}\right)^2$$

puesto que $n > 0$ podemos cancelar el factor $\frac{10^n - 1}{9}$ para llegar a

$$c \cdot (10^n + 1) - b = a^2 \cdot \left(\frac{10^n - 1}{9}\right)$$

Esta ecuación es lineal en 10^n , luego si es cierta para dos valores de n concretos será cierta para cualquier n .

$$\text{Para } n=1 \quad c \cdot (10+1) - b = a^2 \cdot \left(\frac{10-1}{9}\right) \Leftrightarrow 11c - b = a^2$$

$$\text{Para } n=2 \quad c \cdot (100+1) - b = a^2 \cdot \left(\frac{100-1}{9}\right) \Leftrightarrow 101c - b = 11a^2$$

$$\left. \begin{array}{l} 11c - b = a^2 \\ 101c - b = 11a^2 \end{array} \right\} \Rightarrow 101c - b - 11c + b = 11a^2 - a^2 \Leftrightarrow 90c = 10a^2 \Leftrightarrow 9c = a^2$$

$$\left. \begin{array}{l} 11c - b = a^2 \\ 101c - b = 11a^2 \end{array} \right\} \Rightarrow \left. \begin{array}{l} 110c - 10b = 10a^2 \\ 101c - b = 11a^2 \end{array} \right\} \Rightarrow 110c - 10b - 101c + b = 10a^2 - 11a^2$$

$$\Rightarrow 9c - 9b = -a^2 \Rightarrow a^2 - 9b = -a^2 \Rightarrow 9b = -2a^2 \Rightarrow b = \frac{2a^2}{9}$$

$$\text{Queremos encontrar el valor máximo de } a + b + c = a + \frac{2a^2}{9} + \frac{a^2}{9} = a + \frac{a^2}{3}$$

Equivale a encontrar el valor máximo de a .

La ecuación $9c = a^2$ tiene tres soluciones (a, c) : $(3,1)$, $(6,4)$, $(9,9)$.

Pero sustituyendo para encontrar b : $11c - b = a^2 \Rightarrow b = 11c - a^2$

$$(3,1) \Rightarrow b = 11 - 9 = 2, \quad (6,4) \Rightarrow b = 44 - 36 = 8, \quad (9,9) \Rightarrow b = 99 - 81 = 18$$

Vemos que la solución $(9,9)$ no es aceptable pues incumple la condición $b \leq 9$.

Observamos que se cumple siempre para $a=6, b=8, c=4$. Lo podemos ver con un ejemplo:

$$n=4 \rightarrow 6^2 \cdot 1111 = 39996 = 40000 - 4 = 40000 + 4 - 8 = 4(10^4 + 1) - 8$$

Si observamos los números que se obtienen de la forma $a^2 \underbrace{11\dots 1}_n$:

$$\begin{array}{lllll} 1 - 1111 & 2 - 4444 & 3 - 9999 & 4 - 17776 & 5 - 27775 \\ 6 - 39996 & 7 - 54439 & 8 - 71104 & 9 - 89991 & \end{array}$$

Vemos que el siguiente candidato para obtener un número “parecido” a 10^n sería

$$a=9 \rightarrow 9^2 \cdot 11\dots 11 = 89\dots 91$$

Pero vemos que no es aceptable:

$$n=4 \rightarrow 9^2 \cdot 1111 = 89991 = 90000 - 9 = 90000 + 9 - 18 = 9(10^4 - 1) - 18$$

Pero b no puede ser 18.

Luego la solución es $6+8+4=18$ (D).

AMC12B 2018 Enunciados

1

Kate cocina un pan que mide 20 por 18 pulgadas. Este pan se corta en piezas que miden 2 por 2 pulgadas. ¿Cuántas piezas de pan se van a obtener?

(A) 90 (B) 100 (C) 180 (D) 200 (E) 360

2

Sam condujo 96 millas en 90 minutos. Su velocidad media durante los primeros 30 minutos fue 60 mph (millas por hora), y su velocidad media durante los segundos 30 minutos fue de 65 mph. ¿Cuál fue su velocidad media, en mph, durante los últimos 30 minutos?

(A) 64 (B) 65 (C) 66 (D) 67 (E) 68

3

Una recta con pendiente 2 y una recta con pendiente 6 se cortan en el punto (40,30). Determina la distancia entre sus respectivos puntos de corte con el eje X.

(A) 5 (B) 10 (C) 20 (D) 25 (E) 50

4

Supongamos que un círculo tiene una cuerda de longitud 10, y que la distancia del centro del círculo a la cuerda es 5. ¿Cuál es el área del círculo?

(A) 25π (B) 50π (C) 75π (D) 100π (E) 125π

5

Determina el número de subconjuntos de $\{2,3,4,5,6,7,8,9\}$ que contienen al menos un número primo.

(A) 128 (B) 192 (C) 224 (D) 240 (E) 256

6

Supongamos que S latas de soda se pueden ofrecer en una máquina expendedora por Q cuartos. Determina la expresión que describe el número de latas de soda que se pueden ofrecer por D dólares, donde 1 dólar son 4 cuartos.

(A) $\frac{4DQ}{S}$ (B) $\frac{4DS}{Q}$ (C) $\frac{4Q}{DS}$ (D) $\frac{DQ}{4S}$ (E) $\frac{DS}{4Q}$

7

Determina el valor de $\log_3 7 \cdot \log_5 9 \cdot \log_7 11 \cdot \log_9 13 \cdot \dots \cdot \log_{21} 25 \cdot \log_{23} 27$

(A) 3 (B) $3\log_7 23$ (C) 6 (D) 9 (E) 10

8

Sea $\overline{AB}$ el diámetro de una circunferencia con $AB = 24$. Si C es un punto, diferente de A o B , que se mueve por dicha circunferencia, el baricentro de $\triangle ABC$ genera una curva cerrada excepto por dos puntos. Determina el entero positivo más cercano al área de la región rodeada por dicha curva.

- (A) 25 (B) 38 (C) 50 (D) 63 (E) 75

9

Determina $\sum_{i=1}^{100} \sum_{j=1}^{100} (i + j)$

- (A) 100100 (B) 500500 (C) 505000 (D) 1001000 (E) 1010000

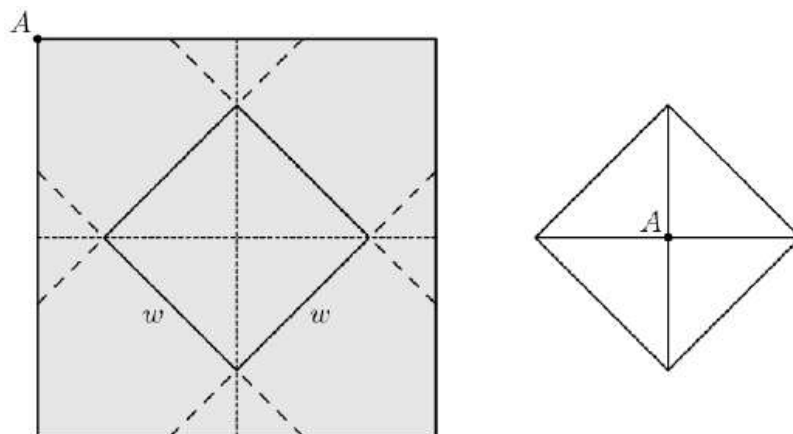
10

Supongamos que una lista de 2018 enteros tiene una única moda, que ocurre exactamente 10 veces. ¿Cuál es el mínimo número de valores distintos que puede haber en esta lista?

- (A) 202 (B) 223 (C) 224 (D) 225 (E) 234

11

Construimos una caja cerrada de base cuadrada doblando y recortando un cartón cuadrado de la siguiente manera: Se trazan dos rectas por el centro del cuadrado, perpendiculares entre ellas y paralelas a los lados. El centro de la caja será el centro del cuadrado, tal y como aparece en la figura de la izquierda. Las cuatro esquinas del cuadrado de cartón se levantan y se juntan en el centro de la cara superior de la caja, el punto A en la figura de la izquierda. La caja tiene una base de longitud w y altura h . Determina el área del cuadrado de cartón.



- (A) $2(w+h)^2$ (B) $\frac{(w+h)^2}{2}$ (C) $2w^2 + 4wh$ (D) $2w^2$ (E) w^2h

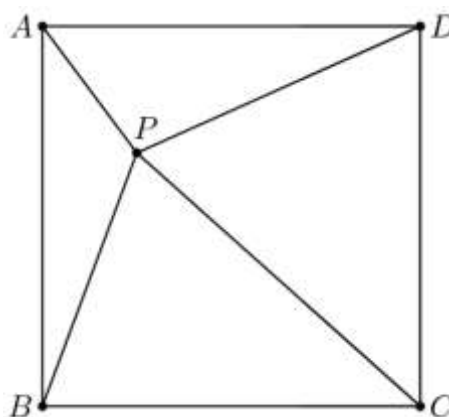
12

El lado $\overline{AB}$ del triángulo $\triangle ABC$ tiene longitud 10. La bisectriz del ángulo A corta el lado $\overline{BC}$ en D, y $CD=3$. El conjunto de todos los posibles valores de AC es un intervalo abierto (m,n) . Determina $m+n$.

- (A) 16 (B) 17 (C) 18 (D) 19 (E) 20

13

Sea ABCD un cuadrado de lado 30. Sea P un punto en su interior cumpliendo $AP=12$ y $BP=26$. Determina el área del cuadrilátero convexo formado por los baricentros de los triángulos $\triangle ABP$, $\triangle BCP$, $\triangle CDP$ y $\triangle DAP$.



- (A) $100\sqrt{2}$ (B) $100\sqrt{3}$ (C) 200 (D) $200\sqrt{2}$ (E) $200\sqrt{3}$

14

Joey, Chloe y su hija Zoe cumplen años el mismo día. Joey es un año mayor que Chloe y Zoe cumple hoy su primer año. Hoy es el primero de los nueve aniversarios en los que la edad de Chloe será múltiple de la edad de Zoe. Determina la suma de los dos dígitos de la edad de Joey que tendrá la próxima vez que sea múltiple de la edad de Zoe.

- (A) 7 (B) 8 (C) 9 (D) 10 (E) 11

15

Determina el número de enteros positivos de 3 cifras que son divisibles entre 3 pero que no contienen el dígito 3.

- (A) 96 (B) 8 (C) 9 (D) 10 (E) 11

16

Las soluciones de la ecuación $(z+6)^8=81$ están conectadas en el plano complejo formando un polígono regular, tres de cuyos vértices denotamos por A, B y C. Determina el área mínima del triángulo $\triangle ABC$.

- (A) $\frac{1}{6}\sqrt{6}$ (B) $\frac{3}{2}\sqrt{2}-\frac{3}{2}$ (C) $2\sqrt{3}-3\sqrt{2}$ (D) $\frac{1}{2}\sqrt{2}$ (E) $\sqrt{3}-1$

17

Sean p y q enteros positivos tales que

$$\frac{5}{9} < \frac{p}{q} < \frac{4}{7}$$

y siendo q tan pequeño como sea posible. Determina $q - p$.

(A) 7 (B) 11 (C) 13 (D) 17 (E) 19

18

Definimos la función f recursivamente por $f(1) = f(2) = 1$ y

$$f(n) = f(n-1) - f(n-2) + n$$

para todos los enteros $n \geq 3$. Determina $f(2018)$.

(A) 2016 (B) 2017 (C) 2018 (D) 2019 (E) 2020

19

Mary toma un número entero par de 4 dígitos n y escribe todos sus divisores en orden creciente, de izquierda a derecha: $1, 2, \dots, \frac{n}{2}, n$. En un cierto momento, Mary escribe el número 323 como divisor de un cierto n . ¿Cuál es el menor entero divisor de n que Mary escribirá a la derecha de 323?

(A) 324 (B) 330 (C) 340 (D) 361 (E) 646

21

Dado un triángulo $\triangle ABC$ con lados $AB = 13$, $AC = 12$ y $BC = 5$, denotamos por O su circuncentro y por I su incentro. Sea M el centro de la circunferencia que es tangente a los lados AC y BC y al circuncírculo de $\triangle ABC$. Determina el área del triángulo $\triangle MOI$.

(A) $5/2$ (B) $11/4$ (C) 3 (D) $13/4$ (E) $7/2$

22

Determina el número de polinomios $P(x)$ de grado menor o igual que 3, cuyos coeficientes son todos elementos del conjunto $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ y que cumplen $P(-1) = -9$.

(A) 110 (B) 143 (C) 165 (D) 220 (E) 286

23

Ajay está situado en el punto A, cerca de Pontianak, Indonesia, latitud 0° y longitud 110° E. Billy está situado en el punto B, cerca de Big Baldy Mountain, Idaho, USA, latitud 45° N longitud 115° O. Suponiendo que la Tierra es una esfera perfecta con centro C, determina la amplitud, en grados del ángulo $\angle ACB$.

(A) 105 (B) $112\frac{1}{2}$ (C) 120 (D) 135 (E) 150

24

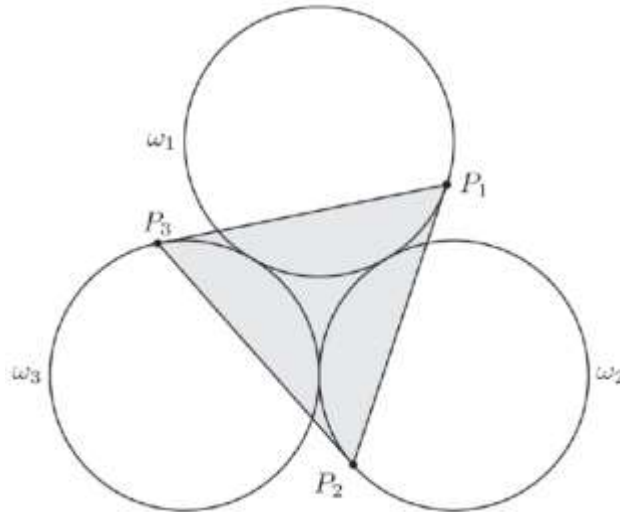
Denotando por $\lfloor x \rfloor$ el mayor entero menor o igual que x , Determina el número de soluciones reales de la ecuación

$$x^2 + 10000 \lfloor x \rfloor = 10000 x$$

(A) 197 (B) 198 (C) 199 (D) 200 (E) 201

25

Sean $\omega_1, \omega_2, \omega_3$ circunferencias de radio 4 situadas en el plano de forma que cada una de ellas es externamente tangente a las otras dos. Sean P_1, P_2, P_3 puntos pertenecientes respectivamente a $\omega_1, \omega_2, \omega_3$ de forma que $P_1P_2 = P_2P_3 = P_3P_1$ y la recta P_iP_{i+1} es tangente a ω_i para cada $i=1,2,3$, con $P_4 = P_1$, tal y como se muestra en la figura. El área del triángulo $\Delta P_1P_2P_3$ se puede escribir de la forma $\sqrt{a} + \sqrt{b}$ para ciertos enteros positivos a, b . Determina $a+b$.



(A) 546 (B) 548 (C) 550 (D) 552 (E) 554

AMC12B 2018 Soluciones (Letra)

1. A
2. D
3. B
4. B
5. D
6. B
7. C
8. C
9. E
10. D
11. A
12. C
13. C
14. E
15. A
16. B
17. A
18. B
19. C
20. C
21. E
22. D
23. C
24. C
25. D

AMC12B 2018 Soluciones desarrolladas

1

Son $20/2=10$ piezas de largo por $18/2=9$ piezas de ancho, en total $10 \cdot 9 = 90$ piezas (A)

2

En la primera media hora:

$$60 = \frac{\text{espacio}}{\text{tiempo}} = \frac{\text{espacio}}{0.5} \Rightarrow \text{espacio} = 60 \cdot 0.5 = 30 \text{ millas en la primera media hora.}$$

En la segunda media hora:

$$65 = \frac{\text{espacio}}{\text{tiempo}} = \frac{\text{espacio}}{0.5} \Rightarrow \text{espacio} = 65 \cdot 0.5 = 32.5 \text{ millas en la segunda media hora.}$$

Luego en la tercera media hora tiene que recorrer $96 - 30 - 32.5 = 33.5$ millas, y por tanto su velocidad media será

$$\text{velocidad} = \frac{\text{espacio}}{\text{tiempo}} = \frac{33.5}{0.5} = 67 \text{ millas/hora (D)}$$

3

Determinamos las rectas y sus respectivos puntos de corte con el eje X.

$$y = 2x + b \Rightarrow 30 = 2 \cdot 40 + b \Rightarrow b = 30 - 80 = -50 \rightarrow y = 2x - 50$$

$$0 = 2x - 50 \Rightarrow x = 25$$

$$y = 6x + b \Rightarrow 30 = 6 \cdot 40 + b \Rightarrow b = 30 - 240 = -210 \rightarrow y = 6x - 210$$

$$0 = 6x - 210 \Rightarrow x = 35$$

Luego la distancia es $35 - 25 = 10$ (B)

4

Sabemos que la distancia entre el centro de la circunferencia y la cuerda es la distancia del centro de la circunferencia al punto medio de la cuerda, generando un triángulo rectángulo de catetos 5 y 5, con lo cual el radio es $r = \sqrt{5^2 + 5^2} = \sqrt{50}$, y el área es $A = \pi r^2 = 50\pi$ (B).

5

Este conjunto contiene 4 números primos y 4 números compuestos. Para evitar repeticiones, organizaremos los subconjuntos válidos por el número de primos que contiene:

Primera versión.

$$\text{Con un número primo: } \binom{4}{1} \cdot 2^4 \quad \text{Con dos números primos: } \binom{4}{2} \cdot 2^4$$

$$\text{Con tres números primos: } \binom{4}{3} \cdot 2^4 \quad \text{Con cuatro números primos: } \binom{4}{4} \cdot 2^4$$

Total:

$$\binom{4}{1} \cdot 2^4 + \binom{4}{2} \cdot 2^4 + \binom{4}{3} \cdot 2^4 + \binom{4}{4} \cdot 2^4 = 2^4 \left[\binom{4}{1} + \binom{4}{2} + \binom{4}{3} + \binom{4}{4} \right] =$$

$$= 2^4 \left[2^4 - \binom{4}{0} \right] = 2^4 (2^4 - 1) = 16 \cdot 15 = 240 \quad (D)$$

En donde hemos aplicado la identidad $\binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{n} = 2^n$.

Segunda versión.

Mucho más rápidamente, pasando al complementario: Hay en total 2^8 subconjuntos, y que no contengan ninguno de los 4 primos son 2^4 , y por tanto los que sí contienen alguno de los 4 primos son $2^8 - 2^4 = 256 - 16 = 240 \quad (D)$.

6

https://artofproblemsolving.com/wiki/index.php/2018_AMC_12B_Problems/Problem_6

7

Aplicando la regla de la cadena de los logaritmos:

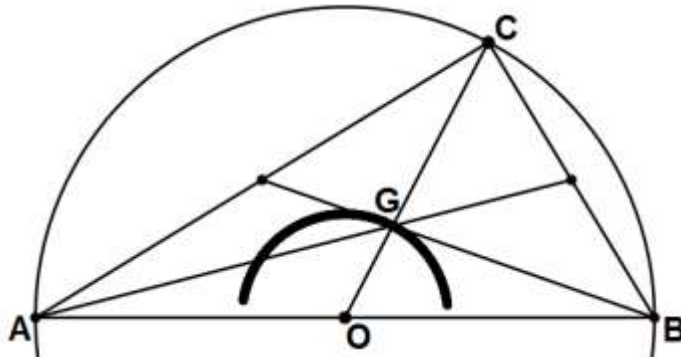
$$\log_3 7 \cdot \log_7 11 \cdot \log_{11} 15 \cdot \log_{15} 19 \cdot \log_{19} 23 \cdot \log_{23} 27 = \log_3 27 = \log_3 3^3 = 3$$

$$\log_5 9 \cdot \log_9 13 \cdot \log_{13} 17 \cdot \log_{17} 21 \cdot \log_{21} 25 = \log_5 25 = \log_5 5^2 = 2$$

Y puesto que hay 11 factores, ya los hemos utilizado todos, y el resultado es $3 \cdot 2 = 6 \quad (C)$

8

Sea G dicho baricentro. Sea O el centro de dicha circunferencia.



Sabemos, por las propiedades del baricentro, que $OG = \frac{1}{3}OC = \frac{1}{3}12 = 4$, así pues, el baricentro describe una circunferencia de radio 4, cuyo área es $\pi \cdot 4^2 \cong 3.1 \cdot 16 = 49.6$, y el entero que mejor se aproxima es 50 (C).

9

Fijando un i tenemos:

$$\sum_{j=1}^{100} (i+j) = i+1+i+2+i+3+\dots+i+100 = 100i+1+2+3+\dots+100 = 100i + \frac{100 \cdot 101}{2} =$$

$$= 100i + \frac{100 \cdot 101}{2} = 100i + 5050$$

$$\sum_{i=1}^{100} \sum_{j=1}^{100} (i+j) = \sum_{i=1}^{100} 100i + 5050 = 5050 \cdot 100 + 100 \sum_{i=1}^{100} i = 505000 + 100 \frac{100 \cdot 101}{2} =$$

$$= 505000 + 100 \cdot 50 \cdot 101 = 505000 + 505000 = 1010000 \quad (E)$$

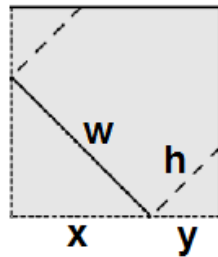
10

Sabemos que existe un único número n que aparece 10 veces, luego el resto de números aparecerán, como mucho, nueve veces.

$$2008 = 223 \cdot 9 + 1$$

No olvidemos que también está n , luego habrá, como mucho, $223+1+1=255$ números distintos (D)

11



Aplicando Pitágoras tenemos que

$$\left. \begin{aligned} w^2 &= 2x^2 \Rightarrow x = \frac{w}{\sqrt{2}} \\ h^2 &= 2y^2 \Rightarrow y = \frac{h}{\sqrt{2}} \end{aligned} \right\} \Rightarrow x+y = \frac{w}{\sqrt{2}} + \frac{h}{\sqrt{2}} = \frac{w+h}{\sqrt{2}} \Rightarrow 2(x+y) = \frac{2(w+h)}{\sqrt{2}} = \sqrt{2}(w+h)$$

$$\text{Y por tanto el área será } (2(x+y))^2 = \frac{2(w+h)^2}{\sqrt{2}} = 2(w+h)^2$$

12

Sean $x=AC$, $y=BD$. Aplicando el Teorema de la bisectriz:

$$\frac{3}{x} = \frac{y}{10} \Leftrightarrow 30 = xy \Leftrightarrow y = \frac{30}{x} \quad (\text{puesto que está claro que } x \neq 0 \text{ pues es un lado del triángulo})$$

Aplicando las desigualdades triangulares:

$$\begin{cases} x < 10+3+y \\ 3+y < x+10 \\ 10 < x+3+y \end{cases} \Leftrightarrow \begin{cases} x < 13+y \\ y < x+7 \\ 7 < x+y \end{cases} \Leftrightarrow \begin{cases} x < 13+30/x \\ 30/x < x+7 \\ 7 < x+30/x \end{cases}$$

$$x < 13 + \frac{30}{x} \Leftrightarrow x^2 < 13x + 30 \Leftrightarrow 0 < x^2 - 13x - 30 = (x - 15)(x + 2) \Leftrightarrow x \in (0, 15)$$

$$\frac{30}{x} < x + 7 \Leftrightarrow 30 < x^2 + 7x \Leftrightarrow 30 - x^2 - 7x < 0 \Leftrightarrow 0 < x^2 + 7x - 30 = (x - 3)(x + 10) \Leftrightarrow x > 3$$

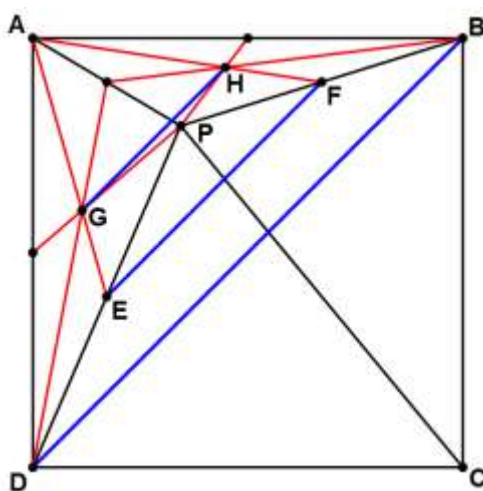
$$7 < x + \frac{30}{x} \Leftrightarrow 7x < x^2 + 30 \Leftrightarrow 7x - x^2 - 30 < 0 \Leftrightarrow 0 < x^2 + 7x + 30$$

Esta última desigualdad se cumple siempre, luego el conjunto solución es el conjunto intersección de las dos primeras, es decir, $(3, 15)$, y la respuesta es $3 + 15 = 18$ (C).

13

Sean G y H los respectivos baricentros de $\triangle DAP$ y $\triangle ABP$.

Sean E y F los respectivos puntos medios de DP y PB.



Aplicando el Teorema del Conector de puntos medios (GA/7.2.1) en el triángulo $\triangle DBP$, tenemos que $EF \parallel BD$ y $EF = BD/2$.

Sabemos, además, que las medianas se cortan en razón 2:1 (GA/11.5.3b), luego por Tales y semejanza de triángulos, $GH \parallel EF$ y $GH = \frac{2}{3} EF$.

Con todo esto llegamos a $GH \parallel BD$ y $GH = \frac{2}{3} \frac{1}{2} BD = \frac{1}{3} BD$

Donde BD es la diagonal de un cuadrado de lado 30, es decir: $BD = \sqrt{30^2 + 30^2} = 30\sqrt{2}$

Y por lo tanto $GH = \frac{1}{3} 30\sqrt{2} = 10\sqrt{2}$.

Esto mismo lo podemos hacer con los otros tres lados del cuadrilátero, obteniendo un paralelogramo con los lados todos iguales a $10\sqrt{2}$.

Los lados de este rombo son paralelos a las diagonales del cuadrado, pero como las diagonales son perpendiculares entre sí, los lados de este rombo serán también perpendiculares entre sí, es decir, el cuadrilátero es un cuadrado, y su área será, por tanto:

$$(10\sqrt{2})^2 = 200 \quad (C)$$

Como curiosidad observamos que las dimensiones de este cuadrado no dependen de la posición del punto P.

14

Sean J , C y Z las edades respectivas de Joey, Chloe y Zoe.

Tenemos que $J = C + 1$ y $Z = 1$

En cada aniversario tenemos la siguiente pauta:

Zoe	1	2	3	4	Z
Chloe	C	C+1	C+2	C+3	C+Z-1
Joey	C+1	C+2	C+3	C+4	C+Z

Está claro que C es múltiplo de 1, y existirán ocho números más $Z_1, Z_2, \dots, Z_8$ tales que

$$Z_1 | C + Z_1 - 1, Z_2 | C + Z_2 - 1, \dots, Z_8 | C + Z_8 - 1$$

Puesto que $Z_i | Z_i$, entonces está claro que $Z_i | C - 1$, es decir, el número $C - 1$ tiene exactamente ocho divisores diferentes de 1. Luego $C - 1$ tiene nueve divisores contando el 1. Aplicando 19.4, tenemos que

$$C - 1 = p_1^{a_1} p_2^{a_2} \dots p_r^{a_r}$$

entonces

$$9 = (a_1 + 1)(a_2 + 1) \dots (a_r + 1)$$

Con un solo factor primo sería imposible, pues como mínimo tendríamos el resultado $C - 1 = 2^8 = 256$, una edad absurda.

Con dos factores primos, la única posibilidad aceptable para una edad es

$$9 = (2 + 1)(2 + 1) \Rightarrow C - 1 = 2^2 3^2 = 36$$

Y por tanto $C = 36 + 1 = 37 \Rightarrow J = C + 1 = 38$

Zoe	1	2	3	4	5	Z
Chloe	37	38	39	40	41	...
Joey	38	39	40	41	42	$C = 37 + Z$

Nos piden determinar el siguiente Z tal que $Z | J \Leftrightarrow Z | 37 + Z \Rightarrow Z | 37$, pero puesto que 37 es primo, la única posibilidad es $Z = 37 \Rightarrow J = 37 + 37 = 74$, y la respuesta correcta es $7 + 4 = 11$ (E).

15

Queremos determinar la cantidad de enteros $\bar{a} \bar{b} \bar{c}$ de forma que

$$a \in \{1, 2, 4, 5, 6, 7, 8, 9\}$$

$$b \in \{0, 1, 2, 4, 5, 6, 7, 8, 9\}$$

$$c \in \{1, 5, 7, 9\}$$

satisfaciendo $a + b + c \equiv 0 \pmod{3}$.

Ordenamos los casos en función del tercer dígito c :

- A) $a + b + 1 \equiv 0 \pmod{3} \Leftrightarrow a + b \equiv 2 \pmod{3}$
 B) $a + b + 5 \equiv 0 \pmod{3} \Leftrightarrow a + b \equiv 1 \pmod{3}$
 C) $a + b + 7 \equiv 0 \pmod{3} \Leftrightarrow a + b \equiv 2 \pmod{3}$
 D) $a + b + 9 \equiv 0 \pmod{3} \Leftrightarrow a + b \equiv 0 \pmod{3}$

Vemos que los grupos B, C y D, juntos, recorren todas las combinaciones posibles de parejas (a, b) con un total de $8 \cdot 9 = 72$ casos.

Luego solo tenemos que añadir los elementos del grupo A, que se pueden contar uno por uno:
 11, 15, 17, 20, 26, 29, 41, 44, 47, 50, 56, 59, 62, 65, 68, 71, 74, 77, 80, 86, 89, 92, 95, 98

En total hay $8 \cdot 3 = 24$ casos de combinaciones posibles (a, b) .

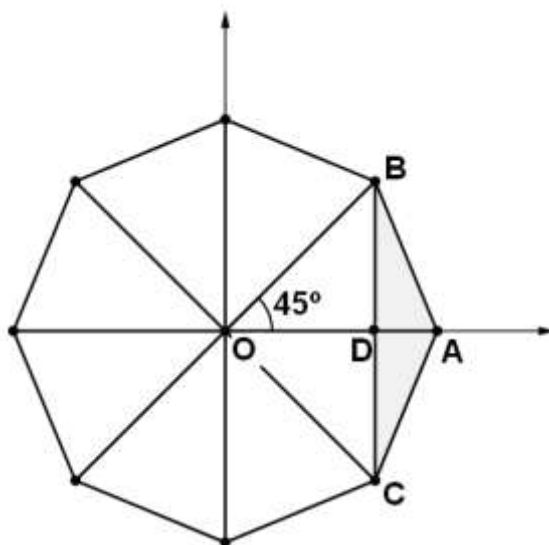
Finalmente, tenemos un total de $72 + 24 = 96$ combinaciones posibles (a, b, c) (A)

16

Las raíces de $(z + 6)^8 = 81$ forman en el plano complejo un octágono de radio

$$\sqrt[8]{81} = 3^{4/8} = \sqrt{3}$$

Este octágono está trasladado seis unidades hacia la izquierda, pero eso no afecta en nada las áreas que determinan sus vértices. Vemos que el área mínima se encuentra cuando los tres vértices son consecutivos, por ejemplo cuando $A = \sqrt{3}$, $B = \sqrt{3}_{45^\circ}$, $C = \sqrt{3}_{315^\circ}$



$$A = (\sqrt{3}, 0), B = \sqrt{3}_{45^\circ} = \left(\frac{\sqrt{3}}{\sqrt{2}}, \frac{\sqrt{3}}{\sqrt{2}} \right), C = \sqrt{3}_{315^\circ} = \left(\frac{-\sqrt{3}}{\sqrt{2}}, \frac{\sqrt{3}}{\sqrt{2}} \right)$$

El triángulo $\triangle ABC$ tiene área:

$$[\triangle ABC] = \frac{1}{2} \left(\frac{2\sqrt{3}}{\sqrt{2}} \right) \left(\sqrt{3} - \frac{\sqrt{3}}{\sqrt{2}} \right) = \frac{\sqrt{3}}{\sqrt{2}} \left(\sqrt{3} - \frac{\sqrt{3}}{\sqrt{2}} \right) = \frac{3\sqrt{2}}{2} - \frac{3}{2} \quad (\text{B})$$

17

Primera versión.

$$\frac{5}{9} < \frac{p}{q} < \frac{4}{7} \Leftrightarrow 35q < 63p < 36q \Leftrightarrow 0 < 63p - 35q < 36q - 35q \Leftrightarrow 0 < 7(9p - 5q) < q$$

De aquí se deduce que $q > 7$.

Para los primeros valores de q tenemos que

$$1 = 9p - 5q \Rightarrow p = \frac{1+5q}{9}, \text{ que no es entero para todos los valores } q \leq 14.$$

Para $q > 14$, también podríamos tener $9p - 5q = 2$, y entonces: $p = \frac{2+5q}{9}$.

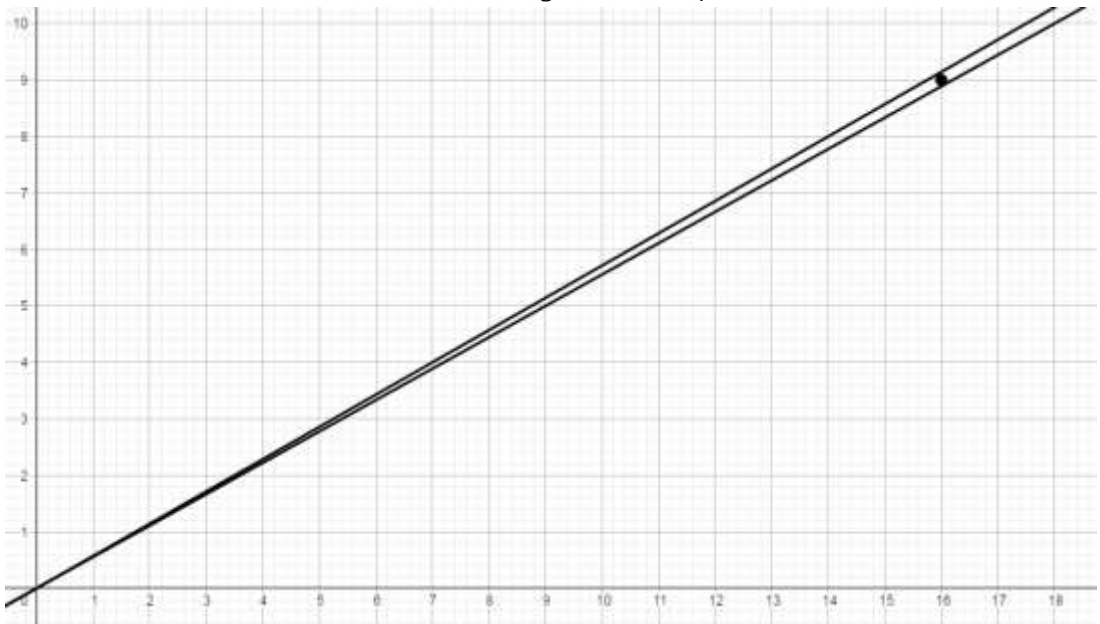
Seguimos probando: El valor $q = 15$ no es aceptable, pero con $q = 16$ tenemos:

$$p = \frac{1+5 \cdot 16}{9} = 9, \text{ resultado entero. Por lo tanto la solución es } 16 - 9 = 7 \text{ (A).}$$

Segunda versión.

$$\frac{5}{9} < \frac{p}{q} < \frac{4}{7} \Leftrightarrow \frac{5}{9}q < p < \frac{4}{7}q$$

Representando las respectivas gráficas $f(q) = \frac{5}{9}q$, $g(q) = \frac{4}{7}q$



Se observa que el primer punto de componentes enteras que está entre las dos gráficas es (16,9), y por tanto la solución es $16 - 9 = 7$ (A).

18

$$f(n) = f(n-1) - f(n-2) + n = f(n-2) - f(n-3) + n - 1 - f(n-2) + n = -f(n-3) + 2n - 1$$

Luego

$$f(n-3) = -f(n-6) + 2(n-3) - 1 = -f(n-6) + 2n - 7$$

Y por tanto:

$$f(n) = -(-f(n-6) + 2n - 7) + 2n - 1 = f(n-6) - 2n + 7 + 2n - 1 = f(n-6) + 6$$

Así pues, por cada vez que puedo restar 6, sumo 6, y puesto que $2018 = 336 \cdot 6 + 2$

Puedo restar 336 veces, y por tanto

$$f(2018) = f(2) + 336 \cdot 6 = 1 + 2016 = 2017 \quad (\text{B})$$

19

Primera versión.

Sabemos que $2 \mid n$ y que $323 \mid n$, y puesto que 2 y 323 son coprimos, se deduce que $646 \mid n$, es decir, $n = 646k$. Puesto que $1000 \leq n < 10000$, está claro que $2 \leq k \leq 15$.

Por otro lado, vemos que $323 = 17 \cdot 19$, luego la lista de todos los números n posibles es:

$$\begin{aligned} &2 \cdot 17 \cdot 19 \cdot 2, 2 \cdot 17 \cdot 19 \cdot 3, 2 \cdot 17 \cdot 19 \cdot 2^2, 2 \cdot 17 \cdot 19 \cdot 5, 2 \cdot 17 \cdot 19 \cdot 2 \cdot 3 \\ &2 \cdot 17 \cdot 19 \cdot 7, 2 \cdot 17 \cdot 19 \cdot 2^3, 2 \cdot 17 \cdot 19 \cdot 3^2, 2 \cdot 17 \cdot 19 \cdot 2 \cdot 5, 2 \cdot 17 \cdot 19 \cdot 11 \\ &2 \cdot 17 \cdot 19 \cdot 3 \cdot 2^2, 2 \cdot 17 \cdot 19 \cdot 13, 2 \cdot 17 \cdot 19 \cdot 2 \cdot 7, 2 \cdot 17 \cdot 19 \cdot 3 \cdot 5 \end{aligned}$$

Tomamos las posibles respuestas que nos ofrece el enunciado y vamos descartando, de menor a mayor:

$324 = 2^2 \cdot 3^4$ y no aparece como divisor de ninguno de los números anteriores.

$330 = 2 \cdot 3 \cdot 5 \cdot 11$ y tampoco es aceptable.

$340 = 2^2 \cdot 5 \cdot 17$ y este valor sí que aparece como divisor de $2 \cdot 17 \cdot 19 \cdot 2 \cdot 5$ luego 340 es la respuesta correcta (C).

Segunda versión.

Sabemos que $323 = 17 \cdot 19 \mid n$, luego el siguiente divisor que buscamos debe ser múltiplo de 17 o de 19, porque en caso contrario obtendríamos un número demasiado grande.

Por ejemplo:

Supongamos que $324 = 2^2 \cdot 3^4 \mid n$. Entonces:

$$\left. \begin{aligned} 324 &= 2^2 \cdot 3^4 \mid n \\ 323 &= 17 \cdot 19 \mid n \end{aligned} \right\} \Rightarrow 2^2 \cdot 3^4 \cdot 17 \cdot 19 \mid n \Rightarrow 104652 \mid n \Rightarrow n \geq 104652 \quad \text{absurdo.}$$

El primer valor de la lista que cumple esta condición es la opción C: $340 = 2^2 \cdot 5 \cdot 17$, y con este número podemos obtener un valor de n aceptable: $2 \cdot 17 \cdot 19 \cdot 2 \cdot 5 = 6460$, así pues, la respuesta es (C).

21

En primer lugar vemos que $13^2 = 12^2 + 5^2$, con lo que $\triangle ABC$ es un triángulo rectángulo con hipotenusa AB.

Resolveremos este problema mediante coordenadas cartesianas.

Sean $C = (0,0)$, $B = (12,0)$ y $A = (0,5)$.

Está claro que el ortocentro es el punto medio de la hipotenusa, es decir: $O = (6, 5/2)$.

Las coordenadas del incentro las podemos deducir de la fórmula $[\Delta ABC] = r \cdot s$ (ver [GA/11.4.8](#))

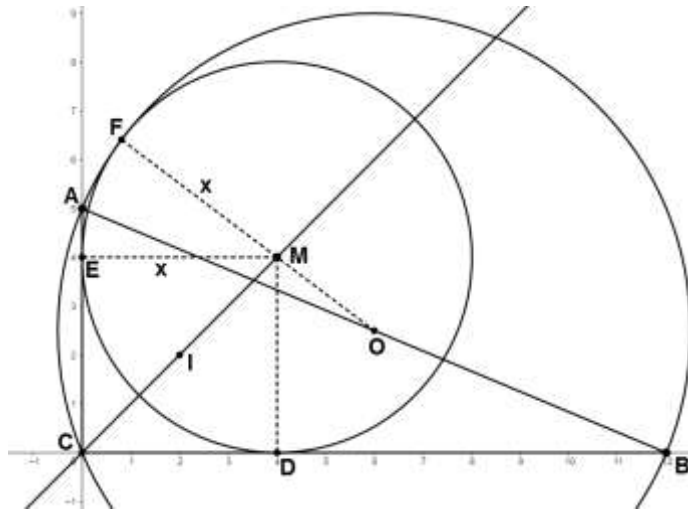
$$\left. \begin{aligned} [\Delta ABC] &= \frac{12 \cdot 5}{2} = 30 \\ s &= \frac{12 + 5 + 13}{2} = 15 \end{aligned} \right\} \Rightarrow r = \frac{[\Delta ABC]}{s} = \frac{30}{15} = 2$$

La bisectriz del ángulo A tiene por ecuación $y = x$, y por tanto $I = (2, 2)$.

El punto M equidista de los lados CB y CA, luego pertenecerá a la bisectriz $y = x$, y podemos escribir $M = (x, x)$ para cierto x que tenemos que determinar.

Sean D, E y F los puntos de tangencia a los lados AC, BC y al circuncírculo de ΔABC , respectivamente.

La clave para encontrar M es observar que $x = EM = DM = MF$



Y por tanto

$$\frac{13}{2} = FO = FM + MO = x + \sqrt{(x-6)^2 + (x-5/2)^2} \Rightarrow$$

$$\frac{13}{2} - x = \sqrt{(x-6)^2 + (x-5/2)^2} \Rightarrow \left(\frac{13}{2} - x\right)^2 = (x-6)^2 + \left(x - \frac{5}{2}\right)^2 \Rightarrow$$

$$\frac{169}{4} + x^2 - 13x = x^2 - 12x + 36 + x^2 + \frac{25}{4} - 5x \Rightarrow 0 = x^2 - 4x = x(x-4) \Leftrightarrow \begin{cases} x=0 \\ x=4 \end{cases}$$

Así pues, llegamos a $M = (4, 4)$ y el área del triángulo lo podemos calcular mediante la “fórmula de la lazada” (Ver [GA/18.6.5](#)):

$$\left. \begin{aligned} I &= (2, 2) \\ M &= (4, 4) \\ O &= (6, 5/2) \end{aligned} \right\} \Rightarrow \begin{pmatrix} 4 & 4 \\ 6 & 5/2 \\ 2 & 2 \\ 4 & 4 \end{pmatrix} = 4 \cdot \frac{5}{2} + 6 \cdot 2 + 2 \cdot 4 - 6 \cdot 4 - 2 \cdot \frac{5}{2} - 2 \cdot 4 =$$

$$= 10 + 12 - 24 - 5 = -7 \Rightarrow [\Delta MOI] = 7/2 \quad (E)$$

22

Primera versión.

Sea $P(x) = ax^3 + bx^2 + cx + d$

$$-9 = P(-1) = a(-1)^3 + b(-1)^2 + c(-1) + d = -a + b - c + d = b + d - (a + c) \Leftrightarrow$$

$$a + c - (b + d) = 9$$

a) $a + c = 9, b + d = 0$

$$a + c = 9 \Rightarrow a = 9, c = 0; a = 8, c = 1; a = 7, c = 2; \dots; a = 0, c = 9 \rightarrow 10 \text{ casos}$$

$$b + d = 0 \Rightarrow b = 0, d = 0 \rightarrow 1 \text{ caso.}$$

$$\text{Total: } 10 \cdot 1 = 10 \text{ casos.}$$

b) $a + c = 10, b + d = 1$

$$a + c = 10 \Rightarrow a = 9, c = 1; a = 8, c = 2; a = 7, c = 3; \dots; a = 1, c = 9 \rightarrow 9 \text{ casos}$$

$$b + d = 1 \Rightarrow b = 1, d = 0; b = 0, d = 1 \rightarrow 2 \text{ casos.}$$

$$\text{Total: } 9 \cdot 2 = 18 \text{ casos.}$$

c) $a + c = 11, b + d = 2$

$$a + c = 11 \Rightarrow a = 9, c = 2; a = 8, c = 3; a = 7, c = 4; \dots; a = 2, c = 9 \rightarrow 8 \text{ casos}$$

$$b + d = 2 \Rightarrow b = 2, d = 0; b = 1, d = 1; b = 0, d = 2 \rightarrow 3 \text{ casos.}$$

$$\text{Total: } 8 \cdot 3 = 24 \text{ casos.}$$

d) $a + c = 12, b + d = 3$

$$a + c = 12 \Rightarrow a = 9, c = 3; a = 8, c = 4; a = 7, c = 5; \dots; a = 3, c = 9 \rightarrow 7 \text{ casos}$$

$$b + d = 3 \Rightarrow b = 3, d = 0; b = 2, d = 1; b = 1, d = 2; b = 0, d = 3 \rightarrow 4 \text{ casos.}$$

$$\text{Total: } 7 \cdot 4 = 28 \text{ casos.}$$

Y así hasta:

$$a + c = 0, b + d = 9 \text{ que es igual que el caso a), con } 1 \cdot 10 = 10 \text{ casos.}$$

Así pues, el número de casos es:

$$10 \cdot 1 + 9 \cdot 2 + 8 \cdot 3 + 7 \cdot 4 + 6 \cdot 5 + 5 \cdot 6 + 4 \cdot 7 + 3 \cdot 8 + 2 \cdot 9 + 1 \cdot 10 =$$

$$= 2(10 \cdot 1 + 9 \cdot 2 + 8 \cdot 3 + 7 \cdot 4 + 6 \cdot 5) = 2 \cdot 110 = 220 \quad (D)$$

Segunda versión.

Llegando a $-a - c + b + d = -9$, haciendo el cambio de variable $-a = a' - 9$ y $-c = c' - 9$, tenemos la ecuación

$$a' + c' + b + d = 9 \text{ con } 0 \leq a', b, c', d \leq 9$$

Y podemos aplicar "Barras y estrellas" (CO/5.7b), para obtener un total de

$$\binom{9+4-1}{9} = \binom{12}{9} = \frac{12!}{9!(12-9)!} = \frac{12 \cdot 11 \cdot 10}{6} = 4 \cdot 11 \cdot 5 = 220$$

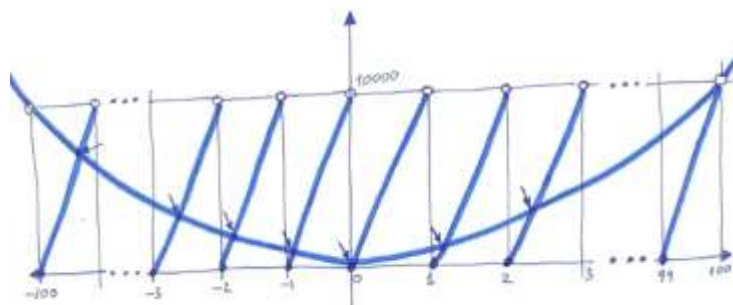
23

https://artofproblemsolving.com/wiki/index.php/2018_AMC_12B_Problems/Problem_23

24

$$x^2 + 10000 \lfloor x \rfloor = 10000 x \Leftrightarrow x^2 = 10000 x - 10000 \lfloor x \rfloor = 10000 \{x\}$$

Donde $\{x\}$ denota la parte fraccionaria de x . Su gráfica tiene forma de diente de sierra, tomando valores entre 0 y 10000, y la gráfica de x^2 es la típica parábola. Representando y comparando estas dos gráficas vemos que se cortarán en 100 valores de x negativos, 98 valores de x positivos y $x=0$:



En total, $100+98+1=199$ (C).

Nota: En https://artofproblemsolving.com/wiki/index.php/2018_AMC_12B_Problems/Problem_24 se presentan hasta 7 versiones diferentes para solucionar este problema.

25

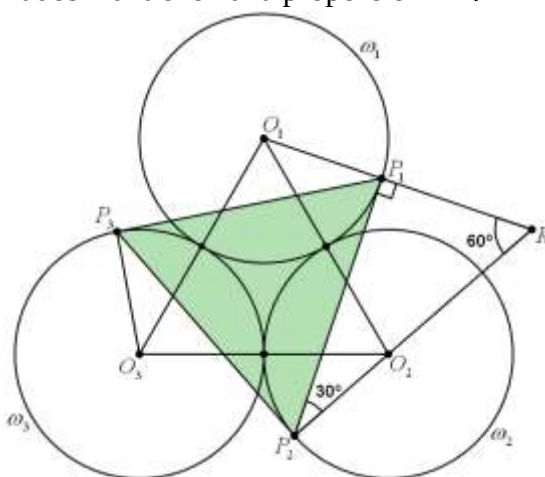
Primera versión.

Sean O_1, O_2, O_3 los respectivos centros de $\omega_1, \omega_2, \omega_3$.

Sea K el punto de intersección entre O_1P_1 y O_2P_2 .

$$\left. \begin{array}{l} \angle P_3P_2P_1 = 60^\circ \\ \angle P_3P_2K = 90^\circ \end{array} \right\} \Rightarrow \angle KP_2P_1 = 30^\circ$$

Y puesto que $\angle KP_1P_2 = 90^\circ$, el triángulo ΔKP_1P_2 es un triángulo 30° - 60° - 90° . Para este triángulo sabemos que sus lados mantienen una proporción $1:\sqrt{3}:2$ (GA/8.5.2).



Así pues, denotando por $d = P_1K$, $2d = P_2K$ y $\sqrt{3}d = P_1P_2$, y aplicando el Teorema del Coseno (GA/9.1.4) en el triángulo ΔO_1KO_2

$$\begin{aligned}
8^2 &= (d+4)^2 + (2d-4)^2 - 2(d+4)(2d-4)\cos 60^\circ \Leftrightarrow \\
8^2 &= d^2 + 4^2 + 8d + 4d^2 - 16d + 16 - 2d^2 - 4d + 16 \Leftrightarrow \\
0 &= 3d^2 - 12d - 16 \Leftrightarrow d = 2 \pm \frac{2}{3}\sqrt{21}
\end{aligned}$$

Tomando el valor positivo $d = 2 + \frac{2}{3}\sqrt{21}$

Obtenemos el lado del triángulo $\Delta P_1P_2P_3$:

$$\sqrt{3}d = \sqrt{3}\left(2 + \frac{2}{3}\sqrt{21}\right) = 2\sqrt{3} + 2\sqrt{7}$$

Y por lo tanto su altura es:

$$\sqrt{(2\sqrt{3} + 2\sqrt{7})^2 - \left(\frac{1}{2}(2\sqrt{3} + 2\sqrt{7})\right)^2} = \sqrt{\frac{3}{4}(2\sqrt{3} + 2\sqrt{7})^2} = \frac{\sqrt{3}}{2}(2\sqrt{3} + 2\sqrt{7})$$

Y, finalmente, su área es:

$$\begin{aligned}
\frac{1}{2} \cdot \frac{\sqrt{3}}{2} (2\sqrt{3} + 2\sqrt{7}) \cdot (2\sqrt{3} + 2\sqrt{7}) &= \frac{\sqrt{3}(2\sqrt{3} + 2\sqrt{7})^2}{4} = \\
&= \frac{\sqrt{3}(4 \cdot 3 + 4 \cdot 7 + 8\sqrt{3}\sqrt{7})}{4} = \sqrt{3}(10 + 2\sqrt{3}\sqrt{7}) = 10\sqrt{3} + 6\sqrt{7} = \\
&= \sqrt{3 \cdot 10^2} + \sqrt{7 \cdot 6^2} = \sqrt{300} + \sqrt{252}
\end{aligned}$$

Por tanto la solución es $300 + 252 = 552$ (D)

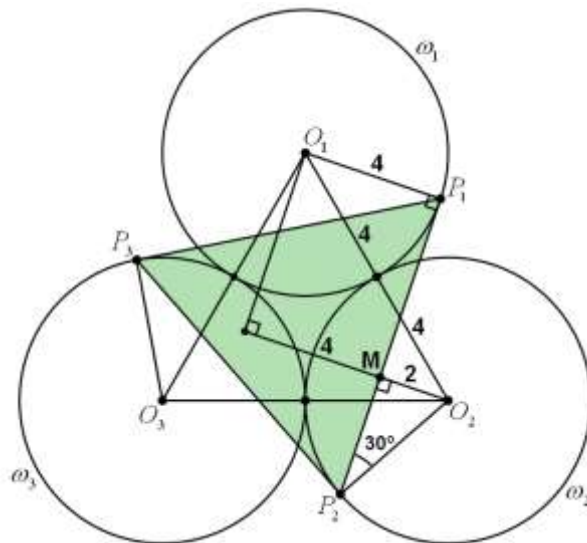
Segunda versión.

Sea M el pie de la perpendicular a P_1P_2 por O_2 .

Como en la primera versión, sabemos que $\angle O_2P_2M = 30^\circ$ y por tanto el triángulo ΔO_2P_2M es un triángulo 30° - 60° - 90° . Puesto que, además, sabemos que $P_2O_2 = 4$, se deduce, de nuevo, que $O_2M = 2$ y $P_2M = 2\sqrt{3}$.

Puesto que, además, $O_1P_1 = 4$ y $O_1O_2 = 8$, podemos deducir que

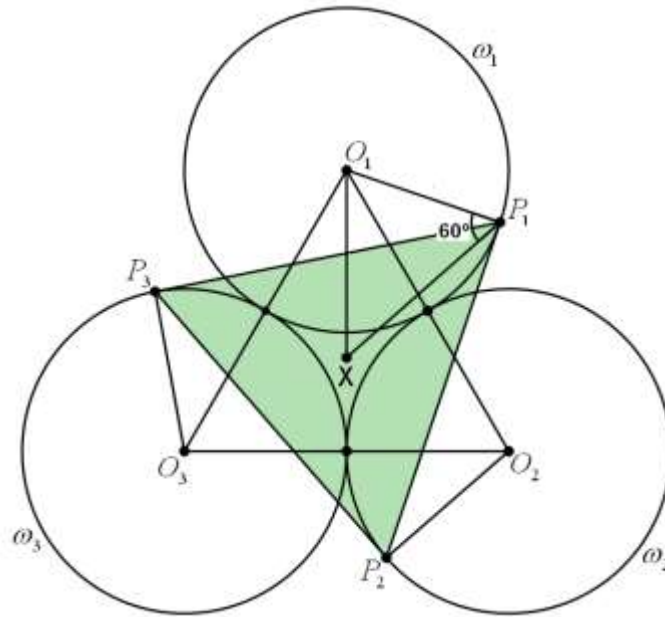
$$P_1M = \sqrt{8^2 - (4+2)^2} = 2\sqrt{7}, \text{ y por lo tanto } P_1P_2 = P_1M + P_2M = 2\sqrt{7} + 2\sqrt{3}$$



Y el área se determina como en la primera versión.

Tercera versión.

Esta tercera versión se basa en el hecho de que los triángulos equiláteros $\Delta O_1O_2O_3$ y $\Delta P_1P_2P_3$ comparten el mismo baricentro X .



El triángulo equilátero $\Delta O_1O_2O_3$ tiene lado 8, luego su altura (y por tanto su mediana) tiene longitud

$$h_1 = \sqrt{8^2 - 4^2} = \sqrt{64 - 16} = 4\sqrt{3}$$

Por la propiedad del baricentro ([GA/11.5.3b](#)), tenemos

$$XO_1 = \frac{2}{3} 4\sqrt{3} = \frac{8}{3}\sqrt{3}$$

Por otro lado, como en las versiones anteriores, sabemos que $\angle O_1P_1X = 60^\circ$, y por tanto podemos aplicar el Teorema del Coseno en el triángulo ΔO_1P_1X . Tomando $x = XP_1$,

$$\left(\frac{8}{3}\sqrt{3}\right)^2 = 4^2 + x^2 - 2 \cdot 4 \cdot x \cos(60^\circ) \Leftrightarrow$$

$$\frac{8^2}{3^2} \cdot 3 = 16 + x^2 - 4x \Leftrightarrow$$

$$\frac{8^2}{3} = 16 + x^2 - 4x \Leftrightarrow x = 2 \pm \frac{2}{3}\sqrt{21}$$

De nuevo, aplicando la propiedad del baricentro, podemos determinar la altura del triángulo $\Delta P_1P_2P_3$:

$$h_2 = \frac{3}{2} XP_1 = \frac{3}{2} \left(2 + \frac{2}{3}\sqrt{21}\right) = 3 + \sqrt{21}$$

y con esta altura podemos calcular su área como en las versiones anteriores.

2021 AMC 12A Problems

2021 AMC 12A (Answer Key)

Printable version: | AoPS Resources (<http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2021>) • PDF (http://www.artofproblemsolving.com/Forum/resources/files/usa/USA-AMC_12-AHSME-2021-44)

Instructions

1. This is a 25-question, multiple choice test. Each question is followed by answers marked A, B, C, D and E. Only one of these is correct.
2. You will receive 6 points for each correct answer, 2.5 points for each problem left unanswered if the year is before 2006, 1.5 points for each problem left unanswered if the year is after 2006, and 0 points for each incorrect answer.
3. No aids are permitted other than scratch paper, graph paper, ruler, compass, protractor and erasers (and calculators that are accepted for use on the test if before 2006. No problems on the test will *require* the use of a calculator).
4. Figures are not necessarily drawn to scale.
5. You will have **75 minutes** working time to complete the test.

1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

What is the value of

$$2^{1+2+3} - (2^1 + 2^2 + 2^3)?$$

- (A) 0 (B) 50 (C) 52 (D) 54 (E) 57

Solution

Problem 2

Under what conditions does $\sqrt{a^2 + b^2} = a + b$ hold, where a and b are real numbers?

- (A) It is never true.
 (B) It is true if and only if $ab = 0$.

(C) It is true if and only if $a + b \geq 0$.

(D) It is true if and only if $ab = 0$ and $a + b \geq 0$.

(E) It is always true.

Solution

Problem 3

The sum of two natural numbers is 17,402. One of the two numbers is divisible by 10. If the units digit of that number is erased, the other number is obtained. What is the difference of these two numbers?

(A) 10,272 (B) 11,700 (C) 13,362 (D) 14,238 (E) 15,462

Solution

Problem 4

Tom has a collection of 13 snakes, 4 of which are purple and 5 of which are happy. He observes that

- all of his happy snakes can add,
- none of his purple snakes can subtract, and
- all of his snakes that can't subtract also can't add.

Which of these conclusions can be drawn about Tom's snakes?

(A) Purple snakes can add.

(B) Purple snakes are happy.

(C) Snakes that can add are purple.

(D) Happy snakes are not purple.

(E) Happy snakes can't subtract.

Solution

Problem 5

When a student multiplied the number 66 by the repeating decimal, $\underline{1}.\underline{a}\underline{b}\underline{a}\underline{b}.\dots = \underline{1}.\overline{\underline{a}\underline{b}}$, where a and b are digits, he did not notice the notation and just multiplied 66 times $\underline{1}.\underline{a}\underline{b}$. Later he found that his answer is 0.5 less than the correct answer. What is the 2-digit number $\underline{a}\underline{b}$?

(A) 15 (B) 30 (C) 45 (D) 60 (E) 75

Solution

Problem 6

A deck of cards has only red cards and black cards. The probability of a randomly chosen card being red is $\frac{1}{3}$. When 4 black cards are added to the deck, the probability of choosing red becomes $\frac{1}{4}$. How many cards were in the deck originally?

(A) 6 (B) 9 (C) 12 (D) 15 (E) 18

Solution

Problem 7

What is the least possible value of $(xy - 1)^2 + (x + y)^2$ for all real numbers x and y ?

(A) 0 (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) 1 (E) 2

Solution

Problem 8

A sequence of numbers is defined by $D_0 = 0$, $D_1 = 0$, $D_2 = 1$ and $D_n = D_{n-1} + D_{n-3}$ for $n \geq 3$. What are the parities (evenness or oddness) of the triple of numbers $(D_{2021}, D_{2022}, D_{2023})$, where E denotes even and O denotes odd?

(A) (O, E, O) (B) (E, E, O) (C) (E, O, E) (D) (O, O, E) (E) (O, O, O)

Solution

Problem 9

Which of the following is equivalent to

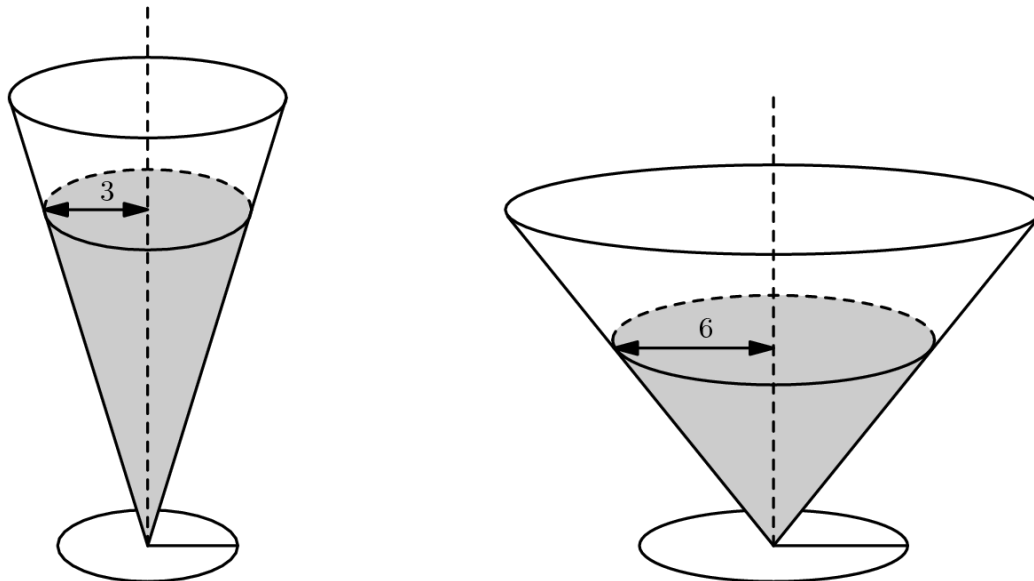
$$(2+3)(2^2+3^2)(2^4+3^4)(2^8+3^8)(2^{16}+3^{16})(2^{32}+3^{32})(2^{64}+3^{64})?$$

- (A) $3^{127} + 2^{127}$ (B) $3^{127} + 2^{127} + 2 \cdot 3^{63} + 3 \cdot 2^{63}$ (C) $3^{128} - 2^{128}$ (D) $3^{128} + 2^{128}$ (E) 5^{127}

Solution

Problem 10

Two right circular cones with vertices facing down as shown in the figure below contains the same amount of liquid. The radii of the tops of the liquid surfaces are 3 cm and 6 cm. Into each cone is dropped a spherical marble of radius 1 cm, which sinks to the bottom and is completely submerged without spilling any liquid. What is the ratio of the rise of the liquid level in the narrow cone to the rise of the liquid level in the wide cone?



- (A) 1 : 1 (B) 47 : 43 (C) 2 : 1 (D) 40 : 13 (E) 4 : 1

Solution

Problem 11

A laser is placed at the point $(3, 5)$. The laser beam travels in a straight line. Larry wants the beam to hit and bounce off the y -axis, then hit and bounce off the x -axis, then hit the point $(7, 5)$. What is the total distance the beam will travel along this path?

- (A) $2\sqrt{10}$ (B) $5\sqrt{2}$ (C) $10\sqrt{2}$ (D) $15\sqrt{2}$ (E) $10\sqrt{5}$

Solution

Problem 12

All the roots of the polynomial $z^6 - 10z^5 + Az^4 + Bz^3 + Cz^2 + Dz + 16$ are positive integers, possibly repeated. What is the value of B ?

- (A) -88 (B) -80 (C) -64 (D) -41 (E) -40

Solution

Problem 13

Of the following complex numbers z , which one has the property that z^5 has the greatest real part?

- (A) -2 (B) $-\sqrt{3} + i$ (C) $-\sqrt{2} + \sqrt{2}i$ (D) $-1 + \sqrt{3}i$ (E) $2i$

Solution

Problem 14

What is the value of

$$\left(\sum_{k=1}^{20} \log_{5^k} 3^{k^2} \right) \cdot \left(\sum_{k=1}^{100} \log_{9^k} 25^k \right)?$$

- (A) 21 (B) $100 \log_5 3$ (C) $200 \log_3 5$ (D) 2,200 (E) 21,000

Solution

Problem 15

A choir director must select a group of singers from among his 6 tenors and 8 basses. The only requirements are that the difference between the numbers of tenors and basses must be a multiple of 4, and the group must have at least one singer. Let N be the number of different groups that could be selected. What is the remainder when N is divided by 100?

- (A) 47 (B) 48 (C) 83 (D) 95 (E) 96

Solution

Problem 16

In the following list of numbers, the integer n appears n times in the list for $1 \leq n \leq 200$.

1, 2, 2, 3, 3, 3, 4, 4, 4, ..., 200, 200, ..., 200

What is the median of the numbers in this list?

- (A) 100.5 (B) 134 (C) 142 (D) 150.5 (E) 167

Solution

Problem 17

Trapezoid $ABCD$ has $\overline{AB} \parallel \overline{CD}$, $BC = CD = 43$, and $\overline{AD} \perp \overline{BD}$. Let O be the intersection of the diagonals $\overline{AC}$ and $\overline{BD}$, and let P be the midpoint of $\overline{BD}$. Given that $OP = 11$, the length of AD can be written in the form $m\sqrt{n}$, where m and n are positive integers and n is not divisible by the square of any prime. What is $m + n$?

- (A) 65 (B) 132 (C) 157 (D) 194 (E) 215

Solution

Problem 18

Let f be a function defined on the set of positive rational numbers with the property that $f(a \cdot b) = f(a) + f(b)$ for all positive rational numbers a and b . Furthermore, suppose that f also has the property that $f(p) = p$ for every prime number p . For which of the following numbers x is $f(x) < 0$?

- (A) $\frac{17}{32}$ (B) $\frac{11}{16}$ (C) $\frac{7}{9}$ (D) $\frac{7}{6}$ (E) $\frac{25}{11}$

Solution

Problem 19

How many solutions does the equation $\sin\left(\frac{\pi}{2} \cos x\right) = \cos\left(\frac{\pi}{2} \sin x\right)$ have in the closed interval $[0, \pi]$?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Solution

Problem 20

Suppose that on a parabola with vertex V and a focus F there exists a point A such that $AF = 20$ and $AV = 21$. What is the sum of all possible values of the length FV ?

- (A) 13 (B) $\frac{40}{3}$ (C) $\frac{41}{3}$ (D) 14 (E) $\frac{43}{3}$

Solution

Problem 21

The five solutions to the equation

$$(z - 1)(z^2 + 2z + 4)(z^2 + 4z + 6) = 0$$

may be written in the form $x_k + y_k i$ for $1 \leq k \leq 5$, where x_k and y_k are real. Let $\mathcal{E}$ be the unique ellipse that passes through the points $(x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4)$, and (x_5, y_5) . The eccentricity of $\mathcal{E}$ can be written in the form $\sqrt{\frac{m}{n}}$ where m and n are relatively prime positive integers. What is $m + n$? (Recall that the eccentricity of an ellipse $\mathcal{E}$ is the ratio $\frac{c}{a}$, where $2a$ is the length of the major axis of $\mathcal{E}$ and $2c$ is the distance between its two foci.)

- (A) 7 (B) 9 (C) 11 (D) 13 (E) 15

Solution

Problem 22

Suppose that the roots of the polynomial $P(x) = x^3 + ax^2 + bx + c$ are $\cos \frac{2\pi}{7}$, $\cos \frac{4\pi}{7}$, and $\cos \frac{6\pi}{7}$, where angles are in radians. What is abc ?

- (A) $-\frac{3}{49}$ (B) $-\frac{1}{28}$ (C) $\frac{\sqrt[3]{7}}{64}$ (D) $\frac{1}{32}$ (E) $\frac{1}{28}$

Solution

Problem 23

Frieda the frog begins a sequence of hops on a 3×3 grid of squares, moving one square on each hop and choosing at random the direction of each hop up, down, left, or right. She does not hop diagonally. When the direction of a hop would take Frieda off the grid, she "wraps around" and jumps to the opposite edge. For example if Frieda begins in the center square and makes two hops "up", the first hop would place her in the top row middle square, and the second hop would cause Frieda to jump to the opposite edge, landing in the bottom row middle square. Suppose Frieda starts from the center square, makes at most four hops at random, and stops hopping if she lands on a corner square. What is the probability that she reaches a corner square on one of the four hops?

- (A) $\frac{9}{16}$ (B) $\frac{5}{8}$ (C) $\frac{3}{4}$ (D) $\frac{25}{32}$ (E) $\frac{13}{16}$

Solution

Problem 24

Semicircle Γ has diameter $\overline{AB}$ of length 14. Circle Ω lies tangent to $\overline{AB}$ at a point P and intersects Γ at points Q and R . If $QR = 3\sqrt{3}$ and $\angle QPR = 60^\circ$, then the area of $\triangle PQR$ equals $\frac{a\sqrt{b}}{c}$, where a and c are relatively prime positive integers, and b is a positive integer not divisible by the square of any prime. What is $a + b + c$?

- (A) 110 (B) 114 (C) 118 (D) 122 (E) 126

Solution

Problem 25

Let $d(n)$ denote the number of positive integers that divide n , including 1 and n . For example, $d(1) = 1$, $d(2) = 2$, and $d(12) = 6$. (This function is known as the divisor function.) Let

$$f(n) = \frac{d(n)}{\sqrt[3]{n}}.$$

There is a unique positive integer N such that $f(N) > f(n)$ for all positive integers $n \neq N$. What is the sum of the digits of N ?

- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

Solution

See also

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1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
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2021 AMC 12A Answer Key

1. B
2. D
3. D
4. D
5. E
6. C
7. D
8. C
9. C
10. E
11. C
12. A
13. B
14. E
15. D
16. C
17. D
18. E
19. C
20. B
21. A
22. D
23. D
24. D
25. E

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2021 AMC 12B Problems

2021 AMC 12B (Answer Key)

Printable version: | AoPS Resources (<http://www.artofproblemsolving.com/Forum/resources.php?c=182&cid=44&year=2021>) • PDF (http://www.artofproblemsolving.com/Forum/resources/files/usa/USA-AMC_12-AHSME-2021-44)

Instructions

1. This is a 25-question, multiple choice test. Each question is followed by answers marked A, B, C, D and E. Only one of these is correct.
2. You will receive 6 points for each correct answer, 2.5 points for each problem left unanswered if the year is before 2006, 1.5 points for each problem left unanswered if the year is after 2006, and 0 points for each incorrect answer.
3. No aids are permitted other than scratch paper, graph paper, ruler, compass, protractor and erasers (and calculators that are accepted for use on the test if before 2006. No problems on the test will *require* the use of a calculator).
4. Figures are not necessarily drawn to scale.
5. You will have **75 minutes** working time to complete the test.

1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

How many integer values of x satisfy $|x| < 3\pi$?

- (A) 9 (B) 10 (C) 18 (D) 19 (E) 20

Solution

Problem 2

At a math contest, 57 students are wearing blue shirts, and another 75 students are wearing yellow shirts. The 132 students are assigned into 66 pairs. In exactly 23 of these pairs, both students are wearing blue shirts. In how many pairs are both students wearing yellow shirts?

- (A) 23 (B) 32 (C) 37 (D) 41 (E) 64

Solution

Problem 3

Suppose

$$2 + \frac{1}{1 + \frac{1}{2 + \frac{2}{3+x}}} = \frac{144}{53}.$$

What is the value of x ?

- (A) $\frac{3}{4}$ (B) $\frac{7}{8}$ (C) $\frac{14}{15}$ (D) $\frac{37}{38}$ (E) $\frac{52}{53}$

Solution

Problem 4

Ms. Blackwell gives an exam to two classes. The mean of the scores of the students in the morning class is 84, and the afternoon class's mean score is 70. The ratio of the number of students in the morning class to the number of students in the afternoon class is $\frac{3}{4}$. What is the mean of the score of all the students?

- (A) 74 (B) 75 (C) 76 (D) 77 (E) 78

Solution

Problem 5

The point $P(a, b)$ in the xy -plane is first rotated counterclockwise by 90° around the point $(1, 5)$ and then reflected about the line $y = -x$. The image of P after these two transformations is at $(-6, 3)$. What is $b - a$?

- (A) 1 (B) 3 (C) 5 (D) 7 (E) 9

Solution

Problem 6

An inverted cone with base radius 12cm and height 18cm is full of water. The water is poured into a tall cylinder whose horizontal base has a radius of 24cm. What is the height in centimeters of the water in the cylinder?

- (A) 1.5 (B) 3 (C) 4 (D) 4.5 (E) 6

Solution

Problem 7

Let $N = 34 \cdot 34 \cdot 63 \cdot 270$. What is the ratio of the sum of the odd divisors of N to the sum of the even divisors of N ?

- (A) 1 : 16 (B) 1 : 15 (C) 1 : 14 (D) 1 : 8 (E) 1 : 3

Solution

Problem 8

Three equally spaced parallel lines intersect a circle, creating three chords of lengths 38, 38, and 34. What is the distance between two adjacent parallel lines?

- (A) $5\frac{1}{2}$ (B) 6 (C) $6\frac{1}{2}$ (D) 7 (E) $7\frac{1}{2}$

Solution

Problem 9

What is the value of

$$\frac{\log_2 80}{\log_{40} 2} - \frac{\log_2 160}{\log_{20} 2}?$$

- (A) 0 (B) 1 (C) $\frac{5}{4}$ (D) 2 (E) $\log_2 5$

Solution

Problem 10

Two distinct numbers are selected from the set $\{1, 2, 3, 4, \dots, 36, 37\}$ so that the sum of the remaining 35 numbers is the product of these two numbers. What is the difference of these two numbers?

- (A) 5 (B) 7 (C) 8 (D) 9 (E) 10

Solution

Problem 11

Triangle ABC has $AB = 13$, $BC = 14$ and $AC = 15$. Let P be the point on $\overline{AC}$ such that $PC = 10$. There are exactly two points D and E on line BP such that quadrilaterals $ABCD$ and $ABCE$ are trapezoids. What is the distance DE ?

- (A) $\frac{42}{5}$ (B) $6\sqrt{2}$ (C) $\frac{84}{5}$ (D) $12\sqrt{2}$ (E) 18

Solution

Problem 12

Suppose that S is a finite set of positive integers. If the greatest integer in S is removed from S , then the average value (arithmetic mean) of the integers remaining is 32. If the least integer in S is also removed, then the average value of the integers remaining is 35. If the greatest integer is then returned to the set, the average value of the integers rises to 40. The greatest integer in the original set S is 72 greater than the least integer in S . What is the average value of all the integers in the set S ?

- (A) 36.2 (B) 36.4 (C) 36.6 (D) 36.8 (E) 37

Solution

Problem 13

How many values of θ in the interval $0 < \theta \leq 2\pi$ satisfy

$$1 - 3\sin\theta + 5\cos 3\theta = 0?$$

- (A) 2 (B) 4 (C) 5 (D) 6 (E) 8

Solution

Problem 14

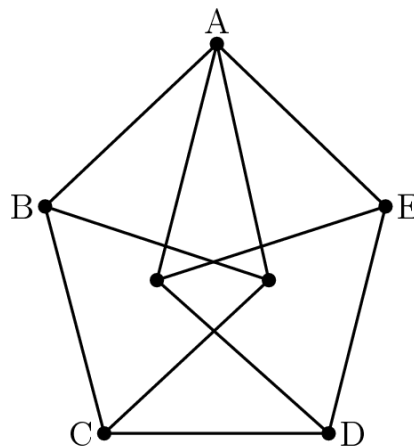
Let $ABCD$ be a rectangle and let $\overline{DM}$ be a segment perpendicular to the plane of $ABCD$. Suppose that $\overline{DM}$ has integer length, and the lengths of $\overline{MA}$, $\overline{MC}$, and $\overline{MB}$ are consecutive odd positive integers (in this order). What is the volume of pyramid $MABCD$?

- (A) $24\sqrt{5}$ (B) 60 (C) $28\sqrt{5}$ (D) 66 (E) $8\sqrt{70}$

Solution

Problem 15

The figure is constructed from 11 line segments, each of which has length 2. The area of pentagon $ABCDE$ can be written as $\sqrt{m} + \sqrt{n}$, where m and n are positive integers. What is $m + n$?



- (A) 20 (B) 21 (C) 22 (D) 23 (E) 24

Solution

Problem 16

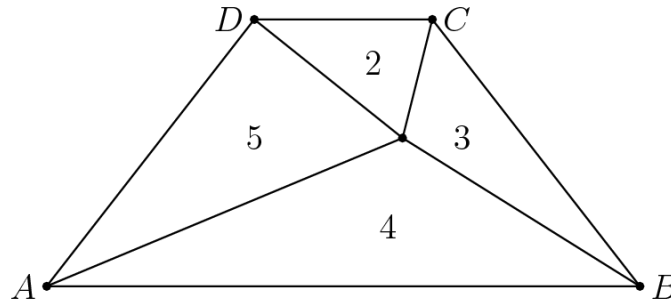
Let $g(x)$ be a polynomial with leading coefficient 1, whose three roots are the reciprocals of the three roots of $f(x) = x^3 + ax^2 + bx + c$, where $1 < a < b < c$. What is $g(1)$ in terms of a , b , and c ?

- (A) $\frac{1+a+b+c}{c}$ (B) $1+a+b+c$ (C) $\frac{1+a+b+c}{c^2}$ (D) $\frac{a+b+c}{c^2}$ (E) $\frac{1+a+b+c}{a+b+c}$

Solution

Problem 17

Let $ABCD$ be an isosceles trapezoid having parallel bases $\overline{AB}$ and $\overline{CD}$ with $AB > CD$. Line segments from a point inside $ABCD$ to the vertices divide the trapezoid into four triangles whose areas are 2, 3, 4, and 5 starting with the triangle with base $\overline{CD}$ and moving clockwise as shown in the diagram below. What is the ratio $\frac{AB}{CD}$?



- (A) 3 (B) $2 + \sqrt{2}$ (C) $1 + \sqrt{6}$ (D) $2\sqrt{3}$ (E) $3\sqrt{2}$

Solution

Problem 18

Let z be a complex number satisfying $12|z|^2 = 2|z+2|^2 + |z^2+1|^2 + 31$. What is the value of $z + \frac{6}{z}$?

- (A) -2 (B) -1 (C) $\frac{1}{2}$ (D) 1 (E) 4

Solution

Problem 19

Two fair dice, each with at least 6 faces are rolled. On each face of each die is printed a distinct integer from 1 to the number of faces on that die, inclusive. The probability of rolling a sum of 7 is $\frac{3}{4}$ of the probability of rolling a sum of 10, and the probability of rolling a sum of 12 is $\frac{1}{12}$. What is the least possible number of faces on the two dice combined?

- (A) 16 (B) 17 (C) 18 (D) 19 (E) 20

Solution

Problem 20

Let $Q(z)$ and $R(z)$ be the unique polynomials such that

$$z^{2021} + 1 = (z^2 + z + 1)Q(z) + R(z)$$

and the degree of R is less than 2. What is $R(z)$?

- (A) $-z$ (B) -1 (C) 2021 (D) $z+1$ (E) $2z+1$

Solution

Problem 21

Let S be the sum of all positive real numbers x for which

$$x^{2\sqrt{2}} = \sqrt{2}^{2^x}.$$

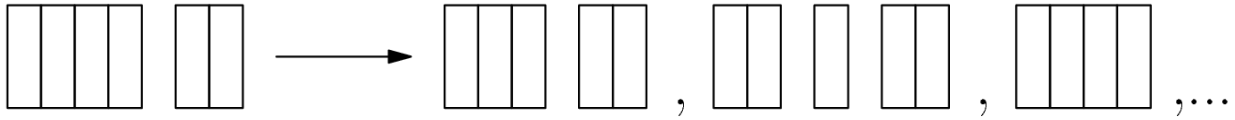
Which of the following statements is true?

- (A) $S < \sqrt{2}$ (B) $S = \sqrt{2}$ (C) $\sqrt{2} < S < 2$ (D) $2 \leq S < 6$ (E) $S \geq 6$

Solution

Problem 22

Arjun and Beth play a game in which they take turns removing one brick or two adjacent bricks from one "wall" among a set of several walls of bricks, with gaps possibly creating new walls. The walls are one brick tall. For example, a set of walls of sizes 4 and 2 can be changed into any of the following by one move: $(3, 2)$, $(2, 1, 2)$, (4) , $(4, 1)$, $(2, 2)$, or $(1, 1, 2)$.



Arjun plays first, and the player who removes the last brick wins. For which starting configuration is there a strategy that guarantees a win for Beth?

- (A) $(6, 1, 1)$ (B) $(6, 2, 1)$ (C) $(6, 2, 2)$ (D) $(6, 3, 1)$ (E) $(6, 3, 2)$

Solution

Problem 23

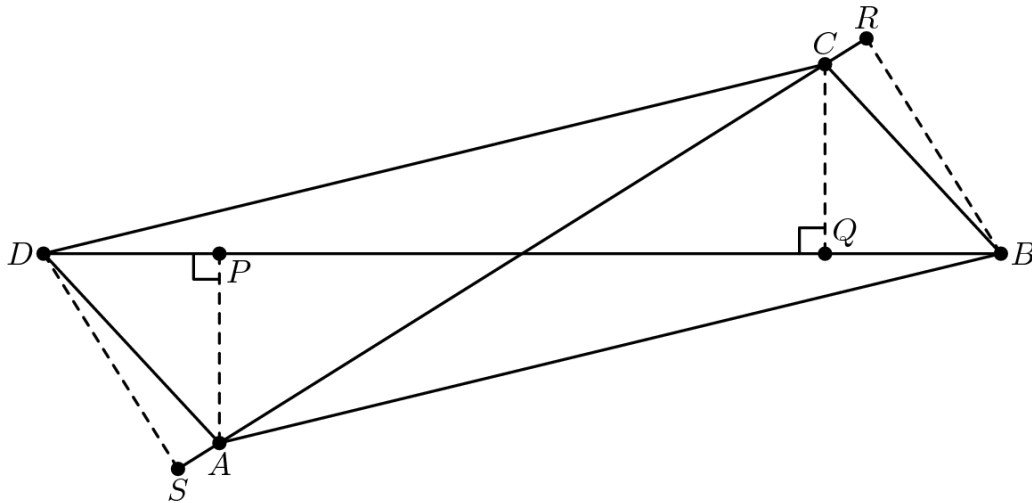
Three balls are randomly and independently tossed into bins numbered with the positive integers so that for each ball, the probability that it is tossed into bin i is 2^{-i} for $i = 1, 2, 3, \dots$. More than one ball is allowed in each bin. The probability that the balls end up evenly spaced in distinct bins is $\frac{p}{q}$, where p and q are relatively prime positive integers. (For example, the balls are evenly spaced if they are tossed into bins 3, 17, and 10.) What is $p + q$?

- (A) 55 (B) 56 (C) 57 (D) 58 (E) 59

Solution

Problem 24

Let $ABCD$ be a parallelogram with area 15. Points P and Q are the projections of A and C , respectively, onto the line BD ; and points R and S are the projections of B and D , respectively, onto the line AC . See the figure, which also shows the relative locations of these points.



Suppose $PQ = 6$ and $RS = 8$, and let d denote the length of $\overline{BD}$, the longer diagonal of $ABCD$. Then d^2 can be written in the form $m + n\sqrt{p}$, where m , n , and p are positive integers and p is not divisible by the square of any prime. What is $m + n + p$?

- (A) 81 (B) 89 (C) 97 (D) 105 (E) 113

Solution

Problem 25

Let S be the set of lattice points in the coordinate plane, both of whose coordinates are integers between $\frac{a}{b}$ and 30, inclusive. Exactly 300 points in S lie on or below a line with equation $y = mx$. The possible values of m lie in an interval of length $\frac{a}{b}$, where a and b are relatively prime positive integers. What is $a + b$?

- (A) 31 (B) 47 (C) 62 (D) 72 (E) 85

Solution

See also

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Preceded by 2021 AMC 12A Problems	Followed by 2022 AMC 12A Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
All AMC 12 Problems and Solutions	

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2021 AMC 12B Answer Key

1. D
2. B
3. A
4. C
5. D
6. A
7. C
8. B
9. D
10. E
11. D
12. D
13. D
14. A
15. D
16. A
17. B
18. A
19. B
20. A
21. D
22. B
23. A
24. A
25. E

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2021 Fall AMC 12A Problems

2021 Fall AMC 12A (Answer Key)

Printable versions: • Fall AoPS Resources (https://artofproblemsolving.com/community/contest/collection/c3415_amc_12/2021) • Fall PDF (https://artofproblemsolving.com/community/contest/download/c3415_amc_12/2021)

Instructions

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1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 •
18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See Also

Problem 1

What is the value of $\frac{(2112 - 2021)^2}{169}$?

- (A) 7 (B) 21 (C) 49 (D) 64 (E) 91

Solution

Problem 2

Menkara has a 4×6 index card. If she shortens the length of one side of this card by 1 inch, the card would have area 18 square inches. What would the area of the card be in square inches if instead she shortens the length of the other side by 1 inch?

- (A) 16 (B) 17 (C) 18 (D) 19 (E) 20

Solution

Problem 3

Mr. Lopez has a choice of two routes to get to work. Route A is 6 miles long, and his average speed along this route is 30 miles per hour. Route B is 5 miles long, and his average speed along this route is 40 miles per hour, except for a $\frac{1}{2}$ -mile stretch in a school zone where his average speed is 20 miles per hour. By how many minutes is Route B quicker than Route A?

- (A) $2\frac{3}{4}$ (B) $3\frac{3}{4}$ (C) $4\frac{1}{2}$ (D) $5\frac{1}{2}$ (E) $6\frac{3}{4}$

Solution

Problem 4

The six-digit number $\underline{20210A}$ is prime for only one digit A . What is A ?

- (A) 1 (B) 3 (C) 5 (D) 7 (E) 9

Solution

Problem 5

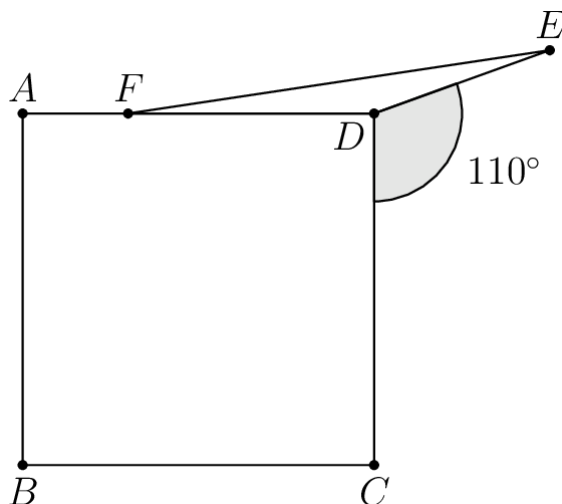
Elmer the emu takes 44 equal strides to walk between consecutive telephone poles on a rural road. Oscar the ostrich can cover the same distance in 12 equal leaps. The telephone poles are evenly spaced, and the 41st pole along this road is exactly one mile (5280 feet) from the first pole. How much longer, in feet, is Oscar's leap than Elmer's stride?

- (A) 6 (B) 8 (C) 10 (D) 11 (E) 15

Solution

Problem 6

As shown in the figure below, point E lies on the opposite half-plane determined by line CD from point A so that $\angle CDE = 110^\circ$. Point F lies on $\overline{AD}$ so that $DE = DF$, and $ABCD$ is a square. What is the degree measure of $\angle AFE$?



- (A) 160 (B) 164 (C) 166 (D) 170 (E) 174

Solution

Problem 7

A school has 100 students and 5 teachers. In the first period, each student is taking one class, and each teacher is teaching one class. The enrollments in the classes are 50, 20, 20, 5, and 5. Let t be the average value obtained if a teacher is picked at random and the number of students in their class is noted. Let s be the average value obtained if a student was picked at random and the number of students in their class, including the student, is noted. What is $t - s$?

- (A) -18.5 (B) -13.5 (C) 0 (D) 13.5 (E) 18.5

Solution

Problem 8

Let M be the least common multiple of all the integers 10 through 30, inclusive. Let N be the least common multiple of $M, 32, 33, 34, 35, 36, 37, 38, 39$, and 40. What is the value of $\frac{N}{M}$?

- (A) 1 (B) 2 (C) 37 (D) 74 (E) 2886

Solution

Problem 9

A right rectangular prism whose surface area and volume are numerically equal has edge lengths $\log_2 x$, $\log_3 x$, and $\log_4 x$. What is x ?

- (A) $2\sqrt{6}$ (B) $6\sqrt{6}$ (C) 24 (D) 48 (E) 576

Solution

Problem 10

The base-nine representation of the number N is $27,006,000,052_{\text{nine}}$. What is the remainder when N is divided by 5?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Solution

Problem 11

Consider two concentric circles of radius 17 and 19. The larger circle has a chord, half of which lies inside the smaller circle. What is the length of the chord in the larger circle?

- (A) $12\sqrt{2}$ (B) $10\sqrt{3}$ (C) $\sqrt{17 \cdot 19}$ (D) 18 (E) $8\sqrt{6}$

Solution

Problem 12

What is the number of terms with rational coefficients among the 1001 terms in the expansion of $(x\sqrt[3]{2} + y\sqrt{3})^{1000}$?

- (A) 0 (B) 166 (C) 167 (D) 500 (E) 501

Solution

Problem 13

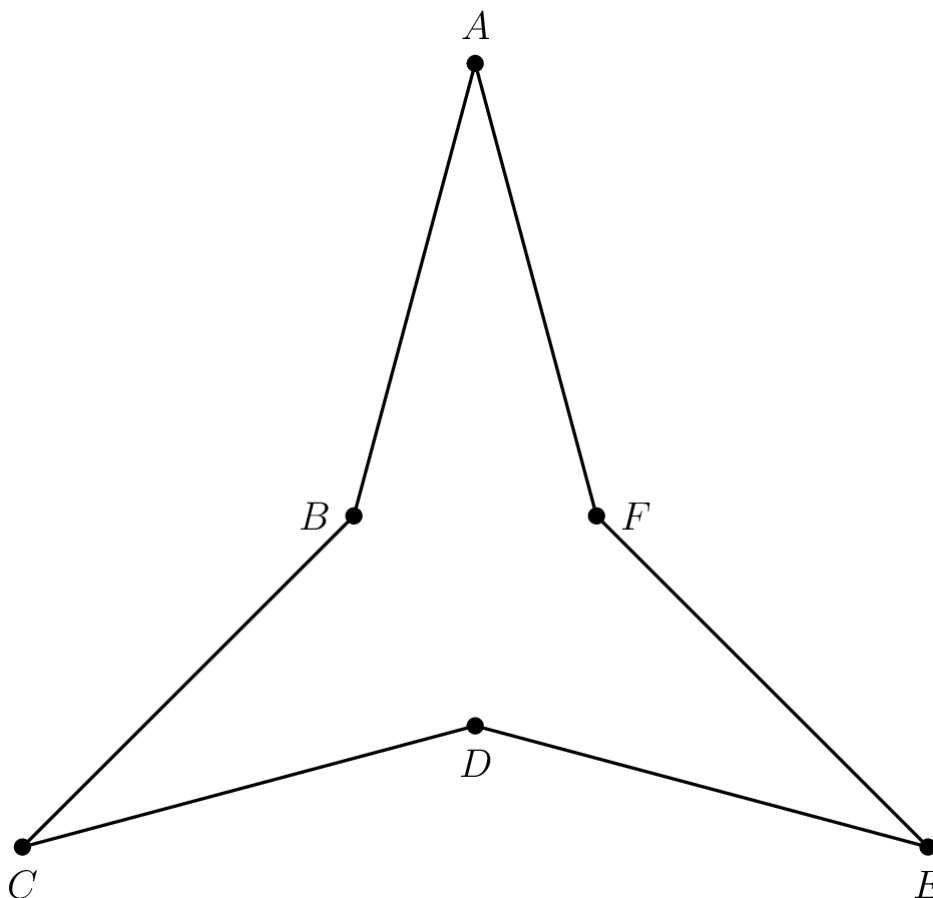
The angle bisector of the acute angle formed at the origin by the graphs of the lines $y = x$ and $y = 3x$ has equation $y = kx$. What is k ?

- (A) $\frac{1 + \sqrt{5}}{2}$ (B) $\frac{1 + \sqrt{7}}{2}$ (C) $\frac{2 + \sqrt{3}}{2}$ (D) 2 (E) $\frac{2 + \sqrt{5}}{2}$

Solution

Problem 14

In the figure, equilateral hexagon $ABCDEF$ has three nonadjacent acute interior angles that each measure 30° . The enclosed area of the hexagon is $6\sqrt{3}$. What is the perimeter of the hexagon?



- (A) 4 (B) $4\sqrt{3}$ (C) 12 (D) 18 (E) $12\sqrt{3}$

Solution

Problem 15

Recall that the conjugate of the complex number $w = a + bi$, where a and b are real numbers and $i = \sqrt{-1}$, is the complex number $\overline{w} = a - bi$. For any complex number z , let $f(z) = 4i\overline{z}$. The polynomial

$$P(z) = z^4 + 4z^3 + 3z^2 + 2z + 1$$

has four complex roots: z_1 , z_2 , z_3 , and z_4 . Let

$$Q(z) = z^4 + Az^3 + Bz^2 + Cz + D$$

be the polynomial whose roots are $f(z_1)$, $f(z_2)$, $f(z_3)$, and $f(z_4)$, where the coefficients A , B , C , and D are complex numbers. What is $B + D$?

- (A) -304 (B) -208 (C) $12i$ (D) 208 (E) 304

Solution

Problem 16

An organization has 30 employees, 20 of whom have a brand A computer while the other 10 have a brand B computer. For security, the computers can only be connected to each other and only by cables. The cables can only connect a brand A computer to a brand B computer. Employees can communicate with each other if their computers are directly connected by a cable or by relaying messages through a series of connected computers. Initially, no computer is connected to any other. A technician

arbitrarily selects one computer of each brand and installs a cable between them, provided there is not already a cable between that pair. The technician stops once every employee can communicate with each other. What is the maximum possible number of cables used?

- (A) 190 (B) 191 (C) 192 (D) 195 (E) 196

Solution

Problem 17

For how many ordered pairs (b, c) of positive integers does neither $x^2 + bx + c = 0$ nor $x^2 + cx + b = 0$ have two distinct real solutions?

- (A) 4 (B) 6 (C) 8 (D) 12 (E) 16

Solution

Problem 18

Each of 20 balls is tossed independently and at random into one of 5 bins. Let p be the probability that some bin ends up with 3 balls, another with 5 balls, and the other three with 4 balls each. Let q be the probability that every bin ends up with 4 balls. What is $\frac{p}{q}$?

- (A) 1 (B) 4 (C) 8 (D) 12 (E) 16

Solution

Problem 19

Let x be the least real number greater than 1 such that $\sin(x) = \sin(x^2)$, where the arguments are in degrees. What is x rounded up to the closest integer?

- (A) 10 (B) 13 (C) 14 (D) 19 (E) 20

Solution

Problem 20

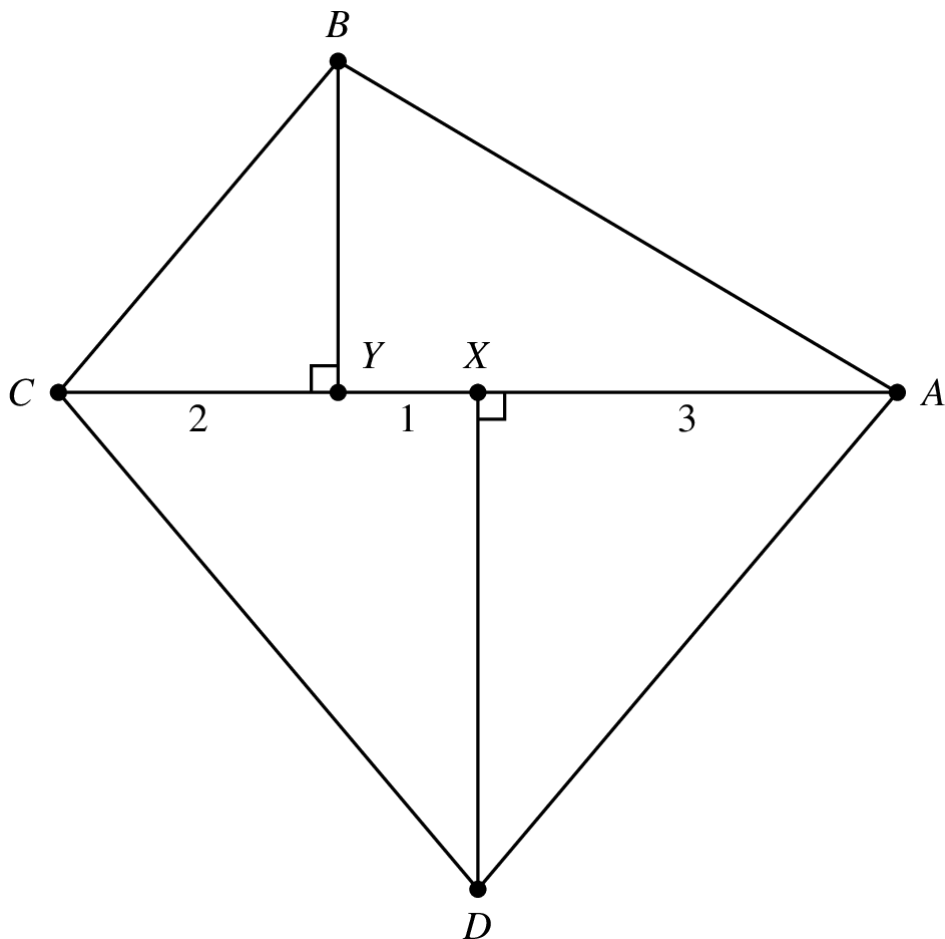
For each positive integer n , let $f_1(n)$ be twice the number of positive integer divisors of n , and for $j \geq 2$, let $f_j(n) = f_1(f_{j-1}(n))$. For how many values of $n \leq 50$ is $f_{50}(n) = 12$?

- (A) 7 (B) 8 (C) 9 (D) 10 (E) 11

Solution

Problem 21

Let $ABCD$ be an isosceles trapezoid with $\overline{BC} \parallel \overline{AD}$ and $AB = CD$. Points X and Y lie on diagonal $\overline{AC}$ with X between A and Y , as shown in the figure. Suppose $\angle AXD = \angle BYC = 90^\circ$, $AX = 3$, $XY = 1$, and $YC = 2$. What is the area of $ABCD$?



- (A) 15 (B) $5\sqrt{11}$ (C) $3\sqrt{35}$ (D) 18 (E) $7\sqrt{7}$

Solution

Problem 22

Azar and Carl play a game of tic-tac-toe. Azar places an X in one of the boxes in a 3-by-3 array of boxes, then Carl places an O in one of the remaining boxes. After that, Azar places an X in one of the remaining boxes, and so on until all boxes are filled or one of the players has three of their symbols in a row—horizontal, vertical, or diagonal—whichever comes first, in which case that player wins the game. Suppose the players make their moves at random, rather than trying to follow a rational strategy, and that Carl wins the game when he places his third O . How many ways can the board look after the game is over?

- (A) 36 (B) 112 (C) 120 (D) 148 (E) 160

Solution

Problem 23

A quadratic polynomial with real coefficients and leading coefficient 1 is called *disrespectful* if the equation $p(p(x)) = 0$ is satisfied by exactly three real numbers. Among all the disrespectful quadratic polynomials, there is a unique such polynomial $\tilde{p}(x)$ for which the sum of the roots is maximized. What is $\tilde{p}(1)$?

- (A) $\frac{5}{16}$ (B) $\frac{1}{2}$ (C) $\frac{5}{8}$ (D) 1 (E) $\frac{9}{8}$

Solution

Problem 24

Convex quadrilateral $ABCD$ has $AB = 18$, $\angle A = 60^\circ$, and $\overline{AB} \parallel \overline{CD}$. In some order, the lengths of the four sides form an arithmetic progression, and side $\overline{AB}$ is a side of maximum length. The length of another side is a . What is the sum of all possible values of a ?

- (A) 24 (B) 42 (C) 60 (D) 66 (E) 84

Solution

Problem 25

Let $m \geq 5$ be an odd integer, and let $D(m)$ denote the number of quadruples (a_1, a_2, a_3, a_4) of distinct integers with $1 \leq a_i \leq m$ for all i such that m divides $a_1 + a_2 + a_3 + a_4$. There is a polynomial

$$q(x) = c_3x^3 + c_2x^2 + c_1x + c_0$$

such that $D(m) = q(m)$ for all odd integers $m \geq 5$. What is c_1 ?

- (A) -6 (B) -1 (C) 4 (D) 6 (E) 11

Solution

See Also

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Preceded by 2021 AMC 12B Problems	Followed by 2021 Fall AMC 12B Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
All AMC 12 Problems and Solutions	

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2021 Fall AMC 12A Answer Key

1. C
2. E
3. B
4. E
5. B
6. D
7. B
8. D
9. E
10. D
11. E
12. C
13. A
14. E
15. D
16. B
17. B
18. E
19. B
20. D
21. C
22. D
23. A
24. E
25. E

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2021 Fall AMC 12B Problems

2021 Fall AMC 12B (Answer Key)

Printable versions: • Fall AoPS Resources (https://artofproblemsolving.com/community/contest/collection/c3415_amc_12/2021) • Fall PDF (https://artofproblemsolving.com/community/contest/download/c3415_amc_12/2021)

Instructions

1. This is a 25-question, multiple choice test. Each question is followed by answers marked A, B, C, D and E. Only one of these is correct.
2. You will receive 6 points for each correct answer, 2.5 points for each problem left unanswered if the year is before 2006, 1.5 points for each problem left unanswered if the year is after 2006, and 0 points for each incorrect answer.
3. No aids are permitted other than scratch paper, graph paper, ruler, compass, protractor and erasers (and calculators that are accepted for use on the test if before 2006. No problems on the test will *require* the use of a calculator).
4. Figures are not necessarily drawn to scale.
5. You will have **75 minutes** working time to complete the test.

1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 •
18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

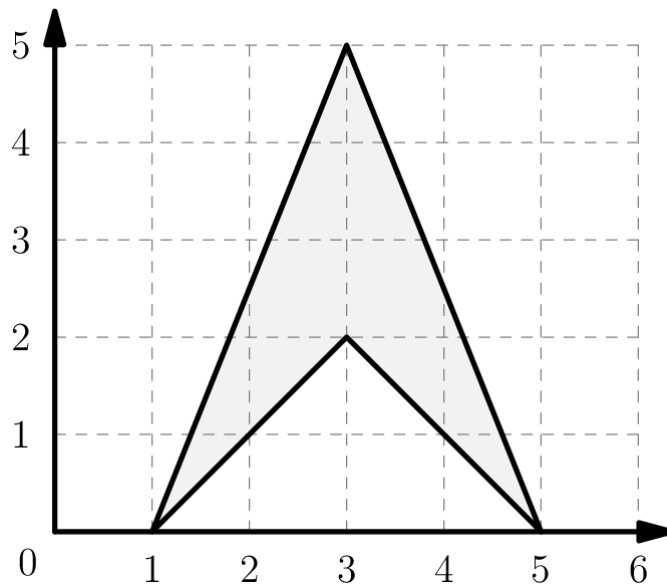
What is the value of $1234 + 2341 + 3412 + 4123$?

- (A) 10,000 (B) 10,010 (C) 10,110 (D) 11,000 (E) 11,110

Solution

Problem 2

What is the area of the shaded figure shown below?



- (A) 4 (B) 6 (C) 8 (D) 10 (E) 12

Solution

Problem 3

At noon on a certain day, Minneapolis is N degrees warmer than St. Louis. At 4:00 the temperature in Minneapolis has fallen by 5 degrees while the temperature in St. Louis has risen by 3 degrees, at which time the temperatures in the two cities differ by 2 degrees. What is the product of all possible values of N ?

- (A) 10 (B) 30 (C) 60 (D) 100 (E) 120

Solution

Problem 4

Let $n = 8^{2022}$. Which of the following is equal to $\frac{n}{4}$?

- (A) 4^{1010} (B) 2^{2022} (C) 8^{2018} (D) 4^{3031} (E) 4^{3032}

Solution

Problem 5

Call a fraction $\frac{a}{b}$, not necessarily in the simplest form, *special* if a and b are positive integers whose sum is 15. How many distinct integers can be written as the sum of two, not necessarily different, special fractions?

- (A) 9 (B) 10 (C) 11 (D) 12 (E) 13

Solution

Problem 6

The largest prime factor of 16384 is 2 because $16384 = 2^{14}$. What is the sum of the digits of the greatest prime number that is a divisor of 16383?

- (A) 3 (B) 7 (C) 10 (D) 16 (E) 22

Solution

Problem 7

Which of the following conditions is sufficient to guarantee that integers x, y , and z satisfy the equation

$$x(x - y) + y(y - z) + z(z - x) = 1?$$

- (A) $x > y$ and $y = z$
(B) $x = y - 1$ and $y = z - 1$
(C) $x = z + 1$ and $y = x + 1$
(D) $x = z$ and $y - 1 = x$
(E) $x + y + z = 1$

Solution

Problem 8

The product of the lengths of the two congruent sides of an obtuse isosceles triangle is equal to the product of the base and twice the triangle's height to the base. What is the measure, in degrees, of the vertex angle of this triangle?

- (A) 105 (B) 120 (C) 135 (D) 150 (E) 165

Solution

Problem 9

Triangle ABC is equilateral with side length 6. Suppose that O is the center of the inscribed circle of this triangle. What is the area of the circle passing through A, O , and C ?

- (A) 9π (B) 12π (C) 18π (D) 24π (E) 27π

Solution

Problem 10

What is the sum of all possible values of t between 0 and 360 such that the triangle in the coordinate plane whose vertices are

$$(\cos 40^\circ, \sin 40^\circ), (\cos 60^\circ, \sin 60^\circ), \text{ and } (\cos t^\circ, \sin t^\circ)$$

is isosceles?

- (A) 100 (B) 150 (C) 330 (D) 360 (E) 380

Solution

Problem 11

Una rolls 6 standard 6-sided dice simultaneously and calculates the product of the 6 numbers obtained. What is the probability that the product is divisible by 4?

- (A) $\frac{3}{4}$ (B) $\frac{57}{64}$ (C) $\frac{59}{64}$ (D) $\frac{187}{192}$ (E) $\frac{63}{64}$

Solution

Problem 12

For n a positive integer, let $f(n)$ be the quotient obtained when the sum of all positive divisors of n is divided by n . For example,

$$f(14) = (1 + 2 + 7 + 14) \div 14 = \frac{12}{7}$$

What is $f(768) - f(384)$?

- (A) $\frac{1}{768}$ (B) $\frac{1}{192}$ (C) 1 (D) $\frac{4}{3}$ (E) $\frac{8}{3}$

Solution

Problem 13

Let $c = \frac{2\pi}{11}$. What is the value of

$$\frac{\sin 3c \cdot \sin 6c \cdot \sin 9c \cdot \sin 12c \cdot \sin 15c}{\sin c \cdot \sin 2c \cdot \sin 3c \cdot \sin 4c \cdot \sin 5c}?$$

- (A) -1 (B) $-\frac{\sqrt{11}}{5}$ (C) $\frac{\sqrt{11}}{5}$ (D) $\frac{10}{11}$ (E) 1

Solution

Problem 14

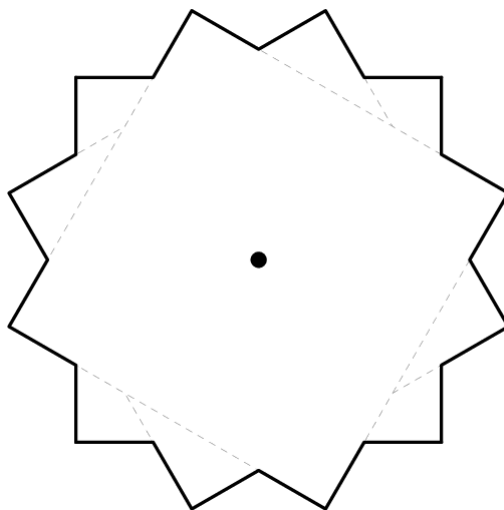
Suppose that $P(z)$, $Q(z)$, and $R(z)$ are polynomials with real coefficients, having degrees 2, 3, and 6, respectively, and constant terms 1, 2, and 3, respectively. Let N be the number of distinct complex numbers z that satisfy the equation $P(z) \cdot Q(z) = R(z)$. What is the minimum possible value of N ?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 5

Solution

Problem 15

Three identical square sheets of paper each with side length 6 are stacked on top of each other. The middle sheet is rotated clockwise 30° about its center and the top sheet is rotated clockwise 60° about its center, resulting in the 24-sided polygon shown in the figure below. The area of this polygon can be expressed in the form $a - b\sqrt{c}$, where a , b , and c are positive integers, and c is not divisible by the square of any prime. What is $a + b + c$?



- (A) 75 (B) 93 (C) 96 (D) 129 (E) 147

Solution

Problem 16

Suppose a , b , c are positive integers such that

$$a + b + c = 23$$

and

$$\gcd(a, b) + \gcd(b, c) + \gcd(c, a) = 9.$$

What is the sum of all possible distinct values of $a^2 + b^2 + c^2$?

- (A) 259 (B) 438 (C) 516 (D) 625 (E) 687

Solution

Problem 17

A bug starts at a vertex of a grid made of equilateral triangles of side length 1. At each step the bug moves in one of the 6 possible directions along the grid lines randomly and independently with equal probability. What is the probability that after 5 moves the bug never will have been more than 1 unit away from the starting position?

- (A) $\frac{13}{108}$ (B) $\frac{7}{54}$ (C) $\frac{29}{216}$ (D) $\frac{4}{27}$ (E) $\frac{1}{16}$

Solution

Problem 18

Set $u_0 = \frac{1}{4}$, and for $k \geq 0$ let u_{k+1} be determined by the recurrence

$$u_{k+1} = 2u_k - 2u_k^2.$$

This sequence tends to a limit; call it L . What is the least value of k such that

$$|u_k - L| \leq \frac{1}{2^{1000}}?$$

- (A) 10 (B) 87 (C) 123 (D) 329 (E) 401

Solution

Problem 19

Regular polygons with 5, 6, 7, and 8 sides are inscribed in the same circle. No two of the polygons share a vertex, and no three of their sides intersect at a common point. At how many points inside the circle do two of their sides intersect?

- (A) 52 (B) 56 (C) 60 (D) 64 (E) 68

Solution

Problem 20

A cube is constructed from 4 white unit cubes and 4 blue unit cubes. How many different ways are there to construct the $2 \times 2 \times 2$ cube using these smaller cubes? (Two constructions are considered the same if one can be rotated to match the other.)

- (A) 7 (B) 8 (C) 9 (D) 10 (E) 11

Solution

Problem 21

For real numbers x , let

$$P(x) = 1 + \cos(x) + i \sin(x) - \cos(2x) - i \sin(2x) + \cos(3x) + i \sin(3x)$$

where $i = \sqrt{-1}$. For how many values of x with $0 \leq x < 2\pi$ does

$$P(x) = 0?$$

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Solution

Problem 22

Right triangle ABC has side lengths $BC = 6$, $AC = 8$, and $AB = 10$.

A circle centered at O is tangent to line BC at B and passes through A . A circle centered at P is tangent to line AC at A and passes through B . What is OP ?

- (A) $\frac{23}{8}$ (B) $\frac{29}{10}$ (C) $\frac{35}{12}$ (D) $\frac{73}{25}$ (E) 3

Solution

Problem 23

What is the average number of pairs of consecutive integers in a randomly selected subset of 5 distinct integers chosen from the set $\{1, 2, 3, \dots, 30\}$? (For example the set $\{1, 17, 18, 19, 30\}$ has 2 pairs of consecutive integers.)

- (A) $\frac{2}{3}$ (B) $\frac{29}{36}$ (C) $\frac{5}{6}$ (D) $\frac{29}{30}$ (E) 1

Solution

Problem 24

Triangle ABC has side lengths $AB = 11$, $BC = 24$, and $CA = 20$. The bisector of $\angle BAC$ intersects $\overline{BC}$ in point D , and intersects the circumcircle of $\triangle ABC$ in point $E \neq A$. The circumcircle of $\triangle BED$ intersects the line AB in points B and $F \neq B$. What is CF ?

- (A) 28 (B) $20\sqrt{2}$ (C) 30 (D) 32 (E) $20\sqrt{3}$

Solution

Problem 25

For n a positive integer, let $R(n)$ be the sum of the remainders when n is divided by 2, 3, 4, 5, 6, 7, 8, 9, and 10. For example, $R(15) = 1 + 0 + 3 + 0 + 3 + 1 + 7 + 6 + 5 = 26$. How many two-digit positive integers n satisfy $R(n) = R(n + 1)$?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Solution

See also

2021 Fall AMC 12A (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/community/c13))	
Preceded by 2021 Fall AMC 12A Problems	Followed by 2022 AMC 12A Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
All AMC 12 Problems and Solutions	

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2021 Fall AMC 12B Answer Key

1. E
2. B
3. C
4. E
5. C
6. C
7. D
8. D
9. B
10. E
11. C
12. B
13. E
14. B
15. E
16. B
17. A
18. A
19. E
20. A
21. A
22. C
23. A
24. C
25. C

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2022 AMC 12A Problems**Problem 1**

What is the value of

$$3 + \frac{1}{3 + \frac{1}{3 + \frac{1}{3} ?}}$$

- (A) $\frac{31}{10}$ (B) $\frac{49}{15}$ (C) $\frac{33}{10}$ (D) $\frac{109}{33}$ (E) $\frac{15}{4}$

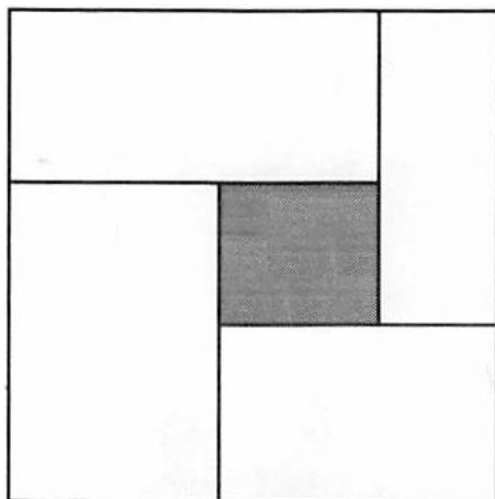
Problem 2

The sum of three numbers is 96. The first number is 6 times the third number, and the third number is 40 less than the second number. What is the absolute value of the difference between the first and second numbers?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Problem 3

Five rectangles, A , B , C , D , and E , are arranged in a square shown below. These rectangles have dimensions 1×6 , 2×4 , 5×6 , 2×7 , and 2×3 , respectively. (The figure is not drawn to scale.) Which of the five rectangles is the shaded one in the middle?



- (A) A (B) B (C) C (D) D (E) E

Problem 4

The least common multiple of a positive integer n and 18 is 180, and the greatest common divisor of n and 45 is 15. What is the sum of the digits of n ?

- (A) 3 (B) 6 (C) 8 (D) 9 (E) 12

Problem 5

The *taxicab distance* between points (x_1, y_1) and (x_2, y_2) in the coordinate plane is given by

$$|x_1 - x_2| + |y_1 - y_2|.$$

For how many points P with integer coordinates is the taxicab distance between P and the origin less than or equal to 20?

- (A) 441 (B) 761 (C) 841 (D) 921 (E) 924

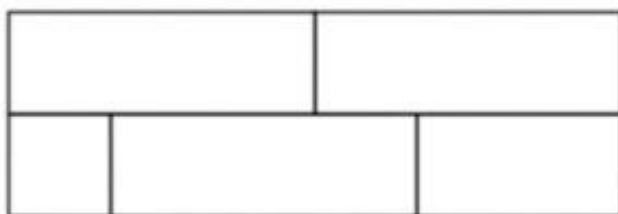
Problem 6

A data set consists of 6 (not distinct) positive integers: 1, 7, 5, 2, 5, and X . The average (arithmetic mean) of the 6 numbers equals a value in the data set. What is the sum of all positive values of X ?

- (A) 10 (B) 26 (C) 32 (D) 36 (E) 40

Problem 7

A rectangle is partitioned into 5 regions as shown. Each region is to be painted a solid color -- red, orange, yellow, blue, or green -- so that regions that touch are painted different colors, and colors can be used more than once. How many different colorings are possible?



- (A) 120 (B) 270 (C) 360 (D) 540 (E) 720

Problem 8

The infinite product

$$\sqrt[3]{10} \cdot \sqrt[3]{\sqrt[3]{10}} \cdot \sqrt[3]{\sqrt[3]{\sqrt[3]{10}}} \dots$$

evaluates to a real number. What is that number?

- (A) $\sqrt{10}$ (B) $\sqrt[3]{100}$ (C) $\sqrt[4]{1000}$ (D) 10 (E) $10\sqrt[3]{10}$

Problem 9

On Halloween 31 children walked into the principal's office asking for candy. They can be classified into three types: Some always lie; some always tell the truth; and some alternately lie and tell the truth. The alternaters arbitrarily choose their first response, either a lie or the truth, but each subsequent statement has the opposite truth value from its predecessor. The principal asked everyone the same three questions in this order.

"Are you a truth-teller?" The principal gave a piece of candy to each of the 22 children who answered yes.

"Are you an alternater?" The principal gave a piece of candy to each of the 15 children who answered yes.

"Are you a liar?" The principal gave a piece of candy to each of the 9 children who answered yes.

How many pieces of candy in all did the principal give to the children who always tell the truth?

- (A) 7 (B) 12 (C) 21 (D) 27 (E) 31

Problem 10

How many ways are there to split the integers 1 through 14 into 7 pairs such that in each pair the greater number is at least 2 times the lesser number?

- (A) 108 (B) 120 (C) 126 (D) 132 (E) 144

Problem 11

What is the product of all real numbers x such that the distance on the number line between $\log_6 x$ and $\log_6 9$ is twice the distance on the number line between $\log_6 10$ and 1?

- (A) 10 (B) 18 (C) 25 (D) 36 (E) 81

Problem 12

Let M be the midpoint of $\overline{AB}$ in regular tetrahedron $ABCD$. What is $\cos(\angle CMD)$?

- (A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{2}{5}$ (D) $\frac{1}{2}$ (E) $\frac{\sqrt{3}}{2}$

Problem 13

Let $\mathcal{R}$ be the region in the complex plane consisting of all complex numbers z that can be written as the sum of complex numbers z_1 and z_2 , where z_1 lies on the segment with endpoints 3 and $4i$, and z_2 has magnitude at most 1. What integer is closest to the area of $\mathcal{R}$?

- (A) 13 (A) 14 (A) 15 (A) 16 (A) 17

Problem 14

What is the value of

$$(\log 5)^3 + (\log 20)^3 + (\log 8)(\log 0.25)$$

Where all logarithms have base 10?

- (A) $\frac{3}{2}$ (B) $\frac{7}{4}$ (C) 2 (D) $\frac{9}{4}$ (E) $\frac{5}{2}$

Problem 15

The roots of the polynomial

$$10x^3 - 39x^2 + 29x - 6$$

are the height, length, and width of a rectangular box (right rectangular prism). A new rectangular box is formed by lengthening each edge of the original box by 2 units. What is the volume of the new box?

- (A) $\frac{24}{5}$ (B) $\frac{42}{5}$ (C) $\frac{81}{5}$ (D) 30 (E) 48

Problem 16

A *triangular number* is a positive integer that can be expressed in the form

$$t_n = 1 + 2 + 3 + \cdots + n,$$

for some positive integer n . The three smallest triangular numbers that are also perfect squares are

$$t_1 = 1 = 1^2, t_8 = 36 = 6^2, \text{ and } t_{49} = 1225 = 35^2.$$

What is the sum of the digits of the fourth smallest triangular number that is also a perfect square?

- (A) 6 (B) 9 (C) 12 (D) 18 (E) 27

Problem 17

Suppose a is a real number such that the equation

$$a \cdot (\sin x + \sin(2x)) = \sin(3x)$$

has more than one solution in the interval $(0, \pi)$. The set of all such a can be written in the form

$$(p, q) \cup (q, r),$$

where p , q , and r are real numbers with $p < q < r$. What is $p + q + r$?

- (A) -4 (B) -1 (C) 0 (D) 1 (E) 4

Problem 18

Let T_k be the transformation of the coordinate plane that first rotates the plane k degrees counterclockwise around the origin and then reflects the plane across the y -axis. What is the least positive integer n such that performing the sequence of transformations $T_1, T_2, T_3, \dots, T_n$ returns the point $(1, 0)$ back to itself?

- (A) 359 (B) 360 (C) 719 (D) 720 (E) 721

Problem 19

Suppose that 13 cards numbered $1, 2, 3, \dots, 13$ are arranged in a row. The task is to pick them up in numerically increasing order, working repeatedly from left to right. In the example below, cards 1, 2, 3 are picked up on the first pass, 4 and 5 on the second pass, 6 on the third pass, 7, 8, 9, 10 on the fourth pass, and 11, 12, 13 on the fifth pass.



For how many of the $13!$ possible orderings of the cards will the 13 cards be picked up in exactly two passes?

- (A) 4082 (B) 4095 (C) 4096 (D) 8178 (E) 8191

Problem 20

Isosceles trapezoid $ABCD$ has parallel sides $\overline{AD}$ and $\overline{BC}$, with $BC < AD$ and $AB = CD$. There is a point P in the plane such that $PA = 1$, $PB = 2$, $PC = 3$, and $PD = 4$. What is $\frac{BC}{AD}$?

- (A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$ (E) $\frac{3}{4}$

Problem 21

Let

$$P(x) = x^{2022} + x^{1011} + 1.$$

Which of the following polynomials is a factor of $P(x)$?

- (A) $x^2 - x + 1$ (B) $x^2 + x + 1$ (C) $x^4 + 1$ (D) $x^6 - x^3 + 1$ (E) $x^6 + x^3 + 1$

Problem 22

Let c be a real number, and let z_1, z_2 be the two complex numbers satisfying the equation

$$z^2 - cz + 10 = 0.$$

Points $z_1, z_2, \frac{1}{z_1}$, and $\frac{1}{z_2}$ are the vertices of (convex) quadrilateral Q in the complex plane. When the area of Q obtains its maximum value, c is the closest to which of the following?

- (A) 4.5 (B) 5 (C) 5.5 (D) 6 (E) 6.5

Problem 23

Let h_n and k_n be the unique relatively prime positive integers such that

$$\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} = \frac{h_n}{k_n}.$$

Let L_n denote the least common multiple of the numbers $1, 2, 3, \dots, n$. For how many integers n with $1 \leq n \leq 22$ is $k_n < L_n$?

- (A) 0 (B) 3 (C) 7 (D) 8 (E) 10

Problem 24

How many strings of length 5 formed from the digits 0, 1, 2, 3, 4 are there such that for each $j \in \{1, 2, 3, 4\}$, at least j of the digits are less than j ? (For example, 02214 satisfies this condition because it contains at least 1 digit less than 1, at least 2 digits less than 2, at least 3 digits less than 3, and at least 4 digits less than 4. The string 23404 does not satisfy the condition because it does not contain at least 2 digits less than 2.)

- (A) 500 (B) 625 (C) 1089 (D) 1199 (E) 1296

Problem 25

A circle with integer radius r is centered at (r, r) . Distinct line segments of length c_i connect points $(0, a_i)$ to $(b_i, 0)$ for $1 \leq i \leq 14$ and are tangent to the circle, where a_i , b_i , and c_i are all positive integers and

$$c_1 \leq c_2 \leq \cdots \leq c_{14}.$$

What is the ratio $\frac{c_{14}}{c_1}$ for the least possible value of r ?

- (A) $\frac{21}{5}$ (B) $\frac{85}{13}$ (C) 7 (D) $\frac{39}{5}$ (E) 17

Answer Key

1. D
2. E
3. B
4. B
5. C
6. D
7. D
8. A
9. A
10. E
11. E
12. B
13. A
14. C
15. D
16. D
17. A
18. A
19. D
20. B
21. E
22. A
23. D
24. E
25. E

2022 AMC 12B Problems

Problem 1

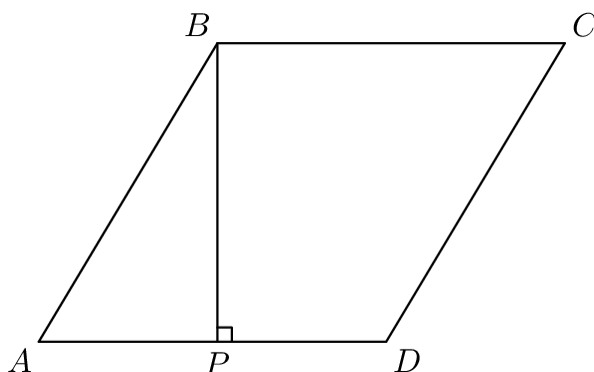
Define $x \diamond y$ to be $|x - y|$ for all real numbers x and y . What is the value of

$$(1 \diamond (2 \diamond 3)) - ((1 \diamond 2) \diamond 3)?$$

- (A) -2 (B) -1 (C) 0 (D) 1 (E) 2

Problem 2

In rhombus $ABCD$, point P lies on segment $\overline{AD}$ such that $BP \perp AD$, $AP = 3$, and $PD = 2$. What is the area of $ABCD$?



- (A) $3\sqrt{5}$ (B) 10 (C) $6\sqrt{5}$ (D) 20 (E) 25

Problem 3

How many of the first ten numbers of the sequence $121, 11211, 1112111, \dots$ are prime numbers?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Problem 4

For how many values of the constant k will the polynomial $x^2 + kx + 36$ have two distinct integer roots?

- (A) 6 (B) 8 (C) 9 (D) 14 (E) 16

Problem 5

The point $(-1, -2)$ is rotated 270° counterclockwise about the point $(3, 1)$. What are the coordinates of its new position?

- (A) $(-3, -4)$ (B) $(0, 5)$ (C) $(2, -1)$ (D) $(4, 3)$ (E) $(6, -3)$

Problem 6

Consider the following 100 sets of 10 elements each:

$$\begin{aligned} &\{1, 2, 3, \dots, 10\}, \\ &\{11, 12, 13, \dots, 20\}, \\ &\{21, 22, 23, \dots, 30\}, \\ &\vdots \\ &\{991, 992, 993, \dots, 1000\}. \end{aligned}$$

How many of these sets contain exactly two multiples of 7?

- (A) 40 (B) 42 (C) 43 (D) 49 (E) 50

Problem 7

Camila writes down five positive integers. The unique mode of these integers is 2 greater than their median, and the median is 2 greater than their arithmetic mean. What is the least possible value for the mode?

- (A) 5 (B) 7 (C) 9 (D) 11 (E) 13

Problem 8

What is the graph of $y^4 + 1 = x^4 + 2y^2$ in the coordinate plane?

- (A) Two intersecting parabolas (B) Two nonintersecting parabolas
(C) Two intersecting circles (D) A circle and a hyperbola
(E) A circle and two parabolas

Problem 9

The sequence $a_0, a_1, a_2, \dots$ is a strictly increasing arithmetic sequence of positive integers such that

$$2^{a_7} = 2^{27} \cdot a_7.$$

What is the minimum possible value of a_2 ?

- (A) 8 (B) 12 (C) 16 (D) 17 (E) 22

Problem 10

Regular hexagon $ABCDEF$ has side length 2. Let G be the midpoint of $\overline{AB}$, and let H be the midpoint of $\overline{DE}$. What is the perimeter of $GCHF$?

- (A) $4\sqrt{3}$ (B) 8 (C) $4\sqrt{5}$ (D) $4\sqrt{7}$ (E) 12

Problem 11

Let

$$f(n) = \left(\frac{-1 + i\sqrt{3}}{2} \right)^n + \left(\frac{-1 - i\sqrt{3}}{2} \right)^n,$$

where $i = \sqrt{-1}$. What is $f(2022)$?

- (A) -2 (B) -1 (C) 0 (D) $\sqrt{3}$ (E) 2

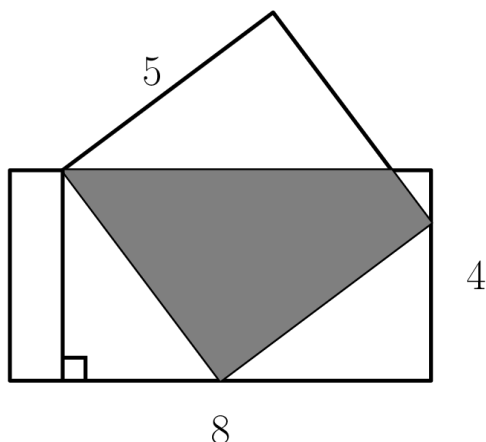
Problem 12

Kayla rolls four fair 6-sided dice. What is the probability that at least one of the numbers Kayla rolls is greater than 4 and at least two of the numbers she rolls are greater than 2?

- (A) $\frac{2}{3}$ (B) $\frac{19}{27}$ (C) $\frac{59}{81}$ (D) $\frac{61}{81}$ (E) $\frac{7}{9}$

Problem 13

The diagram below shows a rectangle with side lengths 4 and 8 and a square with side length 5. Three vertices of the square lie on three different sides of the rectangle, as shown. What is the area of the region inside both the square and the rectangle?



- (A) $15\frac{1}{8}$ (B) $15\frac{3}{8}$ (C) $15\frac{1}{2}$ (D) $15\frac{5}{8}$ (E) $15\frac{7}{8}$

Problem 14

The graph of

$$y = x^2 + 2x - 15$$

intersects the x -axis at points A and C and the y -axis at point B . What is $\tan(\angle ABC)$?

- (A) $\frac{1}{7}$ (B) $\frac{1}{4}$ (C) $\frac{3}{7}$ (D) $\frac{1}{2}$ (E) $\frac{4}{7}$

Problem 15

One of the following numbers is not divisible by any prime number less than 10. Which is it?

- (A) $2^{606} - 1$ (B) $2^{606} + 1$ (C) $2^{607} - 1$ (D) $2^{607} + 1$ (E) $2^{607} + 3^{607}$

Problem 16

Suppose x and y are positive real numbers such that

$$x^y = 2^{64} \quad \text{and} \quad (\log_2 x)^{\log_2 y} = 2^{27}.$$

What is the greatest possible value of $\log_2 y$?

- (A) 3 (B) 4 (C) $3 + \sqrt{2}$ (D) $4 + \sqrt{3}$ (E) 7

Problem 17

How many 4×4 arrays whose entries are 0s and 1s are there such that the row sums (the sum of the entries in each row) are 1, 2, 3, and 4, in some order, and the column sums (the sum of the entries in each column) are also 1, 2, 3, and 4, in some order? For example, the array

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

satisfies the condition.

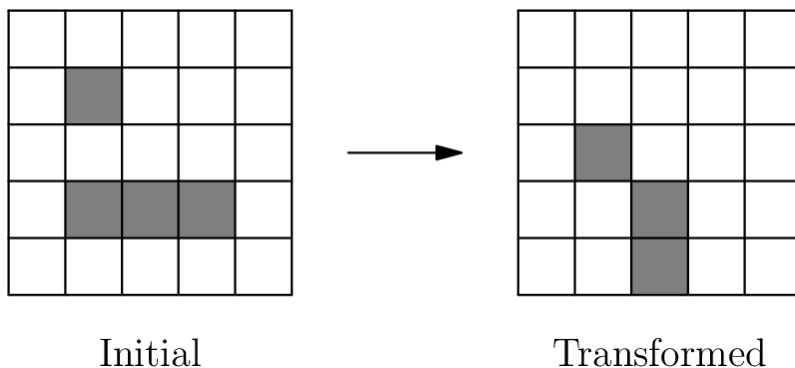
(A) 144 (B) 240 (C) 336 (D) 576 (E) 624

Problem 18

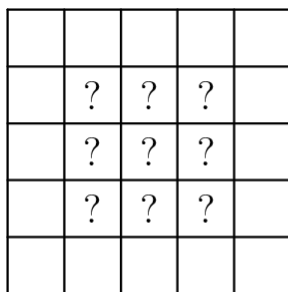
Each square in a 5×5 grid is either filled or empty, and has up to eight adjacent neighboring squares, where neighboring squares share either a side or a corner. The grid is transformed by the following rules:

- Any filled square with two or three filled neighbors remains filled.
- Any empty square with exactly three filled neighbors becomes a filled square.
- All other squares remain empty or become empty.

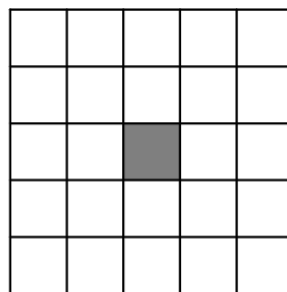
A sample transformation is shown in the figure below.



Suppose the 5×5 grid has a border of empty squares surrounding a 3×3 subgrid. How many initial configurations will lead to a transformed grid consisting of a single filled square in the center after a single transformation? (Rotations and reflections of the same configuration are considered different.)



Initial



Transformed

- (A) 14 (B) 18 (C) 22 (D) 26 (E) 30

Problem 19

In $\triangle ABC$ medians $\overline{AD}$ and $\overline{BE}$ intersect at G and $\triangle AGE$ is equilateral.

Then $\cos(C)$ can be written as $\frac{m\sqrt{p}}{n}$, where m and n are relatively prime positive integers and p is a positive integer not divisible by the square of any prime. What is $m + n + p$?

- (A) 44 (B) 48 (C) 52 (D) 56 (E) 60

Problem 20

Let $P(x)$ be a polynomial with rational coefficients such that when $P(x)$ is divided by the polynomial $x^2 + x + 1$, the remainder is $x + 2$, and when $P(x)$ is divided by the polynomial $x^2 + 1$, the remainder is $2x + 1$. There is a unique polynomial of least degree with these two properties. What is the sum of the squares of the coefficients of that polynomial?

- (A) 10 (B) 13 (C) 19 (D) 20 (E) 23

Problem 21

Let S be the set of circles in the coordinate plane that are tangent to each of the three circles with equations

$$x^2 + y^2 = 4, \quad x^2 + y^2 = 64, \quad \text{and} \quad (x - 5)^2 + y^2 = 3.$$

What is the sum of the areas of all circles in S ?

- (A) 48π (B) 68π (C) 96π (D) 102π (E) 136π

Problem 22

Ant Amelia starts on the number line at 0 and crawls in the following manner. For $n = 1, 2, 3$, Amelia chooses a time duration t_n and an increment x_n independently and uniformly at random from the interval $(0, 1)$. During the n th step of the process, Amelia moves x_n units in the positive direction, using up t_n minutes. If the total elapsed time has exceeded 1 minute during the n th step, she stops at the end of that step; otherwise, she continues with the next step, taking at most 3 steps in all. What is the probability that Amelia's position when she stops will be greater than 1?

- (A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$ (E) $\frac{5}{6}$

Problem 23

Let $x_0, x_1, x_2, \dots$ be a sequence of numbers, where each x_k is either 0 or 1. For each positive integer n , define

$$S_n = \sum_{k=0}^{n-1} x_k 2^k$$

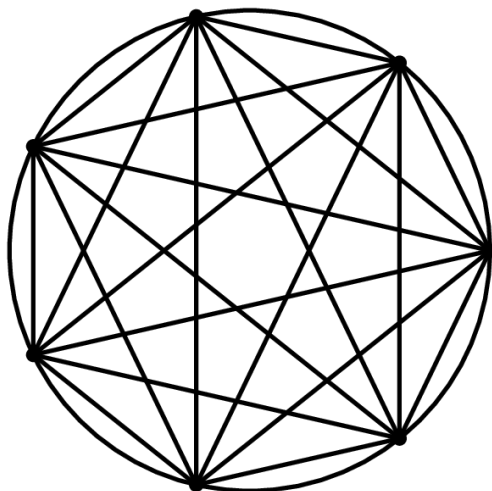
Suppose $7S_n \equiv 1 \pmod{2^n}$ for all $n \geq 1$. What is the value of the sum

$$x_{2019} + 2x_{2020} + 4x_{2021} + 8x_{2022}?$$

- (A) 6 (B) 7 (C) 12 (D) 14 (E) 15

Problem 24

The figure below depicts a regular 7-gon inscribed in a unit circle.

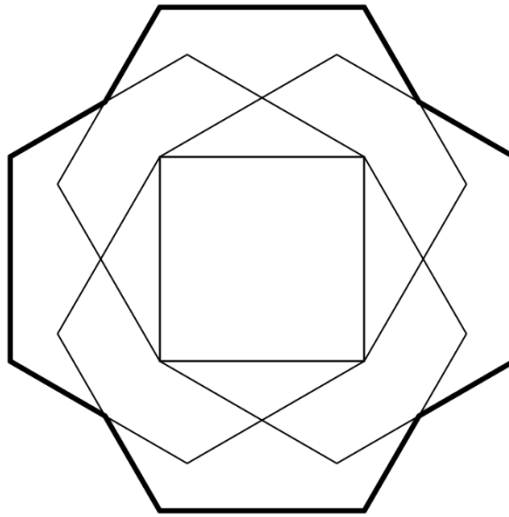


What is the sum of the 4th powers of the lengths of all 21 of its edges and diagonals?

- (A) 49 (B) 98 (C) 147 (D) 168 (E) 196

Problem 25

Four regular hexagons surround a square with a side length 1, each one sharing an edge with the square, as shown in the figure below. The area of the resulting 12-sided outer nonconvex polygon can be written as $m\sqrt{n} + p$, where m , n , and p are integers and n is not divisible by the square of any prime. What is $m + n + p$?



- (A) -12 (B) -4 (C) 4 (D) 24 (E) 32

Answer Key

1. A
2. D
3. A
4. B
5. B
6. B
7. D
8. D
9. B
10. D
11. E
12. D
13. D
14. E
15. C
16. C
17. D
18. C
19. A
20. E
21. E
22. C
23. A
24. C
25. B



2023 AMC 12A Problems

2023 AMC 12A (Answer Key)

Printable versions: • AoPS Resources (https://artofproblemsolving.com/community/contest/collection/c3415_amc_12/2023) • PDF (https://artofproblemsolving.com/community/contest/download/c3415_amc_12/2023)

Instructions

1. This is a 25-question, multiple choice test. Each question is followed by answers marked A, B, C, D and E. Only one of these is correct.
2. You will receive 6 points for each correct answer, 2.5 points for each problem left unanswered if the year is before 2006, 1.5 points for each problem left unanswered if the year is after 2006, and 0 points for each incorrect answer.
3. No aids are permitted other than scratch paper, graph paper, ruler, compass, protractor and erasers (and calculators that are accepted for use on the test if before 2006. No problems on the test will *require* the use of a calculator).
4. Figures are not necessarily drawn to scale.
5. You will have **75 minutes** working time to complete the test.

1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

Cities A and B are 45 miles apart. Alicia lives in A and Beth lives in B . Alicia bikes towards B at 18 miles per hour. Leaving at the same time, Beth bikes toward A at 12 miles per hour. How many miles from City A will they be when they meet?

- (A) 20 (B) 24 (C) 25 (D) 26 (E) 27

Solution

Problem 2

The weight of $\frac{1}{3}$ of a large pizza together with $3\frac{1}{2}$ cups of orange slices is the same weight of $\frac{3}{4}$ of a large pizza together with $\frac{1}{2}$ cups of orange slices. A cup of orange slices weigh $\frac{1}{4}$ of a pound. What is the weight, in pounds, of a large pizza?

- (A) $1\frac{4}{5}$ (B) 2 (C) $2\frac{2}{5}$ (D) 3 (E) $3\frac{3}{5}$

Solution

Problem 3

How many positive perfect squares less than 2023 are divisible by 5?

- (A) 8 (B) 9 (C) 10 (D) 11 (E) 12

Solution

Problem 4

How many digits are in the base-ten representation of $8^5 \cdot 5^{10} \cdot 15^5$?

- (A) 14 (B) 15 (C) 16 (D) 17 (E) 18

Solution

Problem 5

Janet rolls a standard 6-sided die 4 times and keeps a running total of the numbers she rolls. What is the probability that at some point, her running total will equal 3?

- (A) $\frac{2}{9}$ (B) $\frac{49}{216}$ (C) $\frac{25}{108}$ (D) $\frac{17}{72}$ (E) $\frac{13}{54}$

Solution

Problem 6

Points A and B lie on the graph of $y = \log_2 x$. The midpoint of $\overline{AB}$ is $(6, 2)$. What is the positive difference between the x -coordinates of A and B ?

- (A) $2\sqrt{11}$ (B) $4\sqrt{3}$ (C) 8 (D) $4\sqrt{5}$ (E) 9

Solution

Problem 7

A digital display shows the current date as an 8-digit integer consisting of a 4-digit year, followed by a 2-digit month, followed by a 2-digit date within the month. For example, Arbor Day this year is displayed as 20230428. For how many dates in 2023 will each digit appear an even number of times in the 8-digit display for that date?

- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

Solution

Problem 8

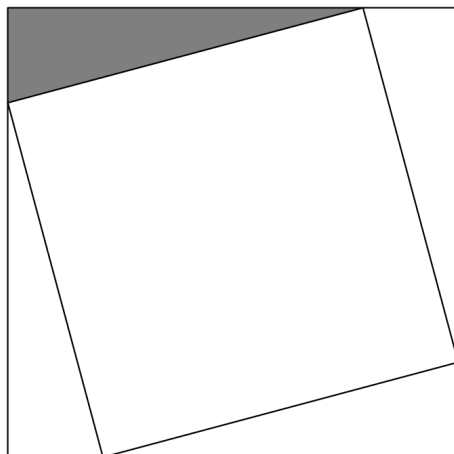
Maureen is keeping track of the mean of her quiz scores this semester. If Maureen scores an 11 on the next quiz, her mean will increase by 1. If she scores an 11 on each of the next three quizzes, her mean will increase by 2. What is the mean of her quiz scores currently?

- (A) 4 (B) 5 (C) 6 (D) 7 (E) 8

Solution

Problem 9

A square of area 2 is inscribed in a square of area 3, creating four congruent triangles, as shown below. What is the ratio of the shorter leg to the longer leg in the shaded right triangle?



- (A) $\frac{1}{5}$ (B) $\frac{1}{4}$ (C) $2 - \sqrt{3}$ (D) $\sqrt{3} - \sqrt{2}$ (E) $\sqrt{2} - 1$

Solution

Problem 10

Positive real numbers x and y satisfy $y^3 = x^2$ and $(y - x)^2 = 4y^2$. What is $x + y$?

- (A) 12 (B) 18 (C) 24 (D) 36 (E) 42

Solution

Problem 11

What is the degree measure of the acute angle formed by lines with slopes 2 and $\frac{1}{3}$?

- (A) 30 (B) 37.5 (C) 45 (D) 52.5 (E) 60

Solution

Problem 12

What is the value of

$$2^3 - 1^3 + 4^3 - 3^3 + 6^3 - 5^3 + \cdots + 18^3 - 17^3?$$

- (A) 2023 (B) 2679 (C) 2941 (D) 3159 (E) 3235

Solution

Problem 13

In a table tennis tournament every participant played every other participant exactly once. Although there were twice as many right-handed players as left-handed players, the number of games won by left-handed players was 40% more than the number of games won by right-handed players. (There were no ties and no ambidextrous players.) What is the total number of games played?

- (A) 15 (B) 36 (C) 45 (D) 48 (E) 66

Solution

Problem 14

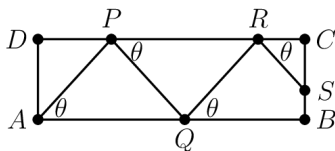
How many complex numbers satisfy the equation $z^5 = \bar{z}$, where $\bar{z}$ is the conjugate of the complex number z ?

- (A) 2 (B) 3 (C) 5 (D) 6 (E) 7

Solution

Problem 15

Usain is walking for exercise by zigzagging across a 100-meter by 30-meter rectangular field, beginning at point A and ending on the segment $\overline{BC}$. He wants to increase the distance walked by zigzagging as shown in the figure below ($APQRS$). What angle θ $\angle PAB = \angle QPC = \angle RQB = \dots$ will produce a length that is 120 meters? (This figure is not drawn to scale. Do not assume that the zigzag path has exactly four segments as shown; there could be more or fewer.)



- (A) $\arccos \frac{5}{6}$ (B) $\arccos \frac{4}{5}$ (C) $\arccos \frac{3}{10}$ (D) $\arcsin \frac{4}{5}$ (E) $\arcsin \frac{5}{6}$

Solution

Problem 16

Consider the set of complex numbers z satisfying $|1 + z + z^2| = 4$. The maximum value of the imaginary part of z can be written in the form $\frac{\sqrt{m}}{n}$, where m and n are relatively prime positive integers. What is $m + n$?

- (A) 20 (B) 21 (C) 22 (D) 23 (E) 24

Solution

Problem 17

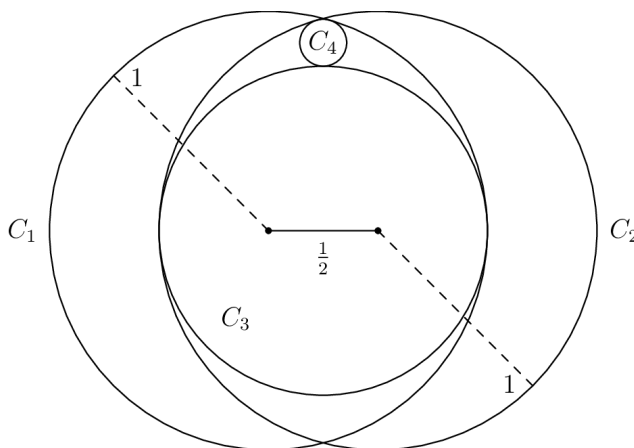
Flora the frog starts at 0 on the number line and makes a sequence of jumps to the right. In any one jump, independent of previous jumps, Flora leaps a positive integer distance m with probability $\frac{1}{2^m}$. What is the probability that Flora will eventually land at 10?

- (A) $\frac{5}{512}$ (B) $\frac{45}{1024}$ (C) $\frac{127}{1024}$ (D) $\frac{511}{1024}$ (E) $\frac{1}{2}$

Solution

Problem 18

Circle C_1 and C_2 each have radius 1, and the distance between their centers is $\frac{1}{2}$. Circle C_3 is the largest circle internally tangent to both C_1 and C_2 . Circle C_4 is internally tangent to both C_1 and C_2 and externally tangent to C_3 . What is the radius of C_4 ?



- (A) $\frac{1}{14}$ (B) $\frac{1}{12}$ (C) $\frac{1}{10}$ (D) $\frac{3}{28}$ (E) $\frac{1}{9}$

Solution

Problem 19

What is the product of all the solutions to the equation

$$\log_{7x} 2023 \cdot \log_{289x} 2023 = \log_{2023x} 2023?$$

- (A) $(\log_{2023} 7 \cdot \log_{2023} 289)^2$ (B) $\log_{2023} 7 \cdot \log_{2023} 289$ (C) 1
(D) $\log_7 2023 \cdot \log_{289} 2023$ (E) $(\log_7 2023 \cdot \log_{289} 2023)^2$

Solution

Problem 20

Rows 1, 2, 3, 4, and 5 of a triangular array of integers are shown below:

					1				
				1		1			
			1		3		1		
		1		5		5		1	
	1		7		11		7		1

Each row after the first row is formed by placing a 1 at each end of the row, and each interior entry is 1 greater than the sum of the two numbers diagonally above it in the previous row. What is the units digit of the sum of the 2023 numbers in the 2023rd row?

- (A) 1 (B) 3 (C) 5 (D) 7 (E) 9

Solution

Problem 21

If A and B are vertices of a polyhedron, define the distance $d(A, B)$ to be the minimum number of edges of the polyhedron one must traverse in order to connect A and B . For example, if $\overline{AB}$ is an edge of the polyhedron, then $d(A, B) = 1$, but if $\overline{AC}$ and $\overline{CB}$ are edges and $\overline{AB}$ is not an edge, then $d(A, B) = 2$. Let Q, R , and S be randomly chosen distinct vertices of a regular icosahedron (regular polyhedron made up of 20 equilateral triangles). What is the probability that $d(Q, R) > d(R, S)$?

- (A) $\frac{7}{22}$ (B) $\frac{1}{3}$ (C) $\frac{3}{8}$ (D) $\frac{5}{12}$ (E) $\frac{1}{2}$

Solution

Problem 22

Let f be the unique function defined on the positive integers such that

$$\sum_{d|n} d \cdot f\left(\frac{n}{d}\right) = 1$$

for all positive integers n , where the sum is taken over all positive divisors of n . What is $f(2023)$?

- (A) -1536 (B) 96 (C) 108 (D) 116 (E) 144

Solution

Problem 23

How many ordered pairs of positive real numbers (a, b) satisfy the equation

$$(1 + 2a)(2 + 2b)(2a + b) = 32ab?$$

- (A) 0 (B) 1 (C) 2 (D) 3 (E) an infinite number

Solution

Problem 24

Let K be the number of sequences $A_1, A_2, \dots, A_n$ such that n is a positive integer less than or equal to 10, each A_i is a subset of $\{1, 2, 3, \dots, 10\}$, and A_{i-1} is a subset of A_i for each i between 2 and n , inclusive. For example, $\{\}, \{5, 7\}, \{2, 5, 7\}, \{2, 5, 7\}, \{2, 5, 6, 7, 9\}$ is one such sequence, with $n = 5$. What is the remainder when K is divided by 10?

- (A) 1 (B) 3 (C) 5 (D) 7 (E) 9

Solution

Problem 25

There is a unique sequence of integers $a_1, a_2, \dots, a_{2023}$ such that

$$\tan 2023x = \frac{a_1 \tan x + a_3 \tan^3 x + a_5 \tan^5 x + \dots + a_{2023} \tan^{2023} x}{1 + a_2 \tan^2 x + a_4 \tan^4 x + \dots + a_{2022} \tan^{2022} x}$$

whenever $\tan 2023x$ is defined. What is a_{2023} ?

- (A) -2023 (B) -2022 (C) -1 (D) 1 (E) 2023

Solution

See also

2023 AMC 12A (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/community/c13))	
Preceded by 2022 AMC 12B Problems	Followed by 2023 AMC 12B Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
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- Mathematics competition resources

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AMC 12A 2023

1

Las ciudades A y B están a 45 millas de separación. Alicia vive en A y Beth vive en B. Alicia se dirige hacia B a 18 millas por hora, y saliendo al mismo tiempo, Beth se dirige hacia A a 12 millas por hora. ¿A cuántas millas de la ciudad A se encontrarán?

(A) 20 (B) 24 (C) 25 (D) 26 (E) 27

2

El peso de $\frac{1}{3}$ de pizza grande junto con $\frac{7}{2}$ de vasos de zumo de naranja es igual al peso de $\frac{3}{4}$ de pizza grande junto con $\frac{1}{2}$ vasos de zumo de naranja. Si un vaso de zumo de naranja pesa $\frac{1}{4}$ de libra, ¿Cuál es el peso, en libras, de una pizza grande?

(A) $9/5$ (B) 2 (C) $12/5$ (D) 3 (E) $18/5$

3

Determina el número de cuadrados perfectos positivos menores de 2023 divisibles entre 5.

(A) 8 (B) 9 (C) 10 (D) 11 (E) 12

4

Determina el número de dígitos que tiene la representación en base 10 del número $8^5 \cdot 5^{10} \cdot 15^5$.

(A) 14 (B) 15 (C) 16 (D) 17 (E) 18

5

Janet lanza un dado de seis caras cuatro veces y va sumando las cantidades obtenidas. Determina la probabilidad de obtener, en algún momento, una suma igual a tres.

(A) $2/9$ (B) $49/216$ (C) $25/108$ (D) $17/72$ (E) $13/54$

6

Los puntos A y B pertenecen a la gráfica de $y = \log_2 x$. El punto medio de $\overline{AB}$ es $(6, 2)$. Determina la diferencia positiva de las x-coordenadas de A y B.

(A) $2\sqrt{11}$ (B) $4\sqrt{3}$ (C) 8 (D) $4\sqrt{5}$ (E) 9

7

Una pantalla digital muestra la fecha actual como un entero de 8 dígitos: Cuatro dígitos correspondiendo al año seguidos de 2 dígitos correspondiendo al mes y seguidos finalmente por 2 dígitos correspondiendo al día del mes. Por ejemplo, 20230428. Determina la cantidad de fechas que hay en el 2023 en las que cada dígito aparezca una cantidad par de veces.

- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

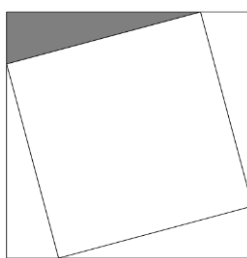
8

Maureen está haciendo un seguimiento de las medias en sus puntuaciones de las pruebas que ha realizado este semestre. Si obtuviera un 11 en la próxima prueba, su media se incrementaría en 1, y si puntuara un 11 en cada uno de las tres próximas pruebas, se incrementaría en 2. ¿Cuál es la media actual de sus puntuaciones?

- (A) 4 (B) 5 (C) 6 (D) 7 (E) 8

9

Un cuadrado de área 2 está inscrito en un cuadrado de área 3, creando cuatro triángulos congruentes, tal y como se muestra en la imagen. Determina la razón del lado corto al lado largo del triángulo sombreado.



- (A) $1/5$ (B) $1/4$ (C) $2 - \sqrt{3}$ (D) $\sqrt{3} - \sqrt{2}$ (E) $\sqrt{2} - 1$

10

Sean x , y números reales positivos satisfaciendo $y^3 = x^2$ y $(y - x)^2 = 4y^2$. Determina $x + y$.

- (A) 12 (B) 18 (C) 24 (D) 36 (E) 42

11

Determina la medida angular en grados del ángulo agudo formado por las rectas de pendiente 2 y $1/3$.

- (A) 30 (B) 37.5 (C) 45 (D) 52.5 (E) 60

12

Determina el valor de

$$2^3 - 1^3 + 4^3 - 3^3 + 6^3 - 5^3 + 6^3 - 5^3 + \dots + 18^3 - 17^3$$

- (A) 2023 (B) 2679 (C) 3941 (D) 3159 (E) 3235

13

En un campeonato de ping-pong cada participante todos los participantes se enfrentan entre ellos exactamente una vez. Aunque hay el doble de diestros que de zurdos, el número de partidas ganadas por los zurdos fue un 40% más que el número de partidas ganadas por los diestros. No se produjeron empates ni había ningún ambidiestro. Determina el total de partidas jugadas.

- (A) 15 (B) 36 (C) 45 (D) 48 (E) 66

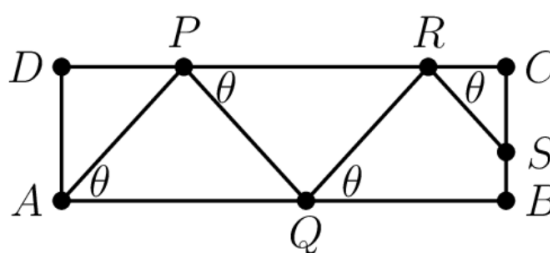
14

Determina la cantidad de números complejos satisfaciendo $z^5 = \bar{z}$, donde $\bar{z}$ es el conjugado del número complejo z .

- (A) 2 (B) 3 (C) 5 (D) 6 (E) 7

15^F

Usain camina en ziga-zaga por un campo rectangular de 100×30 metros, empezando en el punto A y acabando en el segmento $\overline{BC}$. Él quiere incrementar la distancia recorrida tal y como se muestra en la imagen siguiente (APQRS). Determina el ángulo $\theta = \angle PAB = \angle QPC = \angle RQB = \dots$ que producirá una longitud de 120 metros. (Esta imagen no está dibujada a escala. No asuma que este camino en ziga-zaga tenga exactamente cuatro segmentos, tal y como se muestra en la imagen, puede tener más segmentos o menos).



- (A) $\arccos \frac{5}{6}$ (B) $\arccos \frac{4}{5}$ (C) $\arccos \frac{3}{10}$ (D) $\arcsin \frac{4}{5}$ (E) $\arcsin \frac{5}{6}$

16

Consideremos el conjunto de números complejos z satisfaciendo la condición $|1 + z + z^2| = 4$. Si el máximo valor posible que puede obtener la parte imaginaria de z se puede escribir como $\frac{\sqrt{m}}{n}$, donde m y n son enteros positivos coprimos, determina $m + n$.

- (A) 20 (B) 21 (C) 22 (D) 23 (E) 24

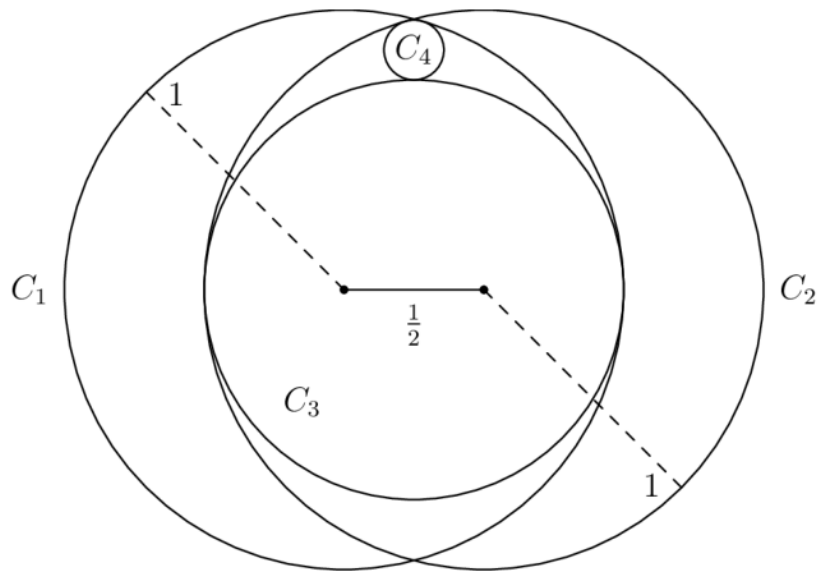
17

La rana Flora está en el 0 de la recta numérica y empieza una sucesión de saltos hacia la derecha. En cada salto, independientemente de los saltos anteriores, Flora salta una distancia entera positiva m con probabilidad $\frac{1}{2^m}$. Determina la probabilidad de que Flora pase por la posición 10.

- (A) $5/512$ (B) $45/1024$ (C) $127/1024$ (D) $511/1024$ (E) $1/2$

18

Las circunferencias C_1 y C_2 tienen ambas radio 1, y la distancia entre sus centros es $1/2$. La circunferencia C_3 es la mayor circunferencia internamente tangente a C_1 y C_2 . La circunferencia C_4 es internamente tangente a C_1 y C_2 y externamente tangente a C_3 . Determina el radio de C_4 .



- (A) $1/14$ (B) $1/12$ (C) $1/10$ (D) $3/28$ (E) $1/9$

19

Determina el producto de todas las soluciones de la ecuación

$$\log_{7x} 2023 \cdot \log_{289x} 2023 = \log_{2023x} 2023$$

- (A) $(\log_{2023} 7 \cdot \log_{2023} 289)^2$ (B) $\log_{2023} 7 \cdot \log_{2023} 289$ (C) 1
(D) $\log_7 2023 \cdot \log_{289} 2023$ (E) $(\log_7 2023 \cdot \log_{289} 2023)^2$

20

Presentamos las filas 1, 2, 3, 4 y 5 de una tabla de enteros:

$$\begin{array}{ccccccccc}
 & & & & 1 & & & & \\
 & & & 1 & & 1 & & & \\
 & & 1 & & 3 & & 1 & & \\
 & 1 & & 5 & & 5 & & 1 & \\
 1 & & 7 & & 11 & & 7 & & 1
 \end{array}$$

En la que cada fila se forma poniendo 1 en los extremos y en las posiciones intermedias sumandos los dos números inmediatamente superiores de la fila anterior y añadiendo 1. Determina la cifra de las unidades de la suma de los 2023 números de la fila 2023.

- (A) 1 (B) 3 (C) 5 (D) 7 (E) 9

21

Si A y B son los vértices de un poliedro, definimos la distancia $d(A, B)$ como el mínimo número de aristas del poliedro que hay que recorrer para conectar A con B . Por ejemplo, si $\overline{AB}$ es una arista del poliedro, entonces $d(A, B) = 1$, pero si $\overline{AC}$ y $\overline{CB}$ son aristas y $\overline{AB}$ no es una arista, entonces $d(A, B) = 2$. Sean Q , R y S tres vértices tomados de forma aleatoria de un icosaedro regular (un poliedro regular formado por 20 triángulos equiláteros). Determina la probabilidad de que $d(Q, R) > d(R, S)$.

- (A) $\frac{7}{22}$ (B) $\frac{1}{3}$ (C) $\frac{3}{5}$ (D) $\frac{5}{12}$ (E) $\frac{1}{2}$

22

Sea f la única función definida para todos los enteros positivos tal que

$$\sum_{d|n} d \cdot f\left(\frac{n}{d}\right) = 1$$

para cualquier entero n , donde esta suma recorre todos los divisores positivos de n .
Determina $f(2023)$.

- (A) -1536 (B) 96 (C) 108 (D) 116 (E) 144

23

Determina el número de pares ordenados de números reales positivos satisfaciendo la ecuación

$$(1+2a)(2+2b)(2a+b)=32ab$$

- (A) 0 (B) 1 (C) 2 (D) 3 (E) una cantidad infinita

24

Sea K el número de secuencias $A_1, A_2, \dots, A_n$ en donde n es un entero positivo menor o igual a 10, cada A_i es un subconjunto de $\{1, 2, 3, \dots, 10\}$ y cada A_{i-1} es un subconjunto de A_i para cada i entre 2 y n , inclusivo. Por ejemplo, $\{\}, \{5, 7\}, \{2, 5, 7\}, \{2, 5, 7\}, \{2, 5, 6, 7, 9\}$ es una de estas secuencias, con $n = 5$. Determina el residuo cuando dividimos K entre 10.

(A) 1 (B) 3 (C) 5 (D) 7 (E) 9

25

Existe una única secuencia de enteros $a_1, a_2, \dots, a_{2023}$ tal que

$$\tan 2023x = \frac{a_1 \tan x + a_3 \tan^3 x + a_5 \tan^5 x + \dots + a_{2023} \tan^{2023} x}{1 + a_2 \tan^2 x + a_4 \tan^4 x + \dots + a_{2022} \tan^{2022} x}$$

Siempre que $\tan 2023x$ está definido. Determina a_{2023} .

(A) -2023 (B) -2022 (C) -1 (D) 1 (E) 2023



2023 AMC 12A Answer Key

1. E
2. A
3. A
4. E
5. B
6. D
7. E
8. D
9. C
10. D
11. C
12. D
13. B
14. E
15. A
16. B
17. E
18. D
19. C
20. C
21. A
22. B
23. B
24. C
25. C

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AMC 12A 2023 Soluciones

1

El espacio recorrido (en millas) por Alicia desde el punto A se puede modelizar por la función $s_1(t) = 18t$

Mientras que el espacio recorrido en millas por Beth desde el mismo punto A se puede modelizar por la función $s_2(t) = 45 - 12t$.

Así pues, se encontrarán en el momento en el que hayan recorrido el mismo espacio:

$$s_1(t) = s_2(t) \Leftrightarrow 18t = 45 - 12t \Leftrightarrow 30t = \frac{45}{30} = \frac{3}{2}$$

En ese momento el espacio recorrido será

$$s_1\left(\frac{3}{2}\right) = 18 \cdot \frac{3}{2} = 27 \text{ millas (E).}$$

2

Hay que resolver la ecuación $\frac{1}{3}p + \frac{7}{2}c = \frac{3}{4}p + \frac{1}{2}c$ sabiendo que $c = \frac{1}{4}p$, y esta ecuación tiene por solución $p = \frac{9}{5}$ (A).

3

Está claro que un entero es divisible entre 5 sí y solo si lo es su cuadrado, luego buscamos todos los enteros positivos múltiplos de 5 de forma que su cuadrado sea menor que 2023.

Probando números observamos que $40^2 = 1600$ y $45^2 = 2025$, luego serán todos los enteros de la forma $5^2, 10^2, 15^2, \dots, 40^2$, es decir, $(5k)^2$ con $1 \leq k \leq 8$, ocho números en total (A).

4

$$8^5 \cdot 5^{10} \cdot 15^5 = (2^3)^5 \cdot 5^{10} \cdot (3 \cdot 5)^5 = 2^{15} \cdot 5^{10} \cdot 3^5 \cdot 5^5 = 2^{15} \cdot 5^{15} \cdot 3^5 = 10^{15} \cdot 3^5 = 243 \cdot 10^{15}$$

Y este número tiene 18 dígitos (E).

5

Si nos imaginamos el árbol de posibilidades, vemos que las “ramas” aceptables son

3XXX con probabilidad $1/6$

12XX y 21XX con probabilidad $2(1/6)^2$

111X con probabilidad $(1/6)^3$

En total tenemos una probabilidad de $\frac{1}{6} + 2\left(\frac{1}{6}\right)^2 + \left(\frac{1}{6}\right)^3 = \frac{49}{216}$ (B).

6

Primera versión.

Sean $A = (x_1, y_1)$ y $B = (x_2, y_2)$. La condición del enunciado equivale a

$$\frac{x_1 + x_2}{2} = 6, \frac{y_1 + y_2}{2} = 2$$

$$\frac{x_1 + x_2}{2} = 6 \Leftrightarrow x_1 + x_2 = 12$$

$$\frac{y_1 + y_2}{2} = 2 \Leftrightarrow y_1 + y_2 = 4 \Leftrightarrow \log_2 x_1 + \log_2 x_2 = 4 \Leftrightarrow \log_2 (x_1 \cdot x_2) = 4$$

$$\Leftrightarrow x_1 \cdot x_2 = 2^4 = 16$$

$$\frac{x_1 + x_2}{2} = 6 \Leftrightarrow x_1 + x_2 = 12$$

Así pues, tenemos el sistema

$$\begin{cases} x_1 \cdot x_2 = 16 \\ x_1 + x_2 = 12 \end{cases}$$

Luego

$$x_1 + x_2 = 12 \Rightarrow x_1 = 12 - x_2 \Rightarrow 16 = (12 - x_2) \cdot x_2 = 12x_2 - x_2^2$$

$$\Leftrightarrow 16 - 12x_2 + x_2^2 = 0 \Leftrightarrow x_2 = \frac{12 \pm \sqrt{12^2 - 4 \cdot 1 \cdot 16}}{2} = \frac{12 \pm \sqrt{12^2 - 4 \cdot 1 \cdot 16}}{2} = \frac{12 \pm \sqrt{80}}{2} =$$

$$= \frac{12 \pm 4\sqrt{5}}{2} = 6 \pm 2\sqrt{5}$$

Puesto que sabemos que pertenece a la gráfica de un logaritmo, podemos descartar la solución negativa.

$$x_2 = 6 + 2\sqrt{5} \Rightarrow x_1 = 12 - 6 - 2\sqrt{5} = 6 - 2\sqrt{5}$$

Y finalmente

$$x_2 - x_1 = 6 + 2\sqrt{5} - (6 - 2\sqrt{5}) = 6 + 2\sqrt{5} - 6 + 2\sqrt{5} = 4\sqrt{5} \quad (D)$$

Observación: Un razonamiento mucho más elegante sería aprovechar la identidad

$$(x_2 - x_1)^2 = (x_2 + x_1)^2 - 4x_2 \cdot x_1 = 12^2 - 4 \cdot 16 = 80 \Rightarrow x_2 - x_1 = \sqrt{80} = 4\sqrt{5}$$

Segunda versión.

Sean x_1, x_2 las componentes x de los puntos A y B. Por construcción de punto medio tendremos

$$\begin{cases} \frac{x_1 + x_2}{2} = 6 \\ \frac{\log_2 x_1 + \log_2 x_2}{2} = 2 \end{cases}$$

$$\frac{x_1 + x_2}{2} = 6 \Rightarrow x_1 + x_2 = 12$$

$$2 = \frac{\log_2 x_1 + \log_2 x_2}{2} \Rightarrow 4 = \log_2 x_1 + \log_2 x_2 = \log_2 (x_1 x_2) \Rightarrow x_1 x_2 = 2^4 = 16$$

Luego, aplicando las fórmulas de Vieta, x_1, x_2 serán las dos soluciones de la ecuación

$$x^2 - 12x + 16 = 0 \Leftrightarrow x = \frac{12 \pm \sqrt{12^2 - 4 \cdot 1 \cdot 16}}{2} = \frac{12 \pm \sqrt{80}}{2}$$

Y por tanto su diferencia será

$$\frac{12 + \sqrt{80}}{2} - \frac{12 - \sqrt{80}}{2} = \frac{12 + \sqrt{80} - (12 - \sqrt{80})}{2} = \frac{12 + \sqrt{80} - 12 + \sqrt{80}}{2} = \sqrt{80} = 4\sqrt{5} \quad (D).$$

7

Tenemos que las fechas se mostrarán de la forma 2023ABCD, y por tanto será necesario en el mes y en el año, al menos un 3, y un cero, y que los demás dígitos aparezcan en cantidad par. Veamos mes a mes:

20230113

20230131

20230223

20230311

20230322

Entre 202304CD y 202309CD no hay combinaciones aceptables.

20231013

20231031

20231103

20231130

202312CD no hay combinaciones aceptables.

En total hay 9 fechas aceptables (E).

8

Sea n el número de pruebas realizadas y S la suma de las puntuaciones de dichas pruebas. Las condiciones del enunciado se pueden escribir mediante las ecuaciones siguientes:

$$\frac{S+11}{n+1} = \frac{S}{n} + 1, \quad \frac{S+33}{n+3} = \frac{S}{n} + 2$$

Desarrollando la primera:

$$\frac{S+11}{n+1} = \frac{S}{n} + 1 \Leftrightarrow \frac{S+11}{n+1} = \frac{S+n}{n} \Leftrightarrow n(S+11) = (n+1)(S+n) \Leftrightarrow$$

$$nS + 11n = nS + n^2 + S + n \Leftrightarrow 10n = n^2 + S \Leftrightarrow S = 10n - n^2$$

Desarrollando la segunda:

$$\frac{S+33}{n+3} = \frac{S}{n} + 2 \Leftrightarrow \frac{S+33}{n+3} = \frac{S+2n}{n} \Leftrightarrow n(S+33) = (n+3)(S+2n) \Leftrightarrow$$

$$nS + 33n = nS + 2n^2 + 3S + 6n \Leftrightarrow 27n = 2n^2 + 3S$$

Luego

$$27n = 2n^2 + 3S = 2n^2 + 3(10n - n^2) = 2n^2 + 30n - 3n^2 = 30n - n^2 \Leftrightarrow$$

$$0 = 3n - n^2 = n(3 - n)$$

Y la única solución aceptable es $n = 3 \Rightarrow S = 10 \cdot 3 - 3^2 = 21 \Rightarrow \frac{S}{n} = 7$ (D).

9

Llamaremos y al lado corto y x al lado largo del triángulo. Queremos determinar y/x .

Aplicando Pitágoras, las condiciones del enunciado equivalen a

$$\begin{cases} (x+y)^2 = 3 \\ \left(\sqrt{x^2+y^2}\right)^2 = 2 \end{cases} \Leftrightarrow \begin{cases} (x+y)^2 = 3 \\ x^2 + y^2 = 2 \end{cases}$$

$$3 = (x+y)^2 = x^2 + y^2 + 2xy = 2 + 2xy \Rightarrow 1 = 2xy \Rightarrow \frac{1}{2x} = y$$

$$(x+y)^2 = 3 \Rightarrow \sqrt{3} = x+y = x + \frac{1}{2x} \Rightarrow 2\sqrt{3}x = 2x^2 + 1 \Rightarrow 2x^2 - 2\sqrt{3}x + 1 = 0 \Rightarrow$$

$$x = \frac{2\sqrt{3} \pm \sqrt{12 - 4 \cdot 2 \cdot 1}}{2 \cdot 2} = \frac{2\sqrt{3} \pm 2}{2 \cdot 2} = \frac{\sqrt{3} \pm 1}{2} \Rightarrow y = \frac{1}{\sqrt{3} \pm 1} \Rightarrow$$

$$\frac{y}{x} = \frac{1/(\sqrt{3} \pm 1)}{(\sqrt{3} \pm 1)/2} = \frac{2}{(\sqrt{3} \pm 1)^2} = \frac{2}{3 \pm 2\sqrt{3} + 1} = \frac{2}{4 \pm 2\sqrt{3}} = \frac{1}{2 \pm \sqrt{3}}$$

Veamos las dos opciones:

$$\frac{y}{x} = \frac{1}{2 + \sqrt{3}} = \frac{2 - \sqrt{3}}{4 - 3} = 2 - \sqrt{3}$$

$$\frac{y}{x} = \frac{1}{2 - \sqrt{3}} = \frac{2 + \sqrt{3}}{4 - 3} = 2 + \sqrt{3}$$

Puesto que $\sqrt{3} \cong 1.74$, solo la primera opción es compatible con $y < x$, y por tanto la solución es C.

$$10 (y - x)^2 = 4y^2 = (2y)^2 \Rightarrow |y - x| = |2y|, \text{ y puesto que } y > 0 \text{ tenemos}$$

$$2y = |y - x|$$

Veamos las dos posibilidades:

a) $2y = y - x \Rightarrow y = -x$ absurdo porque no pueden tener signo diferente.

$$b) 2y = x - y \Rightarrow 3y = x \Rightarrow y^3 = (3y)^2 = 9y^2 \xrightarrow{y>0} y = 9 \Rightarrow x = 27 \Rightarrow x + y = 36 \text{ (D).}$$

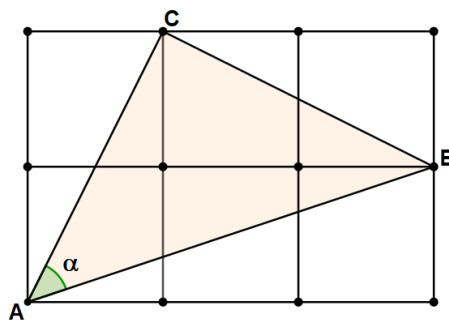
Observación: Una manera elegante de resolver este problema podría ser por diferencia de cuadrados:

$$4y^2 = (y - x)^2 \Leftrightarrow (2y)^2 - (y - x)^2 = 0 \Leftrightarrow (2y - y + x)(2y + y - x) = 0$$

$$\Leftrightarrow (y + x)(3y - x) = 0 \Leftrightarrow \begin{cases} y + x = 0 \Leftrightarrow y = -x \\ 3y - x = 0 \Leftrightarrow 3y = x \Rightarrow \dots \end{cases}$$

11

Primera versión. Aplicando ★39.



$$[\Delta ABC] = \frac{AB \cdot AC \cdot \sin \alpha}{2}$$

$$AB = \sqrt{1^2 + 3^2} = \sqrt{10}$$

$$AC = \sqrt{1^2 + 2^2} = \sqrt{5}$$

$$[\Delta ABC] = 6 - 1 - 1 - \frac{3}{2} = \frac{5}{2}$$

$$\frac{5}{2} = \frac{\sqrt{10} \cdot \sqrt{5} \cdot \sin \alpha}{2} \Rightarrow \sin \alpha = \frac{1}{\sqrt{2}} \Rightarrow \alpha = 45^\circ \quad (C)$$

Segunda versión.

La perpendicular de la recta con pendiente 2 será $-1/2$, que unirá el punto $(1,2)$ con el punto $(3,1)$, formando un triángulo $(0,0), (1,2), (3,1)$, rectángulo en $(1,2)$. Es también isósceles, luego los ángulos en los vértices $(0,0)$ y $(3,1)$ tienen que ser iguales, y por tanto tienen que ser iguales a 45° .

Tercera versión. Mediante trigonometría.

$$\tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x \tan y} = \frac{2 - 1/3}{1 + 2 \cdot 1/3} = \frac{5/3}{5/3} = 1 \Rightarrow x - y = 45^\circ$$

$$\tan x = 2$$

$$\tan y = \frac{1}{3}$$

12

Aplicamos la identidad notable $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$ cuando $b = a-1$ para obtener

$$a^3 - (a-1)^3 = (a-a+1)(a^2 + a^2(a-1) + (a-1)^2) = 3a(a-1) + 1$$

Realizamos estos cálculos de la manera más óptima posible:

$$2 \cdot 1 + 4 \cdot 3 + 6 \cdot 5 + 8 \cdot 7 + 10 \cdot 9 + 12 \cdot 11 + 14 \cdot 13 + 16 \cdot 15 + 18 \cdot 17 =$$

$$2 + 12 + 30 + 56 + 90 + 132 + 182 + 240 + 306 = 1050$$

$$1050 \cdot 3 + 9 = 3159 \quad (D).$$

Observación. Hay formas más elegantes de resolver esta suma, no necesariamente más rápidas. Por ejemplo,

$$\begin{aligned} \sum_{n=1}^9 (2n-1)(2n) &= \sum_{n=1}^9 4n^2 - 2n = 4 \sum_{n=1}^9 n^2 - 2 \sum_{n=1}^9 n = 4 \left(\frac{9(9+1)(18+1)}{6} \right) - 2 \left(\frac{9(9+1)}{2} \right) = \\ &= 2009 + 1140 + 90 = 3159 \end{aligned}$$

13

En primer lugar calculamos la proporción x de partidas ganadas por los diestros.

La ecuación que obtenemos es

$$x + x + 0.4x = 1 \Leftrightarrow 2.4x = 1 \Leftrightarrow x = \frac{1}{2.4} = \frac{5}{12}$$

Y por tanto la proporción de partidas ganadas por zurdos es $1 - \frac{5}{12} = \frac{7}{12}$.

Sea n el número de jugadores zurdos. Entonces el número de jugadores diestros será $2n$.

Los diestros juegan entre ellos $\frac{2n(2n-1)}{2} = n(2n-1)$ partidas.

Los zurdos juegan entre ellos $\frac{n(n-1)}{2}$ partidas

Entre un zurdo y un diestro se juegan $2n \cdot n = 2n^2$ partidas.

Luego el total de partidas es $\frac{3n(3n-1)}{2}$

La proporción de partidas que se juegan entre dos diestros (y que por tanto siempre ganará un diestro) será

$$\frac{\frac{n(2n-1)}{2}}{3n(3n-1)} = \frac{2(2n-1)}{3(3n-1)} = \frac{4n-2}{9n-3} = \frac{4}{9} - \frac{2}{3(9n-3)}$$

Esta función es estrictamente creciente, y vemos que prácticamente siempre será mayor que $5/12$, luego la condición del enunciado no será posible para cualquier número n a partir de un cierto número inicial que pasamos a calcular:

$$\frac{4n-2}{9n-3} = \frac{5}{12} \Leftrightarrow n = 3$$

Luego $n \leq 3$.

Luego el total de partidas puede ser:

$$n = 1 \rightarrow \frac{3 \cdot 1(3 \cdot 1 - 1)}{2} = 3 \text{ partidas.}$$

$$n = 2 \rightarrow \frac{3 \cdot 2(3 \cdot 2 - 1)}{2} = 15 \text{ partidas.}$$

$$n = 3 \rightarrow \frac{3 \cdot 3(3 \cdot 3 - 1)}{2} = 36 \text{ partidas.}$$

Puesto que las proporciones ganadas por diestros y zurdos tienen denominador 12, está claro que el número de partidas debe ser múltiplo de 12, con lo que solo nos queda válida la solución 36 (B).

14

En primer lugar vemos que $z = 0$ satisface la condición del enunciado. Supongamos que $z \neq 0$. Entonces $|z| = 1$. Se puede demostrar de varias maneras, por ejemplo:

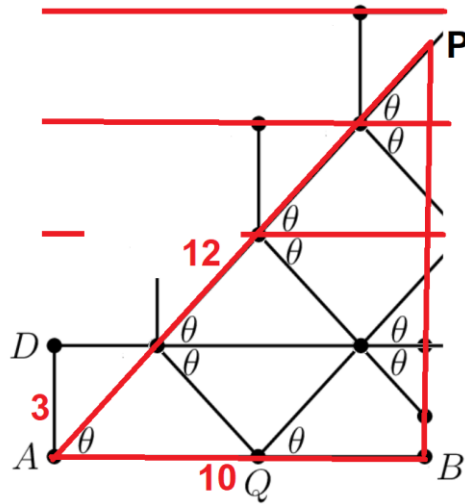
$$z^5 = \bar{z} \Rightarrow |z^5| = |\bar{z}| = |z| \Rightarrow |z|^5 = |\bar{z}| = |z| \stackrel{z \neq 0}{\Rightarrow} |z|^4 = 1 \Rightarrow |z| = 1$$

Luego $z^5 = \bar{z} \Rightarrow z^5 z = \bar{z} z = |z|^2 = 1 \Rightarrow z^6 = 1$, es decir, son las seis raíces complejas de 1. En total tenemos $6+1=7$ soluciones posibles.

15

Si colocamos segmentos 10×3 y nos imaginamos que la trayectoria es en línea recta, sin rebotar, vemos que generará un triángulo rectángulo de hipotenusa 12 y base 10. Si recortamos esta imagen y doblamos estos segmentos observaremos que aparecerá la línea en ziga-zaga con la longitud deseada.

El ángulo será $\theta = \arccos \frac{10}{12} = \arccos \frac{5}{6}$



16

La clave para resolver este problema es escribir la condición del enunciado como

$$\left| \left(z + \frac{1}{2} \right)^2 + \frac{3}{4} \right| = 4$$

Y hacer una interpretación geométrica de esta condición.

Sea $u = \left(z + \frac{1}{2} \right)^2$. El conjunto $\left| u + \frac{3}{4} \right| = 4$ es una circunferencia ω de radio 4 centrada en $\left(-\frac{3}{4}, 0 \right)$.

Sea $v = z + \frac{1}{2}$. Tenemos, por lo tanto, una transformación

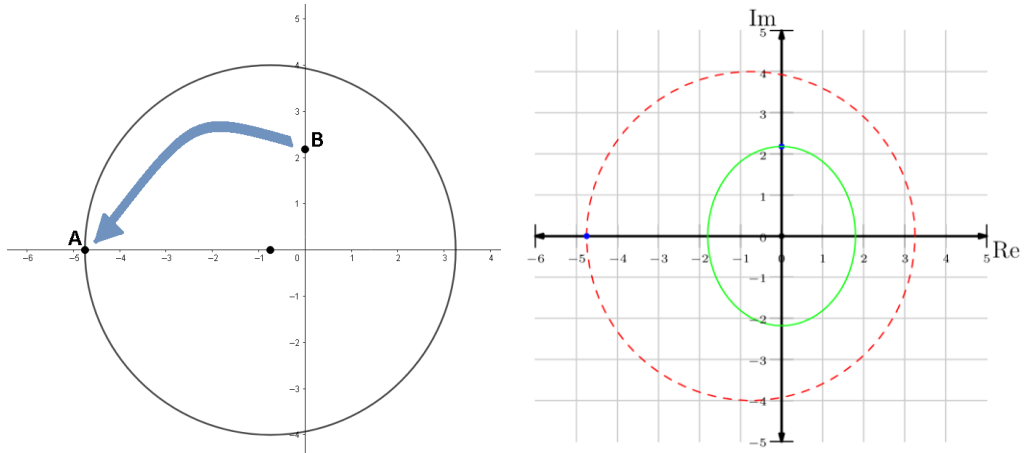
$$f : C \rightarrow C$$

$$v \mapsto v^2$$

Cuya imagen es la circunferencia ω , y queremos saber el valor máximo de la parte imaginaria de v .

El valor con mayor argumento de ω es el punto $A = \frac{-3}{4} - 4 + 0i = \frac{-19}{4} + 0i$, que es imagen del

punto $B = 0 + \frac{\sqrt{19}}{2}i$ mediante la transformación f . Y este punto es el que tiene mayor parte imaginaria.



Finalmente, la transformación

$$C \rightarrow C$$

$$v \mapsto v + \frac{3}{4}$$

No cambia la parte imaginaria del número, por lo tanto el valor máximo de la parte imaginaria continua siendo $\frac{\sqrt{19}}{2}$, y la respuesta correcta del problema es $19 + 2 = 21$ (B).

Fuente de esta solución: https://artofproblemsolving.com/wiki/index.php/2023_AMC_12A_Problems/Problem_16

17

La probabilidad de pasar por la posición 1 es $1/2$, pues es la probabilidad de saltar $0 \rightarrow 1$, que es

$$\frac{1}{2^1} = \frac{1}{2}.$$

Podemos determinar la probabilidad de pasar por la posición 2 mediante el lenguaje de la probabilidad compuesta:

$$P(2) = P(1 \rightarrow 2) \cdot P(1) + P(0 \rightarrow 2) \cdot P(0) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{4} \cdot 1 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Para la posición 3:

$$\begin{aligned} P(3) &= P(2 \rightarrow 3) \cdot P(2) + P(1 \rightarrow 3) \cdot P(1) + P(0 \rightarrow 3) \cdot P(0) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{4} \cdot \frac{1}{2} + \frac{1}{8} \cdot 1 = \\ &= \frac{1}{4} + \frac{1}{8} + \frac{1}{8} = \frac{2+1+1}{8} = \frac{1}{2} \end{aligned}$$

Observamos que podemos especular que $P(n) = \frac{1}{2}$ para toda posición n .

Esto lo podemos demostrar por inducción. Los primeros casos $P(1) = P(2) = P(3) = \frac{1}{2}$ los acabamos de demostrar.

Supongamos que es cierto hasta $n-1$: $P(n-1) = \frac{1}{2}$, y vamos a demostrar que es cierto para n :

$$\begin{aligned} P(n) &= P(0 \rightarrow n) \cdot P(0) + P(1 \rightarrow n) \cdot P(1) + P(2 \rightarrow n) \cdot P(2) + \dots + P(n-1 \rightarrow n) \cdot P(n-1) = \\ &= \frac{1}{2^n} \cdot 1 + \frac{1}{2^{n-1}} \cdot \frac{1}{2} + \frac{1}{2^{n-2}} \cdot \frac{1}{2} + \dots + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2^n} + \frac{1}{2^n} + \frac{1}{2^{n-1}} + \dots + \frac{1}{2^2} = (*) \end{aligned}$$

La suma $\frac{1}{2^n} + \frac{1}{2^{n-1}} + \dots + \frac{1}{2^2}$ es parte de una suma geométrica, cuya suma es una identidad notable (ver última página del [libro de Problemas de Álgebra](#)).

$$\begin{aligned} \frac{1}{2^n} + \frac{1}{2^{n-1}} + \dots + \frac{1}{2^2} &= \frac{1}{2^n} + \frac{1}{2^{n-1}} + \dots + \frac{1}{2^2} + \frac{1}{2^1} + \frac{1}{2^0} - \frac{1}{2} - \frac{1}{2^0} = \\ &= \sum_{k=0}^n \left(\frac{1}{2}\right)^k - \frac{1}{2} - 1 = \frac{(1/2)^{n+1} - 1}{1/2 - 1} - \frac{1}{2} - 1 = -2((1/2)^{n+1} - 1) - \frac{1}{2} - 1 = \\ &= -\left(\frac{1}{2}\right)^n + 2 - \frac{1}{2} - 1 = -\left(\frac{1}{2}\right)^n + \frac{1}{2} \end{aligned}$$

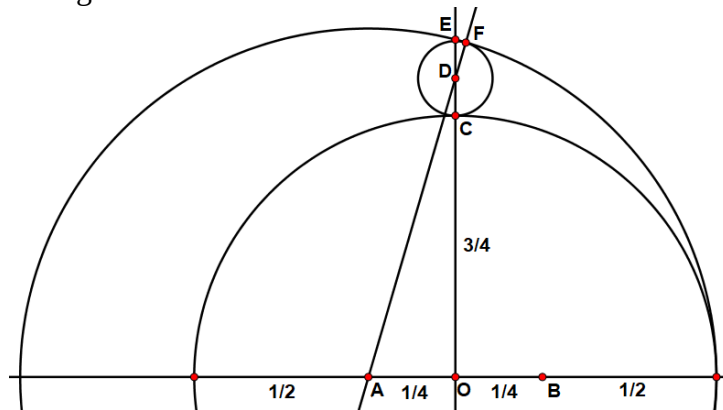
Así pues,

$$(*) = \frac{1}{2^n} - \frac{1}{2^n} + \frac{1}{2} = \frac{1}{2}$$

tal y como queríamos ver.

18

Sean A y B los centros respectivos de C_1 y C_2 . Está claro que C_3 tiene por centro O el punto medio del segmento AB y que su radio es $3/4$. Trazamos la perpendicular a AB por O y sea C su punto de corte con C_3 , D el centro de C_4 y E su punto de corte con $C_1 \cap C_2$, tal y como se muestra en el esquema siguiente:



Sea F el punto de tangencia entre C_4 y C_1 . Por las propiedades básicas de tangencia sabemos que los puntos A, D y F están alineados.

Claramente el radio de C_3 es $OC=3/4$. Sea r el radio de C_4 .

Entonces $AD=AF-DF=1-r$ y $OD=OC+CD=3/4+r$, y aplicando Pitágoras obtenemos la ecuación

$$\begin{aligned} AD^2 &= AO^2 + OD^2 \Leftrightarrow (1-r)^2 = \left(\frac{1}{4}\right)^2 + \left(\frac{3}{4}+r\right)^2 \Leftrightarrow \\ 1-2r+r^2 &= \frac{1}{16} + \frac{9}{16} + \frac{3}{2}r + r^2 \Leftrightarrow 1-2r = \frac{5}{8} + \frac{3}{2}r \Leftrightarrow 1-\frac{5}{8} = \frac{3}{2}r + 2r \Leftrightarrow \\ \Leftrightarrow \frac{3}{8} &= \frac{7}{2}r \Leftrightarrow r = \frac{3}{28} \end{aligned}$$

19

Aplicamos la identidad del cambio de base: $\log_a b = \frac{\log_c a}{\log_c b}$

$$\log_{7x} 2023 \cdot \log_{289x} 2023 = \log_{2023x} 2023 \Leftrightarrow$$

$$\frac{\log_7 2023}{\log_7 7x} \cdot \frac{\log_7 2023}{\log_7 289x} = \frac{\log_7 2023}{\log_7 2023x} \Leftrightarrow$$

$$\log_7 2023x \cdot \log_7 2023 = \log_7 7x \cdot \log_7 289x \Leftrightarrow$$

$$(\log_7 2023 + \log_7 x) \cdot \log_7 2023 = (\log_7 7 + \log_7 x) \cdot (\log_7 289 + \log_7 x)$$

Haciendo un cambio de variable $u = \log_7 x$ la ecuación anterior se convierte en

$$(\log_7 2023 + u) \cdot \log_7 2023 = (1 + u) \cdot (\log_7 289 + u)$$

Que es una ecuación de segundo grado en u:

$$(\log_7 2023)^2 + u \log_7 2023 = \log_7 289 + u + \log_7 289u + u^2 \Leftrightarrow$$

$$(\log_7 2023)^2 + (\log_7 2023 - 1 - \log_7 289)u - \log_7 289 - u^2 = 0$$

Esta ecuación tendrá dos soluciones en u que se convertirán en dos soluciones en x, luego su producto será

$$u = \log_7 x \Leftrightarrow x = 7^u \Rightarrow x_1 \cdot x_2 = 7^{u_1} \cdot 7^{u_2} = 7^{u_1+u_2}$$

Y aplicando las fórmulas de Vieta para la ecuación de segundo grado tenemos que

$$u_1 + u_2 = \log_7 2023 - 1 - \log_7 289 = \log_7 2023 - \log_7 289 - 1 = \log_7 \frac{2023}{289} - 1 =$$

$$= \log_7 7 - 1 = 1 - 1 = 0$$

Y por tanto, finalmente, $x_1 \cdot x_2 = 7^{u_1+u_2} = 7^0 = 1$ (C)

20

Estudiando como se construye esta pirámide de números es fácil ver que la suma S_n de los números de la fila n, es igual a

$$S_n = 2S_{n-1} + n$$

Luego S_n será una función cercana a una potencia de 2. En efecto,

$$S_n = 2^n - n$$

Para determinar la cifra de las unidades de S_{2023} vemos que las potencias de 2 van haciendo un ciclo de 4 valores:

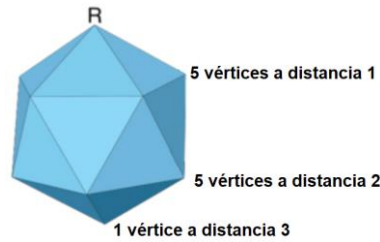
$$2^0 = 1, 2^1 = 2, 2^2 = 4, 2^3 = 8, 2^4 = 16, 2^5 = 32, 2^6 = 64, \dots$$

$$2023 = 4 \cdot 500 + 3$$

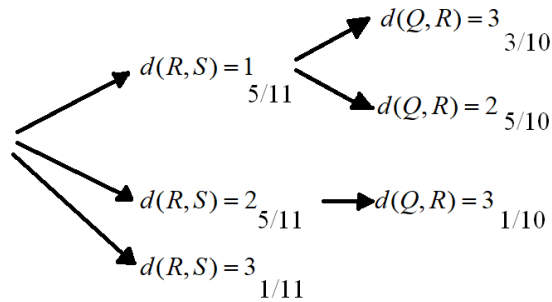
Luego 2^{2023} acaba igual que 2^3 , en ocho, y por tanto S_{2023} acaba igual que $8-3$, es decir, en 5 (C).

21

Un icosaedro tiene 12 vértices. Fijamos el vértice R, luego quedan 11 vértices para elegir los otros dos vértices, Q y S. Está claro que siempre $1 \leq d(A, B) \leq 3$.



Podemos resolver este problema fácilmente mediante probabilidad condicionada:



El caso $d(R, S) = 3$ ya no se contempla, pues entonces el vértice Q ya no podría estar más lejos.

Luego, aplicando la fórmula de la probabilidad total:

$$P(d(Q, R) > d(R, S)) = \frac{5}{11} \cdot \frac{1}{10} + \frac{5}{11} \cdot \frac{5}{10} + \frac{5}{11} \cdot \frac{1}{10} = \frac{7}{22} \quad (A)$$

22

Sea $x = f(2023)$

Sabemos que $2023 = 7 \cdot 17^2$, luego el enunciado nos dice que

$$1 \cdot x + 7 \cdot f\left(\frac{7 \cdot 17^2}{7}\right) + 17 \cdot f\left(\frac{7 \cdot 17^2}{17}\right) + 17^2 \cdot f\left(\frac{7 \cdot 17^2}{17^2}\right) + 7 \cdot 17 \cdot f\left(\frac{7 \cdot 17^2}{7 \cdot 17}\right) + 2023 f(1) = 1$$

Aplicando la identidad del enunciado al caso $n = 1$

$$1 \cdot f(1) = 1 \Rightarrow f(1) = 1$$

Luego

$$x + 7 \cdot f(17^2) + 17 \cdot f(7 \cdot 17) + 17^2 \cdot f(7) + 7 \cdot 17 \cdot f(17) + 2023 = 1$$

Aplicando la identidad del enunciado al caso $n = 17^2$

$$\begin{aligned} 1 \cdot f\left(\frac{17^2}{1}\right) + 17 \cdot f\left(\frac{17^2}{17}\right) + 17^2 \cdot f\left(\frac{17^2}{17^2}\right) &= 1 \Leftrightarrow f(17^2) + 17 \cdot f(17) + 17^2 = 1 \\ \Leftrightarrow 7 \cdot f(17^2) + 7 \cdot 17 \cdot f(17) + 7 \cdot 17^2 &= 7 \end{aligned}$$

Substituyendo en la ecuación anterior:

$$x + 17 \cdot f(7 \cdot 17) + 17^2 \cdot f(7) + 7 - 7 \cdot 17^2 + 2023 = 1$$

Aplicando la identidad del enunciado al caso $n = 7 \cdot 17$

$$\begin{aligned} 1 \cdot f\left(\frac{7 \cdot 17}{1}\right) + 17 \cdot f\left(\frac{7 \cdot 17}{17}\right) + 7 \cdot f\left(\frac{7 \cdot 17}{7}\right) + 7 \cdot 17 \cdot f\left(\frac{7 \cdot 17}{7 \cdot 17}\right) &= 1 \Leftrightarrow \\ 1 \cdot f(7 \cdot 17) + 17 \cdot f(7) + 7 \cdot f(17) + 7 \cdot 17 &= 1 \Leftrightarrow \\ 17 \cdot f(7 \cdot 17) + 17^2 \cdot f(7) &= 17 - 7 \cdot 7 \cdot f(17) - 7 \cdot 17^2 \end{aligned}$$

Substituyendo en la ecuación anterior:

$$x + 17 - 17 \cdot 7 \cdot f(17) - 7 \cdot 17^2 + 7 - 7 \cdot 17^2 + 2023 = 1$$

Aplicando la identidad del enunciado al caso $n = 17$

$$1 \cdot f\left(\frac{17}{1}\right) + 17 \cdot f\left(\frac{17}{17}\right) = 1 \Leftrightarrow 1 \cdot f(17) + 17 = 1 \Leftrightarrow f(17) = 1 - 17 = -16$$

Substituyendo en la ecuación anterior:

$$x + 17 - 17 \cdot 7 \cdot (-16) - 7 \cdot 17^2 + 7 - 7 \cdot 17^2 + 2023 = 1 \Rightarrow$$

$$x = 1 - 17 + 17 \cdot 7 \cdot (-16) + 7 \cdot 17^2 - 7 + 7 \cdot 17^2 - 2023 = 96$$

23

Aplicamos la desigualdad AM-GM en cada uno de los factores de la izquierda:

$$\left. \begin{array}{l} 1 + 2a \geq 2\sqrt{2a} \\ 2 + 2b \geq 2\sqrt{4b} \\ 2a + b \geq 2\sqrt{4b} \end{array} \right\} \Rightarrow (1 + 2a)(2 + 2b)(2a + b) \geq 2\sqrt{2a} \cdot 2\sqrt{4b} \cdot 2\sqrt{4b} = 32ab$$

Y la igualdad solo acontece cuando

$$\left. \begin{array}{l} 1 = 2a \\ 2 = 2b \\ 2a = b \end{array} \right\} \Leftrightarrow a = \frac{1}{2}, b = 1$$

Y por tanto hemos demostrado que existe una única solución aceptable.

Fuente de esta solución:

24

Fijamos un n , y consideremos una sucesión $A_1 \subset A_2 \subset A_3 \subset \dots \subset A_n$.

Cada uno de los diez números del conjunto $\{1, 2, 3, \dots, 10\}$ puede ser colocado por primera vez en una de las $n + 1$ siguientes posiciones: En ningún A_i , en A_1 , en A_2 , ..., o en A_n . Luego hay $(n + 1)^{10}$ maneras de colocarlos.

Así pues, $K = \sum_{n=1}^{10} (n + 1)^{10} = \sum_{n=2}^{11} n^{10} = \sum_{n=1}^{11} n^{10} - 1$, y queremos calcular $K \pmod{10}$

En primer lugar calculamos $K \pmod{5}$. Aquí aplicaremos el PTF: $n^5 \equiv n \pmod{5}$, por lo que $n^{10} = (n^5)^2 \equiv n^2 \pmod{5}$

Y por tanto

$$K = \sum_{n=1}^{11} n^{10} - 1 \equiv \sum_{n=1}^{11} n^2 - 1 \equiv \frac{11 \cdot 12 \cdot 23}{6} - 1 = 11 \cdot 2 \cdot 23 - 1 = 505 \equiv 0 \pmod{5}$$

Ahora calculamos $K \pmod{2}$. Aquí aplicaremos que solo nos interesa la paridad:

$K = \sum_{n=2}^{11} n^{10} = 2^{10} + 3^{10} + 4^{10} + \dots + 11^{10}$, hay 5 pares y 5 impares, que sumados generan un impar,

luego es impar, y por tanto $K \equiv 1 \pmod{2}$

Finalmente, aplicando el Teorema chino del Residuo (o por simplemente por observación)

$$\left. \begin{array}{l} K \equiv 0 \pmod{5} \\ K \equiv 1 \pmod{2} \end{array} \right\} \Rightarrow K \equiv 5 \pmod{10}$$

25

Sea $t = \tan x$. Aplicamos la identidad trigonométrica de la tangente de la suma:

$$\tan \alpha + \beta = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

Luego

$$T_2 = \tan 2x = \frac{\tan x + \tan x}{1 - \tan x \tan x} = \frac{2t}{1 - t^2}$$

y en general

$$T_n = \tan nx = \tan((n-1)x + x) = \frac{\tan(n-1)x + \tan x}{1 - \tan(n-1)x \tan x} = \frac{T_{n-1} + t}{1 - T_{n-1} \cdot t}$$

Así pues, tenemos la siguiente sucesión recursiva:

$$T_1 = t$$

$$T_2 = \frac{2t}{1 - t^2}$$

$$T_n = \frac{T_{n-1} + t}{1 - T_{n-1} \cdot t}$$

Si vamos desarrollando esta fórmula con paciencia podemos ir sacando los sucesivos valores:

$$T_3 = \frac{-t^3 + 3t}{1 - 3t^2}, T_4 = \frac{-4t^3 + 4t}{1 - 6t^2 + t^4}, T_5 = \frac{t^5 - 10t^3 + 5t}{1 - 10t^2 + 5t^4}, T_7 = \frac{7t - 35t^3 + 21t^5 - t^7}{1 - 21t^2 + 35t^4 - 7t^6}, \dots$$

Y podemos ver que en las posiciones impares el coeficiente a_n buscado va alternando entre 1 y -1:

$$a_1 \rightarrow 1, a_3 \rightarrow -1, a_5 \rightarrow 1, a_7 \rightarrow -1 \dots$$

Luego para la posición 2023 será $2023 = 2 \cdot 1011 + 1$, y puesto que 1011 es impar, tendremos

$$a_{2023} = -1 \quad (C)$$

Para una solución completa de este problema, en la página de AoPs correspondiente vemos que este problema en realidad se deduce de la siguiente fórmula:

$$\tan nx = \frac{\binom{n}{1} \tan x - \binom{n}{3} \tan^3 x + \binom{n}{5} \tan^5 x - \binom{n}{7} \tan^7 x + \dots}{1 - \binom{n}{2} \tan^2 x + \binom{n}{4} \tan^4 x - \binom{n}{6} \tan^6 x + \dots}$$

Esta fórmula se puede demostrar utilizando números complejos:

$$\begin{aligned}
\cos 2023x + i \sin 2023x &= (\cos x + i \sin x)^{2023} = \\
&= \cos^{2023} x + \binom{2023}{1} \cos^{2022} x (i \sin x) + \binom{2023}{2} \cos^{2021} x (i \sin x)^2 + \\
&+ \binom{2023}{3} \cos^{2020} x (i \sin x)^3 + \dots + \binom{2022}{2023} \cos x (i \sin x)^{2022} + (i \sin x)^{2023} = \\
&= \cos^{2023} x + i \binom{2023}{1} \cos^{2022} x \sin x - \binom{2023}{2} \cos^{2021} x \sin^2 x - \\
&- i \binom{2023}{3} \cos^{2020} x \sin^3 x + \dots - \binom{2022}{2023} \cos x \sin^{2022} x - i \sin^{2023} x
\end{aligned}$$

Separamos la parte real y la parte imaginaria:

$$\begin{aligned}
\cos 2023x &= \cos^{2023} x - \binom{2023}{2} \cos^{2021} x \sin^2 x + \dots - \binom{2023}{2022} \cos x \sin^{2022} x \\
\sin 2023x &= \binom{2023}{1} \cos^{2022} x \sin x - \binom{2023}{3} \cos^{2020} x \sin^3 x + \dots - \sin^{2023} x
\end{aligned}$$

Y por tanto:

$$\begin{aligned}
\tan 2023x &= \frac{\sin 2023x}{\cos 2023x} = \\
&= \frac{\binom{2023}{1} \cos^{2022} x \sin x - \binom{2023}{3} \cos^{2020} x \sin^3 x + \dots - \sin^{2023} x}{\cos^{2023} x - \binom{2023}{2} \cos^{2021} x \sin^2 x + \dots - \binom{2023}{2022} \cos x \sin^{2022} x} = \\
&= \frac{\binom{2023}{1} \frac{\cos^{2022} x \sin x}{\cos^{2023} x} - \binom{2023}{3} \frac{\cos^{2020} x \sin^3 x}{\cos^{2023} x} + \dots - \frac{\sin^{2023} x}{\cos^{2023} x}}{\frac{\cos^{2023} x}{\cos^{2023} x} - \binom{2023}{2} \frac{\cos^{2021} x \sin^2 x}{\cos^{2023} x} + \dots - \binom{2023}{2022} \frac{\cos x \sin^{2022} x}{\cos^{2023} x}} = \\
&= \frac{\binom{2023}{1} \tan x - \binom{2023}{3} \tan^3 x + \dots - \tan^{2023} x}{1 - \binom{2023}{2} \tan^2 x + \dots - \binom{2023}{2022} \tan^{2022} x}
\end{aligned}$$

Fuente: https://artofproblemsolving.com/wiki/index.php?title=2023_AMC_12A_Problems/Problem_25&oldid=202210

Enlaces a los libros Toomates Coolección:

1	http://www.toomates.net/biblioteca/ProblemasFunciones.pdf	#1.2.3
2	http://www.toomates.net/biblioteca/ProblemasAlgebra.pdf	#7.1.16
3	http://www.toomates.net/biblioteca/Aritmetica.pdf	#6.119
4	http://www.toomates.net/biblioteca/Aritmetica.pdf	#3.73
5	http://www.toomates.net/biblioteca/Probabilidad.pdf	#3.26
6	http://www.toomates.net/biblioteca/ProblemasAlgebra.pdf	#3.2.30
7	http://www.toomates.net/biblioteca/Combinatoria.pdf	#1.7.60
8	http://www.toomates.net/biblioteca/ProblemasAlgebra.pdf	#7.7.11
9	http://www.toomates.net/biblioteca/Geometria.pdf	#5.5.8
10	http://www.toomates.net/biblioteca/ProblemasAlgebra.pdf	#4.3.10
11	http://www.toomates.net/biblioteca/Geometria.pdf	#4.5.15
12	http://www.toomates.net/biblioteca/ProblemasAlgebra.pdf	#2.1.60
13	http://www.toomates.net/biblioteca/Combinatoria.pdf	#1.7.47
14	http://www.toomates.net/biblioteca/ProblemasAlgebra.pdf	#12.1.36
15	http://www.toomates.net/biblioteca/ProblemasTrigonometria.pdf	#2.20
16	http://www.toomates.net/biblioteca/ProblemasAlgebra.pdf	#12.3.9
17	http://www.toomates.net/biblioteca/Probabilidad.pdf	#3.25
18	http://www.toomates.net/biblioteca/Geometria.pdf	#6.4.10
19	http://www.toomates.net/biblioteca/ProblemasAlgebra.pdf	#3.2.31
20	http://www.toomates.net/biblioteca/Aritmetica.pdf	#7.6.29
21	http://www.toomates.net/biblioteca/Probabilidad.pdf	#3.27
22	http://www.toomates.net/biblioteca/Aritmetica.pdf	#6.121
23	http://www.toomates.net/biblioteca/Desigualdades.pdf	#2.1.24
24	http://www.toomates.net/biblioteca/Aritmetica.pdf	#13.1.17
25	http://www.toomates.net/biblioteca/ProblemasAlgebra.pdf	#12.4.2



2023 AMC 12B Problems

2023 AMC 12B (Answer Key)

Printable versions: • AoPS Resources (https://artofproblemsolving.com/community/contest/collection/c3415_amc_12/2023) • PDF (https://artofproblemsolving.com/community/contest/download/c3415_amc_12/2023)

Instructions

1. This is a 25-question, multiple choice test. Each question is followed by answers marked A, B, C, D and E. Only one of these is correct.
2. You will receive 6 points for each correct answer, 2.5 points for each problem left unanswered if the year is before 2006, 1.5 points for each problem left unanswered if the year is after 2006, and 0 points for each incorrect answer.
3. No aids are permitted other than scratch paper, graph paper, ruler, compass, protractor and erasers (and calculators that are accepted for use on the test if before 2006. No problems on the test will *require* the use of a calculator).
4. Figures are not necessarily drawn to scale.
5. You will have **75 minutes** working time to complete the test.

1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

Mrs. Jones is pouring orange juice into four identical glasses for her four sons. She fills the first three glasses completely but runs out of juice when the fourth glass is only $\frac{1}{3}$ full. What fraction of a glass must Mrs. Jones pour from each of the first three glasses into the fourth glass so that all four glasses will have the same amount of juice?

- (A) $\frac{1}{12}$ (B) $\frac{1}{4}$ (C) $\frac{1}{6}$ (D) $\frac{1}{8}$ (E) $\frac{2}{9}$

Solution

Problem 2

Carlos went to a sports store to buy running shoes. Running shoes were on sale, with prices reduced by 20% on every pair of shoes. Carlos also knew that he had to pay a 7.5% sales tax on the discounted price. He had 43 dollars. What is the original (before discount) price of the most expensive shoes he could afford to buy?

- (A) 46 (B) 50 (C) 48 (D) 47 (E) 49

Solution

Problem 3

A $3 - 4 - 5$ right triangle is inscribed in circle A , and a $5 - 12 - 13$ right triangle is inscribed in circle B . What is the ratio of the area of circle A to the area of circle B ?

- (A) $\frac{9}{25}$ (B) $\frac{1}{9}$ (C) $\frac{1}{5}$ (D) $\frac{25}{169}$ (E) $\frac{4}{25}$

Solution

Problem 4

Jackson's paintbrush makes a narrow strip with a width of 6.5 millimeters. Jackson has enough paint to make a strip 25 meters long. How many square centimeters of paper could Jackson cover with paint?

- (A) 162,500 (B) 162.5 (C) 1,625 (D) 1,625,000 (E) 16,250

Solution

Problem 5

You are playing a game. A 2×1 rectangle covers two adjacent squares (oriented either horizontally or vertically) of a 3×3 grid of squares, but you are not told which two squares are covered. Your goal is to find at least one square that is covered by the rectangle. A "turn" consists of you guessing a square, after which you are told whether that square is covered by the hidden rectangle. What is the minimum number of turns you need to ensure that at least one of your guessed squares is covered by the rectangle?

- (A) 3 (B) 5 (C) 4 (D) 8 (E) 6

Solution

Problem 6

When the roots of the polynomial.

$$P(x) = (x - 1)^1(x - 2)^2(x - 3)^3 \cdots (x - 10)^{10}$$

are removed from the number line, what remains is the union of 11 disjoint open intervals. On how many of these intervals is $P(x)$ positive?

- (A) 3 (B) 4 (C) 5 (D) 6 (E) 7

Solution

Problem 7

For how many integers n does the expression

$$\sqrt{\frac{\log(n^2) - (\log n)^2}{\log n - 3}}$$

represent a real number, where $\log$ denotes the base 10 logarithm?

- (A) 900 (B) 2 (C) 902 (D) 2 (E) 901

Solution

Problem 8

How many nonempty subsets B of $\{0, 1, 2, 3, \dots, 12\}$ have the property that the number of elements in B is equal to the least element of B ? For example, $B = \{4, 6, 8, 11\}$ satisfies the condition.

- (A) 256 (B) 136 (C) 108 (D) 144 (E) 156

Solution

Problem 9

What is the area of the region in the coordinate plane defined by

$$||x| - 1| + ||y| - 1| \leq 1?$$

- (A) 4 (B) 8 (C) 10 (D) 12 (E) 15

Solution

Problem 10

In the xy -plane, a circle of radius 4 with center on the positive x -axis is tangent to the y -axis at the origin, and a circle with radius 10 with center on the positive y -axis is tangent to the x -axis at the origin. What is the slope of the line passing through the two points at which these circles intersect?

- (A) $\frac{2}{7}$ (B) $\frac{3}{7}$ (C) $\frac{2}{\sqrt{29}}$ (D) $\frac{1}{\sqrt{29}}$ (E) $\frac{2}{5}$

Solution

Problem 11

What is the maximum area of an isosceles trapezoid that has legs of length 1 and one base twice as long as the other?

- (A) $\frac{5}{4}$ (B) $\frac{8}{7}$ (C) $\frac{5\sqrt{2}}{4}$ (D) $\frac{3}{2}$ (E) $\frac{3\sqrt{3}}{4}$

Solution

Problem 12

For complex numbers $u = a + bi$ and $v = c + di$, define the binary operation $\otimes$ by

$$u \otimes v = ac + bdi.$$

Suppose z is a complex number such that $z \otimes z = z^2 + 40$. What is $|z|$?

- (A) $\sqrt{10}$ (B) $3\sqrt{2}$ (C) $2\sqrt{6}$ (D) 6 (E) $5\sqrt{2}$

Solution

Problem 13

A rectangular box P has distinct edge lengths a , b , and c . The sum of the lengths of all 12 edges of P is 13, the sum of the areas of all 6 faces of P is $\frac{11}{2}$, and the volume of P is $\frac{1}{2}$. What is the length of the longest interior diagonal connecting two vertices of P ?

- (A) 2 (B) $\frac{3}{8}$ (C) $\frac{9}{8}$ (D) $\frac{9}{4}$ (E) $\frac{3}{2}$

Solution

Problem 14

For how many ordered pairs (a, b) of integers does the polynomial $x^3 + ax^2 + bx + 6$ have 3 distinct integer roots?

- (A) 5 (B) 6 (C) 8 (D) 7 (E) 4

Solution

Problem 15

Suppose a , b , and c are positive integers such that

$$\frac{a}{14} + \frac{b}{15} = \frac{c}{210}.$$

Which of the following statements are necessarily true?

I. If $\gcd(a, 14) = 1$ or $\gcd(b, 15) = 1$ or both, then $\gcd(c, 210) = 1$.

II. If $\gcd(c, 210) = 1$, then $\gcd(a, 14) = 1$ or $\gcd(b, 15) = 1$ or both.

III. $\gcd(c, 210) = 1$ if and only if $\gcd(a, 14) = \gcd(b, 15) = 1$.

- (A) I, II, and III (B) I only (C) I and II only (D) III only (E) II and III only

Solution

Problem 16

In Coinland, there are three types of coins, each worth 6, 10, and 15. What is the sum of the digits of the maximum amount of money that is impossible to have?

- (A) 8 (B) 10 (C) 7 (D) 11 (E) 9

Solution

Problem 17

Triangle ABC has side lengths in arithmetic progression, and the smallest side has length 6. If the triangle has an angle of 120° , what is the area of ABC ?

- (A) $12\sqrt{3}$ (B) $8\sqrt{6}$ (C) $14\sqrt{2}$ (D) $20\sqrt{2}$ (E) $15\sqrt{3}$

Solution

Problem 18

Last academic year Yolanda and Zelda took different courses that did not necessarily administer the same number of quizzes during each of the two semesters. Yolanda's average on all the quizzes she took during the first semester was 3 points higher than Zelda's average on all the quizzes she took during the first semester. Yolanda's average on all the quizzes she took during the second semester was 18 points higher than her average for the first semester and was again 3 points higher than Zelda's average on all the quizzes Zelda took during her second semester. Which one of the following statements cannot possibly be true?

- (A) Yolanda's quiz average for the academic year was 22 points higher than Zelda's.
(B) Zelda's quiz average for the academic year was higher than Yolanda's.
(C) Yolanda's quiz average for the academic year was 3 points higher than Zelda's.
(D) Zelda's quiz average for the academic year equaled Yolanda's.

(E) If Zelda had scored 3 points higher on each quiz she took, then she would have had the same average for the academic year as Yolanda.

Solution

Problem 19

Each of 2023 balls is placed in one of 3 bins. Which of the following is closest to the probability that each of the bins will contain an odd number of balls?

- (A) $\frac{2}{3}$ (B) $\frac{3}{10}$ (C) $\frac{1}{2}$ (D) $\frac{1}{3}$ (E) $\frac{1}{4}$

Solution

Problem 20

Cyrus the frog jumps 2 units in a direction, then 2 more in another direction. What is the probability that he lands less than 1 unit away from his starting position?

- (A) $\frac{1}{6}$ (B) $\frac{1}{5}$ (C) $\frac{\sqrt{3}}{8}$ (D) $\frac{\arctan \frac{1}{2}}{\pi}$ (E) $\frac{2 \arcsin \frac{1}{4}}{\pi}$

Solution

Problem 21

A lampshade is made in the form of the lateral surface of the frustum of a right circular cone. The height of the frustum is $3\sqrt{3}$ inches, its top diameter is 6 inches, and its bottom diameter is 12 inches. A bug is at the bottom of the lampshade and there is a glob of honey on the top edge of the lampshade at the spot farthest from the bug. The bug wants to crawl to the honey, but it must stay on the surface of the lampshade. What is the length in inches of its shortest path to the honey?

- (A) $6 + 3\pi$ (B) $6 + 6\pi$ (C) $6\sqrt{3}$ (D) $6\sqrt{5}$ (E) $6\sqrt{3} + \pi$

Solution

Problem 22

A real-valued function f has the property that for all real numbers a and b ,

$$f(a+b) + f(a-b) = 2f(a)f(b).$$

Which one of the following cannot be the value of $f(1)$?

- (A) 0 (B) 1 (C) -1 (D) 2 (E) -2

Solution

Problem 23

When n standard six-sided dice are rolled, the product of the numbers rolled can be any of 936 possible values. What is n ?

- (A) 6 (B) 8 (C) 9 (D) 10 (E) 11

Solution

Problem 24

Suppose that a , b , c and d are positive integers satisfying all of the following relations.

$$abcd = 2^6 \cdot 3^9 \cdot 5^7$$

$$\text{lcm}(a, b) = 2^3 \cdot 3^2 \cdot 5^3$$

$$\text{lcm}(a, c) = 2^3 \cdot 3^3 \cdot 5^3$$

$$\text{lcm}(a, d) = 2^3 \cdot 3^3 \cdot 5^3$$

$$\text{lcm}(b, c) = 2^1 \cdot 3^3 \cdot 5^2$$

$$\text{lcm}(b, d) = 2^2 \cdot 3^3 \cdot 5^2$$

$$\text{lcm}(c, d) = 2^2 \cdot 3^3 \cdot 5^2$$

What is $\text{gcd}(a, b, c, d)$?

- (A) 30 (B) 45 (C) 3 (D) 15 (E) 6

Solution

Problem 25

A regular pentagon with area $\sqrt{5} + 1$ is printed on paper and cut out. The five vertices of the pentagon are folded into the center of the pentagon, creating a smaller pentagon. What is the area of the new pentagon?

- (A) $4 - \sqrt{5}$ (B) $\sqrt{5} - 1$ (C) $8 - 3\sqrt{5}$ (D) $\frac{\sqrt{5} + 1}{2}$ (E) $\frac{2 + \sqrt{5}}{3}$

Solution

See also

2023 AMC 12B (Problems • Answer Key • Resources) (http://www.artofproblemsolving.com/community/c1333)	
Preceded by 2023 AMC 12A Problems	Followed by 2024 AMC 12A Problems
1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
All AMC 12 Problems and Solutions	

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AMC 12B 2023 Enunciados

1

La señora Jones está sirviendo zumo de naranja en cuatro vasos idénticos para sus cuatro hijos. Llena los tres primeros vasos completamente pero al llenar el cuarto se queda sin zumo cuando había llenado $\frac{1}{3}$ del vaso. Determina la fracción de cada uno de los primeros vasos que la señora Jones debe pasar al cuarto vaso para que los cuatro vasos tengan la misma cantidad de zumo.

(A) $\frac{1}{12}$ (B) $\frac{1}{4}$ (C) $\frac{1}{6}$ (D) $\frac{1}{8}$ (E) $\frac{2}{9}$

2

Carlos fue a una tienda de deportes a comprarse unas zapatillas de correr. Están de oferta, con un descuento del 20%, pero debe pagar un impuesto del 7.5% sobre el precio reducido. Tiene 43 dólares. Determina el precio (antes del descuento) de las zapatillas más caras que puede comprar.

(A) 46 (B) 50 (C) 48 (D) 47 (E) 49

3

Un triángulo rectángulo 3-4-5 está inscrito en una circunferencia A, y un triángulo rectángulo 5-12-13 está inscrito en una circunferencia B. Determina la razón entre el área del círculo A y el área del círculo B.

(A) $\frac{9}{25}$ (B) $\frac{1}{9}$ (C) $\frac{1}{5}$ (D) $\frac{25}{169}$ (E) $\frac{4}{25}$

4

Jackson tiene un pincel que pinta una franja de 6.5 milímetros de ancho y tiene suficiente pintura para pintar una tira de 25 metros de largo. Cuantos centímetros cuadrados puede pintar?

(A) 162500 (B) 162.5 (C) 1625 (D) 1625000 (E) 16250

5

Sabemos que un rectángulo 2×1 recubre dos casillas adyacentes, en vertical o en horizontal, de una cuadrícula 3×3 , pero no nos dicen donde está situado. El objetivo del juego es adivinar al menos una de las dos casillas cubiertas por este rectángulo. Cada “turno” consiste en señalar una casilla, y se te informa de si está o no ocupada. Determina el mínimo número de turnos que necesitas para garantizar adivinar al menos una de las dos posiciones ocupadas por el rectángulo.

(A) 3 (B) 5 (C) 4 (D) 8 (E) 6

6

Cuando las raíces del polinomio $P(x) = (x-1)^1(x-2)^2(x-3)^3 \dots (x-10)^{10}$ son eliminadas de la recta real obtenemos la unión de 11 intervalos abiertos disjuntos. ¿En cuántos de estos intervalos $P(x)$ es positivo?

(A) 3 (B) 4 (C) 5 (D) 6 (E) 7

7

Determina la cantidad de enteros n para los que la expresión

$$\sqrt{\frac{\log(n^2) - (\log n)^2}{\log n - 3}}$$

Representa un número real, donde $\log$ denota el logaritmo en base 10.

(A) 900 (B) 2 (C) 902 (D) 2 (E) 901

8

Determina el número de subconjuntos B no vacíos de $\{0,1,2,3,\dots,12\}$ tales que su número de elementos sea igual al menor elemento de B . Por ejemplo $\{4,6,8,11\}$ satisface esta condición.

(A) 256 (B) 136 (C) 108 (D) 144 (E) 156

9

Determina el área de la región en el plano coordenado definida por

$$||x|-1| + ||y|-1| \leq 1$$

(A) 4 (B) 8 (C) 10 (D) 12 (E) 15

10

En el plano cartesiano trazamos una circunferencia de radio 4, centro en el eje X positivo y tangente al eje Y en el origen, y una circunferencia de radio 10, con centro en el eje Y positivo y tangente al eje X en el origen. Determina la pendiente de la recta que pasa por los dos puntos de corte de estas dos circunferencias.

(A) $2/7$ (B) $3/7$ (C) $2/\sqrt{29}$ (D) $1/\sqrt{29}$ (E) $2/5$

11

Determina el área máxima de un trapecio isósceles con lados diagonales de longitud 1 y una de las bases de longitud el doble que la otra.

(A) $5/4$ (B) $8/7$ (C) $5\sqrt{2}/4$ (D) $3/2$ (E) $3\sqrt{3}/4$

12

Dados dos números complejos $u = a + bi$ y $v = c + di$, definimos la operación binaria $\otimes$ por

$$u \otimes v = ac + bdi.$$

Supongamos que z es un número complejo tal que $z \otimes z = z^2 + 40$. Determina $|z|$.

(A) $\sqrt{10}$ (B) $3\sqrt{2}$ (C) $2\sqrt{6}$ (D) 6 (E) $5\sqrt{2}$

13

Una caja rectangular P tiene aristas diferentes a , b y c . La suma de las longitudes de las aristas de P es 13, la suma de las áreas de las 6 caras es $11/2$, y el volumen de P es $1/2$. Determina la longitud de la diagonal interior más larga entre dos vértices de P.

(A) 2 (B) $3/8$ (C) $9/8$ (D) $9/4$ (E) $3/2$

14

Determina la cantidad de pares ordenados (a, b) de enteros para los cuales el polinomio $x^3 + ax^2 + bx + 6$ tiene 3 soluciones enteras distintas.

(A) 5 (B) 6 (C) 8 (D) 7 (E) 4

15

Supongamos que a , b y c son enteros positivos tales que

$$\frac{a}{14} + \frac{b}{15} = \frac{c}{210}$$

¿Cuál de las siguientes afirmaciones es necesariamente cierta?

I. Si $Mcd(a, 14) = 1$ o $Mcd(b, 15) = 1$ o ambos, entonces $Mcd(c, 210) = 1$

II. Si $Mcd(c, 210) = 1$, entonces $Mcd(a, 14) = 1$ o $Mcd(b, 15) = 1$ o ambos.

III. $Mcd(c, 210) = 1$ si y solo si $Mcd(a, 14) = Mcd(b, 15) = 1$.

(A) I, II y III (B) I solamente (C) I y II solamente (D) III solamente (E) II y III solamente

16

En un país llamado Coinland hay tres tipos de moneda, de 6, de 10 y de 15. Determina la suma de los dígitos del mayor valor que es imposible obtener con estas monedas.

(A) 8 (B) 10 (C) 7 (D) 11 (E) 9

17

Sea $\triangle ABC$ un triángulo cuyos lados están en progresión aritmética, y su lado menor tiene longitud 6. Determina el área de este triángulo si sabemos que uno de sus ángulos mide 120° .

(A) $12\sqrt{3}$ (B) $8\sqrt{6}$ (C) $14\sqrt{2}$ (D) $20\sqrt{2}$ (E) $15\sqrt{3}$

18

En el año pasado Yolanda y Zelda tomaron cursos diferentes en los que no realizaron necesariamente el mismo número de exámenes. La media de los exámenes que realizó Yolanda durante el primer semestre fue 3 puntos superior a la media de los exámenes que realizó Zelda en el primer trimestre. La media de las pruebas que realizó Yolanda durante el segundo semestre fue 18 puntos superior a la media de las pruebas que realizó ella misma durante el primer semestre, y fue 3 puntos superior a la media de las pruebas que realizó Zelda durante el segundo trimestre. Determina, entre las siguientes opciones, aquella que seguro no puede ser cierta:

- (A) La media anual de Yolanda fue 22 puntos superior a la media anual de Zelda.
- (B) La media anual de Zelda fue mayor que la media anual de Yolanda.
- (C) La media anual de Yolanda fue 3 puntos superior a la media anual de Zelda.
- (D) Las medias anuales de Zelda y Yolanda son iguales.
- (E) Si Zelda hubiera sacado 3 puntos más en cada prueba, entonces su media anual hubiera sido igual que la de Yolanda.

19

Repartimos 2023 bolas en 3 cajas. De entre las siguientes opciones, determina la que más se aproxima a la probabilidad de que en las tres cajas quede una cantidad impar de bolas.

- (A) $\frac{2}{3}$ (B) $\frac{3}{10}$ (C) $\frac{1}{2}$ (D) $\frac{1}{3}$ (E) $\frac{1}{4}$

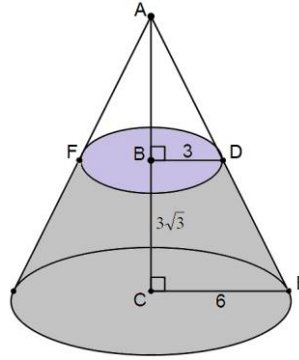
20

La rana Cyrus salta dos unidades en una dirección, y después 2 unidades en otra dirección. Determina la probabilidad que llegue a un punto a menos de una unidad de distancia del punto inicial.

- (A) $\frac{1}{6}$ (B) $\frac{1}{5}$ (C) $\frac{\sqrt{3}}{8}$ (D) $\frac{\arctan \frac{1}{2}}{\pi}$ (E) $\frac{2 \arcsin \frac{1}{4}}{\pi}$

21

Se fabrica una pantalla de lámpara con la forma de una superficie lateral de un tronco de cono circular recto. La altura del cono es $3\sqrt{3}$ pulgadas, el diámetro superior es de 6 pulgadas y el diámetro inferior es de 12 pulgadas. Un insecto está en la base de la pantalla y hay una gota de miel en la parte superior, en el punto más alejado del insecto. Este insecto quiere arrastrarse hasta la miel, pero siempre sin salirse de la superficie de la pantalla. Determina la longitud en pulgadas del camino más corto para llegar a la miel.



- (A) $6+3\pi$ (B) $6+6\pi$ (C) $6\sqrt{3}$ (D) $6\sqrt{5}$ (E) $6\sqrt{3}+\pi$

22

Sea f una función real con la propiedad de que para todo a y b ,

$$f(a+b) + f(a-b) = 2f(a)f(b)$$

¿Cuál de los siguientes valores no puede ser el valor de $f(1)$?

- (A) 0 (B) 1 (C) -1 (D) 2 (E) -2

23

Si sabemos que cuando tiramos n dados convencionales de seis caras, obtenemos 936 posibles productos diferentes de las caras superiores, determina el valor de n .

- (A) 11 (B) 6 (C) 8 (D) 10 (E) 9

24

Supongamos que a , b , c , y d son enteros positivos satisfaciendo las siguientes relaciones: $abcd = 2^6 \cdot 3^9 \cdot 5^7$, $mcm(a,b) = 2^3 \cdot 3^2 \cdot 5^3$, $mcm(a,c) = 2^3 \cdot 3^3 \cdot 5^3$, $mcm(a,d) = 2^3 \cdot 3^3 \cdot 5^3$, $mcm(b,c) = 2^1 \cdot 3^3 \cdot 5^2$, $mcm(b,d) = 2^2 \cdot 3^3 \cdot 5^2$, $mcm(c,d) = 2^2 \cdot 3^3 \cdot 5^2$. Determina $Mcd(a,b,c,d)$

- (A) 30 (B) 45 (C) 3 (D) 15 (E) 6

25

Tenemos un pentágono regular con área $\sqrt{5} + 1$ que imprimimos y recortamos en un papel. Doblamos el papel cinco veces de forma que cada vértice toque el centro del pentágono, creando un pentágono más pequeño en su interior. Determina el área de este nuevo pentágono.

- (A) $4 - \sqrt{5}$ (B) $\sqrt{5} - 1$ (C) $8 - 3\sqrt{5}$ (D) $(\sqrt{5} + 1)/2$ (E) $(2 + \sqrt{5})/3$



2023 AMC 12B Answer Key

1. C
2. B
3. D
4. C
5. C
6. C
7. E
8. D
9. B
10. E
11. D
12. E
13. D
14. A
15. E
16. D
17. E
18. A
19. E
20. E
21. E
22. E
23. E
24. C
25. B

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AMC 12B 2023 Soluciones

1

Tenemos en total $3 + \frac{1}{3} = \frac{10}{3}$ de vaso de zumo de naranja.

Repartido en cuatro vasos tocará a $\frac{10}{3} \div 4 = \frac{10}{12}$ por cada vaso.

En el cuarto vaso hay $\frac{1}{3}$ de zumo, luego faltará $\frac{10}{12} - \frac{1}{3} = \frac{10}{12} - \frac{4}{12} = \frac{6}{12} = \frac{1}{2}$ vaso que repartido entre los otros tres será $\frac{1}{2} \div 3 = \frac{1}{6}$ de vaso (C).

2

Sea x el precio de las zapatillas antes del descuento.

Un descuento del 20% equivale a pagar un 80%, es decir $\frac{80}{100}x = \frac{4}{5}x$

Un aumento del 7.5% implica pagar un $1 + 0.075/100$ del precio, es decir $\frac{1075}{1000} = \frac{43}{40}$, luego el

precio que pagará es $\frac{43}{40} \cdot \frac{4}{5}x = 43 \Rightarrow x = 50$ dólares (B).

3

La clave para resolver este problema es recordar que, aplicando Teorema de Tales (★60), todo triángulo rectángulo está inscrito en una circunferencia cuyo diámetro es la hipotenusa. Luego los radios respectivos son $5/2$ y $13/2$ y por tanto

$$\frac{\pi(5/2)^2}{\pi(13/2)^2} = \frac{\pi \cdot 5^2 / 2^2}{\pi \cdot 13^2 / 2^2} = \frac{5^2}{13^2} = \frac{25}{169} \quad (\text{D})$$

4

$$6.5mm \cdot 25 \times 10^3 mm = 162.5 \times 10^3 mm^2 = 162500 mm^2 = 1625 cm^2 \quad (\text{C})$$

5

La “tentación” inicial es probar primero en la casilla central, con lo que eliminas cuatro posibles posiciones del rectángulo, pero entonces quedan las cuatro esquinas que necesitarán cuatro tiradas más, un total de 5.

2		3
	1	
5		4

Observando más detenidamente las posibles jugadas, vemos que la siguiente sucesión de “tiradas” en el orden indicado es minimal y con ella se garantiza adivinar al menos una casilla ocupada por el rectángulo en cuatro tiradas:

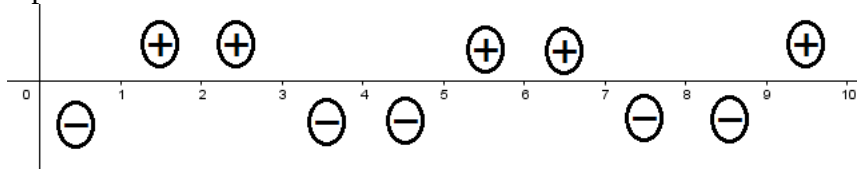
	2	
1		3
	4	

y por tanto la solución es 4 (C).

6

Interpretamos el polinomio como una función, que cortará el eje X en los puntos 1, 2, 3, ... 10. En los puntos pares (2, 4, 6, 8 y 10) la función “rebotará” en el eje, y por lo tanto será un cero pero sin cambiar de signo. En los puntos impares (1, 3, 5, 7 y 9) la función cruzará el eje X, es decir, cambiará de signo.

El signo de la función en $x = 0$ es $P(0) = (-)(+)(-)(+)(-)(+)(-)(+)(-)(+) = (-)$ luego tenemos el siguiente comportamiento:



Y vemos que hay 5 intervalos en los que la función es positiva (C)

7

$\sqrt{\frac{\log(n^2) - (\log n)^2}{\log n - 3}}$ es un número real si y solo si $\frac{\log(n^2) - (\log n)^2}{\log n - 3} \geq 0$, y esto ocurre en tres casos:

$$\text{a) } \begin{cases} \log(n^2) - (\log n)^2 > 0 \\ \log n - 3 > 0 \end{cases} \quad \text{b) } \begin{cases} \log(n^2) - (\log n)^2 < 0 \\ \log n - 3 < 0 \end{cases} \quad \text{c) } \log(n^2) - (\log n)^2 = 0$$

$$\text{a) } \log n - 3 > 0 \Leftrightarrow \log n > 3 \Leftrightarrow n > 10^3 = 1000$$

$$\begin{aligned} \log(n^2) - (\log n)^2 > 0 &\Leftrightarrow \log(n^2) > (\log n)^2 \Leftrightarrow 2 \log n > (\log n)^2 \Leftrightarrow \\ &\Leftrightarrow 2 > \log n \Leftrightarrow n < 10^2 = 100 \end{aligned}$$

Las dos condiciones son incompatibles, luego este caso no se puede dar.

$$\text{b) } \log n - 3 < 0 \Leftrightarrow \log n < 3 \Leftrightarrow n < 10^3 = 1000$$

$$\begin{aligned} \log(n^2) - (\log n)^2 < 0 &\Leftrightarrow \log(n^2) < (\log n)^2 \Leftrightarrow 2 \log n < (\log n)^2 \Leftrightarrow \\ &\Leftrightarrow 2 < \log n \Leftrightarrow 100 = 10^2 < n \end{aligned}$$

Así pues, el rango de valores aceptables es $100 < n < 1000$, es decir, los números naturales entre 101 y 999, que son $999 - 101 + 1 = 899$ valores (A).

c)

$$\begin{aligned} 0 &= \log(n^2) - (\log n)^2 = 2 \log n - (\log n)^2 = \log n (2 - \log n) \Leftrightarrow \\ &\Leftrightarrow \begin{cases} \log n = 0 \Leftrightarrow n = 1 \\ 2 - \log n = 0 \Leftrightarrow 2 = \log n \Leftrightarrow n = 10^2 = 100 \end{cases} \end{aligned}$$

En total hay $899 + 2 = 901$ enteros que satisfacen la ecuación (E).

8

Vamos a ordenar los casos en orden creciente de elementos de B.

a) $\text{Card}(B) = 1$, solo puede ser $B = \{1\}$. Un caso.

b) $\text{Card}(B) = 2$, son de la forma $B = \{2, a\}$ con $3 \leq a \leq 12$. 10 casos.

c) $Card(B) = 3$, son de la forma $B = \{3, a, b\}$ con $4 \leq a < b \leq 12$.

Son el número de grupos de 2 elementos no ordenados, sin repetición, de un total de 9 elementos.

$$\binom{9}{2} = \frac{9!}{2!7!} = \frac{9 \cdot 8}{2} = 36 \text{ elementos.}$$

d) $Card(B) = 4$, son de la forma $B = \{4, a, b, c\}$ con $5 \leq a < b < c \leq 12$.

Son el número de grupos de 3 elementos no ordenados, sin repetición, de un total de 8 elementos.

$$\binom{8}{3} = \frac{8!}{3!5!} = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2} = 56 \text{ elementos.}$$

e) $Card(B) = 5$, son de la forma $B = \{5, a, b, c, d\}$ con $6 \leq a < b < c < d \leq 12$.

Son el número de grupos de 4 elementos no ordenados, sin repetición, de un total de 7 elementos.

$$\binom{7}{4} = \frac{7!}{4!3!} = \frac{7 \cdot 6 \cdot 5 \cdot 4}{4 \cdot 3 \cdot 2} = 35 \text{ elementos.}$$

f) $Card(B) = 6$, son de la forma $B = \{6, a, b, c, d, e\}$ con $7 \leq a < b < c < d < e \leq 12$.

Son el número de grupos de 5 elementos no ordenados, sin repetición, de un total de 6 elementos.

$$\binom{6}{5} = \frac{6!}{5!1!} = 6 \text{ elementos.}$$

Los casos $Card(B) \geq 7$ son inviables. Luego hay

$$1 + 10 + 36 + 56 + 35 + 6 = 144 \text{ casos (D)}$$

9

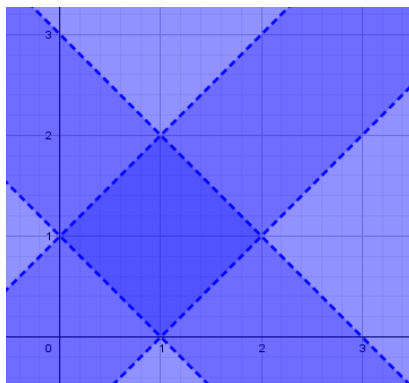
Puesto que x e y están en valor absoluto, podemos reducirnos al caso $x, y \geq 0$ (el primer cuadrante) y después hacer simetría horizontal y vertical, es decir, multiplicar el área por cuatro. Así pues, supongamos que $x, y \geq 0$ y nuestra desigualdad queda

$$|x-1| + |y-1| \leq 1$$

Veamos los cuatro casos posibles:

	$0 \leq x < 1$	$x \geq 1$
$0 \leq y < 1$	$1 - x + 1 - y \leq 1 \Leftrightarrow$ $-x - y \leq -1 \Leftrightarrow$ $x + y \geq 1$	$x - 1 + 1 - y \leq 1 \Leftrightarrow$ $x - y \leq 1$
$y \geq 1$	$1 - x + y - 1 \leq 1 \Leftrightarrow$ $-x + y \leq 1$	$x - 1 + y - 1 \leq 1 \Leftrightarrow$ $x + y \leq 3$

Vemos que son cuatro semiplanos que determinan una región de área 2. Esto mismo se podría deducir directamente viendo que estamos observando un cuadrado de lado $\sqrt{2}$ mediante la “distancia del taxista” (Ver problema 8.1.3).

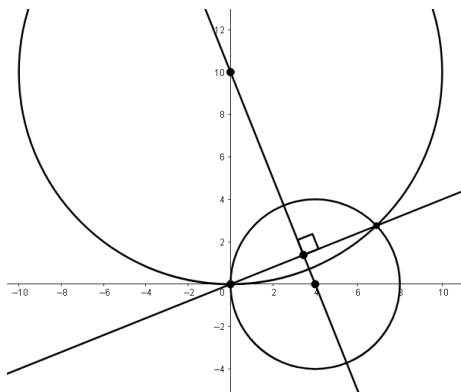


Luego el área total de la figura es $2 \times 4 = 8$ (D).

10

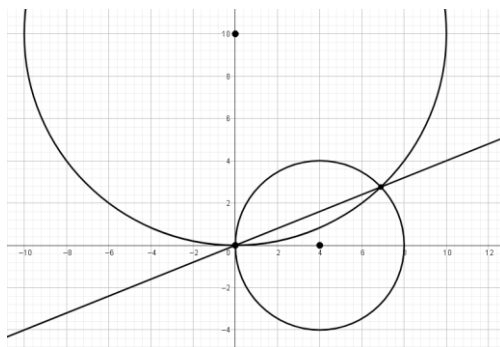
Primera versión.

Con la información del enunciado es fácil ver que son la circunferencia de centro $(4,0)$ y radio 4 y la circunferencia de centro $(0,10)$ y radio 10.



La recta que une sus centros respectivos está claro que tiene pendiente $\frac{-10}{4} = \frac{-5}{2}$ y sabemos que la recta que une los dos puntos de corte es perpendicular a ésta, luego tendrá pendiente $\frac{2}{5}$.

Segunda versión.



Las ecuaciones de las circunferencias son:

$$(x-4)^2 + y^2 = 16 \Rightarrow x^2 - 8x + y^2 = 0$$

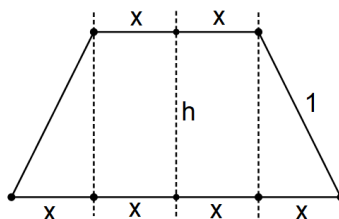
$$x^2 + (y-10)^2 = 100 \Rightarrow x^2 - 20y + y^2 = 0$$

Restando la segunda a la primera deducimos que los puntos de corte pasan por la recta de ecuación

$$-8x + 20y = 0 \Leftrightarrow y = \frac{2}{5}x \text{ que tiene pendiente } 2/5 \text{ (E).}$$

11

Por simetría y puesto que la base mayor es el doble de la base menor podemos suponer un trapecio con bases $4x$ y $2x$ tal y como se muestra en la imagen:



El área de este trapecio será la de seis triángulos rectángulos de área $A = \frac{1}{2}xh$, con la condición $1^2 = x^2 + h^2$.

Pero aplicando 1.1.c, $A = \frac{1}{2}xh \leq \frac{1}{2} \frac{x^2 + h^2}{2} = \frac{1}{4}(x^2 + h^2) = \frac{1}{4}$ y por tanto el área del trapecio siempre será menor que $6 \cdot \frac{1}{4} = \frac{3}{2}$ (D).

12

$$z^2 = (a + bi)^2 = a^2 - b^2 + 2abi$$

$$z \otimes z = z^2 + 40 \Leftrightarrow a^2 + b^2i = a^2 - b^2 + 2abi + 40 \Rightarrow \begin{cases} a^2 = a^2 - b^2 + 40 \Rightarrow b^2 = 40 \\ b^2 = 2ab \Rightarrow a = b/2 \Rightarrow a^2 = b^2 / 40 = 10 \end{cases}$$

$$\text{Finalmente, } |z| = \sqrt{a^2 + b^2} = \sqrt{10 + 40} = \sqrt{50} = 5\sqrt{2} \text{ (E).}$$

13

Primera versión.

Tenemos

$$4a + 4b + 4c = 13 \Rightarrow A = a + b + c = \frac{13}{4}$$

$$2ab + 2ac + 2bc = \frac{11}{2} \Rightarrow B = ab + ac + bc = \frac{11}{4}$$

$$C = abc = \frac{1}{2}$$

Y está claro que la longitud de la diagonal mayor es $\sqrt{a^2 + b^2 + c^2}$.

Este problema es una aplicación directa de las fórmulas de Newton:

$$a^2 + b^2 + c^2 = A^2 - 2B = \left(\frac{13}{4}\right)^2 - 2\frac{11}{4} = \frac{169}{16} - \frac{11}{2} = \frac{169}{16} - \frac{88}{16} = \frac{81}{16} \text{ y por tanto}$$

$$\sqrt{a^2 + b^2 + c^2} = \sqrt{\frac{81}{16}} = \frac{9}{4} \text{ (D)}$$

Segunda versión.

Aplicando las fórmulas de Vieta, observamos que a , b y c son las soluciones del polinomio

$$p(x) = x^3 - \frac{13}{4}x^2 + \frac{11}{4}x - \frac{1}{2}.$$

Multiplicando por 4 este polinomio es equivalente a $q(x) = 4x^3 - 13x^2 + 11x - 2$. Vemos que sus coeficientes suman 0, luego 1 es una raíz de ese polinomio. Mediante la división sintética llegamos a encontrar su factorización:

$p(x) = (x-1)(x-2)(4x-1)$, cuyas raíces son 1, 2 y $1/4$. Finalmente:

$$\sqrt{a^2 + b^2 + c^2} = \sqrt{1 + 2^2 + \left(\frac{1}{4}\right)^2} = \frac{9}{4}$$

Tercera versión.

Otra manera de evitar aplicar mecánicamente las fórmulas de Newton es aplicar la identidad notable

$$a^2 + b^2 + c^2 = (a+b+c)^2 - 2(ab+ac+bc) = \left(\frac{13}{4}\right)^2 - \frac{11}{2} = \frac{81}{16}$$

14

Resolvemos este problema mediante una aplicación directa de las fórmulas de Vieta:

$$a = -(\alpha + \beta + \gamma)$$

$$b = \alpha\beta + \beta\gamma + \alpha\gamma$$

$$6 = -\alpha\beta\gamma$$

De la tercera ecuación se deduce que, más allá de permutaciones que no afectan a (a, b) , tenemos las siguientes cinco posibilidades:

$$\alpha = -1, \beta = 1, \gamma = 6$$

$$\alpha = -1, \beta = 2, \gamma = 3$$

$$\alpha = 1, \beta = -2, \gamma = 3$$

$$\alpha = 1, \beta = 2, \gamma = -3$$

$$\alpha = -1, \beta = -2, \gamma = -3$$

y por tanto la respuesta correcta es (A).

15

Pasando a común denominador la expresión de la izquierda, vemos que $c = 15a + 14b$.

La afirmación I no es cierta. Basta tomar

$$a = 1, b = 3 \Rightarrow c = 15 + 14 \cdot 3 = 57, \text{Mcd}(57, 210) = 3$$

Veamos que III es cierta.

Si $\text{Mcd}(c, 210) = 1$, aplicando el Teorema de Bezout, existirán enteros α, β tales que

$$1 = \alpha(15a + 14b) + \beta 15 \cdot 14 = 15\alpha a + 14\alpha b + \beta 15 \cdot 14 = (15\alpha)a + 14(\alpha b + \beta 15)$$

Luego, de nuevo por el Teorema de Bezout, $\text{Mcd}(a, 14) = 1$.

De la misma forma, tenemos

$$1 = \alpha(15a + 14b) + \beta 15 \cdot 14 = 14\alpha b + 15\alpha a + \beta 15 \cdot 14 = (14\alpha)b + 15(\alpha a + \beta \cdot 14)$$

y por tanto, por el Teorema de Bezout, $\text{Mcd}(b, 15) = 1$.

Una manera de demostrarlo alternativa sería por contrapositivo:

Supongamos que no se cumple “ $Mcd(a,14) = Mcd(b,15) = 1$ ”, es decir, supongamos que $Mcd(a,14) > 1 \Rightarrow a = pa'$ donde $p = 2$ o $p = 7$. Por ejemplo, $p = 2$.

$$c = 15a + 14b = 15 \cdot 2a' + 14b = 2(15a' + 7b) \Rightarrow 2 \mid c \Rightarrow Mcd(c,210) > 1$$

y con razonamientos similares demostramos cualquiera de las otras opciones.

Supongamos que, por el contrario, $Mcd(c,210) > 1$. Entonces existirá un primo p tal que

$$\begin{cases} p \mid 15a + 14b \\ p \mid 210 = 2 \cdot 3 \cdot 5 \cdot 7 \end{cases}$$

Entonces p tiene que ser 2, 3, 5 o 7. Supongamos que $p = 2$. Entonces

$$\begin{cases} p \mid 3 \cdot 5a + 2 \cdot 7b \\ p \mid 2 \cdot 7b \end{cases} \Rightarrow p \mid 3 \cdot 5a \Rightarrow p \mid a$$

y puesto que $p \mid 14$ por tanto $Mcd(a,14) \geq p > 1$.

La afirmación II es cierta, pues III es una versión más fuerte que II.

Luego son ciertas II y III solamente (E).

16

Estamos estudiando todas las combinaciones posibles $6a + 10b + 15c$ con $0 \leq a, b, c$ enteros. Sabemos que si n, m son coprimos, el “número de Frobenius” asociado, es decir, el mayor entero que no se puede representar de la forma $\alpha n + \beta m$, con $0 \leq \alpha, \beta$ es $nm - n - m$ (Ver [Teoría de Números](#) 20.10).

$6a + 10b + 15c = 3(2a + 5c) + 10b$ y puesto que 3 y 10 son coprimos, podemos aplicar la fórmula anterior para garantizar que el máximo entero no representable es $3 \cdot 10 - 3 - 10 = 17$.

Pero ahora tenemos que aplicar la fórmula del número de Frobenius a $2a + 5c$, y su máximo entero no representable es $2 \cdot 5 - 2 - 5 = 3$.

Luego también podemos tener problemas cuando $2a + 5c = 3$. En este caso los valores serán:

$$3(2a + 5c) + 10b = 3 \cdot 3 + 10b = 9 + 10b$$

$b = 2 \rightarrow 9 + 10 \cdot 2 = 29$, y observamos que este número no se puede escribir de ninguna otra forma $6a + 10b + 15c$, luego 29 tampoco es representable.

Sin embargo, si $b \geq 3$, entonces

$$9 + 10b = 9 + 10(b - 3 + 3) = 9 + 10(b - 3) + 30 = 39 + 10(b - 3) = 3 \cdot 13 + 10(b - 3)$$

y 13 sí es representable $2a + 5c = 2 \cdot 4 + 5 \cdot 1$.

Así pues, el valor máximo no representable de la forma $6a + 10b + 15c$ es 29, y la respuesta correcta es $2 + 9 = 11$ (D).

También podría haber problemas en los números de la forma $2a + 5c = 2$, pero entonces serían de la forma

$$3(2a + 5c) + 10b = 3 \cdot 2 + 10b = 6 + 10b, \text{ y estos números son perfectamente representables.}$$

Finalmente, también podría haber problemas en los números de la forma $2a + 5c = 1$, pero entonces serían de la forma

$$3(2a+5c)+10b=3+10b, b \geq 3 \Rightarrow 3+10b=3+10(b-3+3)=3+10(b-3)+30=$$

$$=33+10(b-3)=3 \cdot 11+10(b-3)$$

17

Los lados de este triángulo serán 6, $a+6$ y $2a+6$ para cierto número a . Está claro que 120° es su ángulo mayor, luego estará opuesto al lado mayor, es decir, a $2a+6$.

Aplicamos el Teorema del Coseno para determinar el valor de a :

$$(6+2a)^2 = 6^2 + (6+a)^2 - 2(6+a) \cdot 6 \cos 120^\circ \Leftrightarrow$$

$$0 = -72 - 6a - 3a^2 \Leftrightarrow a = \begin{cases} -6 \\ 4 \end{cases}$$

En el contexto en el que estamos, el único candidato aceptable es $a = -6$, y finalmente, aplicando ★39,

$$[\triangle ABC] = \frac{6 \cdot 10 \cdot \sin 120^\circ}{2} = 15\sqrt{3}$$

18

Sean Y_1, Y_2, Z_1, Z_2 las medias correspondientes a las puntuaciones obtenidas por Yolanda y por Zelda en los trimestres 1 y 2.

Sean n_1, n_2, m_1, m_2 el número de exámenes realizados respectivamente por Yolanda y por Zelda en el trimestre 1 y 2.

Luego las condiciones del enunciado son:

$$Y_1 = Z_1 + 3, Y_2 = Y_1 + 18, Y_2 = Z_2 + 3$$

Luego

$$Y_2 = Z_2 + 3 \Rightarrow Z_2 = Y_2 - 3 = Y_1 + 18 - 3 = Y_1 + 15 = Z_1 + 3 + 15 = Z_1 + 18$$

La media anual de Yolanda es:

$$\frac{Y_1 n_1 + Y_2 n_2}{n_1 + n_2} = \frac{1}{n_1 + n_2} (Y_1 n_1 + Y_2 n_2) = \frac{1}{n_1 + n_2} (Y_1 n_1 + (Y_1 + 18) n_2) = \frac{1}{n_1 + n_2} (Y_1 (n_1 + n_2) + 18 n_2) =$$

$$= Y_1 + \frac{18 n_2}{n_1 + n_2}$$

De la misma manera, la media anual de Zelda es:

$$\frac{Z_1 m_1 + Z_2 m_2}{m_1 + m_2} = \dots = Z_1 + \frac{18 m_2}{m_1 + m_2}$$

Luego la diferencia de medias será

$$\frac{Y_1 n_1 + Y_2 n_2}{n_1 + n_2} - \frac{Z_1 m_1 + Z_2 m_2}{m_1 + m_2} = Y_1 + \frac{18 n_2}{n_1 + n_2} - Z_1 - \frac{18 m_2}{m_1 + m_2} =$$

$$= Y_1 - Z_1 + 18 \left(\frac{n_2}{n_1 + n_2} - \frac{m_2}{m_1 + m_2} \right) = 3 + 18 \left(\frac{n_2}{n_1 + n_2} - \frac{m_2}{m_1 + m_2} \right)$$

Aquí observamos que $n_2 \leq n_1 + n_2 \Rightarrow \frac{n_2}{n_1 + n_2} \leq 1$ y $\frac{m_2}{m_1 + m_2} > 0$ y por tanto

$$\frac{n_2}{n_1 + n_2} + \frac{m_2}{m_1 + m_2} \leq 1$$

Luego

$$\frac{Y_1 n_1 + Y_2 n_2}{n_1 + n_2} - \frac{Z_1 m_1 + Z_2 m_2}{m_1 + m_2} \leq 3 + 18 = 21$$

Y la opción (A) no puede ser cierta nunca.

Fuente: <https://artofproblemsolving.com/>

19

Sean a , b y c el número de bolas que colocamos en cada una de las tres cajas A, B y C.

Está claro que ΔABC

$$0 \leq a \leq 2023$$

$$0 \leq b \leq 2023 - a$$

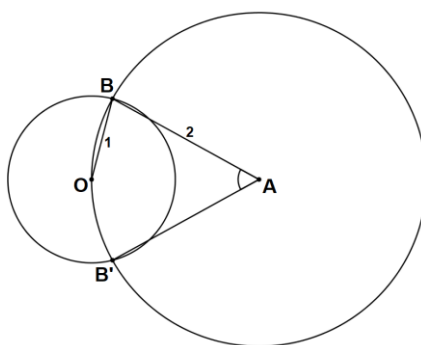
y c queda determinado por a y b : $c = 2023 - a - b$. El valor de c será impar si y solo si lo son a y b , luego nuestro problema se reduce a determinar los valores de a y b que son ambos impares.

Para este segundo problema, si ordenamos los valores de a en vertical en filas y los valores de b en horizontal en columnas, vemos que los “casos favorables” son las casillas correspondientes a fila impar y columna impar, es decir, una de cada cuatro.

Algunos elementos de la diagonal deben ser eliminados, por lo que la probabilidad no será exactamente $1/4$ (el valor exacto es $1012/2023$) pero, puesto que solo nos piden una buena aproximación, no necesitamos entrar en el detalle.

20

En primer lugar observamos que esta propiedad no depende de la primera dirección tomada, por lo que podemos suponer que la rana está en el origen O y en primer lugar salta dos posiciones hacia la derecha, llegando a A . Sean B y B' los puntos de corte entre la circunferencia de centro O y radio 1 y la circunferencia de centro A y radio 2.



Para que la rana esté a menos de una unidad del centro O debe estar en el arco BB' después del segundo salto, y puesto que las longitudes de los arcos de una circunferencia son proporcionales a los ángulos que forman, esta proporcionalidad será

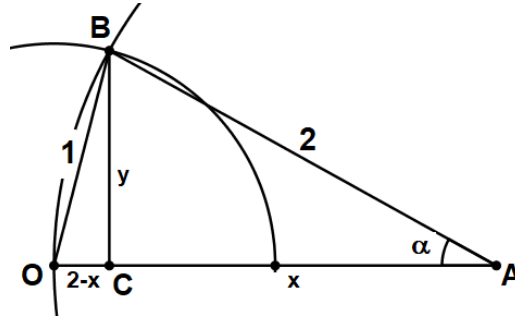
$$\frac{\angle BAB'}{2\pi} = \frac{\angle OAB}{\pi}$$

Luego nuestro problema se reduce a determinar el ángulo $\alpha = \angle OAB$.

Primera versión.

Sea C el punto de corte entre la perpendicular por B y el segmento OA .

Sea $x = AC \Rightarrow OC = 2 - x$, $y = BC$.



Y aplicamos Pitágoras:

$$\left. \begin{array}{l} 2^2 = x^2 + y^2 \\ 1^2 = (2-x)^2 + y^2 \end{array} \right\} \Rightarrow 4 - x^2 = 1^2 - (2-x)^2 \Rightarrow 4 = -3 + 4x \Rightarrow x = \frac{7}{4}$$

Y por tanto $\cos(\alpha) = \frac{7/4}{2} = \frac{7}{8}$

Y por tanto, aplicando la identidad del cuadrado del seno,

$$\sin^2(\alpha/2) = \frac{1 - \cos(\alpha)}{2} = \frac{1 - 7/8}{2} = \frac{1/8}{2} = \frac{1}{4} \Rightarrow \sin(\alpha/2) = \sqrt{\frac{1}{4}} = \frac{1}{2} \Rightarrow$$

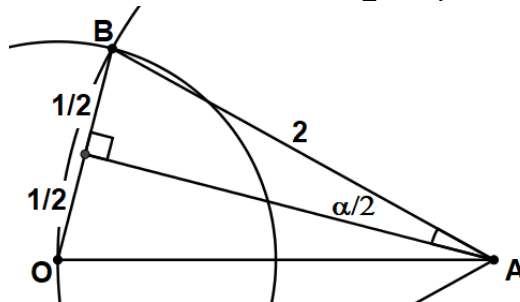
$$\frac{\alpha}{2} = \arcsin\left(\frac{1}{2}\right) \Rightarrow \alpha = 2 \arcsin\left(\frac{1}{2}\right)$$

Y finalmente la probabilidad es

$$\frac{2 \arcsin\left(\frac{1}{2}\right)}{\pi} \quad (\text{E}).$$

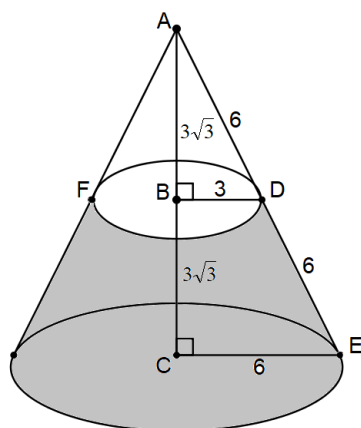
Segunda versión.

Una manera mucho más sencilla y elegante de determinar este ángulo es observando que el triángulo $\triangle OAB$ es isósceles y por tanto $\sin(\alpha/2) = \frac{1/2}{2} = \frac{1}{4}$.

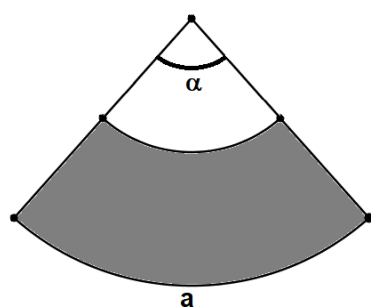


21

Por semejanza de triángulos rectángulos, puesto que un radio es 3 y el otro es 6, es decir uno es el doble que el otro, es fácil ver que la altura BA del cono superior pequeño será también $3\sqrt{3}$ y por tanto la altura total AC del cono será $6\sqrt{3}$. Por Pitágoras deducimos que la apotema AE es $\sqrt{(6\sqrt{3})^2 + 6^2} = 12$ y por tanto $AD = DE = 6$.



Si desplegamos un tronco de cono cualquiera nos quedará un sector de una corona circular:



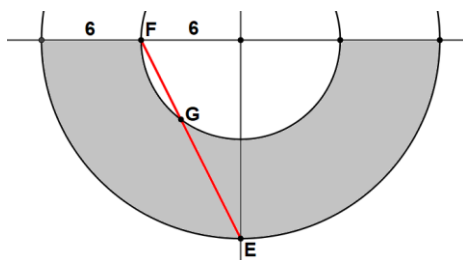
Calculamos el ángulo α :

Radio de la circunferencia grande: 12, luego la longitud de la circunferencia será $2\pi r = 2\pi 12 = 24\pi$.

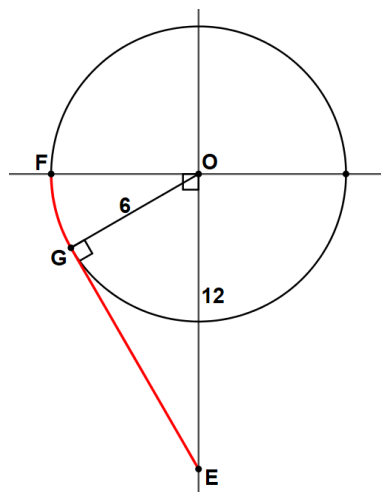
Arco de la circunferencia grande: $a = 2\pi r = 2\pi 6 = 12\pi$

Luego el arco es $\frac{12\pi}{24\pi} = \frac{1}{2}$ de toda la circunferencia, y esto corresponde a $\alpha = \frac{1}{2} 360^\circ = 180^\circ$.

Así pues, tenemos la siguiente figura, en la que queremos ir del punto E al punto F. Aquí la trayectoria es una línea recta:



Y aquí observamos que si trazamos una línea recta de E a F, ¡la trayectoria pasaría por el exterior del tronco del cono!, luego forzosamente debemos dividir nuestra trayectoria óptima en dos partes: Desde E a G en línea recta donde G es el punto de tangencia, y de G a F por el borde de la circunferencia superior del tronco.



Aquí observamos que el triángulo $\triangle OGE$ es un triángulo $1:\sqrt{3}:2$, luego sus ángulos serán 30° - 60° - 90° , es decir, $\angle GOE = 60^\circ$ y por tanto $\angle FOG = 90^\circ - 60^\circ = 30^\circ$. Así pues,

$$\text{arco}FG = 2\pi \cdot 6 \cdot \frac{30^\circ}{360^\circ} = \pi$$

$$GE = \sqrt{12^2 - 6^2} = 6\sqrt{3}$$

y, finalmente, la longitud mínima será $6\sqrt{3} + \pi$ (E)

22

Tomando $b=0$

$$f(a) + f(a) = 2f(a)f(0) \Rightarrow 2f(a) = 2f(a)f(0) \Rightarrow$$

$$\Rightarrow f(a) = f(a)f(0) \Rightarrow f(a) = f(a)f(0)$$

$$\Rightarrow f(a) - f(a)f(0) = 0 \Rightarrow f(a)(1 - f(0)) = 0 \Rightarrow \begin{cases} f(a) = 0 \\ f(0) = 1 \end{cases}$$

Luego, o bien $f=0$, y para esta función se satisface la condición del enunciado, o bien existe un a tal que $f(a) \neq 0$ y por tanto forzosamente $f(0) = 1$.

Supongamos que $f(0) = 1$.

Tomando $b=a$

$$f(2a) + f(0) = 2f(a)f(a) \Rightarrow f(2a) + 1 = 2f(a)^2$$

Tomando $a=1/2$

$$f(1) + 1 = 2f(1/2)^2 \geq 0 \Rightarrow f(1) + 1 \geq 0 \Rightarrow f(1) \geq -1 \text{ luego } f(1) \neq -2 \text{ (E).}$$

23

Sean $0 \leq a, b, c, d, e, f \leq n$ respectivamente el número de “1”, “2”, “3”, ..., “6” obtenidos al tirar n dados. Tenemos que

$$a + b + c + d + e + f = n$$

y queremos determinar la cantidad total de valores diferentes de la forma

$$1^a \cdot 2^b \cdot 3^c \cdot 4^d \cdot 5^e \cdot 6^f$$

Desarrollamos la factorización de este número:

$$1^a \cdot 2^b \cdot 3^c \cdot (2^2)^d \cdot 5^e \cdot (2 \cdot 3)^f = 2^b \cdot 3^c \cdot 2^{2d} \cdot 5^e \cdot 2^f \cdot 3^f = 2^{b+2d+f} \cdot 3^{c+f} \cdot 5^e$$

Por el Teorema fundamental de la aritmética serán números diferentes todos los que tengan asociadas factorizaciones en factores primos diferentes, luego reescribimos nuestro problema como:

Dados $0 \leq b, c, d, e, f \leq n$ cumpliendo $b + c + d + e + f \leq n$, queremos determinar la cantidad de ternas diferentes $(b + 2d + f, c + f, e)$. A este número le llamaremos $F(n)$.

Vemos que el valor de e recorre entre 0 y n , luego podemos dividir esta cantidad en función de e :

$$b + c + d + e + f \leq n \Leftrightarrow b + c + d + f \leq n - e$$

y por tanto

$$F(n) = \sum_{e=0}^n G(n-e)$$

Este sumatorio lo podemos recorrer de izquierda a derecha o de derecha a izquierda, luego

$$F(n) = \sum_{e=0}^n G(n-e) = \sum_{e=0}^n G(e)$$

Donde $G(k)$ es la cantidad de valores diferentes del par $(b + 2d + f, c + f)$ sometido a la condición $b + c + d + f \leq k$

De nuevo tenemos que agrupar los valores de $G(k)$, para ello vamos a utilizar la segunda componente $c + f$ como índice. Sea $i = c + f$.

Está claro que $0 \leq c + f \leq k$, y recorre todos estos valores intermedios. Luego

$$G(k) = \sum_{i=0}^k H(i)$$

donde $H(i)$ es la cantidad de números diferentes de la forma $b + 2d + f$ cumpliendo las condiciones anteriores.

La clave de este problema es observar que, bajo estas condiciones, $b + 2d + f$ recorre todos los números entre 0 y $2k - i$, luego:

$$H(i) = 2k - i + 1$$

Por lo tanto:

$$\begin{aligned} G(k) &= \sum_{i=0}^k H(i) = \sum_{i=0}^k (2k - i + 1) = 2 \sum_{i=0}^k k - \sum_{i=0}^k i + \sum_{i=0}^k 1 = 2k(k+1) - \frac{k(k+1)}{2} + (k+1) = \\ &= (k+1) \left(2k - \frac{k}{2} + 1 \right) = (k+1) \left(\frac{3k}{2} + 1 \right) = \frac{1}{2} (k+1)(3k+2) \end{aligned} \quad Y$$

por tanto:

$$\begin{aligned}
F(n) &= \sum_{e=0}^n G(e) = \sum_{e=0}^n \frac{1}{2} (e+1)(3e+2) = \frac{1}{2} \sum_{e=0}^n (3e^2 + 5e + 2) = \\
&= \frac{1}{2} \left(3 \sum_{e=0}^n e^2 + 5 \sum_{e=0}^n e + \sum_{e=0}^n 2 \right) = \frac{1}{2} \left(3 \frac{n(n+1)(2n+1)}{6} + 5 \frac{n(n+1)}{2} + 2(n+1) \right) = \\
&= \frac{n+1}{2} \left(\frac{n(2n+1)}{2} + \frac{5n}{2} + 2 \right) = \frac{n+1}{2} \left(\frac{2n^2 + n}{2} + \frac{5n}{2} + \frac{4}{2} \right) = \\
&= \frac{n+1}{4} (2n^2 + n + 5n + 4) = \frac{n+1}{4} (2n^2 + 6n + 4) = \frac{n+1}{2} (n^2 + 3n + 2) = \\
&= \frac{(n+1)(n+1)(n+2)}{2} = \frac{(n+1)^2(n+2)}{2}
\end{aligned}$$

Con este resultado tan potente solo queda ver la opción del enunciado que se adapta al resultado 936:

$$\frac{(n+1)^2(n+2)}{2} = 936 = 2^3 \cdot 3^2 \cdot 13 \Leftrightarrow (n+1)^2(n+2) = 2^4 \cdot 3^2 \cdot 13$$

La única opción válida es

$$\left. \begin{aligned} n+2 &= 13 \\ (n+1)^2 &= 2^4 \cdot 3^2 = 4^2 \cdot 3^2 = 12^2 \end{aligned} \right\} \Rightarrow n = 11 \quad (\text{A}).$$

Fuente: Steven Chen en https://artofproblemsolving.com/wiki/index.php?title=2023_AMC_12B_Problems/Problem_23&oldid=207869

24

Está claro que nuestros números serán de la forma

$$a = 2^{a_2} \cdot 3^{a_3} \cdot 5^{a_5}, \quad b = 2^{b_2} \cdot 3^{b_3} \cdot 5^{b_5}, \quad c = 2^{c_2} \cdot 3^{c_3} \cdot 5^{c_5} \quad \text{y} \quad d = 2^{d_2} \cdot 3^{d_3} \cdot 5^{d_5}$$

cumpliendo $a_2 + b_2 + c_2 + d_2 = 6$, $a_3 + b_3 + c_3 + d_3 = 9$ y $a_5 + b_5 + c_5 + d_5 = 7$.

Sabemos que el mínimo común múltiplo de dos números se obtiene mediante el máximo exponente, luego tenemos que ir imponiendo las condiciones. Los exponentes de las potencias de 2, 3 y 5 son independientes, luego las iremos determinando por orden.

En primer lugar, la potencia de 2.

$$a_2 + b_2 + c_2 + d_2 = 6, \quad \max(a_2, b_2) = 3, \quad \max(a_2, c_2) = 3, \quad \max(a_2, d_2) = 3, \quad \max(b_2, c_2) = 1$$

$$\max(b_2, d_2) = 2, \quad \max(c_2, d_2) = 2$$

$$\left. \begin{aligned} \max(b_2, c_2) = 1 &\Rightarrow b_2 \leq 1 \\ \max(b_2, a_2) = 3 \end{aligned} \right\} \Rightarrow a_2 = 3$$

Nos quedan las condiciones siguientes;

$$b_2 + c_2 + d_2 = 3, \quad \max(b_2, c_2) = 1, \quad \max(b_2, d_2) = 2, \quad \max(c_2, d_2) = 2$$

La única combinación aceptable es $b_2 = 1, c_2 = 0, d_2 = 2$ o $b_2 = 0, c_2 = 1, d_2 = 2$, Y el mínimo será 0.

Ahora la potencia de 3.

$$a_3 + b_3 + c_3 + d_3 = 9, \quad \max(a_3, b_3) = 2, \quad \max(a_3, c_3) = 3, \quad \max(a_3, d_3) = 3, \quad \max(b_3, c_3) = 3$$

$$\max(b_3, d_3) = 3, \quad \max(c_3, d_3) = 3$$

$$\left. \begin{aligned} \max(a_3, b_3) = 2 &\Rightarrow b_3 \leq 2 \\ \max(b_3, c_3) = 3 \end{aligned} \right\} \Rightarrow c_3 = 3$$

$$\left. \begin{array}{l} \max(a_3, b_3) = 2 \Rightarrow a_3 \leq 2 \\ \max(b_3, d_3) = 3 \end{array} \right\} \Rightarrow d_3 = 3$$

Nos quedan las condiciones siguientes: $a_3 + b_3 = 3$ y $\max(a_3, b_3) = 2$

Por tanteo la única condición aceptable es que uno de ellos sea 1 y el otro 2. En todo caso el mínimo será 1.

Finalmente, la potencia de 5.

$$a_5 + b_5 + c_5 + d_5 = 7, \max(a_5, b_5) = 3, \max(a_5, c_5) = 3, \max(a_5, d_5) = 3, \max(b_5, c_5) = 2, \max(b_5, d_5) = 2, \max(c_5, d_5) = 2.$$

$$\left. \begin{array}{l} \max(b_5, c_5) = 2 \Rightarrow b_5 \leq 2 \\ \max(a_5, b_5) = 3 \end{array} \right\} \Rightarrow a_5 = 3$$

Nos quedan las condiciones siguientes:

$$b_5 + c_5 + d_5 = 4, \max(b_5, c_5) = 2, \max(b_5, d_5) = 2, \max(c_5, d_5) = 2.$$

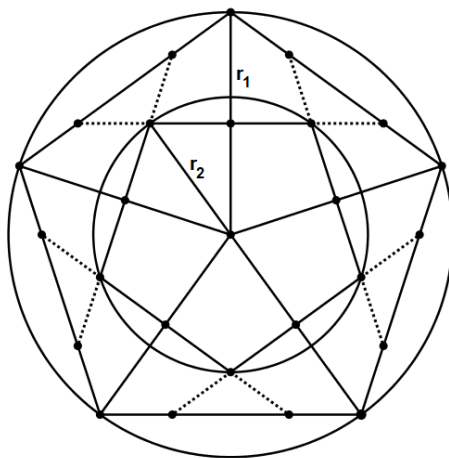
La única opción aceptable es que dos de ellos sean 2 y el otro 0. Luego el mínimo es 0.

Puesto que el máximo común divisor de a, b, c y d se toma con los exponentes mínimos, tenemos que $Mcd(a, b, c, d) = 2^0 \cdot 3^1 \cdot 5^0 = 3$

25

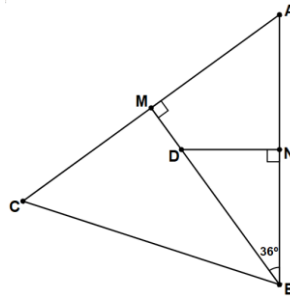
Vemos que estas dos figuras son proporcionales, luego sus áreas serán proporcionales al

cuadrado de los respectivos radios: $\frac{A_2}{A_1} = \left(\frac{r_2}{r_1}\right)^2$



El pentágono está formado por cinco triángulos isósceles $\triangle ABC$ con un ángulo $\angle ABC = 360^\circ / 5 = 72^\circ$, Luego, tomando los puntos medios N de AB y M de AC,

$\angle ABM = 72^\circ / 2 = 36^\circ$, y queremos determinar la razón $\frac{r_2}{r_1} = \frac{BM}{AB}$:



Podemos suponer, sin pérdida de generalidad, $AB=2$, luego al doblar los vértices,

$$AN = NB = AB/2 = 1 \text{ y } BM = AB \cos(36^\circ) = 2 \frac{(1+\sqrt{5})}{4} = \frac{1+\sqrt{5}}{2}.$$

Puesto que $\triangle ABM \approx \triangle DBN$ por ser triángulos rectángulos con un ángulo común en B,

$$\frac{BD}{AB} = \frac{BN}{BM} \Leftrightarrow \frac{BD}{2} = \frac{1}{(1+\sqrt{5})/2} \Leftrightarrow BD = \frac{4}{1+\sqrt{5}}$$

Así pues,

$$\frac{A_2}{A_1} = \left(\frac{r_2}{r_1} \right)^2 = \left(\frac{4/(1+\sqrt{5})}{2} \right)^2 = \left(\frac{2}{(1+\sqrt{5})} \right)^2 = \frac{4}{(1+\sqrt{5})^2}$$

Finalmente,

$$A_2 = \frac{4}{(1+\sqrt{5})^2} A_1 = \frac{4(1+\sqrt{5})}{(1+\sqrt{5})^2} = \frac{4}{1+\sqrt{5}} \frac{1-\sqrt{5}}{1-\sqrt{5}} = \frac{4(1-\sqrt{5})}{1-5} = -(1-\sqrt{5}) = \sqrt{5}-1$$

Observación. En esta solución se supone saber el $\cos(36^\circ) = \frac{1+\sqrt{5}}{4}$. En las soluciones de

[https://artofproblemsolving.com/wiki/index.php/2023 AMC 12B Problems/Problem 25](https://artofproblemsolving.com/wiki/index.php/2023_AMC_12B_Problems/Problem_25) podemos encontrar formas de llegar a este resultado.

Enlaces a los libros Toomates Coolección:

1	http://www.toomates.net/biblioteca/ProblemasAlgebra.pdf	#1.3.35
2	http://www.toomates.net/biblioteca/ProblemasAlgebra.pdf	#1.5.9
3	http://www.toomates.net/biblioteca/Geometria.pdf	#6.1.8
4	http://www.toomates.net/biblioteca/ProblemasAlgebra.pdf	#2.4.1
5	http://www.toomates.net/biblioteca/Combinatoria.pdf	#6.1.12
6	http://www.toomates.net/biblioteca/ProblemasAlgebra.pdf	#4.6.4
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11	http://www.toomates.net/biblioteca/Desigualdades.pdf	#1.10
12	http://www.toomates.net/biblioteca/ProblemasAlgebra.pdf	#12.1.35
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17	http://www.toomates.net/biblioteca/Geometria.pdf	#4.6.1
18	http://www.toomates.net/biblioteca/ProblemasAlgebra.pdf	#6.13
19	http://www.toomates.net/biblioteca/Probabilidad.pdf	#7.6
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22	http://www.toomates.net/biblioteca/ProblemasFunciones.pdf	#5.1.15
23	http://www.toomates.net/biblioteca/Combinatoria.pdf	#7.1.21
24	http://www.toomates.net/biblioteca/Aritmetica.pdf	#5.2.25
25	http://www.toomates.net/biblioteca/Geometria.pdf	#5.4.28

Page

- Article
- Discussion
- View source
- History

Toolbox

- Recent changes
- Random page
- Help
- What links here
- Special pages

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Search

2024 AMC 12A Problems

2024 AMC 12A (Answer Key)

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Instructions

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3. No aids are permitted other than scratch paper, graph paper, ruler, compass, protractor and erasers (and calculators that are accepted for use on the test if before 2006. No problems on the test will *require* the use of a calculator).
4. Figures are not necessarily drawn to scale.
5. You will have **75 minutes** working time to complete the test.

1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18

- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

What is the value of $9901 \cdot 101 - 99 \cdot 10101$?

- (A) 2 (B) 20 (C) 200 (D) 202 (E) 2020

Solution

Problem 2

A model used to estimate the time it will take to hike to the top of the mountain on a trail is of the form $T = aL + bG$, where a and b are constants, T is the time in minutes, L is the length of the trail in miles, and G is the altitude gain in feet. The model estimates that it will take 69 minutes to hike to the top if a trail is 1.5 miles long and ascends 800 feet, as well as if a trail is 1.2 miles long and ascends 1100 feet. How many minutes does the model estimates it will take to hike to the top if the trail is 4.2 miles long and ascends 4000 feet?

- (A) 240 (B) 246 (C) 252 (D) 258 (E) 264

Solution

Problem 3

The number 2024 is written as the sum of not necessarily distinct two-digit numbers. What is the least number of two-digit numbers needed to write this sum?

- (A) 20 (B) 21 (C) 22 (D) 23 (E) 24

Solution

Problem 4

What is the least value of n such that $n!$ is a multiple of 2024?

- (A) 11 (B) 21 (C) 22 (D) 23 (E) 253

Solution

Problem 5

A data set containing 20 numbers, some of which are 6, has mean 45. When all the 6s are removed, the data set has mean 66. How many 6s were in the original data set?

- (A) 4 (B) 5 (C) 6 (D) 7 (E) 8

Solution

Problem 6

The product of three integers is 60. What is the least possible positive sum of the three integers?

- (A) 2 (B) 3 (C) 5 (D) 6 (E) 13

Solution

Problem 7

In $\triangle ABC$, $\angle ABC = 90^\circ$ and $BA = BC = \sqrt{2}$. Points $P_1, P_2, \dots, P_{2024}$ lie on hypotenuse $\overline{AC}$ so that $AP_1 = P_1P_2 = P_2P_3 = \dots = P_{2023}P_{2024} = P_{2024}C$. What is the length of the vector sum

$$\overrightarrow{BP_1} + \overrightarrow{BP_2} + \overrightarrow{BP_3} + \dots + \overrightarrow{BP_{2024}}?$$

- (A) 1011 (B) 1012 (C) 2023 (D) 2024 (E) 2025

Solution

Problem 8

How many angles θ with $0 \leq \theta \leq 2\pi$ satisfy $\log(\sin(3\theta)) + \log(\cos(2\theta)) = 0$?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Solution

Problem 9

Let M be the greatest integer such that both $M + 1213$ and $M + 3773$ are perfect squares. What is the units digit of M ?

- (A) 1 (B) 2 (C) 3 (D) 6 (E) 8

Solution

Problem 10

Let α be the radian measure of the smallest angle in a $3-4-5$ right triangle. Let β be the radian measure of the smallest angle in a $7-24-25$ right triangle. In terms of α , what is β ?

- (A) $\frac{\alpha}{3}$ (B) $\alpha - \frac{\pi}{8}$ (C) $\frac{\pi}{2} - 2\alpha$ (D) $\frac{\alpha}{2}$ (E) $\pi - 4\alpha$

Solution

Problem 11

There are exactly K positive integers b with $5 \leq b \leq 2024$ such that the base- b integer 2024_b is divisible by 16 (where 16 is in base ten). What is the sum of the digits of K ?

- (A) 16 (B) 17 (C) 18 (D) 20 (E) 21

Solution

Problem 12

The first three terms of a geometric sequence are the integers a , 720, and b , where $a < 720 < b$. What is the sum of the digits of the least possible value of b ?

- (A) 9 (B) 12 (C) 16 (D) 18 (E) 21

Solution

Problem 13

The graph of $y = e^{x+1} + e^{-x} - 2$ has an axis of symmetry. What is the reflection of the point $(-1, \frac{1}{2})$ over this axis?

- (A) $(-1, -\frac{3}{2})$ (B) $(-1, 0)$ (C) $(-1, \frac{1}{2})$ (D) $(0, \frac{1}{2})$ (E) $(3, \frac{1}{2})$

Solution

Problem 14

The numbers, in order, of each row and the numbers, in order, of each column of a 5×5 array of integers form an arithmetic progression of length 5. The numbers in positions $(5, 5)$, $(2, 4)$, $(4, 3)$, and $(3, 1)$ are 0, 48, 16, and 12, respectively. What number is in position $(1, 2)$?

$$\begin{bmatrix} \cdot & ? & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & 48 & \cdot \\ 12 & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & 16 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & 0 \end{bmatrix}$$

- (A) 19 (B) 24 (C) 29 (D) 34 (E) 39

Solution

Problem 15

The roots of $x^3 + 2x^2 - x + 3$ are p , q , and r . What is the value of

$$(p^2 + 4)(q^2 + 4)(r^2 + 4)?$$

- (A) 64 (B) 75 (C) 100 (D) 125 (E) 144

Solution

Problem 16

A set of 12 tokens --- 3 red, 2 white, 1 blue, and 6 black --- is to be distributed at random to 3 game players, 4 tokens per player. The probability that some player gets all the red tokens, another gets all the white tokens, and the remaining player gets the blue token can be written as $\frac{m}{n}$, where m and n are relatively prime positive integers. What is $m + n$?

- (A) 387 (B) 388 (C) 389 (D) 390 (E) 391

Solution

Problem 17

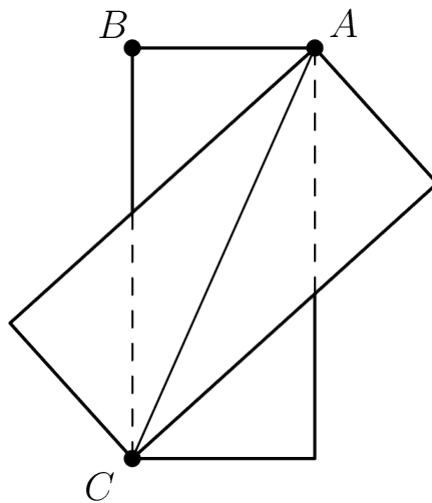
Integers a , b , and c satisfy $ab + c = 100$, $bc + a = 87$, and $ca + b = 60$. What is $ab + bc + ca$?

- (A) 212 (B) 247 (C) 258 (D) 276 (E) 284

Solution

Problem 18

On top of a rectangular card with sides of length 1 and $2 + \sqrt{3}$, an identical card is placed so that two of their diagonals line up, as shown ($\overline{AC}$, in this case).



Continue the process, adding a third card to the second, and so on, lining up successive diagonals after rotating clockwise. In total, how many cards must be used until a vertex of a new card lands exactly on the vertex labeled B in the figure?

- (A) 6 (B) 8 (C) 10 (D) 12 (E) No new vertex will land on B .

Solution

Problem 19

Cyclic quadrilateral $ABCD$ has lengths $BC = CD = 3$ and $DA = 5$ with $\angle CDA = 120^\circ$. What is the length of the shorter diagonal of $ABCD$?

- (A) $\frac{31}{7}$ (B) $\frac{33}{7}$ (C) 5 (D) $\frac{39}{7}$ (E) $\frac{41}{7}$

Solution

Problem 20

Points P and Q are chosen uniformly and independently at random on sides $\overline{AB}$ and $\overline{AC}$, respectively, of equilateral triangle $\triangle ABC$. Which of the following intervals contains the probability that the area of $\triangle APQ$ is less than half the area of $\triangle ABC$?

- (A) $\left[\frac{3}{8}, \frac{1}{2}\right]$ (B) $\left(\frac{1}{2}, \frac{2}{3}\right]$ (C) $\left(\frac{2}{3}, \frac{3}{4}\right]$ (D) $\left(\frac{3}{4}, \frac{7}{8}\right]$ (E) $\left(\frac{7}{8}, 1\right]$

Solution

Problem 21

Suppose that $a_1 = 2$ and the sequence (a_n) satisfies the recurrence relation

$$\frac{a_n - 1}{n - 1} = \frac{a_{n-1} + 1}{(n - 1) + 1}$$

for all $n \geq 2$. What is the greatest integer less than or equal to

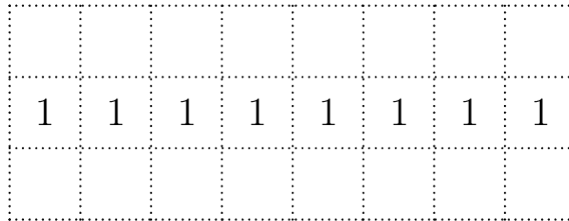
$$\sum_{n=1}^{100} a_n^2?$$

- (A) 338,550 (B) 338,551 (C) 338,552 (D) 338,553 (E) 338,554

Solution

Problem 22

The figure below shows a dotted grid 8 cells wide and 3 cells tall consisting of $1'' \times 1''$ squares. Carl places 1-inch toothpicks along some of the sides of the squares to create a closed loop that does not intersect itself. The numbers in the cells indicate the number of sides of that square that are to be covered by toothpicks, and any number of toothpicks are allowed if no number is written. In how many ways can Carl place the toothpicks?



- (A) 130 (B) 144 (C) 146 (D) 162 (E) 196

Solution

Problem 23

What is the value of

$$\tan^2 \frac{\pi}{16} \cdot \tan^2 \frac{3\pi}{16} + \tan^2 \frac{\pi}{16} \cdot \tan^2 \frac{5\pi}{16} + \tan^2 \frac{3\pi}{16} \cdot \tan^2 \frac{7\pi}{16} + \tan^2 \frac{5\pi}{16} \cdot \tan^2 \frac{7\pi}{16}?$$

- (A) 28 (B) 68 (C) 70 (D) 72 (E) 84

Solution

Problem 24

A *disphenoid* is a tetrahedron whose triangular faces are congruent to one another. What is the least total surface area of a disphenoid whose faces are scalene triangles with integer side lengths?

- (A) $\sqrt{3}$ (B) $3\sqrt{15}$ (C) 15 (D) $15\sqrt{7}$ (E) $24\sqrt{6}$

Solution

Problem 25

A graph is *symmetric* about a line if the graph remains unchanged after reflection in that line. For how many quadruples of integers (a, b, c, d) , where $|a|, |b|, |c|, |d| \leq 5$ and c and d are not both 0, is the graph of

$$y = \frac{ax + b}{cx + d}$$

symmetric about the line $y = x$?

- (A) 1282 (B) 1292 (C) 1310 (D) 1320 (E) 1330

Solution

See also

2024 AMC 12A (Problems • Answer Key • Resources (http://www.artofproblemsolving.com/community/c13))	
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1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25	
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- AMC 12
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Page

- Article
- Discussion
- View source
- History

Toolbox

- Recent changes
- Random page
- Help
- What links here
- Special pages

SEARCH

Search

2024 AMC 12A Answer Key

- 1. A
- 2. B
- 3. B
- 4. D
- 5. D
- 6. B
- 7. D
- 8. A
- 9. E
- 10. C
- 11. D
- 12. E
- 13. D
- 14. C
- 15. D
- 16. C
- 17. D
- 18. A
- 19. D
- 20. D
- 21. B
- 22. C
- 23. B
- 24. D
- 25. B

Retrieved from "https://artofproblemsolving.com/wiki/index.php?title=2024_AMC_12A_Answer_Key&oldid=231917"

Page

- Article
- Discussion
- View source
- History

Toolbox

- Recent changes
- Random page
- Help
- What links here
- Special pages

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Search

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1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18

- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See also

Problem 1

In a long line of people arranged left to right, the 1013th person from the left is also the 1010th person from the right. How many people are in the line?

- (A) 2021 (B) 2022 (C) 2023 (D) 2024 (E) 2025

Solution

Problem 2

What is $10! - 7! \cdot 6!$?

- (A) -120 (B) 0 (C) 120 (D) 600 (E) 720

Solution

Problem 3

For how many integer values of x is $|2x| \leq 7\pi$?

- (A) 16 (B) 17 (C) 19 (D) 20 (E) 21

Solution

Problem 4

Balls numbered $1, 2, 3, \dots$ are deposited in 5 bins, labeled A, B, C, D , and E , using the following procedure. Ball 1 is deposited in bin A , and balls 2 and 3 are deposited in B . The next three balls are deposited in bin C , the next 4 in bin D , and so on, cycling back to bin A after balls are deposited in bin E . (For example, 22, 23, $\dots$, 28 are deposited in bin B at step 7 of this process.) In which bin is ball 2024 deposited?

- (A) A (B) B (C) C (D) D (E) E

Solution

Problem 5

In the following expression, Melanie changed some of the plus signs to minus signs:

$$1 + 3 + 5 + 7 + \cdots + 97 + 99$$

When the new expression was evaluated, it was negative. What is the least number of plus signs that Melanie could have changed to minus signs?

- (A) 14 (B) 15 (C) 16 (D) 17 (E) 18

Solution

Problem 6

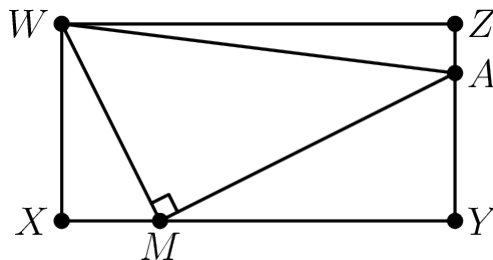
The national debt of the United States is on track to reach $5 \cdot 10^{13}$ dollars by 2033. How many digits does this number of dollars have when written as a numeral in base 5? (The approximation of $\log_{10} 5$ as 0.7 is sufficient for this problem.)

- (A) 18 (B) 20 (C) 22 (D) 24 (E) 26

Solution

Problem 7

In the figure below $WXYZ$ is a rectangle with $WX = 4$ and $WZ = 8$. Point M lies on $\overline{XY}$, point A lies on $\overline{YZ}$, and $\angle WMA$ is a right angle. The areas of $\triangle WXM$ and $\triangle WAZ$ are equal. What is the area of $\triangle WMA$?



- (A) 13 (B) 14 (C) 15 (D) 16 (E) 17

Solution

Problem 8

What value of x satisfies

$$\frac{\log_2 x \cdot \log_3 x}{\log_2 x + \log_3 x} = 2?$$

- (A) 25 (B) 32 (C) 36 (D) 42 (E) 48

Solution

Problem 9

A dartboard is the region B in the coordinate plane consisting of points (x, y) such that $|x| + |y| \leq 8$. A target T is the region where $(x^2 + y^2 - 25)^2 \leq 49$. A dart is thrown and lands at a random point in B . The probability that the dart lands in T can be expressed as $\frac{m}{n} \cdot \pi$, where m and n are relatively prime positive integers. What is $m + n$?

- (A) 39 (B) 71 (C) 73 (D) 75 (E) 135

Solution

Problem 10

A list of 9 real numbers consists of 1, 2, 2, 3, 2, 5, 2, 6, 2, and 7, as well as x, y, z with $x \leq y \leq z$. The range of the list is 7, and the mean and median are both positive integers. How many ordered triples (x, y, z) are possible?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) infinitely many

Solution

Problem 11

Let $x_n = \sin^2(n^\circ)$. What is the mean of $x_1, x_2, x_3, \dots, x_{90}$?

- (A) $\frac{11}{45}$ (B) $\frac{22}{45}$ (C) $\frac{89}{180}$ (D) $\frac{1}{2}$ (E) $\frac{91}{180}$

Solution

Problem 12

Suppose z is a complex number with positive imaginary part, with real part greater than 1, and with $|z| = 2$. In the complex plane, the four points 0, z , z^2 , and z^3 are the vertices of a quadrilateral with area 15. What is the imaginary part of z ?

- (A) $\frac{3}{4}$ (B) 1 (C) $\frac{4}{3}$ (D) $\frac{3}{2}$ (E) $\frac{5}{3}$

Solution

Problem 13

There are real numbers x, y, h and k that satisfy the system of equations

$$x^2 + y^2 - 6x - 8y = h$$

$$x^2 + y^2 - 10x + 4y = k$$

What is the minimum possible value of $h + k$?

- (A) -54 (B) -46 (C) -34 (D) -16 (E) 16

Solution

Problem 14

How many different remainders can result when the 100th power of an integer is divided by 125?

- (A) 1 (B) 2 (C) 5 (D) 25 (E) 125

Solution

Problem 15

A triangle in the coordinate plane has vertices $A(\log_2 1, \log_2 2)$, $B(\log_2 3, \log_2 4)$, and $C(\log_2 7, \log_2 8)$. What is the area of $\triangle ABC$?

- (A) $\log_2 \frac{\sqrt{3}}{7}$ (B) $\log_2 \frac{3}{\sqrt{7}}$ (C) $\log_2 \frac{7}{\sqrt{3}}$ (D) $\log_2 \frac{11}{\sqrt{7}}$ (E) $\log_2 \frac{11}{\sqrt{3}}$

Solution

Problem 16

A group of 16 people will be partitioned into 4 indistinguishable 4-person committees. Each committee will have one chairperson and one secretary. The number of different ways to make these assignments can be written as $3^r M$, where r and M are positive integers and M is not divisible by 3. What is r ?

- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

Solution

Problem 17

Integers a and b are randomly chosen without replacement from the set of integers with absolute value not exceeding 10. What is the probability that the polynomial $x^3 + ax^2 + bx + 6$ has 3 distinct integer roots?

- (A) $\frac{1}{240}$ (B) $\frac{1}{221}$ (C) $\frac{1}{105}$ (D) $\frac{1}{84}$ (E) $\frac{1}{63}$

Problem 18

The Fibonacci numbers are defined by $F_1 = 1$, $F_2 = 1$, and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 3$. What is

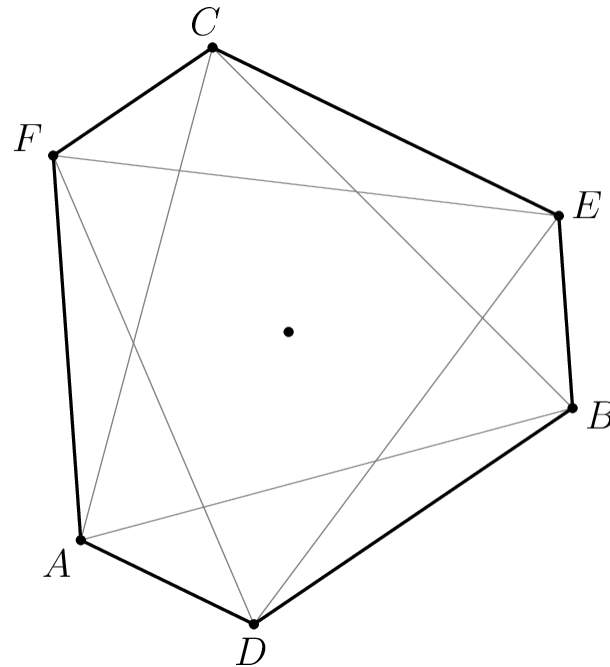
$$\frac{F_2}{F_1} + \frac{F_4}{F_2} + \frac{F_6}{F_3} + \cdots + \frac{F_{20}}{F_{10}}?$$

- (A) 318 (B) 319 (C) 320 (D) 321 (E) 322

Solution

Problem 19

Equilateral $\triangle ABC$ with side length 14 is rotated about its center by angle θ , where $0 < \theta < 60^\circ$, to form $\triangle DEF$. See the figure. The area of hexagon $ADBECF$ is $91\sqrt{3}$. What is $\tan \theta$?



- (A) $\frac{3}{4}$ (B) $\frac{5\sqrt{3}}{11}$ (C) $\frac{4}{5}$ (D) $\frac{11}{13}$ (E) $\frac{7\sqrt{3}}{13}$

Solution

Problem 20

Suppose A , B , and C are points in the plane with $AB = 40$ and $AC = 42$, and let x be the length of the line segment from A to the midpoint of $\overline{BC}$. Define a function f by letting $f(x)$ be the area of $\triangle ABC$. Then the domain of f is an open interval (p, q) and the maximum value r of $f(x)$ occurs at $x = s$. What is $p + q + r + s$?

- (A) 909 (B) 910 (C) 911 (D) 912 (E) 913

Solution

Problem 21

The measures of the smallest angles of three different right triangles sum to 90° . All three triangles have side lengths that are primitive Pythagorean triples. Two of them are $3 - 4 - 5$ and $5 - 12 - 13$. What is the perimeter of the third triangle?

- (A) 40 (B) 126 (C) 154 (D) 176 (E) 208

Solution

Problem 22

Let $\triangle ABC$ be a triangle with integer side lengths and the property that $\angle B = 2\angle A$. What is the least possible perimeter of such a triangle?

- (A) 13 (B) 14 (C) 15 (D) 16 (E) 17

Solution

Problem 23

A right pyramid has regular octagon $ABCDEFGH$ with side length 1 as its base and apex V . Segments $\overline{AV}$ and $\overline{DV}$ are perpendicular. What is the square of the height of the pyramid?

- (A) 1 (B) $\frac{1 + \sqrt{2}}{2}$ (C) $\sqrt{2}$ (D) $\frac{3}{2}$ (E) $\frac{2 + \sqrt{2}}{3}$

Solution

Problem 24

What is the number of ordered triples (a, b, c) of positive integers, with $a \leq b \leq c \leq 9$, such that there exists a (non-degenerate) triangle $\triangle ABC$ with an integer inradius for which a , b , and c are the lengths of the altitudes from A to $\overline{BC}$, B to $\overline{AC}$, and C to $\overline{AB}$, respectively? (Recall that the inradius of a triangle is the radius of the largest possible circle that can be inscribed in the triangle.)

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Problem 25

Pablo will decorate each of 6 identical white balls with either a striped or a dotted pattern, using either red or blue paint. He will decide on the color and pattern for each ball by flipping a fair coin for each of the 12 decisions he must make. After the paint dries, he will place the 6 balls in an urn. Frida will randomly select one ball from the urn and note its color and pattern. The events "the ball Frida selects is red" and "the ball Frida selects is striped" may or may not be independent, depending on the outcome of Pablo's coin flips. The probability that these two events are independent can be written as $\frac{m}{n}$, where m and n are relatively prime positive integers. What is m ? (Recall that two events A and B are independent if $P(A \text{ and } B) = P(A) \cdot P(B)$.)

- (A) 243 (B) 245 (C) 247 (D) 249 (E) 251

Solution

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Page

- Article
- Discussion
- View source
- History

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- Recent changes
- Random page
- Help
- What links here
- Special pages

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Search

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- 1. B
- 2. B
- 3. E
- 4. D
- 5. B
- 6. B
- 7. C
- 8. C
- 9. B
- 10. C
- 11. E
- 12. D
- 13. C
- 14. B
- 15. B
- 16. A
- 17. C
- 18. B
- 19. B
- 20. C
- 21. C
- 22. C
- 23. B
- 24. B
- 25. A

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4. Figures are not necessarily drawn to scale.
5. You will have **75 minutes** working time to complete the test.

1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17
• 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 25.1 Note:
- 26 See also

Problem 1 [\[edit\]](#)

Andy and Betsy both live in Mathville. Andy leaves Mathville on his bicycle at 1:30, traveling due north at a steady 8 miles per hour. Betsy leaves on her bicycle from the same point at 2:30, traveling due east at a steady 12 miles per hour. At what time will they be exactly the same distance from their common starting point?

- (A) 3:30 (B) 3:45 (C) 4:00 (D) 4:15 (E) 4:30

[Solution](#)

Problem 2 [\[edit\]](#)

A box contains 10 pounds of a nut mix that is 50 percent peanuts, 20 percent cashews, and 30 percent almonds. A second nut mix containing 20 percent peanuts, 40 percent cashews, and 40 percent almonds is added to the box resulting in a new nut mix that is 40 percent peanuts. How many pounds of cashews are now in the box?

- (A) 3.5 (B) 4 (C) 4.5 (D) 5 (E) 6

[Solution](#)

Problem 3 [\[edit\]](#)

A team of students is going to compete against a team of teachers in a trivia contest. The total number of students and teachers is 15. Ash, a cousin of one of the students, wants to join the contest. If Ash plays with the students, the average age on that team will increase from 12 to 14. If Ash plays with the teachers, the average age on that team will decrease from 55 to 52. How old is Ash?

- (A) 28 (B) 29 (C) 30 (D) 32 (E) 33

[Solution](#)

Problem 4 [\[edit\]](#)

Agnes writes the following four statements on a blank piece of paper.

- At least one of these statements is true.
- At least two of these statements are true.
- At least two of these statements are false.
- At least one of these statements is false.

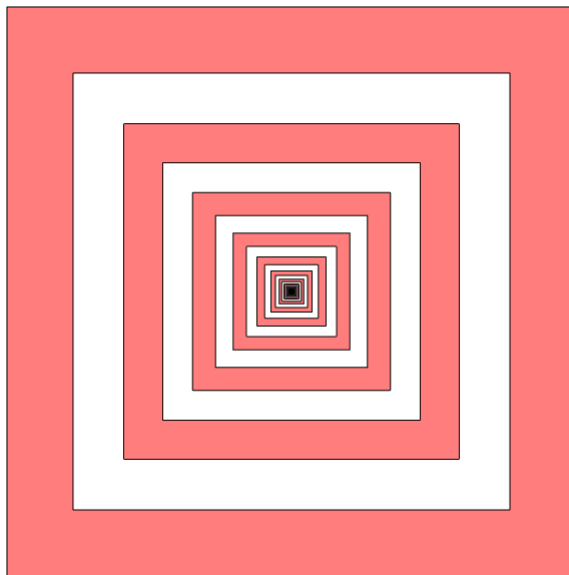
Each statement is either true or false. How many false statements did Agnes write on the paper?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

[Solution](#)

Problem 5 [\[edit\]](#)

In the figure below, the outside square contains infinitely many squares, each of them with the same center and sides parallel to the outside square. The ratio of the side length of a square to the side length of the next inner square is k , where $0 < k < 1$. The spaces between squares are alternately shaded as shown in the figure (which is not necessarily drawn to scale).



The area of the shaded portion of the figure is 64% of the area of the original square. What is k ?

- (A) $\frac{3}{5}$ (B) $\frac{16}{25}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$ (E) $\frac{4}{5}$

[Solution](#)

Problem 6 [\[edit\]](#)

Six chairs are arranged around a round table. Two students and two teachers randomly select four of the chairs to sit in. What is the probability that the two students will sit in two adjacent chairs and the two teachers will also sit in two adjacent chairs?

- (A) $\frac{1}{6}$ (B) $\frac{1}{5}$ (C) $\frac{2}{9}$ (D) $\frac{3}{13}$ (E) $\frac{1}{4}$

[Solution](#)

Problem 7 [\[edit\]](#)

In a certain alien world, the maximum running speed v of an organism is dependent on its number of toes n and number of eyes m . The relationship can be expressed as $v = kn^am^b$ centimeters per hour, where k , a , and b are integer constants. In a population where all organisms have 5 toes, $\log v = 4 + 2 \log m$; and in a population where all organisms have 25 eyes, $\log v = 4 + 4 \log n$, where the logarithms are base 10. What is $k + a + b$?

- (A) 20 (B) 21 (C) 22 (D) 23 (E) 24

[Solution](#)

Problem 8 [\[edit\]](#)

Pentagon $ABCDE$ is inscribed in a circle, and $\angle BEC = \angle CED = 30^\circ$. Let line AC and line BD intersect at point F , and suppose that $AB = 9$ and $AD = 24$. What is BF ?

- (A) $\frac{57}{11}$ (B) $\frac{59}{11}$ (C) $\frac{60}{11}$ (D) $\frac{61}{11}$ (E) $\frac{63}{11}$

[Solution](#)

Problem 9 [edit]

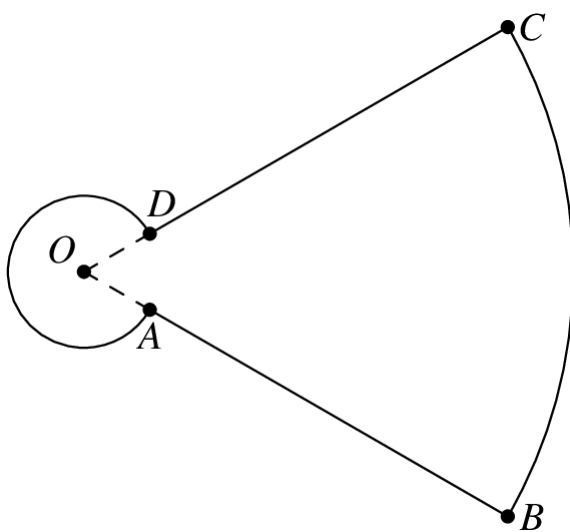
Let w be the complex number $2 + i$, where $i = \sqrt{-1}$. What real number r has the property that r , w , and w^2 are three collinear points in the complex plane?

- (A) $\frac{3}{4}$ (B) 1 (C) $\frac{7}{5}$ (D) $\frac{3}{2}$ (E) $\frac{5}{3}$

[Solution](#)

Problem 10 [edit]

In the figure shown below, major arc $\widehat{AD}$ and minor arc $\widehat{BC}$ have the same center, O . Also, A lies between O and B , and D lies between O and C . Major arc $\widehat{AD}$, minor arc $\widehat{BC}$, and each of the two segments $\overline{AB}$ and $\overline{CD}$ have length 2π . What is the distance from O to A ?



- (A) 1 (B) $1 - \pi + \sqrt{\pi^2 + 1}$ (C) $\frac{\pi}{2}$ (D) $\frac{\sqrt{\pi^2 + 1}}{2}$ (E) 2

[Solution](#)

Problem 11 [edit]

The orthocenter of a triangle is the concurrent intersection of the three (possibly extended) altitudes. What is the sum of the coordinates of the orthocenter of the triangle whose vertices are $A(2, 31)$, $B(8, 27)$, and $C(18, 27)$?

- (A) 5 (B) 17 (C) $10 + 4\sqrt{17} + 2\sqrt{13}$ (D) $\frac{113}{3}$ (E) 54

[Solution](#)

Problem 12 [edit]

The *harmonic mean* of a collection of numbers is the reciprocal of the arithmetic mean of the reciprocals of the numbers in the collection. For example, the harmonic mean of 4, 4, and 5 is

$$\frac{1}{\frac{1}{3}(\frac{1}{4} + \frac{1}{4} + \frac{1}{5})} = \frac{30}{7}.$$

What is the harmonic mean of all the real roots of the 4050th degree polynomial

$$\prod_{k=1}^{2025} (kx^2 - 4x - 3) = (x^2 - 4x - 3)(2x^2 - 4x - 3)(3x^2 - 4x - 3) \dots (2025x^2 - 4x - 3)?$$

- (A) $-\frac{5}{3}$ (B) $-\frac{3}{2}$ (C) $-\frac{6}{5}$ (D) $-\frac{5}{6}$ (E) $-\frac{2}{3}$

[Solution](#)

Problem 13 [\[edit\]](#)

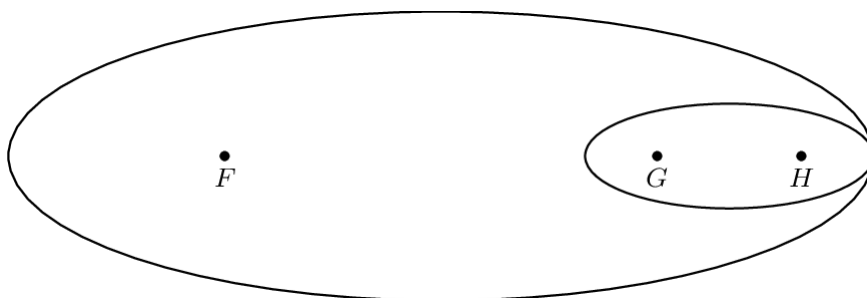
Let $C = \{1, 2, 3, \dots, 13\}$. Let N be the greatest integer such that there exists a subset of C with N elements that does not contain five consecutive integers. Suppose N integers are chosen at random from C without replacement. What is the probability that the chosen elements do not include five consecutive integers?

- (A) $\frac{3}{130}$ (B) $\frac{3}{143}$ (C) $\frac{5}{143}$ (D) $\frac{1}{26}$ (E) $\frac{5}{78}$

[Solution](#)

Problem 14 [\[edit\]](#)

Points F , G , and H are collinear with G between F and H . The ellipse with foci at G and H is internally tangent to the ellipse with foci at F and G , as shown below.



The two ellipses have the same eccentricity e , and the ratio of their areas is 2025. (Recall that the eccentricity of an ellipse is $e = \frac{c}{a}$, where c is the distance from the center to a focus, and $2a$ is the length of the major axis.) What is e ?

- (A) $\frac{3}{5}$ (B) $\frac{16}{25}$ (C) $\frac{4}{5}$ (D) $\frac{22}{23}$ (E) $\frac{44}{45}$

[Solution](#)

Problem 15 [\[edit\]](#)

A set of numbers is called *sum-free* if whenever x and y are (not necessarily distinct) elements of the set, $x + y$ is not an element of the set. For example, $\{1, 4, 6\}$ and the empty set are sum-free, but $\{2, 4, 5\}$ is not. What is the greatest possible number of elements in a sum-free subset of $\{1, 2, 3, \dots, 20\}$?

- (A) 8 (B) 9 (C) 10 (D) 11 (E) 12

[Solution](#)**Problem 16** [\[edit\]](#)

Triangle $\triangle ABC$ has side lengths $AB = 80$, $BC = 45$, and $AC = 75$. The bisector $\angle B$ and the altitude to side $\overline{AB}$ intersect at point P . What is BP ?

- (A) 18 (B) 19 (C) 20 (D) 21 (E) 22

[Solution](#)**Problem 17** [\[edit\]](#)

The polynomial $(z + i)(z + 2i)(z + 3i) + 10$ has three roots in the complex plane, where $i = \sqrt{-1}$. What is the area of the triangle formed by these three roots?

- (A) 6 (B) 8 (C) 10 (D) 12 (E) 14

[Solution](#)**Problem 18** [\[edit\]](#)

How many ordered triples (x, y, z) of different positive integers less than or equal to 8 satisfy $xy > z$, $xz > y$, and $yz > x$?

- (A) 36 (B) 84 (C) 186 (D) 336 (E) 486

[Solution](#)**Problem 19** [\[edit\]](#)

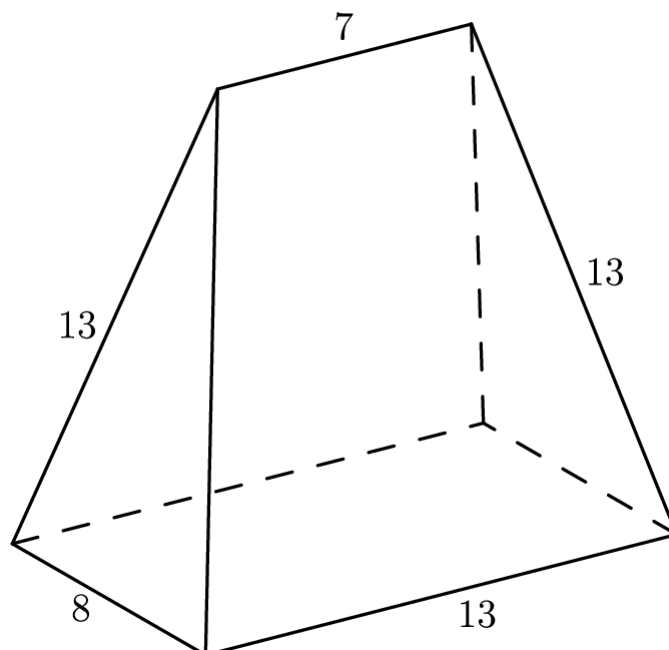
Let a , b , and c be the roots of the polynomial $x^3 + kx + 1$. What is the sum

$$a^3b^2 + a^2b^3 + b^3c^2 + b^2c^3 + c^3a^2 + c^2a^3?$$

- (A) $-k$ (B) $-k + 1$ (C) 1 (D) $k - 1$ (E) k

[Solution](#)**Problem 20** [\[edit\]](#)

The base of the pentahedron shown below is a 13×8 rectangle, and its lateral faces are two isosceles triangles with base of length 8 and congruent sides of length 13, and two isosceles trapezoids with bases of length 7 and 13 and nonparallel sides of length 13.



What is the volume of the pentahedron?

- (A) 416 (B) 520 (C) 528 (D) 676 (E) 832

[Solution](#)

Problem 21 [\[edit\]](#)

There is a unique ordered triple (a, k, m) of nonnegative integers such that

$$\frac{4^a + 4^{a+k} + 4^{a+2k} + \dots + 4^{a+mk}}{2^a + 2^{a+k} + 2^{a+2k} + \dots + 2^{a+mk}} = 964.$$

What is $a + k + m$?

- (A) 8 (B) 9 (C) 10 (D) 11 (E) 12

[Solution](#)

Problem 22 [\[edit\]](#)

Three real numbers are chosen independently and uniformly at random between 0 and 1. What is the probability that the greatest of these three numbers is greater than 2 times each of the other two numbers? (In other words, if the chosen numbers are $a \geq b \geq c$, then $a > 2b$.)

- (A) $\frac{1}{12}$ (B) $\frac{1}{9}$ (C) $\frac{1}{8}$ (D) $\frac{1}{6}$ (E) $\frac{1}{4}$

[Solution](#)

Problem 23 [\[edit\]](#)

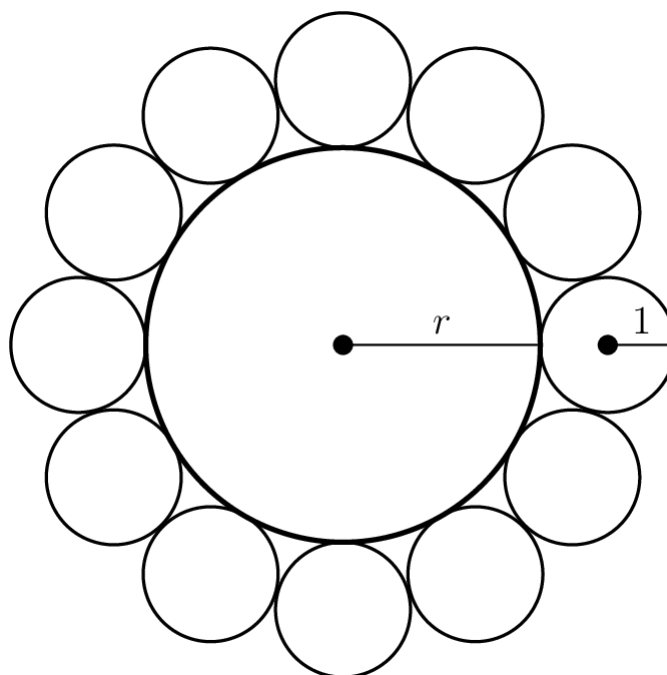
Call a positive integer fair if no digit is used more than once, it has no 0s, and no digit is adjacent to two greater digits. For example, 196, 23, and 12463 are fair, but 1546, 320 and 34321 are not. How many fair positive integers are there?

- (A) 511 (B) 2584 (C) 9841 (D) 17711 (E) 19682

[Solution](#)

Problem 24 [edit]

A circle of radius r is surrounded by 12 circles of radius 1, externally tangent to the central circle and sequentially tangent to each other, as shown. Then r can be written as $\sqrt{a} + \sqrt{b} + c$, where a, b, c are integers. What is $a + b + c$?



- (A) 3 (B) 5 (C) 7 (D) 9 (E) 11

Solution

Problem 25 [edit]

Polynomials $P(x)$ and $Q(x)$ each have degree 3 and leading coefficient 1, and their roots are all elements of $\{1, 2, 3, 4, 5\}$. The function $f(x) = \frac{P(x)}{Q(x)}$ has the property that there exist real numbers $a < b < c < d$ such that the set of all real numbers x such that $f(x) \leq 0$ consists of the closed interval $[a, b]$ together with the open interval (c, d) . How many ordered pairs of polynomials (P, Q) are possible?

(A) 9 (B) 11 (C) 12 (D) 13 (E) 8

Note: [edit]

The original problem asked for the number of functions $f(x)$, which resulted the problem having an answer not present in the answer choices. I have added answer F to this problem, but it was not there in the actual contest.

To make sure there should be only 5, so I removed an answer choice A which... seemed reasonable to guessers

Solution

See also [edit]

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1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17 • 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

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1. E
2. B
3. A
4. B
5. D
6. B
7. C
8. E
9. E
10. B
11. A
12. B
13. D
14. D
15. C
16. D
17. A
18. C
19. E
20. C
21. A
22. E
23. C
24. C
25. None of the official answer choices were correct.



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5. You will have **75 minutes** working time to complete the test.

1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10 • 11 • 12 • 13 • 14 • 15 • 16 • 17
• 18 • 19 • 20 • 21 • 22 • 23 • 24 • 25

Contents

- 1 Problem 1
- 2 Problem 2
- 3 Problem 3
- 4 Problem 4
- 5 Problem 5
- 6 Problem 6
- 7 Problem 7
- 8 Problem 8
- 9 Problem 9
- 10 Problem 10
- 11 Problem 11
- 12 Problem 12
- 13 Problem 13
- 14 Problem 14
- 15 Problem 15
- 16 Problem 16
- 17 Problem 17
- 18 Problem 18
- 19 Problem 19
- 20 Problem 20
- 21 Problem 21
- 22 Problem 22
- 23 Problem 23
- 24 Problem 24
- 25 Problem 25
- 26 See Also

Problem 1 [\[edit\]](#)

The instructions on a 350-gram bag of coffee beans say that proper brewing of a large mug of pour-over coffee requires 20 grams of coffee beans. What is the greatest number of properly brewed large mugs of coffee that can be made from the coffee beans in that bag?

- (A) 16 (B) 17 (C) 18 (D) 19 (E) 20

[Solution](#)

Problem 2 [\[edit\]](#)

Jerry wrote down the ones digit of each of the first 2025 positive squares: 1, 4, 9, 6, 5, 6, What is the sum of all the numbers Jerry wrote down?

- (A) 9025 (B) 9070 (C) 9090 (D) 9115 (E) 9160

[Solution](#)

Problem 3 [\[edit\]](#)

What is the value of $i(i-1)(i-2)(i-3)$, where $i = \sqrt{-1}$?

- (A) $6 - 5i$ (B) $-10i$ (C) $10i$ (D) -10 (E) 10

[Solution](#)

Problem 4 [\[edit\]](#)

The value of the two-digit number $\underline{a}\underline{b}$ in base seven equals the value of the two-digit number $\underline{b}\underline{a}$ in base nine. What is $a + b$?

- (A) 7 (B) 9 (C) 10 (D) 11 (E) 14

[Solution](#)

Problem 5 [\[edit\]](#)

Positive integers x and y satisfy the equation $57x + 22y = 400$. What is the least possible value of $x + y$?

- (A) 10 (B) 11 (C) 13 (D) 14 (E) 15

[Solution](#)

Problem 6 [\[edit\]](#)

Emmy says to Max, "I ordered 36 math club sweatshirts today." Max asks, "How much did each shirt cost?" Emmy responds, "I'll give you a hint. The total cost was $\$ \underline{A} \underline{B} \underline{B} . \underline{B} \underline{A}$, where A and B are digits and $A \neq 0$." After a pause, Max says, "That was a good price." What is $A + B$?

- (A) 7 (B) 8 (C) 11 (D) 14 (E) 15

[Solution](#)

Problem 7 [edit]

What is the value of

$$\sum_{n=2}^{255} \frac{\log_2(1 + \frac{1}{n})}{(\log_2 n)(\log_2(n+1))}$$

?

- (A) $\frac{3}{4}$ (B) $1 - \frac{1}{\log_2 255}$ (C) $\frac{7}{8}$ (D) $\frac{15}{16}$ (E) 1

Solution

Problem 8 [edit]

There are integers a and b such that the polynomial $x^3 - 5x^2 + ax + b$ has $4 + \sqrt{5}$ as a root. What is $a + b$?

- (A) 13 (B) 17 (C) 20 (D) 30 (E) 68

Solution

Problem 9 [edit]

What is the tens digit of 6^{6^6} ?

- (A) 1 (B) 3 (C) 5 (D) 7 (E) 9

Solution

Problem 10 [edit]

The altitude to the hypotenuse of a $30-60-90^\circ$ is divided into two segments of lengths $x < y$ by the median to the shortest side of the triangle. What is the ratio $\frac{x}{x+y}$?

- (A) $\frac{3}{7}$ (B) $\frac{\sqrt{3}}{4}$ (C) $\frac{4}{9}$ (D) $\frac{5}{11}$ (E) $\frac{4\sqrt{3}}{15}$

Solution

Problem 11 [edit]

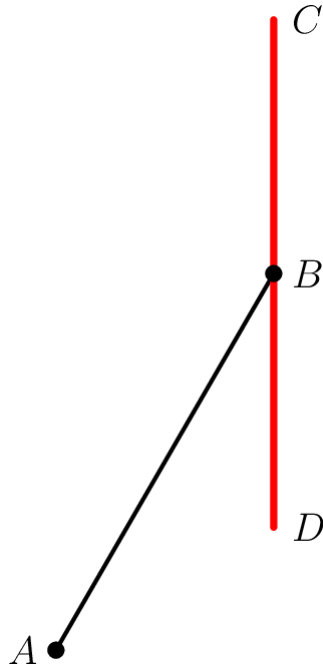
Nine athletes, no two of whom are the same height, try out for the basketball team. One at a time, they draw a wristband at random, without replacement, from a bag containing 3 blue bands, 3 red bands, and 3 green bands. They are divided into a blue group, a red group, and a green group. The tallest member of each group is named the group captain. What is the probability that the group captains are the three tallest athletes?

- (A) $\frac{2}{9}$ (B) $\frac{2}{7}$ (C) $\frac{9}{28}$ (D) $\frac{1}{3}$ (E) $\frac{3}{8}$

Solution

Problem 12 [edit]

The windshield wiper on the driver's side of a large bus is depicted below.



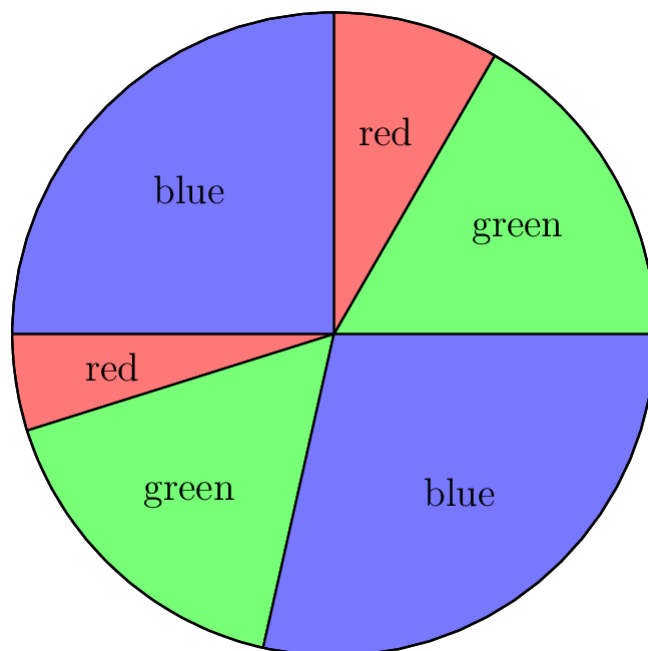
Arm $\overline{AB}$ pivots back and forth around point A , sweeping out an arc of 60° , symmetric about the vertical line through A . The wiper blade $\overline{CD}$ is attached to B at its midpoint and stays vertical as the arm moves. The arm is 3 feet long, and the wiper blade is 3.5 feet tall. What is the area of the windshield cleaned by the wiper, in square feet, to the nearest hundredth? (Assume that the windshield is a flat vertical surface.)

- (A) 9.68 (B) 10.14 (C) 10.50 (D) 11.32 (E) 12.00

[Solution](#)

Problem 13 [edit]

A circle has been divided into 6 sectors of different sizes. Then 2 of the sectors are painted red, 2 painted green, and 2 painted blue so that no two neighboring sectors are painted the same color. One such coloring is shown below.



How many different colorings are possible?

(A) 12 (B) 16 (C) 18 (D) 24 (E) 28

[Solution](#)

Problem 14 [\[edit\]](#)

Consider a decreasing sequence of n positive integers $x_1 > x_2 > \cdots > x_n$ that satisfies the following conditions:

- The average of the first 3 terms in the sequence is 2025.
- For all $4 \leq k \leq n$, the average of the first k terms is 1 less than the average of the first $k - 1$ terms.

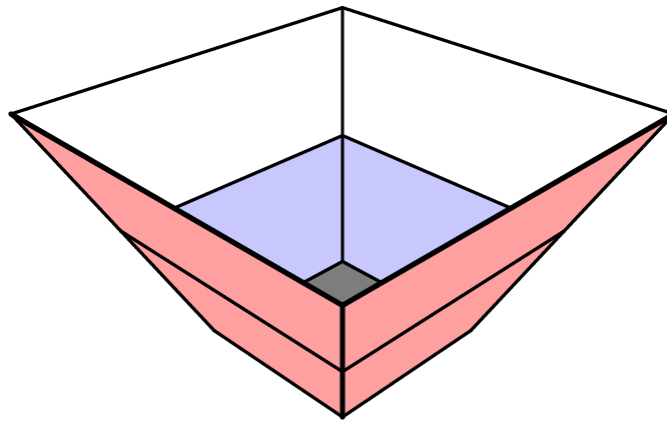
What is the greatest possible value of n ?

(A) 1013 (B) 1014 (C) 1016 (D) 2016 (E) 2025

[Solution](#)

Problem 15 [\[edit\]](#)

A container has a 1×1 square bottom, a 3×3 open square top, and four congruent trapezoidal sides, as shown. Starting when the container is empty, a hose that runs water at a constant rate takes 35 minutes to fill the container up to the midline of the trapezoids.



How many more minutes will it take to fill the remainder of the container?

(A) 70 (B) 85 (C) 90 (D) 95 (E) 105

[Solution](#)

Problem 16 [\[edit\]](#)

An analog clock starts at midnight and runs for 2025 minutes before stopping. What is the tangent of the acute angle between the hour hand and the minute hand when the clock stops?

(A) 0 (B) $\sqrt{2} - 1$ (C) $2 - \sqrt{2}$ (D) $\frac{\sqrt{2}}{2}$ (E) $3 - \sqrt{2}$

[Solution](#)

Problem 17 [\[edit\]](#)

Each of the 9 squares in a 3×3 grid is to be colored red, blue, or yellow in such a way that each red square shares an edge with at least one blue square, each blue square shares an edge with at least one yellow square, and each yellow square shares

an edge with at least one red square. Colorings that can be obtained from one another by rotations and/or reflections are considered the same. How many different colorings are possible?

- (A) 3 (B) 9 (C) 12 (D) 18 (E) 27

[Solution](#)

Problem 18 [\[edit\]](#)

Awnik repeatedly plays a game that has a probability of winning of $\frac{1}{3}$. The outcomes of the games are independent. What is the expected value of the number of games he will play until he has both won and lost at least once?

- (A) $\frac{5}{2}$ (B) 3 (C) $\frac{16}{5}$ (D) $\frac{7}{2}$ (E) $\frac{15}{4}$

[Solution](#)

Problem 19 [\[edit\]](#)

A rectangular grid of squares has 141 rows and 91 columns. Each square has room for two numbers. Horace and Vera each fill in the grid by putting the numbers from 1 through $141 \times 91 = 12,831$ into the squares. Horace fills the grid horizontally: he puts 1 through 91 in order from left to right into row 1, puts 92 through 182 into row 2 in order from left to right, and continues similarly through row 141. Vera fills the grid vertically: she puts 1 through 141 in order from top to bottom into column 1, then 142 through 282 into column 2 in order from top to bottom, and continues similarly through column 91. How many squares get two copies of the same number?

- (A) 7 (B) 10 (C) 11 (D) 12 (E) 19

[Solution](#)

Problem 20 [\[edit\]](#)

A frog hops along the number line according to the following rules:

- It starts at 0.
- If it is at 0, then it moves to 1 with probability $\frac{1}{2}$ and disappears with probability $\frac{1}{2}$.
- For $n = 1, 2, 3$, if it is at n , then it moves to $n + 1$ with probability $\frac{1}{4}$, to $n - 1$ with probability $\frac{1}{4}$, and disappears with probability $\frac{1}{2}$.

What is the probability that the frog reaches 4?

- (A) $\frac{1}{101}$ (B) $\frac{1}{100}$ (C) $\frac{1}{99}$ (D) $\frac{1}{98}$ (E) $\frac{1}{97}$

[Solution](#)

Problem 21 [\[edit\]](#)

Two non-congruent triangles have the same area. Each triangle has sides of length 8 and 9, and the third side of each triangle has integer length. What is the sum of the lengths of the third sides?

- (A) 20 (B) 22 (C) 24 (D) 26 (E) 28

[Solution](#)

Problem 22 [\[edit\]](#)

What is the greatest possible area of the triangle in the complex plane with vertices $2z$, $(1 + i)z$, and $(1 - i)z$, where z is a complex number satisfying $|4z - 2| = 1$?

- (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{9}{16}$ (D) $\frac{3}{4}$ (E) 1

[Solution](#)

Problem 23 [\[edit\]](#)

Let S be the set of all integers $z > 1$ such that for all pairs of nonnegative integers (x, y) with $x < y < z$, the remainder when $2025x$ is divided by z is less than the remainder when $2025y$ is divided by z . What is the sum of the elements of S ?

- (A) 3041 (B) 3542 (C) 3750 (D) 4044 (E) 4319

[Solution](#)

Problem 24 [\[edit\]](#)

How many real numbers satisfy the equation $\sin(20\pi x) = \log_{20}(x)$?

- (A) 199 (B) 200 (C) 398 (D) 399 (E) 400

[Solution](#)

Problem 25 [\[edit\]](#)

Three concentric circles have radii 1, 2, 3. An equilateral triangle with side length s has one vertex on each circle. What is s^2 ?

- (A) 6 (B) $\frac{25}{4}$ (C) $\frac{13}{2}$ (D) $\frac{27}{4}$ (E) 7

[Solution](#)

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[1](#) • [2](#) • [3](#) • [4](#) • [5](#) • [6](#) • [7](#) • [8](#) • [9](#) • [10](#) • [11](#) • [12](#) • [13](#) • [14](#) • [15](#) • [16](#) • [17](#) • [18](#) • [19](#) • [20](#) • [21](#) • [22](#) • [23](#) • [24](#) • [25](#)

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1. B
2. D
3. D
4. A
5. E
6. C
7. C
8. C
9. C
10. A
11. C
12. C
13. D
14. B
15. D
16. B
17. C
18. D
19. C
20. E
21. C
22. C
23. E
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